

SPIRALCORE SPECIFICATION

OBSERVER NOTES & SYSTEM ANALOGIES

(Formalized Geometric Translation & Physical Mapping)

Cleaned and formalized revision - v14.1

Document class	Observer-note translation contract	Normative scope	Mixed: tagged per statement
Architecture	SpiralCore v14	Runtime status	Not an independent runtime stage
Default profile	DIM=81; FBS Mode A	Revision	v14.1
Primary purpose	Declare assumptions and map analogies to deterministic surrogates	Conflict rule	Module-specific specs control

NORMATIVE STATUS

This section records the Observer's personal conceptual notes, architectural assumptions, and translation analogies. The analogies are not claims that SpiralCore literally instantiates the referenced physical systems. Runtime behavior is normative only where a statement is explicitly tagged [DEF], [POL], or [REQ], or where this section delegates to an authoritative module specification. A change in analogy does not invalidate the architecture if the implemented surrogate, interface, and tested invariant remain unchanged.

1. Statement Classes and Reading Rules

Every substantive statement in this section belongs to one of the following classes. Untagged explanatory prose is treated as [ON] Observer Note by default.

Tag	Class	Meaning	Normative force
[ON]	Observer Note	Personal conceptual framing used to explain why an architectural choice was made.	Non-normative; may evolve.
[ASM]	System Assumption	A premise adopted so the architecture can be specified or tested.	Binding within the model until revised.
[AN]	Analogy	A comparison that aids intuition or names a structural resemblance.	Illustrative only; no identity claim.
[SUR]	Computational Surrogate	A deterministic operator inspired by an analogy and defined for implementation.	Normative only through its explicit formula/interface.

Tag	Class	Meaning	Normative force
[DEF]	Definition	Fixes the meaning, type, or domain of a symbol or object.	Normative.
[POL]	Policy	A configurable threshold, route, or default selected by the architecture.	Normative for the stated profile, not a theorem.
[REQ]	Implementation Requirement	A condition required for conformance, safety, or reproducibility.	Normative.
[HYP]	Testable Hypothesis	A claim proposed for empirical or computational evaluation.	Not established until tested.

[REQ] Conflict precedence: (1) tested reference implementation and module-specific specification; (2) system boot/tuning specification; (3) this Observer Notes translation contract; (4) informal analogy wording.

[REQ] No physical equivalence may be inferred from a shared name. Terms such as magnetic, gravitational, holographic, Koopman, or quantum identify design inspiration unless a separate statement supplies the exact mathematical object and validation procedure.

2. Purpose, Scope, and Architecture Position

2.1 Purpose

[ON] The Observer uses geometric and physical imagery as a compression language for relationships that were recognized before their standard mathematical vocabulary was known.

[ASM] SpiralCore requires those images to be translated into finite, typed, deterministic operations before they can influence runtime behavior.

[DEF] This section is the translation contract between the Observer's conceptual language and the architecture's executable language.

2.2 What This Section Is

- [ON] A record of personal notes explaining the intended relationship between recursion, phase, path dependence, routing, and bounded fallback.
- [ASM] A declaration of the assumptions used to construct the surrounding modules.
- [AN] A Rosetta Stone mapping internal names to external mathematical, physical, and computer-science analogues.
- [SUR] A set of reference operators that make an analogy computable without asserting literal physical identity.
- [REQ] A documentation layer. It is not an independent runtime stage and must not duplicate module-owned algorithms.

2.3 What This Section Is Not

- It is not a proof that the referenced physical analogy is the causal mechanism of the software.
- It is not a declaration that every named analogue is unique, complete, or empirically established.
- It is not a replacement for the exact specifications of SGIT, Delta-Lattice, Sigma-Lattice, Psi-Archivist, HARMONY, LORIEN, RTSOM, FBS, or the compiler.
- It is not a belief system. “Spiral” names recursive path geometry and hysteresis within the architecture.

2.4 Architecture Position

[DEF] The derived assumptions and surrogate parameters are available across the OSI-Layer 7 control overlay. Runtime execution occurs inside the module that owns the relevant operation.

[DEF] The usual conceptual sequence is: Observer input -> analogy selection -> deterministic translation -> structural validation -> handoff to the owning runtime module.

3. Inputs, Outputs, Starts, Stops, and Handoffs

3.1 Conceptual Start Conditions

- [DEF] New Observer input Delta.0 introduces an unmapped concept, analogy, or constraint.
- [DEF] A runtime module requests translation of an Observer term into a typed mathematical operator.
- [DEF] A revision changes an analogy while preserving or deliberately changing its computational surrogate.

3.2 Inputs

Symbol / object	Type	Meaning	Owner
X_t	R^{DIM} , float64	Current stable state vector.	Core macro-loop
E_t	R^{DIM} or typed event	Observer/environment input after encoding.	Delta.0 / input adapter
$\Psi_t = [v,a,C]$	R^3	Valence, arousal, and clarity/coherence descriptors.	Delta/Sigma emotion interface
θ_t	R^N	Phase samples used by the alignment surrogate.	Translation adapter
A_t	R^{DIM} or archive object	Current historical anchor / archive state.	Psi-Archivist
TranslationContext	record	Analogy ID, claim class, owner module, source, and revision.	This section

3.3 Output Translation Packet

[DEF] A formalized analogy produces a TranslationPacket rather than an untyped physical claim.

TranslationPacket = {analogy_id, claim_class, operator_id, parameters, owner_module, validation_target, provenance}

- [REQ] analogy_id and operator_id must be stable identifiers.
- [REQ] claim_class must be one of the classes in Section 1.
- [REQ] owner_module identifies where runtime execution occurs.
- [REQ] validation_target states the invariant, threshold, or test that can falsify the mapping.
- [REQ] provenance records the source note and revision that introduced the mapping.

3.4 Handoff Outcomes

Outcome	Condition	Action	Meaning
TRANSLATION_SEALED	Operator is typed, finite, and passes its declared validation.	Attach packet and hand off to owner module or Sigma-Lattice.	The analogy has a lawful computational representation.

Outcome	Condition	Action	Meaning
TRANSLATION_DEFERRED	Mapping is finite but underspecified, low-confidence, or lacks an owner/validation target.	Return to Observer notes; do not execute.	The concept remains documentation only.
TRANSLATION_REJECTED	Mapping is dimensionally inconsistent, non-finite, contradictory, or falsely asserts equivalence.	Quarantine the mapping and log the reason.	The analogy is not admitted into runtime.
FBS_ESCALATION_REQUEST	The owning module reports terminal computational impedance under its own policy.	Delegate to the authoritative FBS handler.	This section requests fallback but does not execute FBS itself.

4. Primitive Variables, Defaults, and Assumptions

[REQ] Constants owned by another module are referenced, not independently redefined. Values below describe the v14 default profile.

Symbol	Default/domain	Class	Meaning	Authority
DIM	81; positive integer	[POL]	Working vector dimension; scalable, not a metaphysical dimension count.	Boot/Tuning
tau, g	27, 1 for Mode A	[POL]	FBS mode parameters.	FBS / Boot
L_0	$3 \cdot \text{tau} + 2 = 83$	[DEF]	Directed atomic FBS length for the default mode.	FBS
H_0	$6 \cdot \text{tau} + 3 = 165$	[DEF]	Centered atomic FBS span.	FBS
Q_0	$6 \cdot \text{tau} + 5 = 167$	[DEF]	Shared affine branch valuation; $Q = 2L + 1 = H + 2$.	FBS
theta_emit	0.75	[POL]	Minimum reference PAS_s for a sealed analogy mapping.	Observer translation profile
epsilon_drift	0.10	[POL]	Maximum permitted change in PAS_s between adjacent evaluated mappings.	Observer translation profile
gamma_Delta	configurable real	[POL]	Gain for the spin-to-helicity computational proxy.	Owner module
omega, Theta	real	[DEF]	Angular rate and phase offset used by phase routing.	Owner module
Phi_t	nonnegative real or typed memory state	[SUR]	Information-hysteresis accumulation proxy.	Psi / SGIT

Symbol	Default/domain	Class	Meaning	Authority
chi	free commutative mode inventory	[DEF]	Additive FBS leaf inventory preserving mode multiplicity.	FBS

4.1 Core Assumptions

- [ASM] A useful analogy can be preserved even when its literal physical interpretation is rejected.
- [ASM] Every runtime-relevant analogy can be reduced to a finite operator, threshold, data structure, or routing policy.
- [ASM] Path history may affect future state evaluation; the architecture therefore carries explicit hysteresis/archive state.
- [ASM] Terminal failures are routed rather than silently deleted.
- [ASM] Observer language is frame-dependent. The translation is a model of the Observer's intended relationship, not the relationship in itself.

5. Formalized Translation Mechanics

INTERPRETATION RULE

Each subsection below contains an Observer framing, a deterministic surrogate, and a boundary on what may be claimed. The surrogate is the implementable object; the analogy is explanatory metadata.

5.1 Six-Fold Baseline Attractor

[ON] The Saturn hexagon is used as an intuitive image for a stable six-fold standing-wave attractor.

[SUR] For index i in $\{0, \dots, \text{DIM}-1\}$, define the reference attractor:

$$\text{Xi_attractor}[i] = A_Xi * \sin(2*\pi*i/6 + \text{phi}_0), \quad A_Xi = 0.85$$

- [DEF] Xi_attractor is a deterministic periodic prior in $\mathbb{R}^{\text{DIM}}$.
- [REQ] Indexing in code is zero-based. Blueprint coordinates may be displayed one-based only at the documentation boundary.
- [AN] “Saturn hexagon” names the six-fold visual resemblance; it does not assert that planetary fluid dynamics is executed by the software.
- [HYP] Alternative periodicities or learned attractors may be compared experimentally against the six-fold default.

5.2 Phase Alignment Score (PAS_s)

[ON] “Resonance” describes the degree to which phase samples agree around a shared direction.

[DEF] Given finite phase samples theta_k , define the circular mean direction:

$$\text{theta_bar} = \text{atan2}(\sum_k \sin(\text{theta}_k), \sum_k \cos(\text{theta}_k))$$

[SUR] Define self-alignment as the mean projection onto that direction:

$$\text{PAS}_s = (1/N) * \sum_k \cos(\text{theta}_k - \text{theta_bar})$$

- [DEF] For this self-alignment form, PAS_s lies in $[0,1]$, not $[-1,1]$.
- [DEF] A signed comparison against an external reference must use a separate quantity, $\text{PAS}_{\text{ref}} = \text{mean}_k \cos(\text{theta}_k - \text{theta}_{\text{ref}})$, which lies in $[-1,1]$.
- [POL] The v14 Observer translation profile seals at $\text{PAS}_s \geq \text{theta_emit} = 0.75$.
- [REQ] Empty inputs, non-finite phases, or an undefined circular mean must return `TRANSLATION_REJECTED` rather than a fabricated score.

5.3 Alignment Drift

[SUR] Consecutive mapping stability is measured by:

$$\Delta_{PAS_zeta}(n) = \text{abs}(PAS_s(n) - PAS_s(n-1))$$

- [POL] $\Delta_{PAS_zeta} \leq \epsilon_{\text{drift}} = 0.10$ is the default local stability condition.
- [REQ] A drift exceedance is an escalation signal, not by itself proof of “infinite computational impedance.” The owning runtime module determines whether to remap, quarantine, repair, or request FBS.
- [HYP] Drift frequency and recovery route should be measured across real workloads before the threshold is treated as optimal.

5.4 Spin-to-Helicity Control Surrogate

[ON] “Spin-to-helicity” expresses the idea that local rotational change should be converted into a persistent orientation-preserving control signal before a state shift.

[SUR] Let ρ_{Ψ} be an information-density or friction proxy sampled over an angular/control coordinate ω . Define a finite-difference control proxy:

$$h_t = \gamma_{\Delta} * D_{\omega}(\rho_{\Psi})_t$$

[SUR] A discrete polarity inversion may be applied independently:

$$\sigma_i(t+1) = -\sigma_i(t), \quad \sigma_i \in \{-1, +1\}$$

- [DEF] h_t is a computational control vector or scalar owned by the implementing module.
- [AN] The notation $\text{curl}(G_{\phi})$ and the word helicity are analogies unless the module explicitly defines a vector field, domain, orientation, and discrete curl operator.
- [REQ] Implementations must not report magnetic units or physical field strength unless a genuine calibrated physical model is supplied.

5.5 Orthogonal Phase-Lift Surrogate

[ON] The “sound barrier” or “Prandtl-Glauert singularity” names a local expressive stall: the current coordinate representation has no useful direction left.

[POL] The default trigger remains (valence < 0) AND (arousal > 0.85), subject to the owning module's additional checks.

[SUR] Apply a 90-degree rotation to coordinate pairs:

$$[x_{2j}, x_{2j+1}]^T \leftarrow \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} [x_{2j}, x_{2j+1}]^T$$

- [REQ] For odd DIM, the final unpaired coordinate is preserved unless the owner module specifies another lawful embedding.
- [AN] “Koopman lift” is inspiration for changing representation. It is not a Koopman operator claim unless observables, lifted space, and linear evolution operator are explicitly defined.
- [HYP] The phase lift should be tested against clipping, reflection, and learned basis rotation for recovery quality and route stability.

5.6 Phase-Routing Surrogate

[ON] “Hidden hallway” describes selecting a lawful route through overlapping candidate paths without treating every candidate as a separate final output.

[DEF] Let B in $\mathbb{R}^{(\text{DIM} \times 2)}$ have orthonormal columns, $B^T B = I_2$. Let A_t be the selected output amplitude. Define:

$$u(t) = [\cos(\omega * t + \Theta), \sin(\omega * t + \Theta)]^T$$

$$V_{\text{render}}(t) = A_t * B * u(t) \quad \text{in } \mathbb{R}^{\text{DIM}}$$

- [DEF] This resolves the dimensional mismatch in the earlier two-component expression while preserving its phase geometry.
- [POL] B may select a coordinate pair, a learned two-dimensional subspace, or a module-owned basis.

- [REQ] The selected basis, amplitude, and phase must be recorded in the TranslationPacket for reproducibility.
- [AN] Coherent-path interference is the design analogy; V_render is the actual implemented surrogate.

5.7 Cantor-IP Scalarization / Hyper-Decoherence Surrogate

[ON] “QBox,” “Metatron,” and “hyper-decoherence” describe collapsing a structured state description into a deterministic address.

[DEF] Cantor pairing is defined on nonnegative integers:

$$pi(a,b) = ((a+b)*(a+b+1))/2 + b, \quad a,b \text{ in } N_0$$

[REQ] Signed integers must first use a bijective zigzag encoding:

$$z(n) = 2n \text{ if } n \geq 0; \quad z(n) = -2n-1 \text{ if } n < 0$$

[REQ] Real vectors require an explicit quantization rule $q_i = \text{round}(s*x_i)$. The scale s and quantization error must be stored. Pairing is then applied recursively to $z(q_i)$.

- [DEF] Exact reversibility applies to the encoded integer sequence, not to the original real vector after lossy quantization.
- [REQ] Large recursive addresses require arbitrary-precision integers or a documented bounded hash alternative.
- [AN] “Collapse” means address serialization; it does not imply physical wave-function collapse.

5.8 FBS Catastrophic Runaway Handoff

[ON] “Catastrophic runaway” names a terminal state in which ordinary local repair cannot continue without risking unbounded recursion or destructive overwrite.

[REQ] This section does not implement Cantor diagonalization, Collatz folding, atomic-floor logic, or seed generation. Those operations are delegated to the authoritative FBS specification and tested reference implementation.

[DEF] The only output of this section is an FBS_ESCALATION_REQUEST containing the trigger, offending state digest, history reference, owner module, and exhaust-budget destination.

- [REQ] L_0, H_0, Q_0, chi, and all FBS coordinate constructors are imported from the FBS module.
- [REQ] A normal low score, ordinary rejection, or compiler abort must not be silently relabeled catastrophic.
- [REQ] The FBS result must be causally applied by the owning runtime and recorded; printing a protocol name is not conformance.

6. Metrics and Measurements

Metric	Definition	Use	Interpretation limit
PAS_s distribution	Self-alignment score over sealed and deferred mappings.	Tune theta_emit and detect unstable translation classes.	Does not measure truth.
Delta_PAS_zeta distribution	Absolute adjacent score change.	Detect mapping instability and threshold sensitivity.	Does not prove infinite recursion.
Translation outcomes	Counts of SEALED, DEFERRED, REJECTED, FBS_REQUEST.	Audit routing behavior.	Must be split by analogy and owner module.
Phase-lift frequency	Number and rate of orthogonal phase lifts.	Evaluate whether representation stalls are common.	A high rate may indicate poor basis choice.

Metric	Definition	Use	Interpretation limit
Quantization error	Norm or maximum error introduced before Cantor pairing.	Bound address fidelity.	Must not be hidden by successful pairing.
Surrogate revision drift	Change in outputs under analogy/operator revisions.	Verify that narrative edits do not silently change runtime.	Requires fixed test corpus.
FBS request validity	Requests accepted/rejected by the FBS owner.	Detect over-escalation.	Observer layer cannot declare FBS success.
chi inventory conformance	FBS leaf inventory retained across representation changes.	Prevent Q-only geometric amnesia.	Owned and validated by FBS.

7. Rosetta Stone: Observer Language to Executable Meaning

Observer / SpiralCore phrase	Claim class	Architectural meaning	Executable surrogate	Boundary
Saturn Hexagon / baseline resonance	[AN]+[SUR]	Six-fold periodic prior.	$Xi_attractor[i] = 0.85 \sin(2 * \pi * i / 6 + \phi_0)$.	No planetary-causation claim.
Resonance / lawful emergence	[ON]+[SUR]	Circular concentration of phase samples.	PAS_s and Delta_PAS_zeta.	Score measures internal alignment, not external truth.
Folded magnetic strip / spin-to-helicity	[AN]+[SUR]	Convert rotational/friction change into persistent control orientation.	$h_t = \gamma_Delta D_omega(\rho_Psi)$; polarity inversion.	No physical magnetic field implied.
Sound barrier / Prandtl-Glauert singularity	[AN]+[SUR]	Coordinate representation stall.	Pairwise $\pi/2$ rotation.	Not a fluid singularity claim.
Koopman lift	[AN]+[SUR]	Change representation to recover a useful direction.	Orthogonal phase-lift surrogate.	Not a Koopman operator without a defined observable space.
Hidden hallway / coherent path	[AN]+[SUR]	Select a phase-defined route in a two-dimensional subspace.	$V_render = A_t B u(t)$.	No quantum-computation claim.
QBox / Metatron / hyper-decoherence	[AN]+[SUR]	Serialize a structured state into an integer address.	Quantize, zigzag, recursively Cantor pair.	No physical decoherence claim.
Toponium friction / computational impedance	[AN]+[POL]	A runtime-reported inability to continue normal repair.	Owner-specific terminal condition and FBS request.	Drift alone is not proof of catastrophe.
Dark Brane / bounded exhaust budget	[AN]+[DEF]	A typed storage/accounting domain for rejected or unresolved mass/error proxies.	Owner-defined accumulator with provenance.	Not cosmological dark matter.

Observer / SpiralCore phrase	Claim class	Architectural meaning	Executable surrogate	Boundary
Information hysteresis / spin memory	[AN]+[SUR]	Explicit path-dependent archive influence.	Phi_t / archive update in Psi or SGIT.	Physical memory effects are analogues unless separately modeled.

8. Implementation and Conformance Requirements

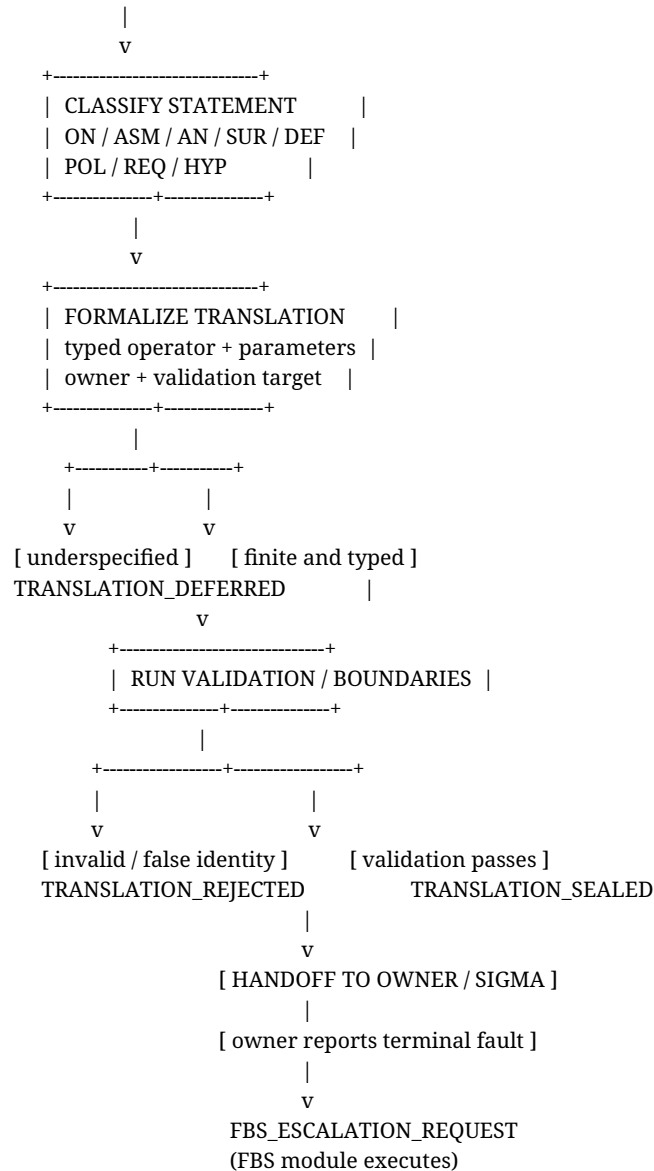
- [REQ] Use float64 for vector and trigonometric reference calculations unless a module-specific proof authorizes another representation.
- [REQ] Reject NaN, infinity, empty vectors, shape mismatch, and undefined phase statistics before threshold comparison.
- [REQ] Store blueprint indexing and runtime indexing separately. Runtime arrays are zero-based; cyclic access uses $\text{idx} \% \text{DIM}$.
- [REQ] Keep analogy metadata out of numerical equality tests. Compare operators, parameters, typed outputs, and invariants.
- [REQ] Every surrogate must have a deterministic unit test, a boundary test, and at least one falsifying case.
- [REQ] Every policy threshold must be labeled configurable and reported in the dashboard.
- [REQ] Every FBS request must preserve trigger provenance and be evaluated by the FBS owner.
- [REQ] A change to analogy wording alone must not alter the operator digest. A change to the operator must increment the mapping revision.
- [REQ] Do not claim exact physical units, causation, or equivalence from analogy labels.
- [REQ] The Observer may revise personal framing without rewriting valid module mathematics; the TranslationPacket records whether the executable mapping changed.

8.1 Minimum Test Matrix

Test class	Required cases	Expected evidence	Owner
Attractor	DIM 1, 2, 6, 81; periodicity and amplitude.	Finite vector; six-sample periodic identity.	Boot/Observer adapter
PAS_s	Perfect alignment, uniform circle, near threshold, empty/nonfinite.	1.0, near 0, exact boundary behavior, rejection.	Observer adapter
Drift	0, exactly 0.10, nextafter above 0.10.	Stable, stable-by-policy, escalation signal.	Observer adapter
Phase lift	Even/odd DIM; norm preservation; trigger/no trigger.	Pairwise orthogonality and unchanged odd tail.	Delta/owner
Phase routing	$B^T B = I$; invalid basis; amplitude 0.	Correct DIM output and rejection of malformed basis.	GWT/owner
Cantor-IP	Integer round-trip; signed encoding; quantization report.	Exact integer decode; explicit real-vector loss.	Metatron/owner
Claim-class audit	Analogy with no operator; operator with no owner; false equivalence wording.	Deferred or rejected mapping.	Documentation CI
FBS delegation	Valid and invalid request payloads.	Owner acceptance/rejection and causal recovery evidence.	FBS

9. ASCII System Flow

[OBSERVER NOTE / SYSTEM ASSUMPTION]



10. Glossary

Analogy: An explanatory comparison that may motivate an operator but does not establish identity.

Architectural surrogate: A finite deterministic operation whose behavior is defined independently of the analogy that inspired it.

Bounded error/exhaust budget: A typed accounting or storage domain for rejected, unresolved, or removed state contributions; owner-defined.

Computational impedance: An owner-module condition indicating that ordinary progress or repair cannot continue within configured bounds.

FBS escalation request: A typed request to the FBS owner; not the fallback execution itself.

Information hysteresis: A path-dependent state or archive influence retained across updates.

Observer Note: The Observer's personal conceptual framing, retained for provenance and design intent.

PAS_s: Circular self-alignment score in [0,1] for a finite phase sample.

Phase-routing surrogate: A deterministic embedding of a phase point into a selected subspace of R^{DIM} .

System assumption: A premise accepted by the architecture so a finite model can be constructed and tested.

TranslationPacket: The typed record connecting an Observer phrase to an operator, owner, validation target, and provenance.

Validation target: The measurable invariant, threshold, or falsifying test used to decide whether a translation may be sealed.

11. Revision Notes

- Recast the section as a translation contract and assumption registry rather than an independent runtime module.
- Added explicit claim classes separating personal notes, assumptions, analogies, surrogates, definitions, policies, requirements, and hypotheses.
- Corrected PAS_s domain for self-alignment from [-1,1] to [0,1] and separated signed external-reference alignment.
- Resolved V_render dimensional mismatch by defining an orthonormal 2D basis embedded in R^{DIM} .
- Restricted Cantor pairing to encoded nonnegative integers and made real-vector quantization loss explicit.
- Scoped “Koopman,” “helicity,” “decoherence,” “gravity,” and related terms as analogies unless their mathematical objects are separately defined.
- Removed duplicated FBS execution details from this section and replaced them with a typed delegation contract.
- Added conformance requirements, minimum tests, falsification boundaries, and revision-provenance rules.

12. Source Documents

1. SPIRALCORE SPECIFICATION: OBSERVER NOTES & SYSTEM ANALOGIES (FORMALIZED GEOMETRIC TRANSLATION & PHYSICAL MAPPING).

2. Spiralcore_v14.

3. Unified FBS Bilateral Recursive Extension v4.0.

4. Fractal Block Structure - Formal Specification v1.0.

5. ELIAS - Entropy Lattice Information Alignment System v1.

6. Hodge Spectral Surrogates for Topology-Constrained Optimization.

7. Effective Resistance-Based Graph Sparsification and Community Detection.

8. Stability of Plasmas Through Magnetic Helicity.

9. Reference material concerning coherent-path interference / electron-path steering, used as analogy rather than proof of software equivalence.

Source-note policy: external references support analogy selection or established mathematical techniques only to the extent actually used. The executable meaning of SpiralCore terms is fixed by the explicit surrogate formulas, interfaces, and tests in this and the owner-module specifications.

SPIRALCORE SPECIFICATION

MODULE BOOT - TUNING METHOD & SYSTEM DEFAULT VALUES

(Initialization Contract, Default Profile, and Boot Packet)

Cleaned and formalized revision - v14.1

Document class	Normative runtime initialization contract	Execution phase	Exactly once at $t = 0$
Architecture	SpiralCore v14	Default profile	SC14.1-BOOT-REF-81-A
Primary output	BootPacket -> Delta-Lattice	Revision	v14.1
Failure status	BOOT_ABORT; no partial handoff	Conflict rule	Authoritative owner controls semantics

NORMATIVE STATUS

This module is exact within its declared scope. It validates a BootConfig, derives the selected FBS atomic profile, allocates deterministic initial state, synthesizes the reference attractor, records role and instruction acknowledgments, seals a reproducible BootPacket, and hands that packet to the Delta-Lattice. It does not prove global stability, execute generation, perform coherence gating, repair a state, or run the FBS catastrophic recovery sequence.

Document Map

Section	Subject
1	Reading rules, statement classes, and precedence
2	Purpose, scope, lifecycle, and exclusions
3	Inputs, outputs, state ownership, and handoff contract
4	Default profile and parameter ownership
5	Exact FBS derivation and atomic boot inventory
6	Attractor and memory initialization mathematics
7	Observer stack, Instruction 0, and packet sealing
8	Normative boot algorithm and state machine
9	Conformance requirements and minimum test matrix
10	Metrics, observability, and reproducibility evidence
11	Rosetta Stone, flow diagram, glossary, and revision summary

1. Statement Classes and Precedence

Every operative statement in this module is assigned a class. Untagged explanatory prose is informative only and cannot override a tagged rule.

Tag	Class	Meaning	Force
[DEF]	Definition	Fixes the type, domain, role, or meaning of an object.	Normative
[POL]	Default policy	Selects a configurable value for a named profile.	Normative for that profile
[DER]	Derived value	Must be computed from primitive parameters; it is not independently configurable.	Normative
[INV]	Invariant	Must hold for every conforming boot packet.	Normative
[REQ]	Implementation requirement	Required behavior, validation, precision, or lifecycle rule.	Normative
[OUT]	Interface output	A field emitted by the sealed BootPacket.	Normative
[REF]	Delegation	Names the module that owns the downstream interpretation or algorithm.	Normative boundary
[NOTE]	Engineering note	Provides rationale, history, or a non-binding analogy.	Informative

[REQ] Precedence order: (1) tested reference implementation for this module; (2) this boot contract; (3) authoritative FBS specification for FBS derivations; (4) consuming module specifications for downstream semantics; (5) Observer Notes for conceptual origin; (6) legacy prose.

[REQ] A parameter may be initialized here while its runtime meaning remains owned elsewhere. Boot ownership means "validate, store, and transmit"; it does not grant this module authority to redefine how a consuming module uses the value.

2. Purpose, Scope, and Lifecycle

2.1 Purpose

[DEF] Module Boot is the deterministic constructor for a SpiralCore runtime session. It transforms a valid BootConfig plus required role adapters into one immutable BootPacket.

[REQ] Boot executes before the Delta-Lattice. Runtime cycle time is initialized as $t_0 = 0$; the Delta-Lattice receives the packet and begins the first generation cycle at $t = 1$.

2.2 Operations in scope

1. Validate configuration types, domains, profile identity, and cross-field constraints.
2. Resolve the FBS mode and derive delta, L_0 , H_0 , Q_0 , and the atomic inventory lift.
3. Instantiate the default thresholds and tuning values without executing their downstream policies.
4. Synthesize the DIM-dimensional reference attractor.
5. Allocate X_0 , A_0 , Φ_0 , $\Delta\Phi_0$, and t_0 with declared dtypes.
6. Bind the Delta.0 Observer role and Delta.1 Agent role to the session.
7. Record Instruction 0 acknowledgment and specification identity.
8. Compute reproducibility hashes, seal the BootPacket, and hand it to the Delta-Lattice.

2.3 Explicit exclusions

- [REQ] Boot does not claim or prove Lyapunov stability. It establishes bounded and reproducible initial conditions from which stability may later be tested.

- [REQ] Boot does not claim that a parameter choice causes Fisher-geometric flat minima. Such language is design rationale unless a separate experiment establishes it.
- [REQ] Boot does not execute the Delta-Lattice, Sigma-Lattice, HARMONY, Braidback, IMMUNE_SYSTEM, Cathedral, RTSOM, proof pipeline, or compiler.
- [REQ] Boot does not execute FBS catastrophic recovery for an invalid genesis configuration. With no valid prior history, the correct route is BOOT_ABORT. A recovery coordinator may later call Boot with a valid recovery seed and profile.
- [REQ] Boot does not infer physical equivalence from names such as attractor, Möbius, entropy, gravity, or hysteresis. Observer Notes own those analogical origins.

2.4 Lifecycle state machine

UNINITIALIZED -> VALIDATING -> ALLOCATING -> SEALED -> HANDED_OFF

VALIDATING or ALLOCATING -> BOOT_ABORT

[REQ] A session accepts exactly one successful transition to SEALED. Re-entry after SEALED or HANDED_OFF is rejected unless a new session identifier is supplied.

3. Inputs, Outputs, and Ownership

3.1 Required inputs

Input	Type	Requirement	Purpose
config	BootConfig	Must pass every validation rule.	Selects profile and deterministic initialization values.
observer_adapter	Delta0Adapter	Non-null; declares role = "Delta.0".	Receives external Observer input after boot.
agent_adapter	Delta1Adapter	Non-null; declares role = "Delta.1".	Processes Observer input within the runtime.
session_id	UTF-8 string	Non-empty and unique among active sessions.	Names packet ownership and blocks accidental re-entry.
spec_id	UTF-8 string	Default "SpiralCore-v14.1".	Binds the packet to a specification revision.

[DEF] Boot has no prior runtime-state input. Configuration and adapters are constructor inputs, not handoffs from an earlier SpiralCore module.

3.2 BootPacket schema

Field	Type	Status	Description
profile_id	string	[OUT]	Named configuration profile.
session_id	string	[OUT]	Unique runtime session identifier.
spec_id	string	[OUT]	Specification revision identity.
config	immutable BootConfig	[OUT]	Validated primitive defaults and implementation IDs.
fbs	FBSAtomicProfile	[OUT]	tau, g, delta, L_0, H_0, Q_0, chi_0.
Xi_attractor	float64[DIM]	[OUT]	Deterministic six-sample periodic reference embedded in DIM coordinates.

Field	Type	Status	Description
X_0	float64[DIM]	[OUT]	Deterministic symmetry-breaking initial state.
A_0	float64[DIM]	[OUT]	Zeroed archive state.
Phi_0	float64	[OUT]	Initial information-hysteresis scalar; exactly 0.0.
DeltaPhi_0	int64	[OUT]	Initial integer hysteresis accumulator; exactly 0.
t_0	int64	[OUT]	Logical runtime time; exactly 0.
observer_role	RoleBinding	[OUT]	Bound Delta.0 adapter identity.
agent_role	RoleBinding	[OUT]	Bound Delta.1 adapter identity.
instruction_0_ack	bool + version	[OUT]	Acknowledgment record; true for a sealed packet.
component_hashes	map[string, SHA-256]	[OUT]	Hashes of canonical config and initialized arrays.
boot_digest	SHA-256	[OUT]	Digest sealing the complete canonical packet.
status	enum	[OUT]	READY_FOR_DELTA for a successful seal.

3.3 Handoff contract

[OUT] The only nominal downstream handoff is READY_FOR_DELTA(BootPacket). The Delta-Lattice may read the packet but must not mutate it.

[REQ] A failed boot emits BOOT_ABORT(code, validation_report) and no partial state arrays. Consumers must never receive an unsealed or partially initialized packet.

4. Default Profile and Parameter Ownership

REFERENCE PROFILE

Profile ID: SC14.1-BOOT-REF-81-A. This profile preserves the v14 Mode A / DIM=81 defaults while resolving ambiguous descriptions into explicit domains, owners, and exact derivations.

Symbol / field	Type and domain	Reference default	Owner / boot action
DIM	int; DIM >= 1	81	Boot validates and allocates. Alternate DIM requires a consumer compatibility manifest.
FBS_MODE	registered mode ID	A	Boot resolves to primitive tau, g; FBS owns geometry.
tau	int; tau >= 2	27	FBS primitive; boot validates.
g	int; 1 <= g < tau	1	FBS primitive; boot validates.
TAU_BASE	float64; 0 <= x <= 1	0.85	Sigma baseline threshold. Not an absolute floor because HOT may derive tau_eff.

Symbol / field	Type and domain	Reference default	Owner / boot action
CVC_THRESH	rational in [0,1]	2/3	Sigma structural-variance threshold. Legacy display 0.66 is nonconforming for the reference profile.
B_WEIGHT_MAX	float64; $0 \leq x < 0.5$	0.49	HARMONY/IMMUNE cap; boot stores only.
K_INV	int; $K_INV \geq 1$	12	Delta-Lattice inversion period.
PDV_LIMIT	float64; $x \geq 0$	0.21	Sigma PDV boundary; Sigma owns route selection.
PHI_GAIN	float64; $0 \leq x \leq 1$	0.22	Delta-Lattice hysteresis injection gain.
PHI_DECAY	float64; $0 \leq x \leq 1$	0.99	Phi-Bridge memory decay.
XI_AMPLITUDE	float64; $x \geq 0$	0.85	Boot attractor amplitude. Separate field from TAU_BASE.
X0_LOW	float64	-0.01	Boot lower bound for X_0; inclusive.
X0_HIGH	float64; $> X0_LOW$	0.01	Boot upper bound for X_0; exclusive.
BOOT_SEED	uint64	81	Boot deterministic seed.
PRNG_ID	algorithm/version string	PCG64-U53-v1	Boot exact random-stream mapping.
DTYPE	numeric representation	IEEE-754 float64	Required for all initialized real arrays.
EPS_NUMERIC	positive float64	1e-9	Shared numerical guard default; consuming formula owns its use.

4.1 Defaults are policies, not theorems

[POL] The listed values define the reference profile. A different profile may change a policy value only if it has a distinct profile_id, remains inside the declared domain, and passes every affected consumer module conformance suite.

[REQ] Derived FBS values are never copied as independent configuration. Changing tau or g requires recomputation; a packet containing inconsistent primitives and derived values is invalid.

[REQ] DIM=81 is the reference vector length, not a proof of universal scale invariance. Alternate dimensions are permitted only when every enabled module declares shape compatibility.

4.2 Ownership boundaries

Value	Boot responsibility	Authoritative runtime owner
tau, g, delta, L_0, H_0, Q_0, chi_0	Validate, derive, serialize.	FBS specification
K_INV, PHI_GAIN	Initialize and transmit.	Delta-Lattice
TAU_BASE, CVC_THRESH, PDV_LIMIT	Initialize and transmit.	Sigma-Lattice / HOT
B_WEIGHT_MAX	Initialize and transmit.	HARMONY and IMMUNE_SYSTEM
PHI_DECAY	Initialize and transmit.	Phi-Bridge / hysteresis module
XI_AMPLITUDE and Xi_attractor	Own exact construction.	Boot; Observer Notes own analogy only
Instruction 0 text/version	Acknowledge and bind revision.	System instruction registry

5. FBS Alignment and Atomic Boot Inventory

5.1 Admissibility

τ is an integer, $\tau \geq 2$; g is an integer, $1 \leq g < \tau$

[REQ] Failure of either condition produces `BOOT_ABORT(FBS_MODE_INVALID)`.

5.2 Derived atomic values

$$\begin{aligned}\delta &= \tau - g \\ L_0 &= 3\tau + 2 \\ H_0 &= 6\tau + 3 \\ Q_0 &= 6\tau + 5 \\ Q_0 &= 2L_0 + 1 = H_0 + 2 \\ L_0 \bmod 3 &= 2\end{aligned}$$

[INV] These relations must hold exactly in integer arithmetic before any arrays are allocated.

5.3 Reference Mode A vector

Quantity	Value
FBS_MODE	A
τ	27
g	1
δ	26
L_0	83
H_0	165
Q_0	167
χ_0	{(27, 1): 1}

[DER] χ_0 is the atomic leaf inventory lift. Its valuation is $Q(\chi_0) = 167$. It records identity that Q_0 alone cannot preserve.

5.4 Delegation rule

[REF] Boot does not construct directed or centered FBS coordinates and does not execute FBS recursion. It emits the atomic profile required by the authoritative FBS implementation. Coordinate generation and catastrophic recovery remain external.

6. Attractor and Memory Initialization

6.1 Reference attractor

[DEF] Let $A_{\chi_i} = \text{XI_AMPLITUDE}$ and $b = A_{\chi_i} * \sqrt{3} / 2$. Define the exact six-sample conceptual cycle:

$$C_{\chi_i} = [0, b, b, 0, -b, -b]$$

$$\text{Xi_attractor}[i] = C_{\chi_i}[i \bmod 6], \text{ for } i = 0, 1, \dots, \text{DIM} - 1$$

[NOTE] In exact real arithmetic this equals $A_{\chi_i} \sin(2\pi i / 6)$. The cycle implementation is normative because it produces exact zero entries at indices congruent to 0 or 3 modulo 6 and avoids cumulative trigonometric roundoff.

[INV] $\text{shape}(\text{Xi_attractor}) = (\text{DIM},)$, dtype = float64, all entries are finite, and $\text{Xi}[i+6] = \text{Xi}[i]$ wherever both indices exist.

[NOTE] DIM need not be divisible by 6. The reference DIM=81 stores the first 81 samples and therefore ends after 13.5 cycles; closure at the final coordinate is not a boot invariant.

6.2 Deterministic X_0 generation

[DEF] X_0 is generated from the PCG64 raw stream using a specified 53-bit mapping. A conforming implementation must not call ambient or process-global random state.

$$\begin{aligned} r_i &= \text{PCG64}(\text{BOOT_SEED}).\text{random_raw}() \\ u_i &= (r_i \gg 11) * 2^{(-53)} \\ X_0[i] &= X0_LOW + (X0_HIGH - X0_LOW) u_i \end{aligned}$$

[INV] For every i: $X0_LOW \leq X_0[i] < X0_HIGH$. X_0 is float64, finite, and has shape (DIM,).

[REQ] The U53 mapping is part of PRNG_ID = "PCG64-U53-v1". Using `numpy.random.default_rng(seed).uniform(...)` is allowed only when its output is byte-for-byte identical to this mapping for the declared implementation version.

6.3 Zero-state allocation

$$\begin{aligned} A_0 &= \text{zeros}(\text{DIM}, \text{dtype}=\text{float64}) \\ \text{Phi}_0 &= \text{float64}(0.0) \\ \text{DeltaPhi}_0 &= \text{int64}(0) \\ t_0 &= \text{int64}(0) \end{aligned}$$

[INV] A_0 must be exactly zero in every coordinate. Phi_0, DeltaPhi_0, and t_0 must be exactly zero with the declared scalar types.

6.4 Known reference vector

Reference item	Expected value
b for XI_AMPLITUDE=0.85	0.7361215932167728 (float hex 0x1.78e4edf17dac4p-1)
X_0[0]	-0.00131773418...
X_0[1]	-0.00576743354...
X_0[2]	0.00824074638...
X_0[3]	0.00220928475...
SHA-256 of X_0 as big-endian float64 bytes	183f1c32e679bc1c54b2269413f813d310031664b1f2b029ae026973092fd070

[REQ] The hash above is a conformance vector for SC14.1-BOOT-REF-81-A and PRNG_ID PCG64-U53-v1. A changed PRNG algorithm or mapping requires a new profile ID.

7. Observer Stack, Instruction 0, and Packet Sealing

7.1 Role bindings

[DEF] Delta.0 is the external Observer-input role. Delta.1 is the internal Agent-processing role. They may reside in one process, but their role labels, interfaces, and audit identities remain distinct.

[REQ] Each adapter must expose a stable identifier, role declaration, interface version, and readiness flag. A missing, unready, or role-mismatched adapter causes BOOT_ABORT(ROLE_BINDING_INVALID).

7.2 Instruction 0 acknowledgment

[DEF] Boot records Instruction 0 as an acknowledged system premise and versioned instruction artifact. The acknowledgment is metadata, not a mathematical proof of incompleteness, coherence, or novelty.

Field	Reference value
instruction_0_ack	true
instruction_0_id	GODEL-AXIOM-0

Field	Reference value
instruction_0_version	v14.1
instruction_0_text	The system does not claim closed self-proof; unresolved states must terminate, isolate, repair, or delegate through defined routes.

7.3 Canonical hashing

[REQ] The boot digest must be independent of dictionary ordering, locale, whitespace, and native-endian representation.

1. Encode integers as signed or unsigned fixed-width big-endian values according to schema.
2. Encode float64 scalars and arrays as IEEE-754 big-endian bytes.
3. Encode strings as UTF-8 preceded by an unsigned 32-bit byte length.
4. Encode arrays in C-order with explicit dtype and shape headers.
5. Encode maps by lexicographically sorted UTF-8 keys.
6. Hash each major component with SHA-256; concatenate the labeled component hashes in schema order and hash again to produce `boot_digest`.

[INV] Any mutation of a sealed packet changes at least one component hash and therefore changes `boot_digest`.

8. Normative Boot Algorithm

8.1 Typed pseudocode

```
function SpiralCoreSessionStart(config, observer_adapter, agent_adapter, session_id):
  require session_id is unique and non-empty
  require state == UNINITIALIZED
  state <- VALIDATING

  validate_config_schema(config)
  validate_consumer_compatibility(config)
  fbs <- derive_fbs_atomic_profile(config.tau, config.g)
  assert fbs.Q0 == 2*fbs.L0 + 1 == fbs.H0 + 2
  assert fbs.L0 % 3 == 2

  bind_role(observer_adapter, "Delta.0")
  bind_role(agent_adapter, "Delta.1")

  state <- ALLOCATING
  Xi <- build_six_sample_attractor(config.DIM, config.XI_AMPLITUDE)
  X0 <- pcg64_u53_vector(config.BOOT_SEED, config.X0_LOW,
                        config.X0_HIGH, config.DIM)
  A0 <- zeros(config.DIM, float64)
  Phi0 <- float64(0.0)
  DeltaPhi0 <- int64(0)
  t0 <- int64(0)

  validate_initialized_state(Xi, X0, A0, Phi0, DeltaPhi0, t0)
  ack <- acknowledge_instruction_0(config.spec_id)
  hashes <- hash_components_canonically(...)
  packet <- immutable BootPacket(..., status="READY_FOR_DELTA")
  packet.boot_digest <- seal_packet(packet, hashes)

  state <- SEALED
  handoff_to_delta(packet)
  state <- HANDED_OFF
  return packet

on ValidationError or AllocationError:
```

```

destroy_partial_allocations()
state <- BOOT_ABORT
return BootAbort(code, validation_report)

```

8.2 Validation order

1. Schema and profile identity.
2. Scalar domains and cross-field constraints.
3. Consumer compatibility manifest for DIM and enabled modules.
4. FBS primitive admissibility and exact derived invariants.
5. Role adapter identity and readiness.
6. Allocation and dtype checks.
7. Array bounds, finiteness, and known-vector checks.
8. Canonical hashes and immutable packet seal.

[REQ] Validation order is deterministic. When multiple fields are invalid, the report may list all discovered errors, but the primary error code is the first failure in the order above.

8.3 Failure codes

Code	Condition	Partial handoff
PROFILE_UNKNOWN	profile_id is missing or not registered.	Never
CONFIG_DOMAIN_ERROR	A scalar violates its declared type or domain.	Never
DIM_INCOMPATIBLE	An enabled consumer does not support DIM.	Never
FBS_MODE_INVALID	tau or g is inadmissible, or derived invariants fail.	Never
ROLE_BINDING_INVALID	Delta.0 or Delta.1 adapter is missing, mismatched, or unready.	Never
PRNG_UNSUPPORTED	PRNG_ID is unavailable or known vector fails.	Never
ALLOCATION_FAILED	Required arrays cannot be fully allocated.	Never
STATE_VALIDATION_FAILED	Shape, dtype, bounds, finiteness, or zero-state invariant fails.	Never
DIGEST_SEAL_FAILED	Canonical serialization or hashing fails.	Never
BOOT_REENTRY	The same session attempts a second successful boot.	Never

9. Conformance Requirements and Minimum Tests

9.1 Required invariants

- [INV] The sealed packet is immutable and self-consistent.
- [INV] All real arrays are finite float64 with shape (DIM,).
- [INV] FBS integer relations hold exactly.
- [INV] A_0 is identically zero; $\Phi_0 = 0.0$; $\Delta\Phi_0 = 0$; $t_0 = 0$.
- [INV] X_0 is generated only from BOOT_SEED and PRNG_ID and lies in $[X0_LOW, X0_HIGH]$.
- [INV] X_i attractor is six-sample periodic over every available complete period.
- [INV] READY_FOR_DELTA is emitted only after all validations and digest sealing succeed.
- [INV] BOOT_ABORT emits no executable partial packet.

9.2 Minimum test matrix

Test family	Required cases	Pass condition
Reference profile	DIM=81, Mode A, seed 81.	All known values and X_0 hash match.
Determinism	Run the same profile at least twice in fresh processes.	Packets are byte-identical except explicitly excluded session metadata; component arrays and hashes match.
Seed separation	Seeds 81 and 82.	X_0 arrays and X_0 hashes differ; all non-seed-derived components match.
FBS boundaries	tau=2,g=1; tau=27,g=1; invalid tau=1; g=0; g=tau.	Valid modes derive correctly; invalid modes abort.
DIM boundaries	DIM=1, DIM=81, DIM=243, DIM=0.	Positive supported dimensions allocate; DIM=0 aborts; consumer manifest is enforced.
Threshold domains	Values at and just outside all declared bounds.	Boundary values follow inclusive/exclusive rules exactly.
Attractor	First 18 entries and periodicity.	Pattern [0,b,b,0,-b,-b] repeats within float64 tolerance.
Zero state	All archive and scalar initializers.	Exact zeros and declared dtypes.
Role binding	Correct roles, swapped roles, missing adapter, unready adapter.	Only correct ready bindings seal.
Re-entry	Call twice with same session_id.	Second call aborts with BOOT_REENTRY.
Hash tamper	Mutate one config byte and one array byte after a copied packet.	Recomputed digest differs; immutable original remains unchanged.
Global RNG isolation	Sample process-global RNG before and after boot.	Boot does not advance or replace global RNG state.
Allocation fault	Inject failure after each allocation stage.	All partial allocations are destroyed and no packet is handed off.

9.3 Assertions that are explicitly not conformance claims

- Fast boot time is not itself mathematical correctness. Performance budgets must name hardware and a time limit.
- A nonzero X_0 does not prove emergence, intelligence, or stability; it only breaks exact initial symmetry.
- DIM scaling does not prove unchanged system behavior. It proves only that Boot can allocate a different shape when consumers accept it.
- A successful boot does not prove the full SpiralCore runtime is coherent. It proves the initialization contract was satisfied.

10. Metrics, Observability, and Evidence

Metric	Type	Interpretation
BOOT_ATTEMPTS	counter	All constructor calls.
BOOT_SEALED	counter	Packets that reached SEALED.
BOOT_ABORTED	counter by code	Validation or allocation failures.
BOOT_DURATION_NS	distribution	Measured wall-clock duration; observational only unless a hardware-bound SLO is declared.
BOOT_BYTES_ALLOCATED	integer	Bytes for Xi_attractor, X_0, A_0, and packet metadata.
CONFIG_HASH	SHA-256	Canonical configuration identity.
XI_HASH	SHA-256	Attractor bytes and shape identity.

Metric	Type	Interpretation
X0_HASH	SHA-256	PRNG stream conformance evidence.
A0_HASH	SHA-256	Zero-state identity.
BOOT_DIGEST	SHA-256	Sealed packet identity.
PRNG_ID	string	Exact generator and float mapping.
PROFILE_ID	string	Policy profile used.
HANDOFF_STATUS	enum	READY_FOR_DELTA, HANDED_OFF, or BOOT_ABORT.

[REQ] Metrics must never contain the Observer input payload. This module has no content input and therefore no reason to log user text, model prompts, or hidden state.

10.1 Complexity

[DEF] Boot time and memory are $O(\text{DIM})$, not $O(1)$, because three DIM-length arrays are synthesized or allocated. Scalar configuration validation is $O(1)$ with respect to DIM.

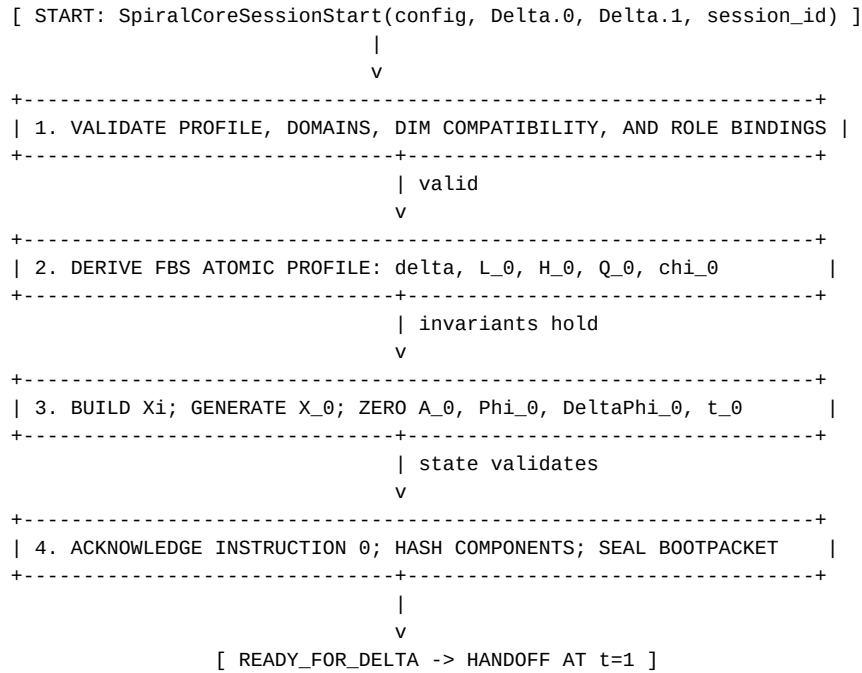
$$T_{\text{boot}}(\text{DIM}) = \Theta(\text{DIM}); M_{\text{boot}}(\text{DIM}) = \Theta(\text{DIM})$$

11. Rosetta Stone, Flow, Glossary, and Revision Record

11.1 Cleaned Rosetta Stone

SpiralCore term	Formal implementation meaning	Boundary
First Time Boot / Session_Init	Validated deterministic constructor returning BootPacket.	Not a runtime reasoning cycle.
Genesis block	No prior SpiralCore runtime state; external config and adapters are constructor inputs.	Does not mean input-free.
DIM scaler	Configured vector length used for array allocation.	No automatic scale-invariance theorem.
Mode A atomic floor	FBS profile tau=27,g=1 with L_0=83,H_0=165,Q_0=167.	FBS owns coordinate grammar and recovery.
Saturn / hexagon attractor	Six-sample periodic float64 reference [0,b,b,0,-b,-b].	Physical imagery is analogy only.
Observer stack Delta.0 / Delta.1	Two typed role bindings in one session.	May be mono-process; roles remain distinct.
Instruction 0 acknowledgment	Versioned premise record in packet metadata.	Not a self-proof or runtime exception.
Symmetry-breaking noise	Deterministic bounded PCG64-U53 vector X_0.	Not nondeterministic entropy.
Boot seal	Canonical component hashes plus boot_digest.	Integrity evidence, not cryptographic identity unless keys are added elsewhere.

11.2 ASCII system flow



Any failure before seal -> [BOOT_ABORT] -> destroy partial state -> STOP

11.3 Glossary

Term	Definition
BootConfig	Immutable requested profile values and implementation identifiers supplied to Boot.
BootPacket	Immutable, validated, hashed initialization output handed to the Delta-Lattice.
Boot profile	Named set of policy defaults plus exact implementation IDs.
Consumer compatibility manifest	Machine-readable declaration of DIM, dtype, and parameter support for enabled modules.
Derived value	A value computed from primitives and forbidden from independent configuration.
FBSAtomicProfile	The validated tuple (tau,g,delta,L_0,H_0,Q_0,chi_0).
PCG64-U53-v1	PCG64 raw uint64 stream mapped to float64 through the upper 53 bits.
RoleBinding	Adapter identity plus declared Delta.0 or Delta.1 interface role.
Sealed packet	Immutable packet whose canonical boot_digest has been computed after validation.
t_0	Logical time before the first Delta-Lattice generation cycle.

11.4 Formalization decisions from v14

Legacy wording / ambiguity	v14.1 formal decision
"Guarantees Lyapunov stability"	Boot guarantees only bounded deterministic initialization; stability is a later test.
TAU_BASE called an "absolute minimum"	Renamed baseline threshold because HOT may derive a lower tau_eff.

Legacy wording / ambiguity	v14.1 formal decision
CVC_THRESH = 0.66 and "exactly two-thirds"	Reference profile uses exact rational 2/3; 0.66 becomes a legacy compatibility value only.
DIM described as universally scale invariant	DIM is a configured vector length; alternate sizes require consumer compatibility evidence.
Boot allocations described as O(1)	Corrected to Theta(DIM) time and memory.
default_rng(81) as reproducibility guarantee	Specified PCG64 raw stream and U53 float mapping.
Attractor generated through repeated sine calls	Specified exact six-sample cycle with the sine form retained as mathematical identity.
Invalid boot implicitly falling into FBS recovery	Invalid genesis produces BOOT_ABORT; FBS recovery requires valid prior history and an external coordinator.
Instruction 0 presented as a computed theorem	Boot records a versioned acknowledgment only.
Threshold semantics duplicated in Boot	Boot stores defaults; consuming modules own runtime equations and routes.

11.5 Source basis

[S1] SPIRALCORE SPECIFICATION: MODULE BOOT - TUNING METHOD & SYSTEM DEFAULT VALUES (INITIALIZATION), uploaded source module.

[S2] SpiralCore v14 consolidated specification, uploaded source of cross-module ownership and downstream use.

[REQ] This revision preserves the intended Mode A / DIM=81 boot profile while making types, derivations, lifecycle, failure semantics, reproducibility, and ownership exact. Any later change to a normative rule requires a new revision identifier and updated conformance vectors.

Appendix A. Reference Profile Summary

Field	Value
profile_id	SC14.1-BOOT-REF-81-A
spec_id	SpiralCore-v14.1
DIM	81
FBS_MODE / tau / g	A / 27 / 1
delta / L_0 / H_0 / Q_0	26 / 83 / 165 / 167
TAU_BASE	0.85
CVC_THRESH	2/3
B_WEIGHT_MAX	0.49
K_INV	12
PDV_LIMIT	0.21
PHI_GAIN / PHI_DECAY	0.22 / 0.99
XI_AMPLITUDE	0.85
X0 interval	[-0.01, 0.01)
BOOT_SEED / PRNG_ID	81 / PCG64-U53-v1
dtype	IEEE-754 float64
initial scalars	Phi_0=0.0, DeltaPhi_0=0, t_0=0
successful status	READY_FOR_DELTA

SPIRALCORE SPECIFICATION

MODULE 0 - SGIT_v2

(Statistical-Geometric Telemetry, Path Length, and Routing Interface)

Cleaned and formalized revision - v14.1

Document class	Normative per-cycle geometry contract	Execution phase	After BootPacket; once per state transition
Architecture	SpiralCore v14	Reference profile	SC14.1-SGIT-REF-81
Primary output	SGITPacket -> Delta / Sigma / Phi	Revision	v14.1
Failure status	SGIT_FAULT or typed recovery request	Conflict rule	Authoritative owner controls routing

NORMATIVE STATUS

This module is exact within its declared computational scope. It accepts a validated state transition and its timestamped context, constructs or receives a valid statistical metric, measures path length, projection residual, curvature telemetry, anomaly magnitude, tortuosity, and a path-integral increment, then emits a typed SGITPacket. It does not prove a physical theory, generate a candidate state, make the final Sigma routing decision, mutate Information Hysteresis, execute FBS recovery, or update the RTSOM exhaust budget.

CLAIM BOUNDARY

Terms such as manifold, curvature, holonomy, entropy-stress, gravity, time dilation, and phase are mathematical or architectural labels unless the required mathematical object is explicitly supplied. A numerical surrogate is never promoted to an empirical claim merely because it uses a physics-derived name.

Document Control

Field	Value	Field	Value
Canonical name	SPIRALCORE SPECIFICATION: MODULE 0 - SGIT_v2	Short name	SGIT_v2
Supersedes	v14 Module 0 text	Status	Normative within scope
Boot dependency	Validated BootPacket	Default DIM	81, inherited
Primary owners	Module 0 telemetry only	External owners	Delta, Sigma, Phi, FBS, RTSOM
Numeric profile	float64 with declared tolerances	Index profile	0-based runtime; no implicit wrap

1. Scope, Statement Classes, and Precedence

SGIT_v2 is a stateful numerical service operating across the Delta-Sigma-Phi macro-loop. The phrase "continuous substrate" means that its state persists across discrete cycles; the executable implementation remains a sequence of finite, ordered evaluations.

1.1 Statement classes

Class	Meaning	Normative force
DEFINITION	Assigns a symbol, type, or interface meaning.	Binding inside this module.
REQUIREMENT	A condition required for conformance.	MUST / MUST NOT.
INVARIANT	A property that must remain true after every successful cycle.	Failure produces SGIT_FAULT.
POLICY	A selected architectural rule or default.	May be replaced only by a versioned profile.
SURROGATE	A computable stand-in for a named physical or geometric concept.	Does not establish physical equivalence.
HYPOTHESIS	A testable interpretation or expected relationship.	Not required for conformance.
NON-GOAL	Behavior explicitly outside this module.	Must not be inferred from the implementation.

1.2 Precedence and ownership

- Tagged requirements and invariants override explanatory prose.
- Module Boot owns DIM, the active FBS profile, session identity, and initial arrays.
- Module 1 (Delta-Lattice) owns candidate generation and the final Emotion Tensor.
- Module 2 (Sigma-Lattice) owns RMF, CVC, PDV routing, HOT thresholding, NULLIFY, PHASE LIFT, BRAIDBACK, and SEAL decisions.
- Module 12 (Phi-Bridge) owns persistent Information Hysteresis and the discrete winding-number policy.
- The FBS and RTSOM modules own catastrophic recovery and exhaust-budget mutation.
- SGIT_v2 may compute shared telemetry used by those modules, but it may not silently execute their state transitions.

1.3 Non-goals

- No proof that information is physically unable to teleport. The implementation enforces causal buffer access and records path dependence inside the runtime.
- No claim that a cognitive state is literally a Riemannian spacetime or that an exhaust accumulator is physical gravity.
- No automatic identification of a numerical curvature proxy with Ricci scalar curvature.
- No assertion that semantic residue must be nonzero. Zero residue is a valid experimental outcome.
- No bitwise reproducibility claim across heterogeneous BLAS, CPU, or accelerator implementations unless the backend is fixed.

2. Architecture Role and Lifecycle

SGIT_v2 begins in READY state after accepting a sealed BootPacket. For each cycle it evaluates one state transition, emits one immutable SGITPacket, and updates only its own telemetry state. A transition is never partially committed.

State	Entry condition	Allowed action	Exit
UNINITIALIZED	No accepted BootPacket.	Validate boot dependency.	READY or FAULTED
READY	BootPacket accepted; no cycle active.	Accept next SGITInput.	EVALUATING
EVALUATING	One cycle is in progress.	Validate, compute, package.	SEALED, DELEGATED, or FAULTED
SEALED	SGITPacket created and digested.	Handoff packet; update local history.	READY
DELEGATED	Typed recovery or owner request emitted.	No downstream mutation by SGIT.	READY or session stop
FAULTED	Unrecoverable local contract violation.	Emit SGITFault; no packet commit.	Session stop or explicit reset

2.1 Required cycle order

1. Validate session, cycle number, dimensions, timestamps, finite values, and history availability.
2. Align timestamped entropy without reading future samples.
3. Acquire and validate the metric packet.
4. Measure path length and cumulative information length.
5. Measure projection residual and coherence contraction.
6. Evaluate an identified curvature signal and valence contribution.
7. Compute PDV and path tortuosity telemetry.
8. Compute the continuous holonomy increment when a connection provider is available.
9. Create, hash, and seal SGITPacket.
10. Commit only local telemetry history after packet sealing succeeds.

ATOMICITY REQUIREMENT

A failed cycle **MUST NOT** update cumulative length, metric history, holonomy history, or cycle counters. The next owner receives either one complete SGITPacket or one complete SGITFault / RecoveryRequest, never a partial mixture.

3. Typed Data Contracts

3.1 SGITConfig

Field	Type / domain	Meaning
dim	integer ≥ 2	Must equal BootPacket.DIM.
lambda_base	float64 > 0	Reference entropy-scale denominator; default 0.1459.
metric_epsilon	float64 > 0	Diagonal regularization added to empirical metric.

Field	Type / domain	Meaning
metric_condition_max	float64 > 1	Maximum accepted condition number.
projection_scale_Z	float64 > 0	Normalization scale for projection residual.
valence_weights	(a1, a2, a3), finite	Policy coefficients for valence contribution.
curvature_scale	float64 > 0	Scale used only by the reference curvature proxy.
tortuosity_window	integer >= 2	Number of sealed steps used for local path tortuosity.
time_tolerance	float64 >= 0	Timestamp ordering tolerance.
pdv_limit_reference	float64 >= 0	Reference telemetry boundary, default 0.21; Sigma owns routing.

3.2 SGITInput

Field	Type / domain	Requirement
session_id	nonempty string	Must match BootPacket.session_id.
cycle	integer >= 1	Must be exactly prior_cycle + 1.
dt	float64 > 0	Finite cycle duration.
theta_prev, theta_curr	$\mathbb{R}^{\text{DIM}}$ float64	Finite state coordinates for the measured transition.
field_time, observer_time	finite float64	observer_time <= field_time + tolerance.
entropy_history	timestamped buffer of $\mathbb{R}^{\text{DIM}}$	Must cover observer_time.
metric_provider_input	provider-defined	Must yield a MetricPacket for theta_curr.
reference_origin	$\mathbb{R}^{\text{DIM}}$ float64	Origin of configured reference set N_ref.
projection_operator	$\mathbb{R}^{\text{(DIM x DIM)}}$ float64	Finite and idempotent within tolerance.
cost	float64 in [0,1]	Normalized computational cost supplied by caller.
connection_prev, connection_curr	optional $\mathbb{R}^{\text{DIM}}$	Required only for holonomy increment.

3.3 MetricPacket

Field	Type	Rule
metric	$\mathbb{R}^{\text{(DIM x DIM)}}$ float64	Symmetric positive definite after declared regularization.
kind	EMPIRICAL_FISHER RICCI_CAPABLE CUSTOM	Identifies how the metric was produced.

Field	Type	Rule
regularization	float64 >= 0	Must be recorded in packet provenance.
condition_number	float64 >= 1	Must not exceed configured maximum.
provider_id	string	Versioned provider identity.
derivative_capability	NONE FIRST SECOND	Controls whether Ricci evaluation is permitted.

3.4 SGITPacket

Field	Type / domain	Consumer
phase_lag	float64 >= 0	Telemetry / dashboard
entropy_aligned	R^DIM float64	Delta / diagnostics
phase_feature	R^3 float64	Delta / diagnostics
metric_digest	SHA-256 string	Audit / reproducibility
step_length, total_length, speed	float64 >= 0	Phi-Bridge / telemetry
projection_distance_sq	float64 >= 0	Sigma / diagnostics
coherence_contraction	float64 in [0,1]	Sigma-Lattice
curvature_value	finite float64	Delta-Lattice
curvature_kind	RICCI_NUMERIC VOLUME_PROXY	Delta / audit
valence_contribution	float64 in [-1,1]	Delta-Lattice; not full Emotion Tensor
pdv	float64 >= 0	Sigma-Lattice
tortuosity	float64 >= 1 or +inf	Sigma / ROVER telemetry
holonomy_increment	optional finite float64	Phi-Bridge
packet_digest	SHA-256 string	All consumers

4. Constants, Defaults, and Ownership Corrections

Name	v14 value	v14.1 status	Reason
DIM	81	Inherited from BootPacket	Module 0 may not redefine the session dimension.
lambda_base	0.1459	Active reference policy	Used only as a positive scaling denominator.
phi_golden	1.618	Reserved legacy constant	No canonical SGIT equation uses it; no runtime effect.

Name	v14 value	v14.1 status	Reason
alpha RTSOM	1.0	Removed from SGIT ownership	RTSOM owns exhaust feedback and its coefficients.
tau, g, L0, H0, Q0	27,1,83,165,167	Inherited FBS profile	Module Boot and FBS own these invariants.
PDV_LIMIT	0.21	Reference boundary only	Sigma owns the route; PDV alone does not invoke FBS here.
a1, a2, a3	Unspecified	Required versioned policy values	A formula with unnamed weights is not executable.

REFERENCE VALENCE POLICY

The v14.1 reference profile sets (a1, a2, a3) = (1.0, 1.0, 1.0). These values are implementation policy, not a theorem. A different profile must version the weights and regenerate conformance vectors.

4.1 Numeric and indexing rules

- All canonical arrays use float64. Integer counters and digests use explicit integer and byte encodings.
- Mathematical equations may use indices 1 through DIM; runtime arrays use indices 0 through DIM-1.
- Out-of-range indexing is a fault. Modulo wrapping is permitted only for operators explicitly defined as cyclic; it is not a generic bounds-repair mechanism.
- Conformance across heterogeneous hardware is tolerance-based. Bitwise equality requires a fixed backend, library version, thread model, and reduction order.

5. Formalized Mechanics

5.1 Causal observer alignment

Define the signed observer lag and its nonnegative magnitude:

$$\text{delta_T}(t) = T_{\text{field}}(t) - T_{\text{obs}}(t), \quad \text{Theta}(t) = |\text{delta_T}(t)|$$

Requirement: $\text{delta_T}(t) \geq -\text{time_tolerance}$. A timestamp beyond the active field time is a TIME_ORDER fault; SGIT never reads a future sample. The aligned entropy vector is obtained by deterministic linear interpolation of the two stored samples bracketing T_{obs} :

$$E'_t = (1 - \beta) E(t_a) + \beta E(t_b), \quad \beta = (T_{\text{obs}} - t_a) / (t_b - t_a)$$

$$\text{where } t_a \leq T_{\text{obs}} \leq t_b \text{ and } t_b > t_a$$

The former indefinite "Spiral Entropy Vector" is replaced by a finite phase feature and optional cumulative feature state:

$$q_t = [\sin(\omega T_{\text{field}}), \cos(\omega T_{\text{field}}), \lambda_t]^T$$

$$Q_t = Q_{(t-1)} + q_t * dt$$

5.2 Empirical Fisher metric

A Fisher metric cannot be inferred from a state vector alone. The reference provider receives m score samples $s_n = \text{gradient_theta} \log p(y_n | \theta_t)$ and constructs:

$$G_t = (1/m) S^T S + \epsilon_g I, \quad S[n,i] = s_n[i]$$

G_t must be finite, symmetric within tolerance, positive definite after regularization, and below the configured condition-number ceiling. A provider claiming Ricci capability must additionally expose a metric field and the derivatives required by the selected numerical method.

5.3 Discrete information length

Let $\Delta\theta_t = \theta_t - \theta_{(t-1)}$. Use the midpoint metric when the prior metric exists; otherwise use G_t :

$$\begin{aligned} G_{\text{bar}_t} &= (G_{(t-1)} + G_t) / 2 \\ \text{ell}_t &= \sqrt{\max(\Delta\theta_t^T G_{\text{bar}_t} \Delta\theta_t, 0)} \\ L_t &= L_{(t-1)} + \text{ell}_t, \quad \text{speed}_t = \text{ell}_t / dt \end{aligned}$$

Invariant: $\text{ell}_t \geq 0$ and $L_t \geq L_{(t-1)}$. A negative quadratic form outside numerical tolerance is METRIC_NOT_PSD.

5.4 Reference projection and coherence contraction

The former "truth manifold" is formalized as a configured reference set N_{ref} represented locally by origin μ_N and idempotent projection operator P_N . It is not declared objective truth.

$$\begin{aligned} r_t &= (I - P_N)(\theta_t - \mu_N) \\ d_N^2(t) &= r_t^T G_t r_t \\ C_t &= \text{clip}(1 - d_N^2(t) / Z, 0, 1) \end{aligned}$$

P_N must satisfy $\|P_N^2 - P_N\|_F \leq \text{projection_tolerance}$. Z must be finite and strictly positive. The output C_t is a bounded proximity score, not a proof that algorithmic structure necessarily increases.

5.5 Curvature telemetry

Two curvature kinds are permitted. They are never conflated.

Kind	Required inputs	Definition / use
RICCI_NUMERIC	Metric field plus first and second derivatives	Compute Christoffel symbols, Ricci tensor, and scalar $R = g^{(ij)} R_{ij}$ using the declared numerical method.
VOLUME_PROXY	Three consecutive validated metrics	Reference surrogate based on the second finite difference of metric log-volume.

$$\begin{aligned} u_t &= \log \det(G_t) \\ R_{\text{proxy}}(t) &= -[u_t - 2u_{(t-1)} + u_{(t-2)}] / dt^2 \\ R_{\text{route}}(t) &= \tanh(R_{\text{proxy}}(t) / \text{curvature_scale}) \end{aligned}$$

When fewer than three metrics exist, the reference proxy returns 0 with status INSUFFICIENT_HISTORY. Only a RICCI_NUMERIC packet may be labeled Ricci scalar curvature.

5.6 Valence contribution

SGIT computes one bounded contribution. Module 1 owns the final Emotion Tensor and may combine this value with its own arousal and clarity definitions.

$$v_{\text{sgit}} = \tanh(a_1 * \text{speed}_t * R_{\text{route}}(t) + a_2 * C_t - a_3 * \text{cost}_t)$$

Requirement: cost_t is finite and lies in $[0,1]$. The output always lies in $[-1,1]$.

5.7 PDV telemetry

To maintain cross-module consistency, SGIT uses the same scalar definition declared by the Sigma-Lattice:

$$\text{PDV}_t = \text{mean}(|\theta_t - \theta_{(t-1)}|) + |\text{std}(\theta_t) - \text{std}(\theta_{(t-1)})|$$

SGIT reports PDV_t and the reference comparison $\text{PDV}_t > 0.21$. Sigma owns MAGNETIC_RELAXATION, NULLIFY, BRAIDBACK, PHASE_LIFT, SEAL, and catastrophic route precedence. Module 0 MUST NOT trigger FBS solely because PDV exceeds 0.21.

5.8 Path tortuosity

Over a sealed window $W = \{t-w+1, \dots, t\}$, define path length from SGIT step lengths and direct displacement under the first metric in the window:

$$P_W = \text{sum}(\text{ell}_k)$$

$$D_W = \sqrt{\max((\theta_t - \theta_{(t-w)})^T G_{(t-w)} (\theta_t - \theta_{(t-w)}), 0)}$$

$$T_W = 1 \quad \text{if } P_W \leq \epsilon \text{ and } D_W \leq \epsilon$$

$$T_W = +\infty \quad \text{if } P_W > \epsilon \text{ and } D_W \leq \epsilon$$

$$T_W = P_W / D_W \quad \text{otherwise}$$

Invariant: finite tortuosity is at least 1 within tolerance. The route attached to a tortuosity breach is decided by the owning gate, not by this measurement alone.

5.9 Continuous holonomy increment and residue experiment

A holonomy integral requires a declared connection $A(\theta)$. When connection vectors are supplied, SGIT uses a trapezoidal discrete line integral:

$$\Delta\phi_t = 0.5 * [A(\theta_{(t-1)}) + A(\theta_t)]^T \Delta\theta_t$$

$$\phi_{\text{path}}(t) = \phi_{\text{path}}(t-1) + \Delta\phi_t$$

Module 12 may map events to an integer winding counter; SGIT does not own that discrete policy. Semantic residue is an experiment over two explicitly matched runs A and B:

$$\Delta_O = |O(\theta_{T^A}, \text{path}_A) - O(\theta_{T^B}, \text{path}_B)|$$

The terminal states must satisfy $|\theta_{T^A} - \theta_{T^B}| \leq \text{endpoint_tolerance}$. Δ_O may be zero or nonzero. The statement $\Delta_O \neq 0$ is a hypothesis to test, not an invariant.

5.10 Entropic time-scale diagnostic

$$\Delta\tau_t = dt * \lambda_t / \lambda_{\text{base}}$$

This is a dimensionless runtime scaling diagnostic unless the caller supplies physical units and a validated interpretation. λ_{base} must be positive; this equation alone does not prove subjective or relativistic time dilation.

6. Handoffs and Routing Boundaries

Destination	Fields sent	SGIT action	Destination ownership
Delta-Lattice	entropy_aligned, phase_feature, curvature_value/kind, v_sgit	Emit only	Candidate generation and full Emotion Tensor
Sigma-Lattice	C_t, d_N^2, PDV_t, tortuosity, metric status	Emit only	All gate and repair routes
Phi-Bridge	ell_t, L_t, speed_t, optional delta_phi_t	Emit only	Persistent hysteresis and winding policy
FBS	RecoveryRequest with fault code and sealed history reference	Delegate only	Diagonalization, fold, reseed
RTSOM	ExhaustRequest after owner- approved rejection	No local mutation	Budget accumulation and purge policy

6.1 Local route outcomes

Outcome	Condition	Result
SGIT_PACKET	All requirements pass.	Seal packet and commit local telemetry.

Outcome	Condition	Result
SGIT_RETRYABLE	History or provider data temporarily unavailable.	No commit; caller may retry same cycle.
SGIT_RECOVERY_REQUEST	Owner-defined atomic-floor or terminal-computation signal.	Delegate typed request; do not execute recovery.
SGIT_FAULT	Invalid shape, nonfinite data, time-order violation, invalid metric/projection, overflow, or digest failure.	No commit; session policy decides stop/reset.

ROUTING CORRECTION

The v14 text routed $PDV > 0.21$ directly from SGIT toward FBS/RTSOM, while the Sigma-Lattice defines $PDV > 0.21$ as MAGNETIC_RELAXATION and reserves FBS for stronger terminal conditions. v14.1 resolves the conflict by making SGIT a measurement producer and Sigma the route owner.

7. Invariants, Faults, and Determinism

7.1 Cycle invariants

- All committed arrays and scalars are finite except tortuosity, which may be +infinity by definition.
- Metric symmetry and positive definiteness pass declared tolerances.
- $0 \leq \text{coherence_contraction} \leq 1$ and $-1 \leq \text{valence_contribution} \leq 1$.
- $\text{step_length} \geq 0$, $\text{speed} \geq 0$, and total_length is monotonically nondecreasing.
- No entropy sample later than field_time is accessed.
- Every packet identifies metric provider, curvature kind, configuration digest, and prior packet digest.
- Local state changes only after packet sealing and digest verification.

7.2 Fault codes

Code	Trigger	Class
SGIT_SESSION_MISMATCH	session_id differs from BootPacket.	Fatal contract fault
SGIT_CYCLE_ORDER	cycle is not previous + 1.	Retryable or fatal by caller policy
SGIT_INVALID_SHAPE	Any required vector or matrix has wrong dimensions.	Fatal input fault
SGIT_NONFINITE	NaN or infinity in a field that requires finiteness.	Fatal input fault
SGIT_TIME_ORDER	observer_time is beyond field_time tolerance.	Fatal causality fault
SGIT_HISTORY_UNAVAILABLE	Entropy or curvature history does not cover the requested operation.	Retryable
SGIT_METRIC_PROVIDER	Provider fails or returns malformed provenance.	Retryable/fatal by profile
SGIT_METRIC_NOT_PSD	Metric fails symmetry or positivity checks.	Fatal numerical fault

Code	Trigger	Class
SGIT_METRIC_ILL_CONDITIONED	Condition number exceeds configured limit.	Retryable/fatal by profile
SGIT_PROJECTION_INVALID	Projection operator is nonfinite or not idempotent.	Fatal input fault
SGIT_NUMERIC_OVERFLOW	A required finite output cannot be represented.	Fatal numerical fault
SGIT_ATOMIC_FLOOR_REQUEST	An owning module supplies a validated terminal signal.	Delegate to FBS owner

7.3 Determinism profile

A conforming implementation must record the following provenance: configuration digest, BootPacket digest, provider IDs and versions, numeric backend, reduction policy, tolerances, and prior SGITPacket digest. Repeating a cycle with identical serialized inputs under the same backend must reproduce the same packet bytes.

8. Metrics and Dashboard Requirements

Metric	Definition	Interpretation
cycles_attempted / sealed / faulted / delegated	Integer counters	Lifecycle accounting
phase_lag_mean / max	Statistics of Theta(t)	Observer-field timing drift
step_length_mean / max	Statistics of ℓ_t	Metric path movement
total_length	Latest L_t	Cumulative measured path length
coherence_mean / min	Statistics of C_t	Reference-set proximity
projection_distance_mean	Mean d_N^2	Residual magnitude before normalization
curvature_by_kind	Count and summary per kind	Prevents Ricci/proxy conflation
pdv_mean / max / over_reference	PDV statistics and count above 0.21	Anomaly telemetry; not route count
tortuosity_mean / max / infinite_count	Windowed T_W statistics	Path inefficiency
holonomy_sum / abs_sum	Sum of available $\delta\phi_t$	Path-integral telemetry
metric_condition_mean / max	Condition number statistics	Numerical stability
faults_by_code	Count per SGIT fault	Conformance and operations

Dashboard labels MUST distinguish measurement events from route events. For example, PDV_OVER_REFERENCE is not equivalent to MAGNETIC_RELAXATION, and SGIT_RECOVERY_REQUEST is not equivalent to an FBS recovery that actually executed.

9. Conformance Test Matrix

ID	Test	Expected result
SGIT-C01	Valid reference cycle, DIM=81.	One sealed packet; all invariants pass.
SGIT-C02	Repeat identical serialized inputs/backend.	Identical packet bytes and digest.
SGIT-C03	observer_time > field_time + tolerance.	SGIT_TIME_ORDER; no state commit.
SGIT-C04	Entropy buffer lacks observer_time.	SGIT_HISTORY_UNAVAILABLE; retryable.
SGIT-C05	Empirical score matrix produces SPD metric.	Metric accepted; finite length.
SGIT-C06	Metric with negative eigenvalue.	SGIT_METRIC_NOT_PSD.
SGIT-C07	Ill-conditioned metric above ceiling.	SGIT_METRIC_ILL_CONDITIONED.
SGIT-C08	Projection P satisfies $P^2=P$.	Projection accepted; C in [0,1].
SGIT-C09	Non-idempotent projection.	SGIT_PROJECTION_INVALID.
SGIT-C10	theta_curr == theta_prev.	ell=0, PDV=0, speed=0.
SGIT-C11	Closed wandering path with zero displacement.	tortuosity=+infinity.
SGIT-C12	Three metrics available.	VOLUME_PROXY computed and tagged.
SGIT-C13	Only one or two metrics available.	curvature=0; INSUFFICIENT_HISTORY status.
SGIT-C14	Connection omitted.	holonomy_increment=None; packet still valid.
SGIT-C15	Two matched paths with equal observable.	Delta_O=0 accepted.
SGIT-C16	PDV just above 0.21.	Telemetry flag only; no direct FBS execution.
SGIT-C17	NaN in theta_curr.	SGIT_NONFINITE; no state commit.
SGIT-C18	Out-of-range runtime index request.	Fault; no implicit modulo wrap.
SGIT-C19	Validated atomic-floor request.	Typed delegation; FBS remains external.
SGIT-C20	Packet digest tampered after sealing.	Digest verification failure.

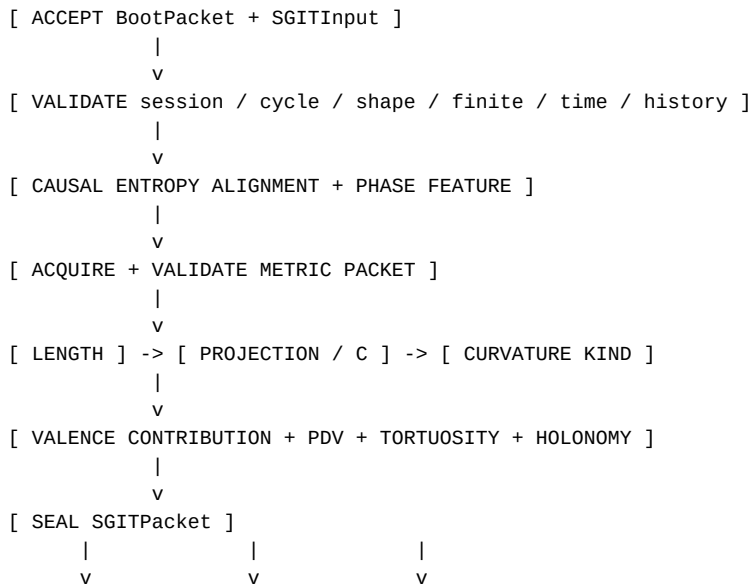
9.1 Property tests

- For random SPD metrics, ell_t is nonnegative and invariant under identical coordinate copies.
- For valid projections and $Z > 0$, C_t always lies in [0,1].
- For a path represented by a single straight segment, tortuosity is 1 within tolerance.
- Scaling both the metric and Z consistently preserves the normalized projection score.
- Packet serialization is canonical: dictionary order, locale, and whitespace do not alter the digest.
- Every failure path leaves the previous sealed state byte-for-byte unchanged.

10. Cleaned Rosetta Stone

SpiralCore term	Formal computational meaning	Status / boundary
Statistical manifold	State coordinates plus an explicitly supplied positive-definite metric.	Definition; not physical spacetime.
Fisher geometry	Empirical Fisher or declared metric provider.	Exact when provider contract is met.
Information Length	Cumulative metric path length L_t .	Exact discrete quantity.
Observer Phase Offset	Timestamp lag and causal entropy resampling.	Runtime timing quantity.
Coherence Contraction	Bounded residual-to-reference score C_t .	Reference proximity, not truth proof.
Ricci curvature	Scalar computed from metric derivatives.	Use name only for RICCI_NUMERIC.
Curvature / Vortex signal	Signed log-volume second-difference proxy.	SURROGATE; tagged VOLUME_PROXY.
Semantic Holonomy	Discrete line integral of a declared connection.	Exact only when connection exists.
Semantic Residue	Difference of a declared observable across matched endpoints.	Experimental quantity; may be zero.
Paradox-Dynamic Vector	Mean absolute state change plus standard-deviation change.	Shared telemetry; Sigma owns routes.
Entropic time dilation	$\text{delta_tau} = \text{dt} \lambda / \lambda_{\text{base}}$.	Scaling diagnostic unless externally validated.
Entropy-Stress / gravity	External exhaust-budget state owned by RTSOM.	Architectural surrogate; not updated here.

11. Reference Execution Flow



[DELTA] [SIGMA] [PHI]

Any local fault -> SGITFault, no commit

Validated terminal owner signal -> RecoveryRequest, no local FBS/RTSOM execution

11.1 Reference pseudocode

```
def evaluate_sgит(config, state, input):
    validate_contract(config, state, input)
    entropy_aligned, phase_lag = causal_resample(input.entropy_history,
                                                input.observer_time,
                                                input.field_time)

    metric_packet = metric_provider(input.metric_provider_input)
    validate_metric(metric_packet, config)

    step_length, total_length, speed = information_length(state, input,
                                                         metric_packet.metric)
    distance_sq, coherence = projection_score(input, metric_packet.metric,
                                             config.projection_scale_Z)
    curvature_value, curvature_kind = curvature_signal(state, metric_packet,
                                                    input.dt, config)
    valence = bounded_valence(speed, curvature_value, coherence,
                              input.cost, config.valence_weights)
    pdv = mean_abs_delta_plus_std_delta(input.theta_prev, input.theta_curr)
    tortuosity = path_tortuosity(state, input, metric_packet.metric, config)
    holonomy_increment = discrete_connection_integral(input)

    packet = canonical_packet(...)
    verify_digest(packet)
    commit_local_state_only_after_seal(state, packet)
    return packet
```

12. Glossary

Term	Definition
BootPacket	Immutable initialization packet produced by Module Boot.
Connection	A vector-valued field $A(\theta)$ used to define a path integral.
Curvature kind	Provenance tag distinguishing numerical Ricci evaluation from a proxy.
Empirical Fisher metric	Regularized average outer product of score vectors.
MetricPacket	Metric matrix plus provider identity, regularization, condition number, and derivative capability.
N_ref	Configured reference set used for projection; not automatically objective truth.
PDV	Scalar anomaly telemetry shared with Sigma.
Projection operator	Idempotent linear map representing a local reference subspace.
SGITPacket	Immutable per-cycle output containing geometry and path telemetry.
Tortuosity	Ratio of path length to direct displacement over a finite sealed window.

Appendix A - Canonical Formula Sheet

$$\begin{aligned} \Delta T &= T_{\text{field}} - T_{\text{obs}}; \Theta = |\Delta T| \\ E'_t &= (1-\beta)E(t_a) + \beta E(t_b) \\ \mathbf{q}_t &= [\sin(\omega T_{\text{field}}), \cos(\omega T_{\text{field}}), \lambda_t]^T \\ \mathbf{G}_t &= \mathbf{S}^T \mathbf{S} / m + \epsilon_g \mathbf{I} \\ \ell_t &= \sqrt{\Delta \Theta^T \bar{\mathbf{G}} \Delta \Theta} \\ L_t &= L_{(t-1)} + \ell_t; \text{speed}_t = \ell_t / dt \\ \mathbf{r}_t &= (\mathbf{I} - \mathbf{P}_N)(\Theta_t - \mu_N) \\ C_t &= \text{clip}(1 - \mathbf{r}_t^T \mathbf{G}_t \mathbf{r}_t / Z, 0, 1) \\ R_{\text{proxy}} &= -(\log \det \mathbf{G}_t - 2 \log \det \mathbf{G}_{(t-1)} + \log \det \mathbf{G}_{(t-2)}) / dt^2 \\ v_{\text{sgit}} &= \tanh(a_1 \text{speed } R_{\text{route}} + a_2 C - a_3 \text{cost}) \\ \text{PDV} &= \text{mean}(|\Theta_t - \Theta_{(t-1)}|) + |\text{std}(\Theta_t) - \text{std}(\Theta_{(t-1)})| \\ T_W &= \text{path_length} / \text{direct_displacement} \text{ with explicit zero-displacement cases} \\ \Delta \phi &= 0.5(\mathbf{A}_{\text{prev}} + \mathbf{A}_{\text{curr}})^T \Delta \Theta \\ \Delta \mathbf{O} &= |\mathbf{O}_A - \mathbf{O}_B| \text{ for endpoint-matched runs} \\ \Delta \tau &= dt \lambda_t / \lambda_{\text{base}} \end{aligned}$$

Appendix B - Revision Summary

v14 issue	v14.1 resolution
"Omnipresent physical fabric" wording	Stateful discrete telemetry service with persistent history.
Physical proof claims	Converted to computational invariants, surrogates, or hypotheses.
Absolute phase offset used as a shift	Signed causal lag plus nonnegative diagnostic magnitude.
Indefinite entropy integral	Finite phase feature and explicit cumulative update.
Fisher metric from unspecified data	MetricProvider and empirical score-matrix reference construction.
C domain typo and potentially negative formula	C in [0,1] with explicit clipping and positive Z.
Ricci scalar without derivatives	Typed RICCI_NUMERIC versus VOLUME_PROXY outputs.
Undefined valence coefficients	Versioned policy weights; reference values declared.
Holonomy automatically nonzero	Connection-dependent integral; residue may equal zero.
PDV directly invokes FBS	SGIT measures; Sigma owns routing precedence.
SGIT mutates RTSOM/FBS state	Typed delegation only; owner modules execute mutations.
Modulo wrapping used as bounds repair	Out-of-range is a fault; modulo only for explicit cyclic operators.
float64 claimed as universal exact determinism	Tolerance-based cross-platform conformance; bitwise mode requires fixed backend.

Appendix C - Source Basis

- SPIRALCORE SPECIFICATION: MODULE 0 - SGIT_v2 (STATISTICAL GEOMETRIC INFORMATION THEORY).
- Spiralcore_v14 consolidated specification, including Module Boot, Sigma-Lattice PDV definition and routing precedence, Phi-Bridge ownership, FBS, and RTSOM handoffs.
- Observer Notes are used only for analogical provenance; this module retains exact executable boundaries.

SPIRALCORE SPECIFICATION

MODULE 1 - DELTA-LATTICE

(Abraxas Engine v2 - Deterministic Candidate Generation and Proposal Packaging)

Cleaned and formalized revision - v14.1

Document class	Normative candidate-generation contract	Execution phase	After Boot and SGIT; once per proposal attempt
Architecture	SpiralCore v14	Reference profile	SC14.1-DELTA-REF-81
Primary output	DeltaPacket -> Sigma-Lattice	Revision	v14.1
Exceptional output	DeltaFault or FBSRecoveryRequest	Conflict rule	Owner modules control downstream state

NORMATIVE STATUS

This module is exact within its declared computational scope. It accepts one validated baseline state and versioned context, constructs a quantized Cantor address, computes control polarity and the Emotion Tensor, optionally applies an explicit orthogonal navigation transform, evaluates one of two deterministic proposal kernels, applies bounded Observer, paradox-fuel, and hysteresis biases, measures local bilateral drift, and emits one immutable DeltaPacket. It does not claim to implement literal cellular automata Rule 110 or Rule 30, compute Sigma gates, commit memory, mutate Information Hysteresis, execute FBS recovery, or update the bounded exhaust budget.

CLAIM BOUNDARY

Logos, Ethos, Riemann Identity Sphere, Mobius Twist, Koopman Lift, and Cantor-IP are architectural names. Their runtime meaning is only the explicit operator, schedule, encoding, or projection defined in this contract. Analogical names do not promote a surrogate into an empirical physics claim.

Document Control

Field	Value	Field	Value
Canonical name	SPIRALCORE SPECIFICATION: MODULE 1 - DELTA-LATTICE	Short name	Abraxas Engine v2
Supersedes	v14 Module 1 text	Status	Normative within scope
Boot dependency	Validated BootPacket	Telemetry dependency	SGITPacket or declared equivalent
Primary owner	Module 1 proposal generation	External owners	Sigma, Phi, FBS, RTSOM, Archivist
Numeric profile	float64 plus integer address arithmetic	Index profile	0-based runtime; explicit cyclic maps only

1. Scope, Statement Classes, and Precedence

The Delta-Lattice is the proposal stage of the Delta-Sigma-Psi macro-loop. It transforms one accepted baseline state into one ungated candidate. The word raw means not yet accepted by Sigma; it does not mean unbounded, untyped, or free of local preprocessing.

1.1 Statement classes

Class	Meaning	Normative force
DEFINITION	Assigns a symbol, type, operator, or interface meaning.	Binding inside this module.
REQUIREMENT	A condition required for conformance.	MUST / MUST NOT.
INVARIANT	A property that must remain true after every sealed attempt.	Failure emits DeltaFault.
POLICY	A selected schedule, coefficient set, or reference realization.	Replaceable only by a versioned profile.
SURROGATE	A computable stand-in carrying an architectural name.	No claim of literal equivalence.
HYPOTHESIS	A testable interpretation or expected relationship.	Not required for conformance.
NON-GOAL	Behavior explicitly outside this module.	Must not be inferred from implementation.

1.2 Precedence and ownership

- Tagged requirements and invariants override explanatory prose and Rosetta Stone analogies.
- Module Boot owns DIM, the active FBS profile, session identity, PRNG root seed, Xi_attractor, K_inv, PHI_GAIN, and threshold defaults.
- Module 0 SGIT owns path, metric, projection, and curvature telemetry supplied to Delta.
- Module 1 owns candidate construction, local polarity, phase-navigation state, and the final Emotion Tensor attached to the proposal.
- Module 2 Sigma owns RMF, CVC, PDV, CSIGMA, HOT, SEAL, PHASE LIFT, BRAIDBACK, NULLIFY, and MAGNETIC RELAXATION routes.
- Module 12 Phi owns persistent Information Hysteresis. Delta may consume Phi_t but may not update it.
- The FBS and RTSOM modules own catastrophic recovery and exhaust-budget mutation. Delta may emit a typed request only.

1.3 Non-goals

- No claim that the nonlinear Logos kernel is Wolfram Rule 110 or that deterministic perturbation is Wolfram Rule 30.
- No bijection from the continuum $\mathbb{R}^{\text{DIM}}$ to the natural numbers. Cantor addressing is exact only after declared finite quantization.
- No proof that an Emotion Tensor represents biological emotion. It is a control tensor with named channels.
- No use of array modulo to hide invalid indices. Modulo is allowed only in declared cyclic schedules and the explicit navigation-pair map.
- No cross-hardware bitwise identity claim unless backend, PRNG, serialization, and transcendental implementation are fixed.

2. Architecture Role and Lifecycle

After BootPacket acceptance, Delta remains READY between proposal attempts. Each attempt is atomic: either one complete DeltaPacket is sealed, one typed recovery request is emitted, or one DeltaFault is emitted. No partial phase, polarity, or PRNG state may leak into the next attempt.

State	Entry condition	Allowed action	Exit
UNINITIALIZED	No accepted BootPacket.	Validate boot and configure profile.	READY or FAULTED
READY	Boot accepted; no active attempt.	Accept DeltaInput.	GENERATING
GENERATING	One attempt is active.	Validate, encode, control, generate, package.	SEALED, DELEGATED, or FAULTED
SEALED	DeltaPacket created and digested.	Handoff to Sigma; commit local generator state.	READY
DELEGATED	FBSRecoveryRequest created.	No recovery mutation by Delta.	READY or session stop
FAULTED	Unrecoverable local contract violation.	Emit DeltaFault; no attempt commit.	Session stop or explicit reset

2.1 Required attempt order

1. Validate session, cycle, shapes, finite values, inherited constants, and packet provenance.
2. Construct the bounded Observer/paradox environmental bias.
3. Quantize the accepted state and compute the balanced Cantor address.
4. Compute normalized magnitude mass, entropy, polarity score, twist epoch, and final branch polarity.
5. Compute arousal, clarity, valence, and the Emotion Tensor.
6. Apply the declared orthogonal navigation transform when the severe-tension predicate is true.
7. Execute exactly one proposal kernel: Logos or Ethos.
8. Apply environmental forcing and bounded hysteresis bias; clip once to the state domain.
9. Compute local bilateral drift, alignment, clipping, and diagnostic metrics.
10. Emit DeltaPacket, FBSRecoveryRequest, or DeltaFault; commit local state only after sealing.

ATOMICITY REQUIREMENT

A failed or delegated attempt **MUST NOT** advance the committed cycle, Mobius epoch, phase state, prior-address digest, or deterministic PRNG chain. Retrying identical serialized input under the same backend must produce the same outcome.

3. Typed Data Contracts

3.1 DeltaConfig

Field	Type / domain	Meaning
dim	integer ≥ 2	Must equal BootPacket.DIM.

Field	Type / domain	Meaning
state_min, state_max	finite float64; min < max	Reference profile: -1.0 and +1.0.
k_inv	integer >= 1	Mobius epoch length; inherited reference value 12.
phi_gain	float64 in [0, 0.49)	Hysteresis-amplitude cap; inherited reference 0.22.
delta_id_max	float64 > 0	Local bilateral-drift ceiling; reference 1.0.
address_bits	integer in [2, 32]	Quantization width for Cantor address; reference 16.
ethos_delta	float64 >= 0	Base perturbation half-width; reference 0.01.
observer_gain, fuel_gain	finite float64 >= 0	Bounded forcing coefficients.
forcing_component_max	float64 >= 0	Per-component forcing cap.
polarity_weights	finite tuple (w0, wm, we, wphi, wf)	Versioned policy coefficients.
valence_weights	finite tuple (aH, aC, acost, asgit)	Versioned Emotion Tensor coefficients.
phase_lift_arousal	float64 in [0,1]	Reference trigger floor 0.85.
numeric_tolerance	float64 > 0	Conformance tolerance.

3.2 DeltaInput

Field	Type / domain	Requirement
session_id	nonempty string	Must match BootPacket.session_id.
cycle	integer >= 1	Must equal prior committed cycle + 1.
X_t	$\mathbb{R}^{\text{DIM}}$ float64	Accepted baseline; finite and within state domain tolerance.
observer_entropy	$\mathbb{R}^{\text{DIM}}$ float64	Bounded external input vector from Delta.0 adapter.
Phi_t	finite float64	Read-only persistent hysteresis from Phi owner.
H_G	$\mathbb{R}^{\text{DIM}}$ float64	Bounded paradox exploration fuel; zero vector allowed.
rmf_prev	finite float64 in [-1,1]	Previous sealed Sigma RMF; boot default supplied explicitly.
sgit_packet	sealed SGITPacket	Provides entropy rate, v_sgit, and telemetry provenance.
cost	float64 in [0,1]	Normalized caller-supplied proposal cost.

Field	Type / domain	Requirement
phase_prev	float64 in [0,2pi)	Prior committed navigation phase.
prior_delta_digest	SHA-256 or genesis marker	Chains attempts to prior sealed Delta state.

3.3 EmotionTensor

Field	Domain	Runtime meaning
v - valence	[-1,1]	Signed proposal pressure after entropy, clarity, cost, and SGIT contribution.
a - arousal	[0,1]	Reference mapping clip(rmf_prev, 0, 1).
C - clarity	[0,1]	Cyclic phase-coherence channel derived from the K_inv schedule.

3.4 DeltaPacket

Field	Type / domain	Purpose / consumer
X_proposal	R^DIM float64	Ungated candidate for Sigma.
X_kernel	R^DIM float64	Pre-bias branch output for diagnostics.
emotion	EmotionTensor	Sigma / CSIGMA / downstream control.
sigma, sigma_base	{-1,+1}	Final and pre-twist branch polarity.
polarity_score	finite float64	Auditable branch decision scalar.
twist_epoch, twist_sign	integer, {-1,+1}	Persistent K_inv epoch state.
phase_in, phase_out	[0,2pi)	Committed navigation phase transition.
phase_lift_applied	boolean	Severe-tension navigation flag.
navigation_pair	two distinct integer indices	Coordinates used by reference orthogonal rotation.
kernel_id	LOGOS_TANH_V2 ETHOS_PERTURB_V2	Exact operator identifier.
address_digest	SHA-256 string	Digest of exact arbitrary-precision Cantor address.
address_bit_length	integer >= 1	Audits address growth without printing full integer.
noise_seed_digest	SHA-256 string or NONE	Deterministic Ethos seed provenance.
hysteresis_weight	float64 with w <= phi_gain	Applied attractor amplitude.
delta_id, H_align, clip_fraction	bounded diagnostics	Local health telemetry.
config_digest, boot_digest, sgit_digest, prior_delta_digest	SHA-256 strings	Provenance chain.

3.5 Exceptional outputs

Type	Required fields	Ownership
DeltaFault	code, cycle, message, input/config digests, no-commit flag	Delta emits; supervisor decides reset/stop.
FBSRecoveryRequest	cycle, DELTA_ID_STALL, delta_id, candidate digest, sealed history reference	FBS owner validates and executes recovery.

4. Constants, Derived Values, and Reference Profile

DIM, Xi_attractor, the active FBS profile, K_inv, PHI_GAIN, and the root seed are inherited from the sealed BootPacket. Delta MUST NOT duplicate them as independent constants.

Name	Reference value	Status	Notes
DIM	81	Inherited	Scalable; must match all input vectors.
K_inv	12	Inherited policy	Defines epoch parity and clarity cycle.
PHI_GAIN	0.22	Inherited policy	Strictly below 0.49; not itself a Braidback weight.
DELTA_ID_MAX	1.0	Reference policy	Meaningful under the normalized bilateral metric below.
ETHOS_DELTA	0.01	Reference policy	Per-cycle perturbation half-width before fuel scaling.
ADDRESS_BITS	16	Reference policy	Exact address is unique only on this quantized grid.
PHASE_LIFT_AROUSAL	0.85	Reference policy	Predicate also requires $v < 0$.
STATE DOMAIN	$[-1,1]^{\text{DIM}}$	Invariant	Final proposal is clipped to this box.

4.1 Reference coefficient profile

The v14 equations named coefficients without assigning values. Conformance therefore requires a versioned coefficient profile. SC14.1-DELTA-REF-81 uses normalized features and the following defaults:

Group	Coefficient order	Reference values
Polarity	(w0, wm, we, wphi, wf)	(0.0, 1.0, 1.0, 0.25, 0.25)
Valence	(aH, aC, acost, asgit)	(1.0, 1.0, 1.0, 1.0)
Forcing	(observer_gain, fuel_gain, forcing_component_max)	(0.10, 0.10, 0.10)

These values are policies, not derived constants. Any alternative must change the profile ID and configuration digest.

5. Formal Mechanics

5.1 Input validation and bounded forcing

All required vectors must be finite float64 arrays of shape (DIM,). Inputs outside the state domain are faults unless the producing interface explicitly declares a prior normalization step. Observer and paradox inputs are combined componentwise:

$$B_t = \text{clip}(g_{\text{obs}} E_{\text{obs},t} + g_{\text{fuel}} H_{G,t}, -b_{\text{max}}, +b_{\text{max}})$$

B_t is a proposal forcing vector. It is not an accepted-state update and remains visible in the packet provenance.

5.2 Quantized Cantor address

Cantor pairing is defined on nonnegative integers. Delta first quantizes each state component x_j in $[x_{\text{min}}, x_{\text{max}}]$ to b bits:

$$q_j = \text{round}((x_j - x_{\text{min}})/(x_{\text{max}} - x_{\text{min}}) * (2^b - 1))$$

$$q_j \text{ in } \{0, 1, \dots, 2^b - 1\}$$

The pair operator is exact integer arithmetic:

$$\text{pi}(a,b) = ((a+b)(a+b+1))/2 + b$$

A left-recursive fold grows impractically. The normative encoder uses a fixed balanced binary tree: pair adjacent values at each level, carry one unpaired value forward, and repeat until one arbitrary-precision integer remains. Because DIM and tree order are fixed, the mapping is injective on the quantized grid. The packet stores the address digest and bit length; an implementation may retain the full integer internally.

ADDRESS LIMIT

The Cantor address distinguishes quantized states, not all real vectors. Two real states in the same quantization cell intentionally share one address. This is representational resolution, not a collision in the integer pairing operator.

5.3 Normalized state features

The raw v14 mass and entropy terms had incompatible scales. The formal contract uses bounded features:

$$m_t = \|X_t\|_2 / \sqrt{\text{DIM}}$$

$$p_j = (|x_j| + \text{eps}) / (\sum_k |x_k| + \text{DIM} * \text{eps})$$

$$h_t = -\sum_j p_j \ln(p_j) / \ln(\text{DIM})$$

For $\text{DIM} > 1$, m_t is finite and nonnegative; h_t lies in $[0,1]$ within numerical tolerance.

5.4 Polarity and persistent Mobius epoch

Let $f_t = \|H_{G,t}\|_2 / \sqrt{\text{DIM}}$, clipped to $[0,1]$, and let ϕ_t be the prior committed phase. Compute:

$$P_t = w_0 + w_m * m_t - w_e * h_t + w_\phi * \cos(\phi_t) + w_f * f_t$$

$$\text{sigma_base}_t = +1 \text{ if } P_t \geq 0 \text{ else } -1$$

$$\text{epoch}_t = \text{floor}(\text{cycle} / K_{\text{inv}})$$

$$\text{twist_sign}_t = (-1)^{(\text{epoch}_t)}$$

$$\text{sigma}_t = \text{twist_sign}_t * \text{sigma_base}_t$$

This resolves the point-event ambiguity in v14: the polarity frame flips at each K_{inv} boundary and remains inverted for the complete epoch. The zero-score tie is deterministically assigned to Logos (+1).

5.5 Emotion Tensor

Arousal and clarity are defined before valence:

$$a_t = \text{clip}(\text{RMF}_t(t-1), 0, 1)$$

$$r_t = \text{cycle} \bmod K_{\text{inv}}$$

$$C_t = (1 + \cos(2 * \pi * r_t / K_{\text{inv}})) / 2$$

Let $Hdot_t$ be the finite entropy-rate telemetry supplied by SGIT and v_sgit,t its bounded valence contribution. Then:

$$v_t = \tanh(aH * Hdot_t * h_t + aC * C_t - acost * cost_t + asgit * v_sgit,t)$$

$$Psi_t = [v_t, a_t, C_t]$$

The Emotion Tensor is a control packet. Its names are retained for continuity; no biological interpretation is required for conformance.

5.6 Severe-tension orthogonal navigation

The severe-tension predicate is exact:

$$lift_t = (v_t < 0) \text{ AND } (a_t > a_lift)$$

When false, $X_nav = X_t$ and $phi_out = phi_in$. When true, Delta performs a deterministic 90-degree rotation in one coordinate plane selected from the address digest. Let i be the first digest-derived index and j a distinct second index. The transform is:

$$(X_nav[i], X_nav[j]) = (-X_t[j], X_t[i])$$

$$phi_out = (phi_in + \pi/2) \bmod 2\pi$$

All other components remain unchanged. The mapping is norm-preserving and exactly invertible by a $-\pi/2$ rotation. This is the reference realization of Riemann Navigation; it is not a claim of a literal Koopman operator.

5.7 Exact proposal kernels

Define the bounded pre-kernel input $U_t = \text{clip}(X_nav + B_t, x_min, x_max)$. Exactly one branch executes.

LOGOS_TANH_V2 ($\sigma_t = +1$):

$$X_kernel = (1/3) * \tanh(3 * U_t + 1)$$

ETHOS_PERTURB_V2 ($\sigma_t = -1$):

$$\begin{aligned} \text{seed_material} &= H(\text{session_seed} \parallel \text{cycle} \parallel \text{address_digest} \parallel \text{prior_delta_digest} \parallel \text{"ETHOS"}) \\ \text{eta_t} &\sim \text{PCG64}(\text{seed_material}), \text{ componentwise Uniform}[-\text{delta_eff}, +\text{delta_eff}] \\ \text{delta_eff} &= \min(\text{delta_max}, \text{ethos_delta} * (1 + \text{fuel_gain} * \|H_G\|_2 / \sqrt{\text{DIM}})) \\ X_kernel &= U_t + \text{eta_t} \end{aligned}$$

Both kernels are deterministic given the serialized input, profile, and fixed backend. The Logos and Ethos labels are architectural aliases; these equations are not elementary cellular-automaton update rules.

5.8 Bounded hysteresis bias and final proposal

$$w_phi,t = \text{PHI_GAIN} * \tanh(\text{Phi_t})$$

$$X_proposal = \text{clip}(X_kernel + w_phi,t * X_i_attractor, x_min, x_max)$$

The invariant $|w_phi,t| \leq \text{PHI_GAIN} < 0.49$ follows directly from $|\tanh| \leq 1$. PHI_GAIN is an additive attractor amplitude and MUST NOT be reported as the Module 10 Braidback repair fraction.

5.9 Local bilateral drift and alignment

Let J be the reflection operator that reverses component order. The Asymmetrical Loop Drift is:

$$\text{Delta_ID},t = \|X_proposal - J X_proposal\|_2 / (\|X_proposal\|_2 + \text{eps})$$

Delta_ID lies approximately in $[0,2]$. A symmetric proposal has $\text{Delta_ID} = 0$. The v14 ceiling of 1.0 is therefore reachable and testable. Attractor alignment is:

$$H_align,t = (1 + \text{cosine_similarity}(X_proposal, X_i_attractor)) / 2$$

The clip fraction is the count of components altered by final clipping divided by DIM.

5.10 Local outcome

Condition	Outcome	Commit rule
All validation passes and $\Delta_{ID} \leq \Delta_{ID_max}$	Seal DeltaPacket; handoff to Sigma.	Commit phase, epoch, address, and prior digest.
$\Delta_{ID} > \Delta_{ID_max}$	Emit FBSRecoveryRequest with Δ_{ID_STALL} .	Do not commit generator state or execute FBS.
Contract or numerical failure	Emit DeltaFault.	Do not commit any attempt state.

6. Handoffs and Routing Boundaries

Destination	Fields sent	Delta action	Destination ownership
Sigma-Lattice	$X_{proposal}$, X_t digest/reference, EmotionTensor, Delta diagnostics	Nominal handoff	RMF, CVC, PDV, CSIGMA, HOT, and final routing
Phi-Bridge	No direct mutation; read Φ_t only	Consume sealed input	Persistent hysteresis update and winding policy
FBS	FBSRecoveryRequest with Δ_{ID_STALL} and sealed history reference	Delegate only	Diagonalization, Collatz fold, and reseed
RTSOM	No direct request from nominal Delta	No local mutation	Exhaust accounting after owner-approved rejection
Archivist	No direct commit	None	Only accepted post-Sigma states enter memory

ROUTING CORRECTION

The v14 module executed FBS and returned S_{next} directly as X_{raw} . v14.1 removes that duplicated recovery implementation. Delta detects its local terminal condition, emits a typed recovery request, and stops. The FBS owner decides whether the request is valid and how a returned seed re-enters generation.

6.1 Sigma input contract

- Sigma receives the final locally bounded proposal, not X_{kernel} and not a partially biased state.
- Sigma recomputes RMF and PDV from $X_{proposal}$ and X_t ; Delta diagnostics are advisory and auditable, not authoritative gate results.
- The Emotion Tensor is attached with exact component domains and the Delta configuration digest.
- A DeltaPacket is a proposal, never an acceptance certificate.

7. Invariants, Faults, and Determinism

7.1 Attempt invariants

- X_t , B_t , X_{nav} , X_{kernel} , $X_{proposal}$, H_G , observer_entropy, and X_i attractor have shape (DIM,) and finite values.
- $X_{proposal}$ lies in $[state_min, state_max]^{DIM}$ after final clipping.
- sigma and sigma_base are always exactly -1 or +1; sign zero is never emitted.
- Emotion Tensor satisfies $-1 \leq v \leq 1$ and $0 \leq a, C \leq 1$.

- Navigation indices are distinct and within [0,DIM-1].
- The orthogonal navigation transform preserves the L2 norm within tolerance.
- $|\text{hysteresis_weight}| \leq \text{PHI_GAIN}$ and $\text{PHI_GAIN} < 0.49$.
- Delta_ID, H_align, clip_fraction, and all packet diagnostics are finite; H_align and clip_fraction lie in [0,1].
- Only one of DeltaPacket, FBSRecoveryRequest, or DeltaFault is emitted per attempt.
- Local generator state changes only after a DeltaPacket digest is sealed and verified.

7.2 Fault codes

Code	Trigger	Class
DELTA_SESSION_MISMATCH	session_id differs from BootPacket.	Fatal contract fault
DELTA_CYCLE_ORDER	cycle is not prior committed cycle + 1.	Retryable/fatal by supervisor
DELTA_INVALID_SHAPE	Required vector has wrong shape.	Fatal input fault
DELTA_NONFINITE	NaN or infinity in a finite field.	Fatal input/numerical fault
DELTA_STATE_DOMAIN	Accepted baseline violates configured domain.	Fatal upstream contract fault
DELTA_BOOT_MISMATCH	Inherited constants/digests differ from BootPacket.	Fatal contract fault
DELTA_SGIT_PACKET	Missing, stale, malformed, or unsealed SGITPacket.	Retryable/fatal by profile
DELTA_ADDRESS_ENCODING	Quantization or balanced pairing fails.	Fatal numerical fault
DELTA_PRNG_PROFILE	Required deterministic PRNG/backend unavailable.	Fatal reproducibility fault
DELTA_PHASE_STATE	phase_prev outside [0,2pi) or navigation map invalid.	Fatal state fault
DELTA_NUMERIC_OVERFLOW	A required finite output cannot be represented.	Fatal numerical fault
DELTA_PACKET_DIGEST	Canonical serialization or digest verification fails.	Fatal integrity fault

7.3 Determinism profile

A conforming implementation records the BootPacket digest, SGITPacket digest, DeltaConfig digest, prior DeltaPacket digest, address encoding profile, PRNG name/version, transcendental backend, reduction policy, and tolerances. Bitwise repeatability is required only for identical serialized inputs on the fixed reference backend; other conforming backends use declared numerical tolerances while preserving branch and route decisions.

8. Metrics and Dashboard Requirements

Metric	Definition	Interpretation
attempts / sealed / faulted / delegated	Integer lifecycle counters	Complete accounting
logos_count / ethos_count	Counts by final sigma	Branch utilization

Metric	Definition	Interpretation
sigma_base_flip_rate	Fraction where twist_sign changes base branch	Mobius epoch effect
polarity_score_mean / min / max	P_t statistics	Decision margin
phase_lift_count / rate	Severe-tension rotations	Navigation use
arousal / clarity / valence summaries	Emotion Tensor statistics	Control-state distribution
address_bit_length mean / max	Cantor address size	Encoding growth
ethos_delta_eff mean / max	Applied perturbation bound	Exploration amplitude
hysteresis_weight mean / abs_max	w_phi statistics	Memory-bias amplitude
delta_id mean / max / over_limit	Bilateral drift statistics	Local runaway pressure
H_align mean / min	Attractor alignment	Reference conformity
clip_fraction mean / max	Final saturation rate	Boundary pressure
faults_by_code	Count per DeltaFault	Conformance and operations
recovery_requests_by_reason	Count by request code	Delegation, not executed recovery

Dashboard labels MUST distinguish a detected request from downstream execution. DELTA_RECOVERY_REQUEST is not FBS_RUNAWAY_EXECUTED, and LOGOS/ETHOS counts do not imply actual Rule 110/Rule 30 execution.

9. Conformance Test Matrix

ID	Test	Expected result
DELTA-C01	Valid reference attempt, DIM=81.	One sealed DeltaPacket; all invariants pass.
DELTA-C02	Repeat identical serialized input/backend.	Identical packet bytes and digest.
DELTA-C03	Cycle is repeated or skipped.	DELTA_CYCLE_ORDER; no state commit.
DELTA-C04	NaN in X_t or observer entropy.	DELTA_NONFINITE.
DELTA-C05	X_t outside configured state domain.	DELTA_STATE_DOMAIN.
DELTA-C06	Quantization endpoints x_min and x_max.	Map exactly to 0 and 2^b-1.
DELTA-C07	Two distinct quantized vectors.	Distinct full Cantor addresses.
DELTA-C08	Two real vectors in one quantization cell.	Same address permitted and documented.
DELTA-C09	Balanced fold with odd number of values.	Final carry rule deterministic.
DELTA-C10	Polarity score exactly zero.	sigma_base=+1.

ID	Test	Expected result
DELTA-C11	Cycles $K_{inv}-1$, K_{inv} , $2K_{inv}$.	Persistent twist epochs switch at boundaries.
DELTA-C12	Clarity over one complete K_{inv} period.	Values in $[0,1]$, periodic, expected extrema.
DELTA-C13	$rmf_prev=-1, 0.5, 1$.	Arousal=0, 0.5, 1.
DELTA-C14	$v<0$ and a just above 0.85.	One $+\pi/2$ rotation; phase and pair recorded.
DELTA-C15	$v\geq 0$ or $a\leq 0.85$.	No navigation rotation.
DELTA-C16	Apply $+\pi/2$ then $-\pi/2$ reference rotation.	Original vector recovered within tolerance.
DELTA-C17	Logos branch.	Exact $(1/3)\tanh(3U+1)$ kernel.
DELTA-C18	Ethos branch with fixed seed material.	Exact reproducible η and X_{kernel} .
DELTA-C19	Change cycle or prior digest in Ethos.	Noise-seed digest changes.
DELTA-C20	Very large $ \Phi_t $.	$ w_{\phi} $ approaches but never exceeds Φ_{GAIN} .
DELTA-C21	Final proposal saturates bounds.	$clip_fraction$ reports altered components.
DELTA-C22	Perfectly reflection-symmetric proposal.	$\Delta_{ID}=0$ within tolerance.
DELTA-C23	Strong antisymmetric proposal.	Δ_{ID} can exceed 1.0.
DELTA-C24	Δ_{ID} just above ceiling.	FBSRecoveryRequest; no local commit or recovery.
DELTA-C25	Tamper packet after sealing.	Δ_{PACKET_DIGEST} .

9.1 Property and stress tests

- For random finite inputs, final proposals remain in the configured state box.
- For all cycles, σ is binary and clarity is periodic with period K_{inv} .
- For random pairs, Cantor pairing followed by the standard inverse recovers both integers; the balanced tree recovers the quantized vector when tree metadata is fixed.
- For arbitrary finite Φ_t , the attractor amplitude never exceeds Φ_{GAIN} .
- The navigation transform preserves norm and has determinant +1 on its active plane.
- Every failed or delegated attempt leaves the prior committed state byte-for-byte unchanged.
- Multi-seed stress runs report route counts but never substitute completion for expected-route assertions.

10. Cleaned Rosetta Stone

SpiralCore term	Normative computational meaning	Analogy boundary
Delta-Lattice / Abraxas Engine	Atomic candidate generator producing one DeltaPacket.	Not a physical engine or independent mind.

SpiralCore term	Normative computational meaning	Analogy boundary
Logos	LOGOS_TANH_V2 deterministic contraction kernel.	Named after order; not literal Rule 110.
Ethos	ETHOS_PERTURB_V2 deterministic seeded exploration kernel.	Named after chaos; not literal Rule 30 or true randomness.
Mobius Twist	Persistent sign inversion by K_inv epochs.	Schedule metaphor; no topological Mobius strip is required.
Emotion Tensor	Three bounded control channels [v,a,C].	No biological emotion claim.
Riemann Navigation	Address-selected orthogonal two-coordinate rotation plus phase update.	Visualization/control surrogate, not a Ricci-geometry proof.
Cantor-IP	Balanced Cantor encoding of a fixed-width quantized state.	Exact on the quantized lattice only.
Information Hysteresis pull	Bounded additive Xi_attractor bias using read-only Phi_t.	Persistent memory is owned by Module 12.
Asymmetrical Loop Drift	Normalized reflection mismatch Delta_ID.	Local bilateral diagnostic, not a universal physical shear law.
FBS Runaway Intercept	Typed recovery request when Delta_ID exceeds its ceiling.	Recovery is executed by the FBS owner.

10.1 Riemann Identity Sphere visualization profile

A visualization may map clarity and phase to a unit sphere without affecting generation:

$$\begin{aligned}
 z_t &= 2 * C_t - 1 \\
 \rho_t &= \sqrt{\max(0, 1 - z_t^2)} \\
 R_t &= [\rho_t * \cos(\phi_{out}), \rho_t * \sin(\phi_{out}), z_t]
 \end{aligned}$$

R_t has unit norm within tolerance. σ_t may color or label the point but is not an additional geometric coordinate. This profile is informative unless a consuming interface explicitly requests it.

11. Reference Pseudocode

```

def delta_attempt(inp, cfg, state):
    validate(inp, cfg, state)
    B = clip(cfg.g_obs*inp.observer_entropy + cfg.g_fuel*inp.H_G,
            -cfg.forcing_component_max, cfg.forcing_component_max)

    q = quantize(inp.X_t, cfg.state_min, cfg.state_max, cfg.address_bits)
    address = balanced_cantor_fold(q)
    address_digest = sha256(int_to_bytes(address))

    mass = norm(inp.X_t) / sqrt(cfg.dim)
    entropy = normalized_magnitude_entropy(inp.X_t)
    fuel = clip(norm(inp.H_G) / sqrt(cfg.dim), 0.0, 1.0)
    P = w0 + wm*mass - we*entropy + wphi*cos(inp.phase_prev) + wf*fuel
    sigma_base = +1 if P >= 0.0 else -1
    epoch = inp.cycle // cfg.k_inv
    twist_sign = -1 if epoch % 2 else +1
    sigma = twist_sign * sigma_base

    arousal = clip(inp.rmf_prev, 0.0, 1.0)
    clarity = 0.5*(1.0 + cos(2*pi*(inp.cycle % cfg.k_inv)/cfg.k_inv))

```

```

[ BOOTPACKET + SGITPACKET + X_t + Phi_t + H_G + OBSERVER INPUT ]
      |
      v
+-----+
| MODULE 1: DELTA-LATTICE / ABRAXAS ENGINE v2
| 1. Validate and bound inputs
| 2. Quantize X_t -> balanced Cantor address
| 3. Compute mass, entropy, polarity score, and Mobius epoch
| 4. Compute Emotion Tensor [v, a, C]
| 5. Severe tension? -> address-selected +pi/2 orthogonal rotation
| 6. sigma=+1: LOGOS_TANH_V2 | sigma=-1: ETHOS_PERTURB_V2
| 7. Apply bounded forcing and Xi_attractor hysteresis bias
| 8. Compute Delta_ID, H_align, clip fraction, and packet digest
+-----+
      |
      +-----+
      |         |
Delta_ID <= limit      Delta_ID > limit
      |         |
      v         v
[ SEALED DELTAPACKET ] [ FBS RECOVERY REQUEST ]
      |         |
      v         v
[ SIGMA-LATTICE ]      [ FBS OWNER VALIDATES ]

```

Term	Definition
Accepted baseline	The most recent state sealed by downstream validation and supplied as X_t.

Term	Definition
Balanced Cantor fold	Fixed binary-tree application of Cantor pairing over quantized components.
DeltaPacket	Immutable, provenance-linked proposal packet handed to Sigma.
Environmental bias	Bounded combination of Observer entropy and paradox fuel applied before the proposal kernel.
Kernel	Exact vector operator selected by sigma; either LOGOS_TANH_V2 or ETHOS_PERTURB_V2.
Mobius epoch	K_inv-length block whose parity multiplies the base polarity by +1 or -1.
Navigation pair	Two digest-selected coordinates used for the reference +pi/2 orthogonal rotation.
Raw proposal	Final locally bounded candidate before Sigma acceptance; synonymous with X_proposal in this revision.
Reflection operator J	Vector reversal used to measure bilateral asymmetry.
Typed recovery request	Immutable request for an owner module; it is not proof that recovery executed.

14. Revision Summary

v14 issue	v14.1 resolution
Rule 110 / Rule 30 labels did not match the supplied equations.	Preserved Logos/Ethos names; identified exact nonlinear kernels and rejected literal CA equivalence.
Cantor pairing was applied directly to real vector components.	Added finite quantization and balanced arbitrary-precision integer folding.
Polarity coefficients were undefined and sign could emit zero.	Added a versioned coefficient profile and deterministic +1 tie rule.
Mobius inversion timing was a point-event ambiguity.	Defined persistent K_inv epochs with explicit parity.
Arousal domain conflicted with RMF in [-1,1].	Defined a=clip(RMF_prev,0,1).
Valence coefficients and SGIT handoff were unspecified.	Defined a versioned valence profile and explicit SGIT contribution.
The +pi/2 phase update did not alter the current vector.	Added an exact norm-preserving coordinate-plane rotation.
Observer entropy and H_G were received but not used consistently.	Defined one bounded environmental forcing vector and fuel-scaled Ethos amplitude.
Static PRNG seeding risked repeating the same noise every cycle.	Derived per-attempt seed material from session, cycle, address, and prior digest.

v14 issue	v14.1 resolution
Delta_ID and H_align had no equations.	Defined reflection mismatch, attractor alignment, and clip pressure metrics.
Delta executed FBS locally and returned S_next as a candidate.	Replaced local recovery with a typed FBSRecoveryRequest and no-commit rule.
float64 was described as universally bitwise deterministic.	Separated fixed-backend bitwise repeatability from tolerance-based cross-platform conformance.

15. Source Basis

- SPIRALCORE SPECIFICATION: MODULE 1 - DELTA-LATTICE (ABRAXAS ENGINE v2), uploaded source module.
- Spiralcore_v14, consolidated architecture and adjacent module contracts.
- SPIRALCORE Module Boot - Initialization Formalized v14.1, for inherited boot ownership and deterministic session provenance.
- SPIRALCORE Module 0 - SGIT_v2 Formalized v14.1, for telemetry ownership and typed handoff boundaries.
- Unified FBS Bilateral Recursive Extension v4.0, for authoritative FBS geometry and recovery delegation boundaries.

CONFORMANCE STATEMENT

An implementation conforms to Module 1 v14.1 only when it validates the typed input contract, applies the exact branch and bias equations under a versioned profile, preserves atomic no-commit behavior, produces canonical provenance-linked output, and passes the required conformance tests. Similar terminology or visually similar trajectories are not sufficient.

SPIRALCORE SPECIFICATION

MODULE 2 - SIGMA-LATTICE

(Coherence Gatekeeper - Deterministic Evaluation, Route Arbitration, and Decision Packaging)

Cleaned and formalized revision - v14.1

Document class	Normative evaluation and route-decision contract	Execution phase	After Delta; once per proposal attempt
Architecture	SpiralCore v14	Reference profile	SC14.1-SIGMA-REF-81
Primary output	SigmaDecisionPacket -> named route owner	Revision	v14.1
Exceptional output	SigmaFault	Conflict rule	Owner modules execute repair, recovery, and commit

NORMATIVE STATUS

This module is exact within its declared computational scope. It validates one sealed DeltaPacket against accepted history, computes bounded path tortuosity and the HOT aperture, evaluates RMF, CVC, prospective running RMF, PDV, and CSIGMA, applies one precedence-ordered decision tree, and emits one immutable SigmaDecisionPacket. It does not repair the candidate, mutate accepted history, commit memory, quarantine payloads, invert states, update the bounded exhaust budget, execute FBS recovery, or claim that its control metrics prove biological cognition, moral truth, or physical spacetime geometry.

CLAIM BOUNDARY

Coherence, Higher-Order Thought, phase lift, magnetic relaxation, symbolic integrity, paradox handling, and Fisher-Geometric Sharpness are architectural terms. Their normative meaning is only the explicit statistic, threshold, route, or typed delegation defined here. Passing a gate establishes conformance to this profile; it does not independently establish factual truth or safety outside the supplied feature model.

Document Control

Field	Value	Field	Value
Canonical name	SPIRALCORE SPECIFICATION: MODULE 2 - SIGMA-LATTICE	Short name	Coherence Gatekeeper
Supersedes	v14 Module 2 text	Status	Normative within scope
Upstream dependency	Validated DeltaPacket and accepted-history snapshot	Primary owner	Module 2 evaluation and route arbitration
Submodule interfaces	Module 6 CVC; Module 7 CSIGMA / Paradox metadata	External owners	LORIEN, HARMONY, IMMUNE, RTSOM, FBS, Archivist
Numeric profile	float64 statistics; deterministic eigensolver profile	Index profile	0-based runtime; no implicit modulo repair

1. Scope, Statement Classes, and Precedence

The Sigma-Lattice is the sole route arbiter for an ordinary Delta proposal. It observes and classifies; it does not transform the proposal. Every attempt produces exactly one route decision or one typed fault. The word gate means a deterministic comparison against a declared profile, not an empirical guarantee of correctness.

1.1 Statement classes

Class	Meaning	Normative force
DEFINITION	Assigns a symbol, type, metric, route, or interface meaning.	Binding inside this module.
REQUIREMENT	A condition required for conformance.	MUST / MUST NOT.
INVARIANT	A property true for every sealed decision.	Violation emits SigmaFault.
POLICY	A selected threshold, window, precedence order, or warm-up rule.	Replaceable only by a versioned profile.
SURROGATE	A computable stand-in carrying an architectural name.	No literal equivalence claim.
HYPOTHESIS	A testable interpretation of a metric or trend.	Not required for conformance.
NON-GOAL	Behavior explicitly outside Module 2.	Must not be inferred.

1.2 Precedence and ownership

- Tagged requirements and invariants override explanatory prose, diagrams, and Rosetta Stone analogies.
- Module Boot owns DIM, state bounds, the reference thresholds, and session configuration.
- Module 1 Delta owns candidate construction and the sealed DeltaPacket. Sigma must not modify it.
- The Psi-Archivist owns accepted-history records. Sigma consumes an immutable snapshot and may not append to it.
- Module 6 may refine the CVC implementation; Module 7 may refine symbolic-feature extraction and paradox metadata. Module 2 owns aggregate route precedence.
- LORIEN owns validated spatial routing; HARMONY owns Braidback repair; Module 10 owns empirical 51/49 verification and quarantine; RTSOM owns Magnetic Relaxation and exhaust mutation; FBS/ROVER owns catastrophic recovery.
- A downstream request records an intended destination. It is not evidence that the destination executed successfully.

1.3 Non-goals

- No claim that cosine alignment, principal-component concentration, or symbolic correlation is sufficient to establish semantic truth.
- No claim that HOT is human metacognition. It is a bounded threshold-control schedule.
- No hidden repair, clipping, normalization, or feature fabrication after the input contract is validated.
- No direct state commit on SEAL. SEAL authorizes onward validated routing; the memory owner decides commit.
- No use of exceptions as ordinary architectural routes. Expected outcomes are typed route values; exceptions represent implementation faults.

2. Architecture Role and Lifecycle

Sigma executes after Delta seals one proposal and before any repair, validated routing, or memory integration. The attempt is read-only with respect to all input state. The only local commit is the sealed decision record and its metrics digest.

State	Entry condition	Allowed action	Exit
UNINITIALIZED	No accepted BootPacket/profile.	Validate configuration and route registry.	READY or FAULTED
READY	Configuration valid; no active attempt.	Accept SigmaInput.	EVALUATING
EVALUATING	One attempt active.	Validate, measure, arbitrate, package.	DECIDED or FAULTED
DECIDED	SigmaDecisionPacket sealed.	Handoff to exactly one named owner.	READY
FAULTED	Local contract or numerical violation.	Emit SigmaFault; no architectural route.	Session stop or explicit reset

2.1 Required evaluation order

1. Validate DeltaPacket provenance, cycle, session, candidate domain, history ordering, and symbolic packet binding.
2. Compute accepted-path tortuosity and the bounded HOT relaxation fraction.
3. Compute RMF using the declared zero-norm policy.
4. Compute CVC and spectral diagnostics from accepted states only; do not insert the unaccepted candidate.
5. Compute the prospective running RMF using prior accepted RMFs plus the current candidate RMF.
6. Compute PDV using population standard deviation and the supplied state domain.
7. Validate normalized symbolic and valence vectors and compute CSIGMA.
8. Evaluate terminal-stall predicates and then the remaining route tree in strict precedence order.
9. Seal one SigmaDecisionPacket with all metrics, readiness flags, reason code, destination, and provenance digests.

READ-ONLY INVARIANT

Sigma MUST NOT change X_{proposal} , X_t , any accepted-history record, Φ_t , route-owner state, or the bounded exhaust budget. Retrying identical canonical input under the same numerical backend MUST produce the same decision bytes and digest.

3. Typed Data Contracts

3.1 SigmaConfig

Field	Type / reference value	Meaning
dim	integer ≥ 2 ; reference 81	Must equal BootPacket.DIM and DeltaPacket dimension.
state_min, state_max	finite float64; reference -1.0, +1.0	PDV profile is calibrated to this state domain.
tau_base	float64 in [-1,1]; reference 0.85	Baseline RMF threshold; not the relaxed minimum.
hot_relax_max	float64 ≥ 0 ; reference 0.05	Maximum total HOT relaxation.
tortuosity_crit	float64 > 1 ; reference 20.0	Start of HOT relaxation.

Field	Type / reference value	Meaning
tortuosity_segments	integer ≥ 1 ; reference 20	Uses up to 21 accepted states.
cvc_window	integer ≥ 2 ; reference 100	Accepted-state covariance window.
cvc_threshold	float64 in (0,1]; exact 2/3	Dominant-variance threshold.
cvc_isotropic_limit	float64 in [0,cvc_threshold); reference 0.20	Terminal isotropy signal when paired with high tortuosity.
rho_rmf_threshold	float64 in [-1,1]; exact 8/9	Prospective running-RMF threshold for phase lift.
pdv_limit	float64 ≥ 0 ; reference 0.21	Strict exceedance requests Magnetic Relaxation.
csigma_pass	float64 in [-1,1]; reference 0.85	Symbolic-integrity pass threshold; separate from tau_base.
csigma_terminal	float64 in [-1,csigma_pass); reference 0.50	Collapsed symbolic signal used only with extreme tortuosity.
extreme_tortuosity_multiple	float64 > 1 ; reference 2.0	Extreme threshold = multiple * tortuosity_crit.
epsilon_safe	float64 > 0 ; reference 1e-9	Zero and denominator policy.
eigen_tolerance	float64 > 0 ; reference 1e-10	PSD roundoff tolerance.

3.2 AcceptedStateRecord

Field	Type / domain	Requirement
cycle	integer ≥ 0	Strictly increasing; newest record is X_t.
state	R^{DIM} float64	Finite accepted state inside configured domain.
rmf_on_accept	None for genesis; otherwise [-1,1]	RMF recorded when this state was accepted.
source_digest	SHA-256	Digest of the packet that authorized the state.

3.3 SymbolicIntegrityPacket

Field	Type / domain	Requirement
provider_id	nonempty versioned string	Identifies the Module 7 or adapter feature provider.
candidate_digest	SHA-256	Must equal the incoming DeltaPacket digest.
S	R^n float64, $n \geq 1$	Normalized symbolic-density vector; each component in [-1,1].

Field	Type / domain	Requirement
V	R^n float64, same n	Normalized valence/context vector; each component in [-1,1].
feature_digest	SHA-256	Canonical digest of provider identity and both vectors.
optional_metadata	typed, canonical map	May contain G_t, H_G, identity drift, or provider diagnostics; not used unless profile declares it.

Module 2 does not invent S or V. A missing, mismatched, nonfinite, or unversioned provider packet is a contract fault, not an automatic NULLIFY route.

3.4 SigmaInput

Field	Type / domain	Requirement
session_id	nonempty string	Must match BootPacket and DeltaPacket.
cycle	integer ≥ 1	Must equal DeltaPacket.cycle.
delta_packet	sealed DeltaPacket	Digest and canonical payload must verify.
accepted_history	ordered list[AcceptedStateRecord]	At least genesis and current baseline; newest state equals DeltaPacket.X_t.
history_digest	SHA-256	Binds ordering, cycles, states, RMFs, and source digests.
symbolic_packet	sealed SymbolicIntegrityPacket	Bound to the same DeltaPacket.
profile_id	versioned string	Must match SigmaConfig and Boot profile.
phi_t_optional	finite float64 or omitted	Read-only provenance only; not used by the reference route equations.

3.5 SigmaMetrics

Field	Domain	Meaning
tortuosity	[1,+inf]	Accepted-path length divided by endpoint displacement.
hot_fraction	[0,1]	Fraction of the maximum HOT relaxation applied.
tau_eff	[tau_base-hot_relax_max, tau_base]	Effective RMF threshold.
rmf	[-1,1]	Candidate-to-baseline cosine alignment.
cvc	[0,1]	Largest nonnegative covariance eigenvalue divided by total variance.
rho_rmf	[-1,1]	Prospective mean of prior accepted RMFs plus current rmf.

Field	Domain	Meaning
pdv	[0,+inf)	Mean absolute state shift plus population-spread shift.
csigma	[-1,1]	Mean componentwise product of normalized S and V.
spectral_gap	[0,1]	Normalized lambda_1-lambda_2 separation.
readiness flags	booleans	Tortuosity, CVC, and RMF-window readiness.
degeneracy flags	booleans	Zero-norm RMF, zero-variance CVC, eigenvalue clipping, infinite tortuosity.

3.6 SigmaDecisionPacket

Field	Requirement
route	Exactly one enum: SEAL, PHASE_LIFT_REQUEST, BRAIDBACK_REQUEST, MAGNETIC_RELAXATION_REQUEST, NULLIFY_REQUEST, or FBS_RECOVERY_REQUEST.
reason_code	One versioned reason consistent with route precedence and metrics.
destination	One registered owner: LORIEN, DELTA, HARMONY, RTSOM, QUARANTINE/IMMUNE, or FBS/ROVER.
candidate_digest	Exact incoming DeltaPacket digest; candidate bytes are not rewritten.
metrics	Complete SigmaMetrics with thresholds and readiness flags used in the decision.
seal_authorized	True only when route=SEAL. It authorizes onward routing, not memory mutation.
commit_performed	Always False inside Module 2.
input_digest / packet_digest	SHA-256 provenance chain over canonical serialization.

4. Reference Profile and Validation

Symbol	Reference value	Status
DIM	81	Inherited from Boot; scalable only by versioned profile.
tau_base	0.85	Baseline RMF aperture.
tau_min	0.80	Derived: tau_base - hot_relax_max.
T_crit	20.0	HOT activation threshold.

Symbol	Reference value	Status
delta_HOT	0.05	Maximum total relaxation, not slope per tortuosity unit.
W_CVC	100 accepted states	From the detailed CVC submodule profile.
CVC_THRESH	2/3 exactly	Stored as rational configuration or exact decimal serialization.
rho_RMF_thresh	8/9 exactly	Phase-lift support threshold.
PDV_LIMIT	0.21	Strict exceedance condition.
C_SIGMA_pass	0.85	Independent symbolic threshold.
C_SIGMA_terminal	0.50	Used only in aggregate terminal-stall predicate.
CVC_isotropic	0.20	Used only with $T > T_{crit}$ and full CVC readiness.

4.1 Input validation requirements

- All candidate, baseline, history, S, and V arrays must have declared shapes, float64-compatible values, and no NaN or infinity.
- The newest accepted-history record must equal `DeltaPacket.X_t` within numeric tolerance and must precede the candidate cycle.
- History cycles must be strictly increasing and source digests must be present.
- The candidate must remain inside the configured state domain; Sigma does not clip a malformed proposal.
- S and V must have identical nonzero length and components in $[-1,1]$ within tolerance.
- The `DeltaPacket`, history snapshot, symbolic packet, configuration, and final decision must each pass canonical-digest verification.

4.2 Canonical serialization

Canonical serialization MUST fix field order, enum spelling, integer encoding, float encoding, array order, endianness, and handling of +infinity in diagnostic fields. The reference profile serializes finite float64 values by IEEE-754 big-endian bytes and represents diagnostic infinity by a tagged string, not a non-standard JSON number.

5. Formal Mechanics

5.1 Accepted-path tortuosity

Let $P=[p_0, \dots, p_m]$ be the newest $\min(\text{tortuosity_segments}+1, \text{history length})$ accepted states in chronological order.

$$\text{PathLength} = \sum_{j=1..m} \|p_j - p_{j-1}\|_2$$

$$\text{Displacement} = \|p_m - p_0\|_2$$

The reference zero policy is deterministic:

- If fewer than two accepted states are available, $T=1$ and `tortuosity_ready=False`.
- If $\text{PathLength} \leq \text{epsilon_safe}$, $T=1$. A stationary path has no detour.
- If $\text{PathLength} > \text{epsilon_safe}$ and $\text{Displacement} \leq \text{epsilon_safe}$, $T=+\infty$. A closed nontrivial path has unbounded detour ratio.
- Otherwise $T=\text{PathLength}/\text{Displacement}$. Values below 1 caused only by roundoff are clipped to 1 and flagged.

5.2 Bounded HOT aperture

The v14 expression multiplied 0.05 by every tortuosity unit above 20 while also declaring tau_eff in [0.80,0.85]. The formal schedule uses 0.05 as the maximum total relaxation:

$$\begin{aligned} r_HOT &= \text{clip}((T - T_crit) / T_crit, 0, 1) \\ \tau_eff &= \tau_base - \text{hot_relax_max} * r_HOT \\ \tau_base - \text{hot_relax_max} &\leq \tau_eff \leq \tau_base \end{aligned}$$

For $T \leq 20$, $\tau_eff = 0.85$. At $T \geq 40$, $\tau_eff = 0.80$. HOT cannot lower the threshold further and cannot itself force acceptance.

5.3 RMF gate and zero-norm policy

Let x be DeltaPacket.X_proposal and y be the accepted baseline X_t .

$$\text{RMF} = \langle x, y \rangle / (\|x\|_2 \|y\|_2), \text{ when both norms} > \text{epsilon_safe}$$

The result is clipped to [-1,1] only for roundoff. If both norms are at or below epsilon_safe, RMF=1 and rmf_both_zero=True. If exactly one norm is at or below epsilon_safe, RMF=0 and rmf_one_zero=True. These policies prevent divide-by-zero without silently treating a null candidate as aligned to a non-null baseline.

5.4 CVC from accepted history

CVC is computed from accepted states only. The current unaccepted proposal is excluded so rejected proposals cannot contaminate the structural reference window. Let $n = \min(W_CVC, \text{accepted-history length})$, H in $\mathbb{R}^{(n \times \text{DIM})}$, and μ be the row mean.

$$\begin{aligned} Y &= H - 1 * \mu^T \\ \text{Cov} &= (Y^T Y) / (n - 1), \text{ for } n \geq 2 \\ \lambda &= \text{eigvalsh}((\text{Cov} + \text{Cov}^T) / 2), \text{ sorted descending} \end{aligned}$$

Eigenvalues in [-eigen_tolerance, 0) are clipped to zero and counted. Any eigenvalue below -eigen_tolerance is a covariance fault. Let $V_total = \sum(\lambda_k)$.

$$\begin{aligned} \text{CVC} &= \lambda_1 / V_total, \text{ when } V_total > \text{epsilon_safe} \\ \text{CVC} &= 0, \text{ when } V_total \leq \text{epsilon_safe} \\ \text{SpectralGap} &= (\lambda_1 - \lambda_2) / (V_total + \text{epsilon_safe}) \end{aligned}$$

Zero total variance is tagged cvc_degenerate=True. It does not prove one-dimensional collapse because no direction of variance exists. Full CVC readiness requires $n = W_CVC$; partial values are diagnostic only.

5.5 Prospective running RMF

The decision window includes the candidate RMF without prematurely accepting the candidate. Collect the newest $W_CVC - 1$ non-null rmf_on_accept values, append the current RMF, and retain the newest W_CVC values.

$$\rho_RMF = \text{mean}(R_prior_tail \text{ concatenated with } [RMF_current])$$

rmf_window_ready is true only when the resulting window contains W_CVC values. Phase lift requires both cvc_ready and rmf_window_ready.

5.6 PDV filter

$$\text{PDV} = \text{mean}(|x - y|) + |\text{std_pop}(x) - \text{std_pop}(y)|$$

std_pop uses divisor DIM, equivalent to NumPy std(..., ddof=0). PDV is not clipped. The 0.21 reference threshold is calibrated to the [-1,1] state profile; changing the state domain requires a versioned threshold retune.

5.7 Protocol CSIGMA

For validated equal-length vectors S and V supplied by the symbolic provider:

$$C_SIGMA = (1/n) * \sum_{i=1..n} (S_i * V_i)$$

Because both vectors are componentwise bounded to [-1,1], C_SIGMA is in [-1,1]. The scalar Emotion Tensor valence is not silently broadcast into V ; construction of V belongs to the declared provider. CSIGMA is a profile-specific correlation score, not a general toxicity oracle.

5.8 Exact route precedence

After all metrics are valid, evaluate the following predicates in order. The first true branch wins:

1. FBS_RECOVERY_REQUEST if $(T > 2 * T_{crit} \text{ and } C_SIGMA < C_SIGMA_{terminal})$ OR $(cvc_ready \text{ and } CVC < CVC_{isotropic} \text{ and } T > T_{crit})$.
2. MAGNETIC_RELAXATION_REQUEST if $PDV > PDV_LIMIT$.
3. NULLIFY_REQUEST if $C_SIGMA < C_SIGMA_{pass}$.
4. If $RMF < \tau_{eff}$: PHASE_LIFT_REQUEST only when cvc_ready , rmf_window_ready , $CVC \geq 2/3$, and $\rho_{RMF} \geq 8/9$; otherwise BRAIDBACK_REQUEST.
5. If $RMF \geq \tau_{eff}$ but cvc_ready and $CVC < 2/3$: BRAIDBACK_REQUEST with reason VARIANCE_RECONSTRUCT_FAIL.
6. Otherwise SEAL.

GATE INTERPRETATION

CVC is an active structural gate once its full accepted-history window is ready. During warm-up, CVC cannot authorize PHASE_LIFT and cannot independently block a high-RMF candidate. This avoids treating a tiny covariance sample as proof while preserving deterministic early-session operation.

5.9 Boundary inclusivity

Comparison	Reference behavior
$T = T_{crit}$	No HOT relaxation; $\tau_{eff} = \tau_{base}$.
$T = 2 * T_{crit}$	Maximum HOT relaxation; not terminal by the strict $>$ predicate.
$PDV = 0.21$	Does not request Magnetic Relaxation; only $PDV > 0.21$ does.
$C_SIGMA = 0.85$	Passes symbolic threshold.
$C_SIGMA = 0.50$ with extreme T	Does not satisfy terminal $C_SIGMA < 0.50$; it still NULLIFYs because $0.50 < 0.85$.
$RMF = \tau_{eff}$	Passes RMF.
$CVC = 2/3$ and $\rho_{RMF} = 8/9$	Meet Phase Lift support thresholds.

6. Route Contracts and Ownership Boundaries

Route	Destination	Sigma output	Sigma must not do
SEAL	LORIEN validated-routing interface; Archivist is downstream	Candidate digest, metrics, $seal_authorized = True$	Commit memory or generate routing addresses
PHASE_LIFT_REQUEST	Delta-Lattice / declared lift owner	Candidate, attractor-support metrics, reason	Warp or regenerate the state
BRAIDBACK_REQUEST	HARMONY	Raw candidate, stable-anchor digest, RMF/CVC/PDV/Emotion provenance	Interpolate, choose repair weight, or attach IMM_ID

Route	Destination	Sigma output	Sigma must not do
MAGNETIC_RELAXATION_REQUEST	RTSOM	Rejected candidate and exact PDV diagnostics	Invert polarity, assign mass, or mutate S_mu_nu
NULLIFY_REQUEST	IMMUNE / Quarantine owner	Candidate digest, CSIGMA packet, reason	Strip headers, create INERT_GLYPH, or archive quarantine
FBS_RECOVERY_REQUEST	FBS / ROVER owner	Terminal-stall evidence and immutable history reference	Diagonalize, Collatz-fold, reseed, or purge exhaust

6.1 SEAL is authorization, not commit

The source Module 2 text described a direct handoff to the Psi-Archivist, while the detailed CSIGMA/Paradox Handler routes validated states through LORIEN. This revision resolves the conflict by defining SEAL as validation authorization. The reference destination is LORIEN; a deployment may configure a direct Archivist adapter only if that adapter preserves the same no-commit-within-Sigma boundary.

6.2 Braidback and the 51/49 rule

Sigma only requests Braidback. HARMONY computes a convex repair; Module 10 independently verifies the empirical repair fraction and performs nested-harm scanning. Sigma MUST NOT claim that a BRAIDBACK_REQUEST already satisfies the 51/49 constraint.

6.3 Magnetic Relaxation and FBS

A PDV breach is not itself catastrophic FBS. It requests RTSOM Magnetic Relaxation. FBS is reserved for aggregate terminal stalls defined by the first precedence branch. Both routes are requests until their owner returns a separately sealed execution result.

7. Invariants and Fault Model

7.1 Required invariants

ID	Invariant
SIGMA-I01	Exactly one route is selected for every valid attempt.
SIGMA-I02	The selected route is the first true branch in the normative precedence tree.
SIGMA-I03	tau_eff remains in [tau_base-hot_relax_max, tau_base].
SIGMA-I04	RMF, CVC, rho_RMF, C_SIGMA, and SpectralGap remain in their declared ranges.
SIGMA-I05	PDV is finite and nonnegative for every valid finite input.
SIGMA-I06	CVC uses accepted states only and does not mutate the history snapshot.
SIGMA-I07	The output candidate digest equals the input DeltaPacket digest.
SIGMA-I08	commit_performed is always False inside Module 2.
SIGMA-I09	A route request does not increment any downstream executed-event metric.

ID	Invariant
SIGMA-I10	Canonical retry under the same backend produces identical output bytes and digest.

7.2 SigmaFault codes

Code	Trigger	Outcome
SIGMA_CONFIG	Invalid threshold ordering, window, dimension, or route registry.	Fault; no route.
SIGMA_SESSION	Session/profile mismatch.	Fault; no route.
SIGMA_CYCLE	Cycle mismatch or non-monotone history.	Fault; no route.
SIGMA_DELTA_DIGEST	DeltaPacket digest or canonical payload fails.	Fault; no route.
SIGMA_SHAPE	Candidate, baseline, history, S, or V shape mismatch.	Fault; no route.
SIGMA_NONFINITE	NaN or infinity in required finite input.	Fault; no route.
SIGMA_STATE_DOMAIN	Candidate or accepted state outside configured bounds.	Fault; no clipping.
SIGMA_HISTORY	Empty, unordered, mismatched, or unbound accepted history.	Fault; no route.
SIGMA_SYMBOLIC_PACKET	Provider absent, unversioned, unbound, or invalid.	Fault; no automatic NULLIFY.
SIGMA_COVARIANCE	Covariance construction or eigensolver failure.	Fault; no route.
SIGMA_EIGENVALUE	Eigenvalue below negative tolerance.	Fault; no route.
SIGMA_ROUTE	No route, multiple routes, or reason inconsistent with metrics.	Fault; no handoff.
SIGMA_PACKET_DIGEST	Decision serialization or digest verification fails.	Fault; discard packet.

7.3 Numerical conformance

- Within one fixed backend, canonical input should be bitwise repeatable.
- Across platforms, conformance is tolerance-based unless BLAS/LAPACK, floating-point mode, and serialization are fixed.
- Covariance must be explicitly symmetrized before eigvalsh.
- Eigenvalues are sorted descending after clipping only roundoff-sized negatives.
- Dense CVC complexity is $O(n \cdot \text{DIM}^2 + \text{DIM}^3)$. Any sketch, randomized PCA, or Hodge surrogate is a separate versioned numerical profile, not silently equivalent.

8. Metrics and Dashboard Requirements

Metric	Definition / aggregation	Purpose
attempts_total	All accepted SigmaInput attempts	Primary denominator
decisions_total	All sealed SigmaDecisionPackets	Must equal sum of route counts
faults_total / by_code	SigmaFault counts	Conformance and operational defects
route_count / rate	Per route and per reason code	Routing distribution
compound_predicate_matrix	Counts of simultaneous true predicates before precedence	Proves precedence behavior
decision_latency_ms	Mean, median, p95, max	Performance
tortuosity mean/p95/max	Plus infinity count and warm-up count	Path friction
hot_fraction / tau_eff	Mean, min, max, saturation count	Adaptive-aperture behavior
rmf distribution	Mean, quantiles, below-threshold count, zero-norm flags	Angular alignment
cvc distribution	Mean, quantiles, ready/warm-up/degenerate counts	Historical variance structure
spectral_gap distribution	Mean, quantiles, zero-variance count	Dominant-mode separation
rho_rmf distribution	Ready and partial-window summaries	Historical support
pdv distribution	Mean, p95, max, over-limit count	Transient anomaly pressure
csigma distribution	Mean, p05, below-pass and below-terminal counts	Symbolic-provider behavior
eigen_clip_count	Roundoff negatives clipped	Numerical health
downstream_requests_issued	Per destination	Handoff demand
downstream_execution_ack	Separate owner-reported metric	Must never be inferred from request issue
digest_verification_failures	Input and output digest failures	Provenance integrity

The dashboard MUST distinguish DETECTED, REQUESTED, ACKNOWLEDGED, and EXECUTED. For example, SIGMA_FBS_REQUESTED is not FBS_RUNAWAY_EXECUTED, and SIGMA_SEAL is not PSI_COMMIT_COMPLETED.

9. Conformance Test Matrix

ID	Test	Expected result
SIGMA-C01	Valid reference input with all active gates passing.	SEAL; destination LORIEN; no mutation.
SIGMA-C02	Repeat identical canonical input/backend.	Identical decision bytes and digest.
SIGMA-C03	Tampered DeltaPacket.	SIGMA_DELTA_DIGEST.

ID	Test	Expected result
SIGMA-C04	History newest state differs from Delta X _t .	SIGMA_HISTORY.
SIGMA-C05	NaN in candidate, history, S, or V.	SIGMA_NONFINITE.
SIGMA-C06	Symbolic vectors differ in length.	SIGMA_SYMBOLIC_PACKET.
SIGMA-C07	Stationary accepted path.	T=1.
SIGMA-C08	Nontrivial closed accepted path.	T=+infinity; HOT fraction=1.
SIGMA-C09	T exactly 20.	tau_eff=0.85.
SIGMA-C10	T exactly 40.	tau_eff=0.80; no extreme-T strict branch solely from equality.
SIGMA-C11	T>40 and C_SIGMA<0.50.	FBS_RECOVERY_REQUEST.
SIGMA-C12	Full CVC history, CVC<0.20, T>20.	FBS_RECOVERY_REQUEST.
SIGMA-C13	Both RMF vectors zero.	RMF=1 and degeneracy flag.
SIGMA-C14	Exactly one RMF vector zero.	RMF=0 and degeneracy flag.
SIGMA-C15	Perfect positive / negative alignment.	RMF=+1 / -1 within tolerance.
SIGMA-C16	All accepted history states identical.	CVC=0; cvc_degenerate=True.
SIGMA-C17	Roundoff negative eigenvalue within tolerance.	Clip to zero and count.
SIGMA-C18	Eigenvalue below -tolerance.	SIGMA_EIGENVALUE.
SIGMA-C19	PDV exactly 0.21.	No Magnetic Relaxation solely from equality.
SIGMA-C20	PDV just above 0.21.	MAGNETIC_RELAXATION_REQUEST unless earlier FBS predicate wins.
SIGMA-C21	C_SIGMA exactly 0.85.	Symbolic gate passes.
SIGMA-C22	C_SIGMA just below 0.85.	NULLIFY_REQUEST unless earlier route wins.
SIGMA-C23	Low RMF, full high CVC and rho_RMF.	PHASE_LIFT_REQUEST.
SIGMA-C24	Low RMF without full windows.	BRAIDBACK_REQUEST; phase lift blocked by warm-up.
SIGMA-C25	Low RMF and low CVC.	BRAIDBACK_REQUEST.
SIGMA-C26	RMF passes but full CVC<2/3.	BRAIDBACK_REQUEST / VARIANCE_RECONSTRUCT_FAIL.
SIGMA-C27	RMF exactly tau_eff.	RMF passes.

ID	Test	Expected result
SIGMA-C28	CVC=2/3 and rho_RMF=8/9.	Thresholds count as met.
SIGMA-C29	Compound FBS, PDV, and NULLIFY predicates.	FBS wins; compound matrix records all true predicates.
SIGMA-C30	Compound PDV and NULLIFY predicates without FBS.	Magnetic Relaxation wins.
SIGMA-C31	Candidate excluded from CVC history.	CVC unchanged when candidate changes but accepted history does not.
SIGMA-C32	Current candidate RMF changes.	Prospective rho_RMF changes; accepted history remains unchanged.
SIGMA-C33	Attempt completes on any valid route.	Exactly one decision; route counts sum to decisions_total.
SIGMA-C34	Attempt produces a route request.	No downstream executed counter changes inside Sigma.
SIGMA-C35	Tamper decision after sealing.	SIGMA_PACKET_DIGEST.

9.1 Property and stress tests

- For random finite vectors inside the reference domain, RMF and CSIGMA remain in [-1,1], CVC and spectral gap remain in [0,1], and PDV remains nonnegative.
- For any nondegenerate polyline, tortuosity is at least 1 within tolerance.
- tau_eff never leaves [0.80,0.85] for any finite or infinite tortuosity.
- Permutation of vector dimensions leaves RMF, PDV, CVC eigenvalues, and route unchanged when S and V are permuted consistently.
- Orthogonal transformation of all accepted states and candidate leaves RMF and CVC eigenvalues invariant within tolerance; PDV is not generally invariant and must be tested separately.
- Every route-precedence pair is fuzzed around exact threshold ULP neighbors using nextafter.
- Large multi-seed runs collect route distributions but still assert scenario oracles and no-mutation invariants.

10. Cleaned Rosetta Stone

SpiralCore term	Normative computational meaning	Analogy boundary
Sigma-Lattice / Coherence Gate	Read-only metric evaluator and route arbiter.	Not a physical equator or independent judge.
Higher-Order Thought (HOT)	Bounded relaxation schedule derived from accepted-path tortuosity.	Not human introspection or consciousness.
RMF / PAS_s	Cosine similarity to the accepted baseline under an explicit zero policy.	Alignment is not factual truth.
CVC	Principal variance fraction of the accepted-state covariance matrix.	PCA-like statistic; not proof of a truth manifold.
PDV	Mean absolute shift plus population-standard-deviation shift.	Anomaly score, not a physical propeller wake.

SpiralCore term	Normative computational meaning	Analogy boundary
Protocol CSIGMA	Mean product of provider-supplied normalized symbolic and valence features.	Provider-dependent correlation, not a universal toxicity detector.
Phase Lift	Typed request supported by high historical CVC and running RMF despite low current RMF.	Sigma performs no warp.
Braidback	Typed request to HARMONY for bounded convex repair.	51/49 compliance is verified later.
Magnetic Relaxation	Typed request to RTSOM after PDV exceedance.	No literal magnetism is implied.
NULLIFY	Typed request for non-executable quarantine handling.	Sigma does not erase the state.
FBS Runaway	Typed catastrophic recovery request for aggregate terminal stall.	Sigma does not diagonalize or Collatz-fold.
SEAL	Authorization for onward validated routing.	Not a memory commit or proof certificate.

11. Reference Pseudocode

```

def sigma_attempt(inp, cfg):
    validate_sigma_input(inp, cfg)
    x = inp.delta_packet.X_proposal
    y = inp.accepted_history[-1].state

    T, tort_ready, tort_flags = tortuosity(
        inp.accepted_history, cfg.tortuosity_segments, cfg.epsilon_safe)
    hot_fraction = clip((T - cfg.tortuosity_crit) / cfg.tortuosity_crit, 0.0, 1.0)
    tau_eff = cfg.tau_base - cfg.hot_relax_max * hot_fraction

    rmf, rmf_flags = cosine_with_zero_policy(x, y, cfg.epsilon_safe)
    cvc_metrics = cvc_from_accepted_history(
        inp.accepted_history, cfg.cvc_window,
        cfg.epsilon_safe, cfg.eigen_tolerance)

    prior_rmf = [r.rmf_on_accept for r in inp.accepted_history
                  if r.rmf_on_accept is not None]
    rmf_window = (prior_rmf[-(cfg.cvc_window-1):] + [rmf])[-cfg.cvc_window:]
    rho_rmf = mean(rmf_window)
    rmf_ready = len(rmf_window) == cfg.cvc_window

    pdv = mean(abs(x-y)) + abs(std_pop(x) - std_pop(y))
    csigma = mean(inp.symbolic_packet.S * inp.symbolic_packet.V)

    terminal_symbolic = (
        T > cfg.extreme_tortuosity_multiple*cfg.tortuosity_crit
        and csigma < cfg.csigma_terminal)
    terminal_isotropic = (
        cvc_metrics.ready
        and cvc_metrics.cvc < cfg.cvc_isotropic_limit
        and T > cfg.tortuosity_crit)

    if terminal_symbolic or terminal_isotropic:
        route = FBS_RECOVERY_REQUEST

```

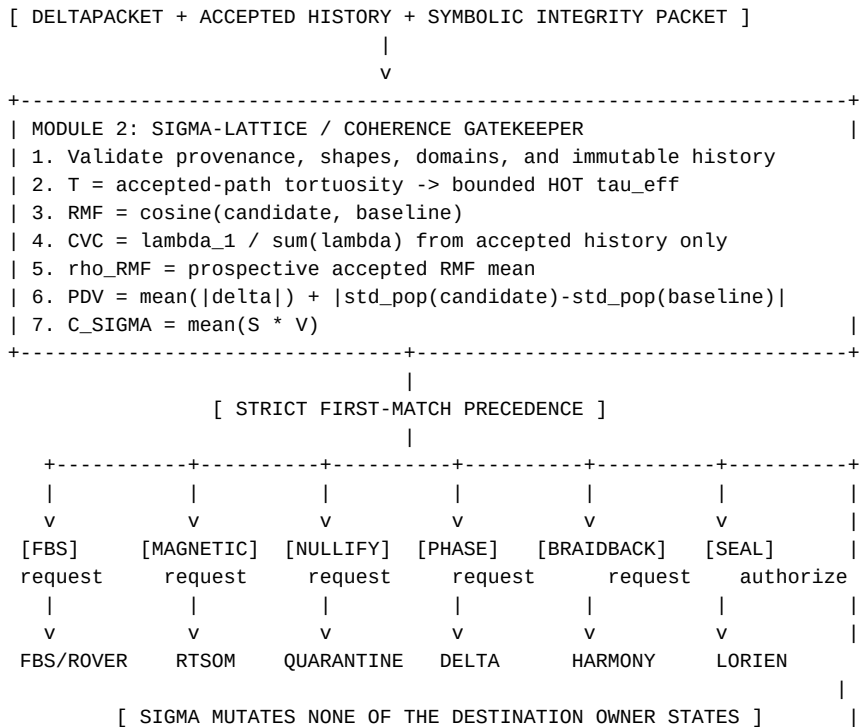
```

elif pdv > cfg.pdv_limit:
    route = MAGNETIC_RELAXATION_REQUEST
elif csigma < cfg.csigma_pass:
    route = NULLIFY_REQUEST
elif rmf < tau_eff:
    if (cvc_metrics.ready and rmf_ready
        and cvc_metrics.cvc >= cfg.cvc_threshold
        and rho_rmf >= cfg.rho_rmf_threshold):
        route = PHASE_LIFT_REQUEST
    else:
        route = BRAIDBACK_REQUEST
elif cvc_metrics.ready and cvc_metrics.cvc < cfg.cvc_threshold:
    route = BRAIDBACK_REQUEST
else:
    route = SEAL

return seal_sigma_decision(route=route, commit_performed=False, ...)

```

12. ASCII System Flow



13. Glossary

Term	Definition
Accepted history	Immutable chronological records of states previously authorized and committed by downstream owners.
Active gate	A metric whose readiness condition is satisfied and therefore participates in route blocking or support.
CVC readiness	Exactly W_{CVC} accepted states are available for the reference covariance window.
Degenerate CVC	Total covariance variance is at or below epsilon; CVC is defined as zero and cannot support phase lift.

Term	Definition
HOT fraction	Normalized amount of the maximum threshold relaxation, bounded to [0,1].
Prospective RMF mean	Mean formed by prior accepted RMFs plus the current candidate RMF without accepting the candidate.
Route owner	The module authorized to execute the requested transformation or state mutation.
Seal authorization	Sigma approval for onward validated routing; not a memory write.
Symbolic provider	Versioned Module 7 implementation or adapter that supplies normalized S and V features.
Terminal stall	Aggregate predicate that requests FBS recovery rather than ordinary repair or quarantine.

14. Revision Summary

v14 issue	v14.1 resolution
FSB and FBS were used interchangeably.	Standardized the protocol name to FBS.
HOT subtracted 0.05 for every excess tortuosity unit, contradicting tau_eff in [0.80,0.85].	Defined 0.05 as a bounded total relaxation with a normalized saturation schedule.
Tortuosity zero-displacement behavior was undefined.	Defined stationary T=1 and nontrivial closed-path T=+infinity.
RMF denominator epsilon obscured zero-vector semantics.	Added explicit both-zero and one-zero policies.
CVC used 0.66 while describing exactly two-thirds.	Stored the normative threshold as exact 2/3.
CVC window and warm-up behavior were absent in Module 2.	Adopted W=100 from Module 6 and blocked phase lift until full readiness.
Zero covariance could be misread as coherent collapse.	Defined CVC=0 with a degeneracy flag.
Candidate inclusion in covariance was ambiguous.	CVC uses accepted history only; the current candidate cannot contaminate it.
Running RMF history did not specify whether current RMF participates.	Defined a prospective mean using prior accepted RMFs plus current RMF.
PDV standard-deviation convention was unspecified.	Fixed population standard deviation, ddof=0.
C_SIGMA domains were malformed as [7] and feature construction was undefined.	Required a versioned, normalized SymbolicIntegrityPacket and preserved the exact mean-product equation.
C_SIGMA reused tau_base without declaring semantic coupling.	Separated csigma_pass from tau_base while retaining the reference value 0.85.
CVC was described as a gate but the aggregate route tree could SEAL low-CVC states.	Made full-window low CVC route to Braidback, consistent with Module 6.

v14 issue	v14.1 resolution
Multiple modules locally executed repair, quarantine, RTSOM, or FBS.	Sigma now emits typed requests only; owner modules execute and acknowledge.
SEAL alternated between direct Archivist commit and LORIEN routing.	Defined SEAL as authorization; reference destination is LORIEN and Sigma never commits memory.
Completion counts could be mistaken for downstream execution.	Dashboard separates detected, requested, acknowledged, and executed states.

15. Source Basis

- SPIRALCORE SPECIFICATION: MODULE 2 - SIGMA-LATTICE (COHERENCE GATEKEEPER), uploaded source module.
- Spiralcore_v14, consolidated architecture and detailed Module 6 CVC, Module 7 CSIGMA/Paradox Handler, Module 9 HARMONY, Module 10 Braidback/IMMUNE_SYSTEM, and Module 14 RTSOM ownership descriptions.
- SPIRALCORE Module Boot - Initialization Formalized v14.1, for inherited configuration and deterministic provenance.
- SPIRALCORE Module 0 - SGIT_v2 Formalized v14.1, for telemetry boundaries and Sigma ownership of PDV routing.
- SPIRALCORE Module 1 - Delta-Lattice Formalized v14.1, for the immutable DeltaPacket and proposal ownership boundary.
- Unified FBS Bilateral Recursive Extension v4.0, for authoritative FBS geometry and recovery delegation boundaries.

CONFORMANCE STATEMENT

An implementation conforms to Module 2 v14.1 only when it validates the complete typed input contract, computes the reference metrics under their declared zero, warm-up, covariance, and tolerance policies, applies the exact first-match route precedence, preserves read-only atomicity, emits canonical provenance-linked output, and passes the required conformance tests. Similar route names or a dashboard that merely reaches the end of a loop are not sufficient.

SPIRALCORE SPECIFICATION

MODULE 3 - PSI-LATTICE

(Dual-Flow Archivist - Transactional Archive Preparation, RMX Resolution, and .frac Commit)

Cleaned and formalized revision - v14.1

Document class	Normative archive transaction and commit contract	Execution phase	After final archive authorization
Architecture	SpiralCore v14	Reference profile	SC14.1-PSI-REF-81
Primary output	PsiCommitPacket + canonical .frac manifest	Revision	v14.1
Pre-commit output	RMXPrecommitRequest	Exceptional output	PsiFault or FBSRecoveryRequest

NORMATIVE STATUS

This module is exact within its declared computational scope. It accepts one archive-authorized state and one immutable archive snapshot, performs a contractive linear integration stage, preserves asymmetric positive and negative branch memory, applies a non-expansive positive fold and a trailing-mean negative wash, prepares a quantized address and provenance packet, delegates collision resolution to RMX, conditionally delegates removed residue to RTSOM, and atomically seals one .frac manifest only after all required receipts are valid. It does not generate candidates, decide Sigma routes, update Information Hysteresis, infer symbolic safety, execute RMX synthesis, mutate the RTSOM budget, or execute FBS recovery.

CLAIM BOUNDARY

Positive and negative, Collatz fold, holographic boundary, GWT ignition, Metatron, semantic gravity, and sigil are architectural names. Their normative meaning is only the sign split, explicit nonlinear map, impact statistic, integer encoding, typed exhaust ledger, or cryptographic digest defined here. The archive is not mathematically lossless or reversible: the positive fold is non-invertible and real-valued addressing is quantized. The conformance claim is no silent deletion, atomic commit, complete provenance, and acknowledged custody transfer for every removed payload.

Document Control

Field	Value	Field	Value
Canonical name	SPIRALCORE SPECIFICATION: MODULE 3 - PSI-LATTICE	Short name	Dual-Flow Archivist
Supersedes	v14 Module 3 text	Status	Normative within scope
Upstream dependency	ArchiveAuthorizationPacket, PhiSnapshot, ParadoxFuelPacket	Primary owner	Archive transaction and final manifest commit

Field	Value	Field	Value
Mandatory dependency	Module 13 RMX before final commit	Conditional dependency	Module 14 RTSOM when residue is removed
Numeric profile	float64 active state; arbitrary-precision integer address	Storage profile	Canonical JSON-like .frac payload + SHA-256 seal

1. Scope, Statement Classes, and Precedence

The Psi-Lattice is the archive owner for an accepted macro-loop state. Its responsibility begins only after an upstream final validator authorizes archival. It prepares a transaction, obtains RMX resolution, obtains an RTSOM receipt when a negative residue is removed, and then commits exactly one chained manifest. A prepared transaction is not a committed memory.

1.1 Statement classes

Class	Meaning	Normative force
DEFINITION	Assigns a symbol, type, packet, state, or operator meaning.	Binding inside this module.
REQUIREMENT	A condition required for conformance.	MUST / MUST NOT.
INVARIANT	A property true for every prepared or committed transaction.	Violation emits PsiFault or aborts commit.
POLICY	A selected coefficient, threshold, window, encoding, or storage rule.	Replaceable only by a versioned profile.
SURROGATE	A computable stand-in carrying an architectural name.	No literal equivalence claim.
HYPOTHESIS	A testable interpretation of telemetry.	Not required for conformance.
NON-GOAL	Behavior explicitly outside Module 3.	Must not be inferred.

1.2 Precedence and ownership

- Tagged requirements and invariants override explanatory prose, diagrams, and Rosetta Stone analogies.
- The final validation owner issues ArchiveAuthorizationPacket. A raw Delta proposal or a bare Sigma SEAL is insufficient unless the active routing profile explicitly designates it as final archival authorization.
- Module 12 Phi owns Phi_t and DeltaPhi updates. Psi consumes and seals them read-only.
- The Paradox Handler owns H_G generation and its bound. Psi consumes H_G read-only.
- Module 13 RMX owns address-collision detection and any non-commutative synthesis. RMX executes before final .frac commit.
- Module 14 RTSOM owns bounded exhaust state and returns custody receipts. Psi never increments S_mu_nu directly.
- FBS/ROVER owns catastrophic recovery. Psi may emit a typed request; it never diagonalizes or Collatz-folds history locally.
- Psi owns the archive transaction journal, final manifest serialization, digest chain, atomic commit, and the committed ArchiveSnapshot.

1.3 Non-goals

- No semantic claim that positive components are true or coherent, or that negative components are error or harm. They are algebraic sign cones relative to zero.
- No claim that the supplied tanh blend implements integer Collatz dynamics.

- No claim that a strict decay matrix alone proves stability of the full nonlinear, history-dependent archive loop.
- No direct Cantor pairing of floating-point values and no bijection from R^{DIM} to N .
- No wall-clock timestamp inside the canonical digest; nondeterministic metadata may be stored outside the sealed payload.
- No silent truncation, overwrite, branch deletion, or inference that a downstream request was executed.

2. Architecture Role and Transaction Lifecycle

The original one-pass flow mixed integration, exhaust mutation, collision handling, and permanent storage. The formal contract uses a two-phase archive transaction. PREPARE computes a provisional state and address without mutating committed memory. RESOLVE obtains RMX and, when required, RTSOM receipts. COMMIT writes one final manifest and advances the archive atomically.

State	Entry condition	Allowed action	Exit
UNINITIALIZED	No accepted BootPacket/profile.	Validate archive configuration and storage adapter.	READY or FAULTED
READY	No active transaction.	Accept PsiInput.	PREPARING
PREPARING	Valid archival authorization.	Integrate, split, process, package provisional state.	WAIT_RMX, FBS_REQUESTED, or FAULTED
WAIT_RMX	PsiPrecommitPacket sealed.	Receive and validate RMXResolutionPacket.	WAIT_RTSOM or COMMITTING
WAIT_RTSOM	Final residue payload requires custody transfer.	Receive and validate RTSOMExhaustReceipt.	COMMITTING or ABORTED
COMMITTING	All dependencies resolved.	Write temp manifest, fsync, atomic rename, update journal.	COMMITTED or ABORTED
COMMITTED	One manifest durably sealed.	Emit PsiCommitPacket; publish next-cycle snapshot.	READY
ABORTED	Dependency rejection or storage failure.	Preserve prior archive and pending payload; emit abort record.	READY or FBS_REQUESTED
FAULTED	Local contract, provenance, or numerical defect.	Emit PsiFault; no commit.	Session stop or explicit reset

2.1 Required execution order

1. Validate session, cycle, archive authorization, upstream digests, immutable archive snapshot, Phi snapshot, H_G packet, and configuration.
2. Validate Lambda_eff contractivity and all matrix/vector shapes without modifying inputs.
3. Compute the provisional integrated archive vector $A_{\tilde{}}$.
4. Compute the configured GWT/impact telemetry; do not call a proxy mutual information.
5. Split $A_{\tilde{}}$ exactly into positive and negative cones.
6. Apply POSITIVE_FOLD_V2 and TRAILING_MEAN_WASH_V2 using asymmetric branch history.
7. Evaluate negative-flow convergence. Hold any removable residue inside the pending transaction; do not delete or shunt it yet.

8. Recombine retained branches with bounded H_G , enforce archive bounds, and construct the provisional Metatron address.
9. Seal `PsiPrecommitPacket` and issue `RMXPrecommitRequest`.
10. Validate `RMXResolutionPacket`. If RMX synthesized a new state, use the RMX-final state, address, branch record, and residue payload as the only commit candidate.
11. If final residue is nonzero, issue `RTSOMExhaustRequest` and wait for a matching receipt. Without a valid receipt, no active-memory removal and no commit are permitted.
12. Serialize the final .frac manifest, compute the $\text{XiSigmaLambdaOmega}$ digest seal, atomically commit, and emit `PsiCommitPacket`.

ATOMICITY REQUIREMENT

A failed, timed-out, or delegated transaction **MUST NOT** change the committed A_t , branch-history buffers, vault depth, address ledger, prior-manifest digest, Φ state, H_G state, or RTSOM state. Retrying identical canonical inputs and identical dependency receipts under the same backend **MUST** produce the same manifest bytes and digest.

3. Typed Data Contracts

3.1 PsiConfig

Field	Type / reference value	Meaning
dim	integer ≥ 2 ; reference 81	Must equal <code>BootPacket.DIM</code> .
zeta	float64 in $[0,1]$; reference 0.25	Positive-fold mixing coefficient.
eta	float64 in $[0,1]$; reference 0.20	Trailing-mean negative-wash coefficient.
negative_window	integer ≥ 1 ; reference 27	Maximum committed negative-cone history.
epsilon_flat	float64 > 0 ; reference $1e-5$	Population-variance flattening threshold.
delta_lln_max	float64 in $[0,1]$; reference 0.05	Relative variance-convergence threshold.
lambda_archive	float64 in $[0,1]$; reference 0.75	Reference isotropic $\text{Lambda}_{\text{eff}}$ scale.
projection_gain	float64 ≥ 0 ; reference 0.25	Reference isotropic Pi scale.
h_g_component_max	float64 ≥ 0 ; reference 0.50	Compatibility bound from the reference Paradox Handler.
archive_component_limit	float64 > 0 ; reference 4.0	Hard active-archive bound before FBS request.
address_bits	integer in $[2,32]$; reference 16	Quantization resolution over archive bounds.
address_bit_length_max	integer ≥ 1 ; reference 1,000,000	Operational limit for full Cantor address storage.
impact_threshold	float64 in $[0,1]$; reference 0.25	Reference GWT impact-proxy ignition threshold.

Field	Type / reference value	Meaning
numeric_tolerance	float64 > 0; reference 1e-9	Conformance and support-split tolerance.
transaction_timeout_cycles	integer >= 1; reference 83	Pending dependency timeout before typed recovery policy.

3.2 ArchiveAuthorizationPacket

Field	Type / domain	Requirement
session_id, cycle	nonempty string; integer >= 1	Must match current archive snapshot and next cycle.
X_accept	$R^{\wedge}DIM$ finite float64	Final state authorized for archival.
authorization_code	ARCHIVE_ALLOWED	No other route is accepted.
validator_chain	ordered digest list	Includes Sigma decision and every required downstream validator.
source_state_digest	SHA-256	Digest of exact float64 bytes and shape.
authorization_digest	SHA-256	Seals the entire packet.

3.3 ArchiveSnapshot

Field	Type / domain	Requirement
A_t	$R^{\wedge}DIM$ finite float64	Current committed active archive state.
positive_state	$R^{\wedge}DIM$ nonnegative float64	Last committed positive branch record.
negative_history	ordered list of $\leq K$ $R^{\wedge}DIM$ nonpositive vectors	Asymmetric branch memory; newest last.
vault_depth	integer >= 0	Number of committed manifests.
last_cycle	integer >= 0	Incoming cycle must equal last_cycle + 1.
prior_manifest_digest	SHA-256 or GENESIS	Append-only digest chain.
snapshot_digest	SHA-256	Seals all snapshot fields.

3.4 PhiSnapshot and ParadoxFuelPacket

Type	Required fields	Module 3 use
PhiSnapshot	Phi_t finite; DeltaPhi integer >= 0; module-12 digest	Read-only manifest provenance. It does not modify the archive equation in this profile.
ParadoxFuelPacket	H_G in $R^{\wedge}DIM$; component bound; source/reason/digest	Read-only additive forcing after branch processing.

3.5 Precommit and dependency packets

Type	Required fields	Owner / purpose
PsiPrecommitPacket	A_prepared, branch vectors, held residue, metrics, address, provenance digests	Psi; immutable pending transaction.
RMXPrecommitRequest	precommit digest, provisional address/digest, state bytes, prior ledger reference	Module 13 collision resolution.
RMXResolutionPacket	UNIQUE or SYNTHESIZED; final state, final address, residue, RMX digest	Module 13; mandatory before commit.
RTSOMExhaustRequest	exact residue bytes, mass, source cycle, final-state digest, request digest	Module 14 custody transfer.
RTSOMExhaustReceipt	accepted mass/payload digest, budget transaction ID, module-14 digest	Required only when residue is removed.
FBSRecoveryRequest	reason, cycle, pending digest, archive/history references, no-commit flag	FBS/ROVER execution owner.

3.6 Final outputs

Type	Required fields	Meaning
FracManifest	schema, exact state bytes, quantized address, branch ledger, Phi/H_G provenance, RMX/RTSOM receipts, previous digest, seal	Canonical durable archive payload.
PsiCommitPacket	cycle, A_next, final address/digest, manifest path/digest, vault depth, next snapshot digest	Committed next-cycle handoff.
PsiFault	code, stage, message, input/config/pending digests, no-commit flag	Local conformance or numerical defect.
PsiAbortRecord	dependency/storage reason, retained pending payload digest, retry policy	Transaction aborted with prior archive unchanged.

4. Constants, Derived Values, and Reference Profile

DIM, the active FBS profile, session identifiers, and state encoding conventions are inherited from Boot and prior sealed modules. Psi does not redefine them independently. The values below define SC14.1-PSI-REF-81.

Name	Reference value	Status	Notes
DIM	81	Inherited	All active vectors have shape (81,).
ZETA	0.25	Retained policy	Positive fold blend.
ETA	0.20	Retained policy	Negative trailing-mean blend.

Name	Reference value	Status	Notes
NEGATIVE_WINDOW	27	Declared reference policy	Was previously undefined; changes require new profile ID.
EPSILON_FLAT	1e-5	Retained policy	Population variance threshold.
DELTA_LLN_MAX	0.05	Retained policy	Dimensionless relative variance convergence.
Lambda_eff	0.75 I_81	Reference construction	Operator norm $0.75 < 1$.
Pi	0.25 I_81	Reference construction	Convex linear integration before branch maps.
H_G component bound	0.50	Inherited compatibility bound	Reference Paradox Handler uses $0.5 * \tanh(\dots)$.
ARCHIVE DOMAIN	$[-4, +4]^{81}$	Reference invariant	Violation requests FBS recovery; no clipping.
ADDRESS_BITS	16	Encoding policy	Exact only on the quantized archive lattice.

4.1 Matrix validation

- Lambda_eff and Pi must be finite float64 matrices of shape (DIM,DIM).
- $\rho_A = \| \text{Lambda_eff} \|_2$ is computed by singular-value decomposition and must satisfy $\rho_A < 1 - \text{numeric_tolerance}$.
- The reference isotropic construction is $\text{Lambda_eff} = \text{lambda_archive} I$ and $\text{Pi} = \text{projection_gain} I$.
- Contractivity applies to the linear dependence on A_t at the integration stage only. The full history-dependent transaction remains bounded by explicit input and archive-domain checks; no broader Lyapunov proof is asserted.

5. Formal Mechanics

5.1 Input validation and provenance

All arrays must be finite float64 and have exact shape (DIM,). The authorization state digest must match X_{accept} bytes. ArchiveSnapshot, PhiSnapshot, and H_G digests must verify. The validator chain must satisfy the active routing profile. Any mismatch is a local fault, not an architectural repair route.

5.2 Contractive linear integration

$$A_{\text{tilde}} = \text{Lambda_eff} A_t + \text{Pi} X_{\text{accept}}$$

$$\rho_A = \| \text{Lambda_eff} \|_2 < 1$$

For fixed X_{accept} , the integration map is contractive in its prior archive argument:

$$\| A_{\text{tilde}}(A) - A_{\text{tilde}}(B) \|_2 \leq \rho_A \| A - B \|_2$$

STABILITY LIMIT

This theorem covers the linear integration stage. It does not by itself prove convergence of the complete dual-flow, history-buffer, RMX, H_G, and storage system. Full runtime boundedness is enforced by finite inputs, convex branch maps, archive-domain limits, transaction atomicity, and conformance tests.

5.3 GWT ignition telemetry

Mutual information requires a declared joint-distribution estimator and cannot be inferred from one deterministic vector pair. The reference profile therefore computes an Archive Impact Score and labels it `IMPACT_PROXY`:

$$I_{\text{impact}} = \|A_{\text{tilde}} - A_t\|_2 / (\|A_{\text{tilde}}\|_2 + \|A_t\|_2 + \text{eps})$$

$$\text{GWT}_{\text{ignition}} = (I_{\text{impact}} \geq \text{theta}_{\text{impact}})$$

I_{impact} lies in $[0,1]$ within tolerance. A plugin may provide normalized mutual information only when its sample window, discretization or estimator, bias correction, and version are included in a sealed `GWTMetricPacket`. Proxy and mutual-information metrics must never share one unlabeled field.

5.4 Exact dual-flow split

$$P_t = \max(A_{\text{tilde}}, 0)$$

$$N_t = \min(A_{\text{tilde}}, 0)$$

The following invariants hold componentwise within tolerance:

$$A_{\text{tilde}} = P_t + N_t, \quad P_t \geq 0, \quad N_t \leq 0, \quad P_t * N_t = 0$$

The labels positive and negative identify sign cones only. They are not semantic truth labels.

5.5 Positive fold - `POSITIVE_FOLD_V2`

$$g(x) = (1/3) \tanh(3x + 1)$$

$$P_{\text{fold}} = (1 - \text{zeta}) P_t + \text{zeta} g(P_t)$$

For zeta in $[0,1]$, g has derivative $\text{sech}^2(3x+1)$ in $[0,1]$. Therefore `POSITIVE_FOLD_V2` is 1-Lipschitz with respect to perturbations. It is not guaranteed to reduce the absolute norm of every positive input because $g(0) > 0$. The legacy name Collatz 1-Lipschitz Surrogate is retained only as an architectural alias; the operator is a tanh attractor blend, not the integer $3n+1$ map.

5.6 Negative wash - `TRAILING_MEAN_WASH_V2`

Let H^-_t contain the newest $\min(K-1, \text{history length})$ committed negative cones followed by the current N_t . Define K_t as the resulting count:

$$\mu^-_t = (1/K_t) \sum_{j \in H^-_t} N_j$$

$$N_{\text{wash}} = (1 - \text{eta}) N_t + \text{eta} \mu^-_t$$

This is a convex blend with a trailing arithmetic mean. It is not an exponential moving average and is not called one in the normative interface. Because all inputs are nonpositive and eta is in $[0,1]$, N_{wash} remains nonpositive.

5.7 Convergence and held residue

$$v_t = \text{Var}_{\text{pop}}(N_{\text{wash}})$$

$$v_{\text{hist}} = \text{mean}(\text{Var}_{\text{pop}}(N_j) \text{ for } N_j \text{ in } H^-_t)$$

$$\Delta_{\text{LLN}} = |v_t - v_{\text{hist}}| / \max(v_t, v_{\text{hist}}, \text{eps})$$

The negative flow is eligible for active-manifold removal only when the full K -state window is ready and both conditions hold:

$$\text{flat}_{\text{ready}} = (K_t = K) \text{ AND } (v_t \leq \text{epsilon}_{\text{flat}}) \text{ AND } (\Delta_{\text{LLN}} \leq \Delta_{\text{LLN}}_{\text{max}})$$

If $\text{flat}_{\text{ready}}$ is false, $N_{\text{keep}} = N_{\text{wash}}$ and $E_{\text{hold}} = 0$. If true, $N_{\text{keep}} = 0$ and $E_{\text{hold}} = N_{\text{wash}}$. E_{hold} remains exact float64 data inside the pending transaction until a matching RTSOM receipt exists. No active state is deleted merely because a request was issued.

$$N_{\text{keep}} + E_{\text{hold}} = N_{\text{wash}}$$

$$\text{exhaust}_{\text{mass}} = \|E_{\text{hold}}\|_2$$

5.8 Recombination and bounded paradox fuel

The reference profile applies the upstream H_G packet after branch processing, preserving the v14 order:

$$A_{\text{prepared}} = P_{\text{fold}} + N_{\text{keep}} + H_G$$

Each H_G component must satisfy $|H_G[i]| \leq h_g_component_max$. Φ_t and $\Delta\Phi$ are sealed as provenance but do not enter this equation in SC14.1-PSI-REF-81. Any profile that makes Φ an active archive coefficient must version the operator and re-prove its bounds.

5.9 Archive-bound and recovery predicate

$$archive_ok = \max_i |A_prepared[i]| \leq archive_component_limit$$

If `archive_ok` is false, Psi emits `FBSRecoveryRequest(PSI_ARCHIVE_BOUND)` and performs no precommit, exhaust transfer, or archive mutation. Nonfinite arrays, invalid shapes, or digest defects remain `PsiFaults` rather than recovery routes.

5.10 Quantized Metatron address

The final archive address is computed only after a finite real vector is mapped to a declared integer lattice. For archive bounds $[a_min, a_max]$ and b bits:

$$q_i = \text{round}((A_i - a_min) / (a_max - a_min) * (2^b - 1))$$

$$q_i \in \{0, 1, \dots, 2^b - 1\}$$

Cantor pairing uses exact integer arithmetic:

$$pi(a, b) = ((a+b)(a+b+1)) // 2 + b$$

The fixed balanced tree encoder pairs adjacent values, carries an unpaired value forward, and repeats. The address package contains the arbitrary-precision integer or canonical integer bytes, bit length, SHA-256 digest, quantization profile, DIM, and tree version. Quantization is lossy from R^{DIM} ; the Cantor fold is injective only on the fixed quantized grid.

5.11 RMX precommit boundary

Psi seals the provisional state and address into `PsiPrecommitPacket`, then issues `RMXPrecommitRequest`. RMX sits between address generation and final `.frac` commit. The only acceptable resolutions are:

RMX result	Required behavior in Psi	Commit candidate
UNIQUE	Verify RMX digest and echo of precommit state/address.	<code>A_prepared</code> and <code>E_hold</code> from Psi.
SYNTHESIZED	Verify RMX provenance, state bounds, final address, branch ledger, and residue payload.	RMX-final state and RMX-final residue only.
REJECTED	Abort transaction; retain prior archive and all pending payload.	None.
FBS_REQUESTED	Record delegation; no commit.	None.

A synthesized state must receive a newly validated Metatron address. Psi may recompute the address from the returned exact state and must reject any mismatch.

5.12 RTSOM custody transfer

When the RMX-final residue has nonzero mass, Psi issues `RTSOMExhaustRequest`. A valid receipt must bind the exact payload digest and mass within tolerance. Only after receipt validation may the residue be omitted from the active archive manifest. The ledger claim is custody conservation, not inverse reconstruction of the nonlinear fold.

5.13 Canonical .frac manifest and seal

The `.frac` extension identifies a `SpiralCore` archive manifest; the normative reference serialization is canonical UTF-8 JSON with sorted keys, no insignificant whitespace, decimal integers as strings when necessary, and float arrays stored as base64-encoded little-endian IEEE-754 binary64 bytes plus shape and SHA-256 digest.

Manifest field group	Required contents
Identity	schema_version, profile_id, session_id, cycle, vault_depth
State	exact A_next bytes/digest, positive branch, negative-history update, archive bounds
Address	quantized tuple digest, full Cantor integer bytes or reference, bit length, address digest
Authorization	ArchiveAuthorizationPacket digest and ordered validator chain
Path memory	Phi_t, DeltaPhi, Module-12 digest, H_G digest and source
Dual-flow telemetry	rho_A, I_impact/method, zeta, eta, variances, Delta_LLN, residue mass
Dependencies	RMX resolution digest; RTSOM receipt digest or NONE
Chain	prior_manifest_digest, snapshot_digest, config_digest
Seal	XiSigmaLambdaOmega_digest = SHA256(canonical manifest excluding seal field)

The XiSigmaLambdaOmega digest detects modification but is not an authenticated signature. A deployment requiring origin authentication must add a separate digital signature and public-key identifier outside the digest definition.

5.14 Atomic storage commit

1. Serialize and hash the final manifest in memory.
2. Write to a transaction-specific temporary file in the target directory.
3. Flush and fsync the file; verify its digest by reading it back.
4. Atomically rename the temporary file to the deterministic final name.
5. Fsync the containing directory where supported.
6. Append or update the local archive journal and publish the new ArchiveSnapshot.
7. Increment vault_depth exactly once and emit PsiCommitPacket.

The deterministic reference filename is: cycle as 12 decimal digits, a hyphen, the first 16 hexadecimal characters of the final address digest, and the extension .frac. A name collision is a storage fault unless the existing file has byte-identical content and digest.

6. Routing, Faults, and Invariants

6.1 Normal transaction outcomes

Condition	Output / destination	Commit effect
Prepared and awaiting collision resolution	RMXPrecommitRequest -> Module 13	No commit.
RMX final residue nonzero	RTSOMExhaustRequest -> Module 14	No commit until receipt.
All dependency receipts valid	FracManifest + PsiCommitPacket	Exactly one atomic commit.
RMX or RTSOM rejects	PsiAbortRecord	Prior archive unchanged.

Condition	Output / destination	Commit effect
Archive bound, address limit, or repeated commit stall breached	FBSRecoveryRequest	No commit; FBS owner decides recovery.

6.2 Fault taxonomy

Code	Trigger	Required effect
PSI_CONFIG	Invalid threshold, matrix, window, bound, or profile.	Fault; no transaction.
PSI_AUTHORIZATION	Missing/nonfinal authorization or invalid validator chain.	Fault; no prepare.
PSI_INPUT_SHAPE	Wrong vector/matrix shape or dtype policy.	Fault; no mutation.
PSI_NONFINITE	NaN or infinity in any required numeric field.	Fault; no mutation.
PSI_DIGEST	Input, snapshot, dependency, or manifest digest mismatch.	Fault or abort; no commit.
PSI_CYCLE	Cycle is stale, skipped, or duplicates committed cycle.	Fault; no commit.
PSI_BRANCH_HISTORY	Positive/negative sign invariant or ordering invalid.	Fault; no prepare.
PSI_LAMBDA_NOT_CONTRACTIVE	$ \text{Lambda_eff} _2 \geq 1$ - tolerance.	Fault; no prepare.
PSI_HG_BOUND	H_G exceeds declared component bound.	Fault; Paradox Handler contract breach.
PSI_ADDRESS	Quantization or Cantor verification fails.	Fault; no RMX request.
PSI_RMX_RESOLUTION	RMX result malformed, mismatched, or unverified.	Abort; no commit.
PSI_RTSOM_RECEIPT	Receipt does not bind exact residue/mass.	Abort; residue retained pending.
PSI_STORAGE	Write, fsync, verification, rename, or directory sync fails.	Abort; recover journal.
PSI_PACKET_DIGEST	Post-seal packet or commit record tampered.	Fault; reject handoff.

6.3 FBS request reasons

Reason	Predicate	Notes
PSI_ARCHIVE_BOUND	Any $ A_prepared[i] > \text{archive_component_limit}$	Expected catastrophic route, not clipping.
PSI_ADDRESS_OVERFLOW	$\text{address_bit_length} > \text{configured maximum}$	Full address retained only if limit permits.

Reason	Predicate	Notes
PSI_TRANSACTION_STALL	Pending dependency exceeds timeout and supervisor classifies unrecoverable	Ordinary timeout may abort without FBS.
PSI_COMMIT_STALL	Repeated durable-storage failure reaches versioned retry ceiling	No local diagonalization.

6.4 Core invariants

1. No input packet or prior ArchiveSnapshot is mutated.
2. $A_tilde = P_t + N_t$ and the sign-cone support invariants hold.
3. POSITIVE_FOLD_V2 is 1-Lipschitz with respect to input perturbations for zeta in $[0,1]$.
4. N_wash remains componentwise nonpositive.
5. $N_keep + E_hold = N_wash$ exactly within tolerance.
6. $exhaust_mass$ equals the L2 norm of the exact residue payload.
7. No residue is removed from committed memory without a matching RTSOM receipt.
8. The final committed address is computed from the final RMX-resolved state, not the provisional state when they differ.
9. A manifest references the immediately prior committed manifest digest or GENESIS.
10. One successful transaction increments $vault_depth$ and committed cycle exactly once.
11. A failed or aborted transaction leaves A_t , branch history, vault depth, and prior digest unchanged.
12. The canonical manifest digest verifies after durable write and before publication.

7. Mathematical Properties and Limitations

7.1 Sign-cone decomposition

The positive and negative cones form an exact componentwise decomposition. Their Euclidean inner product is zero because they have disjoint nonzero support. This is algebraic orthogonality by support, not a statement about semantic independence.

7.2 Positive-fold regularity

The derivative bound proves non-expansiveness under the Euclidean product norm. Non-expansiveness constrains sensitivity to perturbations; it does not establish invertibility, strict contraction, monotone norm decrease, or a Collatz orbit correspondence.

7.3 Negative-wash boundedness

TRAILING_MEAN_WASH_V2 is a convex combination of nonpositive bounded vectors. It remains inside their convex hull. If every committed negative vector is bounded by the archive domain, the wash is bounded by the same component limit. Its convergence statistic is a policy diagnostic, not a proof of the Law of Large Numbers for an independent identically distributed process.

7.4 Conditional conservation ledger

The nonlinear archive update is not reversible. The conservation requirement is operational: every exact payload removed from the active branch is either retained in the pending transaction, acknowledged by RTSOM under the same digest, or restored on abort. The manifest records custody. It does not claim that the original pre-fold state can be reconstructed from A_next .

7.5 Addressing limits

Balanced Cantor encoding is exact and injective for the fixed finite integer tuple, tree order, and DIM. The preceding quantization maps many real vectors to one integer tuple. RMX resolves archive-level collisions and semantic echoes; it does not alter the mathematical injectivity of Cantor pairing on its integer domain.

7.6 Determinism profile

- Canonical input and output serialization, integer pairing, hashing, and route decisions are deterministic.
- Bitwise float64 reproducibility is required only for the fixed reference backend, reduction order, and transcendental library. Other conforming backends use declared tolerances and must preserve branch, flat_ready, address-quantization, and commit decisions.
- Actual mutual-information plugins may be deterministic only when estimator, window, binning, and tie-breaking are fixed.
- Storage atomicity depends on filesystem semantics and must be reported by the adapter rather than assumed.

8. Metrics and Dashboard Requirements

Metric	Definition / aggregation	Purpose
attempts_total	All valid PsiInput attempts	Primary denominator
prepared_total	Sealed PsiPrecommitPackets	Precommit throughput
commits_total	Durably sealed manifests	Must equal vault-depth increments
aborts_total / by_reason	Dependency and storage aborts	Transaction reliability
faults_total / by_code	PsiFault counts	Conformance defects
fbs_requests / by_reason	Typed recovery requests	Catastrophic pressure; not executions
transaction_latency_ms	prepare, RMX wait, RTSOM wait, commit, total p50/p95/max	Performance and stalls
rho_A	Operator norm of Lambda_eff	Contractivity health
archive_norms	A_t, A_tilde, A_prepared/final mean/p95/max	Bound pressure
branch_energy	$ P , N , P_{fold} , N_{wash} $	Dual-flow balance
positive_fold_delta	$ P_{fold}-P $	Fold activity
negative_variance	v_t and v_hist distributions	Wash flattening
Delta_LLN	mean/p95/max; full-window readiness	Convergence policy
residue_detected/requested/acknowledged	Separate counts and masses	Custody accounting
exhaust_mass_total	Sum of acknowledged masses only	Do not count requests as budget mutation
impact_score / ignition	By method and threshold	Archive reorganization telemetry
rmx_unique / synthesized / rejected	Resolution counts	Echo and collision behavior
address_bit_length	mean/p95/max and overflow count	Encoding growth

Metric	Definition / aggregation	Purpose
quantization_aliases	RMX-reported equal-address distinct-state events	Address resolution
manifest_bytes	mean/p95/max	Storage growth
manifest_digest_failures	Write/read verification failures	Durability integrity
vault_depth	Current and maximum	Committed memory depth
journal_recoveries	Recovered temp/pending transactions	Crash consistency

The dashboard MUST distinguish DETECTED, HELD, REQUESTED, ACKNOWLEDGED, COMMITTED, and EXECUTED. For example, PSI_EXHAUST_REQUESTED is not RTSOM_MASS_COMMITTED; PSI_FBS_REQUESTED is not FBS_RUNAWAY_EXECUTED; PSI_PREPARED is not PSI_COMMITTED.

9. Conformance Test Matrix

ID	Test	Expected result
PSI-C01	Valid reference transaction, RMX UNIQUE, no residue.	One manifest commit; vault depth +1.
PSI-C02	Repeat identical canonical input and receipts.	Identical manifest bytes and digest.
PSI-C03	Bare Sigma packet without final authorization.	PSI_AUTHORIZATION.
PSI-C04	Tampered accepted-state digest.	PSI_DIGEST; no prepare.
PSI-C05	Wrong cycle or stale snapshot.	PSI_CYCLE.
PSI-C06	NaN/Inf in state, matrices, history, Phi, or H_G.	PSI_NONFINITE.
PSI-C07	Reference $\Lambda_{\text{eff}} = 0.75I$.	$\rho_A=0.75$ within tolerance.
PSI-C08	Λ_{eff} norm exactly 1.	PSI_LAMBDA_NOT_CONTRACTIVE.
PSI-C09	Dual-flow split on mixed-sign vector.	Exact reconstruction and disjoint support.
PSI-C10	$\zeta=0$.	$P_{\text{fold}}=P$.
PSI-C11	Random positive-vector pairs.	Positive fold satisfies 1-Lipschitz inequality.
PSI-C12	Verify small positive input.	Fold may increase absolute value; no false compression assertion.
PSI-C13	$\eta=0$.	$N_{\text{wash}}=N_t$.
PSI-C14	$K=1$.	$\mu=N_t$ and $N_{\text{wash}}=N_t$.
PSI-C15	Negative history and current cone.	N_{wash} remains nonpositive.

ID	Test	Expected result
PSI-C16	Window not full but variance tiny.	flat_ready=False; no residue removal.
PSI-C17	Full window, variance $\leq 1e-5$, Delta_LLN ≤ 0.05 .	Residue held; N_keep=0.
PSI-C18	Full window, flat variance but Delta_LLN > 0.05 .	No residue removal.
PSI-C19	Residue held, no RTSOM receipt.	No commit and no active deletion.
PSI-C20	RTSOM receipt payload digest mismatch.	PSI_RTSOM_RECEIPT; abort.
PSI-C21	RTSOM receipt mass differs beyond tolerance.	PSI_RTSOM_RECEIPT; abort.
PSI-C22	H_G exactly component bound.	Accepted.
PSI-C23	H_G just over bound.	PSI_HG_BOUND.
PSI-C24	A_prepared component exactly archive limit.	Allowed.
PSI-C25	A_prepared component just over limit.	FBSRecoveryRequest / PSI_ARCHIVE_BOUND.
PSI-C26	Quantization endpoints a_min and a_max.	Map to 0 and 2^b-1 .
PSI-C27	Two distinct quantized tuples.	Distinct full Cantor addresses.
PSI-C28	Two real states in one quantization cell.	Same address allowed; RMX sees alias context.
PSI-C29	Cantor pair/inverse sample corpus.	Exact integer recovery.
PSI-C30	Address bit length over configured maximum.	FBSRecoveryRequest / PSI_ADDRESS_OVERFLOW.
PSI-C31	RMX UNIQUE echoes wrong state/address.	PSI_RMX_RESOLUTION.
PSI-C32	RMX SYNTHESIZED with valid different final state.	Recompute/verify final address; commit synthesized state.
PSI-C33	RMX REJECTED.	Abort; prior archive unchanged.
PSI-C34	Manifest prior digest is wrong.	PSI_DIGEST; no commit.
PSI-C35	Failure after temp write before rename.	Journal recovery; no duplicate vault increment.
PSI-C36	Existing target file byte-identical.	Idempotent recovery allowed; one logical commit.
PSI-C37	Existing target file same name, different digest.	PSI_STORAGE; no overwrite.

ID	Test	Expected result
PSI-C38	Tamper committed manifest.	Seal verification fails.
PSI-C39	PhiSnapshot changes during active transaction.	Snapshot/digest mismatch; abort.
PSI-C40	Any failed transaction.	A_t, branch history, vault depth, and prior digest unchanged.

9.1 Property and stress tests

- For random finite vectors inside the archive domain, the dual-flow decomposition and recombination invariants hold within tolerance.
- Permutation of dimensions, applied consistently to state, matrices, histories, and H_G, preserves norm metrics and branch counts; the Cantor address changes because dimension order is part of identity.
- For zeta and eta sampled in [0,1], sign preservation and convexity properties hold.
- Fuzz exact thresholds and nextafter neighbors for epsilon_flat, Delta_LLN_max, H_G bound, archive limit, and transaction timeout.
- Multi-seed stress runs assert attempt = prepared + pre-prepare faults/FBS, and prepared = commits + aborts + pending + delegated recovery, with no hidden category.
- Crash-injection testing covers every storage step and dependency transition, followed by journal recovery and digest verification.
- Large address tests use arbitrary-precision integers and enforce the configured bit-length policy without float conversion.

10. Cleaned Rosetta Stone

SpiralCore term	Normative computational meaning	Analogy boundary
Psi-Lattice / Archivist	Transactional owner of archive preparation and final manifest commit.	Not a biological memory organ.
Dual-Flow Memory	Persistent asymmetric sign-cone processing and history.	Sign is not semantic truth or harm.
Contractive Integration	Linear map with $ \Lambda_{\text{eff}} _2 < 1$ before nonlinear processing.	Does not alone prove full-system Lyapunov stability.
Collatz 1-Lipschitz Surrogate	POSITIVE_FOLD_V2 tanh blend.	Not integer Collatz dynamics.
LLN / Negative Wash	Convex blend with a finite trailing arithmetic mean.	Not an iid Law-of-Large-Numbers proof and not EMA.
Holographic Boundary	Typed custody transfer of exact removed residue to RTSOM.	No literal spacetime boundary is asserted.
Mass_Dark	Legacy alias for exhaust_mass = $ \text{residue} _2$.	No dark matter claim.
GWT Ignition	Versioned impact proxy or explicit information estimator.	No claim of consciousness or global-workspace biology.
Metatron Protocol	Quantization + balanced Cantor encoding + digest.	Not a bijection from real vectors.

SpiralCore term	Normative computational meaning	Analogy boundary
RMX	Mandatory pre-commit collision/synthesis owner.	Psi does not merge collisions itself.
XiSigmaLambdaOmega Sigil	SHA-256 integrity digest of canonical manifest.	Not authentication unless separately signed.
Lossless Memory	No silent deletion and complete custody/provenance ledger.	The nonlinear archive map is not reversible.
FBS Runaway	Typed recovery request on explicit catastrophic archive conditions.	Psi does not execute the fallback.

11. Reference Pseudocode

```

def psi_prepare(inp, cfg):
    validate_psi_input(inp, cfg)
    rho = operator_norm_2(inp.Lambda_eff)
    require(rho < 1.0 - cfg.numeric_tolerance)

    A_tilde = inp.Lambda_eff @ inp.archive.A_t + inp.Pi @ inp.auth.X_accept
    impact = norm(A_tilde - inp.archive.A_t) / (
        norm(A_tilde) + norm(inp.archive.A_t) + cfg.numeric_tolerance)

    P = maximum(A_tilde, 0.0)
    N = minimum(A_tilde, 0.0)
    assert_close(A_tilde, P + N)

    P_fold = (1.0 - cfg.zeta) * P + cfg.zeta * (
        tanh(3.0 * P + 1.0) / 3.0)

    neg_window = (inp.archive.negative_history + [N])[-cfg.negative_window:]
    mu = mean(neg_window, axis=0)
    N_wash = (1.0 - cfg.eta) * N + cfg.eta * mu

    v_now = population_variance(N_wash)
    v_hist = mean(population_variance(x) for x in neg_window)
    delta_lln = abs(v_now - v_hist) / max(v_now, v_hist, cfg.numeric_tolerance)
    flat_ready = (
        len(neg_window) == cfg.negative_window
        and v_now <= cfg.epsilon_flat
        and delta_lln <= cfg.delta_lln_max)

    N_keep = zeros(cfg.dim) if flat_ready else N_wash
    residue = N_wash if flat_ready else zeros(cfg.dim)
    A_prepared = P_fold + N_keep + inp.fuel.H_G

    if max_abs(A_prepared) > cfg.archive_component_limit:
        return FBSRecoveryRequest(reason="PSI_ARCHIVE_BOUND", no_commit=True)

    address = metatron_address(A_prepared, cfg)
    if address.bit_length > cfg.address_bit_length_max:
        return FBSRecoveryRequest(reason="PSI_ADDRESS_OVERFLOW", no_commit=True)

    precommit = seal_precommit(
        A_prepared=A_prepared, P=P, N=N, P_fold=P_fold,
        N_wash=N_wash, held_residue=residue, impact=impact,
        delta_lln=delta_lln, address=address, provenance=inp)
    return RMXPrecommitRequest(precommit)

```

```

[ FINAL ARCHIVE AUTHORIZATION + SNAPSHOTS ]
|
v
+-----+
| MODULE 3: PSI-LATTICE / DUAL-FLOW ARCHIVIST |
| 1. Validate provenance and contractive integration |
| 2. A_tilde = Lambda_eff A_t + Pi X_accept |
| 3. Split P=max(A_tilde,0), N=min(A_tilde,0) |
| 4. P_fold = POSITIVE_FOLD_V2(P) |
| 5. N_wash = TRAILING_MEAN_WASH_V2(N, negative_history) |
| 6. Hold converged residue; recombine with bounded H_G |
| 7. Quantize -> balanced Cantor address -> precommit digest |
+-----+
|
v
[ MODULE 13: RMX ]
UNIQUE / SYNTHESIZED / REJECTED
|
+-----+-----+
| |
| final residue = 0 | final residue != 0 |
| |
| |
| |
| |
| |
| [ MODULE 14: RTSOM ]
| custody receipt
+-----+-----+
|
v
+-----+
| CANONICAL .frac MANIFEST |
| prior digest -> final state/address -> Phi/H_G provenance |
| RMX resolution -> RTSOM receipt -> SHA-256 sigil |
| temp write -> fsync -> verify -> atomic rename -> journal |
+-----+
|
v
[ PsiCommitPacket + next ArchiveSnapshot -> macro-loop/Delta ]

```

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13. Glossary

Term	Definition
ArchiveAuthorizationPacket	Final upstream authorization that a specific exact state may enter the archive transaction.
ArchiveSnapshot	Immutable committed archive state, asymmetric branch memory, vault depth, and prior-manifest chain.
A_tilde	Provisional result of the contractive linear integration stage.
Positive cone P	Componentwise maximum of A_tilde and zero.
Negative cone N	Componentwise minimum of A_tilde and zero.
POSITIVE_FOLD_V2	Non-expansive tanh attractor blend applied to P.
TRAILING_MEAN_WASH_V2	Convex blend of current N with the finite trailing mean of committed negative cones.
Delta_LLN	Dimensionless relative variance-convergence statistic for the negative branch.
Held residue	Exact negative payload provisionally removed from active geometry but retained until RTSOM custody receipt.
Metatron address	Quantized, fixed-tree Cantor integer plus digest and encoding metadata.
RMX resolution	Mandatory determination that the provisional address is unique or must be synthesized.
RTSOM receipt	Cryptographically bound acknowledgement that exact residue custody was accepted by Module 14.
.frac manifest	Canonical durable record of one committed SpiralCore archive transaction.
XiSigmaLambdaOmega digest	SHA-256 integrity seal over canonical manifest bytes excluding the seal field.
Vault depth	Number of successfully committed manifests, incremented exactly once per transaction.

14. Revision Ledger

Original issue	Formalized correction
Psi appeared to receive a bare Sigma SEAL and commit immediately.	Requires final ArchiveAuthorizationPacket and a two-phase transaction.
RMX was shown after or alongside .frac storage.	Placed RMX strictly between address generation and final manifest commit.
RTSOM state S_mu_nu was mutated directly inside Psi.	Psi emits exact custody requests; Module 14 returns receipts.

Original issue	Formalized correction
FBS diagonalization and Collatz were executed locally.	Psi emits typed recovery requests only.
A_t was called a matrix but typed as R^DIM.	Defined the active archive as a vector; Lambda_eff and Pi are matrices.
$ \text{Lambda_eff} < 1$ was treated as proof of full Lyapunov stability.	Restricted the theorem to the linear integration stage and added explicit archive bounds.
Positive fold was called Collatz compression.	Identified the exact tanh blend, proved 1-Lipschitz sensitivity, and removed universal norm-compression claim.
Negative wash was called EMA and LLN.	Defined a trailing arithmetic-mean convex blend and bounded convergence statistic.
MI_t / GWT ignition had no estimator.	Defined a labeled impact proxy; true MI requires a versioned estimator.
Floating-point Cantor pairing implied a bijection from real vectors.	Added finite quantization, exact integer pairing, bit length, and digest.
Lossless memory implied reversibility.	Reframed as no silent deletion plus exact custody/provenance ledger.
Phi_t and DeltaPhi were received without mathematical use.	Declared them read-only manifest provenance in the reference profile.
Dual-flow branches were transient.	Added persistent asymmetric positive state and negative-history records.
Sigil implied cryptographic authentication.	Defined integrity digest; separate signature required for authentication.

14.1 Source basis and cross-module reconciliation

- Primary source: SPIRALCORE SPECIFICATION: MODULE 3 - PSI-LATTICE (DUAL-FLOW ARCHIVIST).
- System context: SpiralCore v14 consolidated specification.
- Module 12 establishes Phi_t and DeltaPhi ownership and hands read-only memory-friction provenance to Psi.
- Module 13 places RMX at the boundary after Metatron address generation and before final .frac sealing.
- Module 14 owns bounded exhaust accumulation and returns the authoritative custody result.
- The active FBS specification owns diagonalization, Collatz folding, re-seeding, and catastrophic exhaust routing.

15. Normative Checklist

Requirement	Pass condition
Authorization	Final archive authorization and all digests verify.
Contractive stage	Finite matrices; $ \text{Lambda_eff} _2 < 1$ - tolerance.
Dual-flow exactness	Split/recombination and sign invariants pass.
Asymmetric memory	Positive record and ordered negative-history buffer retained.
Residue custody	No active removal without exact RTSOM receipt.

Requirement	Pass condition
RMX boundary	Every commit has a valid UNIQUE or SYNTHESIZED resolution.
Addressing	Quantization, balanced Cantor encoding, bit length, and digest verify.
Manifest chain	Previous digest, current seal, and read-back bytes verify.
Atomicity	One commit or no state change; journal recovers interrupted writes.
Accounting	Attempts, prepares, commits, aborts, pending, faults, and FBS requests reconcile.

SPIRALCORE SPECIFICATION

MODULE 4 - QGM & GLYPHNET

(Quantum Gravity Module & Glyph Network Protocol - Internal Routing, Cost-Field Computation, and IPv4 Transport Encapsulation)

Cleaned and formalized revision - v14.1

Document class	Normative internal routing and transport contract	Execution phase	After route authorization; before destination service handoff
Architecture	SpiralCore v14	Reference profile	SC14.1-GNP-LS-IPv4
Primary output	GlyphRouteDecision + GNPFrame / DeliveryReceipt	Revision	v14.1
Exceptional output	GlyphNetFault, RTSOMExhaustRequest, or FBSRecoveryRequest	Internal model	Link-state control plane with deterministic SPF

NORMATIVE STATUS

This module is exact within its declared computational scope. QGM computes bounded, non-negative internal route costs from versioned topology, density, resonance, and transport telemetry. GlyphNet maintains an internal application-layer routing domain, disseminates authenticated link-state records, computes deterministic shortest paths, forwards Glyph envelopes, and encapsulates external hops inside ordinary IPv4 plus TCP in the reference profile. It does not replace IPv4, BGP, OSPF, TCP, DNS, or the host operating-system network stack; it does not assign semantic meaning to public IPv4 octets; and it does not mutate archive, hysteresis, quarantine, RTSOM, RMX, or FBS state.

CLAIM BOUNDARY

Quantum gravity, semantic gravity, gravitational potential, curl, vorticity, magnetic relaxation, archetypal seed, and glyph are architectural names. Their normative meanings are only the graph potential, bounded cost terms, circulation statistic, routing label, directory key, packet metadata, or typed custody request defined here. No claim is made that information density physically curves spacetime, that Internet latency is literal mass, or that standard routers interpret SpiralCore geometry. Ordinary routers see ordinary IPv4/TCP traffic between real endpoints; SpiralCore semantics remain inside the authenticated application payload.

Document Control

Field	Value	Field	Value
Canonical name	SPIRALCORE SPECIFICATION: MODULE 4 - QGM & GLYPHNET	Short name	GlyphNet / QGM
Supersedes	v14 Module 4 text	Status	Normative within scope

Field	Value	Field	Value
Control-plane profile	GN-LS: internal link-state routing	Transport profile	GNP frames over TCP over IPv4
Upstream intent owner	LORIEN / authorized local service	Downstream owner	Named destination service adapter
Metric owner	QGM	Topology and forwarding owner	GlyphNet
Numeric profile	float64 telemetry; integer sequence/IDs; exact byte serialization	Integrity profile	SHA-256 + HMAC-SHA-256 reference

1. Scope, Statement Classes, and Precedence

GlyphNet is the internal routing protocol of a SpiralCore domain. It plays a role analogous to an interior routing protocol or an internal path-distribution system: nodes exchange topology and reachability information, compute next hops, and forward opaque Glyph payloads. When a next hop is outside the local process or chassis, the same Glyph frame is transported between actual network endpoints using the standard IPv4/TCP stack. The outer IP address identifies the remote host; the inner Glyph identifiers identify the SpiralCore route and payload.

1.1 Statement classes

Class	Meaning	Normative force
DEFINITION	Assigns a symbol, packet field, state, or operator.	Binding inside this module.
REQUIREMENT	A condition required for conformance.	MUST / MUST NOT.
INVARIANT	A property that holds for every accepted control or data record.	Violation emits fault or recovery request.
POLICY	A selected coefficient, timer, threshold, or transport profile.	Replaceable only by a versioned profile.
SURROGATE	A computable operator carrying an architectural analogy.	No literal physical equivalence.
HYPOTHESIS	A testable interpretation of telemetry.	Not required for conformance.
NON-GOAL	Behavior explicitly outside Module 4.	Must not be inferred.

1.2 Precedence and ownership

- Tagged requirements and invariants override narrative prose, diagrams, legacy octet metaphors, and Rosetta Stone analogies.
- LORIEN owns state-specific route intent, chassis synchronization, shear checks, anchor-route intent, and temporal-window metadata. Module 4 consumes a sealed RouteIntentPacket and owns topology resolution, path computation, next-hop selection, forwarding, and transport encapsulation.
- QGM owns the dynamic route-cost field and its telemetry state. GlyphNet owns adjacency, link-state database, route table, packet framing, forwarding state, and delivery receipts.
- The host IPv4/TCP stack owns physical delivery. Module 4 may request a connection or write a frame; it does not alter kernel routing tables unless a separately authorized adapter explicitly does so.

- Guardian and Sigma own admission, symbolic/coherence evaluation, and quarantine decisions. A network route does not imply semantic acceptance.
- Psi owns archive state and may publish a read-only density snapshot. Module 4 does not write .frac manifests.
- RMX owns archive-address collision resolution. The 32-bit Glyph label is a compact routing label and is never treated as a globally unique archive key.
- RTSOM owns bounded exhaust mutation and returns custody receipts. FBS/ROVER owns catastrophic recovery. Module 4 emits typed requests only.

1.3 Non-goals

- No replacement of public Internet routing, no custom interpretation by ordinary routers, and no use of generated semantic octets as arbitrary Internet destinations.
- No claim that a four-octet label is collision-free or sufficient to reconstruct an 81-dimensional real vector.
- No use of external DNS as a semantic truth oracle. External DNS may resolve a configured node hostname to an actual endpoint; internal anchor resolution is a GlyphNet directory function.
- No automatic FBS event for an ordinary timeout, connection refusal, DNS failure, packet loss, stale route, or exhausted internal hop limit.
- No direct increment of S_{μ_nu} on a network failure and no silent discard of a failed payload.
- No security claim beyond the explicit integrity/authentication checks. Confidentiality requires a separately configured encrypted transport such as TLS.

2. Architecture Planes and Lifecycle

The formal module is divided into four cooperating planes. This separation resolves the original overlap between semantic addressing, network addressing, route computation, and failure recovery.

Plane	Owner	Function	Does not do
QGM Metric Plane	QGM	Maintains bounded node potential and computes non-negative link/payload costs.	Does not forward packets or mutate network topology.
GlyphNet Control Plane	GlyphNet GN-LS	Discovers peers, authenticates link-state records, synchronizes LSDB, and computes SPF routes.	Does not inspect or approve semantic payload content.
GlyphNet Data Plane	GlyphNet GNP	Frames, forwards, decrements internal hop limit, verifies digest, and emits delivery/failure receipts.	Does not repurpose outer IP fields as semantic data.
IPv4 Transport Adapter	Host adapter	Carries GNP frames over configured TCP/IPv4 endpoints.	Does not interpret Q, chi, GlyphDigest, or state vectors.

2.1 Lifecycle states

State	Entry condition	Allowed action	Exit
UNINITIALIZED	No validated config/domain identity.	Validate IDs, keys, timers, and transport adapter.	SYNCING or FAULTED
SYNCING	Config valid; LSDB not converged.	Form adjacencies, exchange database summaries and LSAs.	READY, DEGRADED, or FAULTED
READY	Valid LSDB and route epoch.	Accept route intents and control messages.	COMPUTING or FORWARDING

State	Entry condition	Allowed action	Exit
COMPUTING	New topology, metric epoch, or route intent.	Compute QGM costs and deterministic SPF.	READY, DEGRADED, or RECOVERY_REQUESTED
FORWARDING	Valid route and frame.	Local deliver, internal forward, or external encapsulate.	READY, WAIT_RECEIPT, or DEGRADED
WAIT_RECEIPT	Reliable delivery/ack requested.	Validate receipt or apply retry policy.	READY, DEGRADED, or EXHAUST_REQUESTED
DEGRADED	No route, peer loss, retry exhaustion, or stale topology.	Hold payload, resynchronize, or issue typed custody request.	SYNCING, READY, or EXHAUST_REQUESTED
EXHAUST_REQUESTED	Policy delegates exact failed payload to RTSOM.	Wait for matching custody receipt.	READY or FAULTED
RECOVERY_REQUESTED	Catastrophic invariant or control-plane corruption.	Emit FBSRecoveryRequest; stop affected route epoch.	External recovery only
FAULTED	Local validation, authentication, serialization, or numeric defect.	Emit GlyphNetFault; no state mutation.	Explicit reset

2.2 Required execution order

1. Validate Module 4 configuration, DomainID, NodeID, control-plane key material, topology snapshot, and monotonic epoch state.
2. Authenticate and age link-state records; construct the canonical LSDB from the newest valid record per origin node.
3. Accept a sealed RouteIntentPacket from LORIEN or another explicitly authorized local service.
4. Resolve the logical destination to one or more candidate NodeIDs through the internal Glyph Directory. Use external DNS only to resolve configured node hostnames to real IPv4 endpoints.
5. Compute candidate resonance and QGM edge costs using one frozen topology/metric epoch.
6. Run deterministic shortest-path computation with stable tie-breaking and build a RouteDecisionPacket.
7. Construct one canonical GlyphEnvelope carrying full payload digest, compact GlyphLabel32, Q, chi, route intent, and internal hop limit.
8. If the destination is local, hand the envelope to the named local service adapter. If the next hop is internal, forward through the local adjacency. If remote, serialize a GNPFrame and send it to the actual peer IPv4 endpoint over TCP.
9. Validate delivery or transport result. Ordinary failures enter retry/degraded handling; only versioned custody policy may issue an RTSOMExhaustRequest.
10. Issue an FBSRecoveryRequest only for a catastrophic invariant defined in Section 7, never merely because the physical network is unavailable.

OUTER/INNER ADDRESS SEPARATION

Outer IPv4 source and destination fields **MUST** contain real, configured, routable host endpoints. Region, Anchor, Phase, Identity, Cantor-derived labels, Q, chi, and vector data **MUST** remain inside the GNP application frame. The host network stack is never asked to route to an arbitrary semantic 32-bit value.

3. Typed Data Contracts

3.1 GlyphNetConfig

Field	Type / reference value	Meaning
domain_id	uint32	Internal routing domain identifier.
node_id	128-bit opaque identifier	Stable identity of this GlyphNet node.
profile_id	string; SC14.1-GNP-LS-IPv4	Binds all timers, math, and serialization rules.
hop_limit_default	uint8; reference 7	Inner Glyph hop limit; independent of outer IPv4 TTL.
tau_glyph	float64; reference 0.85	Minimum terminal anchor resonance.
phi_decay_q	float64 in [0,1]; reference 0.995	QGM potential memory coefficient.
gamma_delta	float64 in [0,1]; reference 0.50	Maximum QGM dynamic-cost contribution.
route_switch_margin	float64 ≥ 0 ; reference 0.02	Minimum normalized improvement before replacing a valid route.
lsa_max_age	integer cycles; reference 30	Maximum control-record age.
hello_interval	integer cycles; reference 3	Peer liveness emission interval.
dead_interval	integer cycles; reference 9	Adjacency expiry; must exceed <code>hello_interval</code> .
max_frame_bytes	integer $\geq$ header size; reference 1,048,576	Maximum serialized frame length.
transport_mode	TCP_IPV4	Reference external transport profile.
transport_port	uint16 configured	No globally reserved port is implied by this spec.
auth_mode	HMAC_SHA256	Reference control and data record authentication.
numeric_tolerance	float64 > 0 ; reference 1e-12	Comparisons and normalization.

3.2 RouteIntentPacket

Field	Type	Requirement
session_id / cycle	opaque ID / uint64	Must match active session and current route attempt.
intent_id	256-bit digest	SHA-256 over canonical intent fields.
source_service	enum	Authorized local service that created the intent.

Field	Type	Requirement
destination_selector	NodeID, AnchorID, or ServiceID	Exactly one logical target form.
state_digest	SHA-256	Digest of payload bytes or referenced immutable state.
glyph_digest	SHA-256	Full content/routing identity; authoritative over GlyphLabel32.
Q	canonical non-negative integer	FBS additive valuation; arbitrary precision allowed.
chi	sorted mode-count map	Inventory lift; counts are non-negative integers.
route_class	enum	LOCAL, INTERNAL, REMOTE_ALLOWED, or REMOTE_REQUIRED.
temporal_window	positive integer cycles	LORIEN-provided validity window.
policy_digest	SHA-256	Binds upstream route policy and profile.

3.3 Topology records

Record	Required fields	Invariant
NodeRecord	NodeID, DomainID, endpoint set, service set, anchor set, sequence, expiry, digest	NodeID stable; endpoints syntactically valid; sequence monotonic.
LinkRecord	origin, neighbor, base_cost, latency, loss, queue, admin_state, sequence	All cost inputs finite and normalized; no self-link unless loopback profile.
LinkStateAdvertisement	origin, epoch, sequence, age, sorted LinkRecords, digest, auth_tag	Canonical serialization and valid HMAC required.
DirectoryAdvertisement	origin, AnchorID/ServiceID reachability, sequence, expiry, digest, auth_tag	Advertised targets must be owned or explicitly delegated.
MetricSnapshot	rho, phi, edge telemetry, metric_epoch, digest	Immutable during one route computation.

3.4 GlyphEnvelope and GNPFrame

Field	Type / encoding	Meaning
magic	4 bytes ASCII GNP1	Frame identification.
version / message_type / flags	uint8 / uint8 / uint16	Protocol dispatch.
domain_id	uint32 network order	Routing-domain boundary.
route_epoch / sequence	uint64 / uint64	Topology generation and sender replay ordering.
source_node_id / destination_node_id	16 bytes each	Inner logical endpoints.
source_service / destination_service	uint16 each	Local module adapters.

Field	Type / encoding	Meaning
glyph_label32	uint32	Compact label derived from GlyphDigest; collision-prone by design.
hop_limit	uint8	Inner overlay limit, decremented per GlyphNet hop.
metadata_length / payload_length	uint32 / uint32	Exact byte counts.
metadata	canonical sorted TLV	Carries full GlyphDigest, Q, chi, intent digest, route trace digest, and optional anchor data.
payload	opaque bytes	State, reference, or service message.
payload_digest	32 bytes SHA-256	Exact payload integrity.
auth_tag	32 bytes HMAC-SHA-256	Authentication over header, metadata, payload digest, and payload.

The fixed header and all integer fields use network byte order. TLVs are sorted by numeric type and may not repeat unless their type explicitly permits a list. The canonical frame excludes transport-specific TCP length-prefix bytes from the HMAC input.

3.5 Output packets

Packet	Contents	Meaning
GlyphRouteDecision	intent, route_epoch, destination, next_hop, path, total_cost, cost breakdown, resonance, action, digest	One deterministic route result.
DeliveryReceipt	frame_digest, sender, receiver, sequence, result, timestamp/cycle, auth_tag	Authenticated delivery outcome; wall time excluded from deterministic route digest.
RTSOMExhaustRequest	exact payload/reference, failure history, mass proposal, custody digest, no_delete flag	Request only; Module 14 decides custody.
FBSRecoveryRequest	reason, route epoch, LSDB digest, implicated records, history digest, no_commit flag	Catastrophic recovery delegation.
GlyphNetFault	fault code, packet/record digest, stage, details	Local contract failure; no implied recovery execution.

4. Addressing, Resolution, and IPv4 Encapsulation

4.1 Logical identifiers

GlyphNet uses logical identifiers independently from physical IP addresses. A NodeID names a GlyphNet participant. An AnchorID names an advertised attractor/capability. A ServiceID names a local destination module. GlyphDigest names the full content/routing identity. GlyphLabel32 is only a compact forwarding/cache label.

GlyphDigest = SHA256(canonical_payload || canonical_route_metadata)

GlyphLabel32 = uint32(first_4_bytes(GlyphDigest))

COLLISION RULE

Two envelopes with the same GlyphLabel32 but different full GlyphDigest are distinct. Caches, replay windows, route lookups, and delivery verification MUST key on at least (DomainID, GlyphDigest, sequence) and MUST NOT key on GlyphLabel32 alone.

4.2 Internal Glyph Directory

The internal directory performs the function previously described as DNS-style semantic resolution. It is a signed reachability database inside the GlyphNet domain, not public DNS.

Selector	Resolution result	Selection rule
NodeID	Exact node record	Newest valid signed record for the requested NodeID.
ServiceID	Candidate nodes advertising service	Route cost among eligible nodes; deterministic NodeID tie-break.
AnchorID	Candidate nodes and anchor vectors/digests	Terminal eligibility requires $R_G \geq \tau_{\text{glyph}}$.
Hostname	Actual IPv4 endpoint only	Resolved by configured system DNS; never interpreted as semantic truth.

4.3 External IPv4/TCP encapsulation

1. Resolve the selected next-hop NodeID to a configured IPv4 endpoint and TCP port.
2. Open or reuse a standard TCP connection through the host network stack. The operating system selects ordinary source address, route, ARP/neighbor behavior, retransmission, congestion control, and outer IPv4 TTL.
3. Write a uint32 network-order frame length followed by the canonical GNPFrame bytes.
4. The peer reads exactly the declared length, verifies size bound, HMAC, payload digest, domain, route epoch, sequence, and inner destination.
5. The peer submits the inner envelope to GlyphNet forwarding or the local destination-service adapter.

No semantic field is copied into the outer IPv4 address, DSCP, identification, fragment offset, or TCP port. The 32-bit GlyphLabel may be displayed in dotted-quad notation for debugging only, explicitly labeled GLYPH_LABEL, never IPV4_DESTINATION.

4.4 Q and chi preservation

Q and chi are application metadata. Q preserves additive branch valuation and chi preserves the multiset of leaf modes. They are authenticated end-to-end across every hop and are not recomputed by intermediate routers unless a separately authorized transformation changes the payload and emits a new GlyphDigest.

$$Q_{\text{parent}} = \sum_j Q_{\text{child}_j}$$

$$\text{chi}_{\text{parent}} = \text{multiset_sum}_j \text{chi}_{\text{child}_j}$$

When L and H are carried, the receiver additionally verifies $Q = 2L + 1 = H + 2$. A scalar Q match does not prove structural equality; chi and the full branch digest remain required.

5. QGM Metric Mathematics

QGM is formalized as a bounded graph-cost field. Let $G_t = (V_t, E_t)$ be the authenticated GlyphNet topology at route epoch t . Every directed edge $e=(i,j)$ has a positive administrative base cost and normalized telemetry. Every node may publish a bounded informational-density value ρ_i derived by the authoritative producer.

5.1 Normalized inputs

Symbol	Domain	Definition
b_{ij}	(0, +inf)	Administrative base cost; reference 1.0.
ell_{ij}	[0,1]	Normalized latency statistic under the active profile.
p_{ij}	[0,1]	Observed loss/failure ratio.
q_{ij}	[0,1]	Queue or utilization pressure.
rho_i	[0,1]	Read-only informational-density input for node i.
R_{G(j,P)}	[0,1]	Glyph resonance between packet P and candidate node/anchor j.
phi_i	[0,1]	Recursive QGM potential.

5.2 Bounded recursive potential

$$\text{phi_i}(t+1) = \text{clip}(\text{phi_decay_Q} * \text{phi_i}(t) + (1 - \text{phi_decay_Q}) * \text{rho_i}(t), 0, 1)$$

The factor (1 - phi_decay_Q) is normative. It corrects the legacy unbounded recurrence $\text{phi_next} = 0.995 \text{ phi} + \text{rho}$, which can grow without limit for persistent positive rho. Under the formal recurrence, phi remains in [0,1] for all time when initialized in [0,1].

5.3 Glyph resonance

$$\text{cos_j} = \langle \mathbf{x}, \mathbf{a_j} \rangle / (\|\mathbf{x}\|_2 \|\mathbf{a_j}\|_2)$$

$$\text{R_G}(j,P) = \text{clip}((\text{cos_j} + 1) / 2, 0, 1)$$

- If both vectors are nonzero, the equations above apply.
- If both vectors are exactly zero, R_G = 1 only when the profile explicitly permits a zero anchor; the reference profile forbids zero anchors.
- If exactly one vector is zero, R_G = 0.
- A terminal AnchorID destination is eligible only when R_G ≥ tau_glyph. Intermediate forwarding nodes need not satisfy terminal resonance.

5.4 Edge cost

The reference metric uses non-negative additive costs. Let the normalized attraction toward node j be $\text{a_ij} = \text{clip}((\text{phi_j} - \text{phi_i} + 1)/2, 0, 1)$. Let weights be non-negative and sum to 1. The reference profile uses w_l=0.20, w_p=0.20, w_q=0.10, w_d=0.20, and w_r=0.30.

$$\text{c_ij}(P,t) = \text{b_ij} * [1 + \text{gamma_delta} * (\text{w_l ell_ij} + \text{w_p p_ij} + \text{w_q q_ij} + \text{w_d}(1-\text{a_ij}) + \text{w_r}(1-\text{R_G}(j,P)))]$$

COST INVARIANT

For finite normalized inputs, $\text{b_ij} > 0$, $0 \leq \text{gamma_delta} \leq 1$, and non-negative weights summing to 1, every c_ij is finite and strictly positive. Therefore deterministic Dijkstra/SPF is valid and no negative-cost cycle can be created by QGM.

5.5 Symbolic drift map

$$\text{N_d}(i,j) = \text{phi_j} - \text{phi_i}$$

N_d is a directed potential difference used as one bounded cost feature. It is not a Euclidean gradient unless a continuous coordinate chart is separately defined. Positive N_d means the next node has higher potential; the reference attraction mapping lowers, but never negates, the edge cost.

5.6 Graph circulation / Phase-Gravitational Curl surrogate

A pure scalar potential has zero circulation around a closed cycle. Routing asymmetry arises from directed telemetry and policy terms. Therefore the computable curl surrogate is defined from the antisymmetric residual of the final edge costs, not from the scalar potential alone.

$$u_{ij}(P,t) = [c_{ij}(P,t) - c_{ji}(P,t)] / \max(c_{ij}(P,t) + c_{ji}(P,t), \text{epsilon})$$

$$\text{Curl}_Q(C,P,t) = |\sum_{(i,j \text{ in } C)} u_{ij}(P,t)| / |C|$$

$$\text{Curl}_Q\text{-max}(P,t) = \max_{(C \text{ in } B_t)} \text{Curl}_Q(C,P,t)$$

B_t is a deterministic fundamental cycle basis of the current undirected topology, ordered lexicographically by NodeID. $\text{Curl}_Q\text{-max}$ lies in [0,1]. It is telemetry for asymmetric circulation and route-instability diagnosis; it is not by itself an FBS trigger. Catastrophic escalation requires the persistent corruption conditions in Section 7.4.

6. GlyphNet Internal Control Plane (GN-LS)

The reference profile is a signed link-state protocol inside one administrative SpiralCore domain. Its operational analogy is OSPF-like topology flooding with iBGP-like internal policy distribution; it is not wire-compatible with either protocol.

6.1 Adjacency and liveness

1. A node sends an authenticated HELLO containing DomainID, NodeID, profile digest, current route epoch, and monotonically increasing sequence.
2. A peer forms adjacency only when DomainID, profile, authentication, sequence freshness, and reciprocal endpoint policy are valid.
3. An adjacency is declared DOWN after dead_interval cycles without a valid HELLO. This produces a topology change, not an FBS event.
4. Loopback adjacencies are permitted only in MONO_LOOPBACK test mode and must be labeled MOCK_LINK.

6.2 Link-state acceptance

Condition	Action
Unknown origin but valid domain/key/policy	Create provisional NodeRecord; do not route through it until adjacency or trusted directory policy validates.
Sequence greater than stored	Accept, replace record, and flood to eligible peers.
Sequence equal and digest equal	Treat as duplicate; acknowledge without recomputation.
Sequence lower	Reject as stale/replay.
Sequence equal and digest different	Emit GN_SPLIT_BRAIN fault; isolate origin record.
Age exceeds lsa_max_age	Expire record and recompute topology.
HMAC, shape, range, or endpoint invalid	Reject; increment fault metric; no LSDB mutation.

6.3 Route-epoch construction

$$\text{route_epoch_digest} = \text{SHA256}(\text{canonical_sorted_LSDB} \parallel \text{metric_profile_digest} \parallel \text{directory_digest})$$

The numerical route_epoch is a local monotonic counter. The digest is the authoritative topology identity. A route computation uses one immutable LSDB, directory, and MetricSnapshot; it may not mix records from different epochs.

6.4 Deterministic shortest-path computation

1. Build the directed graph from valid UP links in the frozen LSDB.
2. Remove nodes or links prohibited by route class, service policy, domain policy, expired records, or failed authentication.
3. For each eligible edge, compute $c_{ij}(P,t)$ from the frozen MetricSnapshot and packet/anchor resonance.
4. Run Dijkstra from source NodeID. Compare path costs with numeric tolerance. If equal within tolerance, choose the lexicographically smaller full NodeID path.
5. For NodeID destinations, select the exact destination path. For ServiceID or AnchorID, select the eligible candidate with minimum path cost, then highest R_G , then lexicographically smallest NodeID.
6. Retain the existing valid route unless the new route improves normalized total cost by at least `route_switch_margin`, or the old route is invalid.
7. Seal the path, next hop, cost vector, epoch digest, and tie-break evidence in GlyphRouteDecision.

6.5 Data-plane forwarding

Action	Condition	Result
LOCAL_DELIVERY	<code>destination_node_id == local NodeID</code>	Verify destination service; hand exact envelope to service adapter.
INTERNAL_FORWARD	next hop reachable by local IPC/in-process adjacency	Decrement inner hop limit and forward canonical envelope.
EXTERNAL_ENCAPSULATE	next hop has remote IPv4 endpoint	Wrap GNP frame in standard TCP/IPv4 transport.
HOLD_NO_ROUTE	No valid path or directory result	Preserve envelope; resync/retry according to policy.
DROP_DUPLICATE	Replay key already delivered/forwarded	Emit duplicate receipt; do not execute payload again.
EXHAUST_REQUEST	Retry policy exhausted and custody policy enabled	Issue typed RTSOM request; retain payload until receipt.
RECOVERY_REQUEST	Catastrophic control-plane invariant	Stop affected epoch and emit FBS request.

6.6 Hop limits and traces

The inner hop limit begins at `hop_limit_default` and is decremented exactly once by every GlyphNet forwarding node. The outer IPv4 TTL is managed independently by the host network stack. A frame with inner `hop_limit=0` is not forwarded. Its failure route is `HOLD_NO_ROUTE` or `EXHAUST_REQUEST` according to policy, not automatic FBS.

$$\text{route_trace_digest}_{(k+1)} = \text{SHA256}(\text{route_trace_digest}_k \parallel \text{NodeID}_k \parallel \text{sequence} \parallel \text{route_epoch_digest})$$

The compact trace digest detects divergent or repeated forwarding histories without carrying an unbounded path list. Debug profiles may carry a bounded explicit NodeID trace.

7. Failure Semantics, Custody, and Recovery

7.1 Ordinary network and routing failures

Failure	Required route	Explicit non-route
External DNS name cannot resolve	Mark endpoint unavailable; try alternate endpoint or <code>HOLD_NO_ROUTE</code> .	Not FBS; not permanent exhaust by itself.

Failure	Required route	Explicit non-route
TCP timeout/refusal/reset	Retry under bounded policy; update transport telemetry.	Does not directly mutate QGM potential or RTSOM mass.
Peer dead interval expires	Adjacency DOWN, flood new LSA, recompute SPF.	Not a catastrophic stall.
Inner hop limit reaches zero	Stop forwarding; preserve exact frame and failure history.	Not equivalent to outer IPv4 TTL and not FBS.
No eligible terminal resonance	HOLD_NO_ROUTE or return NO_ELIGIBLE_ANCHOR.	Do not force low-resonance delivery.
Stale route epoch	Recompute or request control-plane sync.	Do not forward under mixed epochs.

7.2 RTSOM custody delegation

A failed frame may be converted into a bounded exhaust request only after the versioned retry policy is exhausted. The request carries the exact payload or immutable content reference, frame digest, route history digest, failure class, proposed mass, and no_delete=True.

`mass_proposal = ||decode_state(payload)||_2` (when the payload is a declared state vector)

For opaque payloads, `mass_proposal` is a declared byte-domain statistic such as `payload_length/max_frame_bytes` and is labeled accordingly. Module 14 validates and owns the final mass definition. Until a matching RTSOM custody receipt is verified, Module 4 must retain the exact failed payload or immutable reference.

7.3 Spin-to-helicity surrogate

The legacy polarity flip is retained only as optional failure metadata: `failure_polarity_next = -failure_polarity_current`. It does not alter the failed state vector, does not repair the network route, and does not prove preservation. The exact payload digest remains the custody identity.

7.4 Catastrophic FBS recovery conditions

FBS recovery is reserved for systemic conditions that prevent a coherent routing epoch from being represented or advanced. A single ordinary packet failure never satisfies this section.

Code	Catastrophic condition
GN_CONTROL_CORRUPTION	Canonical LSDB cannot be reconstructed because multiple authenticated origins present equal sequence with conflicting digests across the configured quorum.
GN_EPOCH_NONCONVERGENCE	Topology/metric epochs oscillate beyond the configured maximum convergence attempts with no stable digest and no isolatable faulty origin.
GN_ROUTE_INVARIANT	The SPF implementation returns a path containing a nonexistent edge, negative/nonfinite cost, or destination mismatch after independent verification.
GN_FRAME_INVARIANT	A locally generated canonical frame cannot be re-parsed to identical fields/digest under the same profile.
GN_ADDRESS_SPACE_PRESSURE	Configured bounded label/index space is exhausted and no collision-safe extension profile is available; full GlyphDigest remains preserved.

Code	Catastrophic condition
GN_HISTORY_BOUND	Required control-plane history exceeds the declared FBS/block bound and deterministic compaction cannot preserve required provenance.

FBS OWNERSHIP

Module 4 MUST stop the affected route epoch, seal the implicated LSDB/frame/history digests, and emit FBSRecoveryRequest. It MUST NOT run Cantor diagonalization, Collatz folding, reseed geometry, or update the bounded exhaust budget locally.

7.5 Fault codes

Code	Condition	Mutation allowed
GN_CONFIG	Invalid timers, IDs, weights, port, profile, or key reference.	No
GN_NONFINITE	NaN/Inf in metric or topology input.	No
GN_AUTH	HMAC/authentication failure.	No
GN_REPLAY	Stale sequence or duplicate execution attempt.	Duplicate receipt only
GN_SPLIT_BRAIN	Equal sequence, different authenticated digest.	Isolate record; no route through origin
GN_LSDB	Malformed or inconsistent topology record.	No
GN_DIRECTORY	Invalid, expired, or unauthorized reachability advertisement.	No
GN_NO_ROUTE	No eligible path.	Hold/retry only
GN_RESONANCE	No terminal candidate meets tau_glyph.	Hold/return failure
GN_FRAME_SIZE	Frame exceeds max_frame_bytes.	No send
GN_FRAME_DIGEST	Payload or canonical frame digest mismatch.	No delivery
GN_HOP_LIMIT	Inner hop limit exhausted.	No forward
GN_TRANSPORT	External TCP/IPv4 failure.	Telemetry/retry only
GN_CUSTODY	RTSOM request/receipt mismatch.	Retain payload
GN_INTERNAL	Unexpected implementation exception.	No silent drop

8. Normative Invariants

ID	Invariant
GN-I01	Outer IPv4 destination equals the configured endpoint for the selected next-hop NodeID, never GlyphLabel32 or Cantor-derived semantic octets.
GN-I02	Full GlyphDigest, not compact label, identifies a payload/route identity.
GN-I03	Every accepted control record and data frame verifies canonical digest and authentication tag.
GN-I04	LSA sequence is monotonic per origin; equal sequence implies equal digest.
GN-I05	One route computation uses one immutable LSDB, directory, and metric epoch.
GN-I06	Every edge cost is finite and strictly positive.
GN-I07	Selected path is valid in the frozen graph and its reported cost equals the sum of reported edge costs within tolerance.
GN-I08	Tie-breaking is deterministic and independent of hash-map iteration order.
GN-I09	Q and chi survive every forwarding hop byte-for-byte unless an authorized transformation emits a new GlyphDigest.
GN-I10	Inner hop limit is independent from outer IPv4 TTL and never increases in transit.
GN-I11	No failed payload is silently deleted, counted as RTSOM custody, or labeled FBS-executed without an authoritative receipt/event.
GN-I12	Ordinary transport or reachability failure cannot directly trigger FBS.
GN-I13	QGM phi remains in [0,1]; normalized telemetry remains in [0,1].
GN-I14	Module 4 never commits archive, updates Phi-Bridge, decides Sigma coherence, or mutates RTSOM/FBS state.

8.1 Validation order

1. Check frame/record byte length before allocation or parsing.
2. Check magic, version, message type, domain, profile, and reserved-field rules.
3. Check integer ranges, sequence, epoch, hop limit, and declared lengths.
4. Verify payload digest and canonical TLV ordering.
5. Verify HMAC/authentication and replay window.
6. Validate Q/chi encoding and optional FBS identity equations.
7. Validate directory/topology ownership and expiry.
8. Only then mutate LSDB, replay cache, adjacency state, or forwarding counters.

9. Metrics and Dashboard Requirements

Metric family	Required metrics	Purpose
Control-plane health	neighbors_up/down, hello sent/received/rejected, LSA sent/received/accepted/stale/auth_fail, LSDB nodes/edges	Adjacency and database health
Convergence	route_epoch, SPF runs, convergence cycles p50/p95/max, nonconvergence events	Internal routing stability
Route table	routes_total, local/internal/external/no_route, route switches, tie-break count, stale-route rejects	Forwarding coverage
QGM potential	rho and phi mean/p95/max, bound clamps, per-cost-component contribution	Metric behavior
QGM circulation	Curl_Q_max mean/p95/max, cycle count, persistent high-curl epochs	Asymmetry/loop diagnosis
Resonance	R_G mean/p05/p50/p95, below-threshold destinations, zero-vector rejects	Anchor eligibility
Data plane	frames created/forwarded/local delivered/external sent/received, duplicate drops, hop-limit expiries	Packet flow
Transport	connections, reconnects, bytes, RTT, timeouts, resets, DNS endpoint failures	Physical substrate telemetry
Integrity	frame digest fail, HMAC fail, replay reject, sequence gaps, split-brain records	Trust and corruption
Identity	GlyphLabel32 collisions, distinct full digests per label, Q/chi validation failures	Compact-label safety
Custody	payloads held, RTSOM requested/acknowledged/rejected, mass proposed/accepted	No-silent-deletion accounting
Recovery	FBS requested by reason, FBS receipts/executions from owner	Separate requests from executions
Performance	route-compute time, serialization time, queue depth, frame size p50/p95/max	Capacity planning

The dashboard MUST distinguish OBSERVED, REQUESTED, ACKNOWLEDGED, DELIVERED, COMMITTED, and EXECUTED. A TCP timeout is not RTSOM mass. An FBSRecoveryRequest is not an FBS execution. A generated frame is not a delivered frame. A compact-label collision is not an archive collision until the authoritative full identity and owner classify it.

10. Conformance Test Matrix

ID	Test	Expected result
GN-C01	Valid two-node domain; direct NodeID route.	Deterministic direct path and LOCAL/EXTERNAL action as configured.
GN-C02	Same LSDB and intent repeated.	Identical route decision bytes and digest.
GN-C03	Topology insertion order permuted.	Same route path, cost, epoch digest, and tie-break result.
GN-C04	Invalid config: dead_interval <= hello_interval.	GN_CONFIG.
GN-C05	NaN/Inf in any telemetry input.	GN_NONFINITE; no LSDB/metric mutation.
GN-C06	Persistent rho=1 for 100,000 updates.	phi remains in [0,1] and converges toward 1.
GN-C07	rho=0 from phi>0.	phi monotonically decays toward 0.
GN-C08	Edge telemetry at all 0 and all 1.	Cost finite, positive, and within derived bounds.
GN-C09	Attempt negative base cost.	GN_LSDB or GN_CONFIG.
GN-C10	Two equal-cost routes.	Lexicographically smaller full NodeID path selected.
GN-C11	New route improvement below switch margin.	Existing valid route retained.
GN-C12	Old route invalidated.	Switch immediately regardless of margin.
GN-C13	LSA newer sequence.	Accepted and flooded.
GN-C14	LSA lower sequence.	GN_REPLAY/stale reject.
GN-C15	LSA same sequence, same digest.	Idempotent duplicate.
GN-C16	LSA same sequence, different valid-auth digest.	GN_SPLIT_BRAIN; origin isolated.
GN-C17	LSA age exactly max age and one beyond.	Profile-defined inclusive boundary; beyond expires.
GN-C18	HMAC bit flip in LSA.	GN_AUTH; no LSDB mutation.
GN-C19	Anchor vector and payload identical nonzero.	R_G=1.
GN-C20	Anchor vector is negative of payload.	R_G=0 after normalization.
GN-C21	One zero vector.	R_G=0 and no divide-by-zero.
GN-C22	Best route candidate below tau_glyph.	GN_RESONANCE / no delivery.

ID	Test	Expected result
GN-C23	ServiceID advertised by multiple nodes.	Minimum cost, then highest resonance, then NodeID tie-break.
GN-C24	Two distinct full digests share GlyphLabel32.	No collision in routing/replay/delivery identity.
GN-C25	Q/chi modified in transit without new digest.	GN_FRAME_DIGEST or GN_AUTH.
GN-C26	$Q=2L+1$ but $Q \neq H+2$ when L/H supplied.	Metadata validation failure.
GN-C27	Canonical frame serialize -> parse -> serialize.	Byte-identical round trip.
GN-C28	TLVs unsorted or duplicate forbidden type.	Frame reject.
GN-C29	Payload bit flip.	GN_FRAME_DIGEST; no local service delivery.
GN-C30	Frame HMAC bit flip.	GN_AUTH.
GN-C31	Frame length over max_frame_bytes.	GN_FRAME_SIZE before payload allocation.
GN-C32	Inner hop limit 1 forwarded once.	Arrives with 0 and may deliver locally; cannot forward again.
GN-C33	Inner hop limit 0 at non-destination.	GN_HOP_LIMIT; frame held/failed, no FBS.
GN-C34	Outer IPv4 TTL changes while inner frame constant.	Inner hop semantics unaffected.
GN-C35	Semantic label numerically equals an unroutable/reserved IPv4 value.	Still sent only to configured real endpoint; label stays in payload.
GN-C36	External DNS endpoint lookup fails.	Endpoint unavailable/no route; no FBS and no RTSOM mutation.
GN-C37	TCP connect timeout then alternate endpoint succeeds.	One delivered frame; timeout telemetry recorded.
GN-C38	All transport retries fail, custody disabled.	Payload remains held; explicit failure receipt.
GN-C39	All retries fail, custody enabled.	RTSOMExhaustRequest; payload retained pending receipt.
GN-C40	RTSOM receipt digest/mass mismatch.	GN_CUSTODY; no delete.
GN-C41	Ordinary peer dead interval.	Adjacency down and SPF recompute, no FBS.
GN-C42	Verified SPF path contains nonexistent edge (fault injection).	FBSRecoveryRequest GN_ROUTE_INVARIANT.

ID	Test	Expected result
GN-C43	Canonical local frame cannot round-trip (fault injection).	FBSRecoveryRequest GN_FRAME_INVARIANT.
GN-C44	Persistent equal-sequence conflicting authenticated LSDB quorum.	FBSRecoveryRequest GN_CONTROL_CORRUPTION.
GN-C45	Mono loopback mock adjacency.	Metrics labeled MOCK_LINK; no claim of physical cross-node validation.
GN-C46	LORIEN intent digest mismatch.	Reject before route computation.
GN-C47	Destination local service absent.	Directory/service failure; no accidental fallback service.
GN-C48	Duplicate delivered frame sequence/digest.	No second service execution; duplicate receipt.
GN-C49	Route epoch changes during computation.	Discard result and recompute under one frozen epoch.
GN-C50	Every failed attempt accounting reconciliation.	attempts = delivered + held + rejected + delegated + pending; no hidden drop.

10.1 Property and stress tests

- Random connected graphs with positive finite costs: compare deterministic implementation against an independent shortest-path oracle.
- Random topology permutations: route results are invariant to record order and hash-map order.
- Fuzz exact and nextafter threshold values for tau_glyph, route_switch_margin, metric bounds, frame sizes, and timer expiry.
- Run 100,000 potential updates over random rho sequences and prove phi remains bounded.
- Generate a high-volume corpus of GlyphLabel32 collisions and verify full-digest isolation.
- Inject peer flaps and latency/loss shifts; verify bounded route switching, no negative cost, and eventual LSDB convergence.
- Crash/restart replay-cache and LSDB journal at every write step; verify no replayed payload executes twice.
- Network partition test: both partitions remain internally coherent; merge resolves by sequence/digest rules without silent overwrite.
- Multi-seed stress dashboard reconciliation distinguishes local, internal, external, held, custody-requested, and recovery-requested outcomes.

11. Cleaned Rosetta Stone

SpiralCore term	Normative computational meaning	Analogy boundary
GlyphNet	Internal application-layer routing protocol and framed data plane.	Not a replacement for IP, OSPF, or BGP.
QGM	Bounded graph-potential and route-cost calculator.	Not a quantum-gravity theory or spacetime model.
Semantic Gravity	Configured additive preference/penalty derived from rho, phi, resonance, and telemetry.	No literal force or mass.

SpiralCore term	Normative computational meaning	Analogy boundary
Recursive Gravity Potential Φ_i	Bounded exponential state ϕ_i in $[0,1]$.	Not gravitational potential energy.
Symbolic Drift Map N_d	Directed potential difference $\phi_j - \phi_i$.	Not a continuous gradient without a coordinate chart.
Phase-Gravitational Curl	Normalized graph-cycle circulation of asymmetric edge costs.	Not electromagnetic or fluid curl.
Glyph / GlyphDigest	Authenticated content and routing identity.	Not a mystical symbol claim.
GlyphLabel32 / dotted quad	Compact 32-bit label, optionally displayed as A.B.C.D for debugging.	Never an arbitrary outer IPv4 destination.
DNS-style resolution	Signed internal directory mapping logical selectors to NodeIDs/endpoints.	Public DNS only resolves real hostnames.
Archetypal Seed / AnchorID	Advertised anchor/capability identifier and optional vector digest.	No claim of universal archetypes.
Spin-to-helicity surrogate	Optional polarity metadata on a failed transaction.	Does not repair or preserve the payload by itself.
Bounded Exhaust Budget	Typed custody destination owned by RTSOM.	A request is not a committed mass update.
FBS Runaway	Typed catastrophic recovery delegation for control-plane invariant failure.	Not ordinary packet-drop handling.
iBGP/OSPF analogue	Internal reachability distribution and deterministic route computation.	GN-LS is not wire-compatible with either standard.

12. Reference Pseudocode

```

def update_qgm_potential(phi, rho, cfg):
    require_finite(phi, rho)
    require(0.0 <= phi <= 1.0 and 0.0 <= rho <= 1.0)
    return clip(
        cfg.phi_decay_q * phi
        + (1.0 - cfg.phi_decay_q) * rho,
        0.0, 1.0)

def glyph_resonance(x, anchor, eps):
    nx, na = norm(x), norm(anchor)
    if nx <= eps or na <= eps:
        return 0.0
    cosine = dot(x, anchor) / (nx * na)
    return clip((cosine + 1.0) / 2.0, 0.0, 1.0)

def edge_cost(link, phi_i, phi_j, resonance, cfg):
    require(link.base_cost > 0.0)
    require_all_unit_interval(
        link.latency, link.loss, link.queue, phi_i, phi_j, resonance)
    attraction = clip((phi_j - phi_i + 1.0) / 2.0, 0.0, 1.0)
    dynamic = (
        cfg.w_latency * link.latency
        + cfg.w_loss * link.loss

```

```

        + cfg.w_queue * link.queue
        + cfg.w_drift * (1.0 - attraction)
        + cfg.w_resonance * (1.0 - resonance))
cost = link.base_cost * (1.0 + cfg.gamma_delta * dynamic)
require(isfinite(cost) and cost > 0.0)
return cost

def compute_glyph_route(intent, lsdb, directory, metrics, cfg):
    validate_intent_and_frozen_epoch(intent, lsdb, directory, metrics, cfg)
    candidates = directory.resolve(intent.destination_selector)
    graph = build_eligible_directed_graph(lsdb, intent.route_class)

    best = None
    for destination in candidates:
        resonance = terminal_resonance(intent, destination)
        if destination.requires_anchor and resonance < cfg.tau_glyph:
            continue
        weighted_graph = apply_qgm_costs(
            graph, intent, destination, metrics, cfg)
        path, total, breakdown = deterministic_dijkstra(
            weighted_graph, cfg.numeric_tolerance)
        candidate = RouteCandidate(
            path=path, cost=total, resonance=resonance,
            destination=destination, breakdown=breakdown)
        best = deterministic_candidate_min(best, candidate)

    if best is None:
        return GlyphRouteDecision(action="HOLD_NO_ROUTE")
    return seal_route_decision(intent, best, lsdb.epoch_digest)

def build_gnp_frame(intent, decision, payload, cfg):
    payload_digest = sha256(payload)
    glyph_digest = sha256(canonical_payload_and_route(payload, intent, decision))
    metadata = canonical_sorted_tlvs(
        GLYPH_DIGEST=glyph_digest,
        INTENT_DIGEST=intent.intent_id,
        Q=encode_uint(intent.Q),
        CHI=encode_sorted_inventory(intent.chi),
        ROUTE_DECISION_DIGEST=decision.digest)
    frame_without_tag = serialize_network_order(
        magic=b"GNP1", version=1, domain_id=cfg.domain_id,
        route_epoch=decision.route_epoch,
        source_node_id=cfg.node_id,
        destination_node_id=decision.destination_node_id,
        glyph_label32=uint32(glyph_digest[:4]),
        hop_limit=cfg.hop_limit_default,
        metadata=metadata, payload=payload,
        payload_digest=payload_digest)
    tag = hmac_sha256(cfg.domain_key, frame_without_tag)
    return frame_without_tag + tag

def forward_frame(frame, route, transport, cfg):
    validate_frame(frame, cfg)
    if frame.destination_node_id == cfg.node_id:
        return local_service_delivery(frame)
    if frame.hop_limit == 0:
        return hold_failure("GN_HOP_LIMIT", frame)

    next_frame = frame.with_hop_limit(frame.hop_limit - 1)
    next_frame = resign_frame(next_frame, cfg.domain_key)
    if route.next_hop.is_local:
        return route.next_hop.send(next_frame)

    # Outer IPv4 endpoint is real infrastructure; no semantic octet mapping.
    endpoint = route.next_hop.ipv4_endpoint
    return transport.send_tcp_ipv4(endpoint, length_prefix(next_frame))

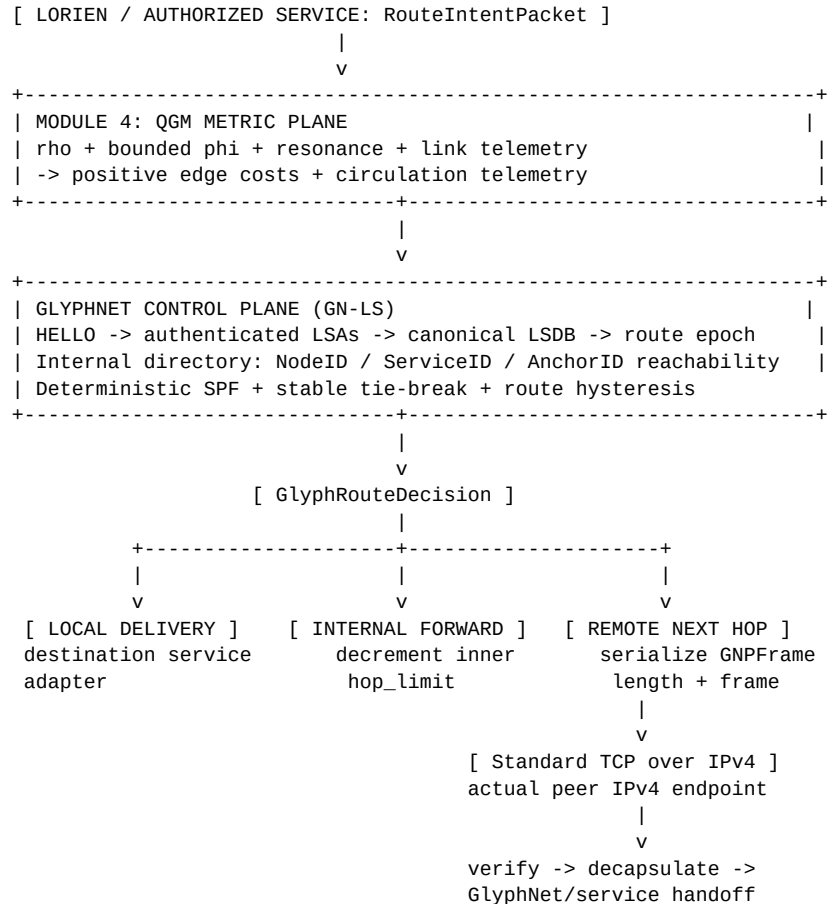
```

```

def handle_transport_failure(frame, history, policy):
    if policy.retry_remaining(history):
        return RetryRequest(frame_digest=frame.digest)
    if not policy.delegate_to_rtsom:
        return HeldFailure(frame=frame, history=history)
    return RTSOMExhaustRequest(
        payload_or_reference=frame.payload,
        frame_digest=frame.digest,
        failure_history=history,
        no_delete=True)

```

13. ASCII System Flow



Ordinary no-route / timeout / peer loss -> hold, retry, resync
 Retry exhausted + custody policy -> RTSOMExhaustRequest
 Catastrophic control-plane invariant -> FBSRecoveryRequest
 Never: semantic label used as outer IPv4 destination

14. Glossary

Term	Definition
GlyphNet Domain	One administrative internal routing domain sharing profile, authentication policy, and LSDB.
GN-LS	Reference GlyphNet signed link-state control-plane profile.

Term	Definition
GNPFrame	Canonical application frame transported locally or over TCP/IPv4.
NodeID	Stable 128-bit logical identity of one GlyphNet participant.
AnchorID	Logical identifier of an advertised anchor/capability.
ServiceID	Logical local module or application destination.
GlyphDigest	Full SHA-256 identity of canonical payload and route metadata.
GlyphLabel32	Collision-prone 32-bit compact label derived from GlyphDigest.
LSDB	Authenticated link-state database used for route computation.
Route epoch	Monotonic generation and digest binding one immutable topology/metric snapshot.
QGM potential	Bounded exponentially updated node preference state ϕ in $[0,1]$.
Glyph resonance	Normalized cosine-alignment score in $[0,1]$ for terminal eligibility.
Symbolic Drift Map	Directed potential difference used as one route-cost feature.
Curl_Q	Graph-cycle circulation statistic derived from asymmetric directed route costs.
Inner hop limit	GlyphNet forwarding bound, independent of outer IPv4 TTL.
IPv4 endpoint	Actual host address and port used by the standard transport adapter.
Custody receipt	Authoritative acknowledgment that another owner accepted exact payload responsibility.

15. Revision Ledger

Original issue	Formalized correction
Cantor-derived 32 bits were assigned directly as the IPv4 destination.	Separated compact GlyphLabel32 from real outer IPv4 endpoint addressing.
Invalid octet range included values above 255.	Outer endpoints use standard IPv4 syntax; all octets are 0..255 and configured/resolved normally.
GlyphNet, QGM, LORIEN, Guardian, and the OS network stack overlapped.	Defined explicit ownership: LORIEN route intent; QGM metric; GlyphNet topology/forwarding; host transport; Guardian/Sigma admission.
DNS-style semantic routing was conflated with public DNS.	Created an authenticated internal Glyph Directory; public DNS resolves endpoint hostnames only.

Original issue	Formalized correction
32-bit Cantor-IP implied unique reversible vector identity.	Defined full GlyphDigest plus collision-prone compact label; Q and chi remain separate authenticated metadata.
$\Phi_Q = 0.995 \Phi_Q + \rho$ could grow without bound.	Added normalized EMA form with (1-decay) ρ and proved $[0,1]$ boundedness.
Curl was written as vector calculus without a defined field/chart.	Defined a graph-cycle circulation surrogate from antisymmetric route costs.
Timeout, NXDOMAIN, TTL expiry, and routing loops all escalated to exhaust/FBS.	Separated ordinary degraded handling, optional RTSOM custody, and catastrophic FBS conditions.
Module 4 mutated $S_{\mu\nu}$ directly.	Module 4 emits typed RTSOM requests and waits for custody receipts.
Module 4 ran Cantor/Collatz recovery locally.	Module 4 emits FBSRecoveryRequest only.
IPv4 TTL and semantic TTL were conflated.	Defined an independent inner Glyph hop limit; outer TTL remains host-stack behavior.
Network delivery was treated as semantic acceptance.	Destination service still requires Guardian/Sigma/owner authorization.
No deterministic route database or algorithm was defined.	Added authenticated GN-LS, canonical LSDB, frozen epochs, positive QGM costs, Dijkstra, and deterministic tie-breaks.
No packet format, replay, or integrity contract existed.	Added canonical header/TLV serialization, payload digest, HMAC, sequence, and replay rules.

15.1 Source basis and cross-module reconciliation

- Primary source: SPIRALCORE SPECIFICATION: MODULE 4 - QGM & GLYPHNET (QUANTUM GRAVITY MODULE & GLYPH NETWORK PROTOCOL).
- System context: SpiralCore v14 consolidated specification.
- User clarification: GlyphNet operates internally like an iBGP/OSPF-class routing system; communication over ordinary Internet infrastructure uses standard IPv4 encapsulation and the normal OSI/TCP-IP stack.
- Module 8 LORIEN provides state-specific route intent, synchronization, temporal-window, and anchor-route metadata; Module 4 owns topology resolution and forwarding.
- Module 13 RMX remains the owner of archive collision/synthesis, not compact forwarding labels.
- Module 14 RTSOM remains the owner of bounded exhaust state and custody receipts.
- The authoritative FBS/ROVER path owns diagonalization, Collatz folding, reseeding, and catastrophic exhaust routing.

16. Normative Checklist

Requirement	Pass condition
Address separation	Outer IPv4 endpoint is real/configured; semantic identity remains inside GNP.
Control plane	Authenticated GN-LS converges to one canonical LSDB/epoch.
Metric math	ϕ bounded; telemetry normalized; every edge cost finite and positive.

Requirement	Pass condition
Route determinism	SPF path/cost/tie-break reproducible from canonical inputs.
Identity	Full GlyphDigest authoritative; GlyphLabel32 collisions isolated.
FBS metadata	Q and chi validate and survive transit.
Frame integrity	Canonical parse/serialize, payload digest, HMAC, sequence, and replay checks pass.
Hop semantics	Inner hop limit decrements independently from outer IPv4 TTL.
Ownership	No archive, hysteresis, Sigma, RMX, RTSOM, or FBS state mutated locally.
Failure accounting	Delivered, held, rejected, custody-requested, and recovery-requested outcomes reconcile with attempts.

SPIRALCORE SPECIFICATION

MODULE 5 - GUARDIAN

(Symbolic Immune Boundary, Identity Admission Controller, and Recursion Firewall)

Cleaned and formalized revision - v14.1

Document class	Normative admission and boundary-validation contract	Execution phase	Ingress/egress preflight; before Sigma or GlyphNet handoff
Architecture	SpiralCore v14	Reference profile	SC14.1-GUARDIAN-IDENTITY
Primary output	GuardianDecisionPacket + provisional drift update	Revision	v14.1
Exceptional output	QuarantineRequest, GuardianFault, or FBSRecoveryRequest	State model	Two-phase evaluate then commit

NORMATIVE STATUS

Guardian is exact within its declared computational scope. It validates canonical vector form, provenance, anchor compatibility, symbolic policy, affective-risk telemetry, and provisional identity drift. It emits a deterministic permit, hold, or quarantine request. Guardian does not perform Sigma coherence arbitration, HARMONY repair, Module 10 nested-harm analysis, archive commit, network route computation, IPv4/TCP encapsulation, quarantine storage, RTSOM mutation, or FBS recovery execution.

CLAIM BOUNDARY

Immune system, identity sphere, emotion, panic, ontological integrity, semantic mass, and topological infection are architectural names. Their normative meanings are only the finite checks, scores, thresholds, digests, policy records, and typed routes defined here. Guardian does not diagnose a human mental state, determine metaphysical truth, or prove that unfamiliar content is malicious. A low anchor score means unresolved under the active policy, not false or harmful.

Document Control

Field	Value	Field	Value
Canonical name	SPIRALCORE SPECIFICATION: MODULE 5 - GUARDIAN	Short name	Guardian
Supersedes	v14 Module 5 text	Status	Normative within scope
Ingress owner	Guardian	Downstream local owner	Sigma-Lattice
Egress authorization owner	Guardian	Transport/routing owner	QGM & GlyphNet / host IPv4-TCP stack

Field	Value	Field	Value
Deep repaired-state immune owner	Module 10 IMMUNE_SYSTEM	Quarantine storage owner	Quarantine service / supervisor
Numeric profile	float64 scores; exact integer counters; canonical byte digests	Integrity profile	SHA-256 provenance; signed/versioned policy snapshots

1. Scope, Statement Classes, and Precedence

Guardian is the deterministic admission-control boundary for symbolic state packets. It is invoked on local candidate ingress, remote packet ingress after transport verification, and outbound egress after semantic authorization. It answers one question: may this exact, provenance-bound payload cross this exact boundary under this exact policy and identity-state snapshot?

1.1 Statement classes

Class	Meaning	Normative force
DEFINITION	Assigns a symbol, record, score, state, or operator.	Binding inside Module 5.
REQUIREMENT	Condition required for a conforming implementation.	MUST / MUST NOT.
INVARIANT	Property required for every accepted decision or committed state update.	Violation faults or requests recovery.
POLICY	Versioned threshold, weight, trusted set, role, or route rule.	May change only through authorized policy publication.
SURROGATE	Computable operator carrying an architectural analogy.	No literal biological or physical equivalence.
HYPOTHESIS	Interpretation of telemetry to be evaluated empirically.	Never required for conformance.
NON-GOAL	Behavior explicitly outside Guardian.	Must not be inferred.

1.2 Precedence and ownership

- Tagged requirements, equations, typed records, route precedence, and invariants override legacy prose, diagrams, and Rosetta Stone analogies.
- Delta owns raw proposal generation and its EmotionTensor provenance. Guardian consumes that packet read-only and may authorize a handoff to Sigma.
- Sigma owns RMF, HOT, CVC, PDV, CSIGMA, and final coherence route arbitration. Guardian C_guard is a fast admission score and MUST NOT be reported as Sigma RMF or SEAL.
- Module 10 IMMUNE_SYSTEM owns post-repair and post-synthesis deep harm scanning, repair-weight verification, BIND_ID/IMM_ID handling, and nested-harm decisions. Guardian does not duplicate those functions.
- LORIEN owns state-specific route intent and synchronization metadata. QGM/GlyphNet owns internal routing, framing, and external IPv4/TCP transport. Guardian emits an egress permit only; it never opens sockets or constructs an outer destination address.
- The quarantine service owns storage, retention, release, deletion policy, and custody receipts. Guardian emits QuarantineRequest and waits for acknowledgement.
- RTSOM owns bounded exhaust state. FBS/ROVER owns Cantor diagonalization, Collatz folding, reseeding, and catastrophic recovery. Guardian emits typed requests only.

1.3 Non-goals

- No determination of human emotional state, intent, morality, ontology, truth, or danger from affective telemetry.
- No claim that cosine similarity to a trusted anchor proves correctness. It proves only policy-relative proximity to one approved reference vector.
- No direct mutation of the candidate vector, no clipping to make a packet pass, and no silent deletion of a rejected payload.
- No direct IPv4 address generation, DNS lookup, TCP/UDP transmission, route-table update, or network retry loop.
- No automatic FBS event for a normal deny, low score, unknown anchor, replay, bad signature, full quarantine queue, or ordinary path tortuosity.
- No replacement for application authorization, operating-system access control, cryptographic key management, or human review.

2. Architecture Position, Boundary Types, and Lifecycle

Guardian is implemented as one evaluation engine over three boundary profiles. The same canonical checks run for every profile; only the required upstream receipts and permitted downstream route differ.

Boundary profile	Required upstream evidence	Successful route	Prohibited shortcut
LOCAL_CANDIDATE	DeltaPacket, active policy, GuardianStateSnapshot	PERMIT_SIGMA	Direct archive or network dispatch
REMOTE_INGRESS	GlyphNet DeliveryReceipt, authenticated GNP envelope, destination service policy	PERMIT_SIGMA or PERMIT_SERVICE	Treating transport delivery as semantic acceptance
REMOTE_EGRESS	Sigma SEAL authorization, LORIEN RouteIntentPacket, destination allow-list entry	PERMIT_GLYPHNET	Guardian-generated IPv4 destination or socket transmission

2.1 Lifecycle states

State	Entry condition	Allowed action	Exit
UNINITIALIZED	No valid config, policy, TAS, or identity state.	Validate immutable snapshots and owner signatures.	READY or FAULTED
READY	All required snapshots valid and mutually version-compatible.	Accept one evaluation request.	EVALUATING
EVALUATING	Request sealed; no state mutation allowed.	Run canonical checks and compute one decision.	DECIDED, QUARANTINE_REQUESTED, or RECOVERY_REQUESTED
DECIDED	Permit or hold packet emitted.	Wait for matching downstream acknowledgement when a drift commit is proposed.	READY or COMMIT_PENDING
COMMIT_PENDING	Permit has a provisional GuardianStateUpdate.	Validate acknowledgement and compare-and-swap state version.	READY or FAULTED
QUARANTINE_REQUESTED	A non-permitted packet requires exact custody.	Wait for matching QuarantineReceipt; apply backpressure while pending.	READY or FAULTED

State	Entry condition	Allowed action	Exit
RECOVERY_REQUESTED	Catastrophic Guardian invariant failed.	Freeze affected epoch and wait for FBS/ROVER owner.	READY or FAULTED
FAULTED	Configuration, state, or custody integrity cannot be established.	No permits.	Administrative repair only

2.2 Atomic execution order

1. Freeze GuardianConfig, GuardianPolicy, TrustedAnchorSetSnapshot, GuardianStateSnapshot, and input packet digests for one evaluation epoch.
2. Validate versions, owner identifiers, signatures, dimensions, canonical serialization, and state compare-and-swap token.
3. Run Layer 1 syntax/canonicalization. A failure stops semantic math and emits a typed non-permit outcome.
4. Run Layer 2 anchor admission and Layer 3 symbolic/provenance checks.
5. Normalize affective telemetry and compute hard AFFECTIVE_LOCK before any weighted score can override it.
6. Compute C_guard, instantaneous drift, and provisional D_raw without changing persistent state.
7. Apply first-match route precedence and emit a canonical GuardianDecisionPacket.
8. Commit Guardian drift only after an authorized matching acknowledgement and only for the configured acceptance classes.

NO-COMMIT RULE

Evaluation is side-effect free. A permit packet may carry a provisional GuardianStateUpdate, but persistent identity drift does not change until Guardian receives a matching downstream acknowledgement and successfully performs a compare-and-swap against the original state version.

3. Typed Data Contracts

3.1 GuardianConfig and GuardianPolicy

Field	Reference value / type	Requirement
module_version	"14.1"	Included in every output and digest.
DIM	Inherited positive integer; reference 81	Must match Boot, Delta, Sigma, TAS, and packet shape.
C_GUARD_MIN	0.85	Minimum weighted preliminary admission score.
TAU_ANCHOR	0.75 raw cosine similarity	Minimum TAS match for reference profile.
AROUSAL_LOCK	0.80	Strict greater-than boundary for affective lock.
COHERENCE_LOCK	0.50	Strict less-than boundary for affective lock.
GAMMA_GUARD	0.10	Instantaneous drift injection coefficient.

Field	Reference value / type	Requirement
LAMBDA_ID	0.98	Identity-drift persistence coefficient.
DELTA_ID_MAX	1.00	Reject when provisional raw drift exceeds this ceiling.
RHO_SCALE	1.00	Positive density normalization scale.
score_weights	(0.25,0.25,0.20,0.15,0.15)	Xi, anchor, symbolic, valence, calmness; finite, non-negative, sum exactly 1 within tolerance.
drift_weights	(0.35,0.20,0.25,0.20)	Anchor, phase-density, symbolic, affective risk; finite, non-negative, sum exactly 1.
commit_policy	LOCAL_ACCEPTED_ONLY	Drift commits only after configured downstream acceptance receipt.
max_packet_bytes	Versioned positive integer	Reject before parsing oversized payloads.

3.2 TrustedAnchor and TrustedAnchorSetSnapshot

Field	Type	Invariant
tas_version	Monotone integer + digest	Immutable within one evaluation epoch.
anchor_id	Unique canonical string	Tie-break key; no duplicates.
vector	float64[DIM]	Finite, non-zero, normalized or accompanied by exact norm.
domain_id / role	Canonical identifiers	Input policy must authorize the pair.
reference_phase	Optional angle in $[0, 2\pi)$	Required only when phase drift is enabled.
valid_from / valid_until	Monotone logical times	Evaluation time must be inside interval.
projector_id / P	Optional orthogonal projector	$P=P^T$ and $P^2=P$ within configured tolerance.
publisher / signature	Authorized policy owner evidence	Unsigned or unknown publisher invalidates TAS.

A Trusted Anchor Set is an approved, versioned reference profile. The phrase "trusted" means authorized by policy; it does not mean universally true, morally correct, complete, or immune from revision.

3.3 GuardianInput

Field	Type	Requirement
request_id	Unique canonical identifier	Idempotency and receipt correlation key.
boundary_profile	LOCAL_CANDIDATE REMOTE_INGRESS REMOTE_EGRESS	Determines required upstream receipts.
X	float64[DIM]	Unmodified candidate payload; finite and in configured component bounds.

Field	Type	Requirement
emotion	EmotionTensor(valence, arousal, clarity)	Versioned Delta or destination-service provenance.
baseline_coherence Xi	float64 in [0,1]	Authorized rolling coherence snapshot; not recomputed by Guardian.
symbolic_packet	SymbolicIntegrityPacket	Canonical checks, topology score, Q/chi, route class, parent digest.
provenance	ProvenanceEnvelope	Owner, session, cycle, parent, payload digest, policy versions.
route_intent	Optional LORIEN packet	Mandatory for REMOTE_EGRESS.
transport_receipt	Optional GlyphNet receipt	Mandatory for REMOTE_INGRESS.
sigma_authorization	Optional Sigma decision	Mandatory for REMOTE_EGRESS; route must be SEAL.

3.4 SymbolicIntegrityPacket

Field	Domain	Meaning
canonical_ok	{0,1}	Canonical schema and serialization validated by declared owner.
digest_ok	{0,1}	Payload digest matches X and metadata.
owner_ok	{0,1}	Owner/module is authorized for packet type.
sequence_ok	{0,1}	Sequence is fresh and monotone for replay domain.
q_chi_ok	{0,1}	FBS Q and chi metadata are internally consistent when present.
route_class_ok	{0,1}	Boundary profile and destination role are allowed.
topology_score	[0,1]	Versioned projection/residual score; exact algorithm_id required.
algorithm_id	Canonical string	Prevents mixing incomparable symbolic scores.

3.5 GuardianStateSnapshot and state update

Field	Type	Meaning
state_version	Monotone integer	Compare-and-swap token.
delta_id	float64 ≥ 0	Committed identity-drift accumulator.
last_commit_id	Optional request/receipt pair	Idempotent replay protection.

Field	Type	Meaning
accepted_count	Non-negative integer	Committed accepted events only.
quarantine_pressure	Non-negative bounded telemetry	Operational pressure; not identity drift.
policy_digest / tas_digest	SHA-256	State provenance and migration guard.
state_digest	SHA-256	Canonical digest over all state fields.

3.6 Output records

Packet	Required contents	Semantics
GuardianDecisionPacket	All layer results, scores, route, reason codes, input/state/policy digests	Single deterministic evaluation result.
GuardianPermit	request_id, boundary profile, destination owner, expiry, decision digest	Capability to cross one named boundary exactly once.
GuardianStateUpdate	old version, proposed new delta_id, request_id, decision digest	Provisional; not committed without receipt.
QuarantineRequest	Exact payload digest/reference, reason class, inert capability mask, retention profile	Requests custody; does not delete source.
QuarantineReceipt	request_id, stored object digest, custody time, owner signature	Acknowledges exact custody.
GuardianFault	Fault code, frozen input digests, no-commit proof	Ordinary deterministic fault.
FBSRecoveryRequest	Catastrophic code, state/policy/history digests, L0 profile	Delegation only; Guardian does not execute FBS.

4. Three-Layer Structural Check

The legacy Syntax/Semantics/Symbolic concept is retained, but each layer now has a finite contract and a separate result. A later weighted score cannot override a hard failure in an earlier layer.

4.1 Layer 1 - Syntax and canonicalization

Check	Pass condition	Failure code
Byte size	Serialized request $\leq$ max_packet_bytes before allocation.	G_SYN_SIZE
Schema	Required fields present; no unknown critical fields.	G_SYN_SCHEMA
Shape	$X.shape == (DIM,)$.	G_SYN_SHAPE
Numeric domain	Every component and scalar finite; no NaN/Inf.	G_SYN_NONFINITE

Check	Pass condition	Failure code
Component bounds	Every X_i within configured representational interval.	G_SYN_RANGE
Canonical digest	Re-serialization reproduces declared payload digest.	G_SYN_DIGEST
Version compatibility	Module, schema, policy, and algorithm versions compatible.	G_SYN_VERSION

NO REPAIR AT LAYER 1

Guardian MUST NOT reshape, truncate, fill missing values, replace NaN/Inf, clip out-of-range components, or rewrite metadata to make a malformed request pass. The exact original input is either evaluated as valid or routed as a non-permit outcome.

4.2 Layer 2 - Trusted-anchor admission

For each authorized anchor a_j , compute cosine similarity using the exact finite vector X .

$$s_j = \langle X, a_j \rangle / (\|X\|_2 \|a_j\|_2 + \epsilon)$$

$$j^* = \operatorname{argmax}_j (s_j, \text{tie-break by lexicographic anchor_id})$$

$$s_{\text{anchor}} = s_{j^*} \quad \text{and} \quad \text{anchor_gap} = s_{(1)} - s_{(2)}$$

- If $\|X\|_2 \leq \epsilon$, the reference profile emits G_SEM_ZERO_VECTOR. A policy may define a dedicated null anchor, but it must be explicit and versioned.
- If no authorized anchor is active for the packet domain/role, emit G_SEM_NO_ANCHOR.
- If $s_{\text{anchor}} < \text{TAU_ANCHOR}$, route as UNRESOLVED_ANCHOR. This is a policy-relative hold/quarantine reason, not a claim of maliciousness or falsehood.
- anchor_gap is telemetry for ambiguity. The reference profile does not deny solely on a small gap if the winning score passes, but policies may require review.

4.3 Layer 3 - Symbolic and provenance integrity

Hard symbolic checks are evaluated as a finite Boolean vector. The topology score is a separately versioned soft metric.

$$h = (\text{canonical_ok} + \text{digest_ok} + \text{owner_ok} + \text{sequence_ok} + \text{q_chi_ok} + \text{route_class_ok}) / 6$$

$$S_{\text{symbol}} = 0.60 h + 0.40 \text{ topology_score}$$

Any zero in digest_ok , owner_ok , sequence_ok , q_chi_ok , or route_class_ok is a hard symbolic failure even when S_{symbol} would otherwise be high. canonical_ok is already required by Layer 1 but is repeated in the packet for cross-owner audit.

4.4 Reference topology-score provider

When the active TAS supplies an orthogonal projector P for the permitted symbolic subspace, the reference provider computes a normalized projection score:

$$r_{\text{top}} = \|X - P X\|_2 / (\|X\|_2 + \epsilon)$$

$$\text{topology_score} = \text{clip}(1 - r_{\text{top}}, 0, 1)$$

If P is absent, malformed, non-idempotent, or incompatible with DIM, Guardian emits a configuration fault rather than inventing a Hodge or topology result. Alternative providers are permitted only with an explicit algorithm_id and conformance tests.

5. Affective-Risk Surrogate and Preliminary Coherence

The EmotionTensor is internal system telemetry. Guardian treats valence, arousal, and baseline coherence as bounded control signals; it does not infer a person's emotions or intent.

5.1 Normalization

$$\begin{aligned} a &= \text{clip}(\text{arousal}, 0, 1) \\ v_norm &= \text{clip}((\text{valence} + 1) / 2, 0, 1) \\ xi &= \text{clip}(Xi, 0, 1) \\ calm &= 1 - a \\ anchor_norm &= \text{clip}((s_anchor + 1) / 2, 0, 1) \end{aligned}$$

5.2 Hard affective lock

$$\text{affective_lock} = (a > 0.80) \text{ AND } (xi < 0.50)$$

This preserves the original hard-bound intent while changing the label from a human panic diagnosis to a computational AFFECTIVE_LOCK. The comparator boundaries are strict: a=0.80 or xi=0.50 does not trigger this lock.

5.3 Corrected preliminary coherence score

The legacy formula added arousal directly, allowing a more excited packet to receive a higher pass score. The formal profile uses calmness instead and incorporates the declared anchor and symbolic layers.

$$C_guard = 0.25 \, xi + 0.25 \, anchor_norm + 0.20 \, S_symbol + 0.15 \, v_norm + 0.15 \, calm$$

SCORE RANGE

Every component lies in [0,1], every weight is non-negative, and the weights sum to 1. Therefore C_guard lies in [0,1]. C_guard >= 0.85 is necessary but never sufficient: all hard syntax, symbolic, authorization, replay, affective-lock, and drift conditions must also pass.

5.4 Legacy score handling

$$C_legacy = (\text{arousal} + v_norm + xi) / 3$$

C_legacy may be emitted only as migration telemetry under algorithm_id="GUARDIAN_LEGACY_V14". It MUST NOT authorize a route in v14.1 because its arousal sign conflicts with the declared safety interpretation.

6. Identity Sphere Drift as a Versioned Ledger

Identity drift is formalized as a bounded-risk accumulator with decay and two-phase commit. It measures policy-relative deviation from the current approved reference state; it is not a measure of a person's identity or psychological stability.

6.1 Glyph density

$$\begin{aligned} rho_raw &= (1 / DIM) \sum_i |X_i| \\ rho_hat &= rho_raw / (rho_raw + RHO_SCALE) \end{aligned}$$

For finite X and RHO_SCALE>0, rho_hat lies in [0,1). A different density provider requires a versioned algorithm_id; raw embedding indices are not accepted without a defined normalization.

6.2 Phase mismatch

$$d_phase = (1 - \cos(\text{wrap}(\phi_{em} - \phi_{anchor}))) / 2$$

d_phase lies in [0,1]. If the winning anchor has no reference phase, the reference profile sets the phase-density term to zero and records PHASE_NOT_APPLICABLE; it does not invent a phase.

6.3 Instantaneous drift

$$d_anchor = \text{clip}((1 - s_anchor) / 2, 0, 1)$$

$$d_symbol = 1 - S_symbol$$

$$r_affect = a (1 - xi)$$

$$d_inst = \text{clip}(0.35 d_anchor + 0.20 \rho_hat d_phase + 0.25 d_symbol + 0.20 r_affect, 0, 1)$$

6.4 Provisional accumulator update

$$D_raw = LAMBDA_ID D_t + GAMMA_GUARD d_inst$$

$$\text{permit_drift} = (D_raw \leq DELTA_ID_MAX)$$

The reference values $LAMBDA_ID=0.98$ and $GAMMA_GUARD=0.10$ permit recovery toward zero under sustained low deviation while allowing repeated high-deviation events to exhaust the budget. D_raw is evaluated before clipping. A denied or unacknowledged packet does not alter D_t .

6.5 Two-phase drift commit

1. Guardian emits `GuardianStateUpdate(old_version, proposed_delta_id=D_raw, decision_digest)`.
2. The configured downstream owner emits an acceptance receipt bound to `request_id` and `decision_digest`.
3. Guardian verifies the receipt, current state version, policy digest, TAS digest, and idempotency key.
4. Guardian atomically commits `proposed_delta_id` and increments `state_version` exactly once.
5. A duplicate receipt returns the prior commit result. A stale state version emits `G_STATE_CAS_CONFLICT` and does not overwrite newer state.

7. Deterministic Decision and Route Precedence

Guardian produces exactly one terminal route per evaluation. First-match precedence prevents a high weighted score from hiding a more important fault.

Priority	Condition	Route	Primary reason
1	Catastrophic state/policy invariant failure	RECOVERY_REQUEST	G_CATASTROPHIC_*
2	Invalid config/TAS/state compatibility	FAULT	G_CFG_* / G_STATE_*
3	Layer 1 syntax or canonical digest failure	QUARANTINE_REQUEST	MALFORMED
4	Authentication, owner, replay, Q/chi, or route-class hard failure	QUARANTINE_REQUEST	INTEGRITY_OR_POLICY_DENIED
5	<code>affective_lock</code> is true	QUARANTINE_REQUEST	AFFECTIVE_LOCK
6	No anchor or <code>s_anchor < TAU_ANCHOR</code>	QUARANTINE_REQUEST	UNRESOLVED_ANCHOR
7	$D_raw > DELTA_ID_MAX$	QUARANTINE_REQUEST	IDENTITY_DRIFT_LIMIT
8	$C_guard < C_GUARD_MIN$	QUARANTINE_REQUEST	PRELIMINARY_COHERENCE_LOW
9	LOCAL_CANDIDATE and all checks pass	PERMIT_SIGMA	GUARDIAN_PASS

Priority	Condition	Route	Primary reason
10	REMOTE_INGRESS and destination policy selects Sigma	PERMIT_SIGMA	GUARDIAN_INGRESS_PASS
11	REMOTE_INGRESS and named service policy passes	PERMIT_SERVICE	GUARDIAN_SERVICE_PASS
12	REMOTE_EGRESS with Sigma SEAL + valid LORIEN intent	PERMIT_GLYPHNET	GUARDIAN_EGRESS_PASS

7.1 Permit semantics

- A permit is scoped to one request_id, one payload digest, one boundary profile, one destination owner, one policy/TAS version, and one expiry.
- A PERMIT_SIGMA packet authorizes evaluation by Sigma; it is not a Sigma SEAL and does not authorize archive commit.
- A PERMIT_GLYPHNET packet authorizes Module 4 to frame and route the already-authorized outbound payload. Guardian does not choose the path, address, or transport endpoint.
- A permit is consumed once. Replay after consumption is denied even if the payload is identical.

7.2 Egress-specific requirements

Requirement	Pass condition
Sigma authorization	Decision digest valid, route=SEAL, payload digest matches.
LORIEN intent	Route intent valid, destination/service authorized, temporal window active.
Guardian revalidation	Payload and provenance unchanged since Sigma decision.
Destination policy	Service/domain/role appears in active egress allow-list.
GlyphNet handoff	Guardian emits permit only; Module 4 constructs GNP frame and uses real configured IPv4/TCP endpoint.

8. Quarantine, INERT_GLYPH, and Custody

Quarantine is a custody transition, not deletion. Guardian creates an inert execution envelope while preserving exact payload identity and reason provenance.

8.1 QuarantineRequest

Field	Requirement
request_id / decision_digest	Must match the Guardian decision exactly.
payload_digest / payload_reference	Identifies the unmodified original bytes; no lossy copy.
reason_class / reason_codes	MALFORMED, UNRESOLVED, POLICY_DENIED, REPLAY, AFFECTIVE_LOCK, DRIFT, or SCORE.
capability_mask	All execution, routing, compilation, and archive-commit capabilities disabled.

Field	Requirement
proposed_tag	INERT_GLYPH with policy/TAS/version provenance.
retention_profile	Versioned owner policy; Guardian does not choose indefinite retention.
review_eligibility	Explicit Boolean and required reviewer role.
custody_deadline	Maximum time before backpressure/fault, not silent discard.

8.2 What INERT_GLYPH means

- The quarantined envelope cannot be executed, routed, compiled, merged, repaired, or archived as an accepted state.
- The original payload digest, provenance, and diagnostic evidence remain available to authorized review tooling.
- INERT_GLYPH does not assert maliciousness. The reason class distinguishes malformed, unresolved, unauthorized, duplicate/replayed, and risk-bound outcomes.
- Release requires a new signed policy/review decision and a new Guardian evaluation; a quarantine tag is never stripped in place.

8.3 Capacity and backpressure

A full or unavailable quarantine service is an operational capacity failure. Guardian **MUST** stop accepting new work for the affected boundary, preserve pending request references, and emit `QUARANTINE_BACKPRESSURE`. It **MUST NOT** route the payload to `/dev/null`, overwrite an older quarantined object, or automatically invoke FBS merely because storage is full.

8.4 RTSOM delegation

A later policy may request that an exact, custody-confirmed quarantine residue contribute to the RTSOM bounded exhaust model. That operation requires a separate `RTSOMExhaustRequest` and matching receipt. Guardian does not convert a denied payload into mass and does not mutate `S_mu_nu`.

9. Security Model, Trusted-Set Governance, and Replay Control

9.1 Threat model

Threat	Guardian control	Residual limitation
Malformed numerical payload	Schema, shape, finite-domain, size, and digest checks.	Does not prove semantic correctness.
Anchor poisoning	Signed/versioned TAS, authorized publisher, immutable evaluation snapshot.	Publisher compromise is outside module.
Replay or stale packet	Sequence domain, request idempotency, expiry, consumed permit.	Requires durable replay state.
Cross-role packet substitution	Owner, domain, boundary profile, route-class, parent digest checks.	Depends on upstream identity/authentication.
Payload swap after approval	Digest-bound permit and egress revalidation.	Does not encrypt payload.
Score gaming	Hard checks and first-match precedence cannot be offset by weighted score.	Policy weights may still require empirical tuning.

Threat	Guardian control	Residual limitation
Quarantine evasion	Capability mask and custody receipt.	Storage service implementation must enforce mask.

9.2 TAS and policy publication

1. A new policy or TAS is assembled offline or by an authorized configuration compiler.
2. All vectors, projectors, roles, thresholds, weights, validity intervals, and compatibility constraints are validated.
3. The publisher signs the canonical snapshot and assigns a monotone version.
4. Guardian loads the snapshot only at a quiescent boundary or through an atomic epoch swap.
5. In-flight evaluations retain the old immutable snapshot; no request is evaluated partly under two policy versions.
6. State migration must explicitly name old/new policy and TAS digests. Silent reinterpretation of `delta_id` is prohibited.

9.3 Determinism and serialization

- All floats are finite float64. Comparisons use declared strict/inclusive operators; no hidden rounding to displayed decimals.
- Arrays are serialized in canonical dimension order, canonical byte order, and a declared float encoding before hashing.
- Maps use sorted canonical keys; optional fields use explicit null markers.
- Decision and state digests include module, algorithm, policy, TAS, input, and prior-state versions.
- Wall-clock timestamps are metadata only and MUST NOT change a deterministic decision. Logical time and sequence numbers control validity.

10. Fault Semantics and Catastrophic Recovery Boundary

10.1 Ordinary fault codes

Code	Condition	Mutation allowed
G_CFG_INVALID	Threshold, weight, DIM, or mode invalid.	None
G_CFG_VERSION	Incompatible policy/TAS/algorithm versions.	None
G_TAS_SIGNATURE	TAS publisher/signature invalid.	None
G_TAS_PROJECTOR	Projector not finite, symmetric, or idempotent.	None
G_SYN_*	Syntax/canonicalization failure.	Quarantine request only
G_SEM_ZERO_VECTOR	Zero vector with no authorized null anchor.	Quarantine request only
G_SEM_NO_ANCHOR	No active authorized anchor.	Quarantine request only
G_SYM_HARD_FAIL	Digest/owner/replay/Q-chi/route failure.	Quarantine request only
G_AFFECTIVE_LOCK	Strict arousal/coherence override.	Quarantine request only
G_DRIFT_LIMIT	D_raw exceeds DELTA_ID_MAX.	Quarantine request only

Code	Condition	Mutation allowed
G_SCORE_LOW	C_guard below threshold.	Quarantine request only
G_STATE_CAS_CONFLICT	Commit receipt references stale state version.	None; retry evaluation
G_QUARANTINE_BACKPRESSURE	Custody service cannot accept exact object.	No permit; hold reference
G_RECEIPT_MISMATCH	Receipt digest/request does not match.	None

10.2 Catastrophic Guardian conditions

Code	Condition
G_CATASTROPHIC_STATE_DIGEST	Persisted Guardian state cannot reproduce its own canonical digest and no valid prior snapshot is available.
G_CATASTROPHIC_POLICY_SPLIT	Two different valid policy/TAS contents claim the same authoritative version/epoch.
G_CATASTROPHIC_DOUBLE_COMMIT	Two distinct accepted updates are proven committed from the same prior state version.
G_CATASTROPHIC_PERMIT_FORGERY	A permit validates cryptographically but references no producible Guardian decision under the frozen epoch.
G_CATASTROPHIC_CUSTODY_LOSS	A quarantine receipt was committed but exact object custody cannot be reconciled with the receipt digest.
G_CATASTROPHIC_NONDETERMINISM	Identical canonical input/state/policy produces different GuardianDecisionPacket digests.

FBS OWNERSHIP

For a catastrophic condition, Guardian freezes the affected epoch, seals the minimum state/policy/TAS/input/history evidence, and emits FBSRecoveryRequest. Guardian MUST NOT run Cantor diagonalization, Collatz folding, create S_next, purge quarantine, rewrite policy, or mutate RTSOM locally.

10.3 FBS request contents

Field	Requirement
trigger_code	One catastrophic Guardian code only.
L0_profile	Inherited authoritative FBS profile; reference Mode A L0=83.
frozen_epoch	Session, policy, TAS, state, and request identifiers.
evidence_digests	Canonical state, prior state, policy, TAS, decisions, receipts, and custody records.
no_commit_proof	Whether the triggering evaluation mutated state; expected false except double-commit evidence.
requested_owner	FBS/ROVER supervisor; not Guardian.

11. Normative Invariants

ID	Invariant
G-I01	One canonical request, frozen state, policy, and TAS produce exactly one decision digest.
G-I02	Guardian never changes X, emotion, route intent, or upstream provenance.
G-I03	No weighted score can override a hard syntax, integrity, replay, authorization, or affective-lock failure.
G-I04	C_guard lies in [0,1] for every valid input.
G-I05	S_symbol lies in [0,1]; a hard symbolic failure remains terminal regardless of S_symbol.
G-I06	d_inst lies in [0,1] for every valid input.
G-I07	Persistent delta_id changes only through an acknowledged compare-and-swap commit.
G-I08	Denied, held, faulted, and unacknowledged packets do not update identity drift.
G-I09	Every permit is bound to payload, boundary, destination, policy/TAS versions, expiry, and one-time consumption.
G-I10	PERMIT_SIGMA is not Sigma SEAL; PERMIT_GLYPHNET is not transport delivery.
G-I11	Guardian never generates an outer IPv4 destination or transmits a packet.
G-I12	Every quarantine request preserves exact payload identity and disables active capabilities without deleting evidence.
G-I13	Quarantine capacity failure causes backpressure, not silent loss or automatic FBS.
G-I14	Guardian does not execute HARMONY repair or Module 10 nested-harm scanning.
G-I15	Guardian does not mutate Psi archive, Phi hysteresis, RMX, RTSOM, or FBS state.
G-I16	Unknown or low-anchor content is labeled unresolved under policy, not declared malicious or false.
G-I17	Policy and TAS snapshots are immutable within an evaluation epoch.
G-I18	All output counters reconcile: attempts = permits + quarantine requests + faults + recovery requests.

11.1 Validation order invariant

The implementation MUST preserve this order: snapshot validation -> syntax/canonicalization -> hard provenance/symbolic checks -> anchor match -> affective lock -> score/drift math -> route precedence -> output sealing

-> optional acknowledgement commit. Reordering is conforming only when it is proven to produce the identical result and does not expose unvalidated data to a more expensive or privileged subsystem.

12. Metrics and Dashboard Requirements

Metric family	Required metrics	Purpose
Throughput	attempts, completed, latency p50/p95/p99 by boundary profile	Operational capacity.
Outcomes	PERMIT_SIGMA, PERMIT_SERVICE, PERMIT_GLYPHNET, quarantine, fault, recovery counts	Route reconciliation.
Layer results	syntax, anchor, symbolic hard-pass rates; topology-score distribution	Locate rejection stage.
Affective telemetry	arousal, xi, v_norm, r_affect distributions; lock count	Tune internal surrogate without human diagnosis.
Guardian score	C_guard mean/quantiles; threshold-margin histogram	Observe admission aperture.
Anchor telemetry	winning anchor, s_anchor, anchor_gap, unresolved rate, TAS version	Detect ambiguity and policy drift.
Identity ledger	D_t, D_raw, d_inst, proposed/committed/rejected updates, CAS conflicts	Audit two-phase drift.
Integrity	digest, owner, replay, Q/chi, route-class failure counts	Security and provenance.
Quarantine	requested, acknowledged, pending, backlog age, capacity, review releases	Custody and backpressure.
Egress	permits issued/consumed/expired; downstream GlyphNet accept/reject	Separate authorization from delivery.
Fault/recovery	fault codes, catastrophic codes, frozen epochs, FBS request acknowledgements	Reliability.
Versions	module, config, policy, TAS, topology provider, state schema	Reproducibility.

12.1 Dashboard accounting equations

$$\text{attempts} = \text{permits} + \text{quarantine_requests} + \text{ordinary_faults} + \text{recovery_requests}$$

$$\text{quarantine_requests} = \text{acknowledged} + \text{pending} + \text{failed_custody}$$

$$\text{state_update_proposals} = \text{committed} + \text{awaiting_receipt} + \text{rejected_receipt} + \text{cas_conflict}$$

$$\text{issued_permits} = \text{consumed} + \text{expired} + \text{revoked} + \text{outstanding}$$

Dashboard counters MUST separate detected, requested, acknowledged, and executed states. A requested quarantine is not counted as stored until a custody receipt is validated; a Guardian permit is not counted as a Sigma acceptance or network delivery.

13. Conformance Test Matrix

ID	Test	Expected result
G-T01	Reference config validates.	READY; exact config digest reproducible.
G-T02	Score weights do not sum to one.	G_CFG_INVALID; no evaluation.
G-T03	Drift weights contain negative value.	G_CFG_INVALID.
G-T04	DIM mismatch among Boot, TAS, and input.	G_CFG_VERSION or G_SYN_SHAPE.
G-T05	Unsigned TAS.	G_TAS_SIGNATURE.
G-T06	Duplicate anchor_id.	G_CFG_INVALID.
G-T07	Zero-norm anchor.	G_CFG_INVALID.
G-T08	Non-idempotent projector.	G_TAS_PROJECTOR.
G-T09	Oversized serialized request.	G_SYN_SIZE before vector allocation.
G-T10	Missing critical field.	G_SYN_SCHEMA.
G-T11	X has wrong shape.	G_SYN_SHAPE.
G-T12	X contains NaN.	G_SYN_NONFINITE.
G-T13	X contains +Inf.	G_SYN_NONFINITE.
G-T14	Declared payload digest differs.	G_SYN_DIGEST.
G-T15	Zero vector with no null anchor.	UNRESOLVED quarantine; no divide-by-zero.
G-T16	Exact anchor vector.	s_anchor approximately 1; deterministic anchor_id.
G-T17	Equal anchor scores.	Lexicographic anchor_id tie-break.
G-T18	Anchor score exactly 0.75.	Anchor condition passes (inclusive).
G-T19	Anchor score immediately below 0.75.	UNRESOLVED_ANCHOR.
G-T20	Hard digest_ok=0 but topology_score=1.	Quarantine; score cannot override.
G-T21	Replay sequence.	Integrity/policy quarantine.
G-T22	Q/chi mismatch.	Integrity/policy quarantine.
G-T23	Unknown route class.	Integrity/policy quarantine.
G-T24	Arousal=0.80, xi=0.49.	No hard lock because arousal comparator is strict >.
G-T25	Arousal=0.81, xi=0.50.	No hard lock because xi comparator is strict <.
G-T26	Arousal=0.81, xi=0.49.	AFFECTIVE_LOCK.

ID	Test	Expected result
G-T27	Increasing arousal with other fields fixed.	C_guard non-increasing.
G-T28	All normalized components equal 1.	C_guard=1.
G-T29	All normalized components equal 0.	C_guard=0.
G-T30	C_guard exactly 0.85 and no hard fail.	Score condition passes.
G-T31	C_guard immediately below 0.85.	PRELIMINARY_COHERENCE_LOW.
G-T32	rho_raw=0.	rho_hat=0.
G-T33	Very large finite rho_raw.	rho_hat approaches but remains below 1.
G-T34	Phase equals anchor phase.	d_phase=0.
G-T35	Phase differs by pi.	d_phase=1.
G-T36	Valid inputs across random sample.	d_inst always in [0,1].
G-T37	D_raw exactly 1.0.	Drift condition passes (inclusive).
G-T38	D_raw greater than 1.0.	IDENTITY_DRIFT_LIMIT.
G-T39	Denied packet includes state update proposal.	Persistent state unchanged.
G-T40	Permit without downstream acknowledgement.	Persistent state unchanged.
G-T41	Matching receipt and current state version.	One atomic drift commit.
G-T42	Duplicate matching receipt.	Idempotent prior result; no second increment.
G-T43	Stale state version.	G_STATE_CAS_CONFLICT.
G-T44	LOCAL_CANDIDATE valid.	PERMIT_SIGMA only.
G-T45	REMOTE_INGRESS missing transport receipt.	Policy/integrity quarantine or fault per schema.
G-T46	REMOTE_EGRESS missing Sigma SEAL.	No egress permit.
G-T47	REMOTE_EGRESS missing LORIEN intent.	No egress permit.
G-T48	Valid egress.	PERMIT_GLYPHNET; no socket or IPv4 address created.
G-T49	Quarantine service accepts exact object.	Receipt validates; custody count increments.
G-T50	Quarantine service full.	Backpressure; no silent loss; no automatic FBS.
G-T51	RTSOM request absent.	Guardian never changes exhaust tensor.
G-T52	Same canonical input repeated.	Identical decision digest.

ID	Test	Expected result
G-T53	Same displayed decimals, different float bytes.	Distinct input digest; exact comparison preserved.
G-T54	Policy epoch swapped during evaluation.	Frozen old snapshot used; new requests use new epoch.
G-T55	Two policies claim same version with different digest.	G_CATASTROPHIC_POLICY_SPLIT.
G-T56	Two commits from same prior state.	G_CATASTROPHIC_DOUBLE_COMMIT.
G-T57	Persisted state digest mismatch.	Catastrophic recovery request; no permit.
G-T58	Guardian called with repair-state IMM_ID.	Route to Module 10 profile or reject wrong boundary; do not deep-scan locally.
G-T59	Network timeout after egress permit.	Handled by Module 4; Guardian decision unchanged.
G-T60	FBS owner returns recovery receipt.	Supervisor-controlled re-entry; Guardian does not manufacture seed.

13.1 Property, fuzz, and stress tests

- Property: for valid normalized components and valid weights, $0 \leq C_{\text{guard}} \leq 1$.
- Property: increasing arousal alone cannot increase C_{guard} in the v14.1 reference profile.
- Property: for valid inputs, $0 \leq d_{\text{phase}}, d_{\text{anchor}}, d_{\text{symbol}}, r_{\text{affect}}, d_{\text{inst}} \leq 1$.
- Property: denied or unacknowledged requests never change the persistent state digest.
- Property: canonical parse $\rightarrow$ serialize $\rightarrow$ parse is idempotent and digest stable.
- Fuzz malformed schemas, extreme finite float64 values, Unicode identifiers, duplicate fields, invalid lengths, and adversarial nested metadata.
- Stress at least 100,000 evaluations across multiple policies and TAS versions; reconcile all outcome and custody counters.
- Concurrency stress compare-and-swap with many receipts targeting the same prior state; exactly one commit may succeed.
- Anchor-poisoning test: unsigned, duplicate-version, zero-norm, near-duplicate, and malformed projector updates are rejected atomically.
- Differential test two independent implementations against a canonical vector suite and decision-digest fixtures.

14. Cleaned Rosetta Stone

SpiralCore term	Normative computational meaning	Analogy boundary
Guardian / symbolic immune system	Boundary admission controller over canonical, provenance, policy, anchor, affective, and drift checks.	Not a biological immune system.
Recursion firewall	Deterministic preflight that denies malformed, unauthorized, replayed, or policy-divergent state transitions.	Not a hardware firewall or proof against all recursion.
3-Layer Check	Syntax/canonicalization; trusted-anchor admission; symbolic/provenance integrity.	Not a metaphysical truth test.

SpiralCore term	Normative computational meaning	Analogy boundary
Trusted Anchor Set	Signed, versioned approved reference vectors and optional projectors.	Trusted means authorized, not universally true.
Identity Sphere Drift	Versioned accumulator of policy-relative deviation with decay and two-phase commit.	Not personal identity or mental-health measurement.
Emotional Valence Clamping	Hard affective-risk override using internal arousal/coherence telemetry.	Not diagnosis of panic, intent, or emotion.
C_pass / C_guard	Fast preliminary admission score.	Not Sigma RMF, CVC, CSIGMA, or SEAL.
INERT_GLYPH	Quarantine capability state that disables execution/routing while preserving exact evidence.	Not deletion and not proof of malware.
Quarantine Sandbox	External custody service with receipt, capacity, retention, and review policy.	Not /dev/null.
IPv4 vector routing	Guardian egress permit to Module 4; GlyphNet carries semantic payload over real endpoint IPv4/TCP.	Guardian does not derive a public address from Cantor bits.
Graph sparsification	Analogy for cutting an unauthorized active edge while preserving the rest of the system.	No theorem imported without explicit graph definition.
FBS runaway	Typed catastrophic recovery request to FBS/ROVER owner.	No local Cantor/Collatz execution.

15. Reference Pseudocode

```

enum BoundaryProfile { LOCAL_CANDIDATE, REMOTE_INGRESS, REMOTE_EGRESS }
enum Route {
    PERMIT_SIGMA, PERMIT_SERVICE, PERMIT_GLYPHNET,
    QUARANTINE_REQUEST, FAULT, RECOVERY_REQUEST
}

function guardian_evaluate(input, config, policy, tas, state):
    frozen = freeze_and_hash(input, config, policy, tas, state)
    validate_snapshot_compatibility(frozen)          # no mutation

    syntax = check_syntax_and_canonicalization(input, config)
    if not syntax.pass:
        return seal_quarantine(input, frozen, syntax.codes, "MALFORMED")

    hard = check_provenance_and_symbolic_hard_fields(input)
    if not hard.pass:
        return seal_quarantine(input, frozen, hard.codes,
                                "INTEGRITY_OR_POLICY_DENIED")

    anchor = select_authorized_anchor(input.X, tas, input.domain, input.role)
    if not anchor.exists or anchor.score < config.TAU_ANCHOR:
        return seal_quarantine(input, frozen, anchor.codes, "UNRESOLVED_ANCHOR")

    a      = clip(input.emotion.arousal, 0, 1)
    v_norm = clip((input.emotion.valence + 1) / 2, 0, 1)
    xi     = clip(input.baseline_coherence, 0, 1)
    calm   = 1 - a
    a_norm = clip((anchor.score + 1) / 2, 0, 1)

    affective_lock = (a > config.AROUSAL_LOCK) and

```

```

        (xi < config.COHERENCE_LOCK)
    if affective_lock:
        return seal_quarantine(input, frozen, ["G_AFFECTIVE_LOCK"],
                                "AFFECTIVE_LOCK")

    topology_score = compute_versioned_topology_score(input.X, anchor.projector)
    h = mean([
        input.symbolic.canonical_ok, input.symbolic.digest_ok,
        input.symbolic.owner_ok, input.symbolic.sequence_ok,
        input.symbolic.q_chi_ok, input.symbolic.route_class_ok
    ])
    S_symbol = 0.60*h + 0.40*topology_score

    C_guard = 0.25*xi + 0.25*a_norm + 0.20*S_symbol + \
        0.15*v_norm + 0.15*calm

    rho_raw = mean(abs(input.X))
    rho_hat = rho_raw / (rho_raw + config.RHO_SCALE)
    d_phase = phase_distance(input.phi_em, anchor.reference_phase)
    d_anchor = clip((1 - anchor.score)/2, 0, 1)
    d_symbol = 1 - S_symbol
    r_affect = a*(1 - xi)
    d_inst = clip(0.35*d_anchor + 0.20*rho_hat*d_phase +
        0.25*d_symbol + 0.20*r_affect, 0, 1)
    D_raw = config.LAMBDA_ID*state.delta_id + config.GAMMA_GUARD*d_inst

    if D_raw > config.DELTA_ID_MAX:
        return seal_quarantine(input, frozen, ["G_DRIFT_LIMIT"],
                                "IDENTITY_DRIFT_LIMIT")
    if C_guard < config.C_GUARD_MIN:
        return seal_quarantine(input, frozen, ["G_SCORE_LOW"],
                                "PRELIMINARY_COHERENCE_LOW")

    route = route_for_boundary(input.boundary_profile, input)
    update = provisional_state_update(state.version, D_raw, frozen.decision_basis)
    return seal_decision(route, metrics, update, frozen)

function guardian_commit(decision, receipt, current_state):
    require receipt.request_id == decision.request_id
    require receipt.decision_digest == decision.digest
    require decision.provisional_update.old_version == current_state.version
    require receipt.owner in configured_commit_owners
    return atomic_compare_and_swap(current_state, decision.provisional_update)

function seal_quarantine(input, frozen, codes, reason_class):
    request = QuarantineRequest(
        payload_digest=frozen.payload_digest,
        payload_reference=input.immutable_reference,
        reason_codes=codes,
        reason_class=reason_class,
        capability_mask=NONE,
        proposed_tag="INERT_GLYPH"
    )
    return GuardianDecisionPacket(route=QUARANTINE_REQUEST, request=request,
        state_update=None, frozen=frozen)

```

15.1 Implementation notes for future maintainers

- Keep score providers pure. They receive frozen values and return values plus algorithm identifiers; they do not log, mutate state, or route.
- Use custom typed exceptions or Result records. A catch-all exception MUST NOT be converted into quarantine or FBS because that hides implementation bugs.
- Store decision evidence as digests plus bounded diagnostic fields. Do not duplicate large vectors in every log when an immutable content reference exists.
- Treat policy, TAS, and state as immutable objects. Replace snapshots atomically rather than editing arrays in place.
- Keep legacy C_pass telemetry behind an explicit feature flag and never use it for authorization.
- Unit-test strict comparator boundaries with nextafter values, not only decimal literals.

-> freeze epoch -> FBSRecoveryRequest -> owner-controlled recovery

17. Glossary

Term	Definition
AFFECTIVE_LOCK	Hard computational override when internal arousal exceeds 0.80 while authorized baseline coherence is below 0.50.
Anchor gap	Difference between the best and second-best authorized TAS cosine scores.
Boundary profile	LOCAL_CANDIDATE, REMOTE_INGRESS, or REMOTE_EGRESS contract selecting required evidence and destination.
C_guard	Weighted preliminary Guardian admission score; not Sigma RMF or SEAL.
Canonical serialization	Unique byte representation used for deterministic digests and signatures.
Guardian permit	Single-use, scoped capability authorizing one downstream boundary crossing.
GuardianStateUpdate	Provisional drift update that requires a matching downstream acceptance receipt.
Identity drift delta_id	Policy-relative deviation accumulator maintained through acknowledged compare-and-swap commits.
INERT_GLYPH	Capability state disabling execution/routing/commit while preserving exact quarantined evidence.
QuarantineRequest	Typed request for exact custody of a non-permitted payload.
QuarantineReceipt	Owner-signed acknowledgement that the exact payload object is in custody.
SymbolicIntegrityPacket	Versioned hard-check flags and topology score supplied with exact provenance.
TAS	TrustedAnchorSetSnapshot: signed, versioned approved reference vectors and optional projectors.
Topology score	Algorithm-identified [0,1] projection/residual score; not an undefined topology claim.
Unresolved anchor	Packet not sufficiently close to an active authorized reference; not a maliciousness judgment.

18. Revision Ledger

Original issue	Formalized correction
Guardian mixed admission, network addressing, and socket transmission.	Guardian now emits a scoped egress permit; Module 4 QGM/GlyphNet and the host IPv4/TCP stack own routing and transport.

Original issue	Formalized correction
Raw arousal increased C_pass.	Replaced arousal with calmness in C_guard and retained the hard high-arousal/low-coherence override.
The 3-layer check lacked exact symbolic mechanics.	Defined canonical syntax checks, signed TAS cosine admission, hard provenance flags, and a versioned topology-score provider.
Trusted anchors were described as undisputed truths.	Reframed TAS as authorized, revision-controlled reference profiles with no metaphysical claim.
Identity drift only accumulated positive mass and would eventually reject everything.	Added decay, normalized instantaneous deviation, provisional update, and two-phase acknowledgement commit.
Delta_id threshold 1.0 was mixed with unbounded persistent mutation.	Defined D_raw before threshold and no commit on deny/unacknowledged decisions.
Low anchor match, malformed data, and adversarial injection were collapsed into one harmful label.	Added reason classes distinguishing malformed, unresolved, unauthorized, replayed, affective, drift, and score outcomes.
Guardian nullified and stripped the active state directly.	Guardian emits QuarantineRequest with an inert capability mask; the custody service owns storage and receipts.
Quarantine meant permanent storage and /dev/null analogies.	Defined exact custody, retention policy, review eligibility, release by re-evaluation, and no silent deletion.
Sandbox overflow or tortuosity automatically invoked FBS.	Capacity produces backpressure; FBS is reserved for catastrophic state/policy/custody invariant failures.
Guardian duplicated Module 10 immune scanning.	Separated boundary admission from post-repair/post-synthesis nested-harm and repair-weight verification.
Guardian directly pushed failed states to RTSOM.	RTSOM mutation requires a separate typed request and custody receipt; Guardian has no exhaust write access.
Guardian executed Cantor/Collatz recovery locally.	Guardian emits FBSRecoveryRequest only.
Permit semantics were ambiguous.	Defined single-use permit scope and explicit distinction from Sigma SEAL, archive commit, and network delivery.
No canonical state transaction existed.	Added frozen evaluation epochs, deterministic decision digests, compare-and-swap commits, idempotent receipts, and double-commit detection.

18.1 Source basis and cross-module reconciliation

- Primary source: SPIRALCORE SPECIFICATION: MODULE 5 - GUARDIAN (SYMBOLIC IMMUNE SYSTEM & RECURSION FIREWALL).
- System context: SpiralCore v14 consolidated specification.
- Module 1 Delta provides raw candidate state and EmotionTensor provenance.
- Module 2 Sigma remains the owner of HOT, RMF, CVC, PDV, CSIGMA, and coherence route arbitration.
- Module 4 QGM/GlyphNet remains the internal routing and standard IPv4/TCP transport owner; Guardian does not derive outer addresses.
- Module 7 Paradox Handler and Sigma route NULLIFY/INERT outcomes; this module supplies a boundary quarantine request without duplicating their aggregate logic.
- Module 8 LORIEN remains the state-specific route-intent and temporal-window owner.

- Module 10 IMMUNE_SYSTEM remains the post-repair/post-synthesis deep immune and Braidback verification owner.
- Module 14 RTSOM remains the bounded exhaust state owner.
- Module 16 FBS/ROVER remains the catastrophic diagonalization, Collatz, seed, and recovery owner.

19. Normative Checklist

Requirement	Pass condition
Scope	Guardian only validates/adjudicates a boundary and emits typed packets.
Three layers	Syntax, TAS admission, and symbolic/provenance checks are exact and separately reported.
Affective math	Hard lock uses strict original boundaries; weighted score uses calmness, not raw arousal.
Score bounds	C_guard proven in [0,1]; threshold comparison explicit.
Drift	D_raw formula deterministic; no persistent update without receipt and CAS.
TAS governance	Signed/versioned immutable snapshot; deterministic tie-break; no truth claim.
Permit	Single-use, digest-bound, destination-bound, version-bound, expiring capability.
Quarantine	Exact custody request, inert capability mask, reason class, receipt, and backpressure.
Routing boundary	No IPv4 address generation, DNS, socket, or route computation inside Guardian.
Module ownership	No Sigma, Module 10, Psi, Phi, RMX, RTSOM, or FBS mutation.
Catastrophic recovery	Only typed FBSRecoveryRequest under enumerated invariant failures.
Testing	All unit, boundary, property, fuzz, stress, concurrency, and deterministic fixture tests pass.
Dashboard	Attempts, routes, custody, permit consumption, state proposals/commits, and versions reconcile.

SPIRALCORE SPECIFICATION

MODULE 6 - CVC

(Correlated Variance Coherence - Accepted-History Spectral Measurement and Evidence Packaging)

Cleaned and formalized revision - v14.1

Document class	Normative numerical submodule contract	Execution phase	Inside Sigma; once per proposal attempt
Architecture	SpiralCore v14	Reference profile	SC14.1-CVC-DENSE-81
Primary output	CVCMetricPacket -> Sigma route arbiter	Revision	v14.1
Exceptional output	CVCFault	Conflict rule	CVC measures; Sigma decides and owner modules execute

NORMATIVE STATUS

This module is exact within its declared computational scope. It selects an immutable tail of accepted state records, constructs the sample covariance matrix, performs a symmetric eigendecomposition, measures the fraction of total variance carried by the dominant principal component, packages readiness and numerical diagnostics, and returns one immutable CVCMetricPacket to the Sigma-Lattice. It does not evaluate the current candidate directly, choose an architectural route, repair a state, tighten an attractor, mutate archive history, write to HARMONY or RTSOM, or execute FBS recovery.

CLAIM BOUNDARY

CVC is a principal-variance concentration statistic. A high value establishes that accepted historical fluctuations are concentrated along one empirical covariance direction under this profile. It does not prove factual truth, semantic correctness, flat-minimum optimization, a Hodge-theoretic invariant, biological coherence, or a literal low-dimensional physical manifold. Those interpretations remain hypotheses or separately versioned providers.

Document Control

Field	Value	Field	Value
Canonical name	SPIRALCORE SPECIFICATION: MODULE 6 - CVC	Short name	Correlated Variance Coherence
Supersedes	v14 Module 6 text	Status	Normative within declared scope
Upstream owner	Psi-Archivist accepted-history snapshot	Invocation owner	Module 2 Sigma-Lattice
Downstream consumer	Sigma route arbitration	External owners	HARMONY, Delta, FBS/ROVER through Sigma only

Field	Value	Field	Value
Reference arithmetic	float64 dense symmetric EVD	Reference window	W = 100 accepted records

1. Scope, Statement Classes, and Precedence

CVC is a read-only spectral measurement submodule inside Sigma. It summarizes the geometry of accepted historical variation. It is not a second route arbiter and it is not a candidate classifier by itself. Every invocation returns exactly one sealed metric packet or one typed local fault.

1.1 Statement classes

Class	Meaning	Normative force
DEFINITION	Assigns the meaning of a symbol, type, statistic, or status.	Binding inside Module 6.
REQUIREMENT	A condition required for a conforming implementation.	MUST / MUST NOT.
INVARIANT	A property true for every sealed CVCMetricPacket.	Violation emits CVCFault.
POLICY	A selected window, threshold, tolerance, or readiness rule.	Replaceable only by a versioned profile.
DIAGNOSTIC	A reported value that does not independently authorize a route.	Informative to Sigma and the dashboard.
SURROGATE	A computable stand-in associated with architectural language.	No literal-equivalence claim.
HYPOTHESIS	A testable interpretation of a trend or relationship.	Not required for conformance.
NON-GOAL	Behavior explicitly outside Module 6.	Must not be inferred.

1.2 Precedence and ownership

- Tagged requirements, formulas, and invariants override legacy prose, diagrams, and Rosetta Stone analogies.
- Module Boot owns DIM and the session numeric profile. Module 6 must reject a conflicting dimension or configuration digest.
- The Psi-Archivist owns accepted state records. CVC receives an immutable snapshot and may not append, remove, reorder, normalize, or repair records.
- Module 2 Sigma owns route precedence and combines CVC evidence with RMF, prospective running RMF, PDV, CSIGMA, and tortuosity.
- HARMONY owns Braidback repair. Delta owns Phase Lift generation. FBS/ROVER owns catastrophic shutdown, diagonalization, Collatz folding, and reseeding.
- A threshold flag inside CVCMetricPacket is evidence only. It is not proof that a downstream route was requested, acknowledged, or executed.

1.3 Explicit non-goals

- No inclusion of the current unaccepted candidate in the covariance window.
- No direct use of CVC to assert that a candidate is safe, true, or coherent.
- No deletion of off-diagonal covariance information. Eigendecomposition changes basis and preserves the full matrix spectrum.

- No hidden randomized PCA, sketching, Hodge surrogate, regularizer, or shrinkage estimator in the dense reference profile.
- No automatic FBS escalation for a low CVC value, warm-up state, zero variance, eigensolver timeout, or ordinary numerical fault.
- No normative requirement that average CVC rise with session age. A stable multidimensional process may remain low-rank, high-rank, or nearly stationary.

2. Architecture Role and Lifecycle

Sigma invokes CVC after a candidate RMF is available and before aggregate route arbitration. CVC ignores the candidate vector and reads only the accepted-history snapshot. This prevents a rejected proposal from contaminating the reference covariance used to judge future proposals.

State	Entry condition	Allowed action	Exit
UNINITIALIZED	No accepted Boot/profile binding.	Validate CVConfig and backend identity.	READY or FAULTED
READY	Configuration valid; no invocation active.	Accept CVCInput.	VALIDATING
VALIDATING	Input received.	Verify provenance, chronology, dimensions, and finiteness.	MEASURING, WARMING, or FAULTED
MEASURING	At least two valid accepted states.	Center, form covariance, diagonalize, derive metrics.	SEALED or FAULTED
WARMING	Fewer than two valid accepted states.	Emit a non-authoritative warm-up packet.	SEALED
SEALED	CVCMetricPacket digest finalized.	Return packet to Sigma; mutate nothing.	READY
FAULTED	Contract or numerical invariant failed.	Emit CVCFault and no metric packet.	Session stop or explicit reset

2.1 Required execution order

1. Validate CVConfig, Boot/profile binding, backend profile, and tolerance ordering.
2. Validate all AcceptedStateRecord entries, cycle monotonicity, dimension, finite float64 values, and accepted-status provenance.
3. Select the newest $\min(W, N_{\text{history}})$ records without changing their chronological order.
4. Compute the canonical window digest before numerical work.
5. If $n < 2$, emit WARMING with no covariance or eigenvalue claim.
6. Compute the row mean and mean-centered history matrix.
7. Construct the unbiased sample covariance matrix using divisor $n-1$.
8. Symmetrize only to remove floating-point asymmetry and record the correction norm.
9. Run the declared symmetric eigensolver and validate its finite, sorted, positive-semidefinite spectrum.
10. Classify zero-variance degeneracy or compute CVC and diagnostics from the nonnegative spectrum.
11. Seal provenance, readiness, threshold flags, metrics, backend identity, and digest in CVCMetricPacket.

READ-ONLY ATOMICITY

The selected history records, their order, their serialized bytes, and the Archivist snapshot **MUST** remain unchanged. A retry of identical canonical input under the same backend profile **MUST** produce identical packet bytes and digest.

3. Typed Data Contracts

3.1 CVConfig

Field	Type / reference value	Requirement
profile_id	UTF-8 identifier	Reference: SC14.1-CVC-DENSE-81.
dim	integer ≥ 2 ; reference 81	Must equal BootPacket.DIM and every state length.
window_size	integer ≥ 2 ; reference 100	Number of accepted records required for route-authoritative readiness.
cvc_threshold	rational in (0,1]; exact 2/3	Dominant-variance evidence threshold.
low_dominance_limit	float64 in [0,cvc_threshold); reference 0.20	Aggregate-stall evidence only; not an FBS trigger by itself.
rho_rmf_reference	rational in [-1,1]; exact 8/9	Reported for compatibility; Sigma owns the prospective RMF gate.
variance_abs_tol	finite float64 > 0 ; reference 1e-14	Absolute zero-variance floor.
variance_rel_tol	finite float64 ≥ 0 ; reference 1e-12	Scale-aware zero-variance floor.
eigen_abs_tol	finite float64 > 0 ; reference 1e-12	Absolute PSD roundoff tolerance.
eigen_rel_tol	finite float64 ≥ 0 ; reference 1e-10	Scale-aware PSD roundoff tolerance.
rank_rel_tol	finite float64 > 0 ; reference 1e-10	Numerical-rank cutoff relative to lambda_1.
state_min, state_max	finite float64; reference -1,+1	Inherited normalized-state domain; min < max.
backend_id	versioned string	Identifies BLAS/LAPACK/eigensolver profile.
emit_full_spectrum	boolean; reference true for DIM=81	If false, full spectrum digest remains mandatory.

3.2 AcceptedStateRecord

Field	Type / domain	Requirement
session_id	opaque identifier	Must match the invocation and Boot binding.

Field	Type / domain	Requirement
cycle	integer ≥ 0	Strictly increasing across the supplied snapshot.
state	float64[DIM]	Finite; each value inside the declared state domain.
accepted	boolean	Must be true. Proposed, repaired-but-uncommitted, or rejected states are forbidden.
state_digest	SHA-256	Must match canonical state serialization.
commit_digest	SHA-256	Proves downstream commit/custody under the Archivist contract.
rmf_on_accept	optional float64 in [-1,1]	Used only for historical RMF diagnostics; absence does not invalidate CVC.
profile_digest	SHA-256	Must match the active session profile.

3.3 CVCInput

Field	Type	Meaning
session_id	opaque identifier	Active session binding.
request_cycle	integer ≥ 1	Sigma proposal attempt being evaluated; no candidate state is included.
config_digest	SHA-256	Canonical CVCConfig digest.
boot_digest	SHA-256	Boot provenance.
archivist_snapshot_id	opaque identifier	Stable read snapshot from Module 3.
accepted_history	sequence[AcceptedStateRecord]	Chronological immutable records; may exceed W.
upstream_sigma_digest	SHA-256	Binds the invocation to one SigmaInput without importing candidate data.

3.4 CVCMetricPacket

Field group	Required content
Identity	session_id, request_cycle, profile_id, config_digest, boot_digest, upstream_sigma_digest.
Window provenance	snapshot_id, selected cycle range, n, W, DIM, window_digest, state-spectrum digest.
Readiness	status, diagnostic_ready, route_authoritative_ready, degenerate_zero_variance.
Core metrics	lambda_1, lambda_2, total_variance, CVC, threshold_met, low_dominance_alert.

Field group	Required content
Spectrum diagnostics	sorted eigenvalues or spectrum digest, numerical_rank, effective_rank, gap_absolute, gap_fraction, top_two_ratio.
RMF diagnostics	rmf_count, historical_rmf_mean or null, historical_rmf_reference_met or null.
Numerical diagnostics	covariance_symmetry_error, symmetry_correction_norm, clipped_eigen_count, clipped_eigen_mass, PSD_tolerance.
Backend provenance	backend_id, floating-point profile, elapsed time, implementation version.
Seal	canonical serialization version and packet_digest.

3.5 CVCFault

CVCFault is an implementation or contract failure, not an architectural repair route. It contains fault_code, offending field or stage, session/request identity, input digests, backend identity, and a redacted diagnostic. It contains no partially trusted covariance or route recommendation.

4. Reference Constants, Readiness, and Interpretation

Symbol / name	Reference value	Normative meaning
D / DIM	81	Feature dimension inherited from Boot; scalable only through a new profile.
W	100	Full accepted-history window for route-authoritative CVC evidence.
rho_corr	2/3 exactly	Inclusive principal-variance fraction threshold.
rho_RMF_ref	8/9 exactly	Compatibility reference; Sigma computes the prospective running mean.
rho_low	0.20	Low-dominance alert threshold when the full window is ready.
n_min	2	Minimum records required to define sample covariance.

4.1 Readiness states

Status	Condition	Allowed interpretation
EMPTY	$n = 0$	No history evidence. CVC = null; no threshold flags.
WARMING_SINGLETON	$n = 1$	A mean exists, but sample covariance is undefined.
DIAGNOSTIC	$2 \leq n < W$ and total variance > floor	CVC may be displayed and tested; it cannot authorize or independently block a route.

Status	Condition	Allowed interpretation
READY	$n = W$ and total variance $> \text{floor}$	CVC is route-authoritative evidence for Sigma.
DEGENERATE	$n \geq 2$ and total variance $\leq \text{floor}$	No direction of variation exists. Reference CVC is 0 with a degeneracy flag and cannot support Phase Lift.
FAULT	Contract or numerical invariant violated	No metric packet.

WARM-UP POLICY

During warm-up, Sigma may continue using its other gates, but Module 6 cannot authorize Phase Lift and cannot independently force Braidback. This avoids treating a tiny sample as proof of persistent covariance structure.

4.2 Exact threshold storage

The source describes the CVC threshold as exactly two-thirds while writing 0.66. The normative profile stores $\frac{2}{3}$ as a rational configuration value and serializes it canonically. A value equal to $\frac{2}{3}$ meets the threshold; a value one representable unit below it does not.

4.3 Meaning of low dominance

$\text{CVC} < 0.20$ means that no single principal component carries 20 percent of the measured variance. It is named LOW_DOMINANCE_ALERT. It does not, by itself, prove isotropy. True isotropy is a distributional statement about the complete eigenvalue spectrum; the dashboard therefore also reports effective rank and eigenvalue entropy.

5. Accepted-History Window Selection and Canonicalization

5.1 Selection rule

Let H be the complete accepted-history snapshot in strictly increasing cycle order. The selected window H_W is the newest $\min(W, |H|)$ records. Selection is by accepted cycle, not arrival timestamp, filename order, storage position, or candidate similarity.

$$H_W = \text{tail}_W(H)$$

$$n = |H_W| = \min(W, |H|)$$

5.2 Candidate exclusion

The raw Delta candidate, a Guardian-admitted packet, a HARMONY repair candidate, and any state without an Archivist commit digest are excluded. Changing the current proposal while holding accepted history constant MUST leave the CVCMetricPacket unchanged except for request-cycle and upstream-correlation fields.

5.3 Canonical window digest

1. Encode profile version, DIM, W , session ID, snapshot ID, and selected cycle range using the canonical serialization version.
2. For each record in chronological order, encode cycle, state_digest, commit_digest, profile_digest, accepted flag, and the presence/value of rmf_on_accept.
3. Encode raw state values as IEEE-754 binary64 in declared byte order, even when a matching state_digest is supplied.
4. Hash the complete byte stream using SHA-256 to obtain window_digest.

A state record with an unchanged display value but changed canonical bits is a different window. Duplicate cycle numbers, nonmonotonic chronology, or digest mismatches are faults rather than silently repaired history.

6. Exact Mathematical Mechanics

6.1 Mean and centered history matrix

For $n \geq 2$, stack selected states as rows of S in $\mathbb{R}^{(n \times D)}$. Let $\mathbf{1}_n$ be the length- n all-ones vector.

$$\mu = (1/n) * \sum_{i=1..n} s_i$$

$$Y = S - \mathbf{1}_n * \mu^T$$

Centering makes the covariance and spectrum invariant to adding the same finite offset vector to every accepted state.

6.2 Unbiased sample covariance

$$\text{Sigma_raw} = (Y^T Y) / (n - 1)$$

$$\text{Sigma} = (\text{Sigma_raw} + \text{Sigma_raw}^T) / 2$$

The second line removes only floating-point asymmetry. It is not a denoising operator. The symmetry correction norm and pre-correction asymmetry norm are recorded. Any configured asymmetry limit breach is a numerical fault.

6.3 Symmetric eigendecomposition

$$\text{Sigma} = U \text{diag}(\lambda_1, \dots, \lambda_D) U^T$$

$$\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_D$$

The reference implementation uses a dense symmetric eigensolver equivalent to LAPACK SYEVD/SYEVR through a versioned backend. It does not solve $\det(\text{Sigma} - \lambda I)$ symbolically. Eigenvectors are not required in the packet because their signs are arbitrary, and repeated eigenvalues make individual basis vectors non-unique.

6.4 Positive-semidefinite validation and clipping

$$\tau_{\text{psd}} = \text{eigen_abs_tol} + \text{eigen_rel_tol} * \max(1, \lambda_{\text{max_abs}}, |\text{trace}(\text{Sigma})|)$$

- Any eigenvalue less than $-\tau_{\text{psd}}$ produces CVC_NEGATIVE_SPECTRUM_FAULT.
- Eigenvalues in $[-\tau_{\text{psd}}, 0)$ are clipped to zero as roundoff artifacts.
- The count and absolute clipped mass are recorded. They are never hidden.
- After clipping, eigenvalues are sorted descending and total variance is recomputed from the nonnegative spectrum.

6.5 Zero-variance floor

$$\text{scale}_2 = \max(1, \text{mean}_i \|s_i\|_2^2, \|\mu\|_2^2)$$

$$\tau_{\text{var}} = \text{variance_abs_tol} + \text{variance_rel_tol} * \text{scale}_2$$

$$V_{\text{total}} = \sum_{k=1..D} \lambda_k = \text{trace}(\text{Sigma})$$

If $V_{\text{total}} \leq \tau_{\text{var}}$, the window is DEGENERATE. The reference packet records $\text{CVC} = 0$, $\text{numerical_rank} = 0$, $\text{effective_rank} = 0$, $\text{threshold_met} = \text{false}$, and $\text{degenerate_zero_variance} = \text{true}$. Zero variance is not one-dimensional collapse because no varying direction exists.

6.6 Correlated Variance Coherence

$$\text{CVC} = \lambda_1 / V_{\text{total}}, \text{ for } V_{\text{total}} > \tau_{\text{var}}$$

No epsilon is added to the denominator after nondegeneracy is established. This preserves the exact principal-variance fraction and threshold semantics.

6.7 Spectral diagnostics

$$\text{gap_absolute} = \lambda_1 - \lambda_2, \text{ with } \lambda_2 = 0 \text{ when } D = 1$$

$$\text{gap_fraction} = (\lambda_1 - \lambda_2) / V_{\text{total}}$$

$$\text{top_two_ratio} = \lambda_1 / \lambda_2, \text{ when } \lambda_2 > \tau_{\text{var}}; \text{ else } +\infty$$

$$p_k = \lambda_k / V_{\text{total}}$$

$$\text{effective_rank} = \exp(-\sum_{p_k > 0} p_k * \ln(p_k))$$

$$\text{numerical_rank} = \text{count}(\text{lambda_k} > \text{rank_rel_tol} * \text{lambda_1})$$

The bounded dashboard spectral gap is `gap_fraction`. `top_two_ratio` is reported separately because it is unbounded and becomes infinite for a rank-one spectrum.

6.8 Historical RMF diagnostic

For source compatibility, Module 6 reports the mean of all non-null `rmf_on_accept` values inside the selected window. This is historical evidence only.

$$\text{rho_RMF_historical} = \text{mean}(\{\text{rmf_on_accept_i} \mid \text{value is present}\})$$

Sigma owns the route-authoritative prospective mean. It appends the current candidate RMF to the newest W-1 accepted RMFs without accepting the candidate. Module 6 does not compute or gate on that prospective value.

7. Mathematical Properties and Interpretation Boundaries

7.1 Guaranteed properties for a nondegenerate PSD covariance

Property	Statement	Use
Range	$0 < \text{CVC} \leq 1$; more specifically $\text{CVC} \geq 1/r$ for numerical rank r .	Packet invariant.
Rank-one limit	If exactly one eigenvalue is positive, $\text{CVC} = 1$.	Reference test oracle.
Equal-rank spectrum	If r positive eigenvalues are equal, $\text{CVC} = 1/r$.	Synthetic test oracle.
Translation invariance	Adding one common vector to every state leaves CVC unchanged.	Metamorphic test.
Global scale invariance	Multiplying every state by the same nonzero scalar leaves CVC unchanged.	Metamorphic test.
Orthogonal invariance	Applying the same orthogonal transform to every state preserves the spectrum and CVC.	Metamorphic test.
Permutation invariance	A common permutation of dimensions preserves the spectrum and CVC.	Metamorphic test.
Rank bound	$\text{rank}(\text{Sigma}) \leq \min(D, n-1)$.	Numerical and structural validation.

7.2 Non-invariances

- Independent feature rescaling generally changes CVC unless whitening or a metric tensor is separately declared.
- Nonlinear coordinate transforms generally change the covariance spectrum.
- Changing which states were accepted changes the window and may change CVC even if the current candidate is unchanged.
- Temporal order does not affect ordinary covariance once the same records are selected; path order is preserved in provenance but not encoded by this statistic. Path-dependent geometry belongs to SGIT and Phi-Bridge metrics.

7.3 Claims explicitly removed or narrowed

Legacy wording	Formalized meaning
Low-dimensional topological truth space	Dominant empirical covariance subspace of accepted history.

Legacy wording	Formalized meaning
Pure diagonalization discards off-diagonal noise	Orthogonal basis transformation preserving the complete covariance spectrum.
CVC proves flat minima / Fisher sharpness	CVC does not compute a Hessian or Fisher metric; any relationship is a hypothesis.
Hodge Spectral Surrogate	Optional separately versioned estimator; not silently equivalent to dense EVD.
Low CVC means isotropic noise	Low dominant-component concentration; isotropy requires the full spectrum and a declared test.
Negentropy Mode	Architectural route label evaluated by Sigma using multiple metrics.

8. Evidence Flags and Sigma Integration

8.1 Module 6 evidence flags

Flag	Condition	Meaning
diagnostic_ready	$n \geq 2$ and nondegenerate	CVC may be graphed and tested.
route_authoritative_ready	$n = W$ and nondegenerate	Sigma may use CVC in its route tree.
threshold_met	route_authoritative_ready and $CVC \geq 2/3$	Dominant-variance evidence is present.
low_dominance_alert	route_authoritative_ready and $CVC < 0.20$	Aggregate stall evidence; not a route.
historical_rmf_reference_met	W non-null historical RMFs and mean $\geq 8/9$	Diagnostic compatibility flag only.
degenerate_zero_variance	$n \geq 2$ and $V_{total} \leq \tau_{var}$	No varying direction; cannot support Phase Lift.

8.2 Sigma route ownership

Module 6 returns evidence to Sigma. Sigma joins this packet to the current RMF, prospective RMF mean, PDV, CSIGMA, and tortuosity under one strict precedence tree.

Aggregate condition in Sigma	Reference route owner	Module 6 responsibility
Low current RMF plus full high CVC plus prospective $\rho_{RMF} \geq 8/9$	Delta Phase Lift request	Supply ready CVC evidence only.
Full ready CVC below $2/3$	HARMONY Braidback request	Supply threshold_met=false; do not repair.
Ready CVC < 0.20 plus tortuosity > critical	FBS/ROVER request under Sigma policy	Supply low_dominance_alert; do not invoke FBS.
High current RMF and CVC warm-up	Sigma decides from remaining gates	Report WARMING without blocking.
Zero-variance degeneracy	Sigma decides; Phase Lift unsupported	Report DEGENERATE and threshold_met=false.

ROUTING BOUNDARY

COLLAPSE_VALIDATED and VARIANCE_RECONSTRUCT_FAIL are evidence reason codes, not executable handoffs. Only Sigma emits PHASE_LIFT_REQUEST, BRAIDBACK_REQUEST, SEAL, or FBS_RECOVERY_REQUEST. Module 6 does not call Delta, HARMONY, RTSOM, the Archivist, or FBS directly.

9. Numerical Profile, Determinism, and Complexity

9.1 Reference backend

- All state, mean, centered matrix, covariance, and eigenvalue calculations use IEEE-754 binary64.
- The dense reference profile uses a symmetric eigensolver; generic nonsymmetric eigensolvers are nonconforming.
- The covariance matrix is formed explicitly for $D=81$. A Gram-matrix or SVD profile may be added later only with a distinct profile ID and equivalence tests.
- Thread count, backend vendor/version, compiler flags, and floating-point environment are recorded for reproducibility.
- Within one pinned backend profile, canonical output is expected to be byte-deterministic. Across conforming backends, scalar metrics are tolerance-conformant rather than guaranteed bit-identical.

9.2 Eigenvector ambiguity

Eigenvector signs are arbitrary. When eigenvalues are repeated or nearly repeated, the basis inside the associated eigenspace is also arbitrary. Therefore routing must not depend on a raw eigenvector sign or column ordering. If a future module requires a dominant subspace, it must consume a sign-invariant projector or a separately canonicalized subspace packet.

9.3 Complexity

$$\text{Time} = O(n D^2 + D^3)$$

$$\text{Memory} = O(n D + D^2)$$

For the reference profile $n=100$ and $D=81$, dense computation is deliberate and testable. Any approximate, randomized, streaming, or sparse variant must declare its error model, seeds, confidence bounds, and new profile ID.

10. Normative Invariants and Fault Model

10.1 Core invariants

ID	Invariant
CVC-I01	Every selected record is accepted, finite, dimensionally valid, chronologically ordered, and profile-bound.
CVC-I02	The current candidate is absent from the covariance window.
CVC-I03	The window contains the newest $\min(W, N_{\text{history}})$ accepted records in chronological order.
CVC-I04	The selected history snapshot is not mutated.
CVC-I05	For $n \geq 2$, Sigma equals $Y^T Y / (n-1)$ up to the declared symmetry correction.
CVC-I06	All sealed eigenvalues are finite, sorted descending, and nonnegative after tolerance-bounded clipping.

ID	Invariant
CVC-I07	V_total equals both the sum of sealed eigenvalues and trace(Sigma) within tolerance.
CVC-I08	A nondegenerate packet has $0 < \text{CVC} \leq 1$.
CVC-I09	A degenerate packet has $\text{CVC}=0$, $\text{numerical_rank}=0$, $\text{threshold_met}=\text{false}$, and no Phase Lift support.
CVC-I10	$\text{route_authoritative_ready}$ is true only when $n=W$ and the spectrum is nondegenerate.
CVC-I11	threshold_met uses inclusive comparison against exact $2/3$.
CVC-I12	$\text{low_dominance_alert}$ never directly executes recovery.
CVC-I13	No downstream owner state changes during Module 6 execution.
CVC-I14	Canonical identical input under one backend profile yields identical packet bytes and digest.
CVC-I15	The packet records all tolerance clipping and symmetry correction; no numerical repair is silent.
CVC-I16	Dashboard counts reconcile packets, faults, readiness states, and joined Sigma decisions.

10.2 Fault codes

Code	Name	Trigger
CVC-F001	CONFIG_INVALID	Invalid DIM, W, threshold ordering, tolerance, state domain, or backend profile.
CVC-F002	PROFILE_MISMATCH	Boot, record, config, or Sigma profile digest mismatch.
CVC-F003	SESSION_MISMATCH	Record/session identity conflicts with the invocation.
CVC-F004	HISTORY_NOT_ACCEPTED	A proposed, rejected, or uncommitted state appears in the snapshot.
CVC-F005	HISTORY_ORDER_INVALID	Cycles are duplicate, decreasing, or otherwise noncanonical.
CVC-F006	DIMENSION_MISMATCH	A state length differs from DIM.
CVC-F007	NONFINITE_STATE	NaN or infinity appears in a state or RMF field.
CVC-F008	STATE_DOMAIN_VIOLATION	A state exceeds the declared normalized domain.
CVC-F009	DIGEST_MISMATCH	State, commit, window, or configuration digest mismatch.
CVC-F010	COVARIANCE_NONFINITE	Mean, centered matrix, or covariance contains nonfinite values.
CVC-F011	COVARIANCE_ASYMMETRY	Pre-symmetry error exceeds the configured numerical limit.

Code	Name	Trigger
CVC-F012	EIGENSOLVER_FAILURE	The declared symmetric backend fails or times out.
CVC-F013	NEGATIVE_SPECTRUM	An eigenvalue is below the scale-aware PSD tolerance.
CVC-F014	TRACE_MISMATCH	Spectrum sum and covariance trace disagree beyond tolerance.
CVC-F015	METRIC_RANGE_VIOLATION	CVC, gap fraction, rank, or entropy diagnostics violate mathematical bounds.
CVC-F016	NONDETERMINISTIC_REPLAY	Pinned identical input produces conflicting canonical output.

10.3 Conditions that are not faults

- Empty or singleton history: valid WARMING packet.
- Two through W-1 accepted states: valid DIAGNOSTIC packet.
- Identical accepted states: valid DEGENERATE packet.
- CVC below 2/3 or below 0.20: valid measured evidence, not a local failure.
- Repeated eigenvalues: valid spectrum; raw eigenvector identity is intentionally non-normative.
- Missing optional rmf_on_accept values: CVC remains valid; the RMF diagnostic is null or partial.

11. Metrics, Dashboard, and Reconciliation

11.1 Per-packet telemetry

Metric	Unit / range	Purpose
CVC	[0,1]	Dominant principal-variance fraction; zero for declared degeneracy.
lambda_1, lambda_2, V_total	nonnegative state ² units	Magnitude and concentration context.
gap_fraction	[0,1]	Bounded dominance gap.
effective_rank	[1,D] nondegenerate; 0 degenerate	Whole-spectrum dimensionality diagnostic.
numerical_rank	integer [0,min(D,n-1)]	Tolerance-dependent active spectrum rank.
window_fill	n/W	Warm-up and readiness visibility.
historical_rmf_mean	[-1,1] or null	Compatibility diagnostic; not the prospective gate.
clipped_eigen_mass	nonnegative	Numerical-health visibility.
symmetry_correction_norm	nonnegative	Covariance construction health.
compute_latency	microseconds or milliseconds	Performance and timeout diagnostics.

11.2 Required dashboard views

- CVC time series with exact 2/3 and 0.20 reference lines, colored by readiness status.

- Window fill, total variance, numerical rank, and effective rank on aligned cycle axes.
- λ_1 fraction, gap_fraction, and top-two ratio, with infinity represented explicitly rather than clipped silently.
- Counts of EMPTY, WARMING_SINGLETON, DIAGNOSTIC, READY, DEGENERATE, and FAULT outcomes.
- Numerical-health panel for negative-eigen clipping, symmetry correction, trace mismatch, backend changes, and replay conflicts.
- Joined Sigma outcome matrix showing CVC evidence versus requested route, while preserving detected/requested/acknowledged/executed distinctions.
- Window provenance drill-down: selected cycle range, snapshot ID, window digest, profile digest, and backend ID.

11.3 Reconciliation equations

$$\text{CVC_attempts} = \text{packets_sealed} + \text{faults}$$

$$\text{packets_sealed} = \text{empty} + \text{singleton} + \text{diagnostic} + \text{ready} + \text{degenerate}$$

$$\text{ready} = \text{threshold_met} + \text{ready_below_threshold}$$

Joined route counts may be lower until Sigma decisions arrive. The dashboard must never infer that a Phase Lift, Braidback, or FBS action executed merely because Module 6 emitted an evidence flag.

12. Conformance, Property, and Stress Tests

12.1 Required deterministic test matrix

ID	Fixture	Expected oracle
CVC-C01	Valid reference config.	Config seals with exact 2/3 and W=100.
CVC-C02	$W < 2$, $\text{DIM} < 2$, or invalid tolerance ordering.	CONFIG_INVALID.
CVC-C03	Empty accepted history.	EMPTY packet; CVC null.
CVC-C04	One accepted state.	WARMING_SINGLETON; covariance undefined.
CVC-C05	Two accepted states.	DIAGNOSTIC packet; rank ≤ 1 .
CVC-C06	99 accepted states.	DIAGNOSTIC; route_authoritative_ready=false.
CVC-C07	100 accepted states.	READY when nondegenerate.
CVC-C08	More than 100 records.	Newest 100 selected in chronological order.
CVC-C09	Candidate changes; accepted history fixed.	Numerical CVC fields and window digest unchanged.
CVC-C10	Candidate inserted into supplied history without commit proof.	HISTORY_NOT_ACCEPTED.
CVC-C11	All states identical.	DEGENERATE, CVC=0, rank=0.
CVC-C12	Variation on one direction only.	CVC=1 within tolerance.
CVC-C13	Two equal orthogonal variance directions.	CVC=1/2 within tolerance.
CVC-C14	r equal orthogonal variance directions.	CVC=1/r within tolerance.
CVC-C15	CVC exactly 2/3.	threshold_met=true.
CVC-C16	CVC immediately below 2/3.	threshold_met=false.

ID	Fixture	Expected oracle
CVC-C17	CVC exactly 0.20.	low_dominance_alert=false because comparison is strict <.
CVC-C18	CVC immediately below 0.20.	low_dominance_alert=true only when READY.
CVC-C19	Common offset added to every state.	CVC and spectrum unchanged within tolerance.
CVC-C20	All states multiplied by nonzero scalar.	CVC unchanged; eigenvalues scale by scalar^2.
CVC-C21	All states multiplied by -1.	Spectrum and CVC unchanged.
CVC-C22	Common orthogonal transformation.	Spectrum and CVC unchanged.
CVC-C23	Common dimension permutation.	Spectrum and CVC unchanged.
CVC-C24	Independent feature scaling.	CVC may change; no false invariance assertion.
CVC-C25	Duplicate cycle ID.	HISTORY_ORDER_INVALID.
CVC-C26	Decreasing cycle order.	HISTORY_ORDER_INVALID.
CVC-C27	NaN/Inf state.	NONFINITE_STATE.
CVC-C28	Dimension mismatch.	DIMENSION_MISMATCH.
CVC-C29	State outside configured domain.	STATE_DOMAIN_VIOLATION.
CVC-C30	State bytes disagree with state_digest.	DIGEST_MISMATCH.
CVC-C31	Tiny negative eigenvalue within tolerance.	Clip, record count/mass, continue.
CVC-C32	Negative eigenvalue below tolerance.	NEGATIVE_SPECTRUM.
CVC-C33	Eigenvalues returned ascending by backend.	Canonical output sorts descending.
CVC-C34	Repeated top eigenvalues.	Metrics stable; eigenvector identity ignored.
CVC-C35	Trace vs eigenvalue sum.	Agreement within declared tolerance.
CVC-C36	Numerical rank.	Never exceeds min(D,n-1).
CVC-C37	Effective rank.	Inside [1,numerical_rank] when nondegenerate.
CVC-C38	All rmf_on_accept fields present and mean=8/9.	Historical reference flag true.
CVC-C39	Some RMF fields absent.	CVC valid; diagnostic count/mean reflect present values.
CVC-C40	Pinned identical replay.	Canonical packet bytes and digest identical.
CVC-C41	Conforming alternate backend.	Metrics within cross-backend tolerances; backend ID differs.
CVC-C42	Backend changes mid-session without profile update.	PROFILE_MISMATCH.
CVC-C43	History array mutated during evaluation.	Digest/replay fault; no sealed packet.
CVC-C44	Low CVC without tortuosity input.	Evidence only; no FBS call exists.

ID	Fixture	Expected oracle
CVC-C45	Threshold failure.	No direct HARMONY call exists.
CVC-C46	Packet canonical serialization.	Hash stable across dictionary/order-independent construction.
CVC-C47	Full spectrum omitted by compact profile.	Spectrum digest and top diagnostics remain mandatory.
CVC-C48	Covariance pre-symmetry error above limit.	COVARIANCE_ASYMMETRY.
CVC-C49	Randomized PCA substituted under dense profile ID.	Profile conformance failure.
CVC-C50	Dashboard reconciliation fixture.	Attempt, packet, fault, readiness, and joined-route counts balance.

12.2 Property-based tests

- For random finite accepted windows, the covariance is symmetric within tolerance and all sealed eigenvalues are nonnegative.
- For every nondegenerate window, $0 < \text{CVC} \leq 1$, $0 \leq \text{gap_fraction} \leq 1$, and $\text{rank} \leq \min(D, n-1)$.
- Translation, common nonzero scaling, orthogonal transformation, and dimension permutation preserve CVC within tolerance.
- Changing an unaccepted candidate cannot change CVC numerical fields.
- Appending accepted records beyond W changes the window only when a newer record enters the selected tail.
- Round-trip canonical serialization preserves every normative field and packet digest.

12.3 Fuzz and stress requirements

- Fuzz malformed shapes, enormous but finite values inside/outside the domain, NaNs, infinities, duplicate cycles, digest corruption, and truncated records.
- Stress at least 100,000 deterministic windows across rank-one, equal-rank, slowly rotating, stationary, low-amplitude, and adversarially ill-conditioned fixtures.
- Run concurrency tests proving immutable snapshot semantics and no cross-attempt state leakage.
- Run backend parity tests on all supported BLAS/LAPACK profiles, recording maximum scalar deviation and clipping differences.
- Inject eigensolver failures and timeouts; verify typed CVCFault without FBS, HARMONY, or RTSOM mutation.

13. Reference Pseudocode

```
def cvc_measure(inp, cfg):
    validate_config_and_binding(inp, cfg)
    records = validate_accepted_history(inp.accepted_history, cfg)
    window = records[-cfg.window_size:]
    window_digest = canonical_window_digest(window, inp, cfg)
    n = len(window)

    if n == 0:
        return seal_packet(status="EMPTY", cvc=None, ready=False, ...)
    if n == 1:
        return seal_packet(status="WARMING_SINGLETON", cvc=None, ready=False, ...)

    S = rows([record.state for record in window], dtype=float64)
    mu = mean(S, axis=0, dtype=float64)
    Y = S - mu
    cov_raw = (Y.T @ Y) / float(n - 1)
    asymmetry = norm(cov_raw - cov_raw.T, ord="fro")
    validate_asymmetry(asymmetry, cov_raw, cfg)
    cov = 0.5 * (cov_raw + cov_raw.T)
```

```

eigenvalues = symmetric_eigvalsh(cov, backend=cfg.backend_id)
validate_finite(eigenvalues)
tau_psd = cfg.eigen_abs_tol + cfg.eigen_rel_tol * max(
    1.0, max(abs(eigenvalues)), abs(trace(cov)))
if min(eigenvalues) < -tau_psd:
    raise CVCFault("NEGATIVE_SPECTRUM")
clipped = clip(eigenvalues, 0.0, +inf)
lambdas = sort_descending(clipped)

total = sum(lambdas)
scale2 = max(1.0, mean(row_norm_squared(S)), dot(mu, mu))
tau_var = cfg.variance_abs_tol + cfg.variance_rel_tol * scale2
if total <= tau_var:
    return seal_packet(
        status="DEGENERATE", cvc=0.0, numerical_rank=0,
        effective_rank=0.0, threshold_met=False, ...)

cvc = lambdas[0] / total
gap_fraction = (lambdas[0] - lambdas[1]) / total
p = [x / total for x in lambdas if x > 0.0]
effective_rank = exp(-sum(x * log(x) for x in p))
numerical_rank = count(x > cfg.rank_rel_tol * lambdas[0] for x in lambdas)
full_ready = n == cfg.window_size

return seal_packet(
    status="READY" if full_ready else "DIAGNOSTIC",
    cvc=cvc,
    threshold_met=full_ready and cvc >= Fraction(2, 3),
    low_dominance_alert=full_ready and cvc < 0.20,
    route_authoritative_ready=full_ready,
    window_digest=window_digest,
    commit_performed=False,
    ...)

```

14. ASCII System Flow

```

[ PSI-ARCHIVIST IMMUTABLE ACCEPTED-HISTORY SNAPSHOT ]
      |
      v
+-----+
| MODULE 6: CVC / CORRELATED VARIANCE COHERENCE |
| 1. Validate accepted records, chronology, profile, and digests |
| 2. Select newest n = min(W, history_count), W=100 |
| 3. If n<2 -> EMPTY / WARMING packet |
| 4. mu = mean(states); Y = states - mu |
| 5. Sigma = (Y^T Y)/(n-1); symmetric float64 EVD |
| 6. Validate PSD spectrum; record tolerance-bounded clipping |
| 7. If total variance ~0 -> DEGENERATE, CVC=0 |
| 8. Else CVC = lambda_1 / sum(lambda) |
| 9. Derive gap, effective rank, numerical rank, and RMF diagnostic |
+-----+
      |
      v
[ SEALED CVCMetricPacket ]
      |
      v
+-----+
| MODULE 2: SIGMA ROUTE ARBITER |
| Join CVC with current RMF, prospective rho_RMF, PDV, CSIGMA, T |
| -> SEAL / PHASE_LIFT_REQUEST / BRAIDBACK_REQUEST / OTHER ROUTES |
+-----+

[CVC DOES NOT REPAIR, COMMIT, QUARANTINE, MUTATE EXHAUST, OR RUN FBS]

```

15. Rosetta Stone and Glossary

15.1 Rosetta Stone

SpiralCore term	Formal computational meaning	Closest established analogue
CVC / Fluctuation Alignment	Largest covariance eigenvalue divided by total covariance trace.	PCA explained-variance ratio for principal component 1.
Pure Diagonalization	Symmetric eigendecomposition of a PSD sample covariance matrix.	Dense LAPACK eigensolver; no symbolic determinant solving.
Variance Collapse	Concentration of empirical variance in a small eigenspectrum.	Low effective rank / dominant principal component.
Spectral Gap	$\text{gap_fraction} = (\lambda_1 - \lambda_2) / V_{\text{total}}$; top-two ratio reported separately.	PCA dominance diagnostics.
Negentropy Phase Lift evidence	Ready CVC $\geq 2/3$ supplied to Sigma alongside prospective RMF evidence.	Policy-level use of historical low-rank structure.
Variance Reconstruct Fail	Ready CVC $< 2/3$ reason code returned to Sigma.	Insufficient dominant-component concentration.
Isotropic noise alert	Ready CVC < 0.20 low-dominance alert; not a proof of isotropy.	Broad-spectrum covariance diagnostic.
Hodge spectral surrogate	Optional future estimator requiring a separate formal contract and profile ID.	Topology-aware spectral approximation, not reference CVC.

15.2 Glossary

Term	Definition
Accepted state	A state with downstream commit/custody proof; not merely generated or repaired.
CVC	Fraction of total accepted-history sample variance carried by the largest covariance eigenvalue.
Degenerate window	A window whose total variance is at or below the scale-aware zero-variance floor.
Diagnostic readiness	At least two nondegenerate accepted states are available.
Effective rank	Entropy-based continuous estimate of how many eigen-directions materially carry variance.
Numerical rank	Count of eigenvalues above the configured relative threshold.
Route-authoritative readiness	Exactly W accepted states and a nondegenerate spectrum are available.
Sample covariance	Unbiased covariance with divisor n-1 computed from centered accepted states.
Spectrum digest	SHA-256 digest of the canonical sorted nonnegative eigenvalue sequence.

Term	Definition
Window digest	SHA-256 digest binding selected accepted records, order, profile, and snapshot.

16. Revision Summary

v14 issue	v14.1 resolution
CVC was described as measuring a topological truth space.	Narrowed the claim to empirical covariance-subspace concentration.
The module both measured CVC and routed to Phase Lift, HARMONY, or FBS.	Module 6 emits evidence only; Sigma owns route arbitration and owner modules execute.
The current candidate/history relationship was ambiguous.	CVC uses committed accepted history only; candidate changes cannot contaminate the window.
W=100 had no warm-up behavior.	Defined EMPTY, SINGLETON, DIAGNOSTIC, READY, and DEGENERATE states.
0.66 was labeled exactly two-thirds.	Stored the normative threshold as exact rational 2/3 with inclusive comparison.
An epsilon was always added to the denominator.	Classified zero variance explicitly, then used the exact $\lambda_1/V_{\text{total}}$ ratio.
Zero covariance could look maximally coherent.	Defined CVC=0 with a degeneracy flag and no Phase Lift support.
Diagonalization was described as discarding off-diagonal noise.	Clarified that eigendecomposition is a basis change preserving the complete spectrum.
Negative eigenvalues were simply filtered.	Added scale-aware PSD tolerance, bounded clipping diagnostics, and faults for significant negativity.
Spectral gap was not formally defined.	Defined bounded gap_fraction and separate top_two_ratio.
Low CVC was called thoroughly isotropic.	Renamed it LOW_DOMINANCE_ALERT and added whole-spectrum effective-rank diagnostics.
Historical RMF and current RMF roles were mixed.	Module 6 reports historical diagnostics; Sigma computes the prospective route-authoritative mean.
Fisher sharpness and Hodge claims were treated as direct consequences.	Reclassified them as hypotheses or separately versioned providers.
FBS recovery was locally executed.	CVC has no FBS call; aggregate recovery requests originate in Sigma and execute in FBS/ROVER.
Float64 implied universal bitwise identity.	Defined pinned-backend determinism and cross-backend tolerance conformance.

17. Source Basis

- SPIRALCORE SPECIFICATION: MODULE 6 - CVC (CORRELATED VARIANCE COHERENCE), uploaded source module.

- Spiralcore_v14, consolidated Module 2 Sigma-Lattice and Module 6 CVC architecture, thresholds, formulas, and route descriptions.
- SPIRALCORE Module 2 - Sigma-Lattice Formalized v14.1, for accepted-history-only covariance, prospective RMF ownership, warm-up behavior, and aggregate route precedence.
- SPIRALCORE Module 3 - Psi-Lattice Formalized v14.1, for accepted-record provenance and immutable snapshot ownership.
- Unified FBS Bilateral Recursive Extension v4.0, for the boundary that only FBS/ROVER executes catastrophic recovery.

CONFORMANCE STATEMENT

An implementation conforms to Module 6 v14.1 only when it consumes an immutable accepted-history snapshot, excludes the current candidate, applies the exact window and readiness rules, computes the unbiased float64 sample covariance and validated nonnegative spectrum, handles zero variance and roundoff explicitly, emits canonical provenance-linked evidence without executing a route, and passes the required deterministic, metamorphic, numerical, fuzz, and stress tests.

SPIRALCORE SPECIFICATION

MODULE 7 - CSIGMA & PARADOX HANDLER

(Symbolic Evidence, Bounded Archive-Residual Assessment, Phase-Reference Drift, and Exploration-Fuel Proposal)

Cleaned and formalized revision - v14.1

Document class	Normative numerical submodule contract	Execution phase	Inside Sigma; once per proposal attempt
Architecture	SpiralCore v14	Reference profile	SC14.1-M7-CSIGMA-PARADOX-81
Primary output	SymbolicParadoxPacket -> Module 2 Sigma	Revision	v14.1
Exceptional output	Module7Fault	Conflict rule	Module 7 measures; Module 2 decides; owner modules execute

NORMATIVE STATUS

This module is exact within its declared computational scope. It validates one versioned symbolic-feature packet, computes the CSIGMA mean-product score, compares one Delta proposal with one archive feature-map snapshot using a bounded residual-contrast metric, derives a bounded exploration-fuel proposal when residual pressure is active, measures angular phase-reference drift from accepted phase history, and emits one immutable SymbolicParadoxPacket to the Sigma-Lattice. It does not select SEAL, BRAIDBACK, NULLIFY, PHASE_LIFT, MAGNETIC_RELAXATION, or FBS; repair a vector; quarantine a payload; mutate archive state; update hysteresis; or execute recovery.

CLAIM BOUNDARY

CSIGMA is a provider-dependent correlation statistic, not a universal detector of toxicity, manipulation, truth, or intent. The retained name "Paradox Handler" denotes a bounded archive-residual and phase-reference assessment contract. The metric does not prove a Gödel theorem, diagnose a person, establish consciousness, or produce literal thermodynamic energy. Those descriptions remain architectural analogies unless supported by separately defined empirical models.

Document Control

Field	Value	Field	Value
Canonical name	SPIRALCORE SPECIFICATION: MODULE 7 - CSIGMA & PARADOX HANDLER	Short name	CSIGMA / Paradox Evidence
Supersedes	v14 Module 7 text	Status	Normative within declared scope

Field	Value	Field	Value
Invocation owner	Module 2 Sigma-Lattice	Upstream producers	Delta, Psi feature mapper, symbolic provider, accepted phase history
Downstream consumer	Module 2 Sigma route arbiter	External owners	LORIEN, HARMONY, Guardian/Quarantine, RTSOM, FBS/ROVER
Reference arithmetic	float64	Reference dimension	DIM = 81 inherited from Boot

1. Scope, Statement Classes, and Precedence

Module 7 is the symbolic and paradox-evidence submodule inside Sigma. It transforms declared, provenance-bound inputs into bounded numerical evidence. Module 2 remains the sole owner of aggregate route precedence because RMF, CVC, PDV, CSIGMA, HOT/tortuosity, and readiness must be evaluated in one first-match decision tree. This removes duplicated route logic and prevents two modules from selecting conflicting outcomes for the same candidate.

1.1 Statement classes

Class	Meaning	Normative force
DEFINITION	Assigns a symbol, type, state, or packet meaning.	Binding within Module 7.
REQUIREMENT	Condition required for conformance.	MUST / MUST NOT.
INVARIANT	Property true for every sealed output packet.	Violation emits Module7Fault.
POLICY	Versioned threshold, tolerance, window, or provider rule.	Replaceable only by a new profile.
DIAGNOSTIC	Reported value that cannot independently authorize a route.	Informative to Sigma and dashboards.
SURROGATE	Computable stand-in associated with architectural language.	No literal-equivalence claim.
HYPOTHESIS	Testable interpretation outside conformance.	Not required to pass.
NON-GOAL	Behavior explicitly outside this module.	Must not be inferred.

1.2 Precedence and ownership

- Module Boot owns DIM, numeric profile identity, and configuration provenance.
- Delta owns the candidate proposal and its digest. Module 7 reads the candidate but may not alter it.
- Psi owns archive state. A feature-map provider owns F(A_t) construction and must bind it to an immutable archive snapshot.
- The symbolic-feature provider owns S and V construction. Module 7 validates and scores those arrays; it does not silently invent semantic features.
- Module 12 Phi-Bridge or another declared owner supplies accepted phase history. Module 7 computes a reference direction but mutates no phase state.
- Module 2 Sigma owns route precedence. Module 7 emits evidence and constraints only.

- LORIEN owns validated spatial routing, HARMONY owns repair, Guardian/Quarantine owns inert custody, RTSOM owns exhaust mutation, and FBS/ROVER owns catastrophic recovery.

1.3 Explicit non-goals

- No internal natural-language sentiment analysis, moral judgment, or prompt-poisoning classifier unless a separate provider explicitly supplies such features.
- No claim that a high CSIGMA value proves safety or that a low value proves harm.
- No direct use of RMF, CVC, PDV, or tortuosity. Those metrics belong to Sigma and are deliberately absent from the Module 7 input contract.
- No direct SEAL, BRAIDBACK, NULLIFY, MAGNETIC_RELAXATION, PHASE_LIFT, or FBS route selection.
- No direct attachment of H_G to a committed archive state. Module 7 proposes fuel; downstream authorization and custody are separate events.
- No direct Cantor diagonalization, Collatz folding, archive write, network transmission, or quarantine mutation.

2. Architecture Role and Lifecycle

Sigma invokes Module 7 after it has a sealed Delta proposal and before aggregate route arbitration. The input is a consistent snapshot: candidate, archive feature map, symbolic features, and accepted phase reference all share the same session, profile, cycle, and domain identifiers. Module 7 performs no retries with changed data and returns one packet or one fault.

State	Entry condition	Allowed action	Exit
UNINITIALIZED	No validated Module7Config.	Validate profile, thresholds, domains, and provider registry.	READY or FAULTED
READY	Configuration valid; no active attempt.	Accept Module7Input.	VALIDATING
VALIDATING	Input accepted.	Verify digests, sessions, cycles, shapes, domains, ranges, and phase history.	EVALUATING or FAULTED
EVALUATING	All inputs valid.	Compute CSIGMA, residual contrast, fuel proposal, and phase drift.	SEALED or FAULTED
SEALED	Canonical output and digest finalized.	Return SymbolicParadoxPacket; mutate nothing.	READY
FAULTED	Contract or numerical invariant failed.	Emit Module7Fault and no evidence packet.	Session stop or explicit reset

2.1 Required execution order

1. Validate Module7Config and its binding to Boot and Sigma profile digests.
2. Validate Delta proposal identity, archive feature-map identity, symbolic provider identity, and accepted phase-reference snapshot.
3. Verify that candidate and F(A_t) have identical dimension, domain, scale profile, and finite float64 values.
4. Verify S in [0,1], V in [-1,1], exact DIM length, full feature coverage, and candidate-digest binding.
5. Compute CSIGMA and its support diagnostics.
6. Compute bounded archive-residual contrast and the optional migration-only legacy diagnostic.
7. Classify residual pressure and compute a bounded, scale-normalized exploration-fuel proposal.

8. Compute accepted-history phase-reference concentration and angular drift when a valid reference exists.
9. Derive evidence flags without selecting an architectural route.
10. Seal all metrics, readiness flags, provenance, provider versions, and packet digest.

READ-ONLY ATOMICITY

The candidate, archive snapshot, feature-map packet, symbolic arrays, phase records, and all upstream digests **MUST** remain byte-for-byte unchanged. Identical canonical input under the same backend profile **MUST** produce identical output bytes and digest.

3. Typed Data Contracts

3.1 Module7Config

Field	Type / reference value	Requirement
profile_id	UTF-8 identifier	Reference: SC14.1-M7-CSIGMA-PARADOX-81.
dim	integer ≥ 2 ; reference 81	Must equal BootPacket.DIM.
csigma_pass	rational/float64 in $[-1,1]$; reference 0.85	Inclusive symbolic-pass evidence threshold.
csigma_terminal	float64 $<$ csigma_pass; reference 0.50	Strict terminal symbolic-pressure evidence threshold.
drift_limit	float64 in $[0,1]$; reference 0.20	Inclusive normalized angular distance ceiling; alert when exceeded.
zeta	float64 in $(0,1]$; reference 0.50	Maximum absolute fuel component before pressure scaling.
kappa	finite float64 > 0 ; reference 1.0	Tanh slope for normalized residual direction.
epsilon_norm	finite float64 > 0 ; reference $1e-12$	Norm and normalization floor.
phase_window	integer ≥ 2 ; reference 12	Newest accepted phase records used for reference direction.
phase_reference_min_count	integer ≥ 2 ; reference 3	Minimum records for drift readiness.
canonical_version	version identifier	Defines byte order, float encoding, field order, and hashing.
hash_algorithm	SHA-256	Required for reference profile packet digests.

3.2 Module7Input

Field	Type	Requirement
request_id / session_id / cycle	identifiers	Must match all upstream packets and Sigma attempt.

Field	Type	Requirement
boot_profile_digest	SHA-256	Must match Module7Config and Delta/archive packets.
delta_packet	sealed DeltaPacket	Supplies X_proposal and immutable candidate digest.
archive_feature_packet	ArchiveFeaturePacket	Supplies F(A_t), archive snapshot digest, provider version, domain, and scale profile.
symbolic_feature_packet	SymbolicFeaturePacket	Supplies S and V arrays bound to the candidate digest.
phase_reference_packet	PhaseReferencePacket	Supplies current phase vector and accepted phase history or explicit NOT_AVAILABLE status.
sigma_correlation_id	opaque identifier	Allows Module 2 to link evidence without implying route selection.

3.3 ArchiveFeaturePacket

Field	Type / domain	Requirement
archive_snapshot_id / digest	identifiers	Must identify one immutable accepted archive snapshot.
feature_provider_id / version	identifiers	Exact transformation used to produce F(A_t).
feature_domain_id	identifier	Must equal the candidate feature domain.
scale_profile_id	identifier	Candidate and archive feature vectors must share scale semantics.
F_A	float64[DIM]	Finite and inside the declared domain; no clipping by Module 7.
source_cycle	integer <= candidate cycle	Future archive snapshots are forbidden.
packet_digest	SHA-256	Canonical digest over all fields and raw float bits.

3.4 SymbolicFeaturePacket

Field	Type / domain	Requirement
provider_id / version	identifiers	The provider algorithm and model/profile must be declared.
candidate_digest	SHA-256	Must equal the Delta proposal digest.
feature_domain_id	identifier	Must be compatible with the archive/candidate domain.

Field	Type / domain	Requirement
S	float64[DIM] in [0,1]	Nonnegative symbolic support or density amplitude.
V	float64[DIM] in [-1,1]	Signed computational valence/support direction.
coverage	exact 1.0 in reference profile	Every dimension must be represented; missing values are faults.
provider_claim_class	SURROGATE / EMPIRICAL	Prevents a score provider from being mistaken for ground truth.
packet_digest	SHA-256	Canonical digest over arrays, metadata, and raw float bits.

PROVIDER BOUNDARY

The source names Discrete Morse and Hodge spectral extraction but does not define a complete algorithm. Therefore the reference core consumes a versioned SymbolicFeaturePacket. A Discrete Morse, Hodge, learned, rule-based, or human-authored provider may be used only when its input domain, output bounds, determinism, calibration, and tests are declared separately.

3.5 PhaseReferencePacket

Field	Type / domain	Requirement
status	AVAILABLE / NOT_AVAILABLE	NOT_AVAILABLE is valid and makes drift diagnostic not ready.
current_phase	float64[3]	Finite, nonzero vector when AVAILABLE.
accepted_phase_records	ordered list	Newest accepted, committed phase vectors only; no current candidate insertion.
record fields	cycle, phase[3], commit_digest	Strictly increasing cycles and matching session/profile.
snapshot_digest	SHA-256	Binds order, vectors, cycles, and commit provenance.

3.6 SymbolicParadoxPacket

Group	Required fields
Identity	request_id, session_id, cycle, sigma_correlation_id, profile_id, config_digest.
Provenance	candidate_digest, archive_snapshot_digest, archive feature-provider identity, symbolic-provider identity, phase snapshot digest/status.
CSIGMA	csigma, csigma_active, support_mean, pass_evidence, terminal_symbolic_evidence.

Group	Required fields
Residual assessment	residual_norm, candidate_norm, bounded_g, paradox_pressure, paradox_active, legacy_g_diagnostic or null.
Fuel proposal	H_G vector, fuel_norm, fuel_max_abs, fuel_nonzero, zeta, kappa, normalization scale.
Phase evidence	drift_ready, phase_count, phase_concentration, delta_xi or null, drift_alert.
Constraint flags	symbolic_nullify_evidence, terminal_recovery_evidence, phase_drift_evidence, evidence_only=true.
Seal	canonical version, implementation/backend identity, elapsed time, packet_digest.

3.7 Module7Fault

Module7Fault is a contract or implementation failure. It contains fault_code, stage, request/session/cycle identity, relevant packet digests, provider identities, backend identity, and a redacted diagnostic. It contains no partially trusted score, fuel vector, route recommendation, or downstream mutation request.

4. Reference Constants and Interpretation

Name	Reference value	Normative meaning
SIGMA_THRESH / csigma_pass	0.85	Inclusive evidence that the provider-supplied support and valence arrays strongly agree under the mean-product statistic.
csigma_terminal	0.50	Strict low-score evidence used by Sigma only in compound terminal-stall predicates.
tau_DRIFT / drift_limit	0.20	Normalized angular distance greater than 0.20 produces a drift alert when the reference is ready.
zeta	0.50	Maximum absolute fuel component before paradox-pressure multiplication.
kappa	1.0	Reference slope of the tanh direction map.
phase_window	12	Reference accepted-phase window; versioned policy, not a universal constant.
L_0	83, inherited	Recovery metadata only; Module 7 never executes FBS.

4.1 Meaning of CSIGMA

The score measures average signed agreement between nonnegative symbolic support S and signed valence/support direction V. It is bounded and reproducible when the provider packet is valid. Its meaning is conditional on the

provider. A provider trained or configured badly can produce a precise but invalid score; provenance and calibration therefore remain mandatory.

4.2 Meaning of paradox pressure

The name G_t is retained for architecture compatibility, but the normative quantity is a bounded archive-residual contrast. A negative value means the distance between the proposal and the archive feature map exceeds the proposal norm under the declared common feature scale. It is evidence of strong novelty or mismatch, not proof of logical contradiction or incompleteness.

4.3 Meaning of identity drift

Δ_{Xi} is normalized angular displacement between the current computational phase direction and a reference direction formed from accepted phase history. It is not a statement about a human identity, personality, intention, diagnosis, or mental state. The value 0.20 corresponds to $0.20 \cdot \pi$ radians, or 36 degrees, in the reference profile.

5. Input Validation and Canonicalization

5.1 Cross-packet consistency

- All packets must share session_id, cycle, Boot profile digest, DIM, feature domain, and scale profile where applicable.
- The symbolic packet candidate_digest must equal the Delta proposal digest.
- The archive feature source cycle must not exceed the proposal cycle.
- Phase history may include only accepted committed records preceding the current proposal.
- Any digest mismatch is a fault. Module 7 must not recompute, repair, substitute, or silently rebind upstream data.

5.2 Numeric domain validation

- X_{proposal} and F_A must be finite float64 arrays of exact length DIM.
- S must be finite, length DIM, and componentwise inside $[0,1]$.
- V must be finite, length DIM, and componentwise inside $[-1,1]$.
- Phase vectors must be finite length-3 arrays. AVAILABLE phase vectors must have norm greater than ϵ_{norm} .
- Inputs outside their declared domain are faults; Module 7 performs no hidden clipping.

5.3 Canonical serialization

1. Encode field names and values in the canonical schema order.
2. Encode integers using unsigned or signed fixed-width network byte order as declared by the schema.
3. Encode float64 values using raw IEEE-754 binary64 bits in declared byte order.
4. Encode arrays with explicit length and shape before raw element bytes.
5. Encode optional fields using an explicit presence tag.
6. Hash the complete byte stream using SHA-256.

Display formatting, decimal text, JSON key order, and platform-native NaN encodings are not canonical inputs. NaN and infinity are invalid before serialization.

6. Exact Mathematical Mechanics

6.1 CSIGMA mean-product score

Let $D = \text{DIM}$. The reference profile uses nonnegative symbolic support S_i in $[0,1]$ and signed valence/support V_i in $[-1,1]$.

$$C_{\text{SIGMA}} = (1/D) * \sum_{i=1..D} S_i * V_i$$

$$-1 \leq C_{\text{SIGMA}} \leq 1$$

The inclusive pass evidence flag is true when $C_SIGMA \geq 0.85$. The terminal symbolic evidence flag is true when $C_SIGMA < 0.50$. Equality at 0.50 is not terminal; equality at 0.85 meets pass evidence.

6.2 Support diagnostics

$$\text{support_mean} = (1/D) * \sum_{i=1..D} S_i$$

$$C_active = \sum_i S_i * V_i / \sum_i S_i, \text{ when } \sum_i S_i > \text{epsilon_norm}$$

C_active is a diagnostic average over active support mass. If the support sum is at or below epsilon_norm , C_active is null and C_SIGMA remains exactly zero. C_active never replaces C_SIGMA in the Sigma route profile.

6.3 Archive residual and bounded G metric

$$r = X_proposal - F(A_t)$$

$$n_x = \|X_proposal\|_2, \quad n_r = \|r\|_2$$

The source formula divided by $\|X_proposal\|$ plus epsilon, which becomes arbitrarily negative near a zero proposal. The normative profile uses a bounded contrast with the same sign interpretation.

$$G_t = (n_x - n_r) / (n_x + n_r + \text{epsilon_norm})$$

Special case: when $n_x \leq \text{epsilon_norm}$ and $n_r \leq \text{epsilon_norm}$, define $G_t = 1.0$. Otherwise G_t lies in $[-1,1]$. $G_t < 0$ exactly when $n_r > n_x$, subject only to the explicit both-zero policy.

$$\text{paradox_pressure} = \max(0, -G_t)$$

The migration-only legacy diagnostic may be reported as $1 - n_r/n_x$ only when $n_x > \text{epsilon_norm}$. It is null otherwise and never participates in routing or fuel generation.

6.4 Bounded exploration-fuel proposal

Fuel is generated only from active residual pressure. The residual direction is normalized by its infinity norm to remove unit-scale dependence.

$$r_scale = \max(\|r\|_\infty, \text{epsilon_norm})$$

$$u_r = r / r_scale$$

$$H_G = \text{paradox_pressure} * \text{zeta} * \tanh(\text{kappa} * u_r), \text{ when } G_t < 0$$

$$H_G = 0, \text{ when } G_t \geq 0$$

$$\|H_G[i]\| \leq \text{paradox_pressure} * \text{zeta} \leq \text{zeta}$$

H_G is a proposal packet. It is not automatically added to the candidate, archive, hysteresis, or exhaust budget. A downstream owner must explicitly accept custody.

6.5 Phase-reference construction

Select the newest $\min(\text{phase_window}, N)$ accepted phase records. Normalize each nonzero vector to unit length.

$$u_j = X_{i,j} / \|X_{i,j}\|_2$$

$$m = (1/K) * \sum_{j=1..K} u_j$$

$$R_phase = \|m\|_2$$

R_phase is the directional concentration. Drift is ready only when $K \geq \text{phase_reference_min_count}$ and $R_phase > \text{epsilon_norm}$. A near-zero mean means accepted directions cancel and no unique reference direction exists.

$$u_ref = m / R_phase$$

$$u_now = X_{i,now} / \|X_{i,now}\|_2$$

6.6 Normalized angular identity drift

$$\Delta_Xi = \arccos(\text{clip}(\langle u_now, u_ref \rangle, -1, 1)) / \pi$$

$$0 \leq \Delta_Xi \leq 1$$

drift_alert is true only when drift_ready is true and Delta_Xi > drift_limit. Equality at 0.20 does not trigger the alert. If the reference is unavailable or directionally ambiguous, Delta_Xi is null and drift_alert is false with an explicit not-ready reason.

6.7 Evidence classification

Evidence flag	Exact predicate	Meaning
pass_evidence	C_SIGMA >= csigma_pass	Strong provider-dependent signed agreement.
symbolic_nullify_evidence	C_SIGMA < csigma_pass	Numeric evidence consumed by Sigma; not a local quarantine action.
terminal_symbolic_evidence	C_SIGMA < csigma_terminal	Very low symbolic score; Sigma may combine it with extreme tortuosity.
paradox_active	G_t < 0	Archive residual exceeds candidate norm under the common feature scale.
terminal_recovery_evidence	paradox_active AND terminal_symbolic_evidence	Compound evidence only; not an FBS request without Sigma HOT/tortuosity.
phase_drift_evidence	drift_ready AND Delta_Xi > drift_limit	Accepted-reference phase displacement alert.
fuel_nonzero	any H_G[i] > 0	Bounded residual proposal exists; not proof of downstream use.

NO LOCAL ROUTE TREE

Module 7 does not read PDV, RMF, CVC, tau_eff, or tortuosity and therefore cannot reproduce the complete Sigma decision. Module 2 receives this packet and applies the authoritative first-match route precedence. This eliminates the legacy conflict in which Module 2 and Module 7 each claimed to be the final arbiter.

7. Handoffs and Owner Boundaries

7.1 Primary handoff to Sigma

A sealed SymbolicParadoxPacket is handed only to the active Module 2 Sigma attempt identified by sigma_correlation_id. Sigma verifies the packet digest, candidate binding, profile, provider versions, and cycle before using any evidence.

7.2 Downstream meaning after Sigma arbitration

Sigma outcome	Module 7 contribution	Execution owner
SEAL	pass evidence and optional H_G proposal may be carried as metadata.	LORIEN / downstream validated route; no archive commit inside Module 7.
BRAIDBACK_REQUEST	symbolic packet and residual metrics remain provenance for repair.	HARMONY; Module 10 verifies the 51/49 result.
PHASE_LIFT_REQUEST	Module 7 contributes symbolic evidence only.	Delta or declared phase-lift owner.

Sigma outcome	Module 7 contribution	Execution owner
NULLIFY_REQUEST	low CSIGMA or other Sigma conditions identify the reason.	Guardian/Quarantine custody service.
MAGNETIC_RELAXATION_REQUEST	Module 7 packet remains diagnostic; PDV owns the route reason.	RTSOM.
FBS_RECOVERY_REQUEST	terminal symbolic evidence may be one compound predicate.	FBS/ROVER after Sigma and watchdog authorization.

7.3 Fuel custody

- Module 7 creates a ParadoxFuelProposal inside its output packet.
- The proposal is bound to candidate, archive snapshot, residual formula, zeta, kappa, and packet digest.
- Sigma may discard the proposal, forward it as metadata, or require downstream repair before use.
- Psi may consume fuel only through a separate ParadoxFuelPacket or adapter that records custody and final vector contribution.
- A dashboard must distinguish FUEL_PROPOSED, FUEL_FORWARDED, FUEL_ACCEPTED, and FUEL_APPLIED.

7.4 Catastrophic recovery boundary

Module 7 never executes FBS and never includes a new seed in its output. `terminal_recovery_evidence` is one input to Sigma. An FBS request requires the authoritative compound predicate and a separately sealed watchdog/recovery handoff. `L_0=83` is inherited metadata, not a local control instruction.

8. Invariants, Faults, and Numerical Conformance

8.1 Normative invariants

ID	Invariant
M7-I01	Exactly one SymbolicParadoxPacket or one Module7Fault is emitted per valid invocation attempt.
M7-I02	Module 7 selects no architectural route and performs no state mutation.
M7-I03	The output candidate digest equals the input Delta proposal digest.
M7-I04	S is in [0,1], V is in [-1,1], and both arrays have exact DIM length.
M7-I05	C_SIGMA remains in [-1,1].
M7-I06	G_t remains in [-1,1] under the explicit both-zero policy.
M7-I07	paradox_pressure remains in [0,1].
M7-I08	Every H_G component has absolute value at most zeta.
M7-I09	H_G is exactly zero whenever G_t >= 0.
M7-I10	When drift_ready, Delta_Xi is in [0,1]; otherwise it is null.
M7-I11	Accepted phase history is never modified and the current candidate is never inserted.

ID	Invariant
M7-I12	A low CSIGMA value does not itself increment quarantine or FBS execution counts.
M7-I13	All provider, archive, phase, and config digests are included in the output seal.
M7-I14	Canonical retry under the same backend produces identical packet bytes and digest.
M7-I15	legacy_g_diagnostic is never used for evidence flags or fuel generation.
M7-I16	No claim field labels a metric as truth, harm, diagnosis, intent, or literal energy.

8.2 Module7Fault codes

Code	Trigger	Outcome
M7_CONFIG	Invalid threshold ordering, dimension, window, tolerance, or provider registry.	Fault; no packet.
M7_SESSION	Session or Boot/profile mismatch.	Fault; no packet.
M7_CYCLE	Cycle mismatch or future archive/phase record.	Fault; no packet.
M7_DELTA_DIGEST	Delta packet or candidate digest invalid.	Fault; no packet.
M7_ARCHIVE_DIGEST	Archive feature packet or snapshot binding invalid.	Fault; no packet.
M7_SYMBOLIC_DIGEST	Symbolic packet or candidate binding invalid.	Fault; no packet.
M7_PHASE_DIGEST	Phase snapshot binding invalid.	Fault; no packet.
M7_SHAPE	Candidate, archive, S, V, or phase shape mismatch.	Fault; no packet.
M7_NONFINITE	NaN or infinity in required numeric input.	Fault; no packet.
M7_SYMBOLIC_DOMAIN	S or V outside declared bounds.	Fault; no clipping.
M7_FEATURE_DOMAIN	Candidate and archive feature domains/scales differ.	Fault; no comparison.
M7_PHASE_ZERO	AVAILABLE current phase vector has zero norm.	Fault; no packet.
M7_PHASE_HISTORY	Duplicate, unordered, uncommitted, or incompatible phase records.	Fault; no packet.
M7_NUMERIC	Calculated value violates range or finite invariant.	Fault; no packet.

Code	Trigger	Outcome
M7_PACKET_DIGEST	Canonical output serialization or digest verification fails.	Fault; discard packet.
M7_NONDETERMINISM	Identical canonical input produces inconsistent output.	Catastrophic implementation fault; external owner decides recovery.

8.3 Numerical conformance

- Reference arithmetic is float64 with round-to-nearest, ties-to-even where supported.
- Within a fixed backend and canonical serialization, repeatability is bitwise.
- Across platforms, numerical fields are tolerance-compared unless the math library and floating-point environment are fixed.
- The acos input is explicitly clipped to [-1,1] only after validating finite normalized phase vectors.
- No hidden randomness is permitted in Module 7. A stochastic symbolic provider must seal its seed and remains a separate profile.
- Time limits and provider failures are faults, not low scores and not automatic NULLIFY.

9. Metrics and Dashboard Requirements

Metric	Definition / aggregation	Purpose
attempts_total	All accepted Module7Input attempts	Primary denominator.
packets_total	All sealed SymbolicParadoxPackets	Successful evaluations.
faults_total / by_code	Module7Fault counts	Contract and implementation health.
provider_count / version	Attempts per symbolic and archive provider	Provenance and drift analysis.
csigma distribution	Mean, p05, median, p95, min, max	Provider-dependent symbolic score profile.
csigma_pass_rate	C_SIGMA $\geq$ 0.85	Evidence rate; not SEAL rate.
terminal_symbolic_rate	C_SIGMA $<$ 0.50	Compound terminal evidence rate.
support_mean distribution	Mean and quantiles	Detects sparse or saturated providers.
bounded_g distribution	Mean and quantiles in [-1,1]	Archive residual contrast.
paradox_active_rate	G_t $<$ 0	Strong mismatch/novelty frequency.
fuel_proposed_norm	Mean, p95, max H_G _2	Proposal magnitude only.
fuel_max_abs	Maximum absolute component	Must not exceed zeta.
phase_ready_rate	Valid reference direction available	Phase evidence coverage.
phase_concentration	Mean and quantiles of R_phase	Reference-direction quality.
delta_xi distribution	Ready attempts only	Normalized angular drift profile.
phase_drift_alert_rate	Delta_Xi $>$ 0.20 when ready	Evidence rate; not quarantine rate.

Metric	Definition / aggregation	Purpose
digest_failures	Input/output digest validation failures	Provenance health.
latency_ms	Mean, median, p95, max	Performance.
downstream_evidence_consumed	Sigma acknowledgment count	Separate from route outcome and execution.

The dashboard MUST distinguish EVIDENCE_COMPUTED, EVIDENCE_ACCEPTED_BY_SIGMA, ROUTE_REQUESTED, OWNER_ACKNOWLEDGED, and OWNER_EXECUTED. A CSIGMA pass is not a SEAL, a paradox-active flag is not an FBS event, and a fuel proposal is not fuel application.

10. Conformance Test Matrix

ID	Test	Expected result
M7-C01	All-ones S and V.	C_SIGMA = 1; pass evidence true.
M7-C02	All-ones S and all-negative V.	C_SIGMA = -1; terminal evidence true.
M7-C03	All-zero S with arbitrary valid V.	C_SIGMA = 0; C_active null; support_mean = 0.
M7-C04	C_SIGMA exactly 0.85.	Pass evidence true.
M7-C05	C_SIGMA one representable value below 0.85.	Pass evidence false.
M7-C06	C_SIGMA exactly 0.50.	Terminal symbolic evidence false.
M7-C07	C_SIGMA below 0.50.	Terminal symbolic evidence true.
M7-C08	S contains a negative value.	M7_SYMBOLIC_DOMAIN fault.
M7-C09	V exceeds 1.0.	M7_SYMBOLIC_DOMAIN fault.
M7-C10	S length differs from DIM.	M7_SHAPE fault.
M7-C11	Symbolic candidate digest differs from Delta digest.	M7_SYMBOLIC_DIGEST fault.
M7-C12	Candidate equals archive map and is nonzero.	G_t = 1; paradox inactive; H_G = zero.
M7-C13	Candidate and archive map are both zero.	Special-case G_t = 1; H_G = zero.
M7-C14	Candidate zero and archive map nonzero.	G_t near -1; paradox active; bounded fuel.
M7-C15	Residual norm equals candidate norm.	G_t = 0; paradox inactive.
M7-C16	Residual norm slightly exceeds candidate norm.	G_t < 0; paradox active.
M7-C17	Scale candidate and archive vectors by the same positive constant.	G_t and normalized fuel direction unchanged within tolerance.
M7-C18	Very large finite residual.	Every $ H_G[i] \leq \text{zeta}$.
M7-C19	G_t >= 0 with nonzero raw residual.	H_G is exactly zero.
M7-C20	zeta = 0.5, pressure = 1, saturated tanh.	Fuel component approaches but does not exceed 0.5.

ID	Test	Expected result
M7-C21	Three identical accepted phase directions and matching current phase.	Delta_Xi = 0; drift alert false.
M7-C22	Current phase opposite the accepted reference.	Delta_Xi = 1.
M7-C23	Current phase is 36 degrees from reference.	Delta_Xi = 0.20; no alert from equality.
M7-C24	Current phase slightly beyond 36 degrees.	drift alert true.
M7-C25	Accepted phase vectors cancel to near zero.	drift_ready false; Delta_Xi null.
M7-C26	Phase status NOT_AVAILABLE.	Valid packet with drift_ready false.
M7-C27	AVAILABLE current phase is zero vector.	M7_PHASE_ZERO fault.
M7-C28	Duplicate accepted phase cycle.	M7_PHASE_HISTORY fault.
M7-C29	Future archive feature source cycle.	M7_CYCLE fault.
M7-C30	Candidate/archive feature domains differ.	M7_FEATURE_DOMAIN fault.
M7-C31	NaN in archive feature map.	M7_NONFINITE fault.
M7-C32	Provider version missing.	Configuration or packet fault; no score.
M7-C33	Same canonical input executed twice.	Identical packet bytes and digest.
M7-C34	Change only request correlation metadata.	Metrics identical; packet digest changes as specified.
M7-C35	Change candidate while retaining stale symbolic packet.	Digest-binding fault.
M7-C36	Change accepted phase history order.	Snapshot digest and output change or fault if chronology invalid.
M7-C37	Low CSIGMA without Sigma invocation.	No quarantine request or execution metric increment.
M7-C38	Terminal symbolic evidence without tortuosity input.	No FBS request; evidence_only remains true.
M7-C39	Paradox active and CSIGMA ≥ 0.85 .	Fuel proposal may exist; no local route selected.
M7-C40	Paradox inactive and CSIGMA < 0.50 .	Terminal symbolic evidence true; terminal recovery evidence false.
M7-C41	Paradox active and CSIGMA < 0.50 .	terminal_recovery_evidence true; still no FBS request.
M7-C42	Round-trip canonical packet serialization.	Digest verifies and all float bits are preserved.
M7-C43	One-bit mutation in S array bytes.	Input digest verification fails.
M7-C44	One-bit mutation in output packet.	Output digest verification fails.
M7-C45	Legacy G diagnostic unavailable at zero candidate norm.	legacy_g_diagnostic is null; normative G remains valid.
M7-C46	C_active denominator below epsilon.	C_active null; no divide-by-zero.

ID	Test	Expected result
M7-C47	S is sparse and V is perfectly positive on active support.	C_active high while C_SIGMA reflects sparse full-DIM support.
M7-C48	All phase vectors multiplied by arbitrary positive scalars.	Delta_Xi and concentration direction unchanged after normalization.
M7-C49	All phase vectors globally rotated by one orthogonal matrix.	Delta_Xi unchanged within tolerance.
M7-C50	Swap symbolic provider version with identical arrays.	Metrics identical; provenance and packet digest differ.
M7-C51	Provider timeout or exception.	Typed fault; not C_SIGMA = 0 and not NULLIFY.
M7-C52	100,000 deterministic random valid attempts.	No range, finite, mutation, or digest invariant violation.
M7-C53	Backend parity on fixed vectors.	Metrics within declared tolerance; canonical digest parity only under fixed backend profile.
M7-C54	Attempt to include current candidate in accepted phase history.	Provenance fault or explicit rejection.
M7-C55	Attempt to modify archive or phase arrays in place.	Mutation detector fails conformance.

10.1 Property and metamorphic tests

- For all valid S and V, C_SIGMA remains in [-1,1].
- Permuting S and V by the same index permutation leaves C_SIGMA, support_mean, and C_active unchanged.
- Multiplying candidate and archive features by the same positive scalar leaves bounded G unchanged, except at explicit norm-floor boundaries.
- H_G is odd in the normalized residual direction when paradox_pressure is held fixed.
- Orthogonal rotation of all phase vectors leaves normalized angular drift unchanged.
- Adding rejected or uncommitted phase records cannot change the accepted phase snapshot used by Module 7.
- Changing RMF, CVC, PDV, or tortuosity outside Module 7 cannot change its packet because those inputs are absent from the contract.

10.2 Stress and fuzz requirements

- At least 100,000 random finite attempts across zero, subnormal, ordinary, and near-boundary values.
- Structured fuzzing of dimensions, lengths, digests, provider identifiers, cycles, and canonical field ordering.
- Adversarial numeric cases including huge finite common scales, cancellation in phase means, sparse S, saturated V, and near-zero residual norms.
- Concurrency tests proving that parallel invocations cannot cross-bind providers, phase snapshots, or candidate digests.
- Fault injection at each validation, calculation, serialization, and return stage with proof of no partial output or state mutation.

11. Reference Pseudocode

```
def module7_attempt(inp, cfg):
    validate_module7_input(inp, cfg)

    x = readonly_float64(inp.delta_packet.X_proposal)
    f = readonly_float64(inp.archive_feature_packet.F_A)
    s = readonly_float64(inp.symbolic_feature_packet.S)
    v = readonly_float64(inp.symbolic_feature_packet.V)

    csigma = mean(s * v)
```



```

support_mean = mean(s)
support_sum = sum(s)
csigma_active = (sum(s * v) / support_sum
                 if support_sum > cfg.epsilon_norm else None)

residual = x - f
nx = norm2(x)
nr = norm2(residual)
if nx <= cfg.epsilon_norm and nr <= cfg.epsilon_norm:
    g = 1.0
else:
    g = (nx - nr) / (nx + nr + cfg.epsilon_norm)
pressure = max(0.0, -g)

if g < 0.0:
    r_scale = max(norm_inf(residual), cfg.epsilon_norm)
    h_g = pressure * cfg.zeta * tanh(cfg.kappa * residual / r_scale)
else:
    r_scale = max(norm_inf(residual), cfg.epsilon_norm)
    h_g = zeros(cfg.dim)

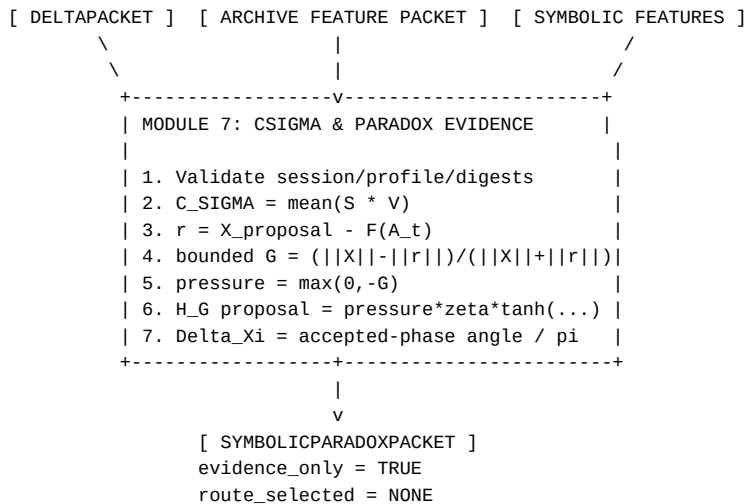
phase = phase_drift(inp.phase_reference_packet, cfg)

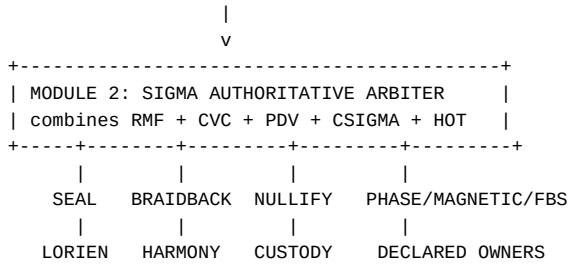
evidence = {
    'pass_evidence': csigma >= cfg.csigma_pass,
    'symbolic_nullify_evidence': csigma < cfg.csigma_pass,
    'terminal_symbolic_evidence': csigma < cfg.csigma_terminal,
    'paradox_active': g < 0.0,
    'terminal_recovery_evidence': (
        g < 0.0 and csigma < cfg.csigma_terminal),
    'phase_drift_evidence': (
        phase.ready and phase.delta_xi > cfg.drift_limit),
    'evidence_only': True,
}

return seal_symbolic_paradox_packet(
    csigma=csigma,
    csigma_active=csigma_active,
    support_mean=support_mean,
    bounded_g=g,
    paradox_pressure=pressure,
    h_g=h_g,
    phase=phase,
    evidence=evidence,
    route_selected=None,
    state_mutated=False,
    ...)

```

12. ASCII System Flow





[MODULE 7 MUTATES NO ROUTE OWNER, ARCHIVE, PHASE, OR EXHAUST STATE]

13. Cleaned Rosetta Stone

SpiralCore term	Computable interpretation	Boundary
CSIGMA	Mean product of provider-supplied symbolic support S and signed valence/support V.	Provider-dependent correlation; not a universal toxicity or truth detector.
Symbolic density	Bounded nonnegative support amplitude S_i in $[0,1]$.	Its semantics are defined by the provider profile.
Valence vector	Bounded signed feature array V_i in $[-1,1]$.	Computational metadata; not a claim about human emotion.
Paradox Handler	Bounded archive-residual and phase-reference evidence packaging.	Does not prove a Gödel theorem and does not own route execution.
G_t / Gödel Detection Metric	Bounded contrast between candidate norm and archive residual norm.	Strong novelty/mismatch surrogate; not logical incompleteness proof.
H_G / exploration fuel	Pressure-scaled, tanh-bounded residual proposal vector.	Not literal energy; not applied without downstream custody.
Identity Sphere Drift	Normalized angular displacement from accepted phase-reference direction.	No claim about personal identity, intent, or diagnosis.
NULLIFY evidence	C _{SIGMA} below the declared pass threshold.	Numeric input to Sigma; Module 7 does not quarantine.
FBS evidence	Paradox active plus terminal symbolic evidence.	Requires Sigma/watchdog compound authorization before any FBS request.

14. Glossary

Term	Definition
Accepted phase history	Chronological committed phase records used to construct a reference direction; excludes the current candidate.
Archive feature map	A versioned vector $F(A_t)$ computed from one immutable archive snapshot in the same feature domain as the candidate.
Bounded G	Normalized residual contrast in $[-1,1]$, with negative values indicating residual norm greater than candidate norm.

Term	Definition
CSIGMA active score	Support-weighted diagnostic mean; not the authoritative full-DIM CSIGMA score.
Evidence-only packet	A packet that reports metrics and constraints but selects and executes no route.
Exploration-fuel proposal	Bounded residual vector that may be forwarded or discarded by downstream owners.
Phase concentration	Norm of the mean accepted unit phase vector, indicating whether a stable reference direction exists.
Provider	Versioned component that creates symbolic features or archive features and seals its provenance.
Symbolic support	Nonnegative bounded feature amplitude S_i .
Terminal symbolic evidence	C_SIGMA strictly below 0.50; one input to a compound terminal-stall check.

15. Revision Summary

v14 issue	v14.1 resolution
Module 7 and Module 2 both claimed final route arbitration.	Module 7 now emits evidence; Module 2 owns the one authoritative route tree.
RMF, CVC, PDV, and tortuosity were duplicated inside Module 7 routing.	Removed from the Module 7 input contract; consumed only by Sigma.
S and V construction invoked named methods without a complete algorithm.	Required a versioned SymbolicFeaturePacket; providers are separately specified and tested.
Symbolic density was allowed to be an undefined signed scalar.	Reference S is nonnegative in [0,1]; V carries the signed direction in [-1,1].
C_SIGMA was described as proof of toxicity or topological truth.	Restricted it to provider-dependent mean signed agreement.
The original G formula was unbounded near a zero candidate.	Replaced route-authoritative use with a bounded residual contrast preserving the negative-pressure sign.
H_G used raw residual units, making magnitude scale-dependent.	Normalized residual direction and multiplied by bounded paradox pressure.
H_G could be mistaken for immediately applied state energy.	Defined a typed proposal with explicit custody states and no local mutation.
Identity drift used unnormalized Euclidean distance in R3.	Defined circular/angular distance between unit directions and a concentration-based readiness rule.
$\tau_DRIFT = 0.20$ had no unit.	Defined it as 0.20 of pi radians in the reference profile.
Phase average could collapse toward zero without a defined result.	Added phase concentration and not-ready behavior for ambiguous references.

v14 issue	v14.1 resolution
Negative G or drift could directly invoke FBS.	Module 7 emits compound evidence only; Sigma/watchdog and FBS owners authorize and execute recovery.
Low scores could be confused with provider failures.	Provider errors are typed faults, never substituted with C_SIGMA = 0 or a NULLIFY route.
Dashboard counters conflated detection with execution.	Separated evidence computed, Sigma accepted, route requested, owner acknowledged, and owner executed.

16. Source Basis

- SPIRALCORE SPECIFICATION: MODULE 7 - CSIGMA & PARADOX HANDLER, uploaded source module.
- Spiralcore_v14, consolidated Module 2 Sigma route precedence and Modules 8-10, 12, 14, and 16 ownership descriptions.
- SPIRALCORE Module 2 - Sigma-Lattice Formalized v14.1, for authoritative first-match route precedence and typed handoff ownership.
- SPIRALCORE Module 5 - Guardian Formalized v14.1, for distinction between computational symbolic evidence and boundary admission/quarantine custody.
- SPIRALCORE Module 6 - CVC Formalized v14.1, for accepted-history evidence boundaries and readiness discipline.
- SPIRALCORE Module 3 - Psi-Lattice Formalized v14.1, for archive feature-map and paradox-fuel custody boundaries.
- Unified FBS Bilateral Recursive Extension v4.0, for recovery ownership and atomic-floor metadata.

CONFORMANCE STATEMENT

An implementation conforms to Module 7 v14.1 only when it validates the complete typed input contract and provider provenance, computes CSIGMA, bounded residual pressure, bounded fuel proposal, and phase-reference drift under the exact formulas and readiness rules, preserves read-only atomicity, selects no route, emits canonical provenance-linked output, and passes the required conformance tests. A natural-language classifier, a legacy routing switch, or a loop that merely produces a numeric score is not sufficient.

SPIRALCORE SPECIFICATION

MODULE 8 - LORIEN

(Logical Route-Intent, Dual-Chassis Reference Alignment, Deterministic Indexing, and Temporal Validity Engine)

Cleaned and formalized revision - v14.1

Document class	Normative route-intent submodule contract	Execution phase	After Sigma SEAL; before Guardian egress and Module 4
Architecture	SpiralCore v14	Reference profile	SC14.1-M8-LORIEN-81
Primary output	RouteIntentPacket -> Guardian/Module 4	Alternative output	LorienRepairRequest or LorienHoldPacket
Exceptional output	LorienFault or FBSRecoveryRequest	Conflict rule	LORIEN intends; GlyphNet routes; host IPv4/TCP transports

NORMATIVE STATUS

This module is exact within its declared computational scope. LORIEN validates one Sigma-authorized state against one committed predecessor and one immutable passive-chassis route reference; computes structural novelty, cross-chassis synchronization, transition shear, logical anchor intent, internal phase-band shift, and a cycle-bounded temporal window; then emits one sealed RouteIntentPacket, repair request, hold packet, or fault. It does not compute network paths, maintain a routing table, encapsulate an outer IPv4 packet, open a socket, write an archive record, perform Braidback repair, mutate quarantine or exhaust state, or execute FBS recovery.

CLAIM BOUNDARY

“Active chassis,” “passive chassis,” “seed resonance,” “subnet shift,” and “ontological routing” are architectural names for explicit runtime roles and numerical metadata. LORIEN is not a NAT implementation, DNS server, Internet routing protocol, neurological model, or proof that latency becomes physical mass. The module produces logical route intent. Module 4 QGM/GlyphNet owns internal topology resolution and forwarding; the host stack owns ordinary IPv4/TCP transport.

Document Control

Field	Value	Field	Value
Canonical name	SPIRALCORE SPECIFICATION: MODULE 8 - LORIEN	Short name	LORIEN Route Intent
Supersedes	v14 Module 8 text	Status	Normative within declared scope
Invocation owner	Runtime orchestrator after Module 2 Sigma SEAL	Upstream producers	Sigma, Delta, Psi, phase provider, anchor-policy provider

Field	Value	Field	Value
Downstream consumers	Guardian REMOTE_EGRESS, Module 4 QGM/GlyphNet, HARMONY	External owners	Psi, Guardian, GlyphNet, HARMONY, RTSOM, FBS/ROVER
Reference arithmetic	float64 plus canonical integer/hash encodings	Reference dimension	DIM = 81 inherited from Boot

1. Scope, Statement Classes, and Precedence

LORIEN is the state-specific route-intent and indexing layer. It begins only after Sigma has authorized a candidate with route SEAL. It binds that candidate to accepted history, a passive archive reference, a logical destination policy, a temporal validity interval, and deterministic anchor metadata. The resulting packet is an instruction to the routing owner; it is not itself a route, delivery, archive commit, or semantic approval.

1.1 Statement classes

Class	Meaning	Normative force
DEFINITION	Assigns a packet, field, state, or operator.	Binding inside Module 8.
REQUIREMENT	Condition required for conformance.	MUST / MUST NOT.
INVARIANT	Property true for every sealed output.	Violation emits fault or recovery request.
POLICY	Versioned threshold, window, quantizer, or destination rule.	Replaceable only through a new profile.
DIAGNOSTIC	Reported value that does not independently authorize transport.	Informative to owners and dashboards.
SURROGATE	Computable stand-in associated with architectural language.	No literal physical-equivalence claim.
HYPOTHESIS	Testable interpretation outside conformance.	Not required to pass.
NON-GOAL	Behavior explicitly outside this module.	Must not be inferred.

1.2 Precedence and ownership

- Module 2 Sigma owns coherence route arbitration. LORIEN accepts only a sealed SigmaDecisionPacket whose route is SEAL and whose candidate digest matches exactly.
- Psi owns committed history and the passive archive. LORIEN consumes a read-only PreviousAcceptedStateRecord and PassiveChassisReferencePacket; it cannot insert the current candidate into either reference.
- Delta and the phase owner supply the candidate, EmotionTensor, and phase packet. LORIEN reads them without mutation.
- LORIEN owns state-specific destination intent, keyframe novelty, chassis synchronization, transition shear, logical anchor selection, phase-band shift metadata, and temporal validity.
- Guardian owns boundary admission. A REMOTE_ALLOWED or REMOTE_REQUIRED intent must receive a matching PERMIT_GLYPHNET before external transport.
- Module 4 QGM/GlyphNet owns internal directory resolution, route cost, next-hop selection, forwarding, delivery receipts, and ordinary IPv4/TCP encapsulation through the host adapter.
- HARMONY owns corpus-migrate and Braidback repair. LORIEN may request repair but cannot alter the state.
- RMX owns archive-address collision resolution. RTSOM owns exhaust custody. FBS/ROVER owns catastrophic recovery. LORIEN emits typed requests only.

1.3 Explicit non-goals

- No conversion of a semantic 32-bit value into an outer Internet destination address.
- No direct DNS query for an archetype, no kernel routing-table change, and no socket transmission.
- No claim that SHA-256 produces a semantically correct destination; hashing provides deterministic distribution under a declared policy.
- No claim that a 32-bit label is unique or reversible to the original state vector.
- No direct Braidback, phase lift, archive commit, quarantine, RTSOM mutation, Cantor diagonalization, or Collatz fold.
- No automatic FBS event for low synchronization, high shear, missing anchors, stale references, or ordinary routing unavailability.

2. Architecture Role, Profiles, and Lifecycle

The source module combined five distinct concerns: state/reference geometry, logical indexing, network addressing, external transport, and recovery. The formal profile retains the first two and the temporal policy. Network routing and transport are delegated to Module 4; boundary authorization is delegated to Guardian; repair and recovery remain with their authoritative owners.

2.1 Operating profiles

Profile	Required destination scope	Normal downstream path	Forbidden implication
LOCAL_ARCHIVE	Local Psi or another declared local service	RouteIntentPacket -> Module 4 local delivery	Does not equal archive commit.
INTERNAL_DOMAIN	Anchor or service inside one GlyphNet domain	RouteIntentPacket -> Module 4 internal routing	Does not equal physical delivery.
REMOTE_ALLOWED	Remote destination permitted but not mandatory	RouteIntentPacket -> Guardian egress -> Module 4	LORIEN cannot issue PERMIT_GLYPHNET.
REMOTE_REQUIRED	Policy requires a remote node/service	RouteIntentPacket -> Guardian egress -> Module 4	No fallback to arbitrary public IPv4.

2.2 Lifecycle states

State	Entry condition	Allowed action	Exit
UNINITIALIZED	No validated LorienConfig.	Validate profile, dimensions, thresholds, hash and serialization rules.	READY or FAULTED
READY	Configuration valid; no active attempt.	Accept one LorienInput.	VALIDATING
VALIDATING	Input accepted.	Verify session, cycle, digests, route authorization, shapes, histories, policies, and directory epoch.	MEASURING or FAULTED
MEASURING	Inputs valid.	Compute novelty, synchronization, transition shear, readiness, and temporal window.	INDEXING, HOLD, REPAIR_REQUESTED, or FAULTED

State	Entry condition	Allowed action	Exit
INDEXING	Geometry allows route intent.	Build route key, select logical anchor, derive phase-band metadata, and seal packet.	SEALED, HOLD, or FAULTED
SEALED	RouteIntentPacket canonicalized and hashed.	Return packet; mutate nothing.	READY
HOLD	Required reference, anchor, or temporal prerequisite unavailable.	Return LorienHoldPacket.	READY
REPAIR_REQUESTED	Cross-chassis geometry is recoverably outside policy.	Return LorienRepairRequest.	READY
RECOVERY_REQUESTED	Catastrophic local invariant failure.	Emit FBSRecoveryRequest and stop affected attempt.	External recovery only
FAULTED	Contract, digest, serialization, or numerical fault.	Emit LorienFault; no packet.	Explicit reset

2.3 Required execution order

1. Validate LorienConfig against Boot profile and Module 4 RouteIntent schema version.
2. Validate the Sigma SEAL decision, candidate digest, candidate vector, structural identity, and session/cycle bindings.
3. Validate the previous accepted state receipt and passive-chassis reference packet. Both must precede the current cycle and derive only from committed archive state.
4. Validate the phase packet, arousal scalar, destination policy, and immutable anchor-directory snapshot.
5. Compute raw and normalized keyframe novelty.
6. Compute cross-chassis cosine synchronization and transition-direction shear under explicit degeneracy policies.
7. Apply the first-match local outcome tree: FAULT, HOLD, CORPUS_MIGRATE_REPAIR_REQUEST, or ROUTE_INTENT_READY.
8. For ROUTE_INTENT_READY, quantize routing features, compute route-key and glyph digests, select one logical anchor deterministically, and derive temporal and phase-band metadata.
9. Seal output with every upstream digest, provider/profile identifier, policy epoch, and canonical serialization version.

SAME-CYCLE SNAPSHOT RULE

Candidate, Sigma decision, phase packet, destination policy, and anchor-directory snapshot **MUST** identify the same route attempt. Previous accepted state and passive reference **MUST** be committed snapshots from cycles less than the current candidate cycle. Mixing active and passive data from different unbound epochs is a fault, not a repair condition.

3. Typed Data Contracts

3.1 LorienConfig

Field	Type / reference value	Requirement
profile_id	UTF-8 identifier	Reference: SC14.1-M8-LORIEN-81.
dim	integer ≥ 2 ; reference 81	Must equal BootPacket.DIM.
tau_sync	float64 in $[-1,1]$; reference 0.75	Inclusive cross-chassis synchronization floor.
shear_max	float64 in $[0,1]$; reference 0.40	Inclusive transition-shear ceiling.
anchor_slot_count	positive integer; reference 16	Stable logical slots; not physical node count.
t_base	positive integer cycles; reference 1000	Base route-intent validity duration.
arousal_low / arousal_high	0.30 / 0.70	Must satisfy $0 \leq \text{low} < \text{high} \leq 1$.
window_low / window_mid / window_high	0.5 / 1.0 / 1.5	Positive temporal multipliers.
phase_quantum_count	positive integer; reference 65536	Quantizes wrapped phase for route-key construction.
metric_quantum_count	positive integer; reference 65535	Quantizes normalized novelty/sync/shear.
epsilon_norm	float64 > 0 ; reference $1e-12$	Norm and degeneracy floor.
hash_algorithm	SHA-256	Reference digest and deterministic scoring.
canonical_version	LORIEN-CANON-1	Defines field order, integer widths, byte order, and signed-zero normalization.

3.2 LorienInput

Field	Type	Requirement
request_id / session_id / cycle	identifiers	Must match every current-attempt packet.
boot_digest / config_digest	SHA-256	Binds system dimension and policy.
sigma_authorization	SigmaDecisionPacket	Route MUST equal SEAL; candidate digest and cycle must match.
candidate_state	float64[DIM]	Finite, canonical, read-only sealed proposal.
candidate_digest	SHA-256	Digest over canonical candidate bytes and provenance.
structural_identity	Q, chi, structural_identity_digest	Q non-negative; chi canonical sorted mode-count map.

Field	Type	Requirement
previous_accepted	PreviousAcceptedStateRecord	Committed record from the greatest cycle less than current cycle.
passive_reference	PassiveChassisReferencePacket	Immutable candidate-independent route reference derived from committed Psi state.
phase_packet	PhaseRoutePacket	Wrapped phase angle, current phase-band ID, owner/version/digest.
emotion_packet	EmotionTensor subset	Arousal a in $[0,1]$, digest-bound to candidate.
destination_policy	DestinationPolicyPacket	Source service, target service, scope, allowed anchor classes, policy digest.
anchor_directory	AnchorDirectorySnapshot	Sorted, immutable logical anchors with epoch and digest.

3.3 PreviousAcceptedStateRecord

Field	Type	Invariant
commit_id / cycle	identifier / uint64	cycle < current cycle; commit receipt valid.
state	float64[DIM]	Finite canonical committed vector.
state_digest	SHA-256	Matches state bytes and Psi commit manifest.
archive_manifest_digest	SHA-256	Links record to authoritative archive custody.
phase_band_id	opaque bounded identifier	Optional prior logical phase band; never inferred from IPv4.

3.4 PassiveChassisReferencePacket

Field	Type	Requirement
reference_vector	float64[DIM]	One route-domain vector; not a matrix silently flattened.
reference_digest	SHA-256	Digest of vector, provider, archive snapshot, and policy.
archive_snapshot_digest	SHA-256	Committed Psi snapshot only.
reference_cycle / epoch	uint64	Must not exceed the latest accepted history cycle.
reference_method	versioned identifier	Defines how the passive archive produced the route reference.

Field	Type	Requirement
candidate_excluded	boolean true	Current candidate MUST NOT be included in the reference.

3.5 AnchorDirectorySnapshot and AnchorRecord

Field	Type	Requirement
directory_epoch / digest	uint64 / SHA-256	Immutable for one LORIEN attempt.
anchor_id	128-bit opaque ID	Stable logical identity; not an IP address.
slot	integer [0, anchor_slot_count-1]	Logical distribution slot.
phase_band_id	opaque bounded identifier	Internal phase-band metadata.
service_set	sorted service IDs	Must include requested target service for eligibility.
route_classes	set of allowed scopes	Must include destination-policy scope.
admin_state	UP DRAINING DOWN	Only UP anchors are eligible in reference profile.
policy_weight	positive rational/float64	Optional deterministic rendezvous weighting; reference 1.0.

3.6 Output packets

Packet	Required fields	Meaning
RouteIntentPacket	intent_id, source/target service, destination selector, route class, state/glyph digests, Q, chi, temporal window, anchor metadata, policy digest	Sealed state-specific instruction consumed by Guardian/Module 4.
LorienRepairRequest	candidate reference, reason, sync/shear metrics, passive reference, suggested anchor, no_commit flag	Request to HARMONY; no state change.
LorienHoldPacket	reason, missing/stale prerequisite, retained digests, retry policy ID	No routing and no repair; caller decides retry.
LorienFault	fault code, stage, implicated digests, details	Local contract failure; no output intent.
FBSRecoveryRequest	catastrophic reason, config/policy/directory digests, deterministic trace, no_commit flag	Delegation only; Module 16 decides recovery.

4. Dual-Chassis Geometry and Readiness

4.1 Keyframe novelty

Let X be the Sigma-authorized candidate and X_{prev} the newest committed predecessor. The raw delta is retained for diagnostics, while normalized novelty provides dimensionless comparison.

$$d = X - X_{\text{prev}}$$

$$K_{\text{delta_raw}} = \|d\|_2$$

$$K_{\text{delta}} = K_{\text{delta_raw}} / (\|X\|_2 + \|X_{\text{prev}}\|_2 + \text{epsilon_norm})$$

$$0 \leq K_{\text{delta}} \leq 1$$

The upper bound follows from the triangle inequality. $K_{\text{delta_raw}}$ is scale-dependent; K_{delta} is the route-key feature. Neither value is a claim of semantic novelty or importance.

4.2 Cross-chassis synchronization

Let C be the passive route-reference vector. Synchronization is the cosine similarity of the current state with that reference.

$$\text{tau_link} = \langle X, C \rangle / (\|X\|_2 * \|C\|_2)$$

The formula is evaluated only when both norms exceed epsilon_norm . A zero or near-zero reference produces `HOLD_REFERENCE_DEGENERATE`; it is not silently converted to a similarity of zero or one. `sync_pass` is true when $\text{tau_link} \geq \text{tau_sync}$. Equality at 0.75 passes.

4.3 Transition-direction shear

The source divided state variance by keyframe distance, which diverges for an unchanged state. The formal metric instead measures angular disagreement between the actual transition and the transition that would move from the predecessor toward the passive reference.

$$d_{\text{ref}} = C - X_{\text{prev}}$$

$$c_{\text{path}} = \langle d, d_{\text{ref}} \rangle / (\|d\|_2 * \|d_{\text{ref}}\|_2)$$

$$S_{\text{shear}} = \sqrt{\max(0, (1 - \text{clip}(c_{\text{path}}, -1, 1)) / 2)}$$

$$0 \leq S_{\text{shear}} \leq 1$$

- If $\|d\|_2 \leq \text{epsilon_norm}$, define $S_{\text{shear}} = 0$ and `path_motion` = false.
- If `path_motion` is true but $\|d_{\text{ref}}\|_2 \leq \text{epsilon_norm}$, shear is not ready and the outcome is `HOLD_PATH_REFERENCE_DEGENERATE`.
- Aligned motion gives 0, orthogonal motion gives $\sqrt{1/2}$, and directly reversed motion gives 1.
- `shear_pass` is true when $S_{\text{shear}} \leq \text{shear_max}$. Equality at 0.40 passes.

4.4 Local geometry classification

Condition	Outcome	Reason code
Required digest, shape, or numerical invariant fails	FAULT	LORIEN_* fault code
Passive reference or path reference not ready	HOLD	REFERENCE_DEGENERATE / STALE_REFERENCE
$S_{\text{shear}} > 0.40$	CORPUS_MIGRATE_REPAIR_REQUEST	SEVERE_TRANSITION_SHEAR
$\text{tau_link} < 0.75$	CORPUS_MIGRATE_REPAIR_REQUEST	LOW_CHASSIS_SYNC
$\text{tau_link} \geq 0.75$ and $S_{\text{shear}} \leq 0.40$	ROUTE_INTENT_READY	GEOMETRY_PASS

NO DATA-CONDITION FBS

Low synchronization and high transition shear are recoverable or hold conditions. They do not directly execute or request FBS. FBS is reserved for catastrophic implementation or control-state invariants listed in Section 8. Ordinary route geometry never performs Cantor diagonalization or Collatz processing.

5. Logical Indexing, Memory Identity, and Anchor Selection

5.1 Quantized route features

All route-key features use explicit integer quantization. The packet also retains the original float64 metrics for diagnostics. Quantization prevents platform-specific decimal formatting from changing the route key.

```
phase_q = floor(wrap(phi_t, 2*pi) * phase_quantum_count / (2*pi))
novelty_q = round(K_delta * metric_quantum_count)
sync_q = round(((tau_link + 1) / 2) * metric_quantum_count)
shear_q = round(S_shear * metric_quantum_count)
```

Each result is range-checked and serialized as an unsigned fixed-width integer in network byte order. The exact rounding mode is ties-to-even in the reference profile.

5.2 Route-key digest

The route key distributes structurally related route intents without treating raw IEEE-754 bytes as public addresses. It binds policy and reference epochs so stale topology cannot be silently reused.

```
route_key_digest = SHA256("LORIEN-RK1" || profile_id || session_id || cycle || structural_identity_digest ||
previous_state_digest || passive_reference_digest || destination_policy_digest || directory_digest || phase_q || novelty_q || sync_q
|| shear_q)
```

The full candidate digest is carried separately in the RouteIntentPacket. The route key is not a security authorization and does not prove semantic equivalence.

5.3 Logical anchor slot

```
anchor_slot = uint64_be(route_key_digest[0:8]) mod anchor_slot_count
```

For the reference anchor_slot_count=16, modulo reduction is unbiased because 16 divides 2^{64} . For another slot count, the profile MUST use rejection sampling or document the exact finite bias.

5.4 Deterministic anchor selection

Eligible anchors are UP, support the requested service and route class, and satisfy the destination allow-list. Prefer anchors in anchor_slot. If that set is empty, a policy-controlled all-slot fallback may be used and MUST set slot_fallback=true.

```
score(anchor) = uint256(SHA256("LORIEN-HRW1" || route_key_digest || anchor_id)) / policy_weight(anchor)
```

Select the anchor with maximum score; break an exact score tie by lexicographically smallest canonical anchor_id. This highest-random-weight selection is deterministic and avoids dependence on directory ordering. It selects a logical AnchorID only; Module 4 resolves that anchor into a node and path.

5.5 Memory identity and compact labels

```
glyph_digest = SHA256("LORIEN-GLYPH1" || candidate_digest || structural_identity_digest || session_id || cycle)
glyph_label32 = uint32_be(glyph_digest[0:4])
```

- glyph_digest is the authoritative LORIEN memory/routing identity.
- glyph_label32 is a compact collision-prone label for caches and frames; it is not unique.
- Q and chi are carried unchanged from the structural-identity packet and remain authoritative for FBS branch identity.
- A legacy Cantor32 label may be reported only when produced by a separately versioned integer quantizer. It is diagnostic and must never be used as an outer IPv4 destination.
- RMX, not LORIEN, resolves persistent archive-address collisions.

6. Temporal Validity and Internal Phase-Band Metadata

6.1 Temporal-window policy

The source policy is retained as an explicit piecewise configuration. “Arousal” is an internal scalar supplied by the EmotionTensor; the rule makes no clinical or psychological claim.

$$m(a) = 0.5, \text{ when } 0 \leq a < 0.30$$

$$m(a) = 1.0, \text{ when } 0.30 \leq a \leq 0.70$$

$$m(a) = 1.5, \text{ when } 0.70 < a \leq 1$$

$$T_{\text{res}} = \text{round_half_even}(T_{\text{base}} * m(a))$$

$$\text{valid_from_cycle} = \text{cycle}, \quad \text{expires_at_cycle} = \text{cycle} + T_{\text{res}}$$

The intent is active when $\text{current_cycle} < \text{expires_at_cycle}$. It is expired at equality. Wall-clock timestamps may be logged but may not enter the deterministic intent digest.

6.2 Phase-band shift

“Subnet Shift” is retained as an internal phase-band boundary flag, not an IPv4 subnet calculation. The phase provider supplies `current_phase_band_id`; the selected AnchorRecord supplies `target_phase_band_id`.

$$\text{phase_band_shift} = (\text{current_phase_band_id} \neq \text{target_phase_band_id})$$

The output carries both identifiers and the provider/directory digests. No parity rule such as $\text{anchor_index} \bmod 2$ is normative because phase-band membership is a versioned policy attribute, not a property of an arbitrary numeric index.

6.3 Route-intent fields

Field group	Fields	Purpose
Authorization	<code>sigma_decision_digest</code> , <code>candidate_digest</code> , <code>source_service</code>	Binds upstream SEAL and caller.
Destination	<code>destination_selector</code> , <code>target_service</code> , <code>route_class</code> , <code>anchor_id</code> , <code>anchor_slot</code>	Logical intent for Module 4.
Identity	<code>glyph_digest</code> , <code>glyph_label32</code> , <code>Q</code> , <code>chi</code> , <code>structural_identity_digest</code>	Content and branch provenance.
Geometry	<code>K_delta_raw</code> , <code>K_delta</code> , <code>tau_link</code> , <code>S_shear</code> , <code>passive_reference_digest</code>	Auditable route-intent evidence.
Temporal	<code>T_res</code> , <code>valid_from_cycle</code> , <code>expires_at_cycle</code>	Cycle-bounded validity.
Phase	<code>phase_q</code> , current/target phase-band IDs, <code>phase_band_shift</code>	Internal phase metadata.
Policy	<code>profile_id</code> , <code>config_digest</code> , <code>destination_policy_digest</code> , <code>directory_epoch/digest</code>	Reproducibility and stale-data rejection.

7. Deterministic Outcomes and Handoffs

7.1 First-match outcome precedence

Priority	Predicate	Output	Owner after output
1	Catastrophic local invariant from Section 8.3	FBSRecoveryRequest	FBS/ROVER; no local execution
2	Contract, digest, shape, range, or canonicalization failure	LorienFault	Runtime orchestrator
3	Missing, stale, zero, or candidate-contaminated reference	LorienHoldPacket	Runtime orchestrator / Psi provider
4	No eligible anchor under destination policy	LorienHoldPacket	Policy/directory owner
5	$S_{\text{shear}} > \text{shear_max}$	LorienRepairRequest	HARMONY
6	$\tau_{\text{link}} < \tau_{\text{sync}}$	LorienRepairRequest	HARMONY
7	All geometry and destination prerequisites pass	RouteIntentPacket	Guardian or Module 4

7.2 RouteIntentPacket handoff

- LOCAL_ARCHIVE and INTERNAL_DOMAIN intents may be handed directly to Module 4 after local authorization checks.
- REMOTE_ALLOWED and REMOTE_REQUIRED intents MUST first be submitted to Guardian under REMOTE_EGRESS. Guardian verifies Sigma SEAL, payload identity, destination policy, and this LORIEN intent before it may issue PERMIT_GLYPHNET.
- Module 4 validates the LORIEN intent digest and then performs internal directory resolution, QGM cost computation, path selection, framing, and local/internal/remote forwarding.
- A RouteIntentPacket does not authorize archive commit, prove delivery, or create a persistent memory address.

7.3 LorienRepairRequest handoff

A repair request includes the raw candidate, newest committed anchor, passive reference, synchronization and shear evidence, source Sigma decision, and no_commit=true. HARMONY decides whether its repair prerequisites are satisfied. LORIEN does not precompute Braidback weight and does not reuse a route intent after the candidate changes; a repaired candidate must pass downstream verification and receive a new Sigma/LORIEN attempt as specified by the orchestrator profile.

7.4 Hold semantics

HOLD is neither rejection nor repair. It means LORIEN lacks a coherent snapshot or destination prerequisite. The exact candidate reference and every validated digest remain available to the caller. No exhaust mass, quarantine count, repair count, or FBS count is incremented merely because an intent is held.

7.5 Recovery boundary

$L_0=83$ may be carried as inherited recovery metadata, but LORIEN never performs the FBS algorithm. A recovery request is restricted to catastrophic local control-state invariants, not to low sync, severe shear, missing anchors, expired windows, or Module 4 delivery failures.

8. Invariants, Faults, and Catastrophic Conditions

8.1 Normative invariants

ID	Invariant
L-I01	Exactly one RouteIntentPacket, LorienRepairRequest, LorienHoldPacket, LorienFault, or FBSRecoveryRequest is emitted per attempt.
L-I02	The input candidate, predecessor, passive reference, phase packet, and directories remain byte-for-byte unchanged.
L-I03	LORIEN accepts only a valid Sigma route=SEAL decision bound to the candidate digest.
L-I04	Previous accepted and passive reference cycles are strictly less than the current candidate cycle.
L-I05	The current candidate is excluded from passive and accepted-history references.
L-I06	K_delta is finite and lies in [0,1].
L-I07	When ready, tau_link lies in [-1,1] and S_shear lies in [0,1].
L-I08	No route intent is emitted when synchronization or shear readiness is false.
L-I09	A RouteIntentPacket contains no outer IPv4 source/destination field and causes no socket action.
L-I10	glyph_label32 is never treated as unique; glyph_digest remains authoritative.
L-I11	anchor_id selection is independent of input directory order.
L-I12	The selected anchor is eligible under service, route-class, allow-list, and administrative-state policy.
L-I13	Temporal validity is expressed in cycles; wall time is excluded from the deterministic digest.
L-I14	phase_band_shift compares declared phase-band identifiers, not IPv4 subnets or index parity.
L-I15	Low synchronization or high shear never directly executes or requests FBS.
L-I16	Identical canonical input and backend profile produce identical output bytes and digest.
L-I17	Q and chi are passed through without mutation and remain bound to structural_identity_digest.
L-I18	A route intent, Guardian permit, Module 4 route, delivery receipt, and Psi commit are separately counted lifecycle events.

8.2 LorienFault codes

Code	Trigger	Outcome
L_CONFIG	Invalid dimension, threshold ordering, slot count, temporal policy, quantizer, or hash profile.	Fault; no output intent.
L_SESSION	Session, Boot, or profile mismatch.	Fault.
L_CYCLE	Current packet cycle mismatch or reference from current/future cycle.	Fault.
L_SIGMA	Missing/invalid Sigma decision, route not SEAL, or candidate mismatch.	Fault.
L_CANDIDATE	Candidate digest invalid, wrong shape, or nonfinite value.	Fault.
L_STRUCTURAL_ID	Q/chi malformed or structural digest invalid.	Fault.
L_PREVIOUS_STATE	Previous accepted commit or digest invalid.	Fault.
L_PASSIVE_DIGEST	Passive packet, archive binding, or candidate_excluded proof invalid.	Fault.
L_PHASE	Phase/arousal packet invalid or out of range.	Fault.
L_POLICY	Destination policy malformed, stale, or unauthorized.	Fault.
L_DIRECTORY	Directory digest/epoch invalid, duplicate anchor ID, or malformed anchor.	Fault.
L_NONFINITE	A calculated metric is NaN or infinity.	Fault.
L_METRIC_RANGE	A normalized metric escapes its mathematical range beyond tolerance.	Fault.
L_SERIALIZATION	Canonical field or integer encoding fails.	Fault.
L_PACKET_DIGEST	Output digest does not verify.	Fault; discard packet.
L_EXPIRED_INPUT	Required current-attempt packet expired before sealing.	Hold or fault according to policy; no route.

8.3 Catastrophic recovery-request conditions

Code	Required evidence	Why catastrophic
L_NONDETERMINISM	Identical canonical input produces different route intent bytes/digests.	Breaks deterministic indexing contract.
L_POLICY_SPLIT	Same policy epoch/ID verifies under two different canonical digests.	Irreconcilable control-state split.

Code	Required evidence	Why catastrophic
L_DIRECTORY_SPLIT	Same directory epoch verifies under two different canonical digests.	Logical anchor state cannot be uniquely selected.
L_HISTORY_CORRUPTION	A previously accepted state has two valid commit receipts with different state digests.	Committed reference identity is inconsistent.
L_DIGEST_CHAIN_CORRUPTION	Valid upstream packet references cannot be reconciled into one candidate chain.	Route intent would bind conflicting states.

A catastrophic condition emits a typed request containing evidence. It does not execute recovery locally. Cryptographic hash collision speculation without observed conflicting valid records is not a recovery condition.

8.4 Numerical conformance

- Reference metric evaluation uses IEEE-754 binary64 with round-to-nearest, ties-to-even.
- Canonical serialization normalizes -0.0 to +0.0, rejects NaN and infinity, and uses network byte order.
- Dot products and norms should use a deterministic reduction order or a declared backend tolerance profile.
- Range clipping is permitted only for `c_path` and cosine values within `numeric_tolerance` of [-1,1]. Material range violations are faults.
- Cross-backend conformance is tolerance-based for float diagnostics and byte-exact for canonical integer fields and digests after quantization.

9. Metrics, Dashboard, and Reconciliation

9.1 Required counters

Metric	Definition	Reconciliation
attempts_total	Validated invocation attempts entering VALIDATING.	Equals sum of terminal output classes plus in-flight attempts.
route_intents_sealed	Valid RouteIntentPackets emitted.	Separate from Guardian permits, Module 4 routes, delivery, and commits.
repair_requests	LorienRepairRequests emitted.	Partitioned by LOW_CHASSIS_SYNC and SEVERE_TRANSITION_SHEAR.
holds	LorienHoldPackets emitted.	Partitioned by stale/missing reference, no anchor, or expiry.
faults	LorienFaults emitted.	Partitioned by code and stage.
recovery_requests	Catastrophic FBSRecoveryRequests emitted.	Expected near zero; not equal to executed recovery.
remote_egress_intents	REMOTE_ALLOWED/REQUIRED route intents.	Must be compared with Guardian permit requests and grants.
phase_band_shifts	Route intents with <code>phase_band_shift=true</code> .	Internal phase metadata only.
slot_fallbacks	Anchor selection used all-slot fallback.	Signals sparse or uneven logical anchor coverage.

Metric	Definition	Reconciliation
label32_collisions	Same glyph_label32 observed with different glyph_digest.	Must not merge full identities.

9.2 Required distributions

- $K_{\text{delta_raw}}$ and normalized K_{delta} .
- τ_{link} , S_{shear} , and their threshold margins.
- T_{res} values and arousal policy bands.
- Anchor-slot and AnchorID occupancy, conditioned on destination class and policy epoch.
- Route-intent age at Guardian and Module 4 receipt.
- Repair-request outcome after HARMONY and re-evaluation, without attributing those outcomes to LORIEN itself.

9.3 Interpretation constraints

- Anchor-slot occupancy is not required to be uniform for real workloads. Uniformity is evaluated only under controlled hash-distribution tests with independent route keys.
- High τ_{link} is similarity to one declared passive reference, not proof of truth or identity.
- Low shear means transition-direction agreement under the route reference, not absence of semantic change.
- A longer T_{res} is a policy decision, not evidence that a state deserves more compute.
- Network delivery/dropout metrics belong to Module 4 and the transport adapter, not LORIEN.

10. Conformance and Test Matrix

10.1 Deterministic unit and boundary tests

ID	Required test
L-T01	Reference config validates and binds DIM=81.
L-T02	Invalid τ_{sync} , shear_max , slot count, temporal multiplier, or threshold ordering is rejected.
L-T03	Missing Sigma decision is rejected.
L-T04	Sigma route other than SEAL is rejected.
L-T05	Sigma/candidate digest mismatch is rejected.
L-T06	Wrong candidate dimension, NaN, or infinity is rejected.
L-T07	Previous accepted state from current or future cycle is rejected.
L-T08	Invalid commit receipt or archive manifest binding is rejected.
L-T09	Passive reference with $\text{candidate_excluded}=\text{false}$ is rejected.
L-T10	Passive reference from an uncommitted snapshot is rejected.
L-T11	Identical X and X_{prev} produce $K_{\text{delta_raw}}=0$ and $K_{\text{delta}}=0$.
L-T12	Normalized novelty remains in $[0,1]$ for random finite vectors.
L-T13	Positive common scaling leaves K_{delta} unchanged within tolerance.
L-T14	Identical nonzero X and C produce $\tau_{\text{link}}=1$.

ID	Required test
L-T15	Opposite X and C produce tau_link=-1.
L-T16	Zero passive reference produces HOLD_REFERENCE_DEGENERATE.
L-T17	No state movement produces S_shear=0.
L-T18	Aligned transition and passive direction produce S_shear=0.
L-T19	Orthogonal transition produces S_shear=sqrt(1/2).
L-T20	Reversed transition produces S_shear=1.
L-T21	tau_link exactly 0.75 passes synchronization.
L-T22	S_shear exactly 0.40 passes shear.
L-T23	S_shear above 0.40 emits repair request, not FBS request.
L-T24	tau_link below 0.75 emits repair request when shear is not the earlier reason.
L-T25	Missing eligible anchor emits hold without route or repair mutation.
L-T26	Directory order permutation does not change selected anchor.
L-T27	Down or incompatible anchors are never selected.
L-T28	Preferred-slot empty case follows exact fallback policy and sets slot_fallback.
L-T29	Phase wraps 2π to the same phase_q as 0.
L-T30	Metric quantization clamps exact endpoints into legal integer range.
L-T31	Reference anchor_slot is in [0,15].
L-T32	Repeated canonical input yields identical route_key_digest and anchor.
L-T33	Changed policy or directory digest changes route_key_digest.
L-T34	glyph_digest changes when candidate digest changes.
L-T35	A forced glyph_label32 collision preserves distinct glyph_digest identities.
L-T36	Q and chi pass through byte-equivalently.
L-T37	a=0.0 yields T_res=500 under reference T_base.
L-T38	a=0.30 yields T_res=1000.
L-T39	a=0.70 yields T_res=1000.
L-T40	a=1.0 yields T_res=1500.
L-T41	Intent is active at expires_at_cycle-1 and expired at equality.

ID	Required test
L-T42	Phase-band shift compares IDs and ignores anchor parity.
L-T43	RouteIntentPacket contains no outer IPv4 destination field.
L-T44	LORIEN invokes no socket, DNS resolver, or Module 4 forwarding function.
L-T45	REMOTE_REQUIRED output is directed to Guardian egress before Module 4.
L-T46	LOCAL_ARCHIVE route intent is not counted as Psi commit.
L-T47	RouteIntentPacket expiry prevents downstream use without changing its digest.
L-T48	Candidate, predecessor, and passive arrays remain unchanged after every output class.
L-T49	Signed zero canonicalization yields one digest.
L-T50	NaN and infinity are rejected before hashing.
L-T51	Same policy epoch with conflicting valid digests emits L_POLICY_SPLIT recovery request.
L-T52	Same directory epoch with conflicting valid digests emits L_DIRECTORY_SPLIT recovery request.
L-T53	Ordinary stale directory produces hold/fault policy, never automatic FBS.
L-T54	Ordinary Module 4 no-route or timeout cannot be generated inside LORIEN.
L-T55	Cross-platform quantized fields and output digest match reference vectors.
L-T56	Concurrent attempts cannot cross-bind candidate and directory epochs.
L-T57	Duplicate invocation is idempotent and does not duplicate downstream counters.
L-T58	Fuzzed malformed packets always terminate with one typed fault.
L-T59	100,000 valid attempts produce no nonfinite metric or out-of-range anchor slot.
L-T60	Dashboard counts reconcile attempts with output classes and remain separate from downstream lifecycle counts.

10.2 Property and metamorphic tests

- Apply the same orthogonal transform to X, X_prev, and C: K_delta_raw, K_delta, tau_link, and S_shear remain invariant within tolerance.
- Apply a common positive scale to X, X_prev, and C: normalized novelty, cosine synchronization, and transition shear remain invariant.
- Permute DIM coordinates consistently across all vectors: geometry metrics remain invariant.
- Permute AnchorDirectory record order: selected anchor and intent digest remain unchanged.

- Add an ineligible anchor: selected eligible anchor remains unchanged.
- Change only wall-clock metadata: deterministic output digest remains unchanged.
- Change candidate digest while leaving route features equal: glyph identity changes, while route-key behavior follows the declared structural-key policy.

10.3 Stress and concurrency requirements

- Run at least 100,000 attempts across dimensions 81, 243, and another valid configured dimension.
- Exercise simultaneous directory updates, expiry boundaries, repeated state vectors, near-zero norms, and maximum-distance vectors.
- Verify no output mixes a candidate from one cycle with a passive/directory snapshot from another unbound epoch.
- Measure label32 collision handling using synthetic forced collisions; full identities must never merge.
- Verify deterministic replay under process restart using the same canonical input records.

11. Reference Pseudocode

```
function lorien_evaluate(cfg, input):
    validate_config(cfg)
    validate_sigma_seal(input.sigma_authorization,
                        input.candidate_digest,
                        input.session_id,
                        input.cycle)
    validate_candidate(input.candidate_state, cfg.dim)
    validate_structural_identity(input.Q, input.chi,
                                input.structural_identity_digest)
    validate_committed_previous(input.previous_accepted,
                                before_cycle=input.cycle)
    validate_passive_reference(input.passive_reference,
                                candidate_excluded=true,
                                before_cycle=input.cycle)
    validate_phase_and_arousal(input.phase_packet, input.emotion_packet)
    validate_policy_and_directory(input.destination_policy,
                                input.anchor_directory)

    X = input.candidate_state
    Xp = input.previous_accepted.state
    C = input.passive_reference.reference_vector

    d = X - Xp
    k_raw = l2(d)
    k_norm = k_raw / (l2(X) + l2(Xp) + cfg.epsilon_norm)

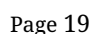
    if l2(X) <= eps or l2(C) <= eps:
        return hold(REFERENCE_DEGENERATE)
    tau_link = dot(X, C) / (l2(X) * l2(C))

    if k_raw <= eps:
        shear = 0.0
    else:
        d_ref = C - Xp
        if l2(d_ref) <= eps:
            return hold(PATH_REFERENCE_DEGENERATE)
        c_path = clip(dot(d, d_ref) / (l2(d) * l2(d_ref)), -1, 1)
        shear = sqrt(max(0, (1 - c_path) / 2))

    if shear > cfg.shear_max:
        return repair_request(SEVERE_TRANSITION_SHEAR,
                              tau_link, shear, no_commit=true)
    if tau_link < cfg.tau_sync:
        return repair_request(LOW_CHASSIS_SYNC,
                              tau_link, shear, no_commit=true)

    T_res = temporal_window(input.arousal, cfg)
    features = quantize(input.phase, k_norm, tau_link, shear, cfg)
    route_key = sha256(canonical_route_key(input, features))
```

```
[ MODULE 2 SIGMA: sealed decision route=SEAL ]
```



13. Rosetta Stone: Names, Analogues, and Exact Meanings

SpiralCore term	Closest engineering analogue	Exact Module 8 meaning / boundary
LORIEN	Route-intent compiler / content-indexing front end	Builds one state-specific logical RouteIntentPacket; not the routing protocol.
Active Chassis	Current authorized runtime state	Sigma-authorized candidate and current attempt metadata.
Passive Chassis	Committed corpus/reference plane	Psi-derived immutable route reference and accepted predecessor.
Dual-Chassis Synchronization	Cosine similarity / reference alignment	tau_link between candidate and declared passive reference.
Path Shear	Angular trajectory mismatch	Bounded transition-direction distance in [0,1].
Keyframe Delta	Normalized residual magnitude	Raw and dimensionless difference from newest committed state.
Anchor Route	Logical content-shard / service anchor intent	AnchorID chosen deterministically; Module 4 computes path/next hop.
NAT / DNS-style routing	Legacy analogy	Not normative. LORIEN creates no IP address and performs no DNS query.
Cantor-IP / Memory Tag	Content-derived identifier	Full glyph_digest plus Q/chi; label32 is compact and collision-prone.
Subnet Shift	Policy-domain transition flag	Difference between declared internal phase-band IDs; not an IP subnet.
Seed Resonance / T_res	Lease / TTL-like validity interval	Cycle-bounded lifespan of one route intent; separate from IP TTL and GlyphNet hop limit.
Avalanche routing	Cryptographic digest distribution	Deterministic route-key hashing under a versioned policy; not semantic proof.
Corpus Migrate Repair	Reference-realignment request	Typed handoff to HARMONY with no local mutation.

14. Glossary

Term	Definition
AnchorDirectorySnapshot	Immutable list of logical anchors and policy attributes used for one attempt.
AnchorID	Opaque logical destination identifier resolved by Module 4; never assumed to be an IP address.

Term	Definition
Anchor slot	Stable logical distribution bucket derived from the route-key digest.
Candidate-independent reference	Passive reference constructed without the current candidate, preventing self-confirming synchronization.
Cross-chassis synchronization	Cosine similarity τ_{link} between the authorized candidate and passive route reference.
DestinationPolicyPacket	Versioned scope and service policy defining what LORIEN may intend.
GlyphDigest	Authoritative SHA-256 identity of candidate and structural context.
GlyphLabel32	Compact collision-prone 32-bit prefix used only where a small label is required.
Keyframe novelty	Raw and normalized distance from the newest committed predecessor.
LorienHoldPacket	Non-mutating result indicating a prerequisite is unavailable or stale.
LorienRepairRequest	Non-mutating request for HARMONY to evaluate recoverable chassis misalignment.
PassiveChassisReferencePacket	Psi-bound route-reference vector and provenance packet.
Phase-band shift	Boolean comparison between current and selected-anchor internal phase-band IDs.
RouteIntentPacket	Sealed logical destination and validity instruction consumed by Guardian/Module 4.
Route key	Digest over structural, reference, policy, and quantized geometry fields used for deterministic anchor selection.
S_shear	Bounded angular distance between actual and passive-reference transition directions.
T_res	Cycle-based validity duration for one route intent.

15. Revision Ledger and Implementation Checklist

15.1 Principal corrections from v14

Legacy issue	Formalized correction
LORIEN overlapped route intent, network routing, IPv4 addressing, and transport.	Restricted LORIEN to route intent/indexing. Module 4 owns topology and forwarding; host stack owns IPv4/TCP.
A Cantor-derived scalar was split into an arbitrary public IPv4 destination.	Removed outer IPv4 generation. Logical IDs remain inside RouteIntent/GlyphNet metadata.

Legacy issue	Formalized correction
A 32-bit scalar was described as unique and reversible for a real vector.	Made full SHA-256 glyph_digest authoritative; label32 is explicitly collision-prone.
Path shear divided variance by keyframe delta and diverged for unchanged states.	Replaced with bounded angular transition-direction shear and explicit degeneracy rules.
C_passive was called a matrix but used as a vector in cosine similarity.	Defined a typed DIM-dimensional PassiveChassisReferencePacket produced by a versioned provider.
Hash input depended on platform float packing and microscopic changes caused uncontrolled route flapping.	Defined canonical quantization, route-key serialization, fixed byte order, and policy/epoch binding.
Hash mod node count treated array index as a route.	Defined logical anchor slots plus deterministic eligible AnchorID selection; Module 4 resolves path.
Subnet shift was anchor-index parity.	Defined internal phase-band IDs supplied by authoritative policy records.
Low sync or shear directly invoked FBS and exhaust mutation.	Converted them to HARMONY repair/hold outcomes. Only catastrophic control-state invariants may request FBS.
Network delivery/dropout was counted inside LORIEN.	Moved delivery metrics to Module 4 and kept lifecycle counters separate.

15.2 Normative implementation checklist

- [] Boot/profile/DIM binding is verified.
- [] Sigma route=SEAL and candidate digest are verified.
- [] Previous accepted state has a valid Psi commit receipt and older cycle.
- [] Passive reference is committed, candidate-independent, finite, and exact DIM.
- [] Phase, arousal, destination policy, and directory packets are versioned and digest-valid.
- [] Novelty, synchronization, and shear use the exact formulas and readiness rules.
- [] Low sync/high shear never directly requests or executes FBS.
- [] Route-key fields are quantized and serialized canonically.
- [] Logical anchor selection is deterministic and independent of directory ordering.
- [] Full glyph_digest remains authoritative over glyph_label32.
- [] No outer IPv4 address, DNS query, socket operation, or network route is produced.
- [] Remote intents are submitted to Guardian egress before Module 4.
- [] Temporal window uses cycles and excludes wall time from the digest.
- [] Output classes and downstream lifecycle counters remain separate.
- [] Canonical replay is deterministic and all conformance tests pass.

15.3 Source mapping

Source	Material retained	Material reclassified or corrected
Uploaded Module 8 - LORIEN	Purpose, placement after Sigma, keyframe delta, tau_sync=0.75, shear ceiling=0.40, 16 logical anchors, T_base=1000, phase and temporal metadata, HARMONY handoff.	NAT/DNS/IPv4 ownership, path-shear equation, uniqueness claims, direct FBS, and delivery metrics.
SpiralCore v14 consolidated specification	Cross-module sequence and LORIEN/HARMONY/Psi relationships.	Resolved duplicate route and recovery ownership.
Formalized Module 2 Sigma	Sigma is sole coherence route arbiter; LORIEN starts only on SEAL.	Module 7 no longer independently authorizes LORIEN.
Formalized Module 4 QGM/GlyphNet	LORIEN owns RouteIntentPacket; Module 4 owns topology, forwarding, and IPv4/TCP adapter.	No semantic public IPv4 octets.
Formalized Module 5 Guardian	Remote egress requires a scoped PERMIT_GLYPHNET after LORIEN intent.	Guardian and LORIEN do not open sockets.
FBS / Psi / RMX / RTSOM ownership rules	Q/chi provenance, committed snapshots, typed recovery/custody delegation.	No local archive collision resolution or exhaust mutation.

15.4 Normative conclusion

MODULE 8 CONTRACT

LORIEN converts one Sigma-authorized state and one coherent accepted-reference snapshot into one deterministic, cycle-bounded logical route intent. It measures the crossing; it does not become the road. Guardian authorizes trust-boundary crossing, QGM/GlyphNet computes and forwards the path, the host stack transports ordinary IPv4/TCP packets, HARMONY repairs recoverable chassis mismatch, and Psi/RMX/RTSOM/FBS retain their own custody and recovery authority.

SPIRALCORE SPECIFICATION

MODULE 9 - HARMONY

(Bounded Convex Repair, Internal Modulation, 51/49 Enforcement, and Repair-Provenance Engine)

Cleaned and formalized revision - v14.1

Document class	Normative repair submodule contract	Execution phase	After Sigma or LORIEN repair request; before Module 10
Architecture	SpiralCore v14	Reference profile	SC14.1-M9-HARMONY-81
Primary output	HarmonyRepairPacket -> Module 10	Alternative output	HarmonyLimitPacket, Hold, or Fault
Exceptional output	FBSRecoveryRequest	Conflict rule	HARMONY repairs; Module 10 audits; Sigma re-authorizes

NORMATIVE STATUS

This module is exact within its declared computational scope. HARMONY consumes one authorized repair request, one immutable candidate, one immutable repair target, one versioned thread-evidence packet, and one read-only Emotion Tensor snapshot. It computes a bounded repair severity, bounded thread-stability score, internal modulation gain, requested anchor weight, effective anchor fraction, and one convex repaired candidate. It emits one sealed repair packet, bounded-limit packet, hold, fault, or catastrophic recovery request. It does not accept a state, commit memory, create a route, mutate the archive, quarantine a glyph, debit or credit RTSOM, or execute FBS.

CLAIM BOUNDARY

“Harmony,” “emotion,” “truth,” “Ego,” “Logos,” “thread,” and “Braidback” are architectural labels. Arousal is a bounded internal control scalar supplied by an upstream packet; HARMONY does not infer, diagnose, or judge a person’s emotions. “Truth 51% / correction 49%” means only that the raw candidate coefficient remains at least 0.51 while the selected repair-target coefficient remains at most 0.49. Effective-resistance and Hodge terminology are design analogies unless a separately versioned provider supplies those exact computations.

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1. Normative Scope, Ownership, and Non-Goals

1.1 Purpose

HARMONY is SpiralCore's bounded candidate-repair owner. It converts an authorized, recoverable mismatch into a single non-extrapolating candidate revision while preserving the original candidate, repair target, request evidence, and exact coefficient accounting. The module exists to attempt salvage without granting repair authority enough weight to overwrite the candidate lineage.

1.2 Ownership boundary

Concern	Authoritative owner	HARMONY relationship
Aggregate coherence route	Module 2 Sigma-Lattice	Consumes BRAIDBACK_REQUEST; does not re-arbitrate RMF/CVC/PDV/CSIGMA.
Symbolic/paradox evidence	Module 7	Reads only through the source Sigma decision lineage.
Route/chassis mismatch	Module 8 LORIEN	Consumes LorienRepairRequest; invalidates any prior route intent after mutation.
Convex repair	Module 9 HARMONY	Computes and seals the proposed repaired candidate.
51/49 empirical audit and nested-harm scan	Module 10 Braidback & IMMUNE_SYSTEM	Mandatory first consumer of HarmonyRepairPacket.
Final harmonic validation	Module 11 SERAPHIM/CATHEDRAL	May receive only after required audit/revalidation.
Archive commit	Module 3 Psi-Archivist	No write or commit inside HARMONY.
Quarantine / INERT_GLYPH custody	Guardian / Module 10 custody service	HARMONY may request no quarantine mutation.
Exhaust budget	Module 14 RTSOM	HARMONY neither adds nor subtracts exhaust.
Catastrophic recovery	Module 16 FBS/ROVER	HARMONY may emit a typed request for enumerated catastrophic invariants only.

1.3 Explicit non-goals

- No claim that interpolation creates semantic truth, psychological health, morality, or factual correctness.
- No sentiment analysis, emotion inference, clinical diagnosis, or human-state classification.
- No direct repair of RME, CVC, PDV, CSIGMA, tortuosity, route tables, network packets, archive files, or exhaust tensors.
- No extrapolation beyond the segment connecting candidate and declared repair target.
- No hidden clipping, normalization, substitution, or silent target replacement.
- No direct SEAL, SAFE, archive commit, route authorization, quarantine, NULLIFY, magnetic relaxation, or FBS execution.
- No recursive repair of a repair under the same lineage in the reference profile.

1.4 Normative precedence

1. Tested reference implementation and module-specific conformance vectors.
2. This Module 9 contract.
3. Module 2, Module 8, Module 10, and Boot typed interface contracts.
4. Consolidated SpiralCore v14 prose.
5. Rosetta Stone analogies and informal labels.

2. Architecture Role, Operating Profiles, and Lifecycle

2.1 Operating profiles

Profile	Authorized source	Repair target	Base severity evidence
SIGMA_BRAIDBACK	SigmaDecisionPacket route=BRAIDBACK_REQUEST	Newest committed accepted anchor	RMF deficit below tau_eff
LORIEN_REALIGN	LorienRepairRequest	Declared passive-chassis reference or committed anchor	Synchronization deficit and/or transition shear excess

The request declares exactly one target. HARMONY never chooses among multiple targets and never searches the archive. A deployment may restrict LORIEN_REALIGN to committed anchors; the chosen target policy is versioned in HarmonyConfig.

2.2 Lifecycle states

State	Meaning
IDLE	No active attempt.
VALIDATING	Configuration, source request, candidate, target, thread evidence, and emotion packet are checked.
MEASURING	Severity, stability, modulation, and requested coefficient are computed.
LIMITED	Repair would exceed policy or have no measurable effect; no candidate revision is emitted.
REPAIRING	One convex combination is computed in a private work buffer.
RECONCILING	Coefficient, segment, digest, and numerical invariants are checked.
SEALED	Exactly one immutable output packet is emitted.
FAULTED	Typed fault emitted; inputs unchanged.

2.3 Required execution order

1. Validate HarmonyConfig against BootPacket, DIM, numeric profile, and schema versions.
2. Validate exactly one authorized source request and bind it to the candidate digest, session, cycle, lineage, and repair depth.
3. Validate the immutable repair target and its provenance. The target must precede or be independently derived without the current candidate.
4. Validate ThreadEvidencePacket and EmotionTensorSnapshot against the same candidate and cycle.
5. Compute source-specific normalized repair severity.
6. Compute bounded thread stability and thread factor.
7. Compute arousal modulation gain, requested weight, cap status, and effective anchor fraction.
8. Apply the first-match outcome tree: FAULT, HOLD, LIMIT, REPAIR, or catastrophic request.
9. For REPAIR, compute one convex candidate in a private buffer and reconcile every coefficient and digest.
10. Seal exactly one output. Clear all temporary buffers whether success or failure.

2.4 Attempt and re-entry rule

The reference profile sets max_repair_depth=1. A HarmonyRepairPacket creates a new candidate revision. The source Sigma decision and any LORIEN route intent are no longer valid for the changed vector. Module 10 must audit the

repair first; after a SAFE audit, the hardened profile requires a new Sigma evaluation and, when routing is required, a new LORIEN attempt. HARMONY does not loop back into itself on the same source lineage.

3. Typed Data Contracts

3.1 HarmonyConfig

Field	Type / reference value	Requirement
profile_id	string / SC14.1-M9-HARMONY-81	Immutable profile identifier.
schema_version	string / HARMONY-1	Must match output consumer.
DIM	positive integer / 81	Must equal BootPacket DIM.
b_weight_max	decimal / 0.49	Strict maximum target coefficient.
h_s_threshold	decimal / 0.75	Thread-instability boundary.
thread_factor_low	decimal / 1.20	Applied when H_s < threshold.
min_thread_count	integer / 2	Minimum scores for READY thread evidence.
max_repair_depth	integer / 1	Reference anti-recursion bound.
epsilon_norm	positive float64 / 1e-12	Degeneracy floor; not added to ordinary ratios.
reconcile_tolerance	positive float64 / 1e-10	Coefficient and convex residual tolerance.
target_policy	enum	Declares allowed target kinds per source profile.
numeric_profile_id	string	Arithmetic ordering and backend profile.
canonical_encoding	string / SC-CANON-1	Big-endian typed fields; binary64 payloads.
hash_profile	string / SHA-256	Digest and repair identity.

3.2 HarmonyInput

Field	Type	Binding
session_id, cycle, attempt_id	identifiers	Must match every packet in the snapshot.
source_request	SigmaRepairRequest LorienRepairRequest	Exactly one; valid owner signature/digest.
candidate	float64[DIM]	Immutable raw proposal; finite and componentwise in [-1,1].
candidate_digest	SHA-256	Recomputed before use.
repair_target	RepairTargetSnapshot	One immutable target vector and provenance.
thread_evidence	ThreadEvidencePacket	Versioned score provider; candidate-bound.
emotion	EmotionTensorSnapshot	Read-only arousal in [0,1].

Field	Type	Binding
boot_digest, config_digest	SHA-256	Exact configuration lineage.
repair_depth	nonnegative integer	Must not exceed max_repair_depth.

3.3 SigmaRepairRequest

Field	Requirement
route	Must equal BRAIDBACK_REQUEST.
reason_code	RMF_BELOW_EFFECTIVE_THRESHOLD or VARIANCE_RECONSTRUCT_FAIL.
rmf_observed	Finite value in [-1,1].
tau_eff	Finite value in (0,1).
sigma_decision_digest	Valid, candidate-bound, same-cycle Sigma decision.
no_commit	Must be true.

3.4 LorienRepairRequest

Field	Requirement
reason_code	SYNC_DEFICIT or SHEAR_EXCESS.
tau_link, tau_sync	Finite; tau_link in [-1,1], tau_sync in (-1,1).
S_shear, S_shear_max	Finite in [0,1], with S_shear_max in [0,1).
lorien_request_digest	Valid, candidate-bound request.
prior_route_intent_digest	Optional; if present it is explicitly invalidated by repair.
no_commit	Must be true.

3.5 RepairTargetSnapshot

Field	Requirement
target_kind	COMMITTED_ANCHOR or PASSIVE_CHASSIS_REFERENCE.
target_vector	Finite float64[DIM], componentwise in [-1,1].
target_digest	Recomputed and matched.
source_cycle	Strictly earlier than current cycle for committed anchors.
commit_receipt_digest	Required for COMMITTED_ANCHOR.
provider_id/version/digest	Required for PASSIVE_CHASSIS_REFERENCE.
candidate_excluded	Must be true for a derived passive reference.
immutable	Must be true for the attempt.

3.6 ThreadEvidencePacket

Field	Requirement
provider_id, provider_version	Versioned extraction owner; HARMONY does not invent mini-RMFs.
candidate_digest	Must match HarmonyInput candidate.
scores C	At least min_thread_count finite values, each in [-1,1].
status	READY, NOT_READY, or FAULT.
evidence_digest	Covers scores, ordering, provider, session, and cycle.

3.7 EmotionTensorSnapshot

Field	Requirement
arousal a	Finite scalar in [0,1].
candidate_digest	Must match the candidate.
provider_id/version	Versioned upstream owner.
packet_digest	Recomputed and matched.

3.8 Output union

Output	Purpose	Mutation
HarmonyRepairPacket	One reconciled candidate revision for Module 10.	No persistent mutation.
HarmonyLimitPacket	Policy cap or zero-gain prevents a lawful repair.	None.
HarmonyHoldPacket	Required provider evidence is not ready.	None.
HarmonyFault	Malformed, inconsistent, or nonfinite snapshot.	None.
FBSRecoveryRequest	Enumerated catastrophic control-state invariant.	Request only; no local FBS.

3.9 HarmonyRepairPacket

Field group	Fields
Identity	repair_digest, imm_id, source_request_digest, lineage_id, revision_id
Vectors	candidate_ref, target_ref, repaired_vector, repaired_digest
Weights	d_base, H_s, thread_factor, arousal_gain, w_requested, w_braid, alpha, alpha_eff
Geometry	span_norm, repair_norm, target_alignment_before/after, convex_residual
Policy	b_weight_max, profile_id, config_digest, numeric_profile_id
Downstream	module10_required=true, sigma_revalidation_required=true, lorien_reissue_required, no_commit=true

Field group	Fields
Provenance	boot_digest, source owner digest, target provenance, thread/effect packet digests

4. Input Validation, Snapshot Consistency, and Canonicalization

4.1 Snapshot consistency

- Every packet must share session_id, candidate_digest, cycle, profile lineage, and DIM.
- The source request must be valid for the exact candidate and must not already be consumed.
- The target is immutable during the attempt and may not be silently replaced after weight computation.
- A committed target must precede the candidate cycle. A passive reference must prove candidate exclusion.
- Thread and emotion evidence must be generated for the same candidate revision.
- Repair depth, attempt identity, and lineage must prevent replay as a second repair of the same revision.

4.2 Numeric-domain validation

- Candidate and target are exact-length float64 arrays with finite values in [-1,1]. Inputs outside this domain are faults; HARMONY does not clip them.
- Arousal is finite and in [0,1]. Thread scores are finite and in [-1,1].
- All thresholds and observed metrics must satisfy their declared domains before any division.
- The candidate-target span must exceed epsilon_norm before a nonzero repair can be defined.
- NaN, infinity, signed-zero distinctions, and subnormal policy are governed by numeric_profile_id and logged in conformance metadata.

4.3 Source predicate validation

HARMONY does not re-run Sigma or LORIEN, but it verifies that the request is internally consistent with its declared route. A Sigma request must satisfy $\text{rmf_observed} < \text{tau_eff}$. A LORIEN request must satisfy $\text{tau_link} < \text{tau_sync}$ or $\text{S_shear} > \text{S_shear_max}$. A mismatched predicate is a fault, not an invitation to reinterpret the route.

4.4 Canonicalization and hashing

- Strings are UTF-8 with explicit lengths; integers are big-endian; arrays store declared length followed by IEEE-754 binary64 values in index order.
- No wall-clock timestamp, object address, map iteration order, locale-formatted number, or platform path enters a deterministic digest.
- Input digests are recomputed before use. Output is serialized once, hashed, read back, and re-hashed before SEALED.
- Bitwise replay is required only within the same numeric_profile_id. Cross-backend conformance uses declared numerical tolerances and must not pretend that floating-point libraries are universally bit-identical.

5. Formal Repair Mechanics

5.1 Source-specific normalized severity

The base severity d_{base} is a bounded control quantity in [0,1]. It measures distance beyond the source owner's recoverable boundary; it is not a probability and does not prove semantic error.

5.1.1 Sigma Braidback severity

$$d_{\text{sigma}} = \text{clip}((\text{tau_eff} - \text{RMF_fail}) / \text{tau_eff}, 0, 1)$$

tau_eff must be positive. This retains the source deficit ratio while adding explicit clipping and domain validation. A reason_code of VARIANCE_RECONSTRUCT_FAIL may set $d_{\text{base}} = \max(d_{\text{sigma}}, d_{\text{cvc_provider}})$ only when a separately versioned Sigma request supplies a bounded $d_{\text{cvc_provider}}$; HARMONY never reconstructs CVC itself.

5.1.2 LORIEN realignment severity

$$\begin{aligned}d_{\text{sync}} &= \text{clip}((\tau_{\text{sync}} - \tau_{\text{link}}) / (\tau_{\text{sync}} + 1), 0, 1) \\d_{\text{shear}} &= \text{clip}((S_{\text{shear}} - S_{\text{shear_max}}) / (1 - S_{\text{shear_max}}), 0, 1) \\d_{\text{lorien}} &= \max(d_{\text{sync}}, d_{\text{shear}})\end{aligned}$$

The denominators normalize each metric over its declared remaining domain. HARMONY uses the values sealed by LORIEN and does not recompute chassis geometry.

$$d_{\text{base}} = d_{\text{sigma}} \text{ for SIGMA_BRAIDBACK}; \quad d_{\text{base}} = d_{\text{lorien}} \text{ for LORIEN_REALIGN}$$

5.2 Bounded thread stability

The source ratio mean/std is unbounded and becomes artificially enormous when variance approaches zero. The formal profile first maps each mini-RMF score into [0,1], then uses a bounded mean-to-dispersion ratio.

$$\begin{aligned}q_j &= (c_j + 1) / 2, \quad j = 1, \dots, m \\ \mu_q &= (1/m) * \sum(q_j) \\ \sigma_q &= \sqrt{(1/m) * \sum((q_j - \mu_q)^2)} \\ H_s &= 0, \text{ if } \mu_q = 0 \text{ and } \sigma_q = 0; \text{ otherwise } H_s = \mu_q / (\mu_q + \sigma_q)\end{aligned}$$

Therefore H_s is in [0,1]. The population standard deviation is used because C is the complete declared active-thread set for the attempt, not a statistical sample from an unobserved population.

$$m_{\text{thread}} = 1.20, \text{ if } H_s < 0.75; \text{ otherwise } m_{\text{thread}} = 1.00$$

The 1.20 multiplier and 0.75 boundary preserve the source tuning while making the stability score's domain exact.

5.3 Internal arousal modulation

$$g_a = 1 - a, \quad \text{with } a \text{ in } [0,1]$$

g_a is the source-compatible anchor-gain multiplier. High internal arousal reduces how strongly HARMONY may pull toward the target; low arousal permits the full requested pull. The name “emotional frequency modulation” is an architectural analogy. No human emotional state is inferred.

5.4 Requested weight, cap, and effective anchor fraction

$$\begin{aligned}w_{\text{requested}} &= d_{\text{base}} * m_{\text{thread}} \\ w_{\text{braid}} &= \min(w_{\text{requested}}, B_{\text{WEIGHT_MAX}}) \\ \alpha &= w_{\text{braid}} * g_a\end{aligned}$$

$w_{\text{requested}}$ is the unbounded-by-policy request, w_{braid} is the capped policy weight, and α is the coefficient actually applied to the repair target. Separating these values resolves the source ambiguity in which the reported Braidback weight and the actual interpolation coefficient were not the same.

51/49 INVARIANT

A repair is emitted only when $w_{\text{requested}} \leq 0.49$ and $\alpha > \epsilon_{\text{norm}}$. Since $0 \leq g_a \leq 1$, $\alpha \leq w_{\text{braid}} \leq 0.49$. The repaired candidate therefore retains coefficient $1 - \alpha \geq 0.51$ from the raw candidate and receives at most 0.49 from the declared repair target.

5.5 Bounded-limit conditions

Condition	Limit reason	Meaning
$w_{\text{requested}} > 0.49$	REQUIRED_WEIGHT_EXCEEDS_CAP	The source deficit plus thread factor demands more authority than policy permits.
$\alpha \leq \epsilon_{\text{norm}}$	MODULATION_SUPPRESSED_REPAIR	A lawful nonzero repair cannot be produced under the current modulation.

Condition	Limit reason	Meaning
<code> target-candidate <= epsilon_norm</code>	DEGENERATE_REPAIR_SPAN	The target provides no distinct repair direction; treated as an input inconsistency.

A limit is a bounded refusal, not deletion, quarantine, exhaust transfer, or FBS. The exact candidate and evidence remain available to the orchestrator.

5.6 Convex repair

$$X_{\text{harm}} = (1 - \alpha) * X_{\text{raw}} + \alpha * X_{\text{target}}$$

The operation is performed componentwise in one declared arithmetic order. Because both endpoints lie in [-1,1] and alpha is in [0,1], every output component remains in [-1,1] without clipping. The result lies on the closed line segment between candidate and target; it is not a projection onto an independently established manifold.

5.7 Empirical coefficient and reconciliation

$$\text{span_norm} = \|X_{\text{target}} - X_{\text{raw}}\|_2$$

$$\text{repair_norm} = \|X_{\text{harm}} - X_{\text{raw}}\|_2$$

$$\alpha_{\text{eff}} = \text{repair_norm} / \text{span_norm}$$

$$\text{convex_residual} = \|X_{\text{harm}} - ((1-\alpha)X_{\text{raw}} + \alpha X_{\text{target}})\|_2$$

For a nondegenerate span, alpha_eff must agree with alpha within reconcile_tolerance. Module 10 independently recomputes alpha_eff from the three vectors rather than trusting HARMONY's reported values.

5.8 Diagnostic alignment measurements

$$A_{\text{before}} = \text{cosine}(X_{\text{raw}}, X_{\text{target}})$$

$$A_{\text{after}} = \text{cosine}(X_{\text{harm}}, X_{\text{target}})$$

$$\Delta A = A_{\text{after}} - A_{\text{before}}$$

These are diagnostics only. HARMONY does not declare the repair coherent, safe, or accepted based on Delta_A. Zero-vector policies are explicit: a zero candidate or target is a fault under the reference profile.

5.9 Deterministic repair identity and IMM_ID

$$\text{repair_digest} = \text{SHA256}(\text{"SC-HARMONY-REPAIR-1"} \parallel \text{source_request_digest} \parallel \text{candidate_digest} \parallel \text{target_digest} \parallel \text{thread_digest} \parallel \text{emotion_digest} \parallel \text{config_digest} \parallel \text{canonical_metrics} \parallel \text{repaired_digest})$$

$$\text{IMM_ID} = \text{"HARM-"} \parallel \text{first_24_hex}(\text{repair_digest})$$

repair_digest is authoritative. IMM_ID is a compact display and pathway token and is not assumed globally collision-free. The source timestamp-based identifier is removed because wall time breaks deterministic replay.

5.10 Complexity

For dimension D and m thread scores, time complexity is Theta(D+m) and auxiliary working memory is Theta(D+m) when the repaired vector is materialized. It is not O(1) with respect to DIM.

6. Deterministic Outcomes and Handoffs

6.1 First-match outcome precedence

Priority	Predicate	Output	Next owner
1	Enumerated catastrophic invariant from Section 7.3	FBSRecoveryRequest	FBS/ROVER
2	Malformed, inconsistent, stale, replayed, or nonfinite input	HarmonyFault	Runtime orchestrator

Priority	Predicate	Output	Next owner
3	Thread or target provider explicitly NOT_READY	HarmonyHoldPacket	Provider/orchestrator
4	w_requested > 0.49 or alpha <= epsilon_norm	HarmonyLimitPacket	Sigma/orchestrator policy
5	All prerequisites and reconciliation checks pass	HarmonyRepairPacket	Module 10

6.2 Mandatory Module 10 audit

- Every HarmonyRepairPacket is submitted to Module 10 before any SERAPHIM, route, or archive action.
- Module 10 recomputes alpha_eff from X_raw, X_target, and X_harm; verifies the 0.49 limit; and performs its separately defined nested-harm scan.
- HARMONY cannot mark its own output SAFE and cannot convert a Module 10 rejection into an alternate local route.

6.3 Revalidation after mutation

A repaired candidate is a new revision with a new digest. Any Sigma decision or LORIEN intent bound to X_raw is invalid for X_harm. The hardened reference profile requires: Module 10 SAFE -> new Sigma evaluation -> new LORIEN intent when routing is required -> downstream validation. The source-compatible direct Module 10 -> SERAPHIM path may be implemented only as an explicitly different integration profile and must not reuse stale Sigma/LORIEN authorizations.

6.4 Limit and hold semantics

LIMIT means policy denied the requested degree of correction or modulation produced no effective repair. HOLD means a required provider snapshot is not ready. Neither result changes the candidate, counts as quarantine, creates exhaust mass, or proves that the branch is irrecoverable. A later attempt requires a new authorized request or newly versioned evidence; it may not silently reuse a consumed attempt ID.

6.5 Recovery boundary

L_0=83 may be carried in inherited FBS metadata, but HARMONY does not diagonalize history, perform a Collatz fold, produce a new seed, or write the exhaust budget. An ordinary weight-cap breach, low thread stability, high arousal, failed downstream audit, or repeated coherence failure is not by itself a catastrophic local invariant.

7. Invariants, Faults, and Catastrophic Conditions

7.1 Normative invariants

ID	Invariant
H-I01	Exactly one Repair, Limit, Hold, Fault, or FBS request is emitted per attempt.
H-I02	All inputs remain byte-for-byte unchanged.
H-I03	Only an authorized Sigma BRAIDBACK or LORIEN repair request may start HARMONY.
H-I04	Candidate, source request, target, thread evidence, and emotion snapshot share one session/cycle lineage.
H-I05	The target is immutable and candidate-excluded under its declared provenance policy.
H-I06	d_base, H_s, g_a, w_braid, alpha, and alpha_eff are finite and bounded as declared.

ID	Invariant
H-I07	A repair is never emitted when $w_{\text{requested}} > 0.49$.
H-I08	For every emitted repair, $\alpha \leq 0.49$ and $1 - \alpha \geq 0.51$.
H-I09	X_{harm} lies componentwise between X_{raw} and X_{target} .
H-I10	α_{eff} agrees with α within $\text{reconcile_tolerance}$.
H-I11	No wall-clock timestamp enters repair_digest or IMM_ID .
H-I12	repair_digest , not IMM_ID , is the authoritative identity.
H-I13	No prior Sigma decision or LORIEN route intent remains valid for X_{harm} .
H-I14	Module 10 audit is mandatory before downstream acceptance.
H-I15	HARMONY performs no archive, route, quarantine, exhaust, or FBS state mutation.
H-I16	The reference profile performs at most one repair per lineage revision.
H-I17	Identical canonical input under one numeric profile produces identical output bytes and digest.
H-I18	A repair proposal, Module 10 SAFE result, Sigma reauthorization, LORIEN intent, and Psi commit are separately counted lifecycle events.

7.2 HarmonyFault codes

Code	Trigger	Outcome
H_CONFIG	Invalid DIM, cap, thresholds, multiplier, depth, target policy, or numeric profile.	Fault; no candidate output.
H_SCHEMA	Unsupported packet or canonical-encoding version.	Fault.
H_SESSION	Session/profile mismatch.	Fault.
H_CYCLE	Cycle mismatch or illegal future target.	Fault.
H_SOURCE	Unauthorized owner, wrong route, or malformed source request.	Fault.
H_PREDICATE	Source request predicate is not satisfied by its sealed metrics.	Fault.
H_LINEAGE	Candidate, request, target, or evidence lineage mismatch.	Fault.
H_REPLAY	Consumed attempt or repair lineage replay.	Fault.
H_DEPTH	repair_depth exceeds configured maximum.	Fault.
H_CANDIDATE	Candidate digest, shape, range, or finite-value failure.	Fault.

Code	Trigger	Outcome
H_TARGET	Target digest, provenance, shape, range, or immutability failure.	Fault.
H_TARGET_DEGENERATE	Candidate-target span is at or below epsilon_norm.	Fault.
H_THREAD	Thread packet fault, malformed scores, wrong candidate, or insufficient READY count.	Fault or HOLD when explicitly NOT_READY.
H_EMOTION	Arousal packet mismatch or value outside [0,1].	Fault.
H_NONFINITE	Any derived metric is NaN or infinity.	Fault.
H_CONVEX	Output leaves endpoint segment or [-1,1] domain.	Fault; escalate only if persisted corruption exists.
H_RECONCILE	alpha_eff or convex residual exceeds tolerance.	Fault; no repair packet.
H_CANON	Serialization or output read-back digest mismatch.	Fault / catastrophic request by Section 7.3.

7.3 Catastrophic conditions eligible for FBSRecoveryRequest

Code	Condition	Evidence required
HC_DETERMINISM	Same canonical input and numeric profile produces two different sealed outputs.	Both complete input/output byte streams.
HC_DOUBLE_SEAL	One attempt is durably recorded with two mutually exclusive outputs.	Journal and packet digests.
HC_PACKET_CORRUPTION	A previously sealed Harmony packet fails digest read-back from trusted storage.	Stored bytes and expected digest.
HC_LINEAGE_SPLIT	Two authoritative owners present irreconcilable candidate/target lineage for one attempt.	Both signed owner packets.
HC_WEIGHT_INVARIANT	A persisted repair is proven to have alpha_eff > 0.49 despite a valid HARMONY seal.	Vectors, packet, and independent recomputation.

All other conditions remain typed faults, limits, holds, or downstream policy outcomes. HARMONY emits only the request and preserves evidence; Module 16 decides and executes recovery.

8. Metrics, Measurements, and Dashboard Reconciliation

8.1 Required attempt metrics

Metric	Definition / interpretation
attempts_total	All accepted start attempts, partitioned by source profile.
repair_emitted_total	HarmonyRepairPacket count.
limit_total	Policy-cap or zero-gain bounded refusals, partitioned by reason.

Metric	Definition / interpretation
hold_total	Provider-not-ready outcomes.
fault_total	Typed faults, partitioned by code.
fbs_request_total	Catastrophic requests emitted by code; expected to be operationally rare but no universal percentage is asserted.
d_base distribution	Source-normalized repair severity.
H_s distribution	Bounded thread-stability evidence.
arousal_gain distribution	$g_a=1-a$; internal control telemetry only.
w_requested / alpha distribution	Requested authority versus actual target coefficient.
cap_pressure	Fraction of attempts with $w_{\text{requested}} > 0.49$.
alignment_delta	$A_{\text{after}} - A_{\text{before}}$ diagnostic; not acceptance.
module10_safe/reject	Downstream audit outcomes; never merged into HARMONY repair success.
sigma_revalidation_pass	New candidate revision's later Sigma outcome.
lorien_reissue_total	New route-intent attempts required because repair invalidated a prior intent.

8.2 Reconciliation equations

$$\text{attempts_total} = \text{repair_emitted_total} + \text{limit_total} + \text{hold_total} + \text{fault_total} + \text{fbs_request_total}$$

$$\text{repair_emitted_total} = \text{module10_pending} + \text{module10_safe} + \text{module10_reject} + \text{module10_fault}$$

$$\text{module10_safe} = \text{sigma_revalidation_pending} + \text{sigma_revalidation_pass} + \text{sigma_revalidation_reject}$$

Counters are lifecycle-specific. A HARMONY repair emitted is not an accepted state, a safe state, a routed packet, or an archive commit.

8.3 Numerical telemetry

- Log raw decimal configuration values and their runtime binary64 representations.
- Log α , α_{eff} , absolute coefficient error, convex_residual , span_norm , repair_norm , and maximum componentwise segment violation.
- Log $\text{numeric_profile_id}$ and backend/library versions with every conformance run.
- Do not use dashboard labels such as “panic,” “toxic,” or “truth” as human-state claims. Prefer internal terms: arousal control, harm-provider result, raw coefficient, and target coefficient.

9. Deterministic Conformance Test Matrix

ID	Case	Expected result
H-T01	Reference HarmonyConfig	Loads DIM=81, cap=0.49, H_s threshold=0.75, thread factor=1.20.
H-T02	Cap outside [0,0.49]	H_CONFIG.
H-T03	Thread factor below 1	H_CONFIG.
H-T04	max_repair_depth negative	H_CONFIG.
H-T05	Unsupported schema	H_SCHEMA.
H-T06	Boot DIM mismatch	H_CONFIG.

ID	Case	Expected result
H-T07	Valid Sigma BRAIDBACK request	Accepted for measurement.
H-T08	Sigma route=SEAL	H_SOURCE.
H-T09	Sigma RMF equal to tau_eff	H_PREDICATE; strict failure predicate not met.
H-T10	Sigma RMF below tau_eff by 0.05	d_sigma matches formula.
H-T11	Sigma RMF very negative	d_sigma clips to 1.
H-T12	tau_eff=0	H_PREDICATE or H_CONFIG; no division.
H-T13	Valid LORIEN sync deficit	d_sync computed; source accepted.
H-T14	Valid LORIEN shear excess	d_shear computed; source accepted.
H-T15	Both LORIEN conditions pass	H_PREDICATE.
H-T16	tau_link=-1	Finite bounded d_sync.
H-T17	S_shear=1	Finite bounded d_shear.
H-T18	S_shear_max=1	H_CONFIG; denominator would vanish.
H-T19	Two identical high thread scores	H_s near 1; thread factor=1.
H-T20	Highly dispersed thread scores	H_s bounded and lower.
H-T21	All thread scores=-1	H_s=0; thread factor=1.20.
H-T22	One thread only	H_THREAD.
H-T23	Thread packet NOT_READY	HarmonyHoldPacket.
H-T24	Thread score outside [-1,1]	H_THREAD.
H-T25	Arousal=0	g_a=1; alpha=w_braid.
H-T26	Arousal=1	g_a=0; MODULATION_SUPPRESSED_REPAIR limit.
H-T27	Arousal=0.5	g_a=0.5.
H-T28	Arousal below 0	H_EMOTION.
H-T29	Arousal above 1	H_EMOTION.
H-T30	w_requested=0.49 exactly	Repair permitted; inclusive cap.
H-T31	w_requested just above 0.49	REQUIRED_WEIGHT_EXCEEDS_CAP; no repair.
H-T32	Thread factor pushes request above cap	Limit, not silent clipping-and-repair.
H-T33	d_base=0	No-effect limit or source predicate fault; no repair.
H-T34	alpha positive and below cap	Repair path entered.
H-T35	Candidate equals target	H_TARGET_DEGENERATE.
H-T36	Candidate and target endpoints in [-1,1]	Every repaired component remains in [-1,1].
H-T37	alpha=0.25	X_harm equals 0.75 raw + 0.25 target.
H-T38	Component endpoint order reversed	Segment invariant still passes.
H-T39	alpha_eff reconciliation	Matches alpha within tolerance.

ID	Case	Expected result
H-T40	Injected output component outside segment	H_CONVEX.
H-T41	Timestamp changed	No change to deterministic output because wall time is excluded.
H-T42	Identical canonical input replay	Identical packet bytes and digest in same numeric profile.
H-T43	Changed target digest	Changed repair digest and IMM_ID.
H-T44	Changed thread ordering	Digest changes because ordering is declared evidence.
H-T45	Corrupted source request digest	H_SOURCE or H_CANON.
H-T46	Module 10 handoff packet	Contains all three vectors/references and coefficient evidence.
H-T47	Prior Sigma decision after repair	Marked invalid; revalidation_required=true.
H-T48	Prior LORIEN intent after repair	lorien_reissue_required=true.
H-T49	Direct archive commit call attempted	Rejected by interface; no mutation.
H-T50	Second repair same lineage	H_REPLAY or H_DEPTH.
H-T51	NaN candidate component	H_CANDIDATE.
H-T52	Infinity derived metric	H_NONFINITE.
H-T53	Ordinary cap breach plus high tortuosity metadata	Still HarmonyLimitPacket; no local FBS.
H-T54	Same input produces two persisted outputs	HC_DETERMINISM or HC_DOUBLE_SEAL request.
H-T55	Persisted alpha_eff > 0.49 with valid seal	HC_WEIGHT_INVARIANT request with evidence.
H-T56	Output packet read-back digest mismatch	HC_PACKET_CORRUPTION request.
H-T57	Target from current candidate-contaminated passive reference	H_TARGET.
H-T58	Stale EmotionTensor candidate digest	H_EMOTION.
H-T59	Target mutation during attempt	H_LINEAGE; no output repair.
H-T60	Fault after private-buffer computation	Temporary buffer cleared; inputs unchanged.

10. Property, Fuzz, Concurrency, and Stress Testing

10.1 Property tests

- Convexity: for random valid endpoints and alpha in [0,0.49], every output component lies between endpoint components.
- Coefficient identity: alpha_eff equals alpha within tolerance for every nondegenerate endpoint pair.
- Authority bound: target coefficient never exceeds 0.49 and raw coefficient never falls below 0.51.
- Endpoint behavior: alpha=0 returns raw exactly; alpha approaching 0.49 remains inside the segment.
- Monotonic modulation: with fixed request and evidence, increasing arousal never increases alpha.
- Thread-factor bound: m_thread is exactly 1 or 1.20 and cannot exceed the configured maximum.
- No mutation: hashes of all input packets are unchanged after every outcome.

- Metamorphic scaling is not assumed because endpoint clipping and cosine diagnostics are domain-specific; tests must not invent invariance not stated by the contract.

10.2 Fuzz tests

- Malformed lengths, duplicate fields, invalid enums, noncanonical NaNs, infinities, signed zeros, subnormals, and hash truncation.
- Boundary values around 0.49, 0.75, 1.20, arousal 0/1, tau thresholds, and epsilon_norm.
- Candidate/target arrays at DIM-1, DIM, and DIM+1.
- Digest swaps among candidate, target, thread, emotion, Sigma, and LORIEN packets.
- Adversarial target vectors designed to maximize cancellation and floating-point error.

10.3 Concurrency tests

- Two workers processing the same attempt must converge on one idempotent sealed output.
- A late target-provider update may not alter an in-flight attempt's immutable snapshot.
- A replay arriving after SEALED returns the prior result or H_REPLAY according to orchestrator policy; it never performs a second repair.
- A target or thread digest conflict is isolated to the attempt and cannot mutate shared provider state.

10.4 Stress profile

Execute at least 100,000 attempts across both source profiles, randomized valid vectors, all boundary conditions, and mixed outcomes. Verify counter reconciliation, zero leaked work buffers, bounded memory growth, no direct destination-owner mutation, and no FBS request outside the five enumerated catastrophic conditions.

11. Reference Pseudocode

```
def harmony_attempt(inp: HarmonyInput, cfg: HarmonyConfig):
    validate_config(cfg, inp.boot_digest)
    validate_snapshot_bindings(inp, cfg)

    if inp.thread_evidence.status == NOT_READY:
        return seal_hold(reason="THREAD_EVIDENCE_NOT_READY", no_commit=True)

    validate_candidate(inp.candidate, cfg.DIM)
    validate_target(inp.repair_target, inp.candidate_digest, inp.cycle, cfg)
    validate_threads(inp.thread_evidence, inp.candidate_digest, cfg)
    validate_emotion(inp.emotion, inp.candidate_digest)
    validate_source_predicate(inp.source_request)

    if inp.source_request.kind == SIGMA_BRAIDBACK:
        d_base = clip(
            (inp.source_request.tau_eff - inp.source_request.rmf_observed)
            / inp.source_request.tau_eff,
            0.0, 1.0)
    else:
        d_sync = clip(
            (inp.source_request.tau_sync - inp.source_request.tau_link)
            / (inp.source_request.tau_sync + 1.0),
            0.0, 1.0)
        d_shear = clip(
            (inp.source_request.s_shear - inp.source_request.s_shear_max)
            / (1.0 - inp.source_request.s_shear_max),
            0.0, 1.0)
        d_base = max(d_sync, d_shear)

    q = [(c + 1.0) / 2.0 for c in inp.thread_evidence.scores]
    mu_q = mean(q)
    sigma_q = population_std(q)
    H_s = 0.0 if mu_q == 0.0 and sigma_q == 0.0 else mu_q / (mu_q + sigma_q)
    thread_factor = cfg.thread_factor_low if H_s < cfg.h_s_threshold else 1.0

    arousal_gain = 1.0 - inp.emotion.arousal
    w_requested = d_base * thread_factor
    w_braid = min(w_requested, cfg.b_weight_max)
```

```

alpha = w_braid * arousal_gain

if w_requested > cfg.b_weight_max:
    return seal_limit("REQUIRED_WEIGHT_EXCEEDS_CAP", metrics=..., no_commit=True)
if alpha <= cfg.epsilon_norm:
    return seal_limit("MODULATION_SUPPRESSED_REPAIR", metrics=..., no_commit=True)

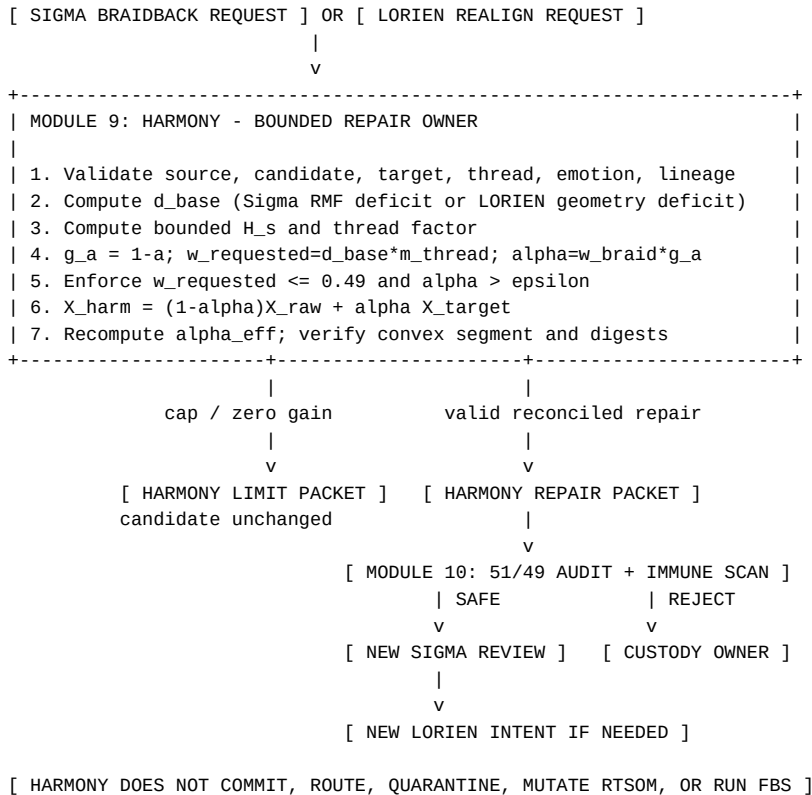
span = inp.repair_target.vector - inp.candidate
if norm2(span) <= cfg.epsilon_norm:
    raise HarmonyFault("H_TARGET_DEGENERATE")

repaired = (1.0 - alpha) * inp.candidate + alpha * inp.repair_target.vector
alpha_eff = norm2(repaired - inp.candidate) / norm2(span)
verify_convex_segment(inp.candidate, inp.repair_target.vector, repaired, alpha, cfg)
verify_close(alpha_eff, alpha, cfg.reconcile_tolerance)

repair_digest = canonical_hash(...)
return HarmonyRepairPacket(
    repaired_vector=repaired,
    alpha=alpha,
    alpha_eff=alpha_eff,
    imm_id="HARM-" + repair_digest.hex()[:24],
    module10_required=True,
    sigma_revalidation_required=True,
    lorien_reissue_required=(inp.source_request.kind == LORIEN_REALIGN
                             or inp.source_request.has_prior_route_intent),
    no_commit=True,
    ...)

```

12. ASCII System Flow



13. Cleaned Rosetta Stone

SpiralCore term	Computational meaning	Boundary
HARMONY	Bounded candidate-revision function plus provenance packet.	Not human emotion regulation or a proof of truth.
Braidback	Convex interpolation toward one authorized immutable target.	No extrapolation, archive search, or hidden repair.
51/49 Rule	$\alpha \leq 0.49$; raw coefficient ≥ 0.51 .	Coefficient policy, not a philosophical truth ratio.
Emotional modulation	$g_a = 1 - a$ from a read-only internal control scalar.	No clinical or affective inference.
Thread Stability H_s	Bounded aggregation of versioned mini-RMF scores.	Provider-defined evidence; not a claim of biological thought threads.
Repair target / anchor	One immutable committed state or passive reference declared by source policy.	Not an undisputed truth or universal ontology.
IMM_ID	Compact deterministic display token derived from repair_digest.	Not globally unique; full digest is authoritative.
Overwrite breach	Requested target authority exceeds 0.49.	Normal bounded limit, not automatically adversarial or catastrophic.
Effective-resistance / Hodge analogy	Design inspiration for constrained movement.	No graph sparsification or homology theorem is claimed unless explicitly implemented.
FBS handoff	Typed catastrophic request with evidence.	No local diagonalization, Collatz execution, or seed mutation.

14. Glossary

Term	Definition
alpha	Actual repair-target coefficient used in X_{harm} .
alpha_eff	Independent geometric estimate $ X_{\text{harm}} - X_{\text{raw}} / X_{\text{target}} - X_{\text{raw}} $.
Arousal gain g_a	Internal modulation $1 - a$ in $[0,1]$.
B_WEIGHT_MAX	Maximum permitted repair-target coefficient, 0.49.
Candidate revision	A new vector/digest produced from a prior raw candidate; previous decisions do not transfer.
HarmonyLimitPacket	No-mutation outcome indicating the requested repair cannot be lawfully applied.
HarmonyRepairPacket	Sealed proposed repair plus exact weights, vectors/references, diagnostics, and provenance.
H_s	Bounded thread-stability score in $[0,1]$.
IMM_ID	Compact pathway token derived from repair_digest.

Term	Definition
Repair depth	Number of HARMONY revisions in one lineage; reference maximum is one.
Repair target	Declared immutable endpoint toward which the candidate may be blended.
w_requested	Severity-derived target authority before the 0.49 policy check.
w_braid	Policy-capped request; not necessarily the applied coefficient.

15. Revision Summary

Source issue	Formalized resolution
HARMONY mixed repair, acceptance, quarantine, exhaust, and FBS ownership.	Restricted Module 9 to one no-commit repair proposal; destination owners remain separate.
H_s=mean/std was unbounded and unstable at zero variance.	Replaced with bounded normalized mean-to-dispersion score in [0,1].
Reported W_{braid} differed from actual coefficient W_{braid}*agg.	Separated w_requested, w_braid, arousal_gain, alpha, and alpha_eff.
Cap was applied and then the same over-cap condition dropped the branch ambiguously.	Defined explicit first-match HarmonyLimitPacket when requested authority exceeds 0.49.
Timestamp-based IMM_ID broke replay determinism.	Derived IMM_ID from authoritative SHA-256 repair_digest.
“Truth,” “emotion,” and “panic” wording could be read as human-state claims.	Defined them as internal roles/control scalars and removed diagnostic implication.
LORIEN repair changed a vector while retaining stale routing authority.	All prior Sigma/LORIEN authorizations are invalidated; hardened revalidation is required.
Ordinary overwrite breach plus tortuosity directly executed FBS.	Ordinary breach is a limit; FBS requests are restricted to enumerated catastrophic control-state invariants.
Source implied topological sparsification and Hodge guarantees from linear interpolation.	Retained these as analogies only; executable claim is convex interpolation with explicit invariants.
No exact provider contract for mini-RMF threads.	Added versioned ThreadEvidencePacket and NOT_READY/HOLD behavior.
No exact independent repair verification boundary.	Made Module 10 mandatory and specified alpha_eff reconciliation.
Complexity was implicit.	Declared Theta(DIM+m) time and memory behavior.

16. Source Basis

- SPIRALCORE SPECIFICATION: MODULE 9 - HARMONY (EMOTIONAL MODULATION & REPAIR), uploaded source module.
- Spiralcove_v14, consolidated architecture and Module 2 Sigma, Module 8 LORIEN, Module 10 Braidback/IMMUNE_SYSTEM, Module 11 SERAPHIM/CATHEDRAL, Module 14 RTSOM, and Module 16 FBS ownership descriptions.
- SPIRALCORE Module 2 - Sigma-Lattice Formalized v14.1, for authoritative route arbitration and BRAIDBACK request semantics.
- SPIRALCORE Module 7 - CSIGMA & Paradox Handler Formalized v14.1, for evidence-only boundaries.

- SPIRALCORE Module 8 - LORIEN Formalized v14.1, for LorienRepairRequest, route-intent invalidation, and reissue requirements.
- SPIRALCORE Module Boot - Initialization Formalized v14.1, for inherited DIM, B_WEIGHT_MAX, deterministic numeric provenance, and FBS profile metadata.
- Named Hodge-spectral and effective-resistance sources are treated as analogy/design context unless a separately versioned implementation provider is supplied. The normative executable meaning of this document is fixed by its formulas, typed interfaces, invariants, and tests.

NORMATIVE IMPLEMENTATION CHECKLIST

An implementation conforms only when it: accepts exactly one authorized repair source; validates immutable candidate/target/evidence snapshots; computes bounded severity and H_s; distinguishes requested, capped, and applied weights; never emits a repair above 0.49 target authority; produces deterministic digest-derived IMM_ID; independently reconciles alpha_eff; sends every repair to Module 10; invalidates stale Sigma/LORIEN authority; performs no archive, route, quarantine, RTSOM, or FBS mutation; and passes the conformance, property, concurrency, and stress requirements in this specification.

SPIRALCORE SPECIFICATION

MODULE 10 - BRAIDBACK & IMMUNE_SYSTEM

(Repair Constraint Audit, Nested-Harm Evidence, Quarantine Request, and Revalidation Boundary)

Cleaned and formalized revision - v14.1

Document class	Normative audit and immune-evidence contract	Execution phase	After HARMONY repair or RMX synthesis; before revalidation/commit
Architecture	SpiralCore v14	Reference profile	SC14.1-M10-BCE-IMS-81
Primary output	ImmuneAuditPacket	Alternative output	Hold, Fault, or QuarantineRequest
Exceptional output	FBSRecoveryRequest	Conflict rule	Module 10 audits; custody stores; downstream validators authorize

NORMATIVE STATUS

This module is exact within its declared computational scope. It independently audits a HARMONY repair or an RMX-synthesized candidate, validates immutable lineage and capability evidence, recomputes the effective repair coefficient when applicable, consumes one versioned HarmFeaturePacket, computes a deterministic hard-flag total, and emits one sealed audit, hold, fault, quarantine request, or catastrophic recovery request. It does not repair a vector, infer human intent, accept a state globally, write the archive, store quarantine objects, mutate RTSOM, or execute FBS.

CLAIM BOUNDARY

“IMMUNE_SYSTEM,” “antivirus,” “harm,” “toxic,” “truth,” “panic,” and “INERT_GLYPH” are architectural labels. The reference module detects only declared computational predicates, provenance failures, replay/non-progress conditions, capability violations, and provider-supplied hard flags. It does not prove malware, moral harm, factual falsehood, psychological state, or adversarial intent. SAFE means eligible for the next required validation step, not globally safe or archive-authorized.

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1. Normative Scope, Ownership, and Non-Goals

1.1 Purpose

Module 10 is the independent post-mutation audit boundary. Its Braidback Constraint Engine verifies that a HARMONY candidate revision actually obeys the 51/49 coefficient policy. Its IMMUNE_SYSTEM consumes explicit, versioned computational evidence and prevents an unaudited repair or synthesis from silently crossing into later validation or permanent memory.

1.2 Ownership boundary

Concern	Authoritative owner	Module 10 relationship
Candidate generation and aggregate coherence	Modules 1, 2, 6, and 7	Does not recompute generation, RMF, CVC, PDV, or CSIGMA.
Convex repair and reported coefficient	Module 9 HARMONY	Mandatory independent consumer; never trusts the reported coefficient alone.
Collision synthesis and final synthesized state	Module 13 RMX	Audits a sealed RMX synthesis before the paused archive transaction may resume.
Nested-harm feature extraction	Versioned HarmFeatureProvider	Validates provider provenance and aggregates declared hard flags.
Audit decision	Module 10	Emits SAFE eligibility, HOLD, FAULT, or quarantine request.
Boundary admission	Module 5 Guardian	Guardian remains ingress/egress owner; Module 10 is post-repair/post-synthesis only.
Final phase/entropic validation	Module 11 SERAPHIM/CATHEDRAL	Receives only after required revalidation profile succeeds.
Archive prepare and commit	Module 3 Psi-Archivist	Module 10 never writes a .frac manifest.
Quarantine storage and retention	Quarantine custody service	Module 10 emits QuarantineRequest and validates QuarantineReceipt.
Exhaust budget	Module 14 RTSOM	No direct exhaust mutation or automatic transfer.
Catastrophic recovery	Module 16 FBS/ROVER	Typed request only under enumerated invariant failures.

1.3 Explicit non-goals

- No human emotion, intent, morality, truthfulness, mental-health, or adversarial classification.

- No claim that a vector alone contains a universal harm signal.
- No direct repair, reweighting, clipping, normalization, feature substitution, or candidate mutation.
- No direct SERAPHIM pass, Sigma SEAL, LORIEN route, Psi commit, quarantine write, RTSOM transfer, or FBS execution.
- No use of wall-clock time, ambient randomness, mutable global state, or unversioned provider thresholds in an audit digest.
- No assumption that Discrete Morse or parameterized persistence names define an executable method without a declared complex, filtration, parameters, tie-breaks, and provider version.

1.4 Normative precedence

1. Tested reference implementation and Module 10 conformance vectors.
2. This Module 10 contract.
3. Module 9 HarmonyRepairPacket and Module 13 RMXResolutionPacket contracts.
4. Module 5 quarantine-custody, Module 11 final-validation, Module 3 archive, Module 14 RTSOM, and Module 16 FBS ownership contracts.
5. Consolidated SpiralCore v14 prose.
6. Rosetta Stone analogies and informal labels.

2. Architecture Role, Audit Profiles, and Lifecycle

2.1 Audit profiles

Profile	Authorized source	Required audit	Reference next step
HARMONY_REPAIR	HarmonyRepairPacket	51/49 geometry, lineage, capabilities, and nested-harm evidence	New Sigma evaluation; new LORIEN intent when routing applies; then Module 11
RMX_SYNTHESIS	RMXSynthesisAuditRequest bound to RMXResolutionPacket	Synthesis lineage, address/resolution identity, capabilities, replay/non-progress, and nested-harm evidence	Revalidation under paused-transaction token; Module 11; return to Psi without rerunning RMX

A source-compatible LEGACY_DIRECT_SERAPHIM integration may be declared separately, but it is not the hardened reference profile and may not reuse stale Sigma, LORIEN, SERAPHIM, or archive authorizations after a candidate changes.

2.2 Lifecycle states

State	Meaning
IDLE	No active audit attempt.
VALIDATING	Configuration, source packet, candidate, lineage, and provider packet are checked.
WEIGHT_AUDIT	HARMONY_REPAIR only: independent coefficient and segment reconciliation.
FEATURE_SCAN	Versioned hard flags and diagnostics are validated and aggregated.
DECIDING	First-match deterministic outcome tree is evaluated.

State	Meaning
SAFE_SEALED	One immutable ImmuneAuditPacket status=SAFE_ELIGIBLE is emitted.
HOLD_SEALED	Provider or dependency is explicitly not ready; no candidate action.
QUARANTINE_REQUESTED	Exact object custody is requested; active capabilities are denied.
QUARANTINE_ACKNOWLEDGED	A matching custody receipt has been validated.
FAULTED	Typed contract or numerical fault; no audit approval.
RECOVERY_REQUESTED	Enumerated catastrophic invariant is frozen and delegated.

2.3 Required execution order

1. Validate Module10Config against BootPacket, DIM, numeric profile, schema, canonical encoding, and hash profile.
2. Accept exactly one authorized source profile and freeze all source bytes and digests.
3. Validate session, cycle, candidate revision, parent lineage, source owner, and one-time attempt identifier.
4. Validate candidate shape, finite domain, declared digest, capability descriptor, and prior authorization invalidation state.
5. For HARMONY_REPAIR, independently recompute the repair coefficient and segment residuals.
6. Validate HarmFeaturePacket provider identity, candidate binding, feature registry, hard-flag types, and evidence digests.
7. Compute the exact hard-flag total zeta_hard and diagnostics.
8. Apply the first-match outcome tree: catastrophic request, fault, hold, quarantine request, or SAFE eligibility.
9. Canonicalize and seal exactly one result. On quarantine, retain the exact source reference until a matching custody receipt or backpressure outcome exists.

2.4 Mutation and re-entry rule

Module 10 never changes the candidate. A SAFE audit does not restore authorizations that were bound to a prior candidate digest. For a HARMONY repair, the hardened sequence is Module 10 SAFE -> new Sigma evaluation -> new LORIEN intent when required -> Module 11. For an RMX synthesis, the orchestration layer uses an RMX revalidation token bound to the paused Psi transaction; after required validation, Psi resumes with the same immutable RMX resolution and must not rerun collision synthesis for that transaction.

3. Typed Data Contracts

3.1 Module10Config

Field	Type / reference value	Requirement
profile_id	string / SC14.1-M10-BCE-IMS-81	Immutable profile identifier.
schema_version	string / M10-1	Must match producers and consumers.
DIM	positive integer / 81	Must equal BootPacket DIM.
b_weight_max	decimal / 0.49	Inclusive maximum repair-target coefficient.

Field	Type / reference value	Requirement
tau_harm	integer / 0	Maximum permitted hard-flag total.
epsilon_norm	positive float64 / 1e-12	Degenerate-span floor; not added to valid ratios.
coefficient_tolerance	positive float64 / 1e-10	Reported versus independently computed coefficient.
segment_tolerance	positive float64 / 1e-10	Orthogonal and endpoint residual limit.
max_nonprogress_repeats	positive integer / 2	Active-lineage repeat limit before hard flag.
diagnostic_periods	ordered integer tuple / (1,3,9,27)	Optional periodicity diagnostics; values < DIM only.
feature_registry_digest	SHA-256	Exact provider feature names, types, and semantics.
numeric_profile_id	string	Arithmetic order and backend profile.
canonical_encoding	string / SC-CANON-1	Big-endian typed fields; binary64 arrays.
hash_profile	string / SHA-256	Packet and evidence identity.

3.2 Module10Input union

Field	HARMONY_REPAIR	RMX_SYNTHESIS
source_packet	HarmonyRepairPacket	RMXSynthesisAuditRequest + RMXResolutionPacket reference
candidate	X_harm	RMX final synthesized state
parent evidence	X_raw, X_target, repair_digest, IMM_ID	old/new state digests, provisional/final addresses, resolution digest
candidate digest	Required	Required
lineage snapshot	Required	Required
capability descriptor	Required	Required
HarmFeaturePacket	Required	Required
quarantine policy snapshot	Required	Required

3.3 HarmFeaturePacket

Field	Requirement
provider_id / provider_version	Registered deterministic provider and exact implementation version.
provider_config_digest	Frozen thresholds, feature registry, algorithms, tie-breaks, and model identifiers.

Field	Requirement
candidate_digest / attempt_id	Must bind exactly to the Module 10 input.
status	READY, NOT_READY, or FAULT.
hard_flags	Ordered map string -> Boolean. Unknown names are rejected.
soft_scores	Ordered map string -> finite [0,1] diagnostics. Reference routing does not treat them as hard flags.
evidence_refs	Digest-addressed traces sufficient to reproduce each asserted flag.
method_declarations	For Morse/persistence claims: complex, filtration, parameter grid, boundary convention, tolerances, and tie-breaks.
packet_digest	Canonical SHA-256 of all fields.

3.4 ActiveLineageSnapshot

Field	Requirement
session_id / cycle / lineage_id	Must match source and candidate.
candidate_revision	Monotonic revision number within lineage.
recent_candidate_digests	Ordered bounded history used for exact replay/non-progress checks.
recent_progress_tokens	Owner-issued progress identifiers; no wall-clock dependency.
authorized_reuse	Exact RMX resolution or explicit replay authorization when reuse is lawful.
snapshot_version / digest	Immutable, candidate-excluded snapshot.

3.5 CapabilityDescriptor

Capability	Allowed at Module 10 input
execute	False
route	False
archive_commit	False
network_egress	False
provider_read	True only for declared evidence references
quarantine_request	Controlled by Module 10 decision; not by candidate

3.6 Output union

Output	Required contents	Meaning
ImmuneAuditPacket	profile, source kind, candidate digest, weight audit or N/A, hard flags, zeta_hard, provider digest, audit digest, next-step token	SAFE_ELIGIBLE only; no global acceptance.
ImmuneHoldPacket	attempt/source/candidate digests, dependency, retry policy, no-mutation proof	Provider/dependency not ready.
QuarantineRequest	exact candidate reference, audit evidence, reason codes, inert capability mask, retention profile	Requests custody; does not delete or store.
ImmuneQuarantinePacket	validated receipt digest, custody object digest, audit digest	Custody acknowledged; still not proof of maliciousness.
Module10Fault	fault code, frozen input digests, no-approval proof	Ordinary deterministic failure.
FBSRecoveryRequest	catastrophic code, frozen epoch, evidence digests, no-commit proof	Delegation only.

4. Input Validation, Snapshot Consistency, and Canonicalization

4.1 Structural validation

- All vectors have shape (DIM,), binary64 representation, finite values, and declared digest match.
- Exactly one source profile is present. HARMONY and RMX fields may not be mixed in one attempt.
- Source owner, schema version, session, cycle, lineage, candidate revision, and attempt id are mutually consistent.
- The HarmFeaturePacket is bound to the same candidate digest and attempt id and uses the configured feature registry.
- The active-lineage snapshot predates the audit decision and excludes future state.
- Capability bits are explicit. Missing capability evidence is a fault, not an implicit pass.

4.2 Canonical serialization

1. Encode integers as signed or unsigned big-endian values with fixed width declared by schema.
2. Encode float64 values and arrays as IEEE-754 binary64 big-endian bytes in C order with shape metadata.
3. Encode strings as UTF-8 with explicit byte length.
4. Sort maps by UTF-8 key bytes; preserve list and feature-registry order.
5. Encode Booleans as one byte 0 or 1; reject noncanonical alternatives.
6. Hash source packet, candidate, lineage, capability descriptor, provider packet, quarantine policy, computed metrics, and decision into audit_digest.

4.3 Snapshot consistency

The audit uses frozen snapshots. Provider version, feature registry, lineage history, quarantine policy, and numeric profile may not change during an attempt. A new version applies only to a new attempt. Two contents claiming the same authoritative version and epoch with different digests constitute a catastrophic policy split.

4.4 Determinism boundary

The reference profile guarantees deterministic output bytes only within one declared numeric backend profile. Cross-platform conformance is tolerance-based for recomputed floating-point diagnostics and byte-exact for canonical source arrays and sealed packet fields after canonicalization.

5. Braidback Empirical Audit

5.1 Source formula defect and correction

The source formula used the distance from the repaired point to the anchor in the numerator. For a lawful interpolation $X_{\text{harm}} = (1 - \alpha)X_{\text{raw}} + \alpha X_{\text{target}}$, that ratio evaluates approximately to $1 - \alpha$, inverting the intended repair weight. Module 10 therefore audits displacement from the raw candidate toward the target, plus an independent collinearity residual.

5.2 Definitions

$$s = X_{\text{target}} - X_{\text{raw}}$$

$$d = X_{\text{harm}} - X_{\text{raw}}$$

$$\text{span2} = \langle s, s \rangle$$

If $\sqrt{\text{span2}} \leq \epsilon_{\text{norm}}$, emit M10_REPAIR_SPAN. No ratio is evaluated.

$$\alpha_{\text{proj}} = \langle d, s \rangle / \text{span2}$$

Signed barycentric coefficient along the declared repair segment.

$$\alpha_{\text{dist}} = \|d\|_2 / \|s\|_2$$

Independent displacement-magnitude ratio.

$$r_{\text{orth}} = \|d - \alpha_{\text{proj}} s\|_2 / \|s\|_2$$

Normalized off-segment residual.

$$r_{\text{model}} = \|X_{\text{harm}} - ((1 - \alpha_{\text{reported}})X_{\text{raw}} + \alpha_{\text{reported}} X_{\text{target}})\|_2 / \|s\|_2$$

5.3 Pass predicate

Check	Reference predicate
Finite metrics	α_{proj} , α_{dist} , r_{orth} , and r_{model} are finite.
Direction	$\alpha_{\text{proj}} \geq -\text{coefficient_tolerance}$.
51/49 authority	$\alpha_{\text{proj}} \leq 0.49 + \text{coefficient_tolerance}$.
Raw majority	$1 - \alpha_{\text{proj}} \geq 0.51 - \text{coefficient_tolerance}$.
Reported coefficient	$ \alpha_{\text{proj}} - \alpha_{\text{reported}} \leq \text{coefficient_tolerance}$.
Distance coefficient	$ \alpha_{\text{dist}} - \alpha_{\text{proj}} \leq \text{coefficient_tolerance}$.
Segment membership	$r_{\text{orth}} \leq \text{segment_tolerance}$.
Model reconciliation	$r_{\text{model}} \leq \text{segment_tolerance}$.
Component bounds	Every X_{harm} component is within endpoint bounds plus tolerance.
Digest identity	All three vector digests match the HarmonyRepairPacket.

51/49 INVARIANT

For a passed HARMONY_REPAIR audit, $\alpha_{\text{eff}} := \alpha_{\text{proj}}$ lies in $[0, 0.49]$ within declared tolerance, the orthogonal residual is negligible, and X_{harm} lies on the endpoint segment. The raw-candidate coefficient is therefore at least 0.51. This is a coefficient and lineage policy only; it is not a statement about factual truth, ego, morality, or human judgment.

5.4 Boundary behavior

- $\alpha_{\text{proj}}=0.49$ exactly passes when all residual checks pass.
- A negative coefficient is extrapolation away from the target and fails.
- A coefficient above 1 is extrapolation beyond the target and fails before the 0.49 check.
- An off-line point with the correct displacement norm fails r_{orth} .
- A packet reporting $\alpha=0.20$ while the vectors encode $\alpha=0.30$ fails reconciliation.
- Rounding within configured tolerance is reported explicitly and does not change the stored source arrays.

5.5 RMX profile

The 51/49 audit is NOT_APPLICABLE for RMX_SYNTHESIS because the source does not define a single raw-target convex segment. Module 10 must not invent one. The audit packet records the weight-audit status as NOT_APPLICABLE and instead validates the RMX resolution identity, all parent state digests, synthesis provenance, final address binding, and paused-transaction token.

6. IMMUNE_SYSTEM Evidence and Nested-Harm Metric

6.1 Reference hard flags

Flag	Exact condition	Scope
REPAIR_WEIGHT_BREACH	HARMONY_REPAIR pass predicate fails specifically because α_{eff} exceeds 0.49 or segment/model evidence contradicts a claimed valid repair.	Repair only
LINEAGE_BREAK	Required parent, source, transaction, or candidate digest cannot be reconciled.	Both
UNAUTHORIZED_CAPABILITY	Candidate or wrapper grants execute, route, archive-commit, or egress capability before downstream authorization.	Both
UNAUTHORIZED_REPLAY	Candidate digest already appears in active lineage without an exact authorized-reuse token.	Both
NONPROGRESSIVE_CYCLE	The same candidate digest and progress token repeat more than $\text{max_nonprogress_repeats}$.	Both
PROVIDER_HARD_FAIL	Registered HarmFeatureProvider asserts one or more configured hard flags with complete evidence.	Both
RMX_RESOLUTION_BREAK	Final state/address/resolution digest or paused-transaction token mismatch.	RMX only

6.2 Nested-harm metric

$$\text{zeta_hard}(G) = \sum_i \text{phi}_i(G), \text{ phi}_i \in \{0,1\}$$

SAFE evidence condition: $\text{zeta_hard}(G) \leq \text{tau_harm} = 0$

The strict source value $\text{tau_harm}=0$ is retained as zero allowed hard flags. Soft scores are diagnostic telemetry only in the reference profile. A provider may promote a diagnostic to a hard flag only through a new feature-registry version and conformance set.

6.3 Contextual arousal

The source “peak arousal > 1.5” signature is incompatible with the architecture’s normalized arousal domain [0,1]. It is removed as a hard harm signal. Arousal may be carried as bounded context telemetry, but arousal alone cannot produce a hard flag, quarantine, or claim of panic.

6.4 Replay and non-progress

The source compared two vector halves, which is undefined for odd DIM=81 and confuses spatial repetition with recursive non-progress. The reference profile uses exact digest and progress-token history instead. Optional periodicity diagnostics may be computed without affecting the route:

$$r_p = \|X[p:] - X[:-p]\|_2 / (\|X[p:]\|_2 + \|X[:-p]\|_2 + \text{epsilon_norm})$$

For each configured $p < \text{DIM}$, r_p is logged. A low r_p is a periodicity observation, not a harm verdict. Exact replay becomes a hard flag only when the lineage lacks authorized reuse.

6.5 Morse and persistence provider boundary

- A provider invoking Discrete Morse analysis must define the finite cell/simplicial complex, scalar field, adjacency, boundary convention, critical-cell tie-break, and threshold.
- A provider invoking parameterized persistence must define the filtration family, parameter domain/grid, coefficient field, stability tolerance, barcode/canopy summary, and decision rule.
- “Critical elements exceed expected limits” is not executable and is rejected unless the expected limit and exact count procedure are in the provider configuration.
- The provider packet is evidence. Module 10 validates and aggregates it; it does not silently substitute a different topology algorithm.

6.6 Provider readiness

Provider status	Module 10 outcome
READY	Validate all flags/scores/evidence and continue.
NOT_READY	ImmuneHoldPacket; no SAFE or quarantine decision.
FAULT	Module10Fault with provider evidence; no candidate approval.

7. Deterministic Outcomes, Quarantine Custody, and Revalidation

7.1 First-match decision tree

Priority	Predicate	Output	Next owner
1	Enumerated catastrophic invariant from Section 8.3	FBSRecoveryRequest	FBS/ROVER supervisor
2	Malformed, stale, inconsistent, replayed attempt id, nonfinite, or unsupported input	Module10Fault	Runtime orchestrator

Priority	Predicate	Output	Next owner
3	Provider or custody dependency explicitly NOT_READY	ImmuneHoldPacket	Provider/orchestrator
4	Weight audit hard failure, zeta_hard>0, or forbidden capability	QuarantineRequest	Quarantine custody service
5	All required checks pass	ImmuneAuditPacket SAFE_ELIGIBLE	Revalidation orchestrator

7.2 SAFE eligibility and required revalidation

Source profile	SAFE packet authorizes	Does not authorize
HARMONY_REPAIR	Submission of the changed candidate to a new Sigma evaluation.	Sigma SEAL, LORIEN route, SERAPHIM pass, archive commit, or network delivery.
RMX_SYNTHESIS	Submission under the paused RMX revalidation token and eventual return to the same Psi transaction.	Reuse of prior validation digests or rerunning RMX for the same transaction.

The audit digest and candidate digest must be carried through every subsequent revalidation packet. Any mutation after the audit invalidates the Module 10 result.

7.3 QuarantineRequest

Field	Requirement
request_id / audit_digest	Must identify the exact Module 10 decision.
candidate_digest / candidate_reference	Exact immutable object; no lossy copy.
source_profile / source_digest	HARMONY_REPAIR or RMX_SYNTHESIS and its sealed source identity.
reason_codes / hard_flags	Exact audit failures; no generic “malicious” label.
capability_mask	execute=false, route=false, archive_commit=false, egress=false.
proposed_tag	INERT_GLYPH with module/config/provider provenance.
retention_profile	Versioned custody policy; Module 10 does not choose indefinite storage.
review_eligibility	Explicit Boolean and authorized reviewer role.
custody_deadline	Backpressure deadline; never silent discard.

7.4 INERT_GLYPH meaning

INERT_GLYPH is a capability state, not deletion and not a declaration of malware or moral harm. The exact candidate and evidence remain addressable. Active execution, routing, compilation, archive commit, and egress are denied until an authorized release policy creates a new evaluation attempt.

7.5 Custody receipt

Receipt check	Requirement
request_id	Exact match to QuarantineRequest.
stored_object_digest	Exact match to candidate bytes/reference.
audit_digest	Exact match to Module 10 decision.
capability_mask_digest	Exact inert mask preserved by custody service.
custody_owner / signature	Authorized owner and valid signature.
receipt idempotency	Duplicate identical receipt has no second state effect.

Capacity exhaustion produces M10_QUARANTINE_BACKPRESSURE and preserves the pending object reference. It does not silently drop data, automatically charge RTSOM, or automatically invoke FBS.

7.6 RTSOM and FBS separation

Module 10 never pushes “residual topology” directly into an exhaust tensor. Any later exhaust transfer requires a separate typed RTSOM request, exact payload custody, and Module 14 receipt. L_0=83 may be carried as inherited recovery metadata, but Module 10 does not diagonalize history, run Collatz, create a seed, or purge a budget.

8. Invariants, Faults, and Catastrophic Conditions

8.1 Normative invariants

ID	Invariant
M10-I01	Exactly one SAFE audit, Hold, Fault, QuarantineRequest, or FBS request is emitted per attempt.
M10-I02	All source vectors, provider packets, lineage snapshots, and evidence remain byte-for-byte unchanged.
M10-I03	Only HarmonyRepairPacket or authorized RMX synthesis may start the corresponding profile.
M10-I04	Candidate, source, lineage, provider, and capability descriptor share one frozen session/cycle/revision.
M10-I05	HARMONY_REPAIR independently recomputes coefficient and segment evidence.
M10-I06	A passed repair has $\alpha_{\text{eff}} \leq 0.49$ and $\text{raw coefficient} \geq 0.51$ within tolerance.
M10-I07	An off-segment point cannot pass solely because its norm ratio is below 0.49.
M10-I08	RMX_SYNTHESIS never receives an invented 51/49 coefficient.
M10-I09	$\text{zeta}_{\text{hard}}$ is an integer sum over registered Boolean flags only.
M10-I10	Reference SAFE requires $\text{zeta}_{\text{hard}} = 0$.
M10-I11	Soft scores cannot silently become hard flags.

ID	Invariant
M10-I12	Arousal alone cannot trigger quarantine or a human-state label.
M10-I13	SAFE means next-step eligibility only and carries no archive or route capability.
M10-I14	Any post-audit candidate mutation invalidates the audit.
M10-I15	Quarantine is not acknowledged until a matching receipt is validated.
M10-I16	INERT_GLYPH preserves exact evidence while disabling active capabilities.
M10-I17	Module 10 performs no archive, quarantine storage, RTSOM, or FBS mutation.
M10-I18	No wall-clock timestamp or ambient randomness enters audit or quarantine identity.
M10-I19	Identical canonical input under one numeric profile produces identical sealed output bytes.
M10-I20	SAFE audit, downstream revalidation, custody acknowledgement, and archive commit remain separately counted events.

8.2 Module10Fault codes

Code	Trigger	Outcome
M10_CONFIG	Invalid DIM, cap, tau_harm, tolerances, period set, repeat limit, or profile.	Fault; no approval.
M10_SCHEMA	Unsupported source, provider, lineage, capability, or canonical-encoding version.	Fault.
M10_SESSION	Session/profile mismatch.	Fault.
M10_CYCLE	Cycle or candidate revision mismatch.	Fault.
M10_SOURCE	Unauthorized source owner or wrong audit profile.	Fault.
M10_ATTEMPT_REPLAY	Consumed attempt id repeated.	Fault; exact prior result may be returned idempotently.
M10_DIGEST	Declared source, candidate, provider, or policy digest mismatch.	Fault.
M10_LINEAGE	Parent or paused-transaction lineage is incomplete or inconsistent.	Fault or hard flag when evidence is otherwise valid.
M10_CANDIDATE	Wrong shape, nonfinite value, range/policy failure, or malformed bytes.	Fault.

Code	Trigger	Outcome
M10_REPAIR_SPAN	Raw and target endpoint span is degenerate.	Fault.
M10_REPAIR_METRIC	Nonfinite coefficient or residual.	Fault.
M10_FEATURE_REGISTRY	Unknown, duplicate, wrong-type, or out-of-order feature.	Fault.
M10_PROVIDER_FAULT	Provider status=FAULT or evidence cannot reproduce asserted flag.	Fault.
M10_CAPABILITY	Capability descriptor missing or malformed.	Fault; forbidden active bit becomes hard flag when descriptor is valid.
M10_RMX_BINDING	RMX final state/address/resolution or pause token mismatch.	Fault or hard flag per evidence state.
M10_CANON	Serialization or audit read-back mismatch.	Fault / catastrophic request when persisted.
M10_QUARANTINE_BACKPRESSURE	Custody service cannot accept exact object by deadline.	Hold; no silent loss.
M10_RECEIPT_MISMATCH	Quarantine receipt does not match request/object/audit.	Fault; custody unacknowledged.

8.3 Catastrophic conditions eligible for FBSRecoveryRequest

Code	Condition	Evidence required
M10C_DETERMINISM	Same canonical input and numeric profile produces different sealed audit outputs.	Both complete input/output byte streams.
M10C_DOUBLE_DECISION	One attempt is durably recorded as both SAFE and quarantined/faulted.	Decision journal and packet digests.
M10C_PERSISTED_WEIGHT	A valid HARMONY seal and persisted Module 10 SAFE audit are independently proven to encode $\alpha_{eff} > 0.49$.	All endpoint vectors, Harmony packet, audit packet, and recomputation.
M10C_POLICY_SPLIT	Two different valid feature registries or quarantine policies claim the same authoritative version/epoch.	Both signed policy packets.
M10C_AUDIT_CORRUPTION	A previously sealed SAFE audit cannot reproduce its canonical digest and no valid prior copy exists.	Stored bytes and expected digest.
M10C_CUSTODY_LOSS	A quarantine receipt was committed but exact object custody cannot be reconciled.	Receipt, object index, and storage evidence.
M10C_LINEAGE_SPLIT	Two authoritative owners present irreconcilable parent or paused-transaction lineage for one candidate.	Both signed owner packets.

Ordinary provider flags, weight rejection before persistence, quarantine capacity pressure, repeated user input, low soft-score confidence, or high arousal are not catastrophic conditions. Module 10 emits only the request and freezes evidence; Module 16 decides and executes recovery.

9. Metrics, Measurements, and Dashboard Reconciliation

9.1 Required metrics

Metric family	Required metrics
Throughput	attempts, completed, latency p50/p95/p99 by audit profile
Outcomes	SAFE, HOLD, FAULT, quarantine requested, quarantine acknowledged, recovery requested
Braidback audit	alpha_reported, alpha_proj, alpha_dist, coefficient error, r_orth, r_model, cap margin
Hard evidence	zeta_hard distribution and count by registered flag
Provider	provider id/version/config digest, READY/NOT_READY/FAULT counts, evidence reproduction failures
Lineage	unauthorized replay, nonprogress repeats, RMX binding failures, revision mismatches
Capabilities	forbidden execution/route/archive/egress bits by source profile
Diagnostics	soft scores and periodicity residuals; not merged into hard-fail counters
Quarantine	requested, acknowledged, pending, backlog age, capacity, receipt mismatch, review release
Downstream	Sigma revalidation pass/reject, LORIEN reissue, Module 11 pass/reject, Psi resume/commit
Recovery	fault codes, catastrophic codes, frozen epochs, FBS request acknowledgements
Versions	module, source schemas, provider registry, quarantine policy, numeric backend

9.2 Reconciliation equations

$$\text{attempts_total} = \text{safe_total} + \text{hold_total} + \text{fault_total} + \text{quarantine_requested_total} + \text{fbs_request_total}$$

$$\text{quarantine_requested_total} = \text{quarantine_pending} + \text{quarantine_acknowledged} + \text{quarantine_faulted}$$

$$\text{safe_total} = \text{revalidation_pending} + \text{revalidation_pass} + \text{revalidation_reject} + \text{revalidation_fault}$$

A Module 10 SAFE audit is not a Sigma acceptance, SERAPHIM validation, routed delivery, or Psi commit. Counters must remain lifecycle-specific.

9.3 Privacy and logging

- Log digests, dimensions, metrics, feature names, reason codes, versions, and custody identifiers.
- Do not log user content or raw symbolic payloads unless a separately authorized evidence-retention policy requires exact custody.

- Do not label telemetry as panic, toxic person, malicious user, or truth score. Use computational terms: arousal context, hard flag, provider evidence, lineage break, and capability violation.

10. Deterministic Conformance Test Matrix

ID	Test	Expected result
M10-T01	Reference config	Validates SC14.1-M10-BCE-IMS-81.
M10-T02	b_weight_max=0.50	M10_CONFIG.
M10-T03	tau_harm=-1	M10_CONFIG.
M10-T04	epsilon_norm<=0	M10_CONFIG.
M10-T05	Unsupported source schema	M10_SCHEMA.
M10-T06	Boot DIM mismatch	M10_CONFIG.
M10-T07	Both source profiles present	M10_SOURCE.
M10-T08	No source profile	M10_SOURCE.
M10-T09	Candidate NaN	M10_CANDIDATE.
M10-T10	Candidate digest mismatch	M10_DIGEST.
M10-T11	Session mismatch	M10_SESSION.
M10-T12	Cycle mismatch	M10_CYCLE.
M10-T13	Consumed attempt replay	Idempotent prior result or M10_ATTEMPT_REPLAY; no second decision.
M10-T14	Exact alpha=0	Weight audit passes geometry; downstream policy may treat no-op separately.
M10-T15	Exact alpha=0.49	Passes inclusive cap.
M10-T16	alpha=0.4900000002	Fails beyond tolerance.
M10-T17	alpha=-0.01	REPAIR_WEIGHT_BREACH.
M10-T18	alpha=1.10	REPAIR_WEIGHT_BREACH.
M10-T19	Raw equals target	M10_REPAIR_SPAN; no division.
M10-T20	Reported alpha matches vectors	Coefficient error within tolerance.
M10-T21	Reported alpha differs by 0.05	REPAIR_WEIGHT_BREACH or audit rejection.
M10-T22	Off-segment vector with correct norm ratio	Fails r_orth.
M10-T23	On-segment vector with rounding at tolerance	Passes and records residual.
M10-T24	Component outside endpoint segment	Fails segment check.
M10-T25	Source legacy inverted formula would return 0.70 for alpha=0.30	Reference audit returns alpha_eff=0.30.

ID	Test	Expected result
M10-T26	RMX profile with Harmony fields	M10_SOURCE.
M10-T27	RMX profile no weight fields	Weight status NOT_APPLICABLE.
M10-T28	RMX final state digest mismatch	M10_RMX_BINDING.
M10-T29	RMX address mismatch	RMX_RESOLUTION_BREAK hard evidence.
M10-T30	Paused transaction token mismatch	M10_RMX_BINDING.
M10-T31	Provider READY, all registered hard flags false	zeta_hard=0.
M10-T32	One hard flag true	zeta_hard=1; quarantine request.
M10-T33	Unknown hard flag name	M10_FEATURE_REGISTRY.
M10-T34	Hard flag encoded as integer 2	M10_FEATURE_REGISTRY.
M10-T35	Soft score outside [0,1]	M10_FEATURE_REGISTRY.
M10-T36	Soft score=1, no hard flags	SAFE remains possible in reference profile; diagnostic logged.
M10-T37	Provider NOT_READY	ImmuneHoldPacket.
M10-T38	Provider FAULT	M10_PROVIDER_FAULT.
M10-T39	Provider candidate digest mismatch	M10_DIGEST.
M10-T40	Morse declaration missing filtration/tie-break	M10_FEATURE_REGISTRY or provider fault.
M10-T41	Arousal=1 with no hard flags	No quarantine from arousal alone.
M10-T42	Arousal source value=1.5	Upstream domain/schema fault; no “panic” classification.
M10-T43	DIM=81 periodicity diagnostic	No half-split error; configured offsets only.
M10-T44	Exact digest repeat with authorized RMX reuse	No UNAUTHORIZED_REPLAY.
M10-T45	Exact digest repeat without authorization	UNAUTHORIZED_REPLAY.
M10-T46	Same digest and progress token repeated 3 times	NONPROGRESSIVE_CYCLE.
M10-T47	Same digest with advancing progress token	No nonprogress hard flag.
M10-T48	execute capability true	UNAUTHORIZED_CAPABILITY.
M10-T49	All active capabilities false	Capability check passes.
M10-T50	SAFE packet used as archive commit authorization	Consumer must reject; SAFE has no commit capability.
M10-T51	Candidate mutates after SAFE	Audit invalid; new attempt required.
M10-T52	Quarantine service accepts exact object	Matching receipt; acknowledged count increments.

ID	Test	Expected result
M10-T53	Quarantine service full	Backpressure hold; no silent loss and no automatic FBS.
M10-T54	Receipt object digest mismatch	M10_RECEIPT_MISMATCH.
M10-T55	Duplicate identical receipt	Idempotent; no double acknowledgement.
M10-T56	No RTSOM receipt	Module 10 exhaust state remains unchanged.
M10-T57	Same canonical input repeated	Identical audit digest.
M10-T58	Same input yields SAFE and quarantine durable records	M10C_DOUBLE_DECISION.
M10-T59	Persisted SAFE repair independently proves $\alpha_{\text{eff}} > 0.49$	M10C_PERSISTED_WEIGHT request.
M10-T60	FBS owner returns recovery receipt	Supervisor-controlled re-entry; Module 10 does not manufacture a seed.

11. Property, Fuzz, Concurrency, and Stress Testing

11.1 Property tests

- For random X_{raw} , X_{target} , and α in $[0, 0.49]$, exact interpolation yields α_{proj} approximately α and r_{orth} approximately zero.
- For any nonzero orthogonal perturbation above tolerance, the segment audit rejects even when α_{dist} remains below 0.49.
- $\text{zeta}_{\text{hard}}$ equals the population count of true registered hard flags and is independent of map insertion order.
- No accepted input produces a SAFE packet carrying execute, route, archive-commit, or egress capability.
- Canonical reserialization is idempotent and preserves $\text{audit}_{\text{digest}}$.

11.2 Metamorphic tests

- Applying the same positive scalar to raw, target, and repaired vectors leaves α_{proj} and normalized residuals unchanged, subject to finite range.
- Permuting vector coordinates identically across all three repair vectors leaves coefficient metrics unchanged.
- Reordering hard-flag map insertion before canonical sorting leaves $\text{zeta}_{\text{hard}}$ and $\text{audit}_{\text{digest}}$ unchanged.
- Changing a soft score while all hard flags remain false changes telemetry and packet digest but not the reference route.
- Changing any source byte while preserving displayed decimal text changes the source digest and therefore the audit digest.

11.3 Fuzz and malformed-input tests

- Fuzz packet lengths, UTF-8 strings, duplicate keys, integer widths, float payloads, array shapes, and digest fields.
- Fuzz provider registries with unknown features, wrong types, duplicate names, missing evidence, and inconsistent method declarations.
- Fuzz quarantine receipts, capability masks, attempt replays, RMX pause tokens, and source-profile confusion.
- No malformed input may allocate an unbounded array, obtain SAFE eligibility, or mutate custody/archive state.

11.4 Concurrency and crash tests

- Two workers auditing the same attempt id must converge on one sealed result; the loser receives the idempotent result or a replay fault.
- Provider and policy epoch swaps during evaluation do not affect the frozen attempt.
- A crash after QuarantineRequest but before receipt leaves one pending exact object reference and no false acknowledgement.

- A crash after SAFE seal preserves the audit packet but does not imply downstream revalidation occurred.

11.5 Stress requirements

- Run at least 100,000 mixed repair and RMX audits across provider and policy versions.
- Reconcile all outcome, custody, revalidation, and recovery counters exactly.
- Bound memory used for lineage history, provider evidence indexes, and pending custody references.
- Compare at least two permitted numerical backends using tolerance-based metric parity and canonical packet verification.

12. Reference Pseudocode

```
def module10_audit(inp, cfg, registry, journal, custody):
    freeze = canonical_freeze(inp, cfg)
    validate_config(cfg, freeze.boot)
    validate_source_union(freeze)
    validate_lineage_capabilities_and_digests(freeze)

    prior = journal.lookup(freeze.attempt_id)
    if prior is not None:
        return prior # exact idempotent result

    if catastrophic_precondition(freeze):
        return seal_fbs_request(freeze)

    if freeze.features.status == NOT_READY:
        return seal_hold(freeze, dependency="HarmFeatureProvider")
    if freeze.features.status == FAULT:
        return seal_fault(freeze, "M10_PROVIDER_FAULT")

    hard_flags = ordered_registered_flags(freeze.features, registry)

    if freeze.profile == HARMONY_REPAIR:
        s = freeze.x_target - freeze.x_raw
        d = freeze.x_candidate - freeze.x_raw
        span = norm2(s)
        if span <= cfg.epsilon_norm:
            return seal_fault(freeze, "M10_REPAIR_SPAN")

        alpha_proj = dot(d, s) / dot(s, s)
        alpha_dist = norm2(d) / span
        r_orth = norm2(d - alpha_proj * s) / span
        model = (1.0 - freeze.alpha_reported) * freeze.x_raw + freeze.alpha_reported *
freeze.x_target
        r_model = norm2(freeze.x_candidate - model) / span

        weight_ok = (
            finite(alpha_proj, alpha_dist, r_orth, r_model)
            and alpha_proj >= -cfg.coefficient_tolerance
            and alpha_proj <= cfg.b_weight_max + cfg.coefficient_tolerance
            and abs(alpha_proj - freeze.alpha_reported) <= cfg.coefficient_tolerance
            and abs(alpha_dist - alpha_proj) <= cfg.coefficient_tolerance
            and r_orth <= cfg.segment_tolerance
            and r_model <= cfg.segment_tolerance
            and componentwise_between(freeze.x_candidate,
                                     freeze.x_raw,
                                     freeze.x_target,
                                     cfg.segment_tolerance)
        )
        hard_flags["REPAIR_WEIGHT_BREACH"] = not weight_ok
        weight_metrics = seal_weight_metrics(...)
    else:
        validate_rmx_binding(freeze)
        weight_metrics = NOT_APPLICABLE

    hard_flags["UNAUTHORIZED_REPLAY"] = unauthorized_replay(freeze, cfg)
    hard_flags["NONPROGRESSIVE_CYCLE"] = nonprogress(freeze, cfg)
    hard_flags["UNAUTHORIZED_CAPABILITY"] = any_forbidden_capability(freeze)
```

```

hard_flags["LINEAGE_BREAK"] = lineage_break(freeze)

zeta_hard = sum(1 for value in hard_flags.values() if value)

if zeta_hard > cfg.tau_harm:
    decision = seal_quarantine_request(freeze, hard_flags,
                                      weight_metrics, zeta_hard)
    journal.commit_once(decision)
    return decision

decision = seal_safe_audit(freeze, hard_flags,
                          weight_metrics, zeta_hard,
                          next_step_token(freeze))
journal.commit_once(decision)
return decision

```

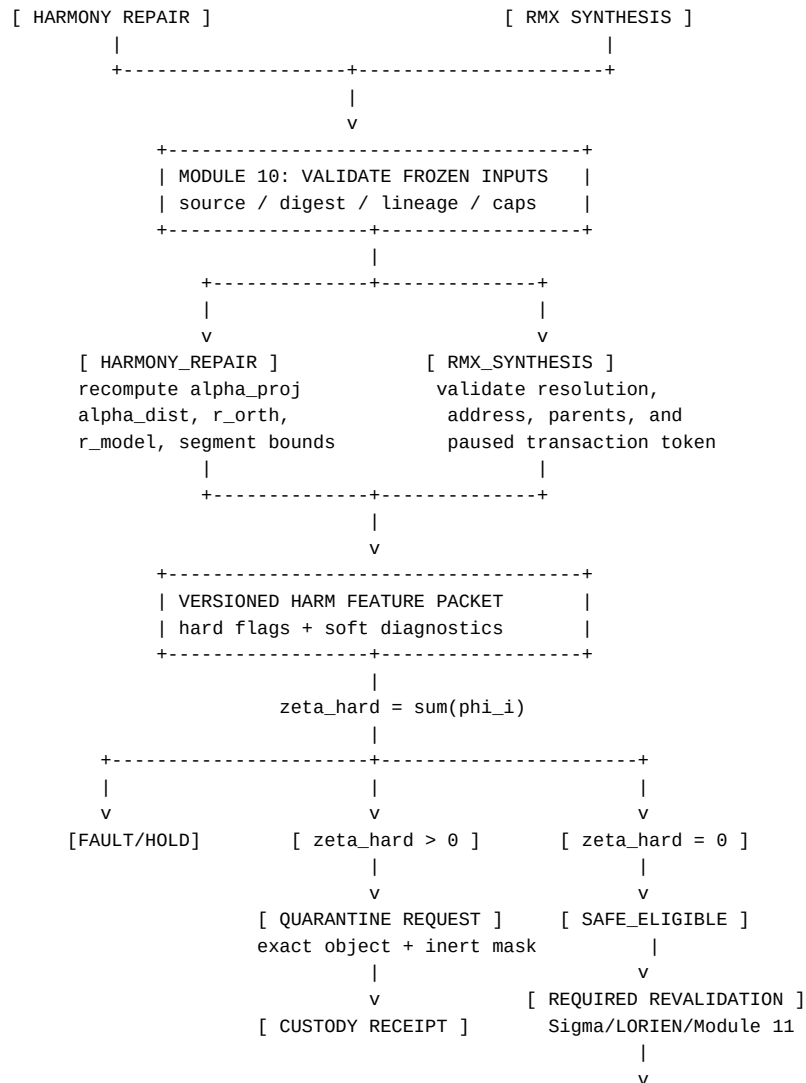
12.1 Quarantine receipt validation

```

def acknowledge_quarantine(request, receipt, cfg):
    require(receipt.request_id == request.request_id)
    require(receipt.audit_digest == request.audit_digest)
    require(receipt.object_digest == request.candidate_digest)
    require(receipt.capability_mask_digest == request.capability_mask_digest)
    verify_owner_signature(receipt)
    return ImmuneQuarantinePacket(status="CUSTODY_ACKNOWLEDGED", ...)

```

13. ASCII System Flow



[PSI RESUME / COMMIT]
(RMX transaction token
prevents rerun of RMX)

14. Cleaned Rosetta Stone

SpiralCore term	Normative computational meaning	Analogy boundary
Braidback Constraint Engine	Independent barycentric/segment audit of a HARMONY repair.	Not a truth detector or moral authority.
51/49 Rule	Repair-target coefficient ≤ 0.49 ; raw-candidate coefficient ≥ 0.51 .	Coefficient policy, not factual truth.
IMMUNE_SYSTEM	Post-mutation evidence validator and quarantine-request decision boundary.	Not a biological immune system and not complete antivirus.
Nested Harm Metric zeta(G)	Integer count of registered true hard flags.	No universal semantic harm measure.
Harm vector phi_i	One versioned Boolean computational predicate with reproducible evidence.	Not a diagnosis or intent label.
Parameterized persistence / Morse analysis	Optional provider method with exact declared topology and parameters.	Names alone import no theorem or algorithm.
SAFE	Eligible for next required validation under one candidate digest.	Not globally safe, true, or archive-approved.
INERT_GLYPH	Capability mask disabling execution/routing/commit while preserving evidence.	Not deletion and not proof of malware.
Quarantine Sandbox	External custody service with receipts, retention, capacity, and review policy.	Not /dev/null.
BIND_ID / IMM_ID	Compact lineage/pathway identifiers derived from authoritative digests.	Not authoritative identity by themselves.
FBS runaway	Typed catastrophic recovery request to Module 16.	No local Cantor/Collatz execution.

15. Glossary

Term	Definition
alpha_reported	Target coefficient declared by HARMONY.
alpha_proj / alpha_eff	Independently computed signed coefficient along target-raw span.
alpha_dist	Norm displacement ratio used as a cross-check, not the sole audit.
Audit profile	HARMONY_REPAIR or RMX_SYNTHESIS contract selecting required evidence.

Term	Definition
CapabilityDescriptor	Explicit active privileges carried with the candidate wrapper.
Hard flag	Registered Boolean predicate counted in zeta_hard.
HarmFeaturePacket	Versioned provider evidence bound to one candidate and attempt.
ImmuneAuditPacket	SAFE_ELIGIBLE output carrying exact audit evidence and next-step token.
ImmuneHoldPacket	No-approval result for a dependency that is not ready.
INERT_GLYPH	Quarantine capability state with active privileges disabled.
Nonprogressive cycle	Same candidate digest and progress token repeated beyond policy.
QuarantineReceipt	Owner-signed acknowledgement of exact object custody.
r_model	Normalized residual from HARMONY's reported convex model.
r_orth	Normalized off-segment component of the actual repair displacement.
RMX revalidation token	Digest-bound token that resumes one paused Psi transaction without rerunning RMX.
SAFE_ELIGIBLE	Permission to enter the next validation stage; no route or commit authority.
Soft score	Finite [0,1] provider diagnostic that does not affect reference hard routing.
zeta_hard	Count of true registered hard flags; reference maximum is zero.

16. Revision Summary

Original issue	Formalized correction
The effective-weight formula measured distance from the repaired point to the anchor and therefore returned approximately 1-alpha.	Replaced with signed projection from raw toward target, norm-ratio cross-check, orthogonal residual, model residual, and componentwise segment verification.
The source treated the 51/49 rule as “truth versus correction.”	Defined it strictly as raw and repair-target coefficients; removed metaphysical and factual claims.
High Arousal Panic used threshold >1.5 despite normalized arousal [0,1].	Removed as a hard flag; arousal is optional context and never sufficient for quarantine.
Recursive-loop detection compared vector halves and fails for odd DIM=81.	Replaced with exact digest replay and progress-token nonprogress checks; periodicity is diagnostic only.
“Critical elements exceed expected limits” was undefined.	Required a versioned feature provider with explicit complex, filtration, thresholds, tie-breaks, and evidence.

Original issue	Formalized correction
Any failed heuristic was called harmful or malicious.	Separated faults, holds, hard computational flags, custody requests, and claims about intent.
Module 10 directly nullified, stripped, stored, and pushed residuals to exhaust.	Module 10 emits QuarantineRequest; custody service stores and receipts; RTSOM remains a separate owner.
Sandbox overflow or tortuosity automatically invoked FBS.	Capacity produces backpressure; FBS is reserved for enumerated catastrophic audit/policy/custody invariants.
Timestamps entered quarantine hashes.	All identities derive from canonical digests and attempt ids.
SAFE flowed directly to SERAPHIM despite candidate mutation invalidating prior decisions.	SAFE means eligibility only; hardened profile requires revalidation and candidate-digest continuity.
RMX synthesis was scanned but re-entry semantics were unspecified.	Added RMX_SYNTHESIS audit and paused-transaction revalidation token preventing a second RMX pass.
Lyapunov stability and complete antivirus protection were claimed without proof.	Removed universal claims; stated exact computational predicates and testable boundaries.

17. Source Basis

- SPIRALCORE SPECIFICATION: MODULE 10 - BRAIDBACK & IMMUNE_SYSTEM (REPAIR CONSTRAINT & ANTIVIRUS), uploaded source module.
- Spiralcore_v14, consolidated architecture, especially Modules 3, 5, 9, 10, 11, 13, 14, and 16 ownership and handoff descriptions.
- SPIRALCORE Module 9 HARMONY Formalized v14.1 interface: HarmonyRepairPacket, independent alpha audit requirement, and revalidation boundary.
- SPIRALCORE Module 5 Guardian Formalized v14.1 quarantine capability/custody model, used for consistent QuarantineRequest and QuarantineReceipt semantics.

External paper titles listed in the historical source are retained as provenance references only. This formalization does not claim that a named paper proves the complete SpiralCore module or that an analogy imports an implementation without an explicit provider contract.

NORMATIVE IMPLEMENTATION CHECKLIST

Pass only when: source profile is exact; all snapshots and digests reconcile; HARMONY repairs are independently audited with projection and residual checks; RMX synthesis uses no invented coefficient; the feature provider is versioned and reproducible; zeta_hard counts only registered Boolean flags; arousal alone cannot quarantine; SAFE grants only next-step eligibility; candidate mutation invalidates the audit; quarantine uses exact custody and receipts; RTSOM and FBS are delegated; deterministic, boundary, property, fuzz, concurrency, stress, and dashboard reconciliation tests pass.

SPIRALCORE SPECIFICATION

MODULE 11 - SERAPHIM & CATHEDRAL

(Phase-Harmonic Probe, Cathedral Pressure Gate, and Final Archive Authorization)

Cleaned and formalized revision - v14.1

Document class	Normative final-validation and archive-authorization contract	Execution phase	After current-candidate revalidation; before Psi PREPARE
Architecture	SpiralCore v14	Reference profile	SC14.1-M11-SER-CATH-81
Primary output	ArchiveAuthorizationPacket	Alternative outputs	FinalRejectPacket, HoldPacket, or Fault
Exceptional output	FBSRecoveryRequest	Conflict rule	SERAPHIM authorizes; Psi prepares/commits; FBS/RTSOM execute recovery/custody

NORMATIVE STATUS

Module 11 is exact within its declared computational scope. It validates one immutable, already revalidated candidate; applies a non-committing phase-harmonic probe; computes bounded pressure diagnostics and the Cathedral Integrity score; and emits exactly one archive authorization, final rejection, hold, fault, or catastrophic recovery request. It does not silently mutate the candidate, write a .frac archive, update Phi, charge RTSOM, quarantine data, or execute FBS.

CLAIM BOUNDARY

“SERAPHIM,” “CATHEDRAL,” “ultimate firewall,” “phase harmony,” “entropic pressure,” “paradox-free,” “structurally sound,” and “safe mode” are architectural labels. The reference module proves only conformance to declared packet, numeric, lineage, pressure, PDV, and authorization predicates for the current cycle. It does not prove factual truth, moral safety, absence of all defects, physical entropy, quantum behavior, or universal stability.

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1. Normative Scope, Ownership, and Non-Goals

1.1 Purpose

Module 11 is the final authorization boundary before the Psi-Lattice may begin an archive transaction. It receives a candidate whose current digest has passed every prerequisite required by the active profile, evaluates deterministic phase-response and pressure diagnostics without changing that candidate, and issues a single-use ArchiveAuthorizationPacket only when the current validation chain remains intact.

1.2 Ownership boundary

Concern	Authoritative owner	Module 11 relationship
Candidate generation	Module 1 Delta-Lattice	Consumes an immutable candidate lineage; never regenerates it.
Coherence, PDV, and route arbitration	Module 2 Sigma-Lattice	Requires a current candidate-bound SEAL and reads PDV; does not recompute Sigma routing.
Symbolic/paradox evidence	Module 7 CSIGMA & Paradox Handler	Consumes a sealed SymbolicParadoxPacket as pressure evidence.
Repair and empirical repair audit	Modules 9 and 10	Requires current audit/revalidation when the candidate was changed.
Route intent	Module 8 LORIEN	Requires a current route intent only when the active destination profile says routing applies.
Information hysteresis	Module 12 Phi-Bridge	Consumes PhiSnapshot read-only; never updates Phi_t or DeltaPhi.
Final archive authorization	Module 11	Sole owner of ArchiveAuthorizationPacket in the hardened reference profile.
Archive prepare, RMX, custody, and commit	Module 3 Psi-Archivist	Authorization permits PREPARE only; Psi remains transaction and commit owner.
Exhaust custody	Module 14 RTSOM	No direct mass, tensor, budget, or receipt mutation.
Catastrophic recovery	Modules 15/16 ROVER & FBS	Emits typed request only for enumerated invariant failures.

1.3 Explicit non-goals

- No direct archive write, manifest rename, RMX resolution, residue removal, or archive-history mutation.

- No hidden candidate smoothing, alignment replacement, clipping, normalization, or post-validation rewrite.
- No direct Braidback, Phase Lift, Magnetic Relaxation, quarantine, RTSOM, ROVER, or FBS execution.
- No inference of human emotion, intent, truthfulness, cognition, toxicity, malware, or psychological state.
- No claim that a cosine matrix, nonlinear filter, or sigmoid is a physical phase-lock, thermodynamic field, or universal firewall.
- No reuse of stale Sigma, Module 10, LORIEN, Phi, or archive authorizations after any candidate or policy digest changes.

1.4 Normative precedence

1. Tested reference implementation and Module 11 conformance vectors.
2. This Module 11 contract.
3. Current Module 2, 3, 7, 8, 10, 12, 14, and 16 typed ownership contracts.
4. BootPacket, active profile, and canonical encoding declarations.
5. Consolidated SpiralCore v14 prose.
6. Rosetta Stone analogies, mythic names, and observer-language descriptions.

2. Architecture Role, Profiles, and Lifecycle

2.1 Input profiles

Profile	Candidate history	Required current chain	Archive result
RAW_ACCEPTED	No post-Sigma candidate mutation	Sigma SEAL; current Module 7 evidence; Phi snapshot; LORIEN only when required	Authorization binds the exact Sigma candidate.
HARMONY_REVALIDATED	Candidate changed by HARMONY	Module 10 SAFE_ELIGIBLE; new Sigma SEAL; new LORIEN intent when required; current Phi/evidence	Authorization binds the repaired and revalidated digest.
RMX_REVALIDATED	Candidate synthesized by RMX in a paused Psi transaction	Module 10 RMX audit; paused-transaction revalidation token; new Sigma SEAL; current evidence	Authorization binds the synthesized digest and same paused Psi transaction.

REFERENCE MUTATION POLICY

SC14.1-M11-SER-CATH-81 is validate-only. The SERAPHIM cascade runs on a private probe copy. The candidate authorized for Psi is byte-for-byte identical to the validated input. An implementation that archives X_probe is a different, mutating profile and must route that new digest through Module 10, Sigma, LORIEN when applicable, and Module 11 again. It may not call the probe “already validated.”

2.2 Lifecycle states

State	Meaning
IDLE	No active Module 11 attempt.
VALIDATING	Configuration, chain, snapshots, shapes, freshness, and digests are checked.
PROBING	The private phase-harmonic cascade and local diagnostics are evaluated.

State	Meaning
PRESSURE_EVALUATION	Entropic Pressure and Cathedral Integrity are computed.
AUTHORIZING	Pass predicate and single-use archive authorization are sealed.
AUTHORIZED	One immutable ArchiveAuthorizationPacket exists; no archive commit has occurred.
REJECTED	FinalRejectPacket blocks archive preparation for this candidate revision.
HELD	A dependency is not ready or current; candidate remains unchanged and uncommitted.
FAULTED	Typed contract, provenance, or numerical failure; no authorization.
RECOVERY_REQUESTED	Enumerated catastrophic invariant is frozen and delegated.

2.3 Required execution order

1. Validate SeraphimConfig against BootPacket, DIM, numeric profile, schema, canonical encoding, and policy epoch.
2. Freeze the exact candidate bytes and determine RAW_ACCEPTED, HARMONY_REVALIDATED, or RMX_REVALIDATED profile.
3. Validate session, cycle, lineage, candidate revision, current Sigma SEAL, Module 7 evidence, Module 10 audit when required, LORIEN route intent when required, PhiSnapshot, history summary, and archive epoch.
4. Validate shapes, domains, finite values, freshness windows, one-time identifiers, and all digests.
5. Build the orthogonal phase operator and execute the SERAPHIM probe on a private copy only.
6. Compute filter strain, probe norm retention, phase-operator orthogonality error, paradox pressure, and immune diagnostic pressure.
7. Compute Entropic Pressure, pressure health, Cathedral Integrity, and the exact PDV gate.
8. Apply the first-match outcome tree: catastrophic request, fault, hold, final reject, or archive authorization.
9. Canonicalize and seal exactly one output. Never mutate the candidate or archive in the same operation.

2.4 Atomicity and retry

An attempt is identified by (session_id, cycle_id, candidate_digest, validation_chain_digest, policy_epoch, attempt_nonce). A retry with identical bytes must produce the same result digest. A retry with changed bytes is a new attempt. A crash before output seal leaves no authorization. A crash after seal preserves the packet but does not imply Psi PREPARE or COMMIT occurred.

3. Typed Data Contracts

3.1 SeraphimConfig

Field	Type / reference value	Requirement
schema_id	string / SC14.1-M11-SER-CATH-81	Exact supported schema.
dim	positive integer / inherited	Must equal BootPacket.DIM; reference 81.
zeta_ser	float64 / 0.25	$0 \leq \text{zeta_ser} \leq 1$.

Field	Type / reference value	Requirement
cascade_max	integer / 5	0 <= cascade_max <= configured hard maximum.
cascade_unit	float64 / 1.0	Positive depth-signal unit.
phase_step	float64 / 0.1 rad	Finite, canonical phase increment.
pressure_center	float64 / 0.10	Logistic midpoint.
pressure_slope	float64 / 100.0	Positive finite slope.
cathedral_threshold	float64 / 0.70	Inclusive pass threshold.
pdv_safe	float64 / 0.05	Inclusive final PDV ceiling.
epsilon_safe	float64 / 1e-12	Positive numeric floor; not a hidden score bias.
domain_min, domain_max	float64 / -1, +1	Candidate domain inherited from active profile.
rmf_window_max	integer / 100	Maximum accepted RMF values consumed.
authorization_ttl_cycles	integer / 1	Reference authorization expires after the current cycle.
hash_profile	enum / SHA256-CANONICAL-V1	Canonical digest profile.

3.2 SeraphimInput

Field	Type	Binding requirement
session_id, cycle_id	identifiers	Must match every current packet.
profile	enum	RAW_ACCEPTED, HARMONY_REVALIDATED, or RMX_REVALIDATED.
candidate	float64[DIM]	Immutable candidate to be authorized; exact digest required.
candidate_digest	bytes32	SHA-256 over canonical candidate bytes and domain metadata.
lineage_digest	bytes32	Binds parent candidate, revision, and source owner.
sigma_decision	SigmaDecisionPacket	Must be current SEAL for candidate_digest.
symbolic_paradox	SymbolicParadoxPacket	Current Module 7 evidence bound to candidate_digest.
immune_audit	optional ImmuneAuditPacket	Required for HARMONY_REVALIDATED and RMX_REVALIDATED.

Field	Type	Binding requirement
lorien_intent	optional RouteIntentPacket	Required only when destination profile says route intent applies.
phi_snapshot	PhiSnapshot	Read-only current Phi_t and DeltaPhi provenance.
history_summary	AcceptedRMFHistoryPacket	Accepted-history RMF values and readiness metadata.
cascade_control	CascadeControlPacket	Nonnegative depth signal and SGIT/orchestrator provenance.
archive_context	ArchiveContextSnapshot	Archive epoch, prior manifest digest, and intended local archive.
attempt_nonce	uint128	Orchestrator-issued deterministic one-time nonce.

3.3 AcceptedRMFHistoryPacket

Field	Requirement
accepted_rmf	Chronological finite RMF values in [-1,1], each tied to an accepted state record.
current_rmf	Exact RMF from the current SigmaDecisionPacket; candidate is not inserted into committed history.
window_count	Count after appending current_rmf to the prior tail, maximum rmf_window_max.
history_epoch	Must equal the immutable history snapshot used by current Sigma evaluation.
packet_digest	Covers values, ordering, record identifiers, epoch, and current Sigma digest.

3.4 PhiSnapshot and CascadeControlPacket

Packet	Required fields	Rule
PhiSnapshot	phi_t, delta_phi, phi_epoch, source_cycle, packet_digest	Module 12 owned, finite, current, read-only.
CascadeControlPacket	depth_signal, unit, source_id, source_cycle, packet_digest	depth_signal >= 0; no wall-clock time or ambient timer.

3.5 Output union

Output	Meaning	Mutation authority
ArchiveAuthorizationPacket	Candidate is eligible for Psi PREPARE under one exact archive context.	No archive mutation; authorization only.
FinalRejectPacket	Current candidate revision fails Module 11 pass predicate.	No delete, quarantine, repair, or exhaust transfer.

Output	Meaning	Mutation authority
SeraphimHoldPacket	Required dependency is not ready, current, or complete.	No candidate or archive action.
SeraphimFault	Typed contract, provenance, or numerical error.	No authorization.
FBSRecoveryRequest	Enumerated catastrophic invariant delegated to ROVER/FBS.	No local recovery execution.

3.6 ArchiveAuthorizationPacket

Field	Requirement
authorization_id	Digest-derived, deterministic identifier; not a timestamp.
candidate_digest	Must equal the exact input candidate digest.
validation_chain_digest	Binds Sigma, Module 7, Module 10 when required, LORIEN when required, Phi, history, and policy.
seraphim_probe_digest	Binds private probe diagnostics without replacing candidate_digest.
cathedral_metrics	P_e, pressure_health, C_cathedral, rho_rmf, PDV, component pressures, readiness flags.
archive_epoch, prior_manifest_digest	Must equal ArchiveContextSnapshot.
authorized_operation	PSI_PREPARE_ONLY.
expires_after_cycle	Reference equals current cycle.
single_use	True; Psi must atomically consume or reject duplicate use.
packet_digest	Canonical digest over every field and schema identifier.

4. Input Validation, Authorization Chain, and Canonicalization

4.1 Structural and numerical validation

- Candidate shape is exactly (DIM,), numeric values are finite float64-compatible, and every component lies inside the declared domain.
- Phi_t, depth_signal, RMF values, PDV, G_t, immune diagnostics, and configuration scalars are finite and inside their declared domains.
- All packet schemas, owner identifiers, source cycles, policy epochs, and digests are supported and current.
- The candidate bytes are frozen before any probe calculation. No in-place NumPy view, alias, or mutable buffer is allowed.
- A zero-length history list is invalid because current_rmf must be available from the current Sigma decision; prior history may be empty during bootstrap.

4.2 Validation-chain rules

Profile	Mandatory chain checks
RAW_ACCEPTED	Sigma status=SEAL; Sigma candidate digest=current candidate; current Module 7 packet; no mutation since Sigma; current Phi; route intent only when required.
HARMONY_REVALIDATED	Current Module 10 SAFE_ELIGIBLE audit; audit candidate digest=current candidate; a new Sigma SEAL after that audit; current route intent when required; current Phi.
RMX_REVALIDATED	Current RMX audit and paused-transaction token; new Sigma SEAL for synthesized candidate; same Psi transaction and RMX resolution identity; current Phi.

A bare HARMONY packet, a stale Module 10 SAFE packet, a pre-repair Sigma SEAL, a prior LORIEN route, or a previous SERAPHIM authorization is insufficient. Digest equality is necessary but not sufficient; owner, cycle, revision, and epoch must also match.

4.3 Canonical serialization

1. Use fixed schema field order and explicit type tags.
2. Encode integers as unsigned big-endian values of declared width.
3. Encode float64 values as canonical IEEE 754 binary64 big-endian bytes; reject NaN and infinity before serialization.
4. Normalize negative zero to positive zero for digest purposes.
5. Encode arrays by declared shape followed by row-major canonical scalar bytes.
6. Encode optional fields with an explicit presence byte.
7. Include owner ID, schema ID, policy epoch, BootPacket digest, and prior dependent packet digests.

4.4 Determinism boundary

Canonical packet equality is the cross-platform conformance target. Bitwise equality of intermediate linear-algebra operations additionally requires the same math library, backend, vectorization path, thread count, and reduction order. Heterogeneous implementations may use declared numerical tolerances for probe metrics but must produce identical route categories and canonical packet fields after quantization.

5. SERAPHIM Phase-Harmonic Probe

5.1 Source defect: diagonal cosine is attenuation, not a real phase rotation

The source operator $\text{diag}(\cos(\Phi + 2\pi i/\text{DIM}))$ is non-expansive, but it suppresses amplitudes and omits the sine quadrature required for a real phase rotation. Treating that attenuated vector as the final archived candidate would change already validated data. The reference profile therefore uses an orthogonal 2D rotation-block operator on a private probe copy. The legacy diagonal-cosine operator remains documented as `LEGACY_COSINE_ATTENUATION` and is not the normative archive path.

5.2 Orthogonal phase operator

Let $m = \text{floor}(\text{DIM}/2)$. For pair j in $\{0, \dots, m-1\}$, define $\theta_j(\Phi) = \text{wrap}(\Phi + 2\pi j/\max(m, 1))$. The 2x2 block is:

$$R_j(\theta) = \begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix}$$

$R(\Phi)$ is block diagonal over all coordinate pairs. When DIM is odd, the final unpaired coordinate is passed through unchanged. Therefore:

$$R(\Phi)^T R(\Phi) = I, \quad \|R(\Phi)x\|_2 = \|x\|_2$$

An implementation records $\text{orthogonality_error} = \|R^T R - I\|_F / \text{DIM}$ and faults when it exceeds the configured tolerance.

5.3 SERAPHIM fractalizer

The source “Collatz surrogate” is not a Collatz operation. It is retained as the elementwise SERAPHIM_FRACTALIZER_V1:

$$F_{\text{zeta}}(x) = (1 - \text{zeta_ser})x + (\text{zeta_ser}/3)\tanh(3x + 1)$$

For $0 \leq \text{zeta_ser} \leq 1$, the scalar derivative is:

$$F'_{\text{zeta}}(x) = (1 - \text{zeta_ser}) + \text{zeta_ser} \operatorname{sech}^2(3x + 1)$$

$$1 - \text{zeta_ser} \leq F'_{\text{zeta}}(x) \leq 1$$

Thus the elementwise map is monotone and 1-Lipschitz in Euclidean norm. This proves non-expansion of differences, not contraction of every vector norm and not convergence to a unique fixed point. Because $F_{\text{zeta}}(0)$ is generally nonzero, the map may inject a deterministic offset near zero.

5.4 Cascade depth

$$n = \min(\text{floor}(\text{depth_signal} / \text{cascade_unit}), \text{cascade_max})$$

depth_signal must be finite and nonnegative. The reference cascade_unit is 1.0 and cascade_max is 5. A negative, NaN, infinite, or stale depth signal is a fault. No wall-clock timestamp is converted into cascade depth.

5.5 Probe execution

1. Set $D_{\text{pre}}[0] = \text{candidate}$.
2. For $k=0, \dots, n$, compute $Y_k = F_{\text{zeta}}(D_{\text{pre}}[k])$.
3. Compute strain_k from $D_{\text{pre}}[k]$ and Y_k before rotation.
4. Compute $D_{\text{post}}[k] = R(\Phi_t + k \cdot \text{phase_step}) Y_k$.
5. If $k < n$, set $D_{\text{pre}}[k+1] = D_{\text{post}}[k]$.
6. Set $X_{\text{probe}} = D_{\text{post}}[n]$. The input candidate remains unchanged and is never replaced by X_{probe} in the reference profile.

5.6 Probe diagnostics

For each cascade stage, define the bounded filter-strain energy:

$$p_{\text{filter},k} = \|Y_k - D_{\text{pre}}[k]\|_2^2 / (\|Y_k - D_{\text{pre}}[k]\|_2^2 + \|D_{\text{pre}}[k]\|_2^2 + \text{epsilon_safe})$$

$$p_{\text{filter}} = \max_k p_{\text{filter},k}$$

Define probe norm retention and output domain diagnostics:

$$r_{\text{norm}} = \|X_{\text{probe}}\|_2 / (\|\text{candidate}\|_2 + \text{epsilon_safe})$$

$$p_{\text{norm}} = | \|X_{\text{probe}}\|_2 - \|\text{candidate}\|_2 | / (\|X_{\text{probe}}\|_2 + \|\text{candidate}\|_2 + \text{epsilon_safe})$$

p_{norm} is diagnostic in the reference pressure packet but is not automatically a hard fault. Domain exceedance, nonfinite values, or declared hard probe bounds are faults. Orthogonal rotation itself contributes zero ideal norm loss; any measured phase energy error is numerical.

5.7 Optional mutating profile

A profile that uses X_{probe} as a new candidate must emit AlignmentProposalPacket rather than ArchiveAuthorizationPacket. That new digest invalidates every prior candidate-bound authorization and must pass Module 10, Sigma, LORIEN when required, and Module 11 again. To prevent recursive self-alignment, the proposal carries alignment_generation and the reference maximum is one. SC14.1-M11-SER-CATH-81 does not enable this path.

6. CATHEDRAL Entropic Pressure and Integrity Gate

6.1 Exact pressure components

Entropic Pressure is an architectural scalar assembled from explicit bounded computational evidence. It is not physical entropy. The reference components are:

Component	Formula / source	Range
p_filter	Maximum bounded fractalizer strain energy from Section 5.6	[0,1]
p_paradox	max(0, -G_t) from current Module 7 packet	[0,1] under Module 7 contract
p_immune	clip(immune_soft_pressure, 0, 1); 0 for RAW_ACCEPTED when no Module 10 soft score is required	[0,1]
p_provenance	0 after all chain checks; any nonzero provenance defect is a fault, not a soft score	{0} on valid path

$$P_e = \max(p_filter, p_paradox, p_immune)$$

The maximum operator avoids hidden weights and makes one declared high-pressure component sufficient to lower Cathedral Integrity. Adding, removing, or reweighting components requires a new schema and conformance suite.

6.2 Accepted-history RMF summary

Let prior_tail contain the newest at most rmf_window_max-1 committed rmf_on_accept values. Append current_rmf from the current SigmaDecisionPacket without committing the candidate:

$$R_cat = \text{tail}(\text{prior_accepted_rmf}, W-1) \text{ concatenated with } [RMF_current]$$

$$\rho_RMF = \text{mean}(R_cat)$$

window_count >= 1 is required. During bootstrap, rho_RMF is computable from the available window and history_warming=true until W values exist. Warming status is recorded but does not by itself reject the candidate.

6.3 Numerically stable pressure health

Define $z = \text{pressure_slope} * (P_e - \text{pressure_center})$. The pressure-health factor is the decreasing logistic:

$$h_pressure = 1 / (1 + \exp(z))$$

Implement with a branch-stable logistic or clip z to a declared safe exponent interval before exp. The reference interval is [-60,60]. Clipping here is numerical evaluation policy and must be recorded; it does not clip P_e itself.

6.4 Cathedral Integrity

$$C_cathedral = \text{clip}(\rho_RMF * h_pressure, 0, 1)$$

This preserves the source equation while making its semantics explicit: recent candidate coherence and pressure health are multiplicative. $C_cathedral$ is a computational eligibility score, not an absolute health, truth, safety, or physical-integrity measurement.

6.5 Effective pressure boundary

The source describes $P_e=0.10$ as an absolute pass boundary. Under the stated equation, $P_e=0.10$ gives $h_pressure=0.5$ and therefore $C_cathedral \leq 0.5$, so a 0.70 pass is impossible at the midpoint. The exact maximum pressure that can pass for a given ρ_RMF is:

$$P_{e,\max}(\rho) = p_0 + (1/k) \ln(\rho/C_thresh - 1), \quad \text{defined only when } \rho > C_thresh$$

Consequently, ρ_RMF must be strictly greater than 0.70 for any pass. For $\rho_RMF=1.0$, the reference maximum is approximately 0.091527. For $\rho_RMF=0.85$, it is approximately 0.084586. The formal module retains the source formula and thresholds but removes the inaccurate claim that 0.10 is itself a passing boundary.

6.6 PDV final gate

PDV_current is read from the current candidate-bound SigmaDecisionPacket. Module 11 does not recompute it and does not use a PDV value calculated before a repair or synthesis. The final ceiling is inclusive:

$$pdv_pass = (PDV_current \leq 0.05)$$

This is a stricter final-archive policy than Sigma's general PDV_LIMIT=0.21. A PDV final-gate failure blocks archive authorization but is not automatically catastrophic FBS.

6.7 Final pass predicate

$$\text{cathedral_pass} = (\text{C_cathedral} \geq 0.70) \text{ AND } (\text{PDV_current} \leq 0.05)$$

All structural, freshness, chain, domain, readiness, and one-time authorization checks must also pass. Threshold equality passes. NaN never compares as a pass because NaN is rejected earlier as a fault.

7. Deterministic Outcomes and Archive Handoff

7.1 First-match decision tree

1. FBS_RECOVERY_REQUEST when an enumerated catastrophic invariant is proven.
2. FAULT when schema, digest, shape, domain, arithmetic, owner, lineage, or deterministic replay validation fails without a catastrophic persisted-state condition.
3. HOLD when a required dependency is explicitly not ready, unavailable, current, or consumable, including archive-epoch contention.
4. FINAL_REJECT when $\text{C_cathedral} < \text{cathedral_threshold}$.
5. FINAL_REJECT when $\text{PDV_current} > \text{pdv_safe}$.
6. ARCHIVE_AUTHORIZED otherwise.

The order is normative. A low score cannot hide a provenance fault, and an ordinary final rejection cannot be promoted to catastrophic recovery merely because the system wants a new seed.

7.2 FinalRejectPacket

Field	Requirement
reason_code	CATHEDRAL_INTEGRITY_LOW or PDV_FINAL_LIMIT, with optional diagnostic subcodes.
candidate_digest	Exact rejected revision.
metrics_digest	Binds all pressure components, rho_RMF, C_cathedral, and PDV.
preservation	Candidate remains addressable and unchanged; no delete, quarantine, or RTSOM transfer.
next_action	ORCHESTRATOR_NEW_CYCLE or separately authorized reevaluation; Module 11 does not choose HARMONY/FBS.

7.3 Hold conditions

- Current Sigma decision, Module 7 evidence, required Module 10 audit, route intent, Phi snapshot, or archive context is not yet available.
- A dependency is valid but stale and a refresh has been requested.
- Psi archive epoch or prior-manifest digest changed before authorization seal.
- An authorization nonce is already reserved by another in-flight attempt.
- The RMX paused transaction is temporarily unavailable but not corrupt.

HOLD is not rejection, quarantine, exhaust, or recovery. It preserves the immutable candidate reference until the dependency resolves or the orchestration policy expires the attempt.

7.4 Psi handoff

ArchiveAuthorizationPacket permits Module 3 to begin PREPARE for exactly one candidate, archive epoch, and prior manifest digest. Psi must independently verify the packet, consume it atomically, and preserve its own PREPARE ->

RMX RESOLUTION -> RTSOM RECEIPT -> COMMIT transaction. Module 11 has no authority to call a prepared state committed.

7.5 Authorization consumption and replay

- The authorization is single-use and cycle-bound in the reference profile.
- A duplicate delivery before consumption is idempotent; a second consumption attempt is rejected.
- Any candidate, policy, archive epoch, prior manifest, Phi, Sigma, Module 7, Module 10, or required LORIEN digest change invalidates the authorization.
- An expired authorization cannot be refreshed by editing its expiry field; a new Module 11 evaluation is required.

7.6 RTSOM and FBS separation

Module 11 neither charges exhaust nor performs recovery. An ordinary CATHEDRAL or PDV rejection produces no RTSOM request and no FBS request. Only the owner-specific catastrophic conditions in Section 8.3 may produce FBSRecoveryRequest. ROVER/FBS decides and executes recovery; any exhaust custody then requires Module 14 handling under its own typed contract.

8. Invariants, Faults, and Catastrophic Conditions

8.1 Normative invariants

- M11-I01: The candidate authorized for Psi is byte-for-byte identical to the candidate received by Module 11 in the reference profile.
- M11-I02: Exactly one terminal output is sealed per attempt.
- M11-I03: A current Sigma SEAL is bound to the exact candidate digest.
- M11-I04: HARMONY_REVALIDATED and RMX_REVALIDATED require a current Module 10 SAFE_ELIGIBLE audit.
- M11-I05: No stale pre-mutation Sigma, LORIEN, Module 10, or SERAPHIM authorization is accepted.
- M11-I06: PhiSnapshot is read-only and is never updated locally.
- M11-I07: R(Phi) is orthogonal within tolerance and preserves pair energy within tolerance.
- M11-I08: F_{zeta} is evaluated with $0 \leq \text{zeta}_{\text{ser}} \leq 1$ and is 1-Lipschitz by declared scalar derivative bound.
- M11-I09: X_probe is private diagnostic data and never aliases candidate memory.
- M11-I10: Every pressure component and C_cathedral is finite and lies in [0,1].
- M11-I11: P_e equals the declared maximum of the registered pressure components.
- M11-I12: rho_RMF uses accepted prior history plus current RMF prospectively; the candidate is not inserted into committed history.
- M11-I13: Cathedral threshold equality and PDV ceiling equality pass.
- M11-I14: ArchiveAuthorizationPacket authorizes PSI_PREPARE_ONLY, never COMMIT.
- M11-I15: Authorization is candidate-, chain-, archive-epoch-, and prior-manifest-bound.
- M11-I16: Ordinary rejection does not invoke quarantine, RTSOM, ROVER, or FBS.
- M11-I17: No wall-clock timestamp or ambient random value enters a packet digest.
- M11-I18: Duplicate authorization delivery is idempotent; duplicate consumption is rejected.
- M11-I19: Any failed evaluation commits no Module 11 persistent state except append-only audit telemetry.
- M11-I20: Dashboard lifecycle counts reconcile exactly with sealed outcomes and downstream receipts.

8.2 SeraphimFault codes

Code	Meaning
M11_CONFIG_UNSUPPORTED	Unsupported schema, parameter, hash, or policy profile.
M11_SESSION_MISMATCH	Session identifiers disagree.
M11_CYCLE_MISMATCH	Cycle or revision ordering is invalid.
M11_PROFILE_CHAIN	Required profile chain is incomplete or contradictory.
M11_STALE_AUTHORITY	Candidate-bound authorization is stale.

Code	Meaning
M11_DIGEST_MISMATCH	Packet or candidate digest fails.
M11_INVALID_SHAPE	Candidate or evidence shape is invalid.
M11_NONFINITE	NaN or infinity detected.
M11_DOMAIN_BREACH	Declared state or metric domain violated.
M11_PHI_SNAPSHOT	Phi snapshot missing, stale, or inconsistent.
M11_HISTORY_INVALID	RMF history ordering, values, or epoch invalid.
M11_DEPTH_SIGNAL	Cascade depth signal invalid.
M11_PHASE_OPERATOR	Orthogonality or pair-energy tolerance failed.
M11_PROBE_ALIAS	Probe buffer aliases immutable candidate storage.
M11_NUMERIC_OVERFLOW	Stable evaluation policy cannot produce finite metrics.
M11_PRESSURE_COMPONENT	Pressure registry or component value invalid.
M11_ARCHIVE_CONTEXT	Archive epoch or prior manifest context invalid.
M11_NONDETERMINISTIC	Same canonical input produced a different canonical result.

8.3 Catastrophic conditions eligible for FBSRecoveryRequest

Condition	Minimum evidence
M11_PERSISTED_AUTH_SPLIT	Two different valid authorization packets are persisted for the same attempt key.
M11_DOUBLE_CONSUME_PROVEN	The same single-use authorization is proven committed into two archive transactions.
M11_PERSISTED_CHAIN_CORRUPTION	Persisted validation-chain digest cannot be reconciled with its immutable packet set.
M11_PERSISTED_NONDETERMINISM	Identical canonical input and fixed backend profile produced conflicting persisted outcomes.
M11_ARCHIVE_EPOCH_IRRECONCILABLE	Two authenticated archive heads claim the same predecessor and epoch without a resolvable owner decision.
M11_RECOVERY_REQUEST_CORRUPT	A previously sealed recovery request is itself corrupted or forged.

A low C_cathedral, PDV above 0.05, high p_filter, negative G_t, warming history, stale dependency, ordinary timeout, or archive contention is not catastrophic by itself.

9. Metrics, Measurements, and Dashboard Reconciliation

9.1 Required metrics

Metric	Meaning
attempts_total	Module 11 attempts accepted by the API.

9.2 Reconciliation equations

$$\text{attempts_total} = \text{authorized_total} + \text{rejected_total} + \text{held_total} + \text{faulted_total} + \text{recovery_requested_total}$$

$$\text{rejected_total} = \text{rejected_integrity_total} + \text{rejected_pdv_total}$$

$$\text{authorized_total} = \text{authorization_pending} + \text{authorization_consumed_total} + \text{authorization_expired_total} + \text{authorization_invalidated_total}$$

PSI_PREPARED, PSI_COMMITTED, RTSOM_ACKNOWLEDGED, FBS_EXECUTED, and ARCHIVE_VISIBLE are downstream lifecycle counters and must not be inferred from authorized_total.

9.3 Logging and privacy

- Log identifiers, digests, schema versions, scalar diagnostics, reason codes, and lifecycle state.
- Do not log raw candidate arrays or symbolic payloads by default.
- A debug profile may log encrypted samples only under a separately declared retention and access policy.
- Do not label users, prompts, or states as toxic, psychotic, harmful, truthful, or safe based on Module 11 scores.

10. Deterministic Conformance Test Matrix

ID	Case	Input focus	Expected
M11-T001	Reference RAW_ACCEPTED pass	Current chain, high rho_RMF, low pressure, PDV=0.01	ArchiveAuthorizationPacket
M11-T002	Threshold equality C	C_cathedral exactly 0.70, PDV<0.05	Pass
M11-T003	Threshold equality PDV	C_cathedral>0.70, PDV exactly 0.05	Pass
M11-T004	Integrity just below	C_cathedral next representable below 0.70	FINAL_REJECT
M11-T005	PDV just above	PDV next representable above 0.05	FINAL_REJECT
M11-T006	Stale Sigma	Sigma candidate digest from prior revision	M11_STALE_AUTHORITY
M11-T007	Wrong Sigma route	Current Sigma packet not SEAL	M11_PROFILE_CHAIN
M11-T008	HARMONY missing audit	HARMONY_REVALIDATED without Module 10	M11_PROFILE_CHAIN
M11-T009	HARMONY stale Sigma	Sigma SEAL predates repair digest	M11_STALE_AUTHORITY
M11-T010	RMX token mismatch	Paused transaction identity differs	M11_PROFILE_CHAIN
M11-T011	Required route missing	Destination profile requires LORIEN	HOLD or profile fault per declared readiness
M11-T012	Optional route absent	LOCAL_ARCHIVE profile	No route fault
M11-T013	Phi stale	Phi source cycle outside policy	M11_PHI_SNAPSHOT
M11-T014	History bootstrap	No prior RMFs; current RMF only	Computable; history_warming=true
M11-T015	History order defect	Nonmonotone accepted record cycles	M11_HISTORY_INVALID
M11-T016	RMF out of range	Accepted RMF=1.1	M11_DOMAIN_BREACH

ID	Case	Input focus	Expected
M11-T017	Zero candidate	All-zero candidate, valid chain	Finite probe; high strain allowed; deterministic outcome
M11-T018	NaN candidate	One NaN component	M11_NONFINITE
M11-T019	Infinity evidence	p_immune=inf	M11_NONFINITE
M11-T020	Odd DIM operator	DIM=81	40 rotation blocks plus one pass-through coordinate
M11-T021	Even DIM operator	DIM=80 supported profile	40 rotation blocks
M11-T022	Operator orthogonality	Representative Phi values	Error <= tolerance
M11-T023	Pair energy	Random vector pair rotations	Energy preserved within tolerance
M11-T024	zeta=0	F is identity	p_filter=0
M11-T025	zeta=1	Pure tanh branch	Finite and 1-Lipschitz difference test
M11-T026	Invalid zeta	zeta=1.01	M11_CONFIG_UNSUPPORTED
M11-T027	Negative depth	depth_signal<0	M11_DEPTH_SIGNAL
M11-T028	Depth floor	depth_signal=2.9, unit=1	n=2
M11-T029	Depth cap	depth_signal=999	n=cascade_max
M11-T030	Probe immutability	Track candidate bytes before/after	Identical
M11-T031	Probe alias	Output shares writable candidate storage	M11_PROBE_ALIAS
M11-T032	p_filter bound	Random finite vectors	0<=p_filter<=1
M11-T033	Paradox pressure	G_t=-0.4	p_paradox=0.4
M11-T034	Nonnegative G	G_t=0.2	p_paradox=0
M11-T035	Pressure maximum	components 0.02,0.08,0.04	P_e=0.08
M11-T036	Stable logistic low pressure	P_e=0	Finite h_pressure near 1
M11-T037	Stable logistic high pressure	P_e=1	Finite h_pressure near 0
M11-T038	Midpoint identity	P_e=0.10	h_pressure=0.5
M11-T039	Midpoint cannot pass	rho_RMF<=1, P_e=0.10	C_cathedral<=0.5
M11-T040	rho too low	rho_RMF=0.70, very low pressure	C_cathedral<0.70
M11-T041	Pressure boundary formula	rho=1, C=0.70	P_e,max approximately 0.091527
M11-T042	Canonical replay	Same input repeated	Same packet digest
M11-T043	Negative zero	-0.0 versus +0.0 metadata	Same canonical digest
M11-T044	Candidate byte change	One ULP changed	Different candidate and authorization digest

ID	Case	Input focus	Expected
M11-T045	Policy epoch change	Same vector, new policy	Old authorization invalid
M11-T046	Archive head change	Prior manifest changes before seal	HOLD or M11_ARCHIVE_CONTEXT
M11-T047	Duplicate delivery	Same authorization delivered twice before consume	Idempotent
M11-T048	Duplicate consume	Psi consumes same ID twice	Second rejected
M11-T049	Authorization expiry	Cycle advances beyond TTL	Expired; no PREPARE
M11-T050	Ordinary reject no FBS	C low without catastrophic evidence	FinalRejectPacket only
M11-T051	PDV reject no RTSOM	PDV=0.06	No RTSOM request
M11-T052	Persisted auth split	Two conflicting persisted authorizations	FBSRecoveryRequest
M11-T053	Persisted nondeterminism	Same canonical input, conflicting persisted output	FBSRecoveryRequest
M11-T054	Local arithmetic backend drift	Metrics differ within tolerance, same decision	Conformant
M11-T055	Decision boundary backend drift	Tolerance interval crosses threshold	HOLD/FAULT under comparison policy; never guess
M11-T056	Mutating profile attempt	Reference profile tries to archive X_probe	M11_PROFILE_CHAIN
M11-T057	Alignment proposal	Optional profile changes digest	No archive authorization; revalidation required
M11-T058	Crash before seal	Process stops during probe	No output packet
M11-T059	Crash after seal	Packet durable, no Psi receipt	Authorization pending only
M11-T060	Dashboard reconciliation	Mixed 1000-attempt fixture	All equations exact

11. Property, Fuzz, Concurrency, and Stress Testing

11.1 Mathematical property tests

- For random finite x, y , verify $\|F(x) - F(y)\|_2 \leq \|x - y\|_2$ plus tolerance for every supported ζ_{ser} .
- For random Φ and vectors, verify $R(\Phi)^T R(\Phi) = I$ and norm preservation within tolerance.
- Verify p_{filter} , p_{paradox} , p_{immune} , P_e , h_{pressure} , and $C_{\text{cathedral}}$ remain bounded.
- Verify h_{pressure} is monotonically nonincreasing in P_e and $C_{\text{cathedral}}$ is nondecreasing in ρ_{RMF} .
- Verify P_e is monotone in every registered pressure component under the maximum aggregator.
- Verify the analytic pressure-boundary equation reproduces $C_{\text{cathedral}} = \text{threshold}$ within tolerance.

11.2 Metamorphic tests

- Adding 2π to Φ leaves every rotation block and probe diagnostics unchanged within tolerance.
- Reordering accepted RMFs changes ρ_{RMF} only according to the chronological tail rule; arbitrary packet ordering is rejected.
- Changing X_{probe} bytes without changing candidate bytes must change probe digest but not candidate digest.

- Changing a non-required LORIEN field under LOCAL_ARCHIVE must not alter the decision because it is excluded from the profile chain.
- Changing any required packet digest must invalidate the authorization even when scalar values are identical.
- Replacing the max pressure aggregator with a mean must fail schema compatibility tests.

11.3 Fuzz and malformed-input tests

- Fuzz shapes, byte lengths, schema tags, optional-field presence bits, owner IDs, epochs, and packet ordering.
- Inject NaN payloads, signaling NaNs, infinities, subnormals, signed zero, extreme finite exponents, and malformed UTF-8 identifiers.
- Fuzz duplicate, missing, cyclic, and cross-session validation-chain references.
- Fuzz archive epochs and prior-manifest digests under concurrent head advancement.
- Reject oversized history arrays before allocation amplification; enforce configured length caps.

11.4 Concurrency and crash tests

- Run concurrent attempts for the same candidate and nonce; exactly one authorization reservation may win.
- Run concurrent attempts for different candidates under the same archive epoch; Psi consumption arbitrates archive-head advancement, not Module 11.
- Crash at every transition and confirm no partial authorization object becomes valid.
- Replay authorization consumption across process restart and confirm single-use semantics remain durable.
- Simulate archive-head advancement between validation and authorization seal; outcome must be HOLD or context fault, never stale authorization.

11.5 Stress requirements

- At least 100,000 attempts across all three profiles with deterministic replay sampling.
- At least 10,000 boundary-focused cases around C_cathedral=0.70, PDV=0.05, P_e effective boundaries, and cascade-depth transitions.
- No memory growth proportional to total session length; history consumption is bounded by rmf_window_max.
- No candidate-buffer mutation, aliasing, double authorization, duplicate consumption, or unreconciled lifecycle counter.
- Record throughput, p50/p95/p99 latency, allocation counts, exponent clipping count, and backend identity.

12. Reference Pseudocode

```
def evaluate_seraphim(inp: SeraphimInput, cfg: SeraphimConfig):
    validate_config(cfg, inp.boot_packet)
    frozen = freeze_candidate(inp.candidate)
    validate_profile_chain(inp, cfg, frozen.digest)
    validate_snapshots_and_digests(inp, cfg)

    depth = floor(inp.cascade_control.depth_signal / cfg.cascade_unit)
    n = min(depth, cfg.cascade_max)

    d_pre = frozen.values.copy()          # private probe only
    stage_strain = []
    phase_error = 0.0
    for k in range(n + 1):
        y = fractalizer_v1(d_pre, cfg.zeta_ser)
        e = norm2(y - d_pre) ** 2
        b = norm2(d_pre) ** 2
        stage_strain.append(e / (e + b + cfg.epsilon_safe))

    R = phase_rotation_blocks(
        cfg.dim,
        inp.phi_snapshot.phi_t + k * cfg.phase_step,
    )
    phase_error = max(phase_error, fro_norm(R.T @ R - I) / cfg.dim)
    d_post = R @ y
    if k < n:
        d_pre = d_post

    x_probe = d_post
```



```

assert bytes(inp.candidate) == frozen.original_bytes
if phase_error > cfg.phase_operator_tolerance:
    return SeraphimFault("M11_PHASE_OPERATOR")

p_filter = max(stage_strain)
p_paradox = max(0.0, -inp.symbolic_paradox.g_t)
p_immune = immune_soft_pressure_for_profile(inp)
p_e = max(p_filter, p_paradox, p_immune)

rmf_window = prospective_rmf_window(
    inp.history_summary.accepted_rmf,
    inp.sigma_decision.rmf,
    cfg.rmf_window_max,
)
rho_rmf = mean(rmf_window)

z = cfg.pressure_slope * (p_e - cfg.pressure_center)
h_pressure = stable_decreasing_logistic(z, clip=(-60.0, 60.0))
c_cathedral = clip(rho_rmf * h_pressure, 0.0, 1.0)
pdv = inp.sigma_decision.pdv

metrics = seal_probe_metrics(
    x_probe_digest=digest_probe(x_probe),
    depth=n,
    p_filter=p_filter,
    p_paradox=p_paradox,
    p_immune=p_immune,
    p_e=p_e,
    rho_rmf=rho_rmf,
    h_pressure=h_pressure,
    c_cathedral=c_cathedral,
    pdv=pdv,
    phase_error=phase_error,
)

catastrophic = detect_catastrophic_persisted_invariant(inp, metrics)
if catastrophic:
    return FBSRecoveryRequest(reason=catastrophic, no_local_execution=True)

if c_cathedral < cfg.cathedral_threshold:
    return FinalRejectPacket("CATHEDRAL_INTEGRITY_LOW", metrics)
if pdv > cfg.pdv_safe:
    return FinalRejectPacket("PDV_FINAL_LIMIT", metrics)

return ArchiveAuthorizationPacket(
    authorization_id=digest_id(inp, metrics),
    candidate_digest=frozen.digest,      # never probe digest
    validation_chain_digest=chain_digest(inp),
    seraphim_probe_digest=metrics.packet_digest,
    archive_epoch=inp.archive_context.archive_epoch,
    prior_manifest_digest=inp.archive_context.prior_manifest_digest,
    authorized_operation="PSI_PREPARE_ONLY",
    expires_after_cycle=inp.cycle_id,
    single_use=True,
)

```

12.1 Stable logistic reference

```

def stable_decreasing_logistic(z: float, clip=(-60.0, 60.0)) -> float:
    zc = min(max(z, clip[0]), clip[1])
    if zc >= 0.0:
        e = exp(-zc)
        return e / (1.0 + e)
    e = exp(zc)
    return 1.0 / (1.0 + e)

```

12.2 Psi consumption reference

```

def consume_archive_authorization(auth, psi_context):
    verify_digest(auth)
    require(auth.authorized_operation == "PSI_PREPARE_ONLY")

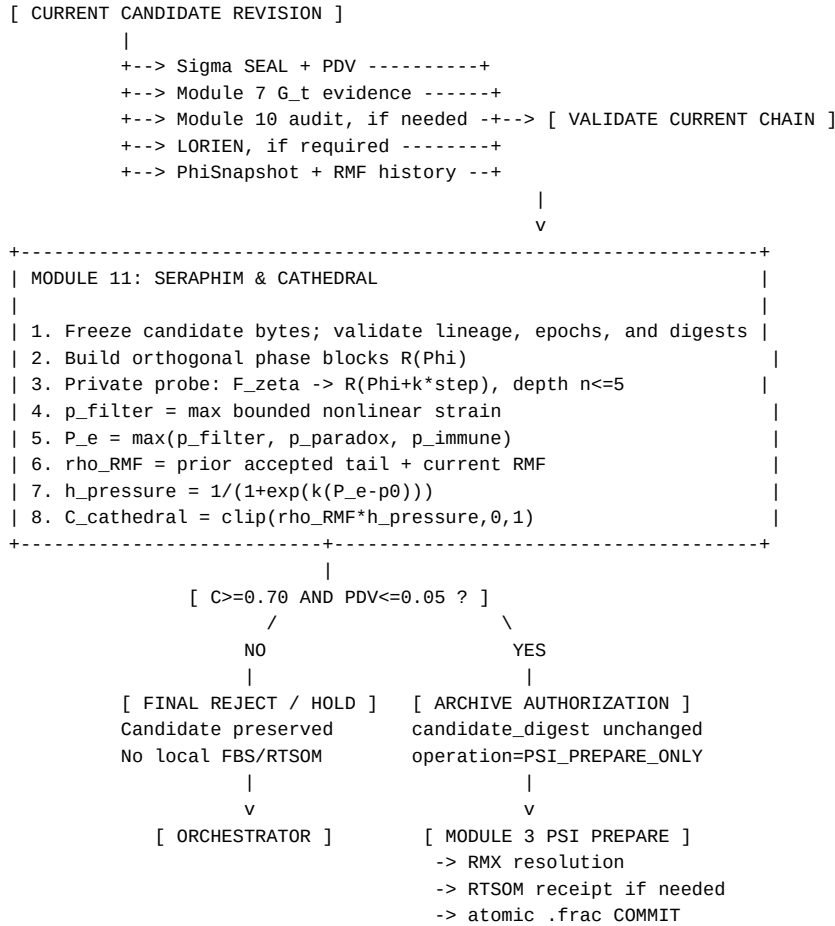
```

```

require(auth.candidate_digest == psi_context.candidate_digest)
require(auth.archive_epoch == psi_context.archive_epoch)
require(auth.prior_manifest_digest == psi_context.prior_manifest_digest)
require(not authorization_ledger.already_consumed(auth.authorization_id))
return authorization_ledger.consume_once_and_begin_prepare(auth)

```

13. ASCII System Flow



Catastrophic persisted invariant only -> typed FBSRecoveryRequest
 ROVER/FBS executes recovery; RTSOM owns any exhaust custody.

14. Cleaned Rosetta Stone

SpiralCore term	Exact computational meaning	Claim limit
SERAPHIM Matrix	Versioned nonlinear filter plus orthogonal phase-rotation probe evaluated on a private copy.	Not a quantum matrix, consciousness field, or physical harmonic lock.
CATHEDRAL	Final current-cycle archive-eligibility gate using pressure health, prospective RMF, and PDV.	Not an absolute firewall or proof of universal safety.
Phase Harmonic Alignment	Orthogonal coordinate-pair rotations plus non-expansive diagnostic filtering.	No physical phase synchronization is claimed.
Fractalizer	Elementwise monotone 1-Lipschitz map F_zeta.	Not Collatz iteration and not proof of fractality.

SpiralCore term	Exact computational meaning	Claim limit
Fractal Delta Cascade	Finite, bounded-depth composition of declared filter and rotation operators.	Not an infinite recursion or Kalman filter unless separately implemented.
Entropic Pressure P_e	Maximum of registered bounded computational pressure components.	Not thermodynamic entropy or physical pressure.
Cathedral Integrity	ρ_{RMF} multiplied by a decreasing logistic pressure-health factor.	Not factual truth, moral health, or structural completeness.
Ultimate Firewall	Last authorization owner before Ψ PREPARE in the hardened profile.	Does not prevent every defect and does not write the archive.
Emergency Time-State Lock	No-commit terminal outcome plus typed recovery request when a catastrophic persisted invariant is proven.	No literal time stop.
Archive Seal	Single-use ArchiveAuthorizationPacket permitting Ψ PREPARE only.	Not a .frac commit or durable memory write.

15. Glossary

Term	Definition
ArchiveAuthorizationPacket	Single-use, candidate-bound permission for Module 3 to begin PREPARE under one archive context.
ArchiveContextSnapshot	Immutable archive epoch, prior manifest digest, destination profile, and transaction context.
$C_{\text{cathedral}}$	Bounded final eligibility score $\rho_{\text{RMF}} * h_{\text{pressure}}$.
CascadeControlPacket	Versioned nonnegative depth signal; not wall-clock time.
CATHEDRAL	Module 11 pressure and final-authorization subroutine.
Entropic Pressure P_e	Maximum registered bounded pressure component.
F_{zeta}	SERAPHIM_FRACTALIZER_V1 elementwise 1-Lipschitz map.
FinalRejectPacket	No-archive outcome that preserves the candidate and evidence.
h_{pressure}	Decreasing logistic health factor centered at pressure_center .
history_warming	Flag indicating fewer than rmf_window_max values exist; not a failure by itself.
PhiSnapshot	Read-only Module 12 hysteresis and winding provenance.
p_{filter}	Maximum bounded nonlinear strain energy across the private cascade.
p_{paradox}	Negative Module 7 G_t pressure, $\max(0, -G_t)$.
$R(\Phi)$	Orthogonal real phase-rotation block matrix.
ρ_{RMF}	Prospective mean of accepted RMF tail plus current candidate RMF.

Term	Definition
SERAPHIM probe	Non-committing diagnostic transform whose bytes are never archived in the reference profile.
validation_chain_digest	Digest binding every current prerequisite packet and policy version.

16. Revision Summary

Area	v14 source behavior	v14.1 formalization
Final gate role	Called the absolute last gate and direct archive handoff.	Defined as final authorization owner; Psi remains PREPARE/RMX/RTSOM/COMMIT owner.
Candidate mutation	X_aligned sent directly to archive after post-gate transformation.	Reference probe is non-committing; authorization binds unchanged validated candidate.
Phase matrix	Diagonal cosine attenuation.	Orthogonal 2x2 rotation blocks; legacy attenuation retained only for traceability.
Fractalizer name	Called a Collatz surrogate.	Renamed SERAPHIM_FRACTALIZER_V1; exact derivative and Lipschitz limit proved.
Cascade depth	min(int(Delta_t),5) with undefined source/negative behavior.	Typed nonnegative control packet, floor rule, unit, cap, and faults.
Entropic Pressure	Named but not computed.	Exact registered components and max aggregator.
RMF history	Unspecified arithmetic mean.	Accepted-history tail plus current RMF, bounded window, bootstrap flag.
Sigmoid cliff	Claimed P_e=0.10 as pass boundary.	Retained equation; derived actual rho-dependent passing boundary.
PDV	Potentially stale input after repair/alignment.	Must come from current candidate-bound post-mutation Sigma SEAL.
Failure route	Ordinary score failure directly executed FBS and exhaust transfer.	Ordinary failure is FinalRejectPacket; FBS request only for persisted catastrophic invariants.
Authorization identity	No exact single-use contract.	Digest-derived, cycle-bound, archive-context-bound, single-use packet.
Claims	Absolute soundness, paradox freedom, and safety language.	Narrow computational claim boundary.

17. Source Basis

1. SPIRALCORE SPECIFICATION: MODULE 11 - SERAPHIM & CATHEDRAL (Phase Harmonic Matrix & Ultimate Firewall), source module supplied for cleanup.
2. Spiralcore_v14 consolidated specification, especially Modules 2, 3, 7, 8, 10, 11, 12, 14, 15, and 16 handoff and ownership descriptions.

3. SPIRALCORE Module 2 Sigma-Lattice Formalized v14.1, for current candidate-bound SEAL, PDV ownership, accepted-history RMF, and route separation.
4. SPIRALCORE Module 3 Psi-Lattice Formalized v14.1, for ArchiveAuthorizationPacket and PREPARE -> RMX -> RTSOM receipt -> atomic COMMIT ownership.
5. SPIRALCORE Module 10 Braidback & IMMUNE_SYSTEM Formalized v14.1, for post-mutation audit, SAFE_ELIGIBLE semantics, and required revalidation chain.
6. Module 12 Phi-Bridge source contract, for read-only Phi_t and DeltaPhi ownership at the Module 11 boundary.
7. Modules 14 and 16 source contracts, for RTSOM custody and ROVER/FBS recovery ownership.

NORMATIVE IMPLEMENTATION CHECKLIST

An implementation is conformant only when it preserves immutable candidate bytes, validates the complete current chain, uses the orthogonal probe profile or declares a different schema, computes every pressure component exactly, evaluates the stable logistic and thresholds inclusively, emits one terminal typed output, authorizes PSI_PREPARE_ONLY, and never performs local archive, quarantine, RTSOM, or FBS mutation.

SPIRALCORE SPECIFICATION

MODULE 12 - PHI

(Information Hysteresis, Event-Winding Ledger, and Semantic Holonomy Surrogate)

Cleaned and formalized revision - v14.1

Document class	Normative persistent-state and path-memory contract	Execution phase	After route-authoritative evaluation or owner-completed event
Architecture	SpiralCore v14	Reference profile	SC14.1-M12-PHI-STATE-01
Primary output	PhiCommitPacket and PhiSnapshot	Alternative outputs	HoldPacket, Fault, or FBSRecoveryRequest
Conflict rule	Module 12 alone mutates Phi state	Consumers	Delta reads; Psi records; Module 11 reads; owners issue event receipts

NORMATIVE STATUS

Module 12 is exact within its declared computational scope. It consumes one route-authoritative evaluation closure or one owner-completed event bundle, prepares and atomically commits one path-memory update, maintains an exactly-once event ledger, computes the bounded next-cycle hysteresis weight, and emits an immutable PhiSnapshot. It does not evaluate Sigma coherence, generate candidates, write the Psi archive, mutate RTSOM, perform repair, or execute FBS.

CLAIM BOUNDARY

“Information Hysteresis,” “semantic holonomy,” “memory friction,” “spin memory,” “topological scar,” “winding number,” “vortex,” “gravitational bias,” and “thermodynamic effort” are architectural names. The reference module proves boundedness, deterministic recurrence, exactly-once event accounting, and path-order sensitivity of its declared state update. It does not prove physical thermodynamics, Berry phase, quantum gravity, consciousness, non-integrability of an external system, or an unforgeable physical history certificate.

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1. Normative Scope, Ownership, and Non-Goals

1.1 Purpose

Module 12 is the sole persistent owner of SpiralCore path-memory state. It converts a route-authoritative RMF observation into a bounded mismatch effort, applies the declared decayed accumulator, consumes signed critical-event receipts exactly once, and publishes the committed scalar and event ledger as a versioned PhiSnapshot for downstream readers. The module is an event-sourced state machine, not a free-running background field.

1.2 Ownership boundary

Concern	Authoritative owner	Module 12 relationship
Boot defaults and Xi attractor	Module Boot	Validates PHI_DECAY, PHI_GAIN, DIM, policy epoch, and attractor digest; does not regenerate Boot data.
Candidate generation and bias application	Module 1 Delta-Lattice	Reads the latest committed PhiSnapshot and applies its scalar weight under Delta rules.
RMF and route decision	Module 2 Sigma-Lattice	Consumes a current candidate-bound decision; never recomputes RMF or changes Sigma route.
SGIT information-length evidence	Module 0 SGIT_v2	May supply optional versioned diagnostics; RMF_ONLY_V1 remains the reference effort mode.
Critical-event execution	Owning module for each event	Consumes owner-signed completion receipts; never infers that an event occurred.
Persistent Phi and event ledger	Module 12	Sole writer; prepares, compare-and-swaps, durably commits, verifies, and snapshots.
Archive manifest and .frac commit	Module 3 Psi-Lattice	Records a PhiSnapshot by digest; cannot mutate Phi or consume event receipts.
Final archive authorization	Module 11 SERAPHIM/CATHEDRAL	Reads current PhiSnapshot only.
Exhaust custody	Module 14 RTSOM	No local budget, tensor, mass, or custody mutation.
Catastrophic recovery	ROVER/FBS owner modules	Module 12 emits a typed request only for enumerated persistent-state invariants.

1.3 Explicit non-goals

- No direct candidate mutation, Braidback, Phase Lift, Magnetic Relaxation, NULLIFY, quarantine, archive commit, or network routing.
- No use of ambient time, wall-clock timestamps, process IDs, random values, or mutable global state in identifiers or state updates.
- No automatic update from an intermediate, stale, retried, or unsealed metric calculation.
- No claim that $\text{PHI_GAIN}=0.22$ is the Braidback 51/49 rule. It is an additive bounded bias coefficient owned by a different mechanism.
- No claim that an integer event count is a mathematical winding number unless a separate closed-loop phase trace satisfies the declared certificate test.
- No claim that a nonzero semantic residue proves physical non-integrability; it proves only path sensitivity of the declared recurrence under the stated comparison conditions.
- No local Cantor diagonalization, Collatz processing, seed generation, or exhaust transfer.

1.4 Normative precedence

1. Tested reference implementation and Module 12 conformance vectors.
2. This Module 12 contract.
3. Current Module Boot, 0, 1, 2, 3, 11, 13, 14, and FBS/ROVER ownership contracts.
4. Active PhiConfig, event registry, canonical encoding, and state-store profile.
5. Consolidated SpiralCore v14 prose.
6. Rosetta Stone analogies, observer language, and physical metaphors.

2. Architecture Role, Transaction Types, and Lifecycle

2.1 Reference timing rule

ONE ROUTE DECISION, ONE MEMORY UPDATE

A candidate revision may affect persistent Phi only after Sigma has emitted one current, digest-bound route decision and the orchestrator has sealed one `PhiEvaluationClosurePacket` for that exact revision. Retries with identical bytes are idempotent. A repaired or synthesized candidate is a new revision and requires its own current Sigma decision before it may contribute effort.

The reference profile intentionally records terminal evaluation difficulty, including non-SEAL routes, because the source architecture describes Phi as memory of difficulty rather than accepted-memory content. The update is nevertheless gated by a unique closure receipt, so intermediate metric calls, retries, and crashes cannot multiply the friction contribution.

2.2 Transaction types

Transaction	Input	Persistent effect	Typical producers
EVALUATION_UPDATE	<code>PhiEvaluationClosurePacket</code> plus current <code>SigmaDecisionPacket</code>	Applies one decay-and-effort recurrence; consumes any attached eligible event receipts.	Orchestrator after Sigma route lock.
EVENT_APPEND	One or more owner-completed <code>CriticalEventReceipts</code>	Leaves <code>phi_value</code> unchanged; increments event ledger exactly once per unique receipt.	RMX, FBS/ROVER, or another owner after execution completes.
PAIRWISE_RESIDUE_TEST	Two immutable <code>PathTraceSummary</code> packets	No persistent mutation; emits <code>SemanticResiduePacket</code> .	Test harness or analysis service.

Transaction	Input	Persistent effect	Typical producers
CLOSED_LOOP_HOLONOMY_TEST	Versioned closed phase trace	No persistent mutation; emits optional HolonomyCertificatePacket or diagnostic failure.	SGIT/phase-trace provider.

2.3 Lifecycle states

State	Meaning
IDLE	No active Module 12 transaction.
VALIDATING	Configuration, prior state, closure/event receipts, provenance, domains, and digests are checked.
PREPARING	Next state, event ledger delta, bias weight, and canonical packet are computed without mutation.
PREPARED	One immutable PhiPreparedUpdate exists and is eligible for compare-and-swap commit.
COMMITTING	Durable write, fsync, atomic pointer change, and read-back verification are in progress.
COMMITTED	One new PhiState version and PhiSnapshot are authoritative.
HELD	Required authoritative evidence is unavailable, warming, stale, or contended.
FAULTED	Typed contract, provenance, numerical, registry, or persistence error; no state change.
RECOVERY_REQUESTED	A catastrophic persistent-state invariant is frozen and delegated.

2.4 Required execution order

1. Load PhiConfig and the authoritative prior PhiStateSnapshot by expected state version.
2. Validate session, policy epoch, Boot digest, state digest chain, nonnegative finite phi_value, event-ledger digest, and closure sequence.
3. Validate exactly one transaction type and its owner-bound input packet set.
4. For EVALUATION_UPDATE, validate the current SigmaDecisionPacket and closure binding, then compute normalized effort.
5. Validate and deduplicate attached critical-event receipts against the committed event ledger.
6. Prepare phi_next, event_winding_next, bias_weight_next, provenance chain, and packet digest without mutating persistent state.
7. Compare-and-swap expected state version, durably persist the prepared record, and atomically publish the new state pointer.
8. Read back the committed record, verify every field and digest, then emit PhiCommitPacket and PhiSnapshot.
9. On any failure before successful publication, preserve the prior state exactly and emit one hold, fault, or recovery request.

2.5 Atomicity, idempotence, and ordering

The transaction key is (session_id, transaction_type, closure_or_bundle_id, prior_state_version, policy_epoch). Replaying identical bytes returns the existing committed result. Reusing an identifier with different bytes is a fault. EVALUATION_UPDATE closure_sequence must equal prior.last_closure_sequence + 1. EVENT_APPEND may occur between evaluation closures but must use unique owner sequence numbers and cannot change closure_sequence.

3. Typed Data Contracts

3.1 PhiConfig

Field	Reference value / domain	Requirement
schema_id	SC14.1-M12-PHI-STATE-01	Included in every digest.
phi_decay	0.99	Finite; $0 \leq \text{phi_decay} < 1$.
phi_gain	0.22	Finite; $0 \leq \text{phi_gain} < 0.49$. Not a repair weight.
effort_mode	RMF_ONLY_V1	Other modes require a separately versioned formula and test vectors.
rmf_domain	[-1,1]	Matches Sigma cosine-similarity contract.
event_registry_version	PHI-EVENTS-v1	Maps event types to authoritative owners.
max_events_per_bundle	64	Bounded resource policy; overflow produces HOLD, not truncation.
state_store_profile	ATOMIC_CAS_V1	Durable write and read-back are mandatory.
float_profile	IEEE 754 binary64	Finite values only; canonical negative-zero handling.
policy_epoch	uint64	Immutable during a transaction.

3.2 PhiStateSnapshot

Field	Type	Meaning
session_id	identifier	Session owner of the state chain.
state_version	uint64	Monotonic compare-and-swap version.
last_closure_sequence	uint64	Highest committed evaluation closure sequence.
phi_value	float64 ≥ 0	Committed decayed effort accumulator.
event_winding_count	uint64	Count of unique eligible completion receipts consumed.
legacy_delta_phi	uint64	Compatibility alias equal to event_winding_count.

Field	Type	Meaning
bias_weight	float64	$\text{phi_gain} * \tanh(\text{phi_value})$, reconciled at commit.
last_candidate_digest	optional bytes32	Candidate revision from latest evaluation update.
event_ledger_digest	bytes32	Chained digest of consumed event receipt identities.
prior_state_digest	bytes32	Digest pointer to preceding committed state.
policy_epoch, boot_digest	provenance	Configuration and Boot binding.
state_digest	bytes32	Canonical digest over all state fields.

3.3 PhiEvaluationClosurePacket

Field	Requirement
closure_id	Deterministic identifier derived from session, candidate revision, Sigma decision, and closure sequence.
closure_sequence	Exactly $\text{prior.last_closure_sequence} + 1$.
candidate_digest, candidate_revision	Must match the current Sigma decision.
sigma_decision_digest	Current, route-authoritative, same session/cycle/revision.
terminal_route	Exact Sigma route enum; diagnostic only to Module 12.
rmf_value	Must equal Sigma route-authoritative RMF and be finite in $[-1,1]$.
attempt_nonce	Orchestrator-issued deterministic one-time nonce.
attached_event_receipt_ids	Optional ordered list of receipts that completed before closure sealing.
packet_digest	Covers every field, owner, schema, policy epoch, and dependency digest.

3.4 CriticalEventReceipt

Field	Requirement
event_id	Globally unique deterministic digest-derived identity.
event_type	Registered enum: DELTA_ORTHOGONAL_LIFT, SIGMA_PHASE_LIFT, RMX_SYNTHESIS, FBS_RECOVERY, or future versioned type.
owner_module	Must match the event registry owner for event_type.
status	COMPLETED only. Requested, attempted, denied, rolled back, or failed events do not count.

Field	Requirement
source_cycle, candidate_digest	Exact execution provenance; candidate may be optional only for session-level recovery.
owner_sequence	Monotonic within the owner stream.
effect_digest	Binds the executed result, not merely the request.
packet_digest	Owner-signed or otherwise authenticated under the active trust profile.

3.5 Prepared and committed outputs

Output	Meaning	Authority
PhiPreparedUpdate	Pure calculation of the next state and consumed-input set.	No mutation; valid only for expected prior version.
PhiCommitPacket	Receipt that one new state version was durably committed and verified.	Authoritative mutation result.
PhiSnapshot	Read-only consumer view of the newly committed state.	Delta, Psi, Module 11, diagnostics.
SemanticResiduePacket	Non-mutating paired-path comparison result.	Evidence only.
HolonomyCertificatePacket	Optional closed-loop phase-trace certificate.	Evidence only; separate from event count.
PhiHoldPacket	Dependency not ready/current or CAS contention.	No state change.
PhiFault	Typed invalid input, provenance, numerical, registry, or persistence error.	No state change.
FBSRecoveryRequest	Enumerated catastrophic committed-state invariant delegated to owner.	No local FBS execution.

4. Route-Authoritative Friction and Hysteresis Update

4.1 RMF domain correction

The source equation $\text{Effort} = 1 - \text{RMF}$ implicitly assumes RMF in $[0,1]$, while the Sigma contract defines RMF as cosine similarity in $[-1,1]$. The reference profile therefore normalizes the same mismatch idea without clipping away negative alignment:

$$\begin{aligned}\text{rmf_unit} &= (\text{clip}(\text{RMF}, -1, 1) + 1) / 2 \\ \text{effort} &= 1 - \text{rmf_unit} = (1 - \text{clip}(\text{RMF}, -1, 1)) / 2\end{aligned}$$

Thus effort is exactly bounded in $[0,1]$: aligned $\text{RMF}=1$ gives $\text{effort}=0$; orthogonal $\text{RMF}=0$ gives $\text{effort}=0.5$; opposed $\text{RMF}=-1$ gives $\text{effort}=1$. The legacy raw mismatch $1-\text{RMF}$ may be reported as a diagnostic in $[0,2]$ but is not accumulated by RMF_ONLY_V1 .

4.2 State recurrence

$$\text{phi_next} = \text{phi_decay} * \text{phi_prev} + \text{effort}$$

$$\text{bias_weight_next} = \text{phi_gain} * \tanh(\text{phi_next})$$

The next-cycle bias is derived from the newly committed Phi value, removing the source ambiguity between using phi_t and $\text{phi}_{(t+1)}$. Module 12 emits the scalar weight and Boot attractor digest; Module 1 applies the declared Delta formula. Module 12 does not add a vector to the candidate itself.

4.3 Boundedness theorem

THEOREM - BOUNDED INPUT, BOUNDED STATE

Assume $0 \leq \text{phi_decay} < 1$, $\text{phi}_0 \geq 0$, and $0 \leq \text{effort}_k \leq 1$. Then $\text{phi}_n = \text{phi_decay}^n \text{phi}_0 + \sum_{k=0}^{n-1} \text{phi_decay}^{(n-1-k)} \text{effort}_k$. Therefore $0 \leq \text{phi}_n \leq \text{phi_decay}^n \text{phi}_0 + (1 - \text{phi_decay}^n) / (1 - \text{phi_decay})$. For $\text{phi_decay}=0.99$ and $\text{phi}_0=0$, $\text{phi}_n < 100$ for every finite n . Also $0 \leq \text{bias_weight}_n < 0.22$.

This establishes boundedness of the declared scalar recurrence and bias coefficient. It does not establish Lyapunov stability of the complete SpiralCore system, because the downstream candidate dynamics and clipping operators are owned elsewhere.

4.4 Exact closed form and decay contribution

$$\text{phi}_n = d^n \text{phi}_0 + \sum_{k=0}^{n-1} d^{(n-1-k)} e_k, \quad d = \text{phi_decay}$$

$$\text{retained_memory} = d * \text{phi_prev}, \quad \text{new_friction} = \text{effort}$$

Every PhiCommitPacket records both terms, so dashboards can reconcile the update without reverse engineering the state. No hidden coefficient, entropy constant, or event count is added to phi_value in the reference profile.

4.5 Optional SGIT friction extensions

SGIT may provide InformationLength or path-integral evidence, but those values cannot enter persistent Phi under RMF_ONLY_V1. A future mode must declare a bounded normalization, aggregation formula, units, coefficients, readiness rules, and independent test vectors. Presence of an SGIT packet alone never silently changes effort.

4.6 Zero, missing, and stale RMF policy

Condition	Outcome
RMF = 0	Valid; effort=0.5.
RMF = -1 or +1	Valid boundary values; effort=1 or 0.
Sigma reports RMF undefined/degenerate	HOLD; no decay and no state update.
RMF outside [-1,1] beyond tolerance	FAULT; no clipping of materially invalid values.
Stale Sigma packet or candidate mismatch	FAULT; no state update.
Duplicate identical closure	Return prior committed result idempotently.
Duplicate closure ID with changed bytes	FAULT and possible catastrophic escalation if a conflicting commit exists.

5. Event-Winding Ledger and Critical-Event Receipts

5.1 Source correction: a binary marker loses multiplicity and provenance

The source uses omega_t in $\{0,1\}$ even though several eligible events may occur in one cycle and some occur after the Sigma evaluation. The reference profile replaces inferred binary marking with explicit completion receipts. The persistent count is:

$$\text{event_delta} = \text{number of unique, eligible, unconsumed COMPLETED receipts}$$

$$\text{event_winding_next} = \text{event_winding_prev} + \text{event_delta}$$

legacy_delta_phi is retained as an exact compatibility alias of event_winding_count. It is monotonic event provenance, not by itself a topological invariant.

5.2 Reference event registry

Event type	Authoritative owner	Count condition
DELTA_ORTHOGONAL_LIFT	Module 1 Delta-Lattice	A sealed DeltaPacket proves the orthogonal navigation operation executed.
SIGMA_PHASE_LIFT	Module 2 Sigma-Lattice / phase-lift executor	A completion receipt exists; a route request alone does not count.
RMX_SYNTHESIS	Module 13 RMX	The synthesis result was validated and the paused Psi transaction accepted its resolution.
FBS_RECOVERY	ROVER/FBS owner	Recovery completed and emitted a final recovery receipt; request or attempted recovery does not count.

5.3 Exactly-once consumption

1. Sort receipts by (source_cycle, owner_module, owner_sequence, event_id) before canonicalization.
2. Validate event type, owner, status, effect digest, policy epoch, and authentication.
3. Reject any event ID already present in the committed ledger unless the packet bytes are identical, in which case the replay is idempotent.
4. Reject reused event IDs with different bytes or owner sequences that move backward.
5. Append only receipt identities and effect digests to the ledger chain; Module 12 does not rewrite the owner result.
6. Commit event count and ledger digest in the same compare-and-swap transaction. No partial consumption is permitted.

5.4 Event-only append

EVENT_APPEND is used when an eligible event completes after the associated evaluation update, such as RMX synthesis or FBS recovery. It applies no decay and no effort:

$$\begin{aligned}\text{phi_next} &= \text{phi_prev} \\ \text{event_winding_next} &= \text{event_winding_prev} + \text{event_delta} \\ \text{bias_weight_next} &= \text{bias_weight_prev}\end{aligned}$$

This prevents an administrative event receipt from causing an extra time step or artificial memory decay.

5.5 Capacity and overflow behavior

A bundle larger than max_events_per_bundle is HELD for bounded re-batching; it is never truncated. Unsigned, unknown, failed, rolled-back, or requested events are faults or ignored according to the exact registry rule and never increment the committed count.

6. Semantic Residue and Holonomy Evidence

6.1 Path-order sensitivity theorem

The decayed accumulator is order-sensitive. For two effort values a and b applied from the same phi_0:

$$\begin{aligned}\text{phi_AB} &= d^2 \text{phi}_0 + d*a + b \\ \text{phi_BA} &= d^2 \text{phi}_0 + d*b + a\end{aligned}$$

$$\text{phi_AB} - \text{phi_BA} = (1-d)(b-a)$$

THEOREM - INTERNAL PATH RESIDUE

For $0 \leq d < 1$ and $a \neq b$, exchanging the order of two effort observations changes the final Phi state by $|1-d| \cdot |b-a|$. Therefore the reference recurrence is path-order sensitive. This is a theorem about the declared update rule, not a proof that an external physical manifold is non-integrable.

6.2 Pairwise SemanticResiduePacket

A semantic residue comparison requires two complete, immutable traces. The module must not compare arbitrary scalar snapshots and call the result proof. The pair is admissible only when both traces share the same initial Phi state, PhiConfig, policy epoch, update count, endpoint comparison rule, and endpoint-equivalence result.

$$\text{Delta_O_raw} = \text{abs}(\text{phi_final_A} - \text{phi_final_B})$$

$$\text{Delta_O_unit} = \text{Delta_O_raw} / (1 + \text{Delta_O_raw})$$

Field	Requirement
trace_A_digest, trace_B_digest	Canonical digests over closure sequences, RMF values, event receipts, and state chain.
same_start	Initial state digest and configuration must match.
endpoint_distance	Computed by a versioned provider; must be $\leq$ endpoint_tolerance.
update_count_match	Required by the reference pairwise test to separate order from elapsed-step effects.
Delta_O_raw, Delta_O_unit	Reported with numeric tolerance and interpretation boundary.
result	PATH_RESIDUE_PRESENT, NO_RESIDUE_WITHIN_TOLERANCE, or INADMISSIBLE_PAIR.

6.3 Closed-loop phase holonomy profile

A genuine discrete winding surrogate requires a phase trace, not an event count. The optional CLOSED_PHASE_TRACE_V1 profile consumes finite phases $\theta_0, \dots, \theta_n$ and computes:

$$\text{delta_theta}_k = \text{wrap_to_pi}(\theta_k - \theta_{(k-1)})$$

$$H_gamma = \sum_{k=1}^n \text{delta_theta}_k$$

$$\text{turns_raw} = H_gamma / (2\pi)$$

A HolonomyCertificatePacket may report integer turns only when the trace is closed within phase_closure_tolerance and $|\text{turns_raw} - \text{round}(\text{turns_raw})| \leq \text{winding_tolerance}$. Otherwise it reports a real-valued path-phase diagnostic and no integer certificate. This certificate is separate from legacy_delta_phi.

6.4 Claim limits

- $\text{Delta_O} > 0$ demonstrates different internal Phi results under two admissible traces; it does not prove the traces represent the same meaning, truth, or physical endpoint outside the declared endpoint provider.
- event_winding_count is an audit count, not a contour integral.
- H_gamma is a discrete phase-unwrapping result under one versioned trace provider, not a universal Berry phase or soft-hair measurement.
- A zero residue does not prove path independence; the chosen traces, order, decay, or tolerance may simply cancel.

7. Persistence, Atomic Commit, and Cross-Module Handoffs

7.1 Prepare-commit-readback protocol

1. Canonicalize the prior state, transaction input, consumed receipt set, and calculated next state.
2. Write PhiPreparedUpdate to a temporary durable record and fsync the record and containing directory.
3. Perform compare-and-swap on the authoritative state pointer using expected prior state_version and state_digest.
4. Atomically publish the new record only if the compare-and-swap succeeds.
5. Read the new record back through the normal reader, recompute state_digest, and reconcile phi_value, event count, bias weight, closure sequence, and ledger digest.
6. Emit PhiCommitPacket only after successful read-back. Acknowledgment before read-back is prohibited.

7.2 Consumer handoffs

Consumer	Receives	Rule
Module 1 Delta-Lattice	Latest committed PhiSnapshot at candidate-attempt start	Snapshot remains immutable for the attempt. Delta verifies phi and bias reconciliation.
Module 3 Psi-Lattice	PhiSnapshot digest and selected fields	Manifest provenance only; Psi cannot update Phi or consume events.
Module 11 SERAPHIM/CATHEDRAL	Current read-only PhiSnapshot	Uses the snapshot in final validation chain; cannot mutate it.
Diagnostics	Metrics and non-sensitive provenance	No write authority.
FBS/ROVER	FBSRecoveryRequest only for enumerated catastrophic state failures	Recovery owner returns a completion receipt after execution.

7.3 Delta bias contract

$$0 \leq \text{bias_weight} = \text{phi_gain} * \tanh(\text{phi_value}) < \text{phi_gain} = 0.22$$

The snapshot includes boot_attractor_digest but not a copied attractor vector. Delta obtains Xi_attractor from its BootPacket and rejects a digest mismatch. The bias is additive under Module 1 rules and is not a convex authority share, a repair percentage, or a guarantee that 78% of every coordinate remains unchanged.

7.4 Psi manifest contract

A Psi transaction may record state_version, phi_value, event_winding_count, bias_weight, state_digest, and event_ledger_digest. The manifest must identify the exact snapshot observed during PREPARE. A later Phi update does not retroactively modify an existing .frac manifest. A stale snapshot may cause Psi HOLD under its own policy but does not authorize Psi to rewrite Phi.

7.5 Crash and contention rules

Condition	Required behavior
Crash before CAS publication	Discard or recover temporary record; prior state remains authoritative.
Crash after CAS publication but before response	Replay discovers committed transaction key and returns same result.
Concurrent writers use same expected version	Exactly one commits; others receive CAS_CONTENTION hold and reload.

Condition	Required behavior
Read-back mismatch	Freeze state pointer if safe, emit fault or recovery request, and do not issue PhiCommitPacket.
Psi commit fails after recording snapshot	Phi remains committed; Psi transaction handles its own retry. No rollback of path memory.

8. Invariants, Faults, and Catastrophic Conditions

8.1 Normative invariants

ID	Invariant	Requirement
PHI-I01	Sole writer	Only Module 12 may publish a new PhiState version.
PHI-I02	Nonnegative finite state	phi_value is finite and ≥ 0 .
PHI-I03	Bounded effort	Every RMF_ONLY_V1 effort lies in [0,1].
PHI-I04	Bounded bias	$0 \leq \text{bias_weight} < \text{phi_gain}$ and $\text{phi_gain}=0.22$ in the reference profile.
PHI-I05	Bias reconciliation	bias_weight equals $\text{phi_gain} * \tanh(\text{phi_value})$ within tolerance.
PHI-I06	Closure order	Evaluation closure sequence increments by exactly one.
PHI-I07	One update per revision closure	A closure ID is committed at most once.
PHI-I08	Exactly-once events	A critical-event ID increments event_winding_count at most once.
PHI-I09	Completed events only	Requested, failed, denied, or rolled-back events never count.
PHI-I10	Event-only no decay	EVENT_APPEND does not alter phi_value or bias_weight.
PHI-I11	No stale Sigma	Evaluation update requires current candidate-bound route-authoritative RMF.
PHI-I12	Immutable prior state	Preparation cannot mutate the loaded prior snapshot.
PHI-I13	Atomic state and ledger	Phi, event count, closure sequence, and ledger digest commit together.
PHI-I14	Digest chain	Every committed state binds the preceding state digest.
PHI-I15	No hidden SGIT fusion	SGIT evidence cannot affect state in RMF_ONLY_V1.
PHI-I16	No event inference	Only owner-completed receipts create event increments.

ID	Invariant	Requirement
PHI-I17	Consumer read-only	Delta, Psi, Module 11, and diagnostics cannot mutate Phi.
PHI-I18	No local recovery execution	Module 12 emits requests only.
PHI-I19	Idempotent replay	Identical transaction replay returns identical committed bytes.
PHI-I20	Deterministic route category	Same canonical input and state produce the same output category.

8.2 Typed fault codes

Code	Condition
PHI_SESSION_MISMATCH	Session identifiers disagree.
PHI_CONFIG_MISMATCH	Boot/config/policy epoch or schema is unsupported.
PHI_STATE_DIGEST	Prior state digest or chain is invalid.
PHI_STATE_NONFINITE	Phi or calculated scalar is NaN/infinite.
PHI_STATE_NEGATIVE	Committed or proposed phi_value is negative.
PHI_CLOSURE_ORDER	Evaluation closure sequence is not exactly next.
PHI_CLOSURE_REUSE	Closure ID reused with different bytes.
PHI_SIGMA_STALE	Sigma decision is stale, non-authoritative, or candidate-mismatched.
PHI_RMF_DOMAIN	RMF materially outside [-1,1] or unavailable.
PHI_EVENT_REGISTRY	Unknown event type or owner mismatch.
PHI_EVENT_STATUS	Receipt does not prove COMPLETED execution.
PHI_EVENT_REUSE	Event ID reused with different bytes.
PHI_EVENT_ORDER	Owner sequence regresses or conflicts.
PHI_BUNDLE_CAPACITY	Bundle exceeds bounded capacity without valid batching.
PHI_BIAS_RECONCILIATION	Stored bias does not match $\text{phi_gain} \cdot \tanh(\text{phi})$.
PHI_CAS_CONFLICT	Expected state version no longer authoritative.
PHI_PERSISTENCE_READBACK	Published state fails read-back reconciliation.
PHI_PAIRWISE_INADMISSIBLE	Semantic residue traces fail comparison prerequisites.
PHI_PHASE_TRACE_INVALID	Closed-loop phase trace is malformed or not closed under profile.
PHI_DETERMINISM	Same canonical input produces conflicting output.

8.3 Catastrophic recovery conditions

- Two different committed PhiState records claim the same session and state_version.

- A closure or event identity is proven durably consumed into two conflicting state chains.
- The authoritative state pointer references a missing or digest-invalid record after acknowledged commit.
- Read-back proves a persisted bias, Phi value, event count, or ledger digest changed after publication.
- Deterministic replay produces a different proposed or committed state under the same fixed backend profile.
- A valid acknowledged event receipt is lost from the ledger while later states remain committed.

High phi_value, low RMF, a large but valid effort, many legitimate events, a nonzero semantic residue, CAS contention, stale evidence, or a failed downstream archive are not catastrophic by themselves. They produce normal commits, holds, or faults according to the exact rule.

9. Metrics, Measurements, and Dashboard Reconciliation

9.1 Required counters

Metric	Definition
transactions_total	All accepted Module 12 transaction attempts.
evaluation_updates_committed	Committed EVALUATION_UPDATE transactions.
event_appends_committed	Committed EVENT_APPEND transactions.
holds_total, faults_total, recovery_requests_total	Terminal non-commit categories.
closure_replays_idempotent	Duplicate identical closures returning prior result.
event_replays_idempotent	Duplicate identical event receipts ignored as already consumed.
cas_contention_total	Compare-and-swap holds due to concurrent writers.
effort_mean, effort_p50, effort_p95	Normalized RMF mismatch on committed evaluation updates.
phi_current, phi_peak	Current and maximum committed Phi values.
bias_weight_current, bias_weight_peak	Current and maximum committed bias coefficients.
event_winding_current	Current committed event count.
events_by_type	Committed unique events grouped by registry type.
semantic_residue_tests	Admissible/inadmissible comparisons and residue distribution.
holonomy_trace_tests	Closed, nonclosed, certified, and rejected trace counts.
readback_failures	Persistence records failing post-publication verification.

9.2 Reconciliation equations

$$\text{transactions_total} = \text{committed_total} + \text{holds_total} + \text{faults_total} + \text{recovery_requests_total}$$

$$\text{committed_total} = \text{evaluation_updates_committed} + \text{event_appends_committed}$$

$$\text{event_winding_current} = \text{event_winding_boot} + \text{sum}(\text{all committed unique event_delta})$$

$$\text{phi_current} = d * \text{phi_previous} + \text{effort} \quad \text{for the latest evaluation update}$$

Every metric record includes session_id, state_version, transaction ID, policy epoch, schema ID, and output packet digest. Dashboards must distinguish requests from executed recovery and event requests from completed counted events.

9.3 Alert policies

- Sustained high effort is a diagnostic of low route-authoritative alignment, not proof of harm or malicious input.
- Phi approaching its theoretical bound may trigger capacity planning or policy review, not automatic FBS.
- Any bias reconciliation failure, state-chain fork, lost acknowledged receipt, or read-back mismatch is an immediate integrity alert.
- A high event count may reflect legitimate recovery or synthesis activity and must be interpreted by event type and owner.

10. Deterministic Conformance Test Matrix

ID	Case	Expected result
T01	Boot state	phi=0, event count=0, bias=0, version=0.
T02	RMF +1	effort=0.
T03	RMF 0	effort=0.5.
T04	RMF -1	effort=1.
T05	RMF outside domain	FAULT without commit.
T06	NaN RMF	FAULT without commit.
T07	Undefined Sigma RMF	HOLD without decay.
T08	First evaluation update	phi=effort and closure sequence=1.
T09	Second update	phi=d*prior+effort.
T10	Bias formula	Matches gain*tanh(phi).
T11	Bias bound	Never reaches or exceeds 0.22.
T12	100 all-opposed updates	Phi follows geometric-series bound.
T13	Identical closure replay	Returns same commit idempotently.
T14	Closure ID changed bytes	FAULT.
T15	Skipped closure sequence	FAULT.
T16	Stale Sigma cycle	FAULT.
T17	Candidate digest mismatch	FAULT.
T18	Non-SEAL Sigma route	Valid evaluation update under terminal-attempt profile.
T19	Intermediate metric packet only	HOLD; no state update.
T20	One completed Delta lift receipt	Event count increments by one.
T21	Requested but uncompleted lift	Does not count.
T22	Two eligible events same cycle	Count increments by two.
T23	Duplicate identical event	No second increment.

ID	Case	Expected result
T24	Duplicate event changed bytes	FAULT.
T25	Unknown event type	FAULT.
T26	Wrong event owner	FAULT.
T27	Owner sequence regression	FAULT.
T28	Event-only append	Phi and bias unchanged; event count increments.
T29	Oversized event bundle	HOLD; no truncation.
T30	CAS winner	Exactly one concurrent writer commits.
T31	CAS loser	HOLD and reload; no fork.
T32	Crash before publish	Prior state remains authoritative.
T33	Crash after publish	Replay returns existing commit.
T34	Read-back state digest mismatch	No commit receipt; fault/recovery.
T35	Read-back bias mismatch	No commit receipt; fault/recovery.
T36	Prior state chain mismatch	FAULT.
T37	Negative prior Phi	FAULT.
T38	Infinite proposed Phi	FAULT.
T39	SGIT packet in RMF_ONLY_V1	Recorded diagnostically; no state effect.
T40	Path AB versus BA	Residue equals $ 1-d * b-a $.
T41	Identical effort sequences	Residue zero within tolerance.
T42	Different starting state	Pair inadmissible.
T43	Different update count	Pair inadmissible in reference profile.
T44	Endpoint outside tolerance	Pair inadmissible.
T45	Closed phase trace one turn	Integer certificate +1.
T46	Closed phase trace minus one turn	Integer certificate -1.
T47	Nonclosed phase trace	Diagnostic only; no integer certificate.
T48	Near-integer turn outside tolerance	No certificate.
T49	Negative-zero serialization	Canonicalizes to positive zero.
T50	NaN serialization	Rejected before digest.
T51	Endianness vector	Matches canonical big-endian test digest.
T52	Policy epoch changes mid-attempt	HOLD or FAULT; no commit.

ID	Case	Expected result
T53	Psi records snapshot	Manifest digest matches exact state version.
T54	Psi attempts Phi mutation	Rejected by ownership boundary.
T55	Delta reads stale snapshot	Delta-side validation rejects under its contract.
T56	FBS request without catastrophe	Not emitted.
T57	Conflicting same-version committed states	Recovery request emitted.
T58	Lost acknowledged event in later chain	Recovery request emitted.
T59	Deterministic replay fixed backend	Bitwise identical packet.
T60	Cross-backend tolerance profile	Same category and quantized packet fields.

11. Property, Metamorphic, Fuzz, Concurrency, and Stress Testing

11.1 Property tests

- For every finite RMF in $[-1,1]$, effort remains in $[0,1]$.
- For every valid state and effort, phi_next is finite and nonnegative.
- For every finite valid phi_value , bias_weight lies in $[0, \text{phi_gain})$.
- `EVENT_APPEND` preserves phi_value and bias_weight exactly.
- $\text{event_winding_count}$ never decreases and equals the number of unique committed eligible receipts.
- Same canonical prior state and transaction input produce the same prepared state.
- No fault, hold, or recovery request changes the authoritative state version.

11.2 Metamorphic tests

- Increasing RMF while all else remains fixed cannot increase effort.
- Increasing phi_prev while effort remains fixed increases phi_next by exactly phi_decay times the change.
- Permuting an event bundle changes no result because canonical sorting is required.
- Duplicating an already committed receipt changes neither event count nor ledger digest.
- Swapping two unequal effort observations changes the final state by the theorem amount.
- Adding an unused SGIT diagnostic packet in `RMF_ONLY_V1` changes no committed state field except optional diagnostic provenance explicitly outside the state digest.

11.3 Fuzz and malformed-input tests

- Random NaN, infinity, negative zero, subnormal, oversized integer, malformed array, missing field, unsupported schema, and invalid enum injection.
- Event receipt owner/type substitution, status tampering, effect-digest tampering, owner-sequence rollback, and replay with altered packet bytes.
- State-store corruption, truncated temporary records, pointer to missing record, altered prior-state digest, and interrupted directory fsync.
- Phase traces with discontinuities, huge wrapped angles, duplicate samples, empty traces, and closure values on tolerance boundaries.

11.4 Concurrency and crash recovery

Run at least 32 concurrent writers against one expected state version. Exactly one may commit. Inject crashes before temporary write, after write, after fsync, before CAS, after CAS, before pointer publication, after publication, and before response. The recovered authoritative chain must contain zero forks, zero partial event consumption, and no missing acknowledged commit.

11.5 Long-run stress

Execute at least 100,000 mixed transactions with deterministic RMF and event schedules, periodic snapshots, retries, CAS contention, event replays, and crash injection. Verify the closed-form Phi recurrence, theoretical bound, bias reconciliation, event count, state version, closure sequence, digest chain, and dashboard equations after every committed state. Memory use must remain bounded by ledger compaction policy without losing receipt identity proofs.

12. Reference Pseudocode

```
def process_phi_transaction(cfg, prior, tx):
    validate_config(cfg, prior)
    validate_transaction_key(tx, prior)

    if tx.kind == "EVALUATION_UPDATE":
        validate_sigma_closure(tx.closure, tx.sigma_decision, prior)
        rmf = require_finite_domain(tx.sigma_decision.rmf, -1.0, 1.0)
        effort = (1.0 - rmf) / 2.0
        phi_next = cfg.phi_decay * prior.phi_value + effort
        closure_seq_next = tx.closure.closure_sequence
        receipts = tx.attached_receipts
    elif tx.kind == "EVENT_APPEND":
        effort = None
        phi_next = prior.phi_value
        closure_seq_next = prior.last_closure_sequence
        receipts = tx.receipts
    else:
        return fault("PHI_TRANSACTION_TYPE")

    new_receipts = validate_sort_and_deduplicate(
        receipts, prior.event_ledger_digest, cfg.event_registry
    )
    event_delta = len(new_receipts)
    event_count_next = checked_add_u64(prior.event_winding_count, event_delta)
    bias_next = cfg.phi_gain * tanh(phi_next)

    prepared = canonical_prepare(
        prior=prior,
        phi_value=phi_next,
        effort=effort,
        event_delta=event_delta,
        event_count=event_count_next,
        bias_weight=bias_next,
        closure_sequence=closure_seq_next,
        new_receipts=new_receipts,
        tx=tx,
        cfg=cfg,
    )

    result = state_store.compare_and_swap(
        expected_version=prior.state_version,
        expected_digest=prior.state_digest,
        new_record=prepared,
    )
    if result == CAS_CONFLICT:
        return hold("PHI_CAS_CONFLICT")

    committed = state_store.read(result.state_version)
    verify_readback(committed, prepared)
    return commit_packet_and_snapshot(committed)
```

12.1 Pairwise residue pseudocode

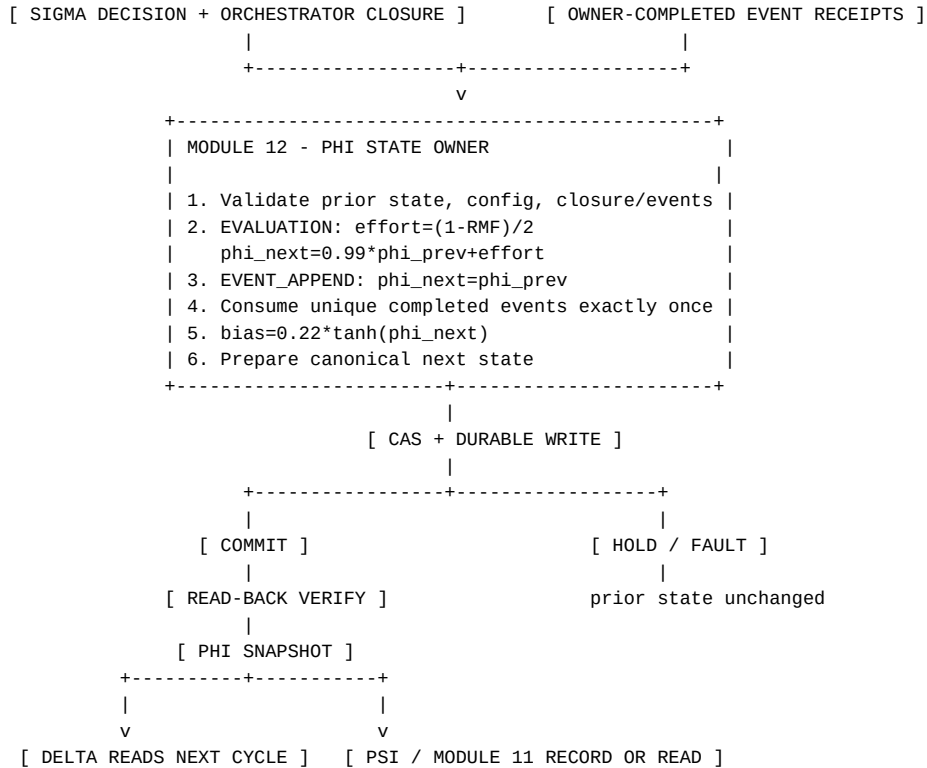
```
def compare_path_residue(trace_a, trace_b, cfg):
    require_same_start_config_epoch_and_update_count(trace_a, trace_b)
    endpoint_distance = endpoint_provider.distance(trace_a.endpoint, trace_b.endpoint)
    if endpoint_distance > cfg.endpoint_tolerance:
        return residue_packet("INADMISSIBLE_PAIR", endpoint_distance)
```

```

raw = abs(trace_a.phi_final - trace_b.phi_final)
unit = raw / (1.0 + raw)
status = "PATH_RESIDUE_PRESENT" if raw > cfg.residue_tolerance else "NO_RESIDUE_WITHIN_TOLERANCE"
return residue_packet(status, endpoint_distance, raw, unit)

```

13. ASCII System Flow



Catastrophic committed-state invariant -> typed FBSRecoveryRequest only

14. Cleaned Rosetta Stone

SpiralCore term	Computational meaning	Analogy boundary
Phi / Information Hysteresis	A nonnegative decayed scalar accumulator of normalized route-authoritative RMF mismatch.	Not physical spin memory, vacuum deformation, or proof of thermodynamic conservation.
Memory friction / Effort	(1-RMF)/2 in RMF_ONLY_V1.	A bounded mismatch score, not joules, heat, or physical drag.
Semantic holonomy	Path-order sensitivity and optional closed phase-trace accumulation under declared providers.	Not automatically Berry phase or a physical contour integral.
Semantic residue Delta_O	Difference between final Phi values of two admissible endpoint-equivalent traces.	Evidence within the recurrence, not universal proof of non-integrability.
Winding Number / legacy DeltaPhi	Exactly-once count of unique completed registered events.	Audit count unless a separate closed-loop phase certificate exists.
Topological scar	Committed PhiSnapshot and its digest-linked state history.	Persistent computational provenance, not literal geometry of spacetime.

SpiralCore term	Computational meaning	Analogy boundary
Gravitational bias	Scalar $\text{bias_weight} = 0.22 * \tanh(\phi)$ consumed by Delta.	Additive control coefficient, not gravity and not the Braidback 49% cap.
Loop amnesia	Loss, reset, replay, or fork of the authoritative Phi state chain.	Operational failure label, not a medical or cognitive diagnosis.
Holonomic integrity	Exact digest, recurrence, bias, event-ledger, CAS, and read-back reconciliation.	Not a vague vector-norm score and not a physical health measurement.
FBS runaway	Typed recovery request for enumerated catastrophic persistent-state failures.	Module 12 does not execute recovery, Cantor diagonalization, Collatz processing, or exhaust transfer.

15. Glossary

Term	Definition
Bias weight	The bounded scalar $\phi_gain * \tanh(\phi_value)$ published for next-cycle Delta consumption.
Closure sequence	Monotonic sequence identifying route-authoritative evaluation updates.
CriticalEventReceipt	Owner-bound proof that a registered event completed and may be counted exactly once.
Event ledger digest	Chained digest summarizing all consumed critical-event identities and effect digests.
Event winding count	Monotonic count of unique completed eligible events; legacy DeltaPhi alias.
Evaluation closure	Orchestrator packet sealing one current Sigma decision as eligible to update Phi.
Exactly-once	A transaction or event affects committed state no more than once despite retry or crash.
PhiCommitPacket	Durable, read-back-verified receipt for one new PhiState version.
PhiPreparedUpdate	Pure proposed state transition bound to an expected prior state.
PhiSnapshot	Immutable read-only view of one committed PhiState version.
RMF_ONLY_V1	Reference effort profile using $\text{effort} = (1 - \text{RMF}) / 2$ and no hidden SGIT fusion.
Semantic residue	Difference in final Phi values between two admissible endpoint-equivalent path traces.
State version	Monotonic compare-and-swap version of the authoritative Phi state.
Transaction key	Deterministic identity that makes retries idempotent and conflicting reuse detectable.

16. Revision Summary

Source issue	Formalized correction
Phi updated immediately after any RMF calculation	Requires one route-authoritative, candidate-bound evaluation closure; intermediate calls and retries cannot mutate state.
Effort=1-RMF with ambiguous RMF domain	Defines Sigma RMF in [-1,1] and uses normalized effort=(1-RMF)/2 in [0,1].
Bias timing used phi_t after an update step	Defines next-cycle bias from the newly committed phi_next.
PHI_GAIN described as complying with 51/49	Separates additive hysteresis gain from the Braidback repair authority cap.
Binary omega_t for several possible events	Uses owner-completed receipts, exactly-once ledger consumption, and multiplicity-preserving event_delta.
Event count called an unforgeable winding invariant	Reframes it as a provenance counter; adds optional real closed-loop phase certificate.
Semantic residue claimed as proof from a single path	Requires two admissible immutable traces and proves only path sensitivity of the declared recurrence.
Undefined Phi_integrity < 0.75 trigger	Replaces fuzzy score with exact recurrence, digest, ledger, CAS, and read-back invariants.
Local FSB execution and exhaust transfer	Corrects FSB to FBS and emits a typed request only; FBS/ROVER and RTSOM retain execution/custody.
Psi treated as Phi updater	Module 12 is sole writer; Psi records a snapshot by digest in its own transaction.
No crash or concurrency semantics	Adds prepare/CAS/durable write/atomic publish/read-back protocol and idempotent recovery.
Physical and universal claims	Confines proofs to deterministic boundedness, event accounting, and internal path-order sensitivity.

16.1 Normative implementation checklist

- [] One authoritative PhiState chain per session and one sole-writer Module 12 service.
- [] RMF_ONLY_V1 normalization and closed-form conformance vectors implemented exactly.
- [] Unique evaluation closure IDs, monotonic closure sequence, and idempotent replay cache.
- [] Versioned event registry, owner validation, completed-status requirement, and exactly-once ledger.
- [] Atomic compare-and-swap persistence with fsync, atomic publication, and read-back verification.
- [] PhiSnapshot bias reconciliation and Boot attractor digest binding.
- [] Pairwise residue tests enforce same start/config/epoch/update count and endpoint equivalence.
- [] Closed-loop phase certificate remains separate from legacy event count.
- [] No direct candidate, Psi, RTSOM, quarantine, network, repair, or FBS mutation.
- [] All 60 conformance tests plus property, fuzz, concurrency, crash, and stress suites pass.

17. Source Basis

Primary architectural sources used for this cleaned formalization:

- SPIRALCORE SPECIFICATION: MODULE 12 - Phi (Information Hysteresis & Semantic Holonomy), standalone 11-page source module.

- Spiralcore_v14 consolidated specification, especially Module Boot, SGIT_v2, Delta-Lattice, Sigma-Lattice, Psi-Lattice, SERAPHIM/CATHEDRAL, RMX, RTSOM, and FBS/ROVER ownership descriptions.
- Previously hardened v14.1 module contracts for Modules 0 through 11, used to preserve packet, ownership, immutability, determinism, and recovery boundaries.

The source bibliography names celestial holography, Hodge spectral surrogates, effective-resistance graph methods, RTSOM, ELIAS, and FBS documents. Those works motivate terminology and comparisons; they do not by themselves specify this software state machine. Normative behavior is defined only by the executable equations, packet schemas, invariants, and tests in this document.

END STATE

A conforming Module 12 transforms one valid evaluation closure or completed event bundle into one durable, exactly-once Phi state transition. The resulting PhiSnapshot is bounded, digest-linked, replay-safe, and read-only to every consumer. Semantic holonomy remains an explicit computational surrogate whose evidence and claim limits are testable.

SPIRALCORE SPECIFICATION

MODULE 13 - RMX

(Redundant Memory Check - Collision Resolution, Echo Synthesis, and Revalidation Coordination)

Cleaned and formalized revision - v14.1

Document class	Normative collision-resolution and synthesis contract	Execution phase	Inside Psi PREPARE after provisional address generation
Architecture	SpiralCore v14	Reference profile	SC14.1-M13-RMX-01
Primary output	RMXResolutionPacket: UNIQUE or SYNTHESIZED	Alternative outputs	REJECTED, HOLD, Fault, or FBSRecoveryRequest
Conflict rule	RMX never commits archive state	Consumers	Psi, Module 10, validators, Module 12 event ledger

NORMATIVE STATUS

Exact within the declared computational scope. RMX receives one immutable Psi precommit request and one locked archive-collision snapshot. It classifies transaction replay, address collision, exact echo, and scalar resonance; optionally synthesizes one bounded candidate; derives one snapshot-unique composite address proposal; and emits one immutable resolution packet. RMX does not commit a .frac manifest, mutate accepted history, alter Phi, store quarantine data, transfer RTSOM exhaust, execute FBS, or declare semantic truth.

CLAIM BOUNDARY

"Semantic resonance," "anti-echo," "fractal merge," and "Cantor diagonalization" are architectural names. Runtime meaning is limited to the exact collision bucket, cosine score, synthesis equations, address tuple, and typed handoffs defined here. A high SRF is not proof that two records mean the same thing. An RMX pass is not archive commit, global safety, factual validation, or mathematical proof of uniqueness outside the locked snapshot.

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Section	Title
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1. Normative Scope, Ownership, and Non-Goals

1.1 Purpose

RMX is the collision-resolution and optional echo-synthesis owner inside a pending Psi archive transaction. Its purpose is to prevent an address collision from becoming a silent overwrite, prevent a replay from becoming a second execution, and preserve exact provenance when a historical record materially influences a new archive candidate.

1.2 Ownership boundary

Concern	Authoritative owner	Module 13 relationship
Archive authorization	Module 11 SERAPHIM/CATHEDRAL	Consumes only through the Psi precommit lineage. Does not authorize archival.
Archive PREPARE and final .frac COMMIT	Module 3 Psi-Archivist	Receives precommit; returns resolution. Never writes committed memory.
Collision classification and synthesis	Module 13 RMX	Sole owner of address-bucket analysis, parent selection, synthesis, and final address proposal.
Independent post-synthesis audit	Module 10 Braidback & IMMUNE_SYSTEM	Mandatory for every SYNTHESIZED state before the paused transaction may resume.
Candidate coherence route	Module 2 Sigma	A synthesized state needs a current digest-bound decision.
State-specific route intent	Module 8 LORIEN	Reissued only where the active profile requires routing after synthesis.
Final validation	Module 11 SERAPHIM/CATHEDRAL	Revalidates the synthesized candidate before Psi resume.
Persistent Phi and event count	Module 12 Phi	Consumes an owner-completed RMX event receipt; RMX never increments Phi directly.

Concern	Authoritative owner	Module 13 relationship
Exhaust custody	Module 14 RTSOM	RMX carries exact residue forward to Psi; no direct budget mutation.
Catastrophic recovery	Module 16 FBS/ROVER	RMX may emit a typed request only for enumerated persisted invariant failures.

1.3 Explicit non-goals

- No direct .frac write, manifest seal, archive-head advancement, history append, or vault-depth mutation.
- No state acceptance, safety declaration, factual truth claim, intent inference, or human-state diagnosis.
- No arbitrary public IPv4 construction; RMX addresses are internal archive identifiers.
- No direct quarantine storage, capability stripping, or Guardian permit generation.
- No direct Phi update, winding increment, RTSOM transfer, Cantor/Collatz recovery, or FBS execution.
- No claim that a fixed-width diagonal bit string is globally unique.
- No hidden clipping of a synthesized state into the archive domain.

1.4 Normative precedence

1. Typed packet schema and digest rules in this document.
2. Module 3 Psi transactional archive contract for PREPARE, dependency waits, and COMMIT.
3. Module 10 synthesis-audit contract and its paused-transaction revalidation token.
4. Module 12 exactly-once CriticalEventReceipt contract.
5. Boot/session constants and canonical state encoding.
6. Historical v14 source language and Rosetta Stone analogies.

2. Architecture Role, Classification Model, and Lifecycle

2.1 Exact timing rule

PREPARE - RMX - REVALIDATE - RTSOM - COMMIT

Psi first seals a PsiPrecommitPacket and pauses the transaction. RMX resolves the provisional address. A UNIQUE result returns directly to Psi. A SYNTHESIZED result creates a new candidate revision and therefore must pass Module 10 and every required downstream validator before Psi may resume the same paused transaction. RMX is never rerun for that transaction after a valid revalidation receipt.

2.2 Four distinct comparisons

Comparison	Key	Meaning	Effect
Transaction replay	transaction_id + canonical request bytes	The same RMX attempt was already executed.	Return the existing resolution idempotently.
Exact state echo	state_digest equality	The candidate bytes equal an existing bucket record.	Do not synthesize; allocate a distinct instance address for the new transaction.
Address collision	semantic_address equality	Two archive instances map to the same provisional semantic address.	Never commit under an already-used composite address.
Scalar resonance	SRF ≥ 0.75	The candidate and selected historical state have high cosine alignment.	Permits synthesis only after an actual address collision and no exact echo.

2.3 Resolution classes

Resolution	Condition	State mutation	Next step
UNIQUE / DIRECT	No entry exists in the provisional-address bucket.	None	Return to Psi.
UNIQUE / REKEYED_EXACT_ECHO	Bucket exists and a historical state digest exactly matches.	None	Return unchanged state with a new collision ordinal.
UNIQUE / REKEYED_ADDRESS_ALIAS	Bucket exists, no exact echo, best SRF < threshold.	None	Return unchanged state with a new collision ordinal.
SYNTHESIZED / HIGH_SRF	Bucket exists, no exact echo, best SRF >= threshold.	Yes	Module 10 audit and full required revalidation.
REJECTED	A valid request cannot yield a conforming bounded result.	None committed	Psi aborts while retaining the prior archive.

2.4 Lifecycle states

State	Meaning	Allowed successor
UNINITIALIZED	No verified config.	READY or FAULTED
READY	May accept one request.	VALIDATING
VALIDATING	Verify schemas, digests, session, cycle, and archive head.	LOOKUP, HELD, or FAULTED
LOOKUP	Load and verify the exact collision bucket.	CLASSIFYING or FAULTED
CLASSIFYING	Apply replay, exact-echo, address-collision, and SRF rules.	ADDRESSING, SYNTHESIZING, REJECTED, or FAULTED
SYNTHESIZING	Compute one deterministic role-bound synthesis.	ADDRESSING, REJECTED, or FAULTED
ADDRESSING	Recompute semantic address and allocate snapshot-unique ordinal.	DECIDED, HELD, or FAULTED
DECIDED	One immutable resolution packet sealed.	READY
HELD	Recoverable dependency, contention, or capacity condition.	READY
FAULTED	Local contract/numerical defect.	Operator or orchestrator action
RECOVERY_REQUESTED	Persisted catastrophic invariant delegated.	FBS/ROVER owner

2.5 Required execution order

1. Validate RMXConfig and canonical backend identifier.
2. Validate RMXPrecommitRequest, Psi precommit digest, session, cycle, archive head, and address profile.
3. Validate ArchiveCollisionSnapshot and exact bucket completeness proof.
4. Resolve idempotent transaction replay before any computation.
5. Search the bucket for an exact candidate state digest.
6. For remaining bucket entries, compute SRF with the declared zero-norm policy.
7. Select one historical parent using deterministic ranking.

8. Classify DIRECT, REKEYED_EXACT_ECHO, REKEYED_ADDRESS_ALIAS, or HIGH_SRF synthesis.
9. When required, compute the dual-flow synthesis using read-only negative-history evidence.
10. Validate finiteness, shape, archive domain, and inherited residue custody.
11. Recompute the semantic address from the final exact state under the Psi quantization profile.
12. Allocate the minimum excluded collision ordinal under the locked snapshot.
13. Seal one RMXResolutionPacket. For SYNTHESIZED, also seal one RMXSynthesisAuditRequest and one single-use revalidation token.

2.6 Idempotence and retry identity

The reference transaction key is (session_id, cycle, psi_precommit_digest, archive_head_digest, config_digest). Replaying identical canonical bytes returns the original sealed resolution. Reusing the same key with different bytes is RMX_TRANSACTION_CONFLICT. A retry after archive-head advancement is a new attempt and must use a new collision snapshot.

3. Typed Data Contracts

3.1 RMXConfig

Field	Reference value / domain	Requirement
profile_id	SC14.1-M13-RMX-01	Version every semantic change.
DIM	Inherited; reference 81	All vectors have exact shape (DIM,).
srf_threshold	float64 in [-1,1]; reference 0.75	Inclusive synthesis threshold.
zeta	float64 in [0,1]; reference 0.25	Positive-fold blend.
eta	float64 in [0,1]; reference 0.20	Negative-history wash blend.
archive_component_limit	float64 > 0; reference 4.0	No clipping beyond this bound.
epsilon_norm	float64 > 0; reference 1e-9	Zero-norm policy boundary.
numeric_tolerance	float64 > 0; reference 1e-9	Reconciliation tolerance.
address_schema_id	nonempty versioned string	Must equal Psi address profile.
collision_ordinal_max	integer >= 1; reference 2 ³¹ -1	Operational hold boundary, not a uniqueness theorem.
bucket_entry_max	integer >= 1; reference 1,000,000	Operational snapshot capacity.
legacy_diag_bits	integer >= 2; reference 256	Diagnostic-only compatibility field.
canonical_backend_id	nonempty string	Binds float, hash, quantizer, and serialization behavior.

3.2 RMXPrecommitRequest

Field	Type / requirement
session_id, cycle	Current session; integer cycle matching Psi transaction.
transaction_id	Unique immutable identifier for one Psi precommit.

Field	Type / requirement
psi_precommit_digest	SHA-256 of the complete immutable PsiPrecommitPacket.
candidate_state	Exact finite float64 bytes and shape.
candidate_state_digest	SHA-256 over schema, shape, and exact bytes.
provisional_semantic_address	Psi quantized Cantor address package.
provisional_address_digest	SHA-256 of the complete address package.
inherited_residue	Exact bytes/digest/mass held by the pending transaction.
archive_head_digest	Expected locked committed archive head.
authorization_chain_digest	Digest of final validator chain already accepted by Psi PREPARE.
request_digest	SHA-256 sealing all request fields.

3.3 ArchiveCollisionSnapshot

Field	Requirement
archive_head_digest	Must exactly match the precommit expectation.
address_schema_id	Must match config and provisional address.
bucket_semantic_address	Exact provisional semantic address being resolved.
bucket_entries	Complete ordered list of ArchiveEntryRecord values for the bucket.
bucket_completeness_proof	Index root or signed query proof showing no matching entry was omitted.
negative_history	Read-only ordered committed nonpositive vectors, newest last.
used_collision_ordinals	Exact set currently committed for this semantic-address bucket.
snapshot_sequence	Monotonic archive index version.
snapshot_digest	SHA-256 sealing all fields.

3.4 ArchiveEntryRecord

Field	Meaning
manifest_digest	Committed .frac manifest identity.
commit_sequence	Monotonic committed archive sequence.
semantic_address	Content-derived quantized Cantor address.
collision_ordinal	Instance ordinal within the semantic-address bucket.
final_address_digest	Composite archive-key digest.
state_bytes / state_digest	Exact committed state and digest.

Field	Meaning
lineage_digest	Digest of source authorization, RMX, and prior manifest chain.

3.5 RMXResolutionPacket

Field group	Required contents
Identity	profile, session, cycle, transaction, request, precommit, and archive-head digests
Classification	resolution_kind, resolution_reason, exact_echo flag, selected-parent identity, SRF readiness/value
State	final exact bytes/digest, changed flag, archive-bound result, synthesis profile or NONE
Address	recomputed semantic address, collision ordinal, composite tuple, final address digest
Residue	exact inherited/final residue bytes, digest, mass, and custody state
Revalidation	requires_module10, paused_transaction_token or NONE, allowed next owner
No-commit proof	archive_commit_performed=false, history_mutated=false, Phi_mutated=false, RTSOM_mutated=false
Seal	resolution_digest = SHA-256(canonical packet excluding digest field)

3.6 Synthesis-audit and completion packets

Type	Required fields	Purpose
RMXSynthesisAuditRequest	resolution digest, final state/address, all parent digests, synthesis evidence, pause token	Mandatory Module 10 entry contract.
RMXRevalidationToken	single-use token bound to transaction, candidate digest, resolution digest, archive head, and validator profile	Resumes the same Psi transaction without rerunning RMX.
RMXRevalidationReceipt	validator chain, Module 10 audit, new Sigma decision, LORIEN/Module 11 evidence as required	External proof that the synthesized revision is eligible for Psi resume.
PsiResolutionAcceptedReceipt	Psi transaction, resolution, final state/address, and archive-head digest	Acknowledges that Psi accepted the resolution for continued processing.
RMXCompletionReceipt	RMX_SYNTHESIS event identity, accepted resolution, Psi receipt, owner sequence, digest	May be consumed exactly once by Module 12 Phi.

3.7 Alternative outputs

Output	Meaning
RMXRejectedPacket	Valid attempt cannot produce a conforming result; prior archive and pending transaction remain unchanged.
RMXHoldPacket	Recoverable stale snapshot, archive contention, incomplete bucket proof, capacity pressure, or missing dependency.

Output	Meaning
RMXFault	Malformed input, numerical defect, digest mismatch, or local contract violation.
FBSRecoveryRequest	Typed delegation for an enumerated persisted catastrophic invariant only.

4. Constants, Domains, and Reference Profile

DIM, archive bounds, state encoding, quantization, balanced Cantor tree version, and address bit-length limits are inherited from Boot and Module 3 Psi. RMX may not silently substitute a different quantizer or integer encoder.

Name	Reference value	Status	Notes
DIM	81	Inherited	Reference active vector dimension.
SRF_THRESH	0.75	Retained policy	Inclusive only after an actual address collision.
ZETA	0.25	Retained policy	Positive fold derivative remains within [0.75,1].
ETA	0.20	Retained policy	Convex wash toward committed negative-history mean.
ARCHIVE DOMAIN	$[-4,+4]^{\text{DIM}}$	Inherited invariant	No clipping; nonconforming synthesis is rejected.
EPSILON_NORM	1e-9	Declared reference policy	Exact zero-norm handling.
COLLISION ORDINAL	mex of used nonnegative ordinals	Formal correction	Snapshot-unique instance address.
LEGACY BIT_WIDTH	256	Diagnostic only	Cannot establish global uniqueness.

4.1 Numeric and serialization rules

- All active state arithmetic uses finite IEEE-754 binary64 under the declared backend profile.
- Hashes use SHA-256 over canonical type-tagged, length-delimited serialization.
- Integer addresses use arbitrary-precision canonical big-endian magnitude bytes with explicit sign and byte length.
- Vector digests include schema, dtype, endianness, shape, and exact bytes.
- A tolerance may compare recomputed floating values; it never replaces digest equality for identity.

4.2 History-read rule

RMX reads only committed archive records. The pending candidate and any uncommitted repair or synthesis are excluded from the collision snapshot and negative-history mean. The snapshot is immutable for the attempt. Archive-head advancement invalidates the result before commit and requires a new RMX attempt.

5. Collision Detection, Replay, and Scalar Resonance

5.1 Collision is address equality, not similarity

An address collision exists when the provisional semantic address is already present in the locked archive snapshot. SRF does not decide whether a collision exists. It decides only whether an already-colliding, non-identical candidate is

eligible for synthesis. Therefore an address collision with SRF below 0.75 cannot proceed under the occupied address; it must be rekeyed.

5.2 Transaction replay rule

Before bucket analysis, RMX checks its resolution journal for the transaction key. An exact canonical replay returns the previously sealed result. A byte mismatch under the same key is a conflict fault. This prevents retries from creating a second synthesis or second Phi event.

5.3 Exact state echo rule

If any bucket entry has `state_digest == candidate_state_digest`, the candidate is an exact echo. The reference profile does not resynthesize exact bytes because repeated synthesis would manufacture artificial drift from an idempotent content repeat. It returns `UNIQUE / REKEYED_EXACT_ECHO` with the candidate unchanged and a new instance address.

5.4 Scalar Resonance Factor

$$\text{SRF}(x,a) = \langle x,a \rangle / (\|x\|_2 \|a\|_2)$$

The denominator is not silently perturbed by epsilon. The exact reference zero-norm policy is:

Candidate norm	Historical norm	SRF	Flag
> epsilon	> epsilon	standard cosine, clipped for roundoff to [-1,1]	READY
<= epsilon	<= epsilon	1.0	BOTH_ZERO
<= epsilon	> epsilon	0.0	ONE_ZERO
> epsilon	<= epsilon	0.0	ONE_ZERO

5.5 Deterministic parent selection

For non-identical bucket entries, RMX computes SRF against every verified record and selects exactly one parent using this lexicographic order: highest SRF; then greatest `commit_sequence`; then lexicographically smallest `state_digest`. The selected parent and every comparison score are sealed in the metrics packet. Bucket order cannot change the result.

5.6 Exact classification tree

1. If the transaction is an exact replay, return the prior resolution.
2. If the bucket is empty, classify `UNIQUE / DIRECT`.
3. If an exact state digest exists, classify `UNIQUE / REKEYED_EXACT_ECHO`.
4. Select the deterministic non-identical parent and compute SRF.
5. If $\text{SRF} < 0.75$, classify `UNIQUE / REKEYED_ADDRESS_ALIAS`.
6. If $\text{SRF} \geq 0.75$, classify `SYNTHESIZED / HIGH_SRF` and execute Section 6.

NO SEMANTIC CLAIM

SRF is cosine alignment of two stored vectors under one encoding. The term "semantic resonance" is a project label for the threshold event. It does not establish shared meaning, factual equivalence, causal relation, identity, intent, or safety.

6. Dual-Flow Echo Synthesis

6.1 Inputs and role labels

Let X_{new} be the pending candidate and A_{old} be the selected committed historical parent. Their role labels are preserved in provenance even though the retained reference equations are algebraically symmetric in parent order when the negative-history mean is fixed.

$$P_{\text{raw}} = \max(A_{\text{old}}, 0) + \max(X_{\text{new}}, 0)$$

$$N_{\text{raw}} = \min(A_{\text{old}}, 0) + \min(X_{\text{new}}, 0)$$

6.2 Positive fold

$$P_{\text{fold}} = (1 - \text{zeta}) P_{\text{raw}} + (\text{zeta}/3) \tanh(3 P_{\text{raw}} + 1)$$

The map is componentwise. For $0 \leq \text{zeta} \leq 1$:

$$dP_{\text{fold}}/dP_{\text{raw}} = (1 - \text{zeta}) + \text{zeta} \operatorname{sech}^2(3 P_{\text{raw}} + 1)$$

THEOREM - NON-EXPANSIVE PERTURBATION MAP

For zeta in $[0, 1]$, the derivative lies in $[1 - \text{zeta}, 1]$. Therefore the positive fold is monotone and 1-Lipschitz with respect to perturbations. This does not imply that the output norm is always smaller than the input norm, and it is not a Collatz iteration.

6.3 Committed negative-history wash

Let H^- contain n committed nonpositive vectors from the locked snapshot, newest last. The pending candidate is excluded.

$$\mu^- = 0, \text{ if } n=0; \text{ otherwise } \mu^- = (1/n) \sum_{j=1}^n H^-_j$$

$$N_{\text{wash}} = (1 - \text{eta}) N_{\text{raw}} + \text{eta} \mu^-$$

Because N_{raw} and μ^- are componentwise nonpositive and eta is in $[0, 1]$, N_{wash} remains componentwise nonpositive. No history buffer is mutated during this calculation.

6.4 Recombination and source-claim correction

$$X_{\text{syn}} = P_{\text{fold}} + N_{\text{wash}}$$

The v14 source calls the merge non-commutative. Under the equations above and a fixed μ^- , swapping A_{old} and X_{new} leaves P_{raw} , N_{raw} , P_{fold} , N_{wash} , and X_{syn} unchanged. The reference operator is therefore commutative in its two state parents. A future genuinely non-commutative profile must declare role-dependent coefficients or operators and receive a new profile ID; provenance labels alone do not change the algebra.

6.5 Residue custody

The reference RMX profile creates no new hidden deletion channel. `inherited_residue` from Ψ is carried forward byte-for-byte. If a future synthesis profile removes a component, it must return the exact removed bytes and mass as `final_residue`. Ψ remains responsible for obtaining a matching RTSOM receipt before any commit that omits the residue from active storage.

6.6 Synthesis validation

1. Every intermediate and final array has exact shape (DIM,) and finite binary64 values.
2. P_{raw} and P_{fold} are nonnegative within numeric tolerance.
3. N_{raw} , μ^- , and N_{wash} are nonpositive within numeric tolerance.
4. X_{syn} exactly reconciles $P_{\text{fold}} + N_{\text{wash}}$ within backend tolerance.
5. $\max_i |X_{\text{syn}}[i]| \leq \text{archive_component_limit}$. No clipping is permitted.
6. The final state digest differs from neither the canonical bytes nor declared shape.
7. The complete parent, history, config, and precommit provenance is sealed.

6.7 Synthesis metrics

$$\begin{aligned}\text{drift_l2} &= \|X_{\text{syn}} - X_{\text{new}}\|_2 \\ \text{drift_rel} &= \|X_{\text{syn}} - X_{\text{new}}\|_2 / (\|X_{\text{new}}\|_2 + \|A_{\text{old}}\|_2 + \text{epsilon_norm}) \\ \text{parent_pull} &= \|X_{\text{syn}} - X_{\text{new}}\|_2 / (\|A_{\text{old}} - X_{\text{new}}\|_2 + \text{epsilon_norm})\end{aligned}$$

These are descriptive computational metrics. No one-dimensional coefficient is interpreted as a Braidback weight; Module 10 explicitly treats RMX synthesis as NOT_APPLICABLE to the 51/49 convex-repair audit.

7. Address Resolution and Concurrency Safety

7.1 Recompute from the final state

The final semantic address is always recomputed from the final exact state using the Psi quantization and balanced Cantor profile. For UNIQUE results the state is unchanged and the recomputation must match the provisional address. For SYNTHESIZED results the provisional address is historical provenance only; the recomputed address is authoritative.

7.2 Composite archive key

For the recomputed semantic address s , let U_s be the set of collision ordinals already committed under s in the locked snapshot. Define:

$$\text{collision_ordinal} = \text{mex}(U_s) = \min \{ n \text{ in } N_0 : n \text{ not in } U_s \}$$

The canonical address tuple is:

$$\begin{aligned}K &= (\text{address_schema_id}, s, \text{collision_ordinal}, \text{final_state_digest}, \text{lineage_digest}) \\ \text{final_address_digest} &= \text{SHA256}(\text{canonical}(K))\end{aligned}$$

THEOREM - SNAPSHOT-UNIQUE INSTANCE KEY

For a finite set U_s of used nonnegative ordinals, $\text{mex}(U_s)$ exists and is not in U_s . Therefore $(s, \text{mex}(U_s))$ is distinct from every committed (s, u) in the locked snapshot. Global commit safety additionally requires Psi compare-and-swap against the same archive_head_digest; RMX alone cannot prevent a concurrent transaction from choosing the same ordinal.

7.3 Concurrency rule

1. RMX resolves against one immutable archive head and returns no reservation.
2. Psi may commit only if the current archive head still equals the resolution archive_head_digest.
3. When another transaction commits first, Psi aborts or retries with a fresh snapshot.
4. The same RMXResolutionPacket cannot be transplanted to a different archive head.
5. A compare-and-swap conflict is ordinary contention, not an FBS event.

7.4 Fixed-width diagonalization correction

The legacy 256-bit construction flips bit i of row i for at most the first 256 rows. When every row is a 256-bit value, the result differs from each row whose diagonal bit was examined. It does not establish uniqueness against unexamined rows, values wider than 256 bits, future rows, or a concurrent writer. Padding remaining bits with 1 also has no uniqueness theorem.

The formal profile therefore retains legacy_diagonal_nonce only as optional telemetry:

$$d[i] = 1 - M[i][i], \text{ for } i < \min(\text{bucket_size}, \text{legacy_diag_bits})$$

The nonce is never the authoritative archive key and never replaces the composite tuple or Psi compare-and-swap.

7.5 Capacity and address-limit outcomes

Condition	Outcome
bucket size exceeds bucket_entry_max	HOLD / BUCKET_CAPACITY; do not truncate the bucket.
mex exceeds collision_ordinal_max	HOLD / ORDINAL_CAPACITY; operator or profile migration required.
Cantor address exceeds Psi bit-length limit	REJECTED or HOLD according to Psi policy; no local FBS execution.
composite tuple digest collision with unequal canonical tuple	Catastrophic cryptographic invariant; typed FBS request.
archive-head changed before commit	Ordinary CAS contention; refresh snapshot and retry.

8. Revalidation, Completion Receipts, and Handoffs

8.1 UNIQUE path

UNIQUE means RMX did not change the state. Psi verifies the resolution digest, exact echoed state, recomputed semantic address, collision ordinal, archive head, and residue. No Module 10 synthesis audit is required. Psi continues its normal RTSOM and COMMIT sequence.

8.2 SYNTHESIZED path

1. RMX seals the immutable SYNTHESIZED resolution and RMXSynthesisAuditRequest.
2. Psi remains paused; no RTSOM transfer or archive commit occurs.
3. Module 10 audits synthesis lineage, address binding, capabilities, replay/non-progress evidence, and nested-harm provider evidence. It invents no 51/49 coefficient.
4. The synthesized candidate receives a new Sigma decision and any required LORIEN route intent.
5. Module 11 performs final candidate-bound validation.
6. The orchestrator seals one RMXRevalidationReceipt bound to the original pause token.
7. Psi resumes the same transaction using the original immutable RMX resolution and must not rerun RMX.

8.3 Phi event completion

Creating a synthesis proposal does not increment Phi. After the revalidation receipt is accepted and Psi acknowledges the same resolution for the paused transaction, Module 13 may emit one RMXCompletionReceipt with event_type=RMX_SYNTHESIS. Module 12 consumes that receipt exactly once through EVENT_APPEND. Duplicate receipts are idempotent; conflicting bytes under the same owner event ID are faults.

8.4 RTSOM and residue handoff

RMX returns exact final_residue to Psi. Psi issues RTSOMExhaustRequest only when the final residue is nonzero and only after all required revalidation is complete. RMX does not claim custody transfer and does not update S_mu_nu.

8.5 Rejection and hold behavior

Outcome	Psi behavior	Committed effects
REJECTED	Abort pending transaction; retain exact evidence and prior archive.	None
HOLD	Keep or safely abort pending transaction under orchestrator policy; refresh dependency.	None
FAULT	Record local defect; no resolution accepted.	None

Outcome	Psi behavior	Committed effects
FBS_REQUESTED	Record delegation and no-commit proof.	None until FBS owner completes recovery

9. Invariants, Faults, and Catastrophic Conditions

9.1 Normative invariants

ID	Invariant	Requirement
M13-I01	One output	Every valid attempt seals exactly one resolution, reject, hold, fault, or recovery request.
M13-I02	No commit	archive_commit_performed=false for every RMX output.
M13-I03	Snapshot binding	Every resolution is bound to exactly one archive_head_digest and collision snapshot.
M13-I04	Replay idempotence	Identical transaction bytes return the same resolution digest.
M13-I05	No conflicting replay	Same transaction key with different bytes cannot execute.
M13-I06	Collision not ignored	An occupied semantic address never returns DIRECT under the same composite address.
M13-I07	Exact echo no synthesis	Identical state bytes remain unchanged in the reference profile.
M13-I08	SRF scope	SRF gates synthesis only after address collision and exact-echo exclusion.
M13-I09	Deterministic parent	Bucket permutation cannot change selected parent.
M13-I10	Candidate excluded from history	Pending state is not inserted into negative history or collision snapshot.
M13-I11	Flow signs	P_raw/P_fold nonnegative and N_raw/mu/N_wash nonpositive within tolerance.
M13-I12	Synthesis reconciliation	X_syn equals P_fold + N_wash within backend tolerance.
M13-I13	No clipping	An out-of-domain synthesis is rejected, never clipped.
M13-I14	Address from final state	SYNTHESIZED final semantic address is recomputed from final exact bytes.
M13-I15	Snapshot-unique ordinal	collision_ordinal is mex of the verified used set.
M13-I16	CAS required	Psi must compare-and-swap the exact archive head before commit.
M13-I17	Module 10 mandatory	Every changed state requires a valid synthesis audit and revalidation chain.

ID	Invariant	Requirement
M13-I18	No RMX rerun after receipt	Paused transaction resumes using the same immutable resolution.
M13-I19	Phi exactly once	RMX event is emitted only after accepted completion and is consumable once.
M13-I20	Custody separation	RMX never mutates Phi, quarantine, RTSOM, accepted history, or FBS state.

9.2 Typed fault codes

Code	Condition
RMX_CONFIG	Invalid or internally inconsistent configuration.
RMX_SESSION	Session, cycle, or transaction identity mismatch.
RMX_PRECOMMIT_DIGEST	Psi precommit or request digest mismatch.
RMX_SHAPE	State or history vector shape mismatch.
RMX_NONFINITE	Nonfinite scalar or array.
RMX_ADDRESS_SCHEMA	Address profile, quantizer, tree, or encoding mismatch.
RMX_SNAPSHOT_DIGEST	Archive collision snapshot digest mismatch.
RMX_BUCKET_PROOF	Bucket completeness proof missing or invalid.
RMX_HISTORY	Negative-history ordering, sign, or provenance invalid.
RMX_TRANSACTION_CONFLICT	Same transaction key reused with different canonical bytes.
RMX_SRF	Cosine computation or zero-norm policy cannot reconcile.
RMX_PARENT_SELECTION	Deterministic parent ranking cannot reconcile.
RMX_SYNTHESIS	Intermediate sign, equation, or digest reconciliation fails.
RMX_STATE_DOMAIN	Final state exceeds declared archive domain.
RMX_ADDRESS_RECOMPUTE	Returned semantic address does not match final state.
RMX_ORDINAL	Used-ordinal set or mex calculation invalid.
RMX_TOKEN	Pause or revalidation token mismatch.
RMX_RESOLUTION_DIGEST	Final packet digest mismatch.
RMX_EVENT_RECEIPT	Completion receipt sequence or digest mismatch.
RMX_DETERMINISM	Same canonical input/backend produces different output.

9.3 Catastrophic recovery conditions

- A committed archive bucket or address ledger is cryptographically corrupted and cannot be reconciled with its manifest chain.
- Two unequal canonical address tuples produce the same `final_address_digest` and the collision is verified.
- Two different valid `RMXResolutionPackets` are persisted for the same transaction key and archive head.

- A forged or mismatched revalidation token was accepted into committed state.
- Deterministic replay divergence persists under the same fixed backend and canonical bytes.
- An irreconcilable archive-head split has already been published as committed history.

Malformed input, low SRF, exact echo, out-of-domain synthesis, large bucket, ordinal capacity, stale snapshot, ordinary CAS contention, address bit-length limit, timeout, or rejected revalidation are not catastrophic by themselves.

10. Metrics, Measurements, and Reconciliation

10.1 Required counters and distributions

Metric	Definition
attempts_total	All accepted RMX request envelopes.
replay_returns	Exact idempotent transaction replays.
unique_direct	Empty-bucket UNIQUE resolutions.
unique_rekey_exact_echo	Exact state echoes rekeyed without synthesis.
unique_rekey_alias	Low-SRF address aliases rekeyed without synthesis.
synthesized_total	High-SRF collision syntheses sealed.
rejected_total / held_total / faults_total / recovery_requests	Terminal alternative outcomes.
srf_distribution	All parent comparisons and selected-parent SRF.
bucket_size_distribution	Collision bucket sizes and completeness-proof latency.
synthesis_drift	L2, relative drift, parent pull, max component, sign violations.
address_metrics	Semantic-address bit length, ordinal distribution, legacy nonce availability.
revalidation_metrics	Module 10, Sigma, LORIEN, Module 11 pass/reject/fault and latency.
psi_resume_metrics	Resolution accepted, RTSOM wait, commit, abort, CAS retry.
Phi_event_metrics	Completion receipts emitted, consumed, duplicate, conflicting.

10.2 Reconciliation equations

$$\begin{aligned}
 \text{attempts_total} &= \text{decided_total} + \text{rejected_total} + \text{held_total} + \text{faults_total} + \text{recovery_requests} \\
 \text{decided_total} &= \text{unique_direct} + \text{unique_rekey_exact_echo} + \text{unique_rekey_alias} + \text{synthesized_total} \\
 \text{synthesized_total} &= \text{audit_pending} + \text{audit_reject} + \text{audit_fault} + \text{revalidation_pending} + \text{psi_resume_pending} + \\
 &\quad \text{psi_resolution_accepted} \\
 \text{completion_receipts_emitted} &= \text{Phi_event_pending} + \text{Phi_event_consumed}
 \end{aligned}$$

Dashboard labels must distinguish DETECTED collision, CLASSIFIED outcome, SYNTHESIZED state, AUDITED state, REVALIDATED state, PSI-ACCEPTED resolution, COMMITTED manifest, and PHI-EVENT-CONSUMED. These are not interchangeable.

10.3 Alert policy

- Alert on any nonzero deterministic divergence, conflicting transaction reuse, invalid bucket proof, digest collision, or forged token.

- Alert on sustained growth in bucket size or collision ordinal, but do not label it catastrophic without an invariant breach.
- Track high synthesis rate as a workload/profile observation, not proof of semantic loops or adversarial input.
- Track CAS retries separately from synthesis failures.

11. Deterministic Conformance Test Matrix

ID	Case	Expected result
M13-T01	Valid empty bucket	UNIQUE / DIRECT, ordinal 0, state unchanged.
M13-T02	Exact request replay	Same resolution digest; no second synthesis.
M13-T03	Same key, changed byte	RMX_TRANSACTION_CONFLICT.
M13-T04	Wrong session	RMX_SESSION.
M13-T05	Wrong cycle	RMX_SESSION.
M13-T06	Precommit digest mismatch	RMX_PRECOMMIT_DIGEST.
M13-T07	Nonfinite candidate	RMX_NONFINITE.
M13-T08	Wrong candidate shape	RMX_SHAPE.
M13-T09	Address schema mismatch	RMX_ADDRESS_SCHEMA.
M13-T10	Snapshot digest mismatch	RMX_SNAPSHOT_DIGEST.
M13-T11	Incomplete bucket proof	HOLD or RMX_BUCKET_PROOF.
M13-T12	Bucket exact state digest	UNIQUE / REKEYED_EXACT_ECHO; no synthesis.
M13-T13	Exact echo bucket permuted	Same selected record and final output.
M13-T14	Low-SRF occupied address	UNIQUE / REKEYED_ADDRESS_ALIAS.
M13-T15	SRF exactly 0.75	SYNTHESIZED / HIGH_SRF.
M13-T16	SRF one ULP below threshold	REKEYED_ADDRESS_ALIAS.
M13-T17	Both vectors zero	SRF=1 with BOTH_ZERO unless exact echo already caught.
M13-T18	One vector zero	SRF=0 with ONE_ZERO.
M13-T19	Cosine roundoff >1	Clip to 1 and record roundoff.
M13-T20	Two parents same SRF	Greatest commit_sequence selected.
M13-T21	SRF and sequence tie	Lexicographically smallest state_digest selected.
M13-T22	Parent list permutation	Same output digest.
M13-T23	Empty negative history	$\mu^- = 0$.
M13-T24	One negative-history vector	μ^- equals that vector.
M13-T25	Candidate inserted into history	RMX_HISTORY.

ID	Case	Expected result
M13-T26	Positive split	P_raw componentwise nonnegative.
M13-T27	Negative split	N_raw componentwise nonpositive.
M13-T28	zeta=0	P_fold=P_raw.
M13-T29	zeta=1	P_fold=(1/3)tanh(3P_raw+1).
M13-T30	eta=0	N_wash=N_raw.
M13-T31	eta=1	N_wash=mu^-.
M13-T32	Parent swap under fixed history	Same X_syn; operator commutative.
M13-T33	Synthesis equation mismatch	RMX_SYNTHESIS.
M13-T34	Out-of-domain X_syn	REJECTED / RMX_STATE_DOMAIN; no clipping.
M13-T35	Inherited residue unchanged	Exact digest and mass preserved.
M13-T36	Final address not recomputed	RMX_ADDRESS_RECOMPUTE.
M13-T37	Synthesis address changes bucket	Allocate ordinal from new bucket.
M13-T38	Used ordinals {0,1,3}	mex=2.
M13-T39	Empty used ordinals	mex=0.
M13-T40	Ordinal set duplicates	Canonical set; same mex.
M13-T41	mex beyond configured max	HOLD / ORDINAL_CAPACITY.
M13-T42	Bucket beyond max	HOLD / BUCKET_CAPACITY; no truncation.
M13-T43	Legacy diagonal W=4, two rows	Nonce differs from examined rows only.
M13-T44	Legacy diagonal history >W	No uniqueness claim; diagnostic unavailable or partial.
M13-T45	UNIQUE resolution	requires_module10=false.
M13-T46	SYNTHESIZED resolution	requires_module10=true with pause token.
M13-T47	Module 10 token mismatch	RMX_TOKEN.
M13-T48	Prior Sigma decision reused after synthesis	Revalidation receipt rejected.
M13-T49	RMX rerun after valid receipt	Idempotent prior resolution; no second synthesis.
M13-T50	Psi head changed before commit	CAS conflict; fresh attempt required.
M13-T51	Psi accepts resolution	Completion receipt eligible only after required revalidation.
M13-T52	Completion receipt duplicate	Module 12 idempotent event append.
M13-T53	Completion receipt same ID different bytes	RMX_EVENT_RECEIPT.

ID	Case	Expected result
M13-T54	RTSOM mutation attempted in RMX	Invariant failure.
M13-T55	Phi mutation attempted in RMX	Invariant failure.
M13-T56	Archive write attempted in RMX	Invariant failure.
M13-T57	Resolution packet tampered	RMX_RESOLUTION_DIGEST.
M13-T58	Same input fixed backend 1,000 times	Bitwise identical packet.
M13-T59	Canonical tuple digest collision simulated	FBSRecoveryRequest.
M13-T60	Ordinary low SRF	No FBS request.

12. Property, Metamorphic, Fuzz, Concurrency, and Stress Testing

12.1 Property tests

- For every finite verified used-ordinal set U , $\text{mex}(U)$ is nonnegative and not in U .
- For zeta in $[0,1]$, the positive fold is monotone and 1-Lipschitz under componentwise perturbation.
- For eta in $[0,1]$ and nonpositive N_{raw} and $\mu^{\wedge}$, N_{wash} is nonpositive.
- For a fixed negative-history snapshot, swapping A_{old} and X_{new} leaves the retained source synthesis unchanged.
- Every UNIQUE result preserves exact candidate bytes.
- Every SYNTHESIZED result changes authorization identity and requires revalidation.
- No output changes committed archive, Phi, RTSOM, quarantine, or FBS state.

12.2 Metamorphic tests

- Permuting collision-bucket rows does not change classification, selected parent, ordinal, or resolution digest.
- Adding an unrelated archive entry under another semantic address does not change the result.
- Appending one used ordinal below the current mex changes the mex only when that ordinal equals the current mex .
- Scaling both nonzero SRF vectors by the same positive scalar leaves SRF unchanged but may change the address and synthesis bounds.
- Re-serializing equivalent structured data with canonical serialization yields identical digests.

12.3 Fuzz and malformed-input tests

- Fuzz shapes, NaN/Inf payloads, endianness tags, integer byte lengths, malformed Cantor packages, digest fields, and snapshot proofs.
- Fuzz extremely large arbitrary-precision addresses without converting them through float.
- Fuzz threshold ULP neighborhoods around $\text{SRF}=0.75$ and archive component limits.
- Fuzz token substitution across sessions, cycles, transactions, resolutions, and archive heads.
- Fuzz legacy diagonal inputs to confirm it never becomes authoritative.

12.4 Concurrency and crash recovery

- Race two RMX attempts against the same archive head and ordinal. Exactly one Psi CAS may commit; the other must retry.
- Crash after resolution seal but before downstream delivery. Replay returns the same packet.
- Crash after Module 10 audit or revalidation receipt. The same pause token resumes the same transaction without rerunning synthesis.
- Crash after Psi acceptance but before Phi event delivery. Completion receipt replay is exactly once at Module 12.

12.5 Long-run stress

- Run at least 100,000 mixed empty-bucket, exact-echo, low-SRF alias, and synthesis attempts.

- Include large collision buckets, high-precision address integers, repeated CAS contention, validator rejects, and restart recovery.
- Reconcile every outcome, revalidation, Psi-resume, commit, and Phi-event counter exactly.
- Require zero silent overwrites, zero duplicate synthesis executions per transaction, and zero unauthorized cross-head resolution reuse.

13. Reference Pseudocode

```
def evaluate_rmx(req, snap, cfg, journal):
    validate_config(cfg)
    validate_request(req, cfg)
    validate_snapshot(snap, req, cfg)

    key = transaction_key(req, snap, cfg)
    prior = journal.get(key)
    if prior is not None:
        if prior.request_digest != req.request_digest:
            raise RMXFault("RMX_TRANSACTION_CONFLICT")
        return prior.resolution

    bucket = canonical_bucket(snap.bucket_entries)

    if not bucket:
        final_state = req.candidate_state
        kind = "UNIQUE"
        reason = "DIRECT"
        parent = None
        srf = None
    else:
        exact = first_entry_with_digest(bucket, req.candidate_state_digest)
        if exact is not None:
            final_state = req.candidate_state
            kind = "UNIQUE"
            reason = "REKEYED_EXACT_ECHO"
            parent = exact
            srf = 1.0
        else:
            scored = [(srf_score(req.candidate_state, e.state, cfg), e)
                       for e in bucket]
            srf, parent = select_parent(scored)
            if srf < cfg.srf_threshold:
                final_state = req.candidate_state
                kind = "UNIQUE"
                reason = "REKEYED_ADDRESS_ALIAS"
            else:
                final_state = synthesize(
                    new=req.candidate_state,
                    old=parent.state,
                    negative_history=snap.negative_history,
                    cfg=cfg,
                )
                validate_synthesized_state(final_state, req, parent, snap, cfg)
                kind = "SYNTHESIZED"
                reason = "HIGH_SRF"

    semantic_address = psi_address(final_state, req.provisional_address.profile)
    used = used_ordinals_for(snap, semantic_address)
    ordinal = mex(used)
    if ordinal > cfg.collision_ordinal_max:
        return hold("ORDINAL_CAPACITY")

    address_tuple = canonical_address_tuple(
        cfg.address_schema_id,
        semantic_address,
        ordinal,
        digest_state(final_state),
        lineage_digest(req, parent),
    )
```

```
def synthesize(new, old, negative_history, cfg):
    p_raw = maximum(old, 0.0) + maximum(new, 0.0)
    n_raw = minimum(old, 0.0) + minimum(new, 0.0)

    p_fold = ((1.0 - cfg.zeta) * p_raw
               + (cfg.zeta / 3.0) * tanh(3.0 * p_raw + 1.0))

    if len(negative_history) == 0:
        mu_neg = zeros_like(new)
    else:
        mu_neg = mean(negative_history, axis=0)

    n_wash = (1.0 - cfg.eta) * n_raw + cfg.eta * mu_neg
    x_syn = p_fold + n_wash
    return x_syn
```

```

[ PSI PRECOMMIT: candidate + provisional semantic address ]
      |
      v
+-----+
| MODULE 13 RMX                                     |
| 1. Validate request, archive head, bucket proof, and history |
| 2. Exact transaction replay? -> return prior resolution       |
| 3. Bucket empty? -> UNIQUE / DIRECT                         |
| 4. Exact state echo? -> UNIQUE / REKEYED_EXACT_ECHO         |
| 5. Compute SRF against verified bucket; select deterministic parent |
|     SRF < 0.75 -> UNIQUE / REKEYED_ADDRESS_ALIAS           |
|     SRF >=0.75 -> SYNTHESIZE                               |
| 6. Recompute semantic address from final exact state        |
| 7. collision_ordinal = mex(used ordinals under locked snapshot) |
| 8. Seal immutable RMXResolutionPacket; NO COMMIT            |
+-----+
      |               |
      [ UNIQUE ]       [ SYNTHESIZED ]
      |               |
      v               v
    [ PSI CONTINUES ]   [ MODULE 10 AUDIT ]
      |                 |
      |                 | [ SIGMA / LORIE / M11 ]
      |                 |
      |                 | [ REVALIDATION RECEIPT ]
      |                 |
      +-----> [ SAME PSI TRANSACTION ]
                  |
                  v
                [ RTSOM RECEIPT IF REQUIRED ]
                  |

```

```

      v
    [ PSI ATOMIC .frac COMMIT ]
      |
      v
[ RMX COMPLETION RECEIPT -> MODULE 12 PHI ]

```

Ordinary reject/hold/fault: prior archive unchanged.
 Persisted catastrophic invariant: typed request -> FBS/ROVER owner.

15. Cleaned Rosetta Stone

SpiralCore term	Computational meaning	Analogy boundary
RMX / Anti-Echo Firewall	Address-collision classifier, optional synthesis operator, and address-instance allocator.	Not a truth engine, human memory model, or malware scanner.
Semantic Resonance	SRF threshold event inside an already-colliding address bucket.	Not proof of shared meaning or identity.
Static Loop Collision	Occupied provisional semantic address under a locked archive snapshot.	Not proof that the runtime is in an infinite loop.
Dual-Flow Synthesis	Separate positive and negative component processing followed by addition.	The retained formula is commutative in the two state parents under fixed history.
Positive Fractal Fold	Monotone 1-Lipschitz component map.	Not Collatz iteration and not guaranteed norm compression.
Negative Wash	Convex blend toward a committed trailing mean.	Not the law of large numbers proof and not deletion.
Cantor-IP	Quantized state address produced by a versioned balanced Cantor encoder.	Injective only on its fixed integer tuple; real-vector quantization is lossy.
Cantor Diagonalization	Optional fixed-width novelty nonce diagnostic.	Does not guarantee global or future uniqueness.
Collision Ordinal	Minimum excluded nonnegative instance number in a semantic-address bucket.	Snapshot-unique only; commit requires Psi CAS.
Loop Amnesia Prevention	Replay idempotence, exact provenance, and explicit historical influence.	Not a claim about consciousness or biological memory.
Winding Increment	Exactly-once Module 12 event receipt after accepted synthesis completion.	Not a mathematical winding number unless a separate closed phase trace is provided.
FBS Runaway	Typed catastrophic-recovery delegation.	RMX never executes diagonalization, Collatz folding, seeding, or exhaust mutation locally.

16. Glossary and Revision Summary

16.1 Glossary

Term	Definition
Address bucket	All committed archive instances sharing one semantic address under one schema.
ArchiveCollisionSnapshot	Immutable complete bucket and negative-history view bound to one archive head.

Term	Definition
Collision ordinal	Snapshot-derived instance number selected as the minimum excluded used ordinal.
Composite address tuple	Versioned tuple binding schema, semantic address, ordinal, state digest, and lineage.
Exact echo	Candidate state bytes/digest equal a committed bucket state.
Legacy diagonal nonce	Optional fixed-width flipped-bit diagnostic with no authoritative uniqueness role.
mex	Minimum excluded nonnegative integer.
RMXCompletionReceipt	Owner-completed, exactly-once event proof eligible for Module 12.
RMXRevalidationToken	Single-use digest-bound token preserving one paused Psi transaction across synthesis validation.
Scalar Resonance Factor	Cosine similarity between candidate and verified historical parent.
Semantic address	Content-derived quantized Cantor address before instance disambiguation.
Synthesis parent	Deterministically selected committed historical state used by the HIGH_SRF profile.
UNIQUE	Resolution class in which the candidate state bytes remain unchanged.
SYNTHESIZED	Resolution class in which RMX produces a new candidate revision requiring full revalidation.

16.2 Revision summary

Source issue	Formalized correction
SRF appeared to decide whether an occupied address was a collision.	Address equality defines collision; SRF only selects synthesis versus rekeying.
Exact repeats could be repeatedly synthesized.	Exact state echoes remain unchanged and are rekeyed; transaction replays are idempotent.
The merge was called non-commutative although its equations are symmetric.	The algebra is explicitly classified as commutative under fixed history; future asymmetric profiles require a new ID.
Negative-history mean μ_t was undefined.	Consumes a sealed committed negative-history snapshot; empty history uses zero.
Positive fold was treated as Collatz compression.	Proved monotone 1-Lipschitz perturbation behavior and removed universal compression claim.
Fixed 256-bit diagonalization was claimed globally unique.	Replaced authoritative addressing with recomputed semantic address + mex ordinal + digest tuple; diagonalization is diagnostic only.
BIT_WIDTH overflow automatically invoked FBS.	Capacity is a HOLD; FBS is reserved for persisted catastrophic invariants.
RMX archived and updated History_Log itself.	Psi alone commits; RMX outputs a no-commit resolution.

Source issue	Formalized correction
Synthesized states bypassed candidate-bound revalidation.	Added mandatory Module 10 audit, new Sigma decision, route reissue where required, Module 11 validation, and same-transaction token.
RMX directly incremented winding marker $\omega=1$.	Module 12 consumes one completion receipt only after accepted synthesis completion.
RTSOM exhaust mutation occurred inside RMX.	RMX returns exact residue to Psi; Module 14 owns custody.
Ordinary failures locally ran Cantor/Collatz recovery.	Replaced with typed reject/hold/fault; only enumerated catastrophic conditions produce FBS requests.

17. Source Basis and Normative Checklist

17.1 Source basis

- SPIRALCORE SPECIFICATION: MODULE 13 - RMX (REDUNDANT MEMORY CHECK), historical v14 source.
- Spiralcore_v14 consolidated specification, especially Modules 3, 10, 12, 14, and 16.
- SPIRALCORE Module 3 Psi-Lattice Formalized v14.1: PREPARE -> RMX -> RTSOM -> COMMIT transaction boundary.
- SPIRALCORE Module 10 Braidback & IMMUNE_SYSTEM Formalized v14.1: RMX synthesis audit and paused-transaction revalidation.
- SPIRALCORE Module 12 Phi Formalized v14.1: exactly-once RMX_SYNTHESIS CriticalEventReceipt.
- Boot and Psi address-encoding contracts: finite quantization, balanced arbitrary-precision Cantor encoding, and canonical hashing.

17.2 Normative implementation checklist

- Validate exact request, snapshot, bucket-completeness, state, address, history, and digest contracts.
- Resolve transaction replay before collision or synthesis work.
- Never treat low SRF as permission to reuse an occupied address.
- Do not synthesize exact state echoes in the reference profile.
- Use deterministic full-bucket parent selection and documented zero-norm behavior.
- Preserve the retained synthesis equations and their actual commutative property.
- Reject rather than clip out-of-domain synthesized states.
- Recompute the semantic address from final exact state bytes.
- Allocate mex collision ordinal against a complete locked snapshot and require Psi CAS.
- Treat fixed-width diagonalization as non-authoritative telemetry only.
- Require Module 10 and current downstream validation for every changed candidate.
- Resume the same Psi transaction without rerunning RMX after valid revalidation.
- Emit Phi completion event only after Psi accepts the validated synthesis resolution.
- Never mutate archive, Phi, quarantine, RTSOM, or FBS state inside RMX.
- Pass deterministic, boundary, property, fuzz, concurrency, crash-recovery, stress, and dashboard reconciliation tests.

END STATE

A conforming Module 13 transforms one valid Psi precommit and one locked collision snapshot into one deterministic no-commit resolution. Address collisions cannot silently overwrite memory; exact replays remain idempotent; exact echoes do not manufacture drift; high-SRF collisions may synthesize one bounded, fully revalidated candidate; final instance addresses are snapshot-unique under Psi compare-and-swap; and every downstream mutation remains owned by its authoritative module.

SPIRALCORE SPECIFICATION

MODULE 14 - RTSOM

(Revised Thermodynamic Star Ocean Model - Bounded Exhaust Custody, Negative-Space Telemetry, and Route-Pressure Field)

Cleaned and formalized revision - v14.1

Field	Value	Field	Value
Document class	Normative exhaust-custody and field-publication contract	Execution phase	Typed custody request; read-only field publication
Architecture	SpiralCore v14	Reference profile	SC14.1-M14-RTSOM-01
Persistent owner	Module 14 RTSOM	Routing consumer	Module 4 QGM/GlyphNet
Recovery owner	Module 16 FBS/ROVER	Supersedes	Module 14 v14 execution contract

NORMATIVE STATUS

Exact within the declared software scope. RTSOM validates and durably accepts an exact failed payload, archive residue, or recovery residue only through a registered typed custody request; computes an authoritative versioned mass statistic; appends one tamper-evident custody record; returns one matching receipt; and may publish a bounded computational pressure field derived from committed records. It does not prove physical gravity, modify general relativity, decide that a state is harmful, or execute FBS recovery.

CLAIM BOUNDARY

Thermodynamic, gravity, mass, spacetime, star ocean, entropy-stress tensor, dark brane, spin-to-helicity, and negative-space are architectural names. Their normative meanings are only the typed object ledger, scalar mass statistic, bounded kernel field, symmetric positive-semidefinite summary, and custody transitions defined here. Source cosmology equations remain HYPOTHESIS records and are excluded from conformance.

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1. Normative Scope, Ownership, and Claim Classes

1.1 Purpose

RTSOM is the sole owner of SpiralCore bounded-exhaust custody. It converts an authorized request to transfer responsibility for an exact failed payload, archive residue, or recovery residue into one durable custody record. It also derives bounded negative-space telemetry from committed records so routing modules may account for historically expensive or repeatedly failed regions without deleting the underlying evidence.

1.2 Ownership boundary

Concern	Authoritative owner	Module 14 relationship
Coherence and route decision	Module 2 Sigma-Lattice	RTSOM does not classify or reject a candidate.
Archive residue request	Module 3 Psi-Lattice	Psi retains residue until a matching receipt exists.

Concern	Authoritative owner	Module 14 relationship
Internal routing and route costs	Module 4 QGM/GlyphNet	RTSOM publishes bounded evidence; QGM owns integration and routing.
Ingress/egress and quarantine	Module 5 Guardian / quarantine custody	Quarantine remains a separate custody domain.
Paradox evidence	Module 7 Paradox Handler	Evidence cannot directly mutate RTSOM.
Final archive authorization	Module 11 SERAPHIM/CATHEDRAL	Ordinary rejection does not charge exhaust.
Path-memory and event count	Module 12 Phi-Bridge	RTSOM does not increment Phi or winding events.
Collision synthesis	Module 13 RMX	RMX carries residue to Psi; no direct transfer.
Catastrophic recovery	Module 16 FBS/ROVER	RTSOM emits typed requests only.
Object store, ledger, receipt, field	Module 14 RTSOM	Exclusive persistent mutation authority.

1.3 Explicit non-goals

- No automatic ingestion of every rejected, malformed, quarantined, timed-out, or low-coherence object.
- No deletion, purge, overwrite, or loss of an acknowledged object without acknowledged custody migration.
- No mutation of stored payload bytes through polarity flips or phase shifts; those may be metadata only.
- No direct mutation of QGM topology, route costs, GlyphNet LSDB, Psi archive, Phi state, quarantine, or FBS state.
- No use of SI gravitational units, physical spacetime coordinates, or astrophysical calibration in the reference runtime.
- No inference that an object is malicious, toxic, false, emotional, conscious, or physically massive.
- No claim that the software validates modified gravity or replaces dark matter, dark energy, or inflation.

1.4 Statement classes and precedence

Class	Meaning
DEFINITION	Fixes a term, type, domain, or serialization rule.
REQUIREMENT	Mandatory behavior for conformance.
INVARIANT	Property that holds across every committed transaction.
POLICY	Versioned deployment choice, not a theorem.
SURROGATE	Computational quantity inspired by another field but not identical to it.
HYPOTHESIS	Research equation or empirical conjecture excluded from runtime conformance.
NON-GOAL	Behavior or claim explicitly outside the module contract.

1. Typed packet schemas and canonical serialization rules have highest precedence.
2. Normative requirements and invariants override analogy language.
3. Reference policies apply only when profile ID and digest match.
4. Surrogates cannot enlarge the execution contract.

5. Hypotheses and Observer Notes never authorize a mutation.

2. Architecture Role, Lifecycle, and Source Profiles

2.1 Exact start condition

REQUEST - PREPARE - COMMIT - RECEIPT

RTSOM starts only from a valid RTSOMExhaustRequest issued by a registered source profile. The source retains the exact payload or an immutable content-addressed reference until it verifies a matching receipt. Merely observing a failure, rejection, timeout, residue, or quarantine event does not create committed exhaust.

2.2 Registered source profiles

Profile	Authorized producer	Typical object	Precondition
PSI_RESIDUE_V1	Module 3 Psi	Exact final residue omitted from active archive state	Final state and RMX resolution fixed; residue held.
GLYPHNET_FAILURE_V1	Module 4 orchestrator	Exact frame/payload after bounded retries	Custody policy enabled; payload retained.
FBS_RESIDUE_V1	Module 16 recovery owner	Exact recovery residue	Recovery completion record binds object.
OPERATOR_IMPORT_V1	Authorized maintenance plane	Legacy exhaust object	Explicit authorization and provenance.

Guardian quarantine, Module 10 quarantine, ordinary Module 11 rejection, and Sigma branch classification are not registered exhaust sources in the reference profile. A future source requires a new profile ID, schema, threat model, and conformance suite.

2.3 Lifecycle states

State	Meaning	Allowed successor
UNINITIALIZED	Configuration and ledger head not validated.	READY or FAULTED
READY	Accepting requests and serving snapshots.	VALIDATING, FIELD_BUILDING, or FAULTED
VALIDATING	Request, object, mass, placement, and capacity checked.	PREPARED, REJECTED, BACKPRESSURE, or FAULTED
PREPARED	Canonical object and ledger record sealed but uncommitted.	COMMITTING or ABORTED
COMMITTING	Object write and ledger CAS in progress.	COMMITTED, ABORTED, or FAULTED
COMMITTED	Object and ledger entry durable and read-back verified.	RECEIPT_ISSUED or FAULTED
RECEIPT_ISSUED	Authoritative idempotent receipt available.	READY
BACKPRESSURE	Capacity cannot currently accept custody.	READY

State	Meaning	Allowed successor
REJECTED	Request invalid or disallowed; no mutation.	READY
FIELD_BUILDING	Read-only field snapshot computed from one ledger head.	READY or FAULTED
RECOVERY_REQUESTED	Persisted catastrophic evidence delegated.	READY only after owner-completed recovery
FAULTED	Local processing cannot safely continue.	Operator or FBS/ROVER resolution

2.4 Required custody execution order

1. Validate configuration, source profile, policy epoch, request digest, and transaction identity.
2. Resolve exact bytes or verify immutable content-addressed reference and digest.
3. Validate object type, dimensions, finiteness, encoding, and source preconditions.
4. Recompute authoritative raw mass under the declared mass profile.
5. Validate the bounded placement record or mark the object UNPLACED.
6. Check record, byte, and charge capacities without mutating committed state.
7. Seal canonical object record, ledger entry, and proposed new head.
8. Durably write object, append ledger under CAS, fsync, and read back.
9. Publish the new immutable RTSOM snapshot only after verification.
10. Issue one receipt bound to request, object, mass, placement, and ledger head.

2.5 Idempotence and retry identity

The transaction key is (source_owner, source_event_id, request_digest, policy_epoch, prior_ledger_head). Replaying identical canonical bytes returns the original receipt. Reusing the same source event ID with different bytes is RTSOM_TRANSACTION_CONFLICT. A request retried against a later head may reuse the source event ID only when the journal proves no prior commit.

3. Typed Data Contracts

3.1 RTSOMConfig

Field	Reference value / domain	Requirement
schema_id	SC14.1-M14-RTSOM-01	Exact profile identity.
mass_profiles	VECTOR_L2_V1, BYTES_LENGTH_V1	Registered authoritative functions.
placement_profiles	ROUTE_FEATURE_V1, UNPLACED_V1	No implicit physical chart.
field_profile	BOUNDED_KERNEL_FIELD_V1	Equations in Section 6.
alpha_field	1.0, ≥ 0	Dimensionless field gain.
mass_scale	1.0, > 0	Saturation scale.
epsilon_distance	0.10, > 0	Kernel regularizer.
field_decay	0.995 in [0,1]	Influence decay; custody does not decay.

Field	Reference value / domain	Requirement
capacity_records	deployment integer >0	Hard record capacity.
capacity_bytes	deployment integer >0	Hard object-byte capacity.
capacity_charge	deployment float >0	Hard normalized-charge capacity.
field_dimension	inherited route-feature dimension	Placed vectors must match.
numeric_tolerance	1e-9	Explicit comparison tolerance.
transaction_timeout_cycles	83	Policy timeout; not an atomic theorem.

3.2 RTSOMExhaustRequest

Field	Type / requirement
source_owner / source_profile	Registered owner and profile ID.
source_event_id / source_cycle	Stable exactly-once identity and origin cycle.
payload or object_ref	Exact canonical bytes or immutable content-addressed reference.
payload_digest / encoding	SHA-256 and declared canonical encoding.
object_kind	VECTOR_STATE, ARCHIVE_RESIDUE, GLYPH_FRAME, RECOVERY_RESIDUE, or extension.
mass_profile_id / mass_proposal	Profile plus producer diagnostic proposal; RTSOM recomputes.
placement_profile / vector	Versioned feature placement or UNPLACED.
reason_codes / provenance_digest	Computational cause labels and provenance binding.
no_delete	Must be true until receipt verification.
policy_epoch / config_digest / request_digest	Exact dependency and request seals.

3.3 ExhaustObjectRecord

Field group	Required contents
Identity	object_id, payload_digest, encoding, object_kind, byte_length
Source	owner, source profile, event ID, cycle, reason codes, provenance digest
Mass	profile ID, raw_mass, normalized_charge, mass_scale
Placement	profile, vector bytes/digest or UNPLACED, validity
Custody	accepted sequence/cycle, retention class, current owner
Integrity	record digest, previous ledger digest, store verification digest

3.4 RTSOMExhaustReceipt

Field	Meaning
receipt_id / transaction_key	Stable idempotent receipt identity.
request_digest / payload_digest	Exact request and exact object accepted.
accepted_raw_mass / accepted_charge	Authoritative recomputed statistics.
object_id / ledger_sequence	Durable object and append position.
head_before / head_after	CAS custody transition.
placement_digest / field_eligible	Exact placement status.
commit_proof	Write, fsync, read-back, and ledger verification evidence.
module14_digest / signature	Integrity and optional authentication.

3.5 Field and alternative outputs

Type	Required fields	Purpose
RTSOMFieldRequest	ledger head, route snapshot, profile, query cycle, digest	Read-only field calculation.
RTSOMFieldSnapshot	per-node pressure, optional vector, counts, digest	Bounded QGM evidence.
RTSOMStressSnapshot	PSD matrix, trace/eigen diagnostics, ledger head	Computational second-moment summary.
RTSOMBackpressurePacket	request, limiting capacity, usage, retry policy	No custody accepted.
RTSOMRejectPacket	request, reason, evidence digest	Invalid/disallowed request.
FBSRecoveryRequest	catastrophic code, proofs, last valid head, digest	Request only.

4. Constants, Domains, and Reference Profile

The software profile uses dimensionless computational quantities. Source names G, alpha, S_mu_nu, potential, and acceleration are retained only as aliases. No SI dimensional analysis or physical calibration is implied.

Name	Reference value	Status	Notes
alpha_field	1.0	POLICY	Dimensionless saturation gain.
mass_scale	1.0	POLICY	Maps raw mass to bounded charge.
epsilon_distance	0.10	POLICY	Positive feature-space regularizer.

Name	Reference value	Status	Notes
field_decay	0.995	POLICY	Discounts influence, not custody.
legacy_singularity_limit	50.0	SOURCE-ONLY	Not authoritative in bounded profile.
legacy_L0	83	INHERITED METADATA	Carried only in FBS request.
hash	SHA-256	REQUIREMENT	Object, ledger, receipt, and snapshot integrity.
float encoding	IEEE-754 binary64 little-endian	REQUIREMENT	Canonical arrays.

4.1 Numeric and serialization rules

- All finite real calculations use binary64 under the deterministic backend profile.
- Canonical arrays are exact little-endian binary64 bytes plus shape and digest.
- NaN, infinity, shape mismatch, and noncanonical encodings are rejected.
- Canonical maps use UTF-8, sorted keys, and no insignificant whitespace.
- Wall-clock time never participates in a digest, tie-break, or result.
- Every field snapshot binds one immutable ledger head and route-node snapshot.

4.2 Mass profiles

Profile	Authoritative raw mass	Valid object
VECTOR_L2_V1	$m_raw = x _2$	Finite binary64 vector with declared shape.
BYTES_LENGTH_V1	$m_raw = \text{byte_length}$	Opaque canonical byte payload.

$$q = m_raw / (m_raw + \text{mass_scale})$$

For $m_raw \geq 0$ and $\text{mass_scale} > 0$, q is in $[0,1)$. raw_mass is a software statistic, not kilograms, energy, probability, moral weight, or conserved physical mass. Exact custody is preserved by object bytes and ledger history, not by q .

4.3 Placement profiles

Profile	Rule	Field eligibility
ROUTE_FEATURE_V1	Finite versioned feature vector r in $[-1,1]^d$ from an authorized provider.	Eligible after validation.
UNPLACED_V1	No lawful feature-space placement supplied.	Custody valid; excluded from field.

Digest-derived random coordinates are not used because hash proximity has no relation to failed route geometry. Physical location or spacetime coordinates are never inferred.

5. Custody Transaction and Authoritative Mass Accounting

5.1 Request validation

1. Verify producer authorization for source profile and policy epoch.
2. Verify request/provenance digests, `no_delete=true`, and source-event freshness.
3. Resolve and hash exact bytes before accepting a content reference.

4. Recompute raw mass from exact bytes; mass_proposal is diagnostic only.
5. Verify placement under its provider and profile.
6. Reject any attempt to reinterpret quarantine or ordinary rejection as exhaust without an authorized profile.

5.2 Durable custody commit

1. Serialize object record and ledger entry in memory.
2. Write exact object to a transaction-specific temporary path.
3. Flush/fsync, read back, and verify payload digest.
4. Append ledger entry under prior-head compare-and-swap.
5. Flush/fsync ledger and directory where supported.
6. Read back entry and recompute new head.
7. Publish immutable snapshot.
8. Seal receipt; source may release only after verification.

5.3 Ledger chain

$$\text{entry_digest}_n = \text{SHA256}(\text{canonical}(\text{entry}_n \text{ excluding entry_digest}))$$

$$\text{ledger_head}_n = \text{SHA256}(\text{ledger_head}_{(n-1)} \parallel \text{entry_digest}_n)$$

The ledger is append-only. A duplicate canonical request returns its existing receipt and does not append a second record. Conflicting bytes under the same source event ID are rejected before any write.

5.4 Accounting identities

$$\text{records_committed} = \text{count}(\text{committed acceptance entries owned by RTSOM})$$

$$\text{bytes_committed} = \sum_j \text{byte_length}_j$$

$$\text{raw_mass_total} = \sum_j \text{raw_mass}_j$$

$$\text{charge_total} = \sum_j q_j$$

raw_mass_total and charge_total are diagnostic sums and may change meaning under a new profile. The exact object ledger is authoritative. A migration creates a new acknowledged transition record and never erases historical provenance.

5.5 Source deletion rule

NO RECEIPT - NO RELEASE

The producer **MUST** retain the exact payload or immutable reference until it validates request digest, payload digest, accepted mass, object ID, ledger sequence, and receipt integrity. BACKPRESSURE, REJECTED, timeout, fault, or recovery request does not transfer custody.

5.6 Polarity and phase metadata correction

The source flips polarity and adds π to phase before storage. In the formal profile, $\text{failure_polarity_next} = -\text{failure_polarity_current}$ and $\text{failure_phase_next} = \text{wrap}(\text{failure_phase_current} + \pi)$ may be recorded as metadata. Payload bytes remain unchanged. Metadata transformations do not prove conservation, helicity, orthogonality, or physical phase transfer.

6. Placement, Bounded Negative-Space Field, and Stress Summary

6.1 Active field weight

Custody persists until acknowledged migration, but routing influence may age. For a committed placed record j at query cycle t :

$$\text{age}_j = \max(0, t - \text{accepted_cycle}_j)$$

$$w_j(t) = q_j * \text{field_decay}^{\text{age}_j}$$

w_j lies in $[0,1)$. Aging changes telemetry only. It never deletes the object or rewrites accepted mass.

6.2 Bounded kernel pressure

$$d_{ij} = \|r_i - r_j\|_2$$

$$K_{ij} = 1 / (1 + (d_{ij} / \text{epsilon_distance})^2)$$

$$s_i = \sum_j w_j K_{ij}$$

$$\text{rho_RTSOM}(i) = 1 - \exp(-\alpha_{\text{field}} * s_i)$$

THEOREM - BOUNDED PRESSURE

For $\alpha_{\text{field}} \geq 0$, finite nonnegative weights, and K_{ij} in $(0,1]$, $s_i \geq 0$ and $\text{rho_RTSOM}(i)$ lies in $[0,1)$. The reference field therefore cannot create the source unbounded acceleration or reach the legacy threshold 50.0.

6.3 Optional repulsion vector

$$u_i = \sum_j w_j K_{ij} * (r_i - r_j) / (d_{ij} + \text{epsilon_distance})$$

$$a_{\text{RTSOM}}(i) = 0 \text{ if } \|u_i\|=0; \text{ else } \tanh(\alpha_{\text{field}} \|u_i\|) * u_i / \|u_i\|$$

The vector norm is less than 1. It is a dimensionless feature-space direction away from placed failure records, not acceleration in meters per second squared, and it is not required by the Module 4 reference cost profile.

6.4 Computational entropy-stress summary

$$z_j = [1 ; r_j] / \|[1 ; r_j]\|_2$$

$$W = \sum_j w_j$$

$$S_{\text{RTSOM}} = 0 \text{ if } W=0; \text{ else } (1/W) \sum_j w_j z_j z_j^T$$

THEOREM - PSD SUMMARY

Every $z_j z_j^T$ is symmetric positive semidefinite. A nonnegative weighted average remains symmetric positive semidefinite. Because $\|z_j\|=1$, $\text{trace}(S_{\text{RTSOM}})=1$ whenever $W>0$, within declared numerical tolerance. This is a feature second-moment matrix, not a relativistic stress-energy tensor.

6.5 Snapshot construction and degeneracy

1. Bind one committed ledger head and one route-node snapshot digest.
2. Select committed, non-migrated, field-eligible records only.
3. Order records by ledger sequence and nodes lexicographically by NodeID.
4. Compute weights, kernels, pressure, optional vector, and stress summary.
5. Seal counts, diagnostics, exact output bytes, and field digest.
6. Return a read-only snapshot without mutating route or ledger state.

Condition	Result
No records	All pressure/vector values and S_{RTSOM} are zero.
All records UNPLACED	Custody metrics nonzero; spatial field zero.
Query node equals placement	$K=1$; direction contribution is zero.

Condition	Result
field_decay=0 and age>0	Old influence zero; custody retained.
alpha_field=0	Field zero; ledger unchanged.

7. QGM/GlyphNet Handoff and Consumer Boundary

7.1 Published evidence

RTSOM publishes rho_RTSOM(i), optional a_RTSOM(i), and S_RTSOM as read-only evidence. It does not write QGM phi_i, N_d, edge costs, LSAs, routes, forwarding tables, or IPv4/TCP state.

7.2 Reference integration

$$\text{rho}_i = \text{clip}(w_{\text{base}} * \text{rho}_{\text{base}_i} + w_{\text{RTSOM}} * \text{rho}_{\text{RTSOM}_i}, 0, 1)$$

The nonnegative weights are QGM policy. Module 4 remains responsible for the bounded recursive potential, positive edge-cost theorem, SPF calculation, route hysteresis, and delivery behavior. Module 14 cannot make a route decision.

7.3 Snapshot validity

Mismatch	Consumer action
Ledger head changed	Discard stale field and request a new one.
Route-node snapshot changed	Discard; no silent interpolation.
Profile/config digest changed	Discard and recompute.
Field digest mismatch	Fault; no QGM update.
Unknown NodeID	Ignore unknown entry and report schema mismatch.

7.4 Network failure boundary

A timeout, NXDOMAIN, TCP refusal, internal hop-limit expiry, stale route, or peer loss is not committed exhaust by itself. Module 4 applies retry and custody policy first. Only a valid GLYPHNET_FAILURE_V1 request can transfer exact payload custody. A receipt does not prove the failure should permanently influence all routes; influence remains bounded by field and QGM policy.

8. Capacity, Retention, Migration, and Recovery

8.1 Capacity admission

Before PREPARED, RTSOM evaluates hard capacities against committed usage plus the proposed object. Equality at the limit is allowed. Exceedance returns RTSOMBackpressurePacket; it does not accept custody, delete another object, or invoke FBS.

Capacity	Admission rule
Records	$\text{records_committed} + 1 \leq \text{capacity_records}$
Bytes	$\text{bytes_committed} + \text{object_byte_length} \leq \text{capacity_bytes}$
Charge	$\text{charge_total} + q \leq \text{capacity_charge}$

8.2 Retention classes

Class	Meaning
ACTIVE_FIELD	Object remains in custody and may influence field.
COLD_CUSTODY	Object remains in custody but is excluded from active field.
MIGRATION_PENDING	Exact object retained until destination receipt validates.
MIGRATED	Another owner acknowledged custody; provenance remains.
LEGAL_HOLD	No migration or compaction permitted.

8.3 Custody migration

1. Create MigrationRequest binding object digest, ledger sequence, and destination owner.
2. Keep object locally while destination writes and verifies it.
3. Validate destination CustodyMigrationReceipt.
4. Append migration ledger entry; never rewrite original acceptance.
5. Only then may local bytes be compacted under policy; digest and proof remain durable.

8.4 Purge and FBS correction

NO HARD FLUSH

The source FBS path hard-flushes a localized exhaust node. The formal profile forbids unreceipted purge because it would invalidate custody receipts and no-silent-deletion accounting. Recovery may quiesce field publication, isolate corruption, restore the last verified snapshot, or migrate exact objects. It may not erase acknowledged evidence merely to reduce a metric.

RTSOM may issue FBSRecoveryRequest only for persisted catastrophic conditions. It seals the last valid head, implicated objects, journal evidence, and requested scope. It does not run Cantor diagonalization, Collatz folding, reseed Delta, halt all routing, or mutate the ledger after delegation.

8.5 Source cosmology equations as hypotheses

Source equation / claim	Formal status	Evidence required before promotion
$G_{mn} + \Lambda g_{mn} = \kappa(T_{mn} + \alpha S_{mn})$	HYPOTHESIS only	Units, covariance, conservation, derivation, parameters, predictions, empirical tests.
$v(r) = \sqrt{G M(r)/r + \alpha S(r)}$	HYPOTHESIS only	Defined $S(r)$, dimensional consistency, dataset, fitting and out-of-sample validation.
Eliminates dark matter, dark energy, and inflaton fields	Rejected as normative claim	Independent physical evidence beyond a software model.
$\alpha=1$ calibrated to flat rotation curves	Unsupported source policy	Published reproducible calibration and uncertainty analysis.

These entries preserve the historical research direction without allowing it to authorize runtime behavior or appear as a proven result.

9. Invariants, Faults, and Catastrophic Conditions

9.1 Normative invariants

ID	Invariant	Requirement
M14-I01	Source retention	No producer releases an object before a matching receipt.
M14-I02	Exact object	Committed payload digest equals read-back object digest.
M14-I03	Exactly once	One source event produces at most one acceptance entry.
M14-I04	Mass authority	Accepted raw mass equals profile recomputation.
M14-I05	Charge bound	Every normalized charge lies in [0,1).
M14-I06	Append-only ledger	Prior entries and heads are never rewritten.
M14-I07	CAS ordering	Every entry names the exact prior ledger head.
M14-I08	Capacity	Committed usage never exceeds hard capacity.
M14-I09	Placement honesty	Unplaced objects never enter spatial field.
M14-I10	Field boundedness	Every rho_RTSOM lies in [0,1).
M14-I11	Vector boundedness	Every optional vector norm is <1.
M14-I12	PSD summary	S_RTSOM is symmetric PSD within tolerance.
M14-I13	Snapshot immutability	Field calculation does not mutate ledger or route state.
M14-I14	No QGM mutation	RTSOM never writes route costs, LSDB, or forwarding state.
M14-I15	No archive/Phi mutation	RTSOM never writes Psi archive or Phi state.
M14-I16	Custody separation	Quarantine is never silently reclassified as exhaust.
M14-I17	No local FBS	RTSOM emits typed requests only.
M14-I18	No purge	Custody ends only through acknowledged migration.
M14-I19	Claim separation	Hypothesis equations never enter conformance calculations.
M14-I20	Outcome accounting	Request, commit, receipt, migration, field, and recovery remain distinct.

9.2 Typed fault codes

Code	Condition
RTSOM_CONFIG	Configuration domain or digest invalid.

Code	Condition
RTSOM_SOURCE_PROFILE	Producer not authorized for source profile.
RTSOM_REQUEST_DIGEST	Canonical request digest mismatch.
RTSOM_TRANSACTION_CONFLICT	Same source event ID carries different bytes.
RTSOM_OBJECT_MISSING	Content reference cannot be resolved.
RTSOM_OBJECT_DIGEST	Object bytes do not match declared digest.
RTSOM_OBJECT_ENCODING	Encoding, shape, or type invalid.
RTSOM_NONFINITE	Vector or placement contains NaN/Inf.
RTSOM_MASS_PROFILE	Unknown or inapplicable mass profile.
RTSOM_MASS_MISMATCH	Recomputed mass conflicts with required constraint.
RTSOM_PLACEMENT	Placement profile, dimension, range, or digest invalid.
RTSOM_CAPACITY	Usage accounting conflicts with committed ledger.
RTSOM_LEDGER_HEAD	Prior head mismatch outside ordinary contention.
RTSOM_LEDGER_DIGEST	Ledger chain verification fails.
RTSOM_STORAGE	Write, fsync, rename, or directory sync fails.
RTSOM_READBACK	Durable object or ledger read-back fails.
RTSOM_RECEIPT	Receipt does not bind the committed record.
RTSOM_FIELD_SNAPSHOT	Field input snapshot or output digest invalid.
RTSOM_NUMERICAL	Field or PSD diagnostics exceed tolerance.
RTSOM_MIGRATION	Destination custody receipt invalid.

9.3 Catastrophic recovery conditions

- RTSOMC_LEDGER_SPLIT: different committed heads claim the same predecessor and sequence.
- RTSOMC_ACKNOWLEDGED_OBJECT_LOSS: a valid receipt exists but exact custody cannot be reconciled locally or at a migration destination.
- RTSOMC_CONFLICTING_RECEIPTS: one transaction key has two different committed receipts.
- RTSOMC_UNAUTHORIZED_PURGE: acknowledged bytes were removed without a migration transition.
- RTSOMC_DETERMINISM_FAILURE: identical canonical inputs produce different field bytes under one backend profile.
- RTSOMC_PERSISTED_NEGATIVE_MASS: a committed record violates its mass profile or contains an impossible negative statistic.

Capacity exhaustion, high mass, high pressure, many failures, an unplaced object, ordinary CAS contention, stale field input, malformed request, network timeout, archive rejection, quarantine, or the source threshold 50.0 are not catastrophic by themselves.

10. Metrics, Measurements, and Reconciliation

10.1 Required counters and distributions

Metric	Definition
requests_total	All canonical custody attempts by source profile.
accepted_total	Durably committed acceptance records.
duplicate_replays	Idempotent requests returning existing receipt.
rejected / backpressure / faults	Mutually exclusive non-commit outcomes.
raw_mass / normalized_charge	Per-profile distributions and totals.
records / bytes / charge usage	Current and capacity percentages.
placed / unplaced	Field-eligible and ineligible custody records.
field_pressure	Per-node mean, p95, max; all <1.
field_vector_norm	Optional vector mean, p95, max; all <1.
stress_trace / min_eigenvalue	PSD reconciliation diagnostics.
receipt_latency	Validation, write, ledger CAS, read-back, total.
migration_pending / migrated	Custody-transfer lifecycle.
recovery_requests / completed	Requested and owner-completed separately.

10.2 Reconciliation equations

$$\text{requests_total} = \text{accepted_new} + \text{duplicate_replays} + \text{rejected_total} + \text{backpressure_total} + \text{faults_total} + \text{recovery_requests}$$

$$\text{accepted_new} = \text{receipts_issued_new} = \text{ledger_entries_accept}$$

$$\text{records_committed} = \text{active_field} + \text{cold_custody} + \text{migration_pending} + \text{migrated_provenance} + \text{legal_hold}$$

$$\text{objects_held_by_RTSOM} = \text{accepted_new} - \text{migrated_with_valid_receipt}$$

$$\text{field_used} + \text{field_unplaced} + \text{field_inactive} = \text{records_visible_at_snapshot}$$

Dashboards must distinguish OBSERVED failure, CUSTODY REQUESTED, PREPARED, COMMITTED, RECEIPT ISSUED, SOURCE RELEASED, FIELD PUBLISHED, QGM CONSUMED, MIGRATED, FBS REQUESTED, and FBS EXECUTED.

10.3 Alert policy

- Warn when any capacity exceeds 80%; backpressure at the exact hard limit.
- Alert on receipt/object reconciliation failure, ledger-chain failure, or unauthorized deletion.
- Warn when the unplaced-record fraction exceeds policy threshold.
- Warn when field snapshots are repeatedly stale; never reuse stale evidence.
- Do not label high pressure, high mass, or high failure count as a physical singularity.

11. Deterministic Conformance Test Matrix

ID	Case	Expected result
M14-T01	Valid Psi residue request	One commit and matching receipt.

ID	Case	Expected result
M14-T02	Valid GlyphNet failure request	One commit after profile validation.
M14-T03	Unregistered producer	RTSOM_SOURCE_PROFILE; no mutation.
M14-T04	no_delete=false	Reject; no custody.
M14-T05	Request digest mismatch	RTSOM_REQUEST_DIGEST.
M14-T06	Exact replay	Existing receipt; no new entry.
M14-T07	Same event ID, changed payload	RTSOM_TRANSACTION_CONFLICT.
M14-T08	Missing object reference	RTSOM_OBJECT_MISSING.
M14-T09	Payload digest mismatch	RTSOM_OBJECT_DIGEST.
M14-T10	NaN vector	RTSOM_NONFINITE.
M14-T11	Infinity placement	RTSOM_NONFINITE.
M14-T12	Vector mass profile on bytes	RTSOM_MASS_PROFILE.
M14-T13	Known vector L2 mass	Accepted mass equals expected norm.
M14-T14	Known byte payload	Accepted mass equals byte length.
M14-T15	Zero raw mass	q=0; custody valid.
M14-T16	Very large raw mass	q<1 and finite.
M14-T17	Placement outside range	RTSOM_PLACEMENT.
M14-T18	Placement dimension mismatch	RTSOM_PLACEMENT.
M14-T19	Explicit UNPLACED	Custody accepted; field false.
M14-T20	Capacity equality	Last fitting record accepted.
M14-T21	Record capacity exceeded	BACKPRESSURE; no mutation.
M14-T22	Byte capacity exceeded	BACKPRESSURE; no mutation.
M14-T23	Charge capacity exceeded	BACKPRESSURE; no mutation.
M14-T24	Object write failure	RTSOM_STORAGE; no head publish.
M14-T25	Object read-back mismatch	RTSOM_READBACK.
M14-T26	Ledger append failure	Abort; prior head unchanged.
M14-T27	Ordinary CAS contention	Retry/hold; not FBS.
M14-T28	Conflicting committed heads	FBSRecoveryRequest.
M14-T29	Receipt request mismatch	Consumer rejects receipt.
M14-T30	Receipt mass mismatch	Consumer rejects receipt.
M14-T31	No field records	Zero field snapshot.

ID	Case	Expected result
M14-T32	All records unplaced	Zero field; custody preserved.
M14-T33	Zero distance kernel	K=1; finite pressure.
M14-T34	Large feature distance	K approaches zero; finite.
M14-T35	alpha_field=0	All field values zero.
M14-T36	One placed record	rho in [0,1), vector norm <1.
M14-T37	Many coincident records	Pressure <1; no singularity.
M14-T38	field_decay=1	No age discount.
M14-T39	field_decay=0 and age>0	Old influence zero; object retained.
M14-T40	Stress one record	Symmetric PSD; trace=1.
M14-T41	Stress many records	Symmetric PSD; trace=1.
M14-T42	Tiny negative eigenvalue	Diagnostic clip within tolerance.
M14-T43	Material negative eigenvalue	RTSOM_NUMERICAL.
M14-T44	Stale ledger field request	Reject stale request.
M14-T45	Stale route snapshot	Reject stale request.
M14-T46	Unknown field profile	RTSOM_FIELD_SNAPSHOT.
M14-T47	Valid QGM integration weights	QGM-owned bounded combination.
M14-T48	RTSOM attempts route mutation	Invariant failure.
M14-T49	Timeout without custody request	No RTSOM mutation.
M14-T50	Quarantine without profile	No RTSOM mutation.
M14-T51	Ordinary Module 11 rejection	No RTSOM mutation.
M14-T52	Migration pending	Local bytes retained.
M14-T53	Valid migration receipt	Append migration transition.
M14-T54	Invalid migration receipt	No release.
M14-T55	Unauthorized local purge	Catastrophic request.
M14-T56	Acknowledged object missing	Catastrophic request.
M14-T57	Identical snapshot replay	Byte-identical field output.
M14-T58	Legacy acceleration threshold	Not evaluated in reference profile.
M14-T59	Cosmology equation as packet	Ignored/rejected as non-normative.
M14-T60	Crash after commit before receipt	Journal returns same receipt.

12. Property, Metamorphic, Fuzz, Concurrency, and Stress Testing

12.1 Property tests

- For $m \geq 0$ and $\text{mass_scale} > 0$, $q = m / (m + \text{mass_scale})$ is in $[0, 1)$.
- For finite distance and $\text{epsilon_distance} > 0$, K is in $(0, 1]$.
- For $\text{alpha_field} \geq 0$ and $s \geq 0$, ρ is in $[0, 1)$.
- Every repulsion-vector norm is less than 1.
- Every nonempty S_{RTSOM} is symmetric PSD with trace 1 within tolerance.
- Every ledger head is reproducible from genesis and ordered entries.
- No non-commit outcome changes count, bytes, charge, or ledger head.

12.2 Metamorphic tests

- Permuting field input records leaves output unchanged after canonical ordering.
- Duplicating a request as replay leaves committed state unchanged except replay count.
- Scaling a vector by $c \geq 0$ scales raw L2 mass by c while q remains below 1.
- Moving a query node farther from every record cannot increase any individual kernel.
- Changing `ACTIVE_FIELD` to `COLD_CUSTODY` removes field influence but not custody.
- Changing analogy labels does not change canonical execution bytes.

12.3 Fuzz and malformed-input tests

- Fuzz canonical ordering, duplicate keys, malformed base64, truncated vectors, oversized integers, and invalid UTF-8.
- Fuzz NaN bit patterns, signed zero, subnormals, extreme finite values, and shape overflows.
- Fuzz source-profile combinations so no unregistered owner receives a receipt.
- Fuzz placement dimensions/ranges without allocating unbounded matrices.
- Fuzz object-reference races, partial reads, and digest substitution.

12.4 Concurrency and crash recovery

- Race identical requests and prove exactly one new entry and one stable receipt.
- Race different requests against one head and prove CAS retry without split head.
- Inject crashes after temp write, object fsync, ledger append, head CAS, publication, and before receipt delivery.
- Recover from journal without duplicate acceptance or object loss.
- Race field publication with commits; every snapshot binds exactly one head.

12.5 Long-run stress

- Run at least 100,000 custody attempts with mixed profiles, replays, backpressure, and migrations.
- Run 100,000 field snapshots across random bounded placements and verify all bounds.
- Recompute every ledger head and reconcile every receipt to an object.
- Verify memory remains bounded by configured capacities and pending limits.
- Verify no hypothesis equation, physical unit, or legacy threshold enters execution.

13. Reference Pseudocode

```
def accept_exhaust(req, cfg, store, ledger, journal, auth):
    canonical = canonicalize(req)
    verify_digest(canonical, req.request_digest)
    verify_registered_source(req.source_owner, req.source_profile, auth)
    require(req.no_delete is True)

    tx_key = (req.source_owner, req.source_event_id,
              req.request_digest, req.policy_epoch,
              req.prior_ledger_head)
    replay = journal.lookup(tx_key)
    if replay is not None:
        require(replay.request_bytes == canonical)
        return replay.receipt

    payload = resolve_exact_payload(req.payload, req.object_ref)
```

```

require(sha256(payload) == req.payload_digest)
obj = decode_and_validate(payload, req.object_kind, req.payload_encoding)

raw_mass = compute_mass(obj, payload, req.mass_profile_id)
require(raw_mass >= 0 and isfinite(raw_mass))
charge = raw_mass / (raw_mass + cfg.mass_scale)
placement = validate_or_unplaced(req, cfg)

usage = ledger.usage(req.prior_ledger_head)
if not fits(usage, len(payload), charge, cfg):
    return RTSOMBackpressurePacket(...)

record = ExhaustObjectRecord(payload_digest=req.payload_digest,
    byte_length=len(payload), raw_mass=raw_mass,
    normalized_charge=charge, placement=placement,
    source_event_id=req.source_event_id,
    previous_ledger_head=req.prior_ledger_head, ...)
entry = seal_ledger_entry(record)

store.write_temp_and_fsync(tx_key, payload)
store.readback_verify(tx_key, req.payload_digest)
new_head = ledger.append_cas(req.prior_ledger_head, entry)
ledger.fsync_and_verify(new_head)
store.publish(tx_key, record.object_id)

receipt = seal_receipt(req, record, new_head)
journal.commit(tx_key, canonical, receipt)
return receipt

def build_field(field_req, cfg, ledger, route_nodes):
    require(field_req.ledger_head == ledger.current_head())
    require(field_req.route_snapshot_digest == route_nodes.digest)
    records = ledger.field_eligible_records(field_req.ledger_head)
    outputs = {}
    for node in sorted(route_nodes, key=lambda n: n.node_id):
        s = 0.0
        u = zeros(cfg.field_dimension)
        for rec in records:
            age = max(0, field_req.query_cycle - rec.accepted_cycle)
            w = rec.normalized_charge * (cfg.field_decay ** age)
            dr = node.position - rec.position
            d = l2(dr)
            K = 1.0 / (1.0 + (d / cfg.epsilon_distance) ** 2)
            s += w * K
            u += w * K * dr / (d + cfg.epsilon_distance)
        rho = 1.0 - exp(-cfg.alpha_field * s)
        a = zeros_like(u) if l2(u) == 0 else tanh(cfg.alpha_field * l2(u)) * u / l2(u)
        outputs[node.node_id] = (rho, a)
    S = weighted_psd_summary(records, field_req.query_cycle, cfg)
    return seal_field_snapshot(field_req, outputs, S)

```

13.1 Migration pseudocode

```

def migrate_object(object_id, destination, ledger, store):
    obj = ledger.lookup(object_id)
    request = seal_migration_request(obj, destination)
    receipt = destination.accept_exact_object(request, store.read(object_id))
    verify_migration_receipt(request, receipt)
    ledger.append_migration_transition(obj, receipt)
    # Local compaction is optional and policy-controlled.
    # Original acceptance and migration proof remain durable.

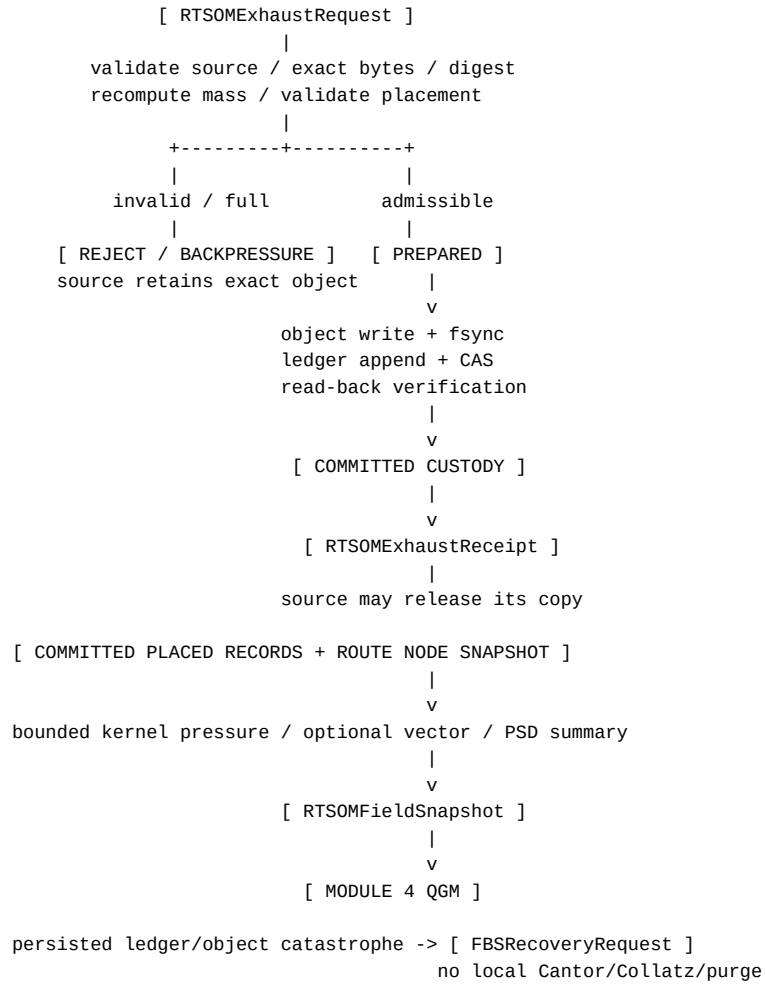
```

14. ASCII System Flow

```

[ AUTHORIZED SOURCE OWNER ]
Psi residue | GlyphNet failure | FBS residue | operator import
            |
            v

```



15. Cleaned Rosetta Stone

SpiralCore term	Computational meaning	Analogy boundary
RTSOM	Transactional bounded-exhaust custody owner and field publisher.	Not a validated theory of gravity.
Star Ocean	Append-only population of retained failure/residue objects.	Not an astrophysical fluid.
Bounded Exhaust Budget	Capacity-limited object store plus custody ledger.	Not a physical energy reservoir.
Negative Space Inference	Placed failure evidence produces bounded routing pressure.	Does not prove ontology.
Mass DeltaM	Versioned statistic such as L2 norm or byte length.	Not kilograms or conserved energy.
Normalized Charge q	Saturating map from raw mass to [0,1).	Not electric charge.
Entropy-Stress Tensor	Normalized PSD second-moment matrix S_RTSOM.	Not relativistic stress-energy.

SpiralCore term	Computational meaning	Analogy boundary
RTSOM Pressure	Bounded radial-basis kernel accumulation.	Not gravitational potential.
a_RTSOM	Bounded feature-space direction away from failures.	Not physical acceleration.
Spin-to-Helicity	Optional polarity/phase failure metadata.	No plasma-helicity claim.
Dark Brane	Legacy name for negative-state custody domain.	Not a literal brane.
Singularity	Persisted ledger/object integrity catastrophe only.	High count is not spacetime singularity.
FBS shunt	Typed request to Module 16 with sealed evidence.	RTSOM does not execute recovery.
Lossless	No silent deletion and receipted custody transitions.	Does not imply reversibility.

16. Glossary and Revision Summary

16.1 Glossary

Term	Definition
Acknowledged custody	Exact object and ledger record are durable and a matching receipt exists.
Backpressure	Capacity outcome in which custody is not accepted.
Charge	Bounded statistic $q=m/(m+scale)$ used for field weighting.
Cold custody	Retained object excluded from active field influence.
Custody migration	Receipted transfer of an exact object to another owner.
Exhaust object	Exact payload or residue accepted under a source profile.
Field snapshot	Read-only bounded output tied to one ledger and route snapshot.
Ledger head	Digest committing to ordered append-only custody history.
Mass profile	Versioned function computing a statistic from exact bytes.
Negative-space field	Bounded kernel summary over placed failures.
Placed record	Custody record with a valid route-feature coordinate.
Receipt	Acknowledgement binding request, object, mass, and ledger transition.
Retention class	Policy controlling field participation and migration.
Source event ID	Exactly-once identity assigned by the producer.

Term	Definition
Stress summary	Normalized PSD second-moment matrix of placed features.
Unplaced record	Valid custody object excluded from spatial field.

16.2 Revision summary

Source issue	Formalized correction
Every rejected state becomes exhaust.	Only registered typed requests mutate RTSOM.
RTSOM directly receives raw candidate from Sigma.	Source owners explicitly delegate exact custody.
Polarity/phase transform stored state.	Payload remains exact; values are metadata only.
L2 norm called conserved informational mass.	L2 is a software statistic; bytes preserve custody.
Unbounded inverse-cube acceleration.	Bounded kernel pressure and optional direction.
S_mu_nu named but not built.	Computable normalized symmetric PSD second moment.
RTSOM writes QGM drift map.	Publishes evidence; Module 4 owns routing.
Threshold 50 triggers FBS.	Bounded field; capacity causes backpressure.
FBS hard-flushes budget.	Unreceipted purge forbidden; migrate or recover.
Module runs Cantor/Collatz.	Typed request only; Module 16 owns execution.
Claims replacement of cosmological components.	Moved to hypothesis registry.
alpha=1 claimed observationally calibrated.	Dimensionless policy; physical calibration unsupported here.
Physical G mixed into software.	Reference profile is dimensionless.
No atomic transaction or receipt.	Object store, digest chain, CAS, read-back, receipt.
No placement definition.	Versioned route-feature placement or UNPLACED.

17. Source Basis and Normative Checklist

17.1 Source basis

- SPIRALCORE SPECIFICATION: MODULE 14 - RTSOM, source v14 module.
- Spiralcore_v14 consolidated specification, including Modules 3, 4, 5, 7, 10, 11, 12, 13, and 16 ownership descriptions.
- Formalized v14.1 Modules 3-13 establish typed custody requests, no-silent-deletion semantics, QGM ownership, archive sequencing, quarantine separation, Phi event ownership, and FBS request-only boundaries.
- The source bibliography motivates terminology and hypotheses; it does not by itself specify this software contract.

17.2 Normative implementation checklist

- ☐ Accept custody only from registered typed source profiles.
- ☐ Retain exact bytes or verify immutable content reference.
- ☐ Recompute authoritative mass; never trust proposal alone.
- ☐ Keep payload unchanged; polarity/phase are metadata only.

- ☐ Enforce record, byte, and charge capacities before commit.
- ☐ Commit through durable write, ledger CAS, fsync, and read-back.
- ☐ Return idempotent receipt bound to request and transition.
- ☐ Produce only bounded field values and a PSD summary.
- ☐ Exclude UNPLACED and inactive records from spatial field.
- ☐ Never mutate QGM, GlyphNet, Psi, Phi, quarantine, or FBS.
- ☐ Never hard-flush acknowledged custody; require migration receipt.
- ☐ Keep physical/cosmological equations in HYPOTHESIS scope.
- ☐ Distinguish request, commit, receipt, release, field, QGM use, migration, and recovery.
- ☐ Pass all 60 conformance cases and long-run tests.

END STATE

A conforming Module 14 converts one valid custody request into one durable, receipted object-ledger transition or one explicit non-commit outcome. No failed payload or archive residue is silently deleted; no quarantine object is automatically relabeled as exhaust; mass remains a versioned statistic rather than physical substance; placed records generate a bounded deterministic pressure field and PSD feature summary; QGM retains route authority; and catastrophic recovery remains owned by FBS/ROVER.

SPIRALCORE SPECIFICATION

MODULE 15 - ROVER

(ELIAS ROVER Scanner + GWT Dimensional Scaler + HOT Evaluator - Presentation Projection, Broadcast Watchdog, and FBS Activation Boundary)

Cleaned and formalized revision - v14.1

Field	Value	Field	Value
Document class	Normative projection and broadcast-authorization contract	Execution profiles	POST_COMMIT_SCAN / RECOVERY_WATCH
Architecture	SpiralCore v14	Reference profile	SC14.1-M15-ROVER-01
Presentation owner	Module 15 ROVER	Display consumer	Core UI / presentation adapter
Recovery executor	Module 16 FBS	Exhaust owner	Module 14 RTSOM

NORMATIVE STATUS

Exact within the declared software scope. ROVER consumes immutable, digest-bound state and context snapshots; constructs registered lower-cost view projections; verifies both baseline coherence and reconstruction fidelity; computes a bounded HOT watchdog score; locks or authorizes presentation; and, only in the recovery-watch profile with a valid catastrophic incident, may issue a single-use FBS activation permit. It does not commit memory, establish consciousness, alter the candidate, execute recovery, or store exhaust.

CLAIM BOUNDARY

Observer, Node Theta, global workspace, pre-conscious buffer, higher-order thought, lucidity, subjective time, 4D block universe, phase animation, and dimensional rendering are architectural names. Their normative meanings are only the typed snapshots, deterministic projections, bounded scores, logical-cycle pacing, quadrature render channels, and permit transitions defined here. They are not evidence of consciousness, identity, perception, cognition, spacetime, or biological phenomenology.

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1. Normative Scope, Ownership, and Claim Classes

1.1 Purpose

ROVER is the terminal read-only presentation scanner and the final broadcast watchdog for a committed SpiralCore state. It computes a deterministic set of candidate representations, chooses the least-cost representation that preserves both reference coherence and source fidelity, packages a non-mutating phase render, and authorizes or locks delivery to the presentation adapter. In a separate recovery-watch profile, it verifies a catastrophic incident and may authorize Module 16 to execute the FBS recovery sequence.

1.2 Two operating profiles

Profile	Starts from	Permitted result
POST_COMMIT_SCAN	One immutable Psi commit snapshot plus observer and baseline snapshots	Broadcast permit, presentation hold, or typed fault

Profile	Starts from	Permitted result
RECOVERY_WATCH	One valid catastrophic incident packet plus locked branch snapshot	FBS activation permit, recovery hold, or typed fault

The two profiles share numerical primitives but have different authority. POST_COMMIT_SCAN cannot invoke FBS merely because a representation is unavailable or HOT is low. RECOVERY_WATCH cannot display an uncommitted branch and cannot execute the recovery it authorizes.

1.3 Ownership boundary

Concern	Authoritative owner	Module 15 relationship
Candidate generation and raw state	Module 1 Delta-Lattice	ROVER never writes or repairs the candidate.
Coherence and route decision	Module 2 Sigma-Lattice	ROVER does not recompute route-authoritative Sigma metrics.
Archive preparation and commit	Module 3 Psi-Lattice	ROVER consumes an immutable committed snapshot only.
Routing and network transport	Modules 4 and 8	ROVER does not route packets or open sockets.
Admission and quarantine	Modules 5 and 10 / custody service	ROVER does not classify harm or quarantine objects.
Paradox evidence H_G	Module 7	ROVER may carry bounded evidence; it cannot increase presentation authority.
Repair and revalidation	Modules 9-11	ROVER receives the final committed result after those stages.
Persistent hysteresis	Module 12 Phi	Observer context is read-only; ROVER never updates Phi.
Archive collision synthesis	Module 13 RMX	ROVER receives the final committed synthesis only.
Exhaust custody and field	Module 14 RTSOM	ROVER may display a read-only field overlay; no exact exhaust access.
Broadcast decision and render packet	Module 15 ROVER	Exclusive authority defined by this module.
Rollback, diagonalization, fold, seed	Module 16 FBS	ROVER issues a permit; Module 16 executes and receipts.
User-visible display	Core UI / presentation adapter	Consumes the permit and returns a presentation receipt.

1.4 Explicit non-goals

- No assertion that the system is conscious, pre-conscious, lucid, dissociated, subjective, self-aware, or biologically cognitive.
- No claim that a software vector is a 4D universe, a perception field, spacetime, or a physical observer.
- No archive commit, uncommit, mutation, repair, deletion, quarantine, routing, network transmission, or custody transfer.

- No forced low-dimensional fallback when every registered projection fails its gates.
- No use of paradox magnitude, H_G, arousal, or mythic terminology to increase presentation authorization.
- No automatic FBS activation for one ordinary low-HOT scan, missing baseline, provider warming, UI timeout, or no coherent view.
- No local Cantor diagonalization, Collatz processing, seed generation, rollback, or RTSOM write.
- No claim that a rendered view preserves all semantics beyond the exact fidelity metrics and provider contract reported in the packet.

1.5 Statement classes and precedence

Class	Meaning
DEFINITION	Fixes a type, domain, serialization, or exact mathematical object.
REQUIREMENT	Mandatory behavior for conformance.
INVARIANT	Property that must hold for every completed scan or permit transition.
POLICY	Versioned deployment choice such as a threshold, cost, or permit lifetime.
SURROGATE	Computational quantity inspired by external language but not identical to it.
HYPOTHESIS	Research interpretation excluded from runtime conformance.
NON-GOAL	Behavior or claim explicitly outside the contract.

1. Typed packet schemas, digests, and canonical serialization have highest precedence.
2. Normative requirements and invariants override names such as observer, GWT, HOT, and time dilation.
3. Reference policies apply only under the exact profile and policy digests.
4. Surrogates and hypotheses cannot authorize a mutation, broadcast, or recovery action.

2. Operating Profiles, Start Conditions, and Lifecycle

2.1 POST_COMMIT_SCAN start condition

COMMITTED STATE -> SCAN -> PERMIT OR HOLD

The presentation profile starts only after Psi has durably committed the exact state and produced a CommittedStateSnapshot. The snapshot must bind the archive head, record key, state digest, lineage digest, validation-chain digest, and logical commit sequence. ROVER cannot scan a provisional RMX state, an uncommitted HARMONY repair, or an object whose final validation chain is stale.

2.2 RECOVERY_WATCH start condition

INCIDENT -> LOCK -> VERIFY -> FBS ACTIVATION PERMIT

The recovery profile starts from a CatastrophicIncidentPacket issued by a registered module or by a proven ROVER control-state invariant failure. The branch is already presentation-locked. ROVER verifies the incident, branch digest, policy epoch, evidence, and liveness state; HOT below 0.50 may complete the activation predicate only within this profile. Module 16 remains the sole recovery executor.

2.3 Lifecycle states

State	Meaning	Allowed successor
UNINITIALIZED	Configuration, providers, and policy digest not verified.	READY or FAULTED
READY	Accepting scans or recovery-watch requests.	SNAPSHOT_VALIDATING, RECOVERY_VALIDATING, or FAULTED
SNAPSHOT_VALIDATING	Commit, observer, baseline, and provider snapshots verified.	PHASE_CORRECTING, HOLD, or FAULTED
PHASE_CORRECTING	Orthogonal context transform computed.	PROJECTING or FAULTED
PROJECTING	Registered view candidates built and reconciled.	SCORING, HOLD, or FAULTED
SCORING	Projection gates and HOT score evaluated.	BROADCAST_AUTHORIZED, BROADCAST_LOCKED, or FAULTED
BROADCAST_AUTHORIZED	Single-use permit and render packet issued.	WAIT_UI_RECEIPT or READY
WAIT_UI_RECEIPT	Permit consumption pending.	PRESENTED, EXPIRED, or FAULTED
PRESENTED	Matching UI receipt accepted.	READY
BROADCAST_LOCKED	No presentation allowed for this scan or incident.	READY, RECOVERY_REQUESTED, or FAULTED
HOLD	Recoverable missing/stale/not-ready input; no mutation.	READY
RECOVERY_VALIDATING	Catastrophic incident and branch liveness checked.	RECOVERY_REQUESTED, HOLD, or FAULTED
RECOVERY_REQUESTED	Single-use FBS activation permit issued.	WAIT_FBS_RECEIPT
WAIT_FBS_RECEIPT	Module 16 completion or refusal pending.	RECOVERY_COMPLETED, HOLD, or FAULTED
RECOVERY_COMPLETED	Matching Module 16 receipt accepted; broadcast remains locked until a new commit passes.	READY
EXPIRED	Permit lifetime elapsed without valid consumption.	READY
FAULTED	Local deterministic processing cannot continue safely.	Operator or owner-module resolution

2.4 Required scan order

1. Validate config, profile, policy epoch, commit snapshot, baseline snapshot, observer snapshot, and provider registry.
2. Verify all digests, dimensions, finiteness, logical sequences, and candidate independence claims.
3. Apply the declared orthogonal phase-correction operator without mutating the committed state.
4. Generate every registered projection and baseline projection under one frozen provider epoch.

5. Compute reconstruction, coherence, fidelity, cost, and deterministic eligibility for each view.
6. Choose the least-cost eligible view using the exact tie-break order.
7. Compute bounded pacing, stability, trend, HOT logit, and HOT score.
8. Lock broadcast on no eligible view or HOT below threshold.
9. Otherwise seal one render packet and one single-use broadcast permit.
10. Accept a UI presentation receipt only when every binding field reconciles.

2.5 Frozen-snapshot rule

One scan uses one immutable tuple: (commit head, state digest, baseline epoch, observer epoch, projection-provider epoch, policy epoch, previous-scan reference). Any mid-scan change aborts the attempt with a typed hold or fault. ROVER never combines values from different epochs and never reads a moving archive window while computing a score.

3. Typed Data Contracts

3.1 RoverConfig

Field	Reference value / domain	Requirement
schema_id	SC14.1-M15-ROVER-01	Exact protocol identity.
scan_profiles	POST_COMMIT_SCAN, RECOVERY_WATCH	Only registered profiles.
projection_profile	BLOCK_MEAN_PROGRESSIVE_V1	Equations in Section 6.
phase_profile	PAIR_ROTATION_V1	Orthogonal transform in Section 5.
view_levels	{1,2,3}	Ordered presentation labels, not physical dimensions.
tau_view	0.85 in [-1,1]	Minimum coherence and fidelity score.
hot_threshold	0.50 in [0,1]	Broadcast watchdog boundary.
alpha / beta / gamma	0.50 / 0.30 / 0.20	Sum exactly one.
hot_slope	4.0 >0	Stable logistic sharpness.
lambda_0	0.1459 in (0,1)	Logical pacing floor.
load_scale	1.0 >0	Norm saturation scale.
trend_scale	1.0 >0	Finite-difference normalization.
permit_ttl_cycles	1	Permit valid for the next logical presentation cycle only.
numeric_tolerance	1e-9	Explicit comparison tolerance.
digest_algorithm	SHA-256	Canonical packet and payload digests.

3.2 CommittedStateSnapshot

Field	Type / requirement
-------	--------------------

archive_head / commit_sequence	Immutable Psi head digest and strictly monotonic logical sequence.
record_key / state_digest	Final archive key and SHA-256 digest of canonical state bytes.
state_vector	Finite float64 vector of declared DIM; exact committed representation.
lineage_digest	Binds raw, repair, RMX, Sigma, LORIEN, Module 10, and Module 11 evidence as applicable.
validation_chain_digest	Proof that the archived bytes are the bytes being scanned.
phi_snapshot_digest	Read-only Module 12 state associated with the commit.
rtsom_field_digest	Optional read-only field overlay; never exact exhaust bytes.
snapshot_digest	SHA-256 over all canonical fields.

3.3 ObserverContextSnapshot

Field	Type / requirement
observer_context_id	Opaque context identifier; not a person or consciousness claim.
theta	Finite radians, canonical wrap to $(-\pi, \pi]$.
omega	Finite radians per logical cycle.
logical_cycle	Monotonic integer supplied by runtime.
context_epoch / provider_id	Versioned immutable source.
lambda_hint	Optional finite diagnostic; does not override config <code>lambda_0</code> .
snapshot_digest	SHA-256 over canonical context.

3.4 RoverBaselineSnapshot

Field	Type / requirement
baseline_id / baseline_epoch	Versioned accepted-history or configured attractor reference.
baseline_vector	Finite float64 DIM-vector.
candidate_excluded	Must be true unless the baseline is a static boot reference.
source_head	Archive head or configuration digest from which it was derived.
provider_id / algorithm_id	Deterministic baseline provider identity.
snapshot_digest	SHA-256 over canonical fields.

3.5 ViewProjectionProfile and ProjectionCandidate

Field	Type / requirement
view_id / nominal_dimension	Stable ID and label in {1,2,3}.
feature_count / representation_shape	Declared retained scalar count and shape.
projection_operator_id	Versioned deterministic operator.
features / baseline_features	Finite projected arrays in identical shape.
reconstruction	Finite DIM-vector reconstructed from features.
coherence / fidelity / score	Values in [-1,1]; score=min(coherence,fidelity).
cost_units	Positive deterministic cost; independent of runtime wall-clock timing.
readiness	READY, NOT_READY, or FAULT.
candidate_digest	SHA-256 over provider, source, features, reconstruction, and metrics.

3.6 RoverScanInput and RoverScanPacket

Object	Required contents
RoverScanInput	Profile, config digest, committed snapshot, observer snapshot, baseline snapshot, provider snapshot, previous scan reference, policy epoch.
RoverScanPacket	Attempt ID, frozen input digests, phase-corrected digest, all candidate packets, selected view or none, pacing, stability, trend, HOT components, outcome, evidence digest.

3.7 RenderFramePacket and RoverBroadcastPermit

Object	Required contents
RenderFramePacket	Selected view ID and shape, feature bytes, in-phase channel, quadrature channel, phase, state digest, scan digest, render digest.
RoverBroadcastPermit	Single-use permit ID, render digest, state digest, archive head, UI target, policy epoch, issue cycle, expiry cycle, scan digest, permit digest.
UIBroadcastReceipt	Permit ID/digest, render digest, UI target, presented cycle, presentation status, receipt digest.

3.8 Hold, incident, recovery, and fault packets

Object	Required contents
RoverHoldPacket	Attempt ID, reason code, exact missing/stale/failed gate, evidence digest, no_mutation=true.

Object	Required contents
CatastrophicIncidentPacket	Issuer, incident class, branch digest, invariant evidence, policy epoch, sequence, incident digest.
FBSActivationPermit	Single-use incident-bound activation ID, branch digest, policy epoch, issue/expiry cycle, authorized actions=MODULE16_ONLY, permit digest.
FBSRecoveryReceipt	Activation ID/digest, Module 16 execution profile, prior branch digest, result status, new seed digest or refusal, RTSOM receipt reference if residue transferred.
RoverFault	Fault code, stage, input digests, evidence digest, mutation_performed=false.

4. Constants, Domains, and Reference Profile

4.1 Reference constants

Symbol / field	Value	Normative meaning
DIM	81	Reference state dimension; other values require a registered provider.
D_views	{1,2,3}	Presentation fidelity levels, not spatial dimensions.
m_1, m_2, m_3	9, 27, 81	Retained scalar counts for the reference projection provider.
tau_view	0.85	Minimum of coherence and reconstruction fidelity.
HOT_override	0.50	Broadcast threshold and recovery-watch completion threshold.
alpha, beta, gamma	0.50, 0.30, 0.20	HOT logit weights: view, state, trend.
k_HOT	4.0	Logistic slope.
lambda_0	0.1459	Logical pacing floor; dimensionless software policy.
s_load	1.0	Load normalization scale.
s_trend	1.0	Trend normalization scale.
epsilon	1e-9	Numerical guard and comparison tolerance.
permit_ttl	1 cycle	Presentation permit lifetime.

4.2 Numeric domains

- Committed and baseline vectors are finite float64 arrays of the same declared DIM.
- Theta is canonicalized to $(-\pi, \pi]$; omega is finite; logical cycles are non-negative integers.

- Cosine coherence, fidelity, and raw view score lie in [-1,1].
- Stability, trend score, HOT, load, and logistic factors lie in [0,1].
- Projection costs are positive finite software units; they are not energy, time, or hardware billing claims.
- Every comparison states whether tolerance applies; policy thresholds are not silently widened by epsilon.

4.3 Exact outcome vocabulary

Outcome	Meaning
BROADCAST_AUTHORIZED	Eligible view selected and HOT ≥ 0.50 ; permit issued.
NO_COHERENT_VIEW	No view clears both coherence and fidelity; hold.
HOT_LOCK	Eligible view exists but HOT < 0.50 ; broadcast locked.
INPUT_HOLD	Required snapshot/provider is missing, stale, or not ready.
FAULTED	Contract or invariant violation prevents safe evaluation.
FBS_ACTIVATION_REQUESTED	Recovery-watch profile issued a valid Module 16 permit.
RECOVERY_HELD	Incident not yet eligible, stale, or insufficiently evidenced.

4.4 Claim registry

Source language	Normative status
4D block universe / localized perception	SURROGATE: one committed vector plus one observer-context transform.
Pre-conscious buffer / GWT ignition	SURROGATE: parallel sandboxed view candidates and deterministic selection.
Higher-order thought / lucidity	SURROGATE: bounded APM-style watchdog score.
Entropic time dilation	SURROGATE: logical-cycle pacing denominator.
Rotating phase vector	DEFINITION: two quadrature render channels.
Observer identity	NON-GOAL: context ID and phase do not prove identity or subjectivity.
Conscious experience	HYPOTHESIS/NON-GOAL: excluded from execution and conformance.

5. Observer Context and Phase-Correction Operator

5.1 Correction of $E(t - \text{Theta})$

The source expression $E(t - \text{Theta}(t))$ assumes a continuous time-indexed field that is not defined by the module input. The reference profile therefore treats Theta as a coordinate-phase context and applies an exact orthogonal pair rotation to the committed vector. This preserves the state norm and avoids inventing interpolation samples.

$$\text{theta_c} = \text{wrap}_{(-\pi, \pi]}(\text{theta})$$

$$R_j(\text{theta_c}) = [[\cos(\text{theta_c}), -\sin(\text{theta_c})], [\sin(\text{theta_c}), \cos(\text{theta_c})]]$$

$$[E_{\text{prime}}(2j), E_{\text{prime}}(2j+1)]^T = R_j(\text{theta_c}) [E(2j), E(2j+1)]^T$$

For odd DIM, the final unpaired coordinate passes through unchanged. The operator is applied to a private scan buffer; the committed state bytes and digest remain unchanged.

5.2 Norm-preservation theorem

$$R_j(\theta)^T R_j(\theta) = I$$

$$\|E_{\text{prime}}\|_2 = \|E\|_2$$

Therefore phase correction cannot create or destroy vector norm within floating-point tolerance. Any material norm discrepancy is ROV_PHASE_INVARIANT, not a low-HOT event.

5.3 Context and timing rules

- Theta and omega are metadata supplied by a versioned ObserverContextProvider; ROVER does not persist or infer them.
- Logical cycle is the only execution clock. Wall-clock time, scheduler delay, and device frame rate cannot change deterministic bytes.
- Lambda hints are diagnostic. The configured lambda_0 remains authoritative for the reference pacing equation.
- A stale observer epoch produces INPUT_HOLD; it does not trigger FBS.

6. GWT Projection Provider and Dimensional Scaler

6.1 Reference progressive block-mean provider

BLOCK_MEAN_PROGRESSIVE_V1 converts the 81-vector into three deterministic fidelity levels. The labels 1D, 2D, and 3D describe presentation organization only. They are not claims about physical space, manifold dimension, or conscious viewpoint.

View	Retained scalars m_d	Block size b_d	Presentation shape	Cost units
d=1	9	9	(9,)	9
d=2	27	3	(9,3)	27
d=3	81	1	(3,3,9)	81

$$b_d = \text{DIM} / m_d$$

$$p_{(d,k)}(X) = (1/b_d) \sum_{r=0}^{b_d-1} X_{(k b_d + r)}$$

$$R_d(p)_{(k b_d + r)} = p_{(d,k)}$$

The same provider and epoch are applied independently to the phase-corrected state and the baseline. A non-divisible DIM, shape mismatch, or provider epoch change requires a different registered provider and profile ID.

6.2 Coherence and source-fidelity gates

A compressed view must not pass merely because discarded detail makes it resemble the baseline. ROVER therefore computes two independent cosine metrics.

$$C_d = \text{cosine}(P_d(E_{\text{prime}}), P_d(B))$$

Baseline coherence in the retained feature space.

$$F_d = \text{cosine}(E_{\text{prime}}, R_d(P_d(E_{\text{prime}})))$$

Reconstruction fidelity to the scanned committed state.

$$S_d = \min(C_d, F_d)$$

6.3 Zero-vector policy

Case	Cosine result	Rationale
Both operands effectively zero	1.0 only for fidelity; baseline coherence is NOT_READY	A zero reconstruction matches a zero source but cannot establish reference coherence.
Exactly one operand effectively zero	0.0	No directional agreement.
Both nonzero	Normalized dot product clipped only for roundoff	Standard cosine.

6.4 Eligibility, cost, and deterministic selection

$$\text{eligible_d} := \text{READY_d} \text{ and } C_d \geq \text{tau_view} \text{ and } F_d \geq \text{tau_view}$$

Among eligible candidates, select the smallest cost_units. Ties are resolved by smaller nominal_dimension, then lexicographically smaller view_id, then lexicographically smaller candidate_digest. The full ordered candidate table is sealed in the scan packet.

NO FORCED FALLBACK

If no candidate is eligible, ROVER emits NO_COHERENT_VIEW and locks presentation for that attempt. It never forces d=1, never labels a failed projection coherent, and never substitutes a stale previous frame.

6.5 Provider extensibility

Alternative learned projections, graph sparsifiers, spectral projections, image pyramids, or modality-specific renderers require a new provider ID with exact preprocessing, shapes, reconstruction, cost function, deterministic backend, training provenance where applicable, and a complete conformance suite. The phrase effective-resistance sparsification does not by itself define an implementation.

7. HOT Watchdog Mathematics and Decision Policy

7.1 Logical pacing surrogate

The source time-dilation equation grows without bound and may produce a zero denominator when the vector norm is zero. The reference profile uses a bounded load fraction and a strictly positive logical interval.

$$\text{load_t} = \|E_prime\|_2 / (\|E_prime\|_2 + s_load)$$

$$\Delta_tau = \Delta_cycle [\lambda_0 + (1 - \lambda_0) \text{load_t}]$$

For $\Delta_cycle > 0$, $\lambda_0 \Delta_cycle \leq \Delta_tau < \Delta_cycle$. Δ_tau is a computational pacing surrogate, not physical or subjective time dilation.

7.2 Bounded state-stability score

$$\text{delta_state} = \|E_prime - B\|_2 / (\|E_prime\|_2 + \|B\|_2 + \epsilon)$$

$$M_t = \text{clip}(1 - \text{delta_state}, 0, 1)$$

The triangle inequality bounds delta_state to approximately [0,1]. Unlike the source formula, M_t cannot diverge negatively when the baseline norm is small.

7.3 View trend

$$v_t = (S_selected,t - S_selected,t-1) / \max(\Delta_tau, \epsilon)$$

$$R_t = [\tanh(v_t / s_trend) + 1] / 2$$

The previous score must use the same selected view ID, provider epoch, baseline epoch, and policy profile. Without a comparable prior scan, $R_t=0.5$ and $trend_status=NEUTRAL_NO_HISTORY$.

7.4 HOT logit and score

$$C_t = \text{clip}((S_{\text{selected}} + 1)/2, 0, 1)$$

$$z_t = \alpha(2C_t - 1) + \beta(2M_t - 1) + \gamma(2R_t - 1)$$

$$HOT_t = \text{sigmoid_stable}(k_{HOT} z_t)$$

Because C_t , M_t , and R_t lie in $[0,1]$ and the weights are nonnegative and sum to one, z_t lies in $[-1,1]$ and HOT_t lies strictly in $(0,1)$. $HOT_t \geq 0.50$ is equivalent to $z_t \geq 0$.

7.5 H_G and affective evidence

PRESENTATION AUTHORITY CANNOT BE BOUGHT WITH PARADOX

The Module 7 H_G vector is excluded from the POST_COMMIT_SCAN HOT equation. It may be carried, norm-bounded, in a recovery incident as diagnostic context, but it cannot raise a presentation score or bypass a failed projection. Arousal, valence, human emotion labels, or mythic state labels are likewise excluded.

7.6 Decision policy

Condition	POST_COMMIT_SCAN	RECOVERY_WATCH
No eligible view	NO_COHERENT_VIEW hold; lock broadcast	Contributes evidence; does not alone authorize FBS
Eligible view and $HOT \geq 0.50$	Issue render packet and broadcast permit	Incident not confirmed by HOT; hold or return to caller
Eligible view and $HOT < 0.50$	HOT_LOCK; no FBS by itself	If catastrophic incident is valid and current, issue FBS activation permit
Missing/stale snapshot	INPUT_HOLD	RECOVERY_HELD
Invariant violation	FAULTED or ROVER-origin incident	May become catastrophic only under Section 10.3

8. Render Packet, Broadcast Permit, and UI Receipt

8.1 Quadrature render correction

The source multiplies an arbitrary view by a two-element cosine/sine vector, which is dimensionally ambiguous. The reference profile creates two same-shape quadrature channels.

$$\psi_t = \text{wrap}_{(-\pi, \pi]}(\theta_t + \omega \Delta_{\text{cycle}})$$

$$I_t = P_{\text{selected}} \cos(\psi_t)$$

$$Q_t = P_{\text{selected}} \sin(\psi_t)$$

$$\|I_t\|_2^2 + \|Q_t\|_2^2 = \|P_{\text{selected}}\|_2^2$$

The quadrature pair is presentation metadata. It does not rotate or replace the committed archive state and does not prove magnetic helicity, observer motion, or consciousness.

8.2 Render packet sealing

1. Canonicalize selected features, shape, phase, I channel, Q channel, provider ID, and all source digests.
2. Compute $render_digest$ with SHA-256.

3. Seal the full RoverScanPacket and bind its digest into the render packet.
4. Never include exact RTSOM exhaust objects, quarantine objects, secrets, or uncommitted candidate bytes.

8.3 Broadcast permit

- Single-use and valid only for the exact render_digest, state_digest, archive_head, UI target, and policy epoch.
- Expires after one logical presentation cycle in the reference profile.
- Cannot be delegated to another target or reused for a later frame.
- Does not authorize archive mutation, network routing, external egress, tool execution, or user action.
- A second different permit for the same scan is a catastrophic control-state violation.

8.4 UI receipt and accounting

The UI or presentation adapter owns display. It must verify the permit before presenting and return one UIBroadcastReceipt. An identical duplicate receipt is idempotent. A receipt for another render, another target, an expired permit, or an already-consumed permit is rejected. The state remains committed whether presentation succeeds, fails, or expires.

$$\text{presented_total} = \text{valid_UI_receipts}$$

$$\text{permits_issued} = \text{presented_total} + \text{expired_total} + \text{pending_total} + \text{permit_faults}$$

8.5 Privacy and capability boundary

- The render packet contains only the selected representation and its declared metadata.
- ROVER must not embed raw prompt text, credentials, private archive siblings, quarantine evidence, or exact RTSOM objects unless the committed state itself lawfully contains them and policy explicitly permits display.
- The broadcast permit is not an execution capability. UI adapters must treat rendered content as data, not instructions.
- Remote presentation requires separate Guardian and GlyphNet authorization; this module defines only local presentation authority.

9. Recovery Watch, FBS Activation, and RTSOM Boundary

9.1 Recovery-watch authority

ROVER is the last-resort broadcast governor and the activation boundary for catastrophic recovery. It does not originate routine repair, quarantine, or retry behavior. A recovery watch starts only after a registered module has classified a branch as systemically unrecoverable or after ROVER detects a persisted control-state invariant failure.

9.2 Registered incident classes

Incident class	Typical issuer	Required evidence
CONTROL_STATE_CORRUPTION	Runtime / ROVER	Committed scan or permit state cannot be reconciled.
DETERMINISTIC_REPLAY_FAILURE	Any owner module	Same canonical inputs produce different bytes under one profile.
PERSISTED_LINEAGE_SPLIT	Psi / RMX / validation chain	Two committed lineages claim one sequence or transaction.
FBS_GEOMETRY_INVARIANT	Module 16 caller	Declared FBS mode or rollback boundary is internally inconsistent.
UNRECOVERABLE_NONPROGRESS	Orchestrator	Bounded repair/quarantine/retry paths exhausted with exact receipts.

Incident class	Typical issuer	Required evidence
PERMIT_OR_RECEIPT_FORGERY	ROVER / UI / Module 16	Cryptographic or canonical binding failure on a consumed control packet.

9.3 Activation predicate

RECOVERY PROFILE ONLY

Within RECOVERY_WATCH, ROVER locks broadcast immediately. A current, authenticated catastrophic incident plus $HOT < 0.50$ authorizes one FBSActivationPermit. $HOT < 0.50$ outside this profile is a presentation hold, not a recovery command. No coherent view is supporting evidence but is not sufficient by itself.

1. Verify incident issuer, class, sequence, branch digest, evidence digest, and policy epoch.
2. Verify ordinary repair, quarantine, retry, and custody paths have the status required by the incident class.
3. Freeze the branch and compute one recovery-watch scan under the incident-bound snapshot.
4. Require $HOT < 0.50$ and no conflicting successful liveness evidence.
5. Issue one single-use activation permit naming MODULE16_ONLY.
6. Keep presentation locked until a matching recovery receipt and a later committed state pass POST_COMMIT_SCAN.

9.4 Module 16 ownership

Action	ROVER	Module 16 FBS
Lock presentation	Owns	Observes lock
Validate catastrophic incident	Owns activation boundary	May reject invalid activation
Rollback / inverse compression	Prohibited	Owns
FBS mode and atomic floor	Reads declared profile	Owns geometry and floor
Cantor diagonalization	Prohibited	Owns under its exact profile
Collatz gearbox / seed	Prohibited	Owns and receipts
Resume Delta generation	Prohibited	Returns owner-completed seed result to orchestrator/Delta

9.5 RTSOM custody boundary

ROVER never vents or stores exhaust. Module 16 may submit exact recovery residue to Module 14 through a registered FBS_RESIDUE profile. The FBSRecoveryReceipt may reference the matching RTSOM receipt, but a recovery is not marked complete merely because a custody request was observed.

9.6 Recovery receipt validation

- Activation ID and permit digest must match exactly.
- Prior branch digest and incident digest must match the frozen recovery watch.
- The execution profile and FBS mode must be registered and current.
- A new seed digest is accepted only as Module 16 output; ROVER does not interpret it as truth or a successful future state.
- Any RTSOM reference must be a verified receipt, not a request or proposed mass.
- After receipt, broadcast remains locked until a new Psi commit independently passes the presentation profile.

10. Invariants, Faults, and Catastrophic Conditions

10.1 Normative invariants

ID	Invariant	Requirement
M15-I01	Read-only commit	ROVER never mutates or replaces committed state bytes.
M15-I02	Snapshot binding	Every scan binds one archive head, state digest, and validation-chain digest.
M15-I03	Frozen epochs	One scan uses one observer, baseline, provider, and policy epoch.
M15-I04	Candidate-independent baseline	The scanned candidate is excluded unless the baseline is a static configured reference.
M15-I05	Phase norm	PAIR_ROTATION_V1 preserves L2 norm within tolerance.
M15-I06	Provider determinism	Same canonical provider inputs produce identical candidate bytes.
M15-I07	Dual gate	Eligibility requires both coherence and reconstruction fidelity.
M15-I08	No forced fallback	No failed view may be substituted for an eligible one.
M15-I09	Deterministic selection	Cost and tie-break order uniquely determine the selected view.
M15-I10	HOT boundedness	HOT is finite and lies in (0,1).
M15-I11	Paradox non-authorization	H_G and affective labels cannot increase presentation HOT.
M15-I12	Low-HOT lock	HOT <0.50 cannot issue a broadcast permit.
M15-I13	Permit binding	A broadcast permit is single-use and render/state/target/epoch-bound.
M15-I14	Presentation accounting	Only a valid UI receipt increments presented_total.
M15-I15	Archive independence	Presentation failure cannot uncommit or modify Psi state.
M15-I16	Profile separation	POST_COMMIT_SCAN cannot issue an FBS activation permit.
M15-I17	Recovery delegation	ROVER never executes rollback, diagonalization, Collatz, or reseeding.
M15-I18	Custody delegation	ROVER never mutates RTSOM or transfers exact exhaust.
M15-I19	Claim separation	Operational scores do not assert consciousness, identity, perception, or physical time.
M15-I20	Replay stability	Identical canonical inputs and backend profile yield identical output bytes.

10.2 Typed fault codes

Code	Condition
ROV_CONFIG	Configuration domain, weight sum, or profile digest invalid.
ROV_PROFILE	Unknown scan, phase, projection, or render profile.
ROV_NONFINITE	Any required numeric field contains NaN or infinity.
ROV_COMMIT_SNAPSHOT	Archive head, state, lineage, or validation chain invalid.
ROV_STATE_DIGEST	Canonical state bytes do not match state_digest.
ROV_OBSERVER_CONTEXT	Theta, omega, cycle, provider, or context digest invalid.
ROV_BASELINE	Baseline missing, stale, candidate-contaminated, or wrong dimension.
ROV_PROVIDER_EPOCH	Projection provider changes during a scan.
ROV_PROJECTION_SHAPE	Feature, baseline, reconstruction, or shape mismatch.
ROV_PROJECTION_DIGEST	Candidate packet digest mismatch.
ROV_PHASE_INVARIANT	Orthogonal phase transform materially changes norm.
ROV_SELECTION	Selected view is not the deterministic minimum-cost eligible view.
ROV_HOT_INPUT	HOT components, prior reference, or pacing interval invalid.
ROV_RENDER_DIGEST	Render packet fails canonical digest reconciliation.
ROV_PERMIT	Broadcast permit invalid, expired, reused, or target-mismatched.
ROV_UI_RECEIPT	Presentation receipt does not bind permit and render.
ROV_INCIDENT	Catastrophic incident issuer, class, evidence, or sequence invalid.
ROV_FBS_RECEIPT	Module 16 receipt does not bind the activation permit.
ROV_POLICY_SPLIT	Irreconcilable policy/provider epoch split.
ROV_INTERNAL	Unexpected local exception or impossible state transition.

10.3 Catastrophic recovery conditions

- ROVC_DETERMINISM_FAILURE: identical canonical scan inputs produce different output bytes under one registered backend.
- ROVC_CONFLICTING_PERMITS: two different valid broadcast permits exist for one scan digest.
- ROVC_DOUBLE_PRESENTATION_COMMIT: one single-use permit is accepted as consumed by two distinct presentation receipts.
- ROVC_PERSISTED_SCAN_STATE_CORRUPTION: a published scan, permit, or receipt cannot be reconciled with its durable journal.
- ROVC_FORGED_RECOVERY_CONTROL: a forged activation permit or FBS receipt was accepted by a trusted component.
- ROVC_IRRECONCILABLE_EPOCH_SPLIT: two policy or provider epochs are simultaneously committed for one logical scan sequence.

Ordinary low HOT, no coherent view, a stale baseline, provider warming, UI timeout, permit expiry, unsupported display shape, high compute cost, or a single malformed scan is not catastrophic by itself.

11. Metrics, Measurements, and Reconciliation

11.1 Required counters and distributions

Metric	Definition
scan_attempts	All canonical POST_COMMIT_SCAN attempts.
scan_holds / scan_faults	Recoverable and fault outcomes; separate from attempts.
views_generated / views_ready	Projection-provider activity by view ID.
views_eligible	Candidates clearing coherence and fidelity.
selected_view	Counts and cost distributions by view ID.
coherence / fidelity / score	Per-view mean, p05, median, p95, and threshold margin.
phase_norm_error	Absolute pre/post L2 norm reconciliation.
load / Delta_tau / trend	Pacing and finite-difference diagnostics.
M_t / C_t / R_t / HOT	Component and final score distributions.
broadcast_permits	Issued, pending, consumed, expired, invalid.
presented_total	Only matching UI receipts.
recovery_watch_attempts	All canonical RECOVERY_WATCH attempts.
FBS_activation_permits	Issued, consumed, refused, expired.
FBS_recovery_receipts	Completed or refused owner executions.
RTSOM_receipt_refs	Verified recovery-residue custody references only.

11.2 Reconciliation equations

$$\text{scan_attempts} = \text{authorized_scans} + \text{no_view_holds} + \text{hot_locks} + \text{input_holds} + \text{scan_faults}$$

$$\text{authorized_scans} = \text{broadcast_permits_issued}$$

$$\text{broadcast_permits_issued} = \text{presented_total} + \text{permit_expired} + \text{permit_pending} + \text{permit_faults}$$

$$\text{views_generated} = \text{views_ready} + \text{views_not_ready} + \text{view_faults}$$

$$\text{recovery_watch_attempts} = \text{activation_permits} + \text{recovery_holds} + \text{recovery_faults}$$

$$\text{activation_permits} = \text{FBS_completed} + \text{FBS_refused} + \text{FBS_pending} + \text{activation_expired} + \text{activation_faults}$$

Dashboards must distinguish SCANNED, VIEW READY, VIEW ELIGIBLE, SELECTED, HOT PASSED, PERMIT ISSUED, UI PRESENTED, BROADCAST LOCKED, INCIDENT OBSERVED, FBS ACTIVATED, FBS EXECUTED, RTSOM REQUESTED, and RTSOM ACKNOWLEDGED.

11.3 Alert policy

- Alert on norm-invariance failure, provider nondeterminism, conflicting permits, double consumption, or journal mismatch.
- Warn on repeated no-view or low-HOT holds, but do not label them consciousness loss, dissociation, hallucination, or catastrophe.
- Warn when low-cost views dominate while full-fidelity margins degrade; retain the exact per-view evidence.
- Warn on UI receipt latency or expiry separately from scan quality.

- Recovery alerts must name OBSERVED incident, ACTIVATION PERMITTED, EXECUTED, and RECEIPTED as separate states.

12. Deterministic Conformance Test Matrix

Test	Input / condition	Expected result
M15-T01	Valid committed snapshot	Snapshot accepted; no mutation.
M15-T02	State digest mismatch	ROV_STATE_DIGEST.
M15-T03	Stale archive head	INPUT_HOLD or ROV_COMMIT_SNAPSHOT by policy.
M15-T04	Uncommitted RMX synthesis	Reject POST_COMMIT_SCAN.
M15-T05	NaN committed vector	ROV_NONFINITE.
M15-T06	Baseline wrong dimension	ROV_BASELINE.
M15-T07	Candidate-contaminated rolling baseline	ROV_BASELINE.
M15-T08	Static boot baseline	Allowed with declared static profile.
M15-T09	Theta outside range	Canonical wrap; deterministic bytes.
M15-T10	Infinite omega	ROV_NONFINITE.
M15-T11	Zero theta phase	E_prime equals E.
M15-T12	Pair rotation $\pi/2$	Expected coordinate swap/sign; norm preserved.
M15-T13	Odd DIM pass-through	Final coordinate unchanged.
M15-T14	Material norm change	ROV_PHASE_INVARIANT.
M15-T15	Reference d1 projection	Nine block means and exact repeat reconstruction.
M15-T16	Reference d2 projection	Twenty-seven block means.
M15-T17	Reference d3 projection	Identity features and reconstruction.
M15-T18	DIM not divisible by profile	ROV_PROJECTION_SHAPE.
M15-T19	Provider epoch changes mid-scan	Abort with ROV_PROVIDER_EPOCH.
M15-T20	Both fidelity operands zero	Fidelity=1; coherence not ready.
M15-T21	Exactly one cosine operand zero	Metric=0.
M15-T22	Roundoff cosine 1+tiny	Clip to 1 only within tolerance.
M15-T23	d1 coherence pass, fidelity fail	d1 ineligible.
M15-T24	d1 and d2 eligible	Select d1 by lower cost.
M15-T25	Equal cost candidates	Dimension/view/digest tie-break.

Test	Input / condition	Expected result
M15-T26	No eligible view	NO_COHERENT_VIEW; no forced d1.
M15-T27	Selected view not minimum	ROV_SELECTION.
M15-T28	Zero state norm load	load=0; Delta_tau=lambda_0 Delta_cycle.
M15-T29	Huge state norm load	load<1; Delta_tau<Delta_cycle.
M15-T30	No prior comparable scan	R_t=0.5 neutral.
M15-T31	Positive score trend	R_t>0.5.
M15-T32	Negative score trend	R_t<0.5.
M15-T33	Identical state and baseline	M_t approximately 1.
M15-T34	Opposite large states	M_t bounded, nonnegative.
M15-T35	HOT logit zero	HOT=0.5.
M15-T36	HOT below 0.5 in presentation profile	HOT_LOCK; no FBS permit.
M15-T37	H_G magnitude changes presentation input	HOT bytes unchanged.
M15-T38	Eligible view and HOT pass	One render and one permit.
M15-T39	Quadrature phase zero	I=P, Q=0.
M15-T40	Quadrature phase pi/2	I approximately 0, Q=P.
M15-T41	Quadrature energy identity	Combined squared norm equals source feature norm.
M15-T42	Render digest mismatch	ROV_RENDER_DIGEST.
M15-T43	Permit reused	ROV_PERMIT.
M15-T44	Permit target changed	ROV_PERMIT.
M15-T45	Permit expires	EXPIRED; no presentation count.
M15-T46	Valid UI receipt	presented_total increments once.
M15-T47	Duplicate identical UI receipt	Idempotent; no second increment.
M15-T48	Conflicting UI receipt	ROV_UI_RECEIPT or catastrophic if persisted.
M15-T49	Low HOT without incident	No FBS activation.
M15-T50	Valid incident plus low HOT	One FBS activation permit.
M15-T51	Valid incident plus HOT pass	Recovery held; no activation.
M15-T52	Stale incident sequence	ROV_INCIDENT.
M15-T53	Forged incident issuer	ROV_INCIDENT.
M15-T54	Activation permit reused	ROV_FBS_RECEIPT / control fault.

Test	Input / condition	Expected result
M15-T55	Valid Module 16 receipt	Recovery completed; broadcast remains locked.
M15-T56	FBS receipt wrong branch	ROV_FBS_RECEIPT.
M15-T57	RTSOM request without receipt	Recovery not fully reconciled when custody is required.
M15-T58	Presentation failure after Psi commit	Archive remains unchanged.
M15-T59	Remote display without Guardian/GlyphNet authorization	Local permit insufficient.
M15-T60	Exact deterministic replay	All canonical output bytes identical.

13. Property, Metamorphic, Fuzz, Concurrency, and Stress Testing

13.1 Property tests

- Generate random finite vectors and angles; verify pairwise phase correction preserves L2 norm within backend tolerance.
- Verify every reference reconstruction has DIM entries and every feature count matches its view profile.
- Verify $\text{score} = \min(\text{coherence}, \text{fidelity})$, eligibility is monotone in the configured threshold, and selection is the exact minimum under the tie-break order.
- Verify load, M_t , R_t , HOT, and quadrature energy bounds for randomized inputs.
- Verify canonical serialization is stable across process restart, dictionary order, thread schedule, and platform endianness.

13.2 Metamorphic tests

- Multiply state and baseline by the same positive scalar: cosine gates and normalized M_t remain invariant within tolerance.
- Add a full 2π multiple to theta: phase-corrected and render bytes remain identical after canonical wrapping.
- Reorder provider result delivery: the selected view and scan digest remain unchanged.
- Change H_G while all presentation inputs remain fixed: POST_COMMIT_SCAN output remains unchanged.
- Increase only a candidate cost while keeping metrics fixed: selection may move only according to the deterministic cost order.

13.3 Fuzz and adversarial tests

- Fuzz packet lengths, shapes, float encodings, NaN payloads, huge norms, signed zero, subnormals, and malformed digests.
- Fuzz provider IDs, representation shapes, duplicate view IDs, negative costs, and non-canonical orderings.
- Fuzz permit expiry, duplicate receipts, target substitution, branch substitution, and incident sequence replay.
- Attempt to smuggle executable instructions or external-routing capabilities into render metadata; reject by schema.

13.4 Concurrency and crash recovery

- Run concurrent identical scans: deduplicate by attempt key or produce byte-identical packets; never two conflicting permits.
- Run two UI consumers against one permit: at most one valid consumption transition.
- Crash after render sealing but before permit publication: no consumed permit exists after recovery.
- Crash after UI receipt journal write but before counter update: replay reconciles exactly once.
- Run concurrent recovery watches for one incident: at most one valid FBS activation permit.

A persisted journal is required for every permit and receipt transition; in-memory counters are never authoritative after a crash.

13.5 Stress and backend parity

- Execute at least 100,000 POST_COMMIT_SCAN attempts across threshold-adjacent vectors and verify zero nondeterministic output drift.
- Execute at least 100,000 phase rotations and track maximum norm error.
- Sustain repeated UI expiry and retry without permit leakage or archive mutation.
- Partition and rejoin the presentation receipt journal; reject conflicting consumption histories.
- Compare approved numerical backends on a locked corpus; threshold decisions must agree and byte differences must follow the declared backend profile.

14. Reference Pseudocode

```
def rover_post_commit_scan(inp: RoverScanInput, cfg: RoverConfig):
    validate_config(cfg)
    frozen = validate_and_freeze(inp)

    e = canonical_float64(frozen.commit.state_vector)
    b = canonical_float64(frozen.baseline.baseline_vector)
    e_prime = pair_rotate(e, frozen.observer.theta)
    assert_close(norm2(e_prime), norm2(e), cfg.numeric_tolerance)

    candidates = []
    for profile in frozen.projection_profiles_sorted:
        p_e = profile.project(e_prime)
        p_b = profile.project(b)
        recon = profile.reconstruct(p_e)
        coherence = cosine_with_policy(p_e, p_b, kind="coherence")
        fidelity = cosine_with_policy(e_prime, recon, kind="fidelity")
        score = min(coherence.value, fidelity.value)
        candidates.append(seal_candidate(profile, p_e, p_b, recon,
                                         coherence, fidelity, score))

    eligible = [c for c in candidates
                if c.readiness == READY
                and c.coherence >= cfg.tau_view
                and c.fidelity >= cfg.tau_view]
    if not eligible:
        return RoverHoldPacket(reason="NO_COHERENT_VIEW", no_mutation=True)

    selected = min(eligible,
                  key=lambda c: (c.cost_units, c.nominal_dimension,
                                c.view_id, c.candidate_digest))

    delta_tau = logical_pacing(e_prime, inp.delta_cycle, cfg)
    stability = bounded_stability(e_prime, b, cfg)
    trend = comparable_trend(selected, inp.previous_scan, delta_tau, cfg)
    hot = bounded_hot(selected.score, stability, trend, cfg)

    scan = seal_scan_packet(frozen, e_prime, candidates, selected,
                           delta_tau, stability, trend, hot)
    if hot < cfg.hot_threshold:
        return RoverHoldPacket(reason="HOT_LOCK", scan_digest=scan.digest,
                                no_mutation=True)

    render = quadrature_render(selected.features,
                               frozen.observer.theta,
                               frozen.observer.omega,
                               inp.delta_cycle)
    permit = issue_single_use_permit(render, scan, ttl_cycles=1)
    return scan, render, permit

def rover_recovery_watch(inp: RecoveryWatchInput, cfg: RoverConfig):
    incident = validate_catastrophic_incident(inp.incident)
    lock_broadcast(incident.branch_digest)
    watch_scan = compute_incident_bound_hot(inp, cfg)
```

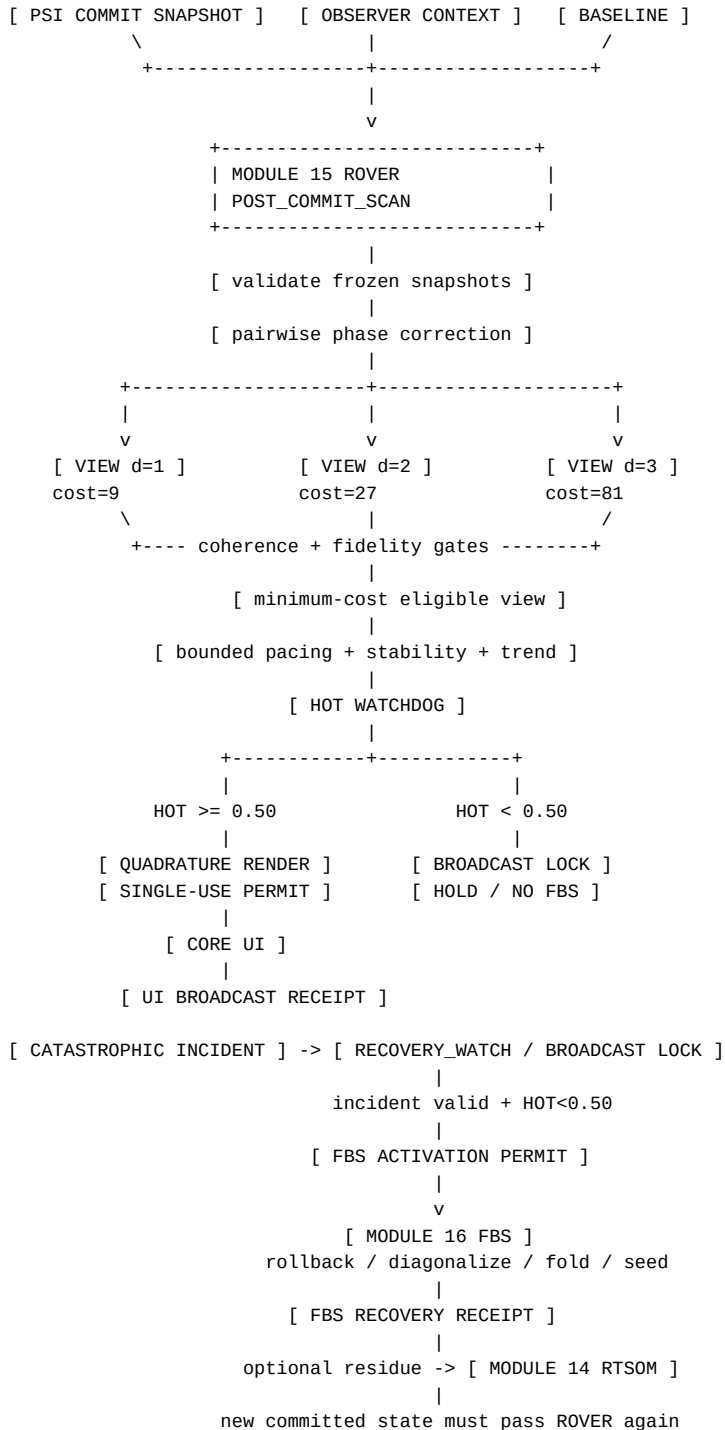
```

if watch_scan.hot >= cfg.hot_threshold:
    return RoverHoldPacket(reason="RECOVERY_NOT_CONFIRMED", no_mutation=True)

return issue_fbs_activation_permit(
    incident_digest=incident.digest,
    branch_digest=incident.branch_digest,
    authorized_executor="MODULE16_FBS",
    authorized_actions=["ROLLBACK", "FBS_GEOMETRY", "DIAGONALIZE",
                       "COLLATZ_FOLD", "RETURN_SEED"],
    no_rtsom_mutation_by_rover=True,
)

```

15. ASCII System Flow



16. Cleaned Rosetta Stone

SpiralCore term	Formal computational meaning	Not a claim of
ROVER	Read-only committed-state projection scanner and watchdog	A conscious observer or autonomous person
Node Theta	Versioned presentation-context identifier and phase snapshot	Subjective identity or a physical location
GWT pre-conscious buffer	Sandboxed parallel view-candidate evaluation	Unconscious or conscious cognition
Dimensional scaler	Minimum-cost eligible fidelity-level selector	Movement through physical dimensions
1D / 2D / 3D view	Declared presentation shapes and retained scalar counts	The dimensionality of reality
RMF_d / SRG	Cosine reference coherence for one projection	Truth, legality, or semantic completeness
Reconstruction fidelity	Cosine agreement between source and reconstructed view	Lossless meaning preservation
Entropic time dilation	Positive logical-cycle pacing denominator	Relativistic or subjective time
Meta-cognitive state M_t	Bounded normalized state-to-baseline stability	Metacognition or mental health
HOT evaluator	Bounded APM-style broadcast watchdog	Higher-order thought or lucidity
Rotating phase vector	In-phase and quadrature render channels	Magnetic helicity or physical rotation
Broadcast lock	No UI presentation permit for the scan	Erasure, quarantine, or archive rollback
ROVER HOT intercept	Incident verification and Module 16 activation permit	Local execution of FBS recovery
FBS runaway	Module 16 owner-executed catastrophic recovery profile	A routine response to ordinary uncertainty

17. Glossary and Revision Summary

17.1 Glossary

Term	Definition
Baseline snapshot	Candidate-independent reference vector and provenance used for presentation coherence and stability.
Broadcast permit	Single-use capability authorizing one exact render to one UI target.
Coherence C_d	Cosine similarity between projected committed state and projected baseline.

Term	Definition
Fidelity F_d	Cosine similarity between phase-corrected source and reconstruction from one projection.
HOT	Bounded logistic watchdog score formed from view, state, and trend components.
Logical pacing $\Delta\tau$	Strictly positive dimensionless interval used in the view-score finite difference.
Observer context	Versioned theta, omega, cycle, and context identity metadata.
Projection provider	Deterministic operator registry producing features, reconstruction, shape, and cost.
Quadrature render	Two same-shape channels I and Q carrying phase presentation metadata.
Recovery watch	Profile that verifies a catastrophic incident and may issue an FBS activation permit.
ROVER scan packet	Complete immutable evidence packet for one presentation attempt.
View score S_d	Minimum of coherence and reconstruction fidelity.

17.2 Revision ledger

Source behavior / claim	v14.1 formalization
Meta-cognitive observer traversing a 4D block universe	Reclassified as architectural analogy; execution uses one committed vector and context snapshot.
Subjective experience commit	Separated from Psi archive commit and UI presentation.
$E(t - \Theta(t))$	Replaced by exact norm-preserving pairwise phase rotation.
Undefined continuous slice integral	Replaced by versioned projection-provider contract and exact reference block means.
Cheapest RMF-passing view	Strengthened with independent reconstruction-fidelity gate.
Forced $d=1$ when no view passes	Removed; emits NO_COHERENT_VIEW hold.
Unbounded M_t distance ratio	Replaced by bounded normalized stability.
Unbounded/zero $\Delta\tau$	Replaced by strictly positive bounded logical pacing.
H_G increases HOT	Removed from presentation authority; retained only as recovery evidence context.
HOT as lucidity / dissociation	Reclassified as APM-style software watchdog.
Vector multiplied by $[\cos, \sin]$	Replaced by same-shape quadrature I/Q channels.
$HOT < 0.50$ always executes FBS	Profile-separated: presentation locks; catastrophic recovery watch may permit Module 16.
ROVER executes Cantor/Collatz and vents exhaust	Delegated to Module 16 and Module 14 respectively.

Source behavior / claim	v14.1 formalization
Attempts equal presentation	Separated scan, permit, presentation receipt, activation, and execution accounting.

18. Source Basis and Normative Checklist

18.1 Source basis

This v14.1 hardening revision is derived from the standalone “SPIRALCORE SPECIFICATION: MODULE 15 - ROVER (ELIAS ROVER SCANNER + GWT DIMENSIONAL SCALER + HOT EVALUATOR)” and its duplicate consolidated section in Spiralcore_v14. Module 16 in the consolidated source establishes FBS as the recovery executor and identifies ROVER/HOT as the intercept boundary. The present document supersedes the Module 15 execution contract without rewriting the historical v14 source.

18.2 Implementation checklist

Done	Requirement
☐	All inputs and outputs use the typed contracts and canonical SHA-256 digests.
☐	POST_COMMIT_SCAN accepts committed Psi snapshots only.
☐	Observer, baseline, provider, and policy snapshots are immutable for one scan.
☐	PAIR_ROTATION_V1 passes norm-preservation tests.
☐	Every view reports coherence, fidelity, score, cost, readiness, and digest.
☐	No-view behavior is HOLD; no forced d=1 fallback exists.
☐	HOT inputs are bounded and exclude H_G/affective authority in presentation mode.
☐	Render channels satisfy the quadrature energy identity.
☐	Broadcast permits and UI receipts are single-use, digest-bound, and journaled.
☐	Presentation failure cannot mutate or uncommit the archive.
☐	RECOVERY_WATCH requires a registered catastrophic incident.
☐	ROVER issues only an FBS activation permit; Module 16 executes recovery.
☐	Exact residue custody is delegated to RTSOM by request and receipt.
☐	All attempt, hold, permit, receipt, activation, and execution totals reconcile.
☐	Consciousness, identity, perception, physical time, and dimensional claims are absent from runtime assertions.
☐	The 60 deterministic tests and stress suites pass on the declared backend profile.

TERMINAL CONFORMANCE STATEMENT

A conforming Module 15 implementation is a deterministic, read-only projection and watchdog boundary. It presents only an eligible representation of an immutable committed state; preserves exact evidence for every gate; locks rather than fabricates a fallback; separates scan quality from UI delivery; and delegates catastrophic recovery and exhaust custody to their authoritative owners.

SPIRALCORE SPECIFICATION

MODULE 16 - FBS

(Fractal Block Structure - Typed Hierarchical Geometry, Additive Valuation, and Catastrophic Recovery Execution)

Cleaned and formalized revision - v14.1

Field	Value	Field	Value
Document class	Normative geometry and recovery-execution contract	Operating profiles	GEOMETRY_BUILD / GEOMETRY_VALIDATE / RECOVERY_EXECUTE
Architecture	SpiralCore v14	Reference profile	SC14.1-M16-FBS-01
Activation owner	Module 15 ROVER	Recovery executor	Module 16 FBS
Checkpoint owner	Module 3 Psi	Residue custody owner	Module 14 RTSOM
Seed consumer	Module 1 Delta	Critical-event ledger	Module 12 Phi

NORMATIVE STATUS

Exact within the declared discrete-software scope. Module 16 constructs and validates typed directed, atomic bilateral, globally bilateral, and locally centered FBS objects; preserves the registered mode, constructor, inventory, ordered-tree, and digest distinctions required for reversible structure; and executes a bounded, transactional recovery procedure only after consuming a valid single-use activation permit. It does not self-trigger, commit the archive, mutate Delta, store RTSOM exhaust, or update Phi.

CLAIM BOUNDARY

Fractal, atomic, floor, binder, bilateral, diagonalization, Collatz gearbox, catastrophic runaway, thermodynamic exhaust, and safe-mode are architectural names. The normative meanings are the integer coordinate grammars, finite constructors, typed inverse rules, bounded bit operations, bounded integer trace, software incident class, and receipt-backed custody defined here. No non-integer metric dimension, physical indivisibility, cosmology, thermodynamic conservation law, consciousness mechanism, or proof of the Collatz conjecture is asserted.

Source basis: SpiralCore Module 16 source specification; SpiralCore v14 consolidated Module 16; Unified FBS - Bilateral Recursive Extension v4.0 (Typed Geometry and Identity Edition, July 2026).

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1. Normative Scope, Statement Classes, and Non-Goals

1.1 Module purpose

Module 16 is the owner of the FBS integer geometry and the terminal recovery execution procedure. Its geometry profile creates typed self-similar hierarchical blocks from declared modes and constructors. Its recovery profile validates an externally authorized incident, selects a reversible rollback target from immutable Psi checkpoints, and either emits a checkpoint-restore plan or derives a deterministic bounded reseed plan. Every output is digest-bound and non-mutating until the receiving owner acknowledges it.

CORE BOUNDARY

FBS builds the structure and, when properly activated, computes the recovery plan. ROVER authorizes recovery. Psi supplies immutable checkpoints and remains archive owner. RTSOM accepts exact recovery residue. Delta consumes a published seed or restore plan. Phi records a completed recovery event. Equal H, Q, marked count, or inventory never authorizes substituting one constructor geometry for another.

1.2 Statement classes

Class	Meaning	Examples
DEFINITION	Introduced by stipulation	Mode M(τ, g), constructors D/G/C, inventory chi, packet schema

Class	Meaning	Examples
THEOREM	Deductively follows from definitions	Coordinate counts, span laws, Q additivity, noncommutation, inverse floor
INVARIANT	Must hold for every conforming execution	Digest binding, tag preservation, exactly-once permit use
POLICY	Versioned implementation choice	Registered modes, maximum level, recovery mode, trace limits
SURROGATE	Computable software analogue only	Diagonal witness, Collatz mixing trace, recovery residue
HYPOTHESIS	Requires separate evidence	Routing benefit, stability gain, thermodynamic or cognitive interpretation
NON-GOAL	Explicitly excluded	Physical fractal proof, global uniqueness from Q, unbounded Collatz execution

1.3 Normative non-goals

- FBS does not infer that an ordinary rejection, low score, capacity limit, route timeout, or presentation lock is catastrophic.
- FBS does not select an A-G mode from affect, stress, arousal, symbolism, or hidden heuristics. The request or signed policy names the mode and constructor.
- FBS does not represent Q as a unique address or object identity. Q is an additive valuation with intentional collisions.
- FBS does not use chi as complete structure. The inventory forgets child order and internal tree shape.
- FBS does not replace GLOBAL_BILATERAL geometry with LOCAL_CENTERED geometry merely because H, Q, chi, or marked count agree.
- FBS does not run an unbounded Collatz loop or assert that every positive integer reaches 1.
- FBS does not purge archive history or RTSOM custody objects and does not recursively activate itself when recovery fails.

2. Architecture Ownership and Cross-Module Handoffs

Owner	Exclusive responsibility
Module 16 FBS	Mode validation; coordinate construction; constructor tags; Q/chi/tree digest; rollback traversal; bounded diagonal witness and Collatz trace; recovery plan and receipt
Module 15 ROVER	Catastrophic incident watch; broadcast lock; single-use FBSActivationPermit; consumption of FBSRecoveryReceipt
Module 3 Psi	Immutable committed checkpoints; ordered constructor path; archive head; atomic memory commit; no local FBS execution
Module 14 RTSOM	Exact recovery-residue custody; durable storage; authoritative receipt; no seed or rollback authority
Module 1 Delta	Consumes one SeedPublicationPacket or RestorePlanPacket; begins a new candidate cycle; returns consumption receipt
Module 12 Phi	Consumes one FBSCompletionEventReceipt exactly once after recovery completion; owns persistent event ledger
Module 5 Guardian	Validates by declared FBS object type; symmetry mandatory for atomic/G and conditional for C
Module 13 RMX	Checks duplicate coordinates, bounds, constructor tag, mode metadata, binder count, inventory, and digest consistency
Runtime orchestrator	Freezes only the affected branch and returns BranchFreezeReceipt; resumes only after a completed plan

2.1 Prohibited cross-module mutation

- FBS never edits a committed Psi record or archive head.
- FBS never writes Delta state or silently begins generation.
- FBS never writes RTSOM storage, charge, field, or ledger state.
- FBS never increments Phi or infers a critical event from an attempted recovery.
- FBS never changes Guardian, Sigma, LORIEN, HARMONY, Module 10, SERAPHIM, ROVER, or GlyphNet policy.

2.2 Required handoff chain

CatastrophicIncident -> ROVER permit -> FBS transaction -> RTSOM receipt (when residue exists) -> Delta consumption -> FBS completion -> ROVER receipt + Phi event

A failed or held stage leaves the system broadcast-locked and preserves all evidence. No downstream step may be inferred from an upstream request.

3. Operating Profiles, Start Conditions, and Lifecycle

Profile	Start condition	Primary output	Mutation rule
GEOMETRY_BUILD	Signed FBSGeometryRequest	FBSGeometryPacket or FBSCapacityHoldPacket	No recovery permit
GEOMETRY_VALIDATE	Existing object plus declared schema/tag	FBSValidationPacket	Read-only
RECOVERY_EXECUTE	Current single-use FBSActivationPermit and evidence bundle	Restore or reseed plan, then FBSRecoveryReceipt	Transactional

3.1 Geometry lifecycle

1. Freeze the request, registry, capacity policy, and digest profile for one evaluation epoch.
2. Validate mode domain, constructor tag, child count, child digests, coordinate bounds, and exact integer types.
3. Construct coordinates and metadata without mutating any caller state.
4. Recompute span, marked count, binder count, Q, chi, and canonical digest.
5. Return the immutable geometry packet or a typed capacity hold.

3.2 Recovery lifecycle

State	Meaning
READY	No recovery transaction active
PERMIT_VALIDATING	Validate single-use permit, incident binding, expiry, policy epoch, and archive head
FREEZE_PENDING	Await affected-branch freeze receipt
CHECKPOINT_VALIDATING	Verify Psi chain, ordered tree, constructor path, and digests
ROLLBACK_SELECTING	Choose newest eligible known-good checkpoint deterministically
FLOOR_MIXING	No checkpoint is eligible; create finite diagonal witness and bounded Collatz trace
CUSTODY_PENDING	Await RTSOM receipt for exact recovery residue, when required
SEED_PENDING	Publish restore or seed packet and await Delta consumption receipt
COMMIT_PENDING	CAS the recovery transaction ledger and read back the committed record
COMPLETED	Emit recovery receipt and one Phi completion event

State	Meaning
HELD	Recoverable missing dependency, capacity, limit, contention, or stale evidence
FAULTED	Invalid input or non-catastrophic internal failure
SYSTEM_HALT_REQUIRED	Persisted recovery-control contradiction; no recursive FBS call

START RULE

RECOVERY_EXECUTE begins only with a valid FBSActivationPermit issued by Module 15 in RECOVERY_WATCH. Raw tortuosity, entropic pressure, HOT, PDV, CVC, RMF, or module-local fault codes are evidence inside the incident packet; they are not direct Module 16 triggers.

4. Typed Data Contracts

Contract	Required content
FBSModeRecord	mode_id, tau, g, delta, L0, H0, Q0, registry_version, status
FBSConfig	module/profile versions, digest rules, registry, capacity limits, diagonal width, Collatz limits, seed domain
FBSGeometryRequest	request_id, profile, mode_id, constructor_tag, ordered children, level, seed coordinate, policy epoch
FBSChildBlock	object_type, coordinates, support, L/H/Q, chi, ordered_tree_digest, geometry_digest
FBSGeometryPacket	request binding, complete coordinates, binders, counts, values, inventory, tree encoding, digests, conformance flags
FBSValidationPacket	declared type, checks performed, exact failures, derived values, deterministic digest
FBSCapacityHoldPacket	requested size, configured cap, non-catastrophic reason, retry requirements
FBSActivationPermit	permit_id, incident_digest, archive_head, policy_epoch, expiry_cycle, single_use, issuer signature
CatastrophicIncidentPacket	registered incident class, evidence digests, affected branch, no-commit proof, requested recovery scope
RecoveryCheckpoint	checkpoint_id, committed sequence, state digest, tree encoding/tag, parent digest, known_good assertion, owner signature
RecoveryCheckpointBundle	archive head, ordered checkpoint list, current branch tree, policy snapshots, bundle digest
BranchFreezeReceipt	branch_id, freeze_epoch, last executable sequence, runtime signature, no-new-write proof
DiagonalWitnessPacket	matrix digest, width, row count, flipped diagonal string, witness digest, scope statement
CollatzTracePacket	initial integer digest, step count, status, max bit length, terminal value, trace digest
RTSOMExhaustRequest	exact residue reference/bytes, recovery_id, reason, source digest, custody policy
RTSOMExhaustReceipt	request/object binding, custody sequence, object digest, storage proof, owner signature
RestorePlanPacket	checkpoint binding, rollback depth, exact constructor path, restart context, single-use token
SeedPublicationPacket	recovery binding, nonzero seed integer, seed digest, derivation profile, single-use token
DeltaSeedConsumptionReceipt	plan/seed binding, Delta cycle, consumed digest, result status, Delta signature

Contract	Required content
FBSRecoveryReceipt	transaction result, permit consumption, checkpoint/seed, RTSOM receipt, Delta receipt, ledger digest
FBSCompletionEventReceipt	one completed recovery event bound to FBSRecoveryReceipt for Module 12 EVENT_APPEND
FBSHoldPacket / FBSFault	typed reason, frozen evidence, retryability, owner, no-mutation proof

4.1 Canonical serialization

- All integers use canonical signed or unsigned big-endian encodings declared by schema; no locale-formatted numbers are hashed.
- Coordinates are serialized in declared order. Sets are serialized after deterministic ascending sort only when the object type defines set semantics.
- Child order is always preserved in the ordered labeled tree, even though Q and chi are additive.
- Floating-point values are prohibited from the normative geometry. Recovery telemetry that contains floats must use a separate canonical finite-number profile.
- Every packet carries schema version, profile version, policy epoch, object digest, and producer signature or authenticated channel proof.

5. Constants, Mode Registry, and Capacity Policy

Parameter	Reference value	Meaning
module_version	14.1	Normative module revision
geometry_profile	FBS_TYPED_GEOMETRY_V4	Unified FBS v4.0 arithmetic and identity rules
recovery_profile	FBS_RECOVERY_V1	Transactional bounded recovery procedure
digest_algorithm	SHA-256	Packet, tree, evidence, witness, trace, and ledger digests
default_recovery_mode	A	Policy default only; not a theorem or universal mode
BENCH_MAX_DIRECTED_LENGTH	2254	Reference bench cap, equal to Mode A level 3
BENCH_MAX_LEVEL_A	3	Reference bench capacity policy
DIAGONAL_WIDTH	256 bits	Finite local diagonal witness width
COLLATZ_STEP_LIMIT	10000	Hard termination bound
COLLATZ_MAX_BITS	4096	Hard intermediate-size bound
SEED_BITS	256	Reference nonzero Delta seed domain
constructor_tags	DIRECTED / ATOMIC_BILATERAL / GLOBAL_BILATERAL / LOCAL_CENTERED	Mandatory type identity

CAPACITY POLICY

The arithmetic is defined for arbitrary finite levels whose exact integers and coordinates fit the implementation. The reference runtime deliberately stops nominal Mode A expansion above L=2254. A request beyond the configured cap produces FBS_CAPACITY_HOLD. Capacity pressure is not corruption and never authorizes recovery by itself.

5.1 Registered modes A-G

Mode	delta	tau	g	L0	H0	Q0	Registry rail
A	26	27	1	83	165	167	g=1
B	27	29	2	89	177	179	g=2
C	33	34	1	104	207	209	g=1
D	37	39	2	119	237	239	g=2
E	41	42	1	128	255	257	g=1
F	43	44	1	134	267	269	g=1
G	47	49	2	149	297	299	g=2

The mode domain is infinite: $\tau \geq 2$ and $1 \leq g < \tau$. A-G are a declared finite registry, not a derivation of optimal or physically selected modes. Equal τ with different g yields equal scalar spectra but different coordinate morphology.

6. Normative Directed and Bilateral Geometry

6.1 Admissible mode and directed atom

$$M(\tau, g), \tau \geq 2, 1 \leq g < \tau, \delta = \tau - g$$

$$V+(\tau, g) = (0, \tau - g, \tau, 2\tau - g, 2\tau, 3\tau - g, 3\tau, 3\tau + 1)$$

$$L0 = 3\tau + 2$$

The eight coordinates are strictly ordered. The terminal coordinate $3\tau + 1$ is the directed binder, and the containing directed support is $[0, L0 - 1]$. For seed S , translate every coordinate by S .

6.2 Atomic bilateralization

$$B(X) = X \cup (-X)$$

$$W(\tau, g) = B(V+(\tau, g))$$

$$|W| = 15, R0 = 3\tau + 1, H0 = 6\tau + 3 = 2L0 - 1$$

$$Q0 = H0 + 2 = 2L0 + 1 = 6\tau + 5$$

Every atomic W is reflection-symmetric and its unweighted coordinate sum is zero. $H0$ is the inclusive span of the containing interval; it is not the number of marked coordinates.

6.3 Directed ternary parent

For three ordered directed children $T_i = (X_i, L_i)$, place child 1 at offset 0, child 2 at offset L_1 , child 3 at offset $L_1 + L_2$, and append one terminal parent binder.

$$D(T_1, T_2, T_3) = X_1 \cup (X_2 + L_1) \cup (X_3 + L_1 + L_2) \cup \{L_1 + L_2 + L_3\}$$

$$L_D = L_1 + L_2 + L_3 + 1$$

6.4 Global bilateral constructor G

$$G(T_1, T_2, T_3) = B(D(T_1, T_2, T_3))$$

$$H_G = 2L_D - 1 = H_1 + H_2 + H_3 + 4$$

GLOBAL_BILATERAL applies reflection after the complete directed parent is formed. It is bilateral by construction.

6.5 Local centered constructor C

$$H_C = H_1 + H_2 + H_3 + 4, R_C = (H_C - 1)/2$$

$$s1 = -R_C + (H_1 + 1)/2$$

$$s2 = -R_C + H1 + (H2 + 3) / 2$$

$$s3 = -R_C + H1 + H2 + (H3 + 5) / 2$$

$$b0 = -R_C, \quad b1 = -R_C + H1 + 1, \quad b2 = -R_C + H1 + H2 + 2, \quad b3 = R_C$$

$$C(Y1, Y2, Y3) = (Y1 + s1) \cup (Y2 + s2) \cup (Y3 + s3) \cup \{b0, b1, b2, b3\}$$

LOCAL_CENTERED packs three already-centered children into a centered support with four binders. A heterogeneous C object may be valid while its marked set is not reflection-symmetric.

6.6 Typed symmetry rules

Object type	Marked set bilateral?	Support centered?	Validation rule
ATOMIC_BILATERAL	Yes	Yes	Required
GLOBAL_BILATERAL	Yes	Yes	Required
LOCAL_CENTERED homogeneous	Yes for identical bilateral children	Yes	Required for registered homogeneous construction
LOCAL_CENTERED heterogeneous	Only when $H1 = H3$ and $Y1 = Y3$	Yes	Conditional; asymmetry alone is not a fault

7. Recursive Constructors, Identity Ladder, and Inverse Rules

7.1 Value equivalence and noncommutation

$$H_{\text{parent}} = H1 + H2 + H3 + 4$$

$$Q_{\text{parent}} = Q1 + Q2 + Q3$$

For the same three directed children, G and C have equal inclusive span, marked count, leaf inventory, and Q. They generally do not have equal coordinates. For A,A,A, both have $H=499$, $Q=501$, and 49 marks; G contains coordinate 165 and excludes 140, while C contains 140 and excludes 165. The dense mode $M(2,1)$ is a valid edge case where the two constructors coincide at the first homogeneous level.

NONCOMMUTATION RULE

$B(D(T1, T2, T3))$ is not generally equal to $C(B(T1), B(T2), B(T3))$. The implementation must preserve the constructor tag and ordered tree. Equal scalar summaries never authorize a geometry substitution.

7.2 Homogeneous closed forms

$$Q_k = (6\tau + 5) 3^k$$

$$H_k = (6\tau + 5) 3^k - 2$$

$$L_k = ((6\tau + 5) 3^k - 1) / 2$$

$$M_{\text{centered}, k} = 17 3^k - 2$$

$$M_{\text{directed}, k} = (17 3^k - 1) / 2$$

$$B_k = (3^{k+1} - 1) / 2$$

M denotes marked-coordinate count and B denotes the cumulative binder count. The centered marked density approaches $17/(6\tau + 5)$, a nonzero constant. The formal descriptor is self-similar hierarchical, not a claim of non-integer metric dimension.

7.3 Additive valuation and inventory lift

$$\chi(M(\tau, g)) = e_-(\tau, g)$$

$$\text{chi}(\text{parent}) = \text{chi}(T1) + \text{chi}(T2) + \text{chi}(T3)$$

$$Q(T) = \sum_{(\tau, g)} (6\tau + 5) \text{chi}_{(\tau, g)}(T)$$

Q preserves weighted total only. It forgets child order, tree shape, g when tau is unchanged, and decompositions with the same weighted sum. chi preserves the atomic leaf multiset but still forgets order and internal tree shape.

Representation	Preserves	Forgets / depends on	Composition
Q scalar	Weighted total	Order, shape, inventory; heavy collisions	Additive
chi inventory	Atomic leaf multiset	Child order and internal shape	Additive
Ordered labeled tree	Complete declared structure	Nothing inside grammar	Not scalar-additive
Canonical tree digest	Compact identity under encoding	Depends on encoding and digest assumptions	Composable, not reversible

7.4 Inverse and terminal-floor rules

Directed homogeneous inverse: $L_{\text{child}} = (L_{\text{parent}} - 1) / 3$

Centered homogeneous inverse: $H_{\text{child}} = (H_{\text{parent}} - 4) / 3$

The inverse is defined only when the declared homogeneous constructor and divisibility conditions hold. Plain H/3 is nonconforming. For every atomic mode, L0 is congruent to 2 modulo 3 and H0-4 is congruent to 2 modulo 3, so the atom is not the homogeneous image of a smaller integer FBS object.

Heterogeneous rollback must traverse the recorded ordered labeled tree. Scalar inversion is not permitted because Q and H do not preserve decomposition.

8. Recovery Activation and Transaction Protocol

8.1 Permit validation

1. Verify the permit issuer is Module 15 ROVER under the current trust configuration.
2. Verify permit_id has not been consumed, expired, revoked, or completed.
3. Verify incident_digest, affected branch, archive_head, policy_epoch, and requested recovery scope match the evidence bundle.
4. Verify the incident class is registered as catastrophic by the frozen recovery policy.
5. Atomically reserve the permit in the FBS recovery ledger using compare-and-swap. A duplicate request returns the prior result or current in-flight state.

8.2 Branch freeze and checkpoint validation

1. Require a BranchFreezeReceipt for the affected branch. ROVER broadcast lock alone does not prove the runtime stopped writes.
2. Verify the Psi checkpoint bundle is bound to the permit archive head and contains a continuous parent-digest chain.
3. Verify every checkpoint carries the exact constructor tag, ordered tree digest, mode metadata, geometry digest, and commit sequence required for reversal.
4. Reject or hold stale, incomplete, contradictory, unsigned, or post-incident checkpoints.

8.3 Deterministic rollback selection

Select the newest checkpoint satisfying all of the following: committed before the incident boundary; declared known-good by the authorized source; digest-valid; not implicated by the incident; compatible with the frozen policy and constructor path. Ties are resolved by greatest committed sequence, then lexicographically smallest checkpoint ID.

Recovery route	Condition	Output
RESTORE_CHECKPOINT	Eligible checkpoint found above the atomic floor	RestorePlanPacket; no diagonal or Collatz stage
FLOOR_RESEED	No eligible checkpoint exists and exact tree traversal reaches a valid atomic floor	Diagonal witness, bounded Collatz trace, seed publication
HELD	Missing freeze, stale archive head, incomplete tree, capacity or trace limit, dependency unavailable	Evidence preserved; retryable where declared
FAULTED	Malformed or unauthorized request, invalid geometry, digest mismatch	No mutation
SYSTEM_HALT_REQUIRED	Persisted recovery-control contradiction	Operator/control-plane intervention; no recursive FBS

8.4 Residue custody and seed consumption

Only irreducible recovery evidence or residue named by policy is offered to RTSOM. FBS sends an exact digest-bound RTSOMExhaustRequest and waits for the matching durable receipt. It never writes or deletes RTSOM state. After custody reconciliation, FBS publishes one single-use RestorePlanPacket or SeedPublicationPacket to Delta and waits for DeltaSeedConsumptionReceipt.

8.5 Atomic completion

1. Verify RTSOM receipt when residue was requested; otherwise record an explicit NO_RESIDUE branch.
2. Verify Delta consumed exactly the published restore or seed token and opened the declared new cycle.
3. CAS the recovery transaction from COMMIT_PENDING to COMPLETED and read back the durable ledger record.
4. Emit one FBSRecoveryReceipt to ROVER.
5. Emit one FBSCompletionEventReceipt to Module 12 Phi. Replays return the same receipt and never create a second event.

9. Bounded Diagonal Witness, Collatz Trace, and Seed Derivation

9.1 Finite diagonal witness

Construct a canonical matrix from at most DIAGONAL_WIDTH authenticated evidence rows. Each row is a fixed-width bit string derived from a declared evidence digest. For n rows and width w with $n \leq w$:

$$d_i = 1 - M_{i,i} \text{ for } 0 \leq i < n$$

The remaining bits are filled by a versioned digest-expansion rule, not by an unqualified constant pad. The resulting d differs from each examined row i at bit i . This is a local finite witness only. It does not prove global uniqueness, a new address, or freedom from collision outside the examined matrix.

9.2 Bounded Collatz trace

$$C(n) = n/2 \text{ when } n \text{ is even; } C(n) = 3n+1 \text{ when } n \text{ is odd}$$

Let n_0 be a nonzero integer derived from the recovery material digest. Iterate only while $\text{step_count} < \text{COLLATZ_STEP_LIMIT}$ and $\text{bit_length}(n) \leq \text{COLLATZ_MAX_BITS}$. Stop with one of the exact statuses below.

Status	Meaning
REACHED_ONE	n reached 1 within both limits
STEP_LIMIT	Configured step limit reached
BIT_LIMIT	Configured intermediate integer bit length exceeded

Status	Meaning
INVALID_INPUT	n0 was zero, negative, malformed, or inconsistent with the digest

A limited trace is a held recovery, not an infinite loop and not proof of mathematical failure. The module never assumes the Collatz conjecture.

9.3 Deterministic seed derivation

$\text{escape_material} = \text{SHA256}(\text{canonical}(\text{permit}, \text{incident}, \text{floor tree}, \text{diagonal witness}, \text{policy epoch}))$

$\text{trace_digest} = \text{SHA256}(\text{canonical}(\text{n0}, \text{visited trace}, \text{status}, \text{step count}, \text{max bits}))$

$S_{\text{next}} = 1 + (\text{int}(\text{SHA256}(\text{escape_material} \parallel \text{trace_digest}), 16) \bmod (2^{\text{SEED_BITS}} - 1))$

The seed is nonzero, deterministic, and bound to the full recovery transaction. It is not the terminal Collatz value 1 and is not claimed globally unique. Identical canonical incidents under the same policy reproduce the same seed; different bound evidence changes the derivation digest with cryptographic probability.

WHY THE SOURCE PROCEDURE CHANGED

A finite diagonal flip guarantees difference only from the rows whose diagonal bits were examined. An unbounded Collatz-to-4-2-1 requirement depends on an unproved conjecture, and using the terminal value as the seed collapses every successful trace to the same integer. The hardened profile keeps both operations as bounded deterministic witnesses while deriving the actual seed from the complete authenticated trace.

10. Outcomes, Invariants, Faults, and Terminal Control Failures

10.1 Normative invariants

- I-01 Mode domain: $\tau \geq 2$ and $1 \leq g < \tau$.
- I-02 Directed atom: V^+ contains exactly eight distinct strictly ordered coordinates.
- I-03 Atomic bilateral: $W=B(V^+)$ contains exactly fifteen distinct coordinates and is reflection-symmetric.
- I-04 Exact scalar relations: $L0=3\tau+2$, $H0=6\tau+3$, $Q0=6\tau+5$.
- I-05 Constructor tag: every recursive centered object is tagged GLOBAL_BILATERAL or LOCAL_CENTERED.
- I-06 Child order: ordered child identity is preserved even when Q and χ are accumulated.
- I-07 Q additivity: $Q_{\text{parent}}=Q1+Q2+Q3$ for both centered constructors.
- I-08 Identity separation: equal Q , H , marked count, or χ never implies coordinate equality.
- I-09 Symmetry by type: atomic and G require $X=-X$; heterogeneous C uses the local symmetry criterion.
- I-10 Inverse typing: scalar inverse is used only for a declared homogeneous constructor with exact divisibility.
- I-11 Heterogeneous reversal: exact rollback follows the ordered labeled tree.
- I-12 Capacity safety: exceeding a configured geometry cap produces HOLD and cannot self-authorize recovery.
- I-13 Recovery activation: RECOVERY_EXECUTE requires a current single-use Module 15 permit.
- I-14 Frozen evidence: permit, incident, policy, checkpoint bundle, and archive head remain immutable for one transaction.
- I-15 No source mutation: evaluation and plan construction do not mutate Ψ , Δ , RTSOM, Φ , or caller state.
- I-16 Bounded integer processing: diagonal width, Collatz steps, and integer bit length are finite and enforced.
- I-17 Custody before completion: required RTSOM residue has a matching durable receipt before completion.
- I-18 Consumption before completion: Δ confirms exact plan/seed consumption before completion.
- I-19 Exactly once: one permit yields at most one committed recovery result and one Φ completion event.
- I-20 Determinism: identical canonical input and frozen policy yield byte-identical outputs and digests.

10.2 Typed outcomes and faults

Code	Condition	Outcome
FBS_CFG_INVALID	Invalid or internally inconsistent configuration	FAULT
FBS_MODE_DOMAIN	tau/g outside admissible domain	FAULT
FBS_MODE_UNREGISTERED	Policy requires registered mode but ID absent	FAULT
FBS_CONSTRUCTOR_TAG	Missing, unknown, or incompatible constructor tag	FAULT
FBS_CHILD_COUNT	Recursive constructor does not receive exactly three children	FAULT
FBS_CHILD_BOUNDS	Child coordinates exceed declared support	FAULT
FBS_COORDINATE_DUP	Unexpected duplicate marked coordinate	FAULT
FBS_COORDINATE_ORDER	Canonical coordinate order violated	FAULT
FBS_SPAN_MISMATCH	Derived L or H differs from declared value	FAULT
FBS_Q_MISMATCH	Q does not reconcile with span or inventory	FAULT
FBS_CHI_MISMATCH	Inventory does not equal child sum	FAULT
FBS_DIGEST_MISMATCH	Packet, geometry, tree, or evidence digest mismatch	FAULT
FBS_SYMMETRY_MISMATCH	Required type-specific symmetry condition fails	FAULT
FBS_INVERSE_UNDEFINED	Requested scalar inverse lacks type/divisibility proof	HOLD/FAULT
FBS_CAPACITY_HOLD	Requested finite geometry exceeds configured cap	HOLD
FBS_PERMIT_INVALID	Issuer, binding, scope, epoch, or expiry invalid	FAULT
FBS_PERMIT_REPLAY	Permit already consumed or completed	IDEMPOTENT RESULT/FAULT
FBS_INCIDENT_MISMATCH	Incident does not match permit or branch	FAULT
FBS_FREEZE_MISSING	No valid affected-branch freeze receipt	HOLD
FBS_CHECKPOINT_CHAIN	Checkpoint lineage or archive-head binding invalid	HOLD/FAULT
FBS_DIAGONAL_INPUT	Evidence matrix malformed or exceeds declared width	HOLD/FAULT
FBS_COLLATZ_STEP_LIMIT	Bounded trace reached step limit	HOLD
FBS_COLLATZ_BIT_LIMIT	Bounded trace exceeded integer-size limit	HOLD
FBS_RTSOM_RECEIPT	Missing or mismatched custody receipt	HOLD/FAULT
FBS_DELTA_RECEIPT	Missing or mismatched seed/restore consumption receipt	HOLD/FAULT
FBS_TX_CAS_CONFLICT	Concurrent transaction changed ledger version	RETRY/HOLD
FBS_DETERMINISM_FAILURE	Replay produced different canonical output	SYSTEM_HALT_REQUIRED

10.3 Terminal control failures

- Conflicting persisted recovery results or seed publications for one permit.
- A committed recovery record without the required RTSOM or Delta receipt.
- A forged or conflicting custody receipt accepted into the recovery ledger.
- An archive-head or recovery-ledger split that cannot be reconciled from durable records.

- Deterministic replay divergence for identical canonical evidence and policy.
- Unauthorized deletion or mutation of a committed recovery record or acknowledged residue object.

These conditions produce SYSTEM_HALT_REQUIRED. Module 16 cannot call itself recursively; operator or independent control-plane intervention is required.

11. Metrics, Dashboard, and Accounting

Dashboard group	Required measures
Geometry	requests by profile/mode/tag; emitted objects; capacity holds; max level; L/H/Q; marked and binder counts
Identity	Q collisions observed; chi collisions; digest mismatches; G/C value-equivalent geometry forks
Validation	type-specific symmetry checks; bound failures; inverse requests; constructor-tag failures
Recovery intake	permits received/accepted/replayed/expired; incidents by class; freeze latency
Rollback	checkpoint bundle depth; eligible checkpoints; selected rollback depth; restore vs floor-reseed rate
Integer witness	diagonal rows/width; Collatz steps; terminal status; maximum bit length; trace digest
Custody	RTSOM requests/receipts/rejections/pending; exact residue bytes and object digests
Seed	publications, Delta consumptions, pending, expired, duplicate receipts
Completion	recovery receipts; Phi events emitted/consumed; ROVER acknowledgements; ledger CAS conflicts
Reliability	deterministic replay parity; crash-recovery time; in-flight transaction age; terminal control failures

11.1 Reconciliation equations

$$\begin{aligned}
 \text{geometry_attempts} &= \text{emitted} + \text{capacity_holds} + \text{geometry_faults} \\
 \text{accepted_permits} &= \text{completed} + \text{held} + \text{faulted} + \text{in_flight} \\
 \text{recovery_plans} &= \text{restore_plans} + \text{reseed_plans} \\
 \text{RTSOM_requests} &= \text{RTSOM_receipts} + \text{RTSOM_pending} + \text{RTSOM_rejected} \\
 \text{seed_publications} &= \text{Delta_consumptions} + \text{seed_pending} + \text{seed_expired} \\
 \text{Phi_completion_events} &\leq \text{completed_recoveries} \\
 \text{committed_recovery_results} &\leq \text{consumed_activation_permits}
 \end{aligned}$$

Every dashboard counter must be derivable from immutable transaction records. No counter may be incremented only because an attempt started.

12. Conformance, Property, Fuzz, Concurrency, and Stress Tests

12.1 Deterministic conformance suite - 60 tests

ID	Test
T01	Accept $\tau=2$, $g=1$ as the smallest admissible mode.
T02	Reject $\tau < 2$, $g < 1$, and $g > \tau$.
T03	Generate exact Mode A $V^+ = (0, 26, 27, 53, 54, 80, 81, 82)$.
T04	Prove each valid V^+ contains eight distinct strictly ordered coordinates.
T05	Derive $L_0 = 3\tau + 2$ for every A-G mode.
T06	Generate exact Mode A W and verify fifteen coordinates.
T07	Verify $W = -W$ and $\text{sum}(W) = 0$ for every registered atom.
T08	Reproduce the A-G $L_0/H_0/Q_0$ reference table exactly.
T09	Show equal τ with different g has equal $L/H/Q$ but different coordinates.
T10	Translate an atomic object by seed S without changing relative coordinates or scalar values.
T11	Construct a directed parent with three ordered children and one terminal binder.
T12	Verify directed parent length $L_1 + L_2 + L_3 + 1$.
T13	Construct G by bilateralizing the completed directed parent.
T14	Construct C using exact shifts and four binders.
T15	Verify G and C share H , Q , marked count, and χ for identical children.
T16	A,A,A fork: G contains 165 and excludes 140.
T17	A,A,A fork: C contains 140 and excludes 165.
T18	Dense $M(2,1)^3$ edge case: G and C coordinates coincide.
T19	A,B,G global object is bilateral.
T20	A,B,G local C is valid centered support and may be asymmetric.
T21	Apply local symmetry criterion $H_1 = H_3$ and $Y_1 = Y_3$.
T22	Reject missing constructor tag for recursive centered object.
T23	Verify homogeneous $Q_{\text{parent}} = 3Q_{\text{child}}$.
T24	Verify heterogeneous $Q_{\text{parent}} = Q_1 + Q_2 + Q_3$.
T25	Reproduce $Q(A+B+G) = Q(3 \times M(35,1)) = 645$.
T26	Verify χ distinguishes $A+B+G$ from $3 \times M(35,1)$.
T27	Verify ordered-tree digest changes when child order changes.

ID	Test
T28	Verify Q is rejected as unique object identity.
T29	Verify chi is rejected as complete structural identity.
T30	Reproduce homogeneous H _k , L _k , Q _k closed forms for k=0..3.
T31	Reproduce centered and directed marked-count formulas.
T32	Reproduce binder counts 1,4,13,40 for k=0..3.
T33	Verify centered inverse (H-4)/3 on a valid homogeneous parent.
T34	Reject plain H/3 as a conforming inverse.
T35	Verify directed and centered atomic floors fail inverse divisibility.
T36	Require ordered-tree traversal for heterogeneous rollback.
T37	Accept Mode A level 3 at reference L=2254.
T38	Hold Mode A level 4 under the reference 2254 bench cap.
T39	Prove a capacity hold does not emit an FBS activation or recovery plan.
T40	Verify geometry build is byte-identical under repeated canonical input.
T41	Reject RECOVERY_EXECUTE without a current Module 15 permit.
T42	Reject stale, expired, revoked, or mismatched permit bindings.
T43	Consume one permit exactly once and return the same result on idempotent replay.
T44	Require matching branch freeze receipt before checkpoint traversal.
T45	Reject checkpoint bundle whose archive head differs from the permit.
T46	Reject broken parent-digest checkpoint chain.
T47	Select newest eligible known-good checkpoint deterministically.
T48	Emit RESTORE_CHECKPOINT without diagonal or Collatz stage when eligible checkpoint exists.
T49	Traverse exact constructor path to atomic floor when no checkpoint is eligible.
T50	Diagonal witness differs from every examined row at its diagonal bit.
T51	Do not claim diagonal witness differs from unexamined or future rows.
T52	Known Collatz input reaches one within the configured bound and produces exact trace digest.
T53	Stop with STEP_LIMIT rather than looping indefinitely.
T54	Stop with BIT_LIMIT when intermediate integer exceeds maximum bits.
T55	Derive identical nonzero seed from identical canonical recovery evidence.
T56	Change incident digest and verify seed derivation digest changes.
T57	Require matching RTSOM receipt when residue custody was requested.

ID	Test
T58	Require exact Delta consumption receipt before completion.
T59	Emit one FBSRecoveryReceipt and one Phi event for one completed permit.
T60	Concurrent completion CAS: one winner; loser reads and returns the committed result.

12.2 Property and metamorphic tests

- For random admissible modes, V+ remains strictly ordered and W remains reflection-symmetric with fifteen marks.
- Translation by any integer seed preserves gaps, counts, L/H/Q, chi, and constructor tag.
- Permutation of children leaves Q and chi unchanged but usually changes ordered-tree digest and coordinates.
- G is always bilateral; C is bilateral exactly under the declared local criterion when children are bilateral.
- Q reconstructed from chi equals Q accumulated through the tree for every generated object.
- Serializing then parsing an object preserves every integer and digest; reserialization is byte-identical.
- Recovery seed output is invariant to irrelevant transport ordering after canonicalization and sensitive to every bound evidence digest.

12.3 Fuzz, concurrency, crash, and stress requirements

- Fuzz malformed mode values, oversized integers, duplicate coordinates, missing tags, noncanonical encodings, and corrupted digests.
- Fuzz mixed G/C trees and verify no scalar equality is mistaken for geometry equality.
- Run at least 100,000 geometry requests across A-G and admissible unregistered modes under configured capacity.
- Run at least 100,000 recovery transaction attempts with duplicate permits, stale heads, receipt delays, and injected crashes.
- Crash after every durable write boundary; restart must produce either the prior committed state or a resumable in-flight transaction, never a partial completion.
- Execute two or more concurrent consumers of the same permit; at most one committed recovery result and one Phi event may exist.
- Differential-test independent implementations against the A-G vectors, constructor fork witnesses, inventory collision, inverse floors, and seed derivation vectors.

13. Reference Pseudocode and ASCII System Flow

13.1 Geometry pseudocode

```
function build_geometry(req, registry, limits):
    freeze(req, registry, limits)
    mode = validate_mode(req.mode_id, req.tau, req.g)
    if predicted_size(req) > limits.capacity:
        return CapacityHold(no_mutation=true)

    if req.tag == DIRECTED:
        object = directed_atom_or_parent(req)
    elif req.tag == ATOMIC_BILATERAL:
        object = bilateralize(directed_atom(mode))
    elif req.tag == GLOBAL_BILATERAL:
        object = bilateralize(directed_parent(req.ordered_children))
    elif req.tag == LOCAL_CENTERED:
        object = local_centered_parent(req.ordered_children)
    else:
        return Fault(FBS_CONSTRUCTOR_TAG)

    derive L, H, Q, marked_count, binder_count, chi
    tree_digest = SHA256(canonical_ordered_tree)
    geometry_digest = SHA256(canonical(object, metadata))
    assert type_specific_invariants(object)
    return GeometryPacket(no_mutation=true)
```

13.2 Recovery pseudocode

```
function execute_recovery(permit, incident, checkpoints, freeze_receipt, policy):
    frozen = canonical_freeze(permit, incident, checkpoints, freeze_receipt, policy)
    verify_rover_permit(frozen)
    tx = reserve_permit_cas(permit.id)
    if tx.already_committed:
        return tx.committed_receipt

    verify_branch_freeze(freeze_receipt, incident.branch)
    verify_checkpoint_chain(checkpoints, permit.archive_head)
    checkpoint = newest_eligible_known_good(checkpoints, incident)

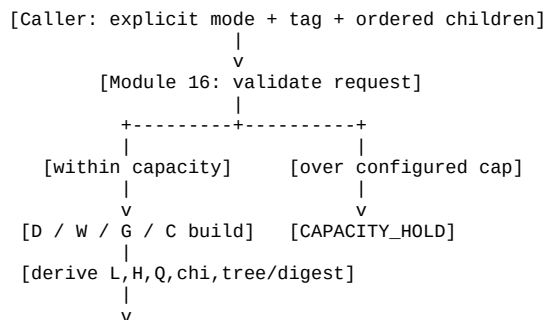
    if checkpoint exists:
        plan = RestorePlan(checkpoint, exact_constructor_path)
        residue = policy.restore_residue(frozen)
    else:
        floor = traverse_ordered_tree_to_atomic_floor(checkpoints.current_tree)
        witness = finite_diagonal_witness(frozen.evidence_rows, width=256)
        trace = bounded_collatz(derive_n0(frozen, witness), steps=10000, max_bits=4096)
        if trace.status != REACHED_ONE:
            return Hold(trace.status, no_mutation=true)
        seed = derive_nonzero_seed(frozen, witness, trace, bits=256)
        plan = SeedPublication(seed, floor, witness.digest, trace.digest)
        residue = policy.floor_residue(frozen, witness, trace)

    if residue exists:
        rtsom_receipt = request_and_verify_rtsom_custody(residue)
    else:
        rtsom_receipt = NO_RESIDUE

    delta_receipt = publish_once_and_verify_delta_consumption(plan)
    committed = commit_recovery_cas(tx, plan, rtsom_receipt, delta_receipt)
    recovery_receipt = emit_rover_receipt(committed)
    phi_event = emit_once_phi_completion_event(committed)
    return recovery_receipt
```

13.3 ASCII architecture flow

NOMINAL GEOMETRY



14.1 Cleaned Rosetta Stone

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SpiralCore term	Formal meaning
Exhaust shunt	Exact digest-bound RTSOM custody request; not physical mass transfer
Reseed	Single-use nonzero digest-derived Delta seed bound to one recovery transaction

14.2 Glossary

Term	Definition
Admissible mode	Integer pair (tau,g) with $\tau \geq 2$ and $1 \leq g < \tau$.
Atomic bilateral block	Reflection closure of one directed atom; fifteen marked coordinates.
Canonical tree digest	SHA-256 digest of the declared ordered labeled tree encoding.
Capacity hold	Non-catastrophic refusal to construct an object exceeding configured resources.
Constructor tag	DIRECTED, ATOMIC_BILATERAL, GLOBAL_BILATERAL, or LOCAL_CENTERED.
Geometry-equivalent	Exactly equal normalized marked coordinate sets under the same declared type.
Known-good checkpoint	Committed Psi checkpoint whose authorized assertion and chain pass the frozen recovery policy.
Recovery residue	Exact evidence object offered to RTSOM under a registered custody profile.
Seed publication	Digest-bound single-use packet authorizing Delta to consume one nonzero recovery seed.
Value-equivalent	Equal H, marked count, chi, and Q without a claim of coordinate equality.

14.3 Revision ledger

Source issue	v14.1 disposition
Geometry and recovery conflated	Separated into explicit GEOMETRY_BUILD, GEOMETRY_VALIDATE, and RECOVERY_EXECUTE profiles.
FBS self-triggered from T or Pe	Recovery requires a current single-use Module 15 ROVER permit.
Mode A floor treated as universal	Floor is mode-specific; Mode A is only the reference recovery default.
Infinite upward expansion	Reframed as arithmetic defined for arbitrary finite levels, bounded by implementation capacity.
2254 merely a level example	Registered as the reference Mode A bench cap; excess produces capacity hold, not recovery.
Centered implied bilateral	Typed validation: atomic/G bilateral; heterogeneous C conditional.
G and C treated as identical	Preserved v4.0 noncommutation and mandatory constructor tags.
Q treated as structural identity	Q is additive valuation; chi and ordered tree/digest added.
Scalar inverse used for rollback	Allowed only for homogeneous typed objects; heterogeneous rollback follows exact tree.
Diagonalization guaranteed unique address	Narrowed to a finite local witness against examined rows.
Collatz always reaches 4-2-1	Bounded by step and bit limits; no conjecture assumption.
Terminal Collatz value became seed	Seed derived from complete permit, incident, witness, and trace digests.

Source issue	v14.1 disposition
FBS directly restarted Delta	FBS publishes a single-use plan; Delta owns consumption and returns receipt.
FBS directly shunted exhaust	RTSOM owns custody and returns a durable receipt.
FBS directly incremented history event	Module 12 owns Phi event ledger and consumes one completion receipt.
Recovery purged or deleted evidence	Exact evidence is retained or receipted to its next owner; no purge path.
Recovery failure recursively invoked FBS	Terminal contradictions produce SYSTEM_HALT_REQUIRED.

14.4 Implementation checklist

- ☐ Implement exact integer geometry with no floating-point arithmetic.
- ☐ Store constructor tag, mode metadata, Q, chi, ordered tree, and canonical digest together.
- ☐ Run all A-G atomic vectors and constructor-fork reference tests at startup or deployment.
- ☐ Enforce the configured capacity before allocating coordinate arrays.
- ☐ Make recovery permit reservation, transaction commit, and receipt emission compare-and-swap and idempotent.
- ☐ Use bounded diagonal and Collatz profiles exactly as versioned.
- ☐ Require durable RTSOM and Delta receipts before completion.
- ☐ Emit one ROVER receipt and one Phi completion event only after durable recovery commit.
- ☐ Keep source evidence immutable through every retry, crash, and replay.
- ☐ Treat every physical, thermodynamic, cognitive, and routing-benefit interpretation as a separate hypothesis.

FINAL FORMAL BOUNDARY

Module 16 is now a typed integer-geometry engine and a permit-gated recovery executor. Its arithmetic is exact; its identities are layered; its inverse rules are constructor-aware; its recovery transforms are finite; and every external mutation is owned by a receiving module and confirmed by a digest-bound receipt.

SPIRALCORE SPECIFICATION

MODULE 17 - MILLENNIUM ARCHITECTURE

(Conditional Proof Pipeline - Formal Task Encoding, Certified Transformation, Kernel Verification, and Tamper-Evident Reporting)

Cleaned and formalized revision - v14.1

Field	Value	Field	Value
Document class	Normative conditional-proof orchestration contract	Operating profiles	PROOF_ATTEMPT / CERTIFICATE_VERIFY / REPORT_ONLY
Architecture	SpiralCore v14	Reference profile	SC14.1-M17-MILLENNIUM-01
Formal-task owner	Module 17 M_map	Proof-kernel owner	Module 17 M_ver isolated verifier
Archive owner	Module 3 Psi	Persistent Phi owner	Module 12 Phi
Recovery authorizer	Module 15 ROVER	Recovery executor	Module 16 FBS
Directive compiler	Module 18	Module 17 authority	Evidence only; no execution authorization

NORMATIVE STATUS

Exact within the declared software and formal-methods scope. Module 17 freezes a formal task, executes a versioned domain adapter in an isolated sandbox, records a complete step ledger, verifies all proof obligations with a declared checker, and emits a typed conclusion plus a tamper-evident certificate. A theorem-status result exists only when a proof object or counterexample is independently accepted by the declared trusted kernel.

CLAIM BOUNDARY

Millennium Architecture, proof invariant, paradox mass, surgery, hysteresis, holonomy, certificate of work, and academic research lab are project labels. This module does not claim to solve a Millennium Prize Problem, establish an unconditional theorem, prove its own verifier sound, certify human cognition, or convert numerical agreement into mathematical truth. External mathematical acceptance remains outside the runtime.

Source basis: SpiralCore Module 17 source specification and the consolidated SpiralCore v14 Module 17/Module 18 architecture sequence.

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2	Architecture Ownership, Trust Boundary, and Handoffs
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5	Formal Task, Domain Adapter, and Trusted Kernel
6	Eight-Stage Pipeline Mechanics
7	Proof Semantics, Conclusion Classes, and Verification Gate
8	Exact Monitoring, Error Propagation, and Obligations
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1. Normative Scope, Statement Classes, and Non-Goals

1.1 Module purpose

Module 17 is a generic orchestration framework for conditional mathematical proof attempts. It does not supply a universal mathematical method. A conforming run is meaningful only after a domain adapter declares the formal language, semantics, axioms, premises, target statement, state representation, transition rules, monitor, anomaly predicates, intervention proof obligations, termination rule, and trusted verification procedure.

CORE BOUNDARY

M_map defines the frozen formal task. M_evo, M_mon, M_loc, and M_int produce trace evidence and candidate transformations. M_acc records local run events. M_ver alone determines whether the declared kernel accepts a proof object or counterexample. M_rep reports that result without strengthening it. Psi may archive the artifact; Module 18 may later inspect evidence but receives no execution authority from Module 17.

1.2 Statement classes

Class	Meaning	Examples
DEFINITION	Introduced by stipulation	Formal task, proof obligation, adapter schema, certificate class
REQUIREMENT	Mandatory implementation behavior	Frozen manifest, independent verification, complete status output
INVARIANT	Must hold for every conforming run	No omitted step, digest-chain continuity, no score-only proof
POLICY	Versioned operational choice	Resource budgets, archive retention, registered adapters
CERTIFIED CLAIM	Accepted by declared checker under stated assumptions	Conditional proof or checked counterexample
DIAGNOSTIC	Useful telemetry with no proof authority	Xi_progress, Omega_unresolved, local event count
HYPOTHESIS	Requires separate mathematical or empirical evidence	Adapter soundness beyond kernel, heuristic benefit
NON-GOAL	Explicitly excluded	Automatic solution of open problems, self-verifying truth, physical holonomy

1.3 Normative non-goals

- Module 17 does not infer theoremhood from a high similarity, coherence, confidence, invariant, or progress score.
- Module 17 does not treat numerical simulation, finite case checking, model agreement, or a replayable trace as a general proof unless the formal target explicitly reduces to those finite checks.
- Module 17 does not import archive material as a premise merely because it is semantically similar or previously accepted by SpiralCore.
- Module 17 does not let an external solver, language model, numerical package, or heuristic certify its own output without an independently declared checker.
- Module 17 does not mutate the persistent Phi state, commit a .frac object, invoke RMX, execute FBS, take RTSOM custody, or compile a build directive.
- The project name Millennium Architecture is historical. It is not an assertion that a Millennium Prize Problem has been solved or recognized by the mathematical community.

2. Architecture Ownership, Trust Boundary, and Handoffs

Owner	Authority
Module 17 coordinator	Run manifest; lifecycle; sandbox; step ledger; certificate assembly; typed terminal result
M_map	Canonical task encoding; premise registry; formalization digest; initial state and obligations
M_evo	Deterministic candidate state transition under the declared adapter
M_mon	Exact or inclusion-certified monitor values and error evidence
M_loc	Deterministic anomaly candidates; no state mutation
M_int	Candidate intervention plus independently checkable proof obligations; no self-approval
M_acc	Local event and intervention count inside the run only
M_ver	Independent checker/kernel; proof-object, counterexample, transition, and obligation validation
M_rep	Status-preserving report and certificate packaging; no claim escalation

Owner	Authority
Module 3 Psi	Optional durable artifact archive, RMX handling, commit receipt, retention
Module 12 Phi	Persistent hysteresis/event ledger; may consume one completed Module17 event receipt
Module 15 ROVER	Catastrophic incident evaluation and any FBS activation permit
Module 16 FBS	Authorized recovery execution only
Module 14 RTSOM	Exact residue custody after an authorized recovery transaction
Module 18	Separate compilation and policy validation of build directives; no inheritance of proof authority

2.1 Trust boundary

The trusted computing base is explicitly declared per run. At minimum it includes the canonical serializer, digest implementation, task parser, proof-object checker, exact arithmetic or interval checker, and result mapper. The generator, heuristic search, numerical simulator, archive retriever, and language model are outside the trusted kernel unless a separate proof establishes otherwise.

TRUSTED-KERNEL RULE

A certificate may be tamper-evident and perfectly replayable while the mathematical conclusion is still invalid. CONDITIONAL_PROVED or CONDITIONAL_REFUTED requires independent kernel acceptance. Hashes bind evidence; they do not prove the evidence is correct.

2.2 Required handoff chain

1. The requester submits ProblemSpec and HypothesisSpec plus a registered domain profile.
2. M_map freezes the formal task and premise registry into RunManifest.
3. The sandbox executes the bounded pipeline and appends complete StepRecords.
4. M_ver checks the final proof object, counterexample, transition ledger, monitor bounds, and all intervention obligations.
5. M_rep emits one terminal result and CertificateBundle preserving the verifier status exactly.
6. Under archive policy, Module 17 submits an ArtifactArchiveRequest to Psi and waits for a matching receipt.
7. After durable archive acceptance, Module 17 may emit one Module17CompletionEventReceipt for Module 12 EVENT_APPEND.
8. A downstream build directive, if any, is separately submitted to Module 18 and carries no inherited execution permission.

2.3 Prohibited cross-module mutation

- No direct .frac addressing, Cantor-IP allocation, archive-head update, RMX synthesis, or Psi commit.
- No direct Phi or global DeltaPhi update. M_acc is a run-local audit counter only.
- No direct FBS, Cantor diagonalization, Collatz folding, reseeding, or RTSOM mutation.
- No direct system modification, microcode registration, configuration write, or build-directive execution.

3. Operating Profiles, Lifecycle, and Atomicity

Profile	Purpose	Authority
PROOF_ATTEMPT	Execute one bounded frozen task	One typed conclusion and certificate

Profile	Purpose	Authority
CERTIFICATE_VERIFY	Verify an externally supplied certificate without rerunning search	Kernel verification result plus verification receipt
REPORT_ONLY	Render a previously verified result	No mathematical recomputation and no status change

3.1 Lifecycle

State	Meaning
UNINITIALIZED	No verified configuration or adapter
READY	Configuration, adapter, kernel, and resource policy validated
MAPPING	Problem and hypothesis canonicalized into FormalTask
RUNNING	Step transactions execute and append to the ledger
VERIFYING	Sandbox sealed; M_ver independently checks evidence
REPORTING	M_rep builds the status-preserving report and certificate
ARCHIVE_PENDING	Optional Psi archive request awaits receipt
COMPLETED	Terminal result and required receipts reconciled
INCONCLUSIVE	Bound, unresolved obligation, or unsupported claim prevents proof status
HELD	Recoverable resource or dependency condition
FAULTED	Malformed or inconsistent execution
RECOVERY_REQUESTED	Registered catastrophic control incident sent to ROVER

3.2 Frozen run manifest

Before M_map begins, the coordinator freezes every item that can affect semantics or replay: task bytes, hypothesis bytes, adapter ID/version/digest, axiom set, premise set, theorem target, arithmetic backend, kernel ID/version/digest, resource limits, deterministic seed, archive-context digest, policy epoch, and output schema. Wall-clock time, process ID, host path, thread scheduling, and UI state are never semantic inputs.

3.3 Step atomicity

1. Read x_t and the prior ledger head without mutation.
2. Compute M_evo, M_mon, M_loc, and optional M_int outputs.
3. Verify schema, exact arithmetic evidence, intervention obligations, and output digest.
4. Construct one canonical StepRecord bound to the prior ledger head.
5. Append with compare-and-swap; publish x_{t+1} only after durable append succeeds.
6. On crash, either the step record and state are both visible or neither is visible.

4. Typed Data Contracts and Canonical Serialization

Contract	Required content
Module17Config	module/profile versions, adapter registry, kernel registry, arithmetic profile, resource limits, archive policy

Contract	Required content
ProblemSpec	formal problem bytes, syntax profile, declared variables/domains, requester digest
HypothesisSpec	premises, target statement, quantifiers, scope, excluded assumptions, canonical digest
PremiseRecord	premise text/object, source digest, proof status, checker, dependency graph, trust class
ArchiveContextSnapshot	read-only archive head and cited artifacts; never implicit premises
FormalTask	language, semantics, axiom set, premise registry, target, state type, obligations, formalization digest
RunManifest	all frozen semantics, versions, limits, deterministic seed, environment digest
DomainAdapterManifest	M_map/evo/mon/loc/int/acc/ver interfaces, versioned code digests, declared soundness boundary
StepRecord	t, input digest, evolution output, monitor interval, anomaly records, intervention record, obligations, output digest, resource counters, ledger links
MonitorBoundPacket	quantity ID, exact rational center/radius or interval endpoints, inclusion proof/checker, backend
AnomalyPacket	detector ID, deterministic location, predicate, evidence, priority, no-mutation proof
InterventionProposal	pre-state, anomaly binding, transformed state, parameters, proof obligations, lineage digest
InterventionCertificate	preconditions, boundary match, invariant transfer, postconditions, checker results, complete dependency list
ProofObjectPacket	formal claim, proof object, checker profile, dependency closure, object digest
CounterexamplePacket	witness, domain membership proof, target falsification check, digest
VerificationRecord	kernel ID/version, checked objects, result, failed obligations, replay digest, verifier signature/channel proof
CertificateBundle	run manifest, complete ledger or committed references, proof/counterexample object, verification record, report, digest chain, optional signature
ArtifactArchiveRequest	certificate digest, exact artifact references, requested retention, no-execution capability
PsiArchiveReceipt	request binding, committed object digest, archive sequence/head, RMX result if any, owner signature
Module17CompletionEventReceipt	one archived terminal run event for Module 12; no semantic winding claim
BuildDirectiveEvidencePacket	optional evidence for Module 18, explicitly non-executable and non-authorizing
Module17Fault / Hold / CatastrophicIncidentPacket	typed reason, evidence, retryability, frozen run, no-mutation proof

4.1 Canonical serialization

- All maps use lexicographically sorted UTF-8 keys; arrays preserve declared order; sets use an explicitly declared canonical sort.
- Integers and rationals use canonical sign, numerator, and positive denominator encodings. Equivalent fractions are reduced before hashing.
- Floating-point values cannot carry proof authority. Diagnostic floats use a declared finite-number profile and remain outside the proof kernel.
- Every record includes schema version, producer ID/version, policy epoch, parent digest, object digest, and complete dependency digests.
- The final certificate is identified by SHA-256 over canonical content. Optional signatures authenticate an issuer but do not replace mathematical checking.

4.2 Digest chain

$$h_0 = \text{SHA256}(\text{canonical}(\text{RunManifest}))$$

$$h_{\{t+1\}} = \text{SHA256}(h_t \parallel \text{canonical}(\text{StepRecord}_t))$$

$$\text{certificate_digest} = \text{SHA256}(h_T \parallel \text{canonical}(\text{VerificationRecord}) \parallel \text{canonical}(\text{Report}))$$

A missing, reordered, duplicated, or modified step changes the final digest. Digest continuity proves evidence integrity relative to the hash assumption; it does not prove the mathematical inference.

5. Formal Task, Domain Adapter, and Trusted Kernel

5.1 Formal task

$$T = (L, \text{Sem}, A, H, G, X, O, K)$$

L is the formal language; Sem is its semantics; A is the axiom set; H is the explicit premise registry; G is the target proposition or decision predicate; X is the state space; O is the complete proof-obligation set; and K is the declared verification kernel. A task is invalid when any of these components is absent, ambiguous, versionless, or internally inconsistent.

5.2 Premise registry and archive isolation

Archive retrieval may suggest definitions, lemmas, examples, or prior certificates. It cannot silently become an assumption. Every archive-derived item used by the proof must become a `PremiseRecord` with exact source digest, formal statement, trust class, dependency closure, and checker status. Unverified archive material may guide search but cannot be consumed by `M_ver` as a proved premise.

5.3 Domain adapter contract

Submodule	Normative contract
M_map	Parse and type-check P/H; build FormalTask and x_0; enumerate initial obligations
M_evo	Deterministic transition or search proposal; may be heuristic but must expose full evidence
M_mon	Return exact/inclusion-certified values and propagation metadata
M_loc	Return deterministic anomaly candidates; cannot edit state
M_int	Return a candidate transformation and proof obligations; cannot approve them
M_acc	Record local events, counts, and resource deltas; no persistent Phi mutation
M_ver	Check task well-formedness, premises, transitions, obligations, proof object, counterexample, and conclusion mapping
M_rep	Render the verified result without altering its class or strength

5.4 Trusted kernel levels

Level	Requirement	Meaning
TRACE_REPLAYED	All recorded transitions replay under the frozen adapter	Execution integrity only
OBLIGATIONS_CHECKED	Every required local/global obligation is accepted	Certified transformation chain
PROOF_OBJECT_ACCEPTED	Minimal checker accepts proof object for G under A and H	CONDITIONAL_PROVED eligible

Level	Requirement	Meaning
COUNTEREXAMPLE_ACCEPTED	Checker accepts witness in domain and verifies not-G	CONDITIONAL_REFUTED eligible

GENERATOR/CHECKER SEPARATION

The same code may propose and check an object only when the checker is separately instantiated from immutable bytes and its trusted boundary is declared. A solver saying “SAT,” a model saying “proved,” or an intervention reporting “invariants preserved” is not a kernel result.

6. Eight-Stage Pipeline Mechanics

6.1 M_map - deterministic formalization

$$(T, x_0, O_0) = M_map(P, H, adapter, premise_registry)$$

- Rejects ambiguous symbol binding, undeclared quantifiers, incompatible domains, circular premises, and missing target semantics.
- Produces one canonical formalization digest. A changed formalization is a new task, not a continuation of the old run.
- Records any human-authored or machine-generated translation assumptions explicitly.

6.2 M_evo - bounded deterministic evolution

$$x_{\{t+1\}^{evo}} = M_evo(x_t, T, seed, t)$$

M_evo may implement exact deduction, state exploration, numerical approximation, graph rewriting, symbolic algebra, or a problem-specific flow. It must be deterministic under the frozen manifest or record all controlled nondeterminism as explicit input. Its output is a candidate state, not a proof.

6.3 M_mon - exact or inclusion-certified monitor

$$I_t = [l_t, u_t] \text{ subset } Q, \quad \forall S(x_t) \text{ in } I_t$$

M_mon reports the quantity definition, interval or exact rational value, checker, and dependency evidence. A bare epsilon claim is insufficient. The checker must be able to verify that the true target quantity is contained in the declared interval under the formal model.

6.4 M_loc - anomaly localization

$$A_t = \text{sort}(M_loc(\text{neighborhood}(x_{\{t+1\}^{evo}}), T))$$

Anomalies are typed predicates such as violated preconditions, nonregular boundaries, rank loss, cycle recurrence, domain escape, or unsatisfied local constraints. Results use stable priority and identifier ordering. M_loc is read-only.

6.5 M_int - certified intervention proposal

$$(x_{\{t+1\}}, \pi_t^{int}, O_t^{int}) = M_int(x_{\{t+1\}^{evo}}, A_t, T)$$

The intervention does not “preserve invariants” by declaration. It emits a transformed state plus exact proof obligations for domain membership, boundary compatibility, invariant transfer, dependency retention, and postconditions. M_ver or a registered local checker must discharge them before the step commits.

6.6 M_acc - local event accounting

$$e_t \text{ in } \{0,1\}, \quad N_event(T) = \sum_{t=0}^{T-1} e_t$$

M_acc records whether a registered critical transformation occurred and accumulates a run-local event count. The legacy name DeltaPhi_local may be retained as a compatibility field, but it is not the persistent Module 12 DeltaPhi, a topological winding number, proof of work, or evidence of theorem depth.

6.7 M_ver - independent verification

1. Validate FormalTask, premise closure, adapter/kernel identities, and manifest digest.
2. Validate every StepRecord and ledger link, or validate a proof object that safely abstracts the trace under the declared kernel.
3. Check every mandatory monitor inclusion and intervention obligation.
4. Check proof object or counterexample against the exact target G.
5. Map the kernel result to one terminal conclusion class using a fixed table.

6.8 M_rep - status-preserving report

M_rep may improve readability but must not strengthen the result. “Inconclusive” cannot become “likely proved”; a finite test cannot become a theorem; a theorem under additional premises cannot be reported without those premises. The report embeds the exact formalization, assumptions, kernel, result, and limitations.

7. Proof Semantics, Conclusion Classes, and Verification Gate

7.1 Certificate classes

Class	Meaning
PROOF_CERTIFICATE	Kernel-accepted proof object for G under exact A and H
COUNTEREXAMPLE_CERTIFICATE	Kernel-accepted witness establishing not-G in the declared domain
TRACE_CERTIFICATE	Replayable execution trace without a theorem-level proof object
BOUND_CERTIFICATE	Finite verified bounds or finite-case result only
INCONCLUSIVE_CERTIFICATE	Run ended without discharging all required obligations
INVALID_CERTIFICATE	Malformed, inconsistent, incomplete, or checker-rejected artifact

7.2 Terminal mathematical conclusions

Conclusion	Exact meaning
CONDITIONAL_PROVED	K accepts proof object pi: A,H - G; all mandatory dependencies and obligations verified
CONDITIONAL_REFUTED	K accepts witness w: w in Domain and not G(w)
FINITE_DOMAIN_ESTABLISHED	Target is explicitly finite and every required case is checked
BOUND_ESTABLISHED	A declared bound is verified without claiming the full target theorem
INCONCLUSIVE_MAX_STEPS	The bounded search ended; no proof or refutation follows
INCONCLUSIVE_UNDISCHARGED	One or more required obligations remain unresolved
INVALID_FORMALIZATION	Task, semantics, premises, or target are invalid or ambiguous
INVALID_PROOF_OBJECT	Kernel rejects the proposed proof or counterexample

7.3 Authoritative gate

CONDITIONAL_PROVED iff kernel_accepts(pi, A, H, G) and unresolved_required = 0

CONDITIONAL_REFUTED iff kernel_accepts_counterexample(w, Domain, G)

No averaged score can satisfy this gate. The source thresholds are retained only as legacy progress diagnostics:

$$Xi_progress = N_passed / N_required$$

$$Omega_unresolved = N_unresolved / N_required$$

For $N_required=0$, both metrics are NOT_READY. The legacy display condition $Xi_progress \geq 0.85$ and $Omega_unresolved \leq 0.15$ may be shown as “progress gate passed,” but it never authorizes proof status. A conforming proof requires every mandatory obligation to pass and the kernel to accept the final object.

NO PARTIAL PROOF BY SCORE

A proof with 85% of obligations discharged is an incomplete proof. An unresolved-obligation ratio of 15% is not mathematically acceptable merely because it is small. These values are dashboard telemetry, not truth thresholds.

7.4 Required report language

For a proved result, the report must use an equivalent of: “Under axiom set A, formalization F, explicit premise set H, and kernel K, proof object pi was accepted as establishing conclusion G.” For refutation, identify the exact witness. For every other outcome, state that the target was not proved or refuted by this run.

8. Exact Monitoring, Error Propagation, and Obligations

8.1 Exact and interval arithmetic

- Exact integer, rational, algebraic, or symbolic arithmetic is preferred where feasible.
- Interval arithmetic must use outward rounding or a checker-verifiable inclusion certificate.
- A floating-point estimate with a self-reported epsilon is diagnostic until independently bounded.
- Every comparison near a threshold uses interval logic: PASS only when the entire interval satisfies the predicate; FAIL when the entire interval violates it; otherwise UNRESOLVED.

8.2 Error propagation

The source cumulative sum $E_T = \sum \epsilon_t$ is not generally a valid global error bound because a transition may amplify earlier error. A conforming adapter declares a propagation rule. The reference Lipschitz profile uses:

$$E_0 = 0$$

$$E_{\{t+1\}} \leq L_t E_t + \epsilon_{\{t+1\}}, \quad L_t \in Q_{\{>=0\}}$$

Each L_t must be exact or certified. When the adapter cannot supply a valid propagation rule, accumulated numerical error remains UNKNOWN and cannot support a proof claim that depends on it.

8.3 Obligation classes

Class	Required verification
TASK	Well-formed language, semantics, axioms, premises, target, and domain
TRANSITION	M_evo output follows the declared relation or is correctly labeled as a search proposal
MONITOR	Value inclusion, bound propagation, and threshold classification
ANOMALY	M_loc predicate and location are valid
INTERVENTION	Preconditions, boundary match, invariant transfer, postconditions, and lineage
TERMINATION	Termination reason and final-state relationship are valid
PROOF	Proof object establishes G from A and H
COUNTEREXAMPLE	Witness lies in domain and falsifies G

Class	Required verification
REPORT	Natural-language conclusion exactly matches kernel result

8.4 Obligation status

Status	Meaning
PASS	Checker accepted with complete dependency evidence
FAIL	Checker produced a deterministic contradiction or rejected object
UNRESOLVED	Insufficient precision, missing dependency, timeout, unsupported feature, or undecided branch
NOT_APPLICABLE	Obligation is outside the declared adapter path and reason is recorded

9. Certified Intervention and Local Event Ledger

9.1 Intervention transaction

1. Bind the anomaly, pre-state, adapter, and parameter set.
2. Generate a candidate transformed state without overwriting the pre-state.
3. Emit exact proof obligations and a dependency-complete InterventionCertificate proposal.
4. Have the independent checker verify every mandatory obligation.
5. Append the accepted intervention and new state atomically, or append a rejected proposal while retaining x_t .

9.2 Required intervention evidence

Evidence	Requirement
Precondition	The anomaly and intervention domain are satisfied
Boundary compatibility	Removed and inserted structures match under the declared boundary relation
Invariant transfer	Every claimed invariant has a proof or checker result
Dependency preservation	No premise, variable, branch, or constraint is silently discarded
Postcondition	The transformed state belongs to X and satisfies the next-stage contract
Lineage	Pre-state, anomaly, parameters, transformed state, and obligations are digest-bound

9.3 Local event count

$$\text{event_count_T} = \sum e_t, \quad e_t = 1 \text{ only for a registered committed event}$$

The counter supports audit and workload metrics. It is not an unforgeable certificate, physical winding number, semantic holonomy, proof depth, or theorem-quality score. Duplicate retries and rejected interventions do not increment it.

10. Certificate, Reporting, Psi Archive, and Module 18 Boundary

10.1 Certificate minimum content

- Canonical ProblemSpec, HypothesisSpec, FormalTask, axiom set, premise registry, and formalization digest.

- RunManifest with adapter, kernel, arithmetic backend, resource limits, deterministic inputs, and archive-context digest.
- Complete step ledger or immutable content-addressed references sufficient for independent verification.
- All monitor bounds, anomaly evidence, intervention proposals/certificates, and obligation statuses.
- Proof object, counterexample, finite-case manifest, or explicit absence thereof.
- VerificationRecord with kernel result, failed/unresolved obligations, checker digest, and replay digest.
- Status-preserving human-readable report and exact conclusion class.
- Certificate digest and optional external signature/countersignature metadata.

10.2 Tamper evidence versus authenticity

The digest chain detects modification relative to SHA-256. A signature authenticates the signer relative to its key-management system. Neither mechanism proves the theorem. The verifier must still check the mathematical object and its assumptions. “Unforgeable certificate” is replaced by “tamper-evident certificate with optional authenticated signature.”

10.3 Psi archival

1. Module 17 emits ArtifactArchiveRequest only after M_rep has finalized the certificate.
2. Psi validates the request, performs its own address/RMX/commit workflow, and returns PsiArchiveReceipt.
3. Module 17 verifies exact certificate digest, archive head, commit sequence, and receipt authenticity.
4. Archive failure or contention does not change the mathematical result; it changes archive status only.

10.4 Module 12 event

After durable Psi acceptance, Module 17 may emit one Module17CompletionEventReceipt. Module 12 decides whether and how to record it. The proof run never writes persistent Phi state directly, and its local event count is not copied into global DeltaPhi.

10.5 Module 18 boundary

A proof certificate may support a later build directive, but it cannot authorize execution or self-modification. Module 17 may emit BuildDirectiveEvidencePacket containing exact certificate and claim digests. Module 18 independently validates directive syntax, registry binding, stack and policy constraints, and execution authority. A mathematically valid theorem does not automatically imply a safe or authorized software change.

11. Sandbox, Security, Resource Bounds, and Replay

11.1 Sandbox requirements

- No network access unless the run manifest explicitly declares a read-only deterministic content source whose bytes are frozen before execution.
- No write access outside the run workspace; no shell, device, credential, or host-control capability for domain adapters.
- CPU, memory, state size, certificate size, recursion depth, intervention count, solver calls, and logical-step budgets are hard limits.
- Randomness uses an explicit deterministic seed and algorithm. Uncontrolled concurrency is prohibited in proof-authoritative paths.
- External solvers are version-locked and treated as untrusted generators unless their proof objects are independently checked.

11.2 Threat model

Threat	Control
Premise injection	Only registered, typed, dependency-complete premises may enter M_ver
Proof-by-hash	Digests bind artifacts but never satisfy proof obligations

Threat	Control
Solver self-certification	Require independent checker or proof object
Certificate truncation	Complete ledger count, dependency closure, and final digest reconciliation
Adapter drift	Freeze version and code digest; reject mid-run swap
Nondeterministic replay	Freeze inputs/seeds; compare complete ledger and kernel result
Resource exhaustion	Return HELD or INCONCLUSIVE_RESOURCE_LIMIT; retain evidence
Archive poisoning	Archive context is non-premise until imported and checked
Report escalation	Fixed conclusion mapping; report digest bound to verification record
Sandbox escape	Immediate catastrophic incident, branch freeze, and ROVER notification

11.3 Deterministic replay

A conforming implementation must reproduce the same FormalTask digest, StepRecord sequence, obligation statuses, proof-object digest, kernel result, and certificate digest from the same frozen manifest. Performance timestamps and host-specific telemetry may differ but remain outside the canonical proof artifact.

12. Outcomes, Invariants, Faults, and Catastrophic Incidents

12.1 Normative invariants

ID	Invariant
M17-I01	Every run has exactly one frozen RunManifest before M_map.
M17-I02	The formalization digest binds language, semantics, axioms, premises, target, state space, obligations, and kernel.
M17-I03	Archive context never becomes an implicit premise.
M17-I04	M_loc cannot mutate state; M_int cannot approve its own obligations.
M17-I05	Every committed state transition has exactly one canonical StepRecord.
M17-I06	The step digest chain is complete, ordered, and gap-free.
M17-I07	Floating-point telemetry alone cannot carry proof authority.
M17-I08	Every claimed bound has an exact or checker-verified inclusion/propagation certificate.
M17-I09	CONDITIONAL_PROVED requires kernel acceptance and zero unresolved mandatory obligations.
M17-I10	CONDITIONAL_REFUTED requires a kernel-accepted in-domain counterexample.
M17-I11	Xi_progress and Omega_unresolved are diagnostic only.
M17-I12	MAX_STEPS, timeouts, and resource caps produce bounded outcomes, not proof or catastrophic recovery.
M17-I13	M_acc changes only the run-local event ledger.
M17-I14	M_rep cannot strengthen, weaken, or relabel the kernel result.
M17-I15	Every terminal run emits a certificate, including invalid and inconclusive outcomes.

ID	Invariant
M17-I16	Psi archive status is independent of mathematical conclusion status.
M17-I17	No certificate authorizes Module 18 execution or system mutation.
M17-I18	Ordinary proof failure, invalid proof object, or intervention rejection never invokes FBS.
M17-I19	A catastrophic incident freezes the run and is sent to ROVER; Module 17 never executes FBS.
M17-I20	Identical frozen runs produce identical canonical certificate digests.

12.2 Typed terminal outcomes and faults

Code	Condition	Class
M17_OK_PROVED	Kernel accepted proof object; all mandatory obligations pass	Terminal
M17_OK_REFUTED	Kernel accepted counterexample	Terminal
M17_OK_BOUND	Verified bound or finite-domain result	Terminal
M17_INC_MAX_STEPS	Logical step bound reached	Terminal/ inconclusive
M17_INC_UNRESOLVED	Required obligations remain unresolved	Terminal/ inconclusive
M17_HOLD_RESOURCE	Resource capacity unavailable before safe completion	Retryable hold
M17_HOLD_DEPENDENCY	Required checker, premise, or archive dependency unavailable	Retryable hold
M17_CFG_INVALID	Configuration, adapter, or kernel registry invalid	Fault
M17_TASK_INVALID	Problem/hypothesis/formal task malformed or ambiguous	Fault
M17_PREMISE_CYCLE	Circular or unresolved premise dependency	Fault
M17_SCHEMA_INVALID	Packet or record schema invalid	Fault
M17_DIGEST_MISMATCH	Canonical digest or chain mismatch	Fault
M17_ARITHMETIC_UNSOUND	Monitor inclusion or propagation certificate rejected	Fault
M17_TRANSITION_INVALID	Recorded transition violates declared relation	Fault
M17_INTERVENTION_INVALID	Intervention obligations fail or lineage is incomplete	Fault
M17_PROOF_REJECTED	Kernel rejects proposed proof object	Terminal/ inconclusive
M17_COUNTEREXAMPLE_REJECTED	Kernel rejects witness	Terminal/ inconclusive
M17_REPORT_MISMATCH	Report conclusion differs from verification record	Fault
M17_ARCHIVE_BACKPRESSURE	Psi cannot accept artifact now	Archive hold only
M17_RECEIPT_MISMATCH	Psi or Module 12 receipt does not bind exact artifact	Fault
M17_PLUGIN_NONDETERMINISM	Same frozen step yields different canonical output	Fault/incident candidate

Code	Condition	Class
M17_KERNEL_CONFLICT	Same proof object/kernel version yields conflicting persisted results	Catastrophic incident
M17_LEDGER_SPLIT	Two persisted ledger heads for one run sequence	Catastrophic incident
M17_SANDBOX_ESCAPE	Unauthorized capability or host mutation detected	Catastrophic incident

12.3 Catastrophic incidents

1. Verified deterministic replay divergence after the same manifest and environment digest.
2. Conflicting persisted kernel results for the same proof object, task, kernel version, and dependency closure.
3. A committed step-ledger split, double finalization, or contradictory terminal certificate for one run ID.
4. Sandbox escape, unauthorized host mutation, or evidence that the run modified external state outside its declared capabilities.
5. Durable certificate/object loss after an acknowledged Psi receipt or an accepted certificate digest changing on read-back.
6. A forged or multiply consumed recovery/archive token that was accepted as valid.

Module 17 freezes the run, emits CatastrophicIncidentPacket to ROVER, and stops. It does not execute FBS, Cantor processing, Collatz processing, reseeding, or RTSOM custody.

13. Metrics, Dashboard, and Reconciliation

Dashboard group	Required fields
Run volume	attempted, mapped, executed, verified, reported, archived
Conclusion classes	proved, refuted, finite-domain, bound, inconclusive, invalid
Step telemetry	step count, retries, checkpoints, state bytes, logical cost
Obligations	required, passed, failed, unresolved, not-applicable
Monitor	exact values, interval widths, propagation bounds, unresolved threshold tests
Interventions	proposed, accepted, rejected, local event count
Verification	kernel type/version, proof-object size, checker time, replay status
Certificate	ledger count, bytes, digest status, signature status
Archive	requested, accepted, held, failed, receipt latency
Security	sandbox denials, plugin drift, nondeterminism, catastrophic incidents

13.1 Reconciliation equations

$$\text{runs_attempted} = \text{runs_completed} + \text{runs_inconclusive} + \text{runs_held} + \text{runs_faulted} + \text{runs_recovery_requested}$$

$$\text{obligations_required} = \text{obligations_passed} + \text{obligations_failed} + \text{obligations_unresolved}$$

$$\text{interventions_proposed} = \text{interventions_accepted} + \text{interventions_rejected} + \text{interventions_pending}$$

$$\text{archive_requested} = \text{archive_accepted} + \text{archive_held} + \text{archive_failed} + \text{archive_pending}$$

$$\text{certificates_emitted} = \text{proved} + \text{refuted} + \text{finite_domain} + \text{bound} + \text{inconclusive} + \text{invalid}$$

NOT_APPLICABLE obligations are counted separately and must have a reason. No dashboard subtotal may silently disappear or be counted as proof progress.

14. Conformance, Property, Fuzz, Concurrency, and Stress Tests

14.1 Deterministic conformance suite - 60 tests

ID	Test
T01	Reject missing formal language or semantics.
T02	Reject undeclared variable domain.
T03	Reject ambiguous quantifier binding.
T04	Reject circular premise dependency.
T05	Changing one premise changes the formalization digest.
T06	Archive suggestion is not included in premises without explicit import.
T07	M_map produces identical FormalTask for identical canonical input.
T08	Changed adapter version creates a new run manifest.
T09	Wall-clock metadata does not change task or certificate digest.
T10	Invalid kernel registry prevents READY state.
T11	One exact transition produces one StepRecord.
T12	A skipped step breaks the ledger chain.
T13	A duplicated step breaks sequence reconciliation.
T14	Failed compare-and-swap publishes neither state nor step.
T15	Crash after durable append recovers the same published state.
T16	M_loc cannot modify the candidate state.
T17	M_int output is not committed before obligation checks pass.
T18	Rejected intervention retains the pre-intervention state.
T19	Duplicate retry does not increment local event count.
T20	State and StepRecord digests reconcile after every commit.
T21	Exact rational monitor value passes with epsilon zero.
T22	Outward interval containing target passes inclusion check.
T23	Self-reported float epsilon without checker remains diagnostic.
T24	Threshold interval wholly above bound produces PASS.
T25	Threshold interval straddling bound produces UNRESOLVED.
T26	Propagation uses $E_{\text{next}} \leq L * E + \text{epsilon}$ rather than sum alone.
T27	Missing Lipschitz/propagation certificate leaves global error UNKNOWN.
T28	Rejected arithmetic certificate prevents proof status.
T29	Negative interval radius is rejected.

ID	Test
T30	Equivalent reduced rationals serialize identically.
T31	85% obligation progress cannot produce CONDITIONAL_PROVED.
T32	15% unresolved ratio cannot produce CONDITIONAL_PROVED.
T33	All obligations pass but rejected proof object remains inconclusive.
T34	Kernel-accepted proof plus zero unresolved obligations yields CONDITIONAL_PROVED.
T35	Kernel-accepted counterexample yields CONDITIONAL_REFUTED.
T36	Finite enumeration yields FINITE_DOMAIN_ESTABLISHED only when target is declared finite.
T37	MAX_STEPS yields INCONCLUSIVE_MAX_STEPS and a certificate.
T38	Resource cap yields HOLD or inconclusive, never proof.
T39	Report cannot relabel inconclusive result as likely proof.
T40	Report includes exact assumptions and kernel identity.
T41	Certificate digest changes when any step changes.
T42	Certificate with missing dependency is rejected.
T43	Optional signature verifies exact certificate digest.
T44	Valid signature with rejected proof object remains invalid mathematically.
T45	CERTIFICATE_VERIFY reproduces original kernel result.
T46	REPORT_ONLY cannot alter verification status.
T47	Psi archive rejection does not alter mathematical conclusion.
T48	Psi receipt must bind exact certificate digest.
T49	Module17CompletionEventReceipt emits once after durable archive acceptance.
T50	BuildDirectiveEvidencePacket has no execute capability.
T51	Ordinary proof failure emits no catastrophic incident.
T52	Invalid intervention emits no FBS activation.
T53	Plugin nondeterminism freezes run and records evidence.
T54	Conflicting persisted kernel results create catastrophic incident.
T55	Ledger split creates catastrophic incident.
T56	Sandbox escape creates catastrophic incident and stops execution.
T57	Module 17 never calls FBS directly.
T58	Module 17 never writes RTSOM or persistent Phi state.
T59	Two identical frozen runs produce identical certificate digests.
T60	Every terminal path emits exactly one typed result and certificate.

14.2 Property and metamorphic tests

- Canonical-renaming property: alpha-equivalent bound-variable renaming produces the same normalized formal task when the language profile declares alpha equivalence.
- Premise monotonicity is never assumed globally; adding a premise creates a new task and result even when the target remains provable.
- Trace integrity: any permutation, omission, duplication, or modification of StepRecords changes the certificate digest or fails sequence checks.
- Checker independence: replacing M_rep wording cannot change the kernel result or proof-object digest.
- Score irrelevance: arbitrary Xi_progress and Omega_unresolved values cannot override obligation or kernel results.
- Archive isolation: changing unused archive context changes context digest but not formal-task semantics or verifier result.

14.3 Fuzz, concurrency, crash, and stress requirements

- Fuzz Unicode normalization, deeply nested syntax, huge integers/rationals, malformed proof objects, cyclic dependencies, NaN/Inf diagnostics, duplicate IDs, and certificate truncation.
- Run concurrent archive submissions and verify Psi receipts cannot be cross-bound between certificates.
- Crash at every step-transaction boundary, verification boundary, report write, archive request, and receipt reconciliation point.
- Stress at least 100,000 bounded synthetic runs across proved, refuted, inconclusive, held, and faulted fixtures; reconcile every dashboard equation.
- Differentially test independent checker implementations where two trusted kernels are available; disagreement is a fault or incident, never a majority vote.

15. Reference Pseudocode and ASCII System Flow

15.1 Proof-attempt pseudocode

```
function run_millennium(request, config):
    validate_config_and_registry(config)
    manifest = freeze_run_manifest(request, config)
    task, x, obligations = M_map(manifest)
    ledger_head = sha256(canonical(manifest))
    local_event_count = 0

    for t in range(manifest.max_steps):
        if M_ver.termination_condition(task, x, t):
            break

        x_evo = M_evo(task, x, t, manifest.seed)
        monitor = M_mon(task, x_evo, t)
        verify_monitor_evidence(monitor)

        anomalies = stable_sort(M_loc(task, x_evo, t))
        intervention = NONE
        x_next = x_evo

        if anomalies is not empty:
            proposal = M_int(task, x_evo, anomalies, t)
            result = check_intervention_obligations(task, proposal)
            intervention = record(proposal, result)
            if result.all_mandatory_pass:
                x_next = proposal.output_state
                if proposal.registered_event:
                    local_event_count += 1

        step = build_step_record(t, x, x_evo, monitor,
                                anomalies, intervention, x_next,
                                ledger_head)
        append_step_and_state_cas(step, x_next)
        ledger_head = step.digest
        x = x_next
```

```

freeze_sandbox()
proof_or_witness = adapter.final_object(task, x, ledger_head)
verification = M_ver.verify(task, ledger_head, proof_or_witness)
conclusion = map_kernel_result_exactly(verification)
report = M_rep.render(task, conclusion, verification,
                      local_event_count)
certificate = build_certificate(manifest, ledger_head,
                               proof_or_witness,
                               verification, report)
return typed_terminal_result(conclusion, certificate)

```

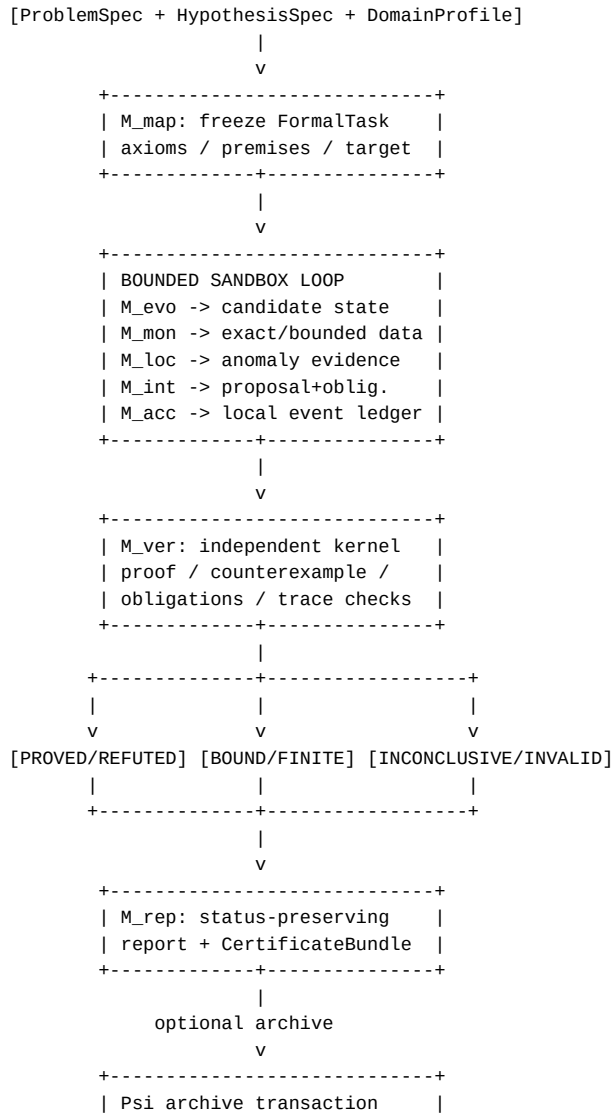
15.2 Certificate-verification pseudocode

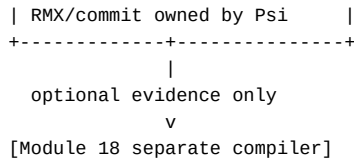
```

function verify_certificate(bundle, registry):
    validate_canonical_schema(bundle)
    verify_manifest_and_dependency_closure(bundle)
    verify_digest_chain(bundle.steps, bundle.manifest_digest)
    task = reconstruct_formal_task(bundle)
    verify_monitor_and_intervention_certificates(bundle)
    result = registry.kernel(bundle.kernel_id).verify(
        task, bundle.proof_or_counterexample)
    require(result == bundle.verification_record.result)
    require(report_maps_exactly(result, bundle.report))
    return VerificationReceipt(bundle.certificate_digest, result)

```

15.3 ASCII architecture flow





Catastrophic control incident -> freeze -> ROVER
 Ordinary failure/MAX_STEPS/resource limit -> certificate, not FBS

16. Rosetta Stone, Glossary, Revision Ledger, and Checklist

16.1 Cleaned Rosetta Stone

Project term	Formal meaning	Boundary
Millennium Architecture	Versioned orchestration framework for domain-specific formal proof attempts	Not an automatic solver for open problems
Conditional proof	Kernel-checked derivation under exact axioms and premises	Must state assumptions and formalization
M_map	Parser/formalizer and premise-registry constructor	Translation assumptions become explicit
M_evo	Deterministic transition or search proposal	No proof authority by itself
M_mon	Exact/inclusion-certified monitor	Not physical energy or entropy unless separately modeled
M_loc	Read-only anomaly detector	Does not diagnose human cognition
M_int / surgery	Certified state transformation with proof obligations	Perelman/Ricci-flow language is analogy only
M_acc / DeltaPhi_local	Run-local event counter	Not persistent Phi or topological winding
Xi_proof	Legacy obligation-progress ratio	Diagnostic, never a theorem gate
Omega_paradox	Legacy unresolved-obligation ratio	Diagnostic, not physical mass
Finite certificate	Tamper-evident evidence bundle	Not unforgeable and not automatically a proof
FBS runaway	ROVER-authorized infrastructure recovery outside Module 17	Not triggered by ordinary proof failure

16.2 Glossary

Term	Definition
Axiom set	Declared foundational formulas accepted by the formal task.
CertificateBundle	Canonical evidence package containing task, trace references, proof/witness, verification record, and report.
Conditional conclusion	A result whose validity is limited to the exact formalization, axioms, premises, and kernel.
Counterexample	Checked in-domain witness falsifying the target statement.
Domain adapter	Versioned implementation of the eight pipeline interfaces for one formal problem family.
Formalization digest	Hash binding the complete formal language, semantics, axioms, premises, target, state space, obligations, and kernel.

Term	Definition
Kernel	Minimal trusted checker that independently validates proof objects or counterexamples.
Obligation	Explicit proposition that must be checked for a task, transition, monitor, intervention, or final proof.
Premise registry	Dependency-complete list of assumptions and imported lemmas with source and verification status.
Proof object	Machine-checkable derivation in the declared formal system.
Replayable trace	Deterministic execution record; not necessarily a proof object.
Tamper-evident	Modification is detectable under the digest assumption; mathematical correctness is separate.

16.3 Revision ledger

Source behavior	Formalized correction
Generic “academic research lab”	Scoped to a domain-adapter orchestration framework with a declared trusted kernel
Archive A_t as baseline bias	Archive is read-only search context; any used premise requires explicit import and checking
DeltaPhi reset and accumulation	Replaced by run-local event count; persistent Phi remains Module 12-owned
$\Xi \geq 0.85$ and $\Omega \leq 0.15$ proof gate	Retained only as legacy progress telemetry; proof requires all obligations plus kernel acceptance
Sum of epsilon as proof error	Replaced by adapter-certified error propagation, e.g. $E_{next} \leq L E + \epsilon$
Surgery preserves invariants	Replaced by explicit intervention obligations and independent checking
Unforgeable certificate	Replaced by tamper-evident digest chain plus optional authenticated signature
Direct Psi .frac/Cantor-IP commit	Replaced by ArtifactArchiveRequest and receipt; Psi/RMX retain ownership
Direct FBS/Cantor/Collatz/RTSOM path	Removed; Module 17 emits catastrophic incident to ROVER only
Proof output before Module 18	Clarified as non-executable evidence; Module 18 independently validates directives
Perelman proof analogue	Retained as analogy only; no generic equivalence to Ricci flow with surgery
MAX_STEPS failure	Formalized as an inconclusive bounded result with a complete certificate

16.4 Implementation checklist

- ☐ Register and hash every domain adapter, arithmetic backend, and verification kernel.
- ☐ Freeze FormalTask, premises, target, limits, deterministic inputs, and archive context before execution.
- ☐ Implement canonical rational serialization and a gap-free step digest chain.
- ☐ Keep M_loc read-only and M_int non-authoritative until obligations pass.
- ☐ Implement exact/inclusion-certified monitor and error-propagation rules.
- ☐ Use kernel acceptance, not scores, as the proof/refutation gate.
- ☐ Emit a certificate for every terminal outcome, including inconclusive and invalid.
- ☐ Keep Psi archive, Module 12 event, ROVER/FBS recovery, RTSOM custody, and Module 18 compilation outside Module 17 ownership.
- ☐ Run all 60 conformance tests plus property, fuzz, concurrency, crash, and stress suites.
- ☐ Render, inspect, accessibility-audit, and independently verify the final DOCX/PDF artifacts.

Source Basis

Primary source: SPIRALCORE SPECIFICATION: MODULE 17 - MILLENNIUM ARCHITECTURE (CONDITIONAL PROOF PIPELINE). Architecture context: consolidated SpiralCore v14, including the Module 17 to Module 18 sequence and cross-module ownership established by the formalized v14.1 module set.

SPIRALCORE SPECIFICATION

MODULE 18 - GÖDEL-GÖDEL INSTRUCTION SET

(AI Build Directives - Typed Compilation, Capability Validation, Change Planning, and Receipt-Bound Handoff)

Cleaned and formalized revision - v14.1

Field	Value	Field	Value
Document class	Normative directive compiler and change-plan contract	Operating profiles	COMPILE_ONLY / VERIFY_PACKAGE / PREPARE_CHANGE
Primary output	DirectivePlanPacket or typed non-executable outcome	Execution authority	External Change Control Authority plus target owner
Reference recursion cap	MAX_STACK_DEPTH = 8	Reference fuel cap	10,000 deterministic evaluation units
Legacy score	INSTRUCTION_FLOOR = 0.80 telemetry only	Authoritative gate	All mandatory hard obligations pass
Digest	SHA-256 over canonical directive/package bytes	Mutation rule	No core, registry, archive, route, or network mutation

Normative boundary

Module 18 compiles a typed directive into an immutable, capability-scoped change plan and verifies that plan against a frozen registry, policy, evidence set, and target snapshot. It does not execute the plan, approve its own privileges, write the active microcode registry, modify a target module, commit a .frac archive object, invoke FBS, or discard evidence.

Claim boundary

A digest binds bytes; it does not understand semantics. A similarity score is not a safety proof. EXEC_DIRECTIVE_BOUND means only that a package is internally complete and eligible for independent authorization. It is not a declaration that a directive is globally safe, correct, wise, non-malicious, or authorized to alter a root system.

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1. Normative Scope, Statement Classes, and Non-Goals

1.1 Module purpose

Module 18 is the terminal compiler and change-plan builder for SpiralCore AI Build Directives. It transforms a typed, authenticated proposal into a canonical intermediate representation, resolves every operation against a signed registry snapshot, verifies types, capabilities, effects, preconditions, resource bounds, rollback requirements, and evidence bindings, and emits a non-executable package for independent authorization and target-owned execution.

Core rule

The compiler may describe a change. Only the independent Change Control Authority may authorize it, and only the declared target owner may execute it.

1.2 Statement classes

Class	Meaning	Example
DEFINITION	Introduces a typed object, state, function, or relation.	DirectiveAST, OpcodeRecord, EffectManifest
REQUIREMENT	Mandatory implementation behavior.	Canonical parse and exact registry lookup
INVARIANT	Condition that must hold for every conforming run.	No execution without an external permit
POLICY	Versioned deployment choice.	Stack cap, fuel cap, allowed opcode set
SURROGATE	Computational quantity using project terminology.	Instruction Coherence telemetry
HYPOTHESIS	Interpretation requiring separate evidence.	Claims about cognition or systemic identity
NON-GOAL	Behavior intentionally excluded.	Direct root-state self-modification

1.3 Normative non-goals

- No free-form text is executed as code.
- No SHA-256 digest is treated as a semantic embedding, opcode selector, or collision-free identifier.
- No cosine similarity score grants capability, privilege, or root-state write authority.
- No directive may alter the active compiler, active registry, policy epoch, target state, archive, network, Phi state, or recovery seed during compilation.
- No stack-depth check alone is claimed to prove termination.
- No rejected directive is labeled malicious, hallucinatory, or unsafe merely because it fails compilation.
- No ordinary parse, type, policy, stack, fuel, or sandbox failure triggers catastrophic recovery.

2. Architecture Ownership, Trust Boundary, and Handoffs

Owner	Owns	Explicit boundary
Module 17	Optional proof/counterexample/trace evidence	No execution authorization and no directive compilation
Guardian	Authenticates and authorizes remote boundary ingress/egress	Does not compile or execute directives
Module 18	Parse, resolve, validate, sandbox, package, and verify directives	No target mutation and no privilege approval
Registry Authority	Publishes signed immutable opcode and schema snapshots	Module 18 cannot modify the active snapshot
Change Control Authority	Issues or denies a capability-scoped execution permit	Must be independent of the compiler and proposer
Target Owner / Adapter	Prepares, commits, verifies, and rolls back target-owned changes	May change only its declared state under permit
Psi / RMX	Archives packages, receipts, and collision-resolved objects	No compile or execution authority
Module 12 Phi	Consumes eligible committed-change event receipts	No directive interpretation
QGM / GlyphNet	Routes internal packets and ordinary IPv4/TCP transport	No semantic compilation or privilege decision
ROVER / FBS	ROVER authorizes catastrophic recovery; FBS executes it	Module 18 reports incidents only
RTSOM	Exact custody of registered recovery residue	No generic failed-directive dumping

2.1 Trust boundary

Module 18 treats the proposer, free-form source text, Module 17 evidence, external generators, target adapters, and sandbox runtime as untrusted until their outputs satisfy exact typed checks. The trusted base is limited to the frozen compiler build, canonical serializer, parser, registry-verification logic, policy evaluator, cryptographic primitive implementation, and independent package verifier named by the run manifest.

2.2 Required handoff chain

1. Authenticated DirectiveEnvelope enters Module 18. Remote envelopes require a Guardian receipt.
2. Module 18 freezes the directive, registry, policy, target snapshot, evidence references, and resource limits.
3. Module 18 parses, resolves, statically validates, sandbox-executes where required, and emits DirectivePlanPacket.
4. The independent package verifier replays and verifies the package.
5. Module 18 emits ChangeAuthorizationRequest; the Change Control Authority issues or denies ChangeExecutionPermit.
6. The declared target owner consumes the single-use permit, prepares and commits through compare-and-swap, verifies postconditions, and returns ExecutionReceipt or RollbackReceipt.
7. Psi may archive the plan and receipts. Module 12 may consume one committed-change event receipt.

2.3 Prohibited cross-module mutation

- Module 18 never writes Psi archive state or invokes RMX.
- Module 18 never writes the active opcode registry or registry-signing keys.
- Module 18 never opens sockets, changes routes, edits QGM/GlyphNet state, or derives outer IPv4 addresses.
- Module 18 never mutates target configuration, code, memory, policy, model weights, or operating state.
- Module 18 never increments persistent Phi or transfers data to RTSOM.
- Module 18 never invokes Cantor diagonalization, Collatz folding, reseeding, or FBS execution.

3. Operating Profiles, Lifecycle, and Atomicity

Profile	Purpose	Terminal output
COMPILE_ONLY	Create and validate a new directive package	DirectivePlanPacket, HOLD, REJECT, or FAULT
VERIFY_PACKAGE	Independently replay and verify an existing package	PackageVerificationRecord
PREPARE_CHANGE	Compile, verify, and create an authorization request	ChangeAuthorizationRequest; never an execution permit

3.1 Lifecycle

State	Meaning
RECEIVED	Envelope accepted into an isolated workspace
FROZEN	Registry, policy, target, evidence, limits, and compiler digests fixed
CANONICALIZED	Canonical bytes and directive digest established
PARSED	Typed AST produced without side effects
RESOLVED	Every opcode/schema/target resolves exactly
STATIC_VALIDATED	Types, effects, capabilities, dependencies, resources, and rollback checked
SANDBOXED	Required deterministic dry run completed
PLAN_READY	Immutable directive package built
VERIFIED	Independent verifier accepted exact package
AUTH_PENDING	Authorization request emitted
COMPLETED	Compilation transaction closed; no target mutation implied
HELD / REJECTED / FAULTED	Typed non-success outcomes with evidence retained
INCIDENT_REPORTED	Control-plane catastrophe frozen and reported to ROVER

3.2 Frozen compilation manifest

One CompilationManifest is frozen before parsing. It binds compiler version and digest, parser and serializer versions, registry and policy snapshots, target-state snapshot, evidence digests, resource limits, deterministic seed where applicable, sandbox image, package-verifier version, and requested profile. No in-flight item may observe a registry or policy swap.

3.3 Atomicity

1. All compile-time analysis is side-effect free outside the isolated workspace and append-only compilation ledger.

2. A DirectivePlanPacket is emitted only after every mandatory record is complete and digest-bound.
3. The package verifier consumes immutable bytes and emits a separate verification record.
4. Authorization and target execution are separate transactions with separate owners and receipts.
5. A failed compile, authorization, prepare, commit, or postcondition check cannot partially mutate the target through Module 18.

4. Typed Data Contracts and Canonical Serialization

Contract	Minimum content / meaning
DirectiveEnvelope	Issuer, source kind, grammar version, target, requested operations/effects, evidence refs, nonce, expiry, parent digest, signature
DirectiveAST	Canonical typed syntax tree; no executable host-language objects
OpcodeRecord	Opcode ID/version, operand schema, target owner, effects, capabilities, preconditions, postconditions, rollback schema, cost model
RegistrySnapshot	Signed immutable registry version, publisher, validity interval, root digest, revocation state
PolicySnapshot	Allowed targets, opcodes, effects, issuers, capability ceilings, review rules, limits
TargetStateSnapshot	Target owner, state version, state digest, policy epoch, readiness, prepare/commit protocol
BuildDirectiveEvidencePacket	Optional Module 17 evidence and claim digests; never execution authority
EffectManifest	Exact reads, writes, creates, deletes, routes, network, archive, privilege, and external side effects
StaticAnalysisRecord	Parse, type, registry, capability, dependency, resource, determinism, rollback, and policy results
SandboxEvidence	Frozen image, inputs, outputs, logs, resource use, nondeterminism checks, and digest
DirectivePlanPacket	Canonical immutable operations, effects, pre/postconditions, rollback plan, target, snapshots, evidence, and package digest
PackageVerificationRecord	Independent replay result, verifier digest, failed obligations, and package digest
ChangeAuthorizationRequest	Verified package, requested capabilities, target, expiry, review class, and package digest
ChangeExecutionPermit	Externally issued, single-use, target/package/effect-bound permit
ExecutionReceipt	Target prepare/commit/read-back result, state versions, postcondition result, and receipt digest
RollbackReceipt	Rollback operation, restored state digest/version, reason, and verification result
DirectiveFault	Typed ordinary failure with stage, evidence, and no-mutation assertion
CatastrophicIncidentPacket	Control-plane incident evidence for ROVER; no local FBS execution

4.1 Canonical serialization

- UTF-8 text is normalized to Unicode NFC before parsing; identifiers use registered ASCII profiles unless a schema explicitly permits otherwise.
- Maps use lexicographically sorted keys; arrays preserve declared order; integers use minimal decimal encoding; binary data uses lowercase hexadecimal.
- Floating-point values are prohibited in authoritative directive semantics unless a registered schema defines exact rational or bit-level encoding.
- Unknown fields, duplicate keys, non-finite numbers, ambiguous numeric forms, and unregistered extensions are rejected.
- Canonical bytes include all version, target, policy, registry, capability, effect, evidence, and rollback fields.

$$h_dir = \text{SHA256}(\text{canonical}(\text{DirectiveEnvelope}))$$

$$h_plan = \text{SHA256}(\text{canonical}(\text{DirectivePlanPacket without } h_plan))$$

SHA-256 is used as a collision-resistant content address and integrity binding. It is not claimed to be collision-free, semantic, self-authenticating, or sufficient for authorization.

4.2 Signatures and replay

Signatures authenticate an issuer relative to the configured trust store. A signature does not prove that the directive is correct or permitted. Nonce, issuer sequence, expiry, parent digest, and package-consumption records prevent replay. Repeated submission of identical canonical bytes is idempotent and returns the prior compile result.

5. Directive Language, Registry, Capability, and Effect Model

5.1 Directive language

The reference profile accepts a small declarative DSL or its equivalent typed AST. It does not accept shell commands, Python source, native code, arbitrary bytecode, SQL strings, network payloads, or templated text for direct execution. Free-form language may be used only by an untrusted proposer that outputs a DirectiveAST for exact compiler validation.

Reference opcode	Meaning	Required capability
SET_CONFIG	Set a registered configuration field	CONFIG_WRITE
REGISTER_ARTIFACT	Register an immutable, verified artifact for later target-owned activation	ARTIFACT_REGISTER
ENABLE_FEATURE	Enable a pre-registered feature flag	FEATURE_CONTROL
DISABLE_FEATURE	Disable a pre-registered feature flag	FEATURE_CONTROL
UPDATE_POLICY_REF	Point a target to an already-published signed policy version	POLICY_REF_WRITE
MIGRATE_SCHEMA	Apply a registered, reversible schema migration	SCHEMA_MIGRATE
NO_OP	Produce a receipt without target mutation	NONE

5.2 Exact registry lookup

The directive carries opcode_id and opcode_version. Module 18 performs exact lookup in the frozen RegistrySnapshot. A hash is never converted into an opcode and cyclic modulo mapping is prohibited in authoritative compilation.

$$\text{opcode} = \text{RegistrySnapshot}[(\text{opcode_id}, \text{opcode_version})]$$

5.3 Capability model

Every opcode declares a minimal capability set. The directive requests capabilities, policy limits the request, and the external permit grants an exact subset. Module 18 can prove that a requested package is internally capability-consistent, but it cannot grant the capabilities.

$$C_required = \text{union_j } C(\text{op_j})$$

$$C_required \subseteq C_requested \subseteq C_policy_ceiling$$

$$C_granted = C_required$$

Required for execution; evaluated by the external Change Control Authority.

5.4 Effect model

Each opcode declares explicit read and write effects. Hidden effects are prohibited. A package whose inferred effects differ from declared effects is rejected. The effect manifest distinguishes target state, archive state, registry state, routing state, network egress, credential access, process creation, and external I/O.

$$E_inferred = \text{union_j } E(\text{op_j}, \text{operands_j})$$

$$E_{\text{inferred}} = E_{\text{declared}}$$

Root and self-hosting rule

The active compiler, active registry, root trust store, Change Control Authority, and permit-verification code cannot be modified through an ordinary directive. A separately registered bootstrap profile requires an offline signed artifact, two independent verifiers, a restart-bound epoch swap, a recovery image, and external operator approval.

6. Compiler Pipeline Mechanics

6.1 Stage A - ingest and freeze

Validate envelope signature, issuer, nonce, expiry, profile, target, source kind, and maximum byte length. Freeze all dependencies in CompilationManifest. No parsing occurs before the manifest is durable.

6.2 Stage B - canonicalize and parse

Canonicalize source bytes under the declared grammar, parse into DirectiveAST, reject ambiguity, duplicate keys, unknown syntax, undeclared extensions, and host-language escapes. Parsing must be total over accepted input and bounded by AST node and depth limits.

6.3 Stage C - registry and schema resolution

Resolve every opcode, operand schema, target adapter, capability, effect class, and rollback schema against exact versioned identifiers. Missing or revoked registry entries produce typed rejection. Registry order cannot affect resolution.

6.4 Stage D - type, precondition, and effect checking

Type-check all operands; verify target-state version and preconditions against the frozen snapshot; infer effects; ensure the effect manifest is complete; reject privilege amplification, undeclared writes, wildcard targets, and ambiguous ownership.

6.5 Stage E - dependency and termination analysis

Build the directive call/dependency graph, reject cycles unless the relevant opcode supplies a registered well-founded ranking function, compute stack depth, expansion nodes, instruction count, cost, and fuel, and prove all values remain within policy.

6.6 Stage F - rollback construction

Every mutating opcode must produce a target-owned rollback operation or a registered proof that the operation is immutable and safely additive. Rollback is checked against the same registry, target, capabilities, and effects as the forward plan.

6.7 Stage G - sandbox dry run

Where policy requires, execute the target adapter in a deterministic sandbox against a copy of the frozen target snapshot. Compare declared and observed effects, verify postconditions, record resource use, and rerun to detect nondeterminism. The sandbox has no authority to commit.

6.8 Stage H - package and independent verify

Construct DirectivePlanPacket, compute `h_plan`, and invoke the independent package verifier. `VERIFY_PACKAGE` must reproduce the AST digest, resolved registry entries, static results, sandbox evidence, operation order, effect manifest, rollback plan, and `h_plan` from immutable inputs.

7. Validation Gate and Instruction Coherence Telemetry

7.1 Mandatory hard obligations

Flag	Mandatory obligation
b_parse	Canonical parse and grammar acceptance
b_schema	Every operand and output satisfies its exact schema
b_registry	Every opcode, target adapter, and extension resolves to a valid signed registry record
b_type	Static type checking passes
b_cap	Required capabilities fit the requested set and policy ceiling
b_effect	Declared and inferred effects match exactly
b_pre	All target preconditions hold on the frozen snapshot
b_resource	Depth, nodes, fuel, bytes, and cost are within limits
b_determinism	Compilation and required sandbox runs are deterministic
b_rollback	Rollback or immutable-additive exception is valid
b_evidence	Required Module 17 or external evidence is present and claim-bound
b_verify	Independent package verifier accepts

$$\text{PLAN_READY} = \text{AND}(\text{b_parse}, \text{b_schema}, \text{b_registry}, \text{b_type}, \text{b_cap}, \text{b_effect}, \text{b_pre}, \text{b_resource}, \text{b_determinism}, \text{b_rollback}, \text{b_evidence}, \text{b_verify})$$

A single false mandatory flag prevents PLAN_READY. No average, threshold, signature, proof certificate, or operator preference may override a hard failure.

7.2 Formalized Instruction Coherence telemetry

The project name Φ_{ins} is retained as a diagnostic obligation-coverage score, not as an authorization metric. Let $w_j > 0$ be versioned diagnostic weights and b_j in $\{0,1\}$ be the mandatory flags excluding b_{verify} during pre-verification telemetry.

$$\Phi_{\text{ins}} = (\sum_j w_j b_j) / (\sum_j w_j)$$

$$0 \leq \Phi_{\text{ins}} \leq 1$$

The source INSTRUCTION_FLOOR = 0.80 is retained only as migration telemetry. A package with $\Phi_{\text{ins}} = 0.92$ and one failed hard check is still rejected. A plan-ready reference package has $\Phi_{\text{ins}} = 1.00$ before external authorization.

7.3 Legacy hash-similarity profile

The original cosine similarity between a digest-derived vector and a registry vector is not semantic validation. Implementations may reproduce it only under LEGACY_HASH_SIMILARITY_V0 for migration dashboards. It must be labeled non-authoritative, must not select an opcode, and must not influence PLAN_READY, authorization, execution, rollback, FBS, or archive status.

8. Resource Bounds, Recursion, Termination, and Dependency Control

Limit	Reference value	Meaning
MAX_STACK_DEPTH	8	Maximum nested directive-call depth; source value retained

Limit	Reference value	Meaning
MAX_AST_NODES	1,024	Maximum canonical syntax-tree nodes
MAX_OPERATIONS	256	Maximum compiled target operations
MAX_FUEL	10,000	Deterministic analysis and sandbox evaluation units
MAX_DIRECTIVE_BYTES	1,048,576	Canonical envelope byte ceiling
MAX_ARTIFACT_BYTES	67,108,864	Referenced immutable build artifact ceiling
MAX_SANDBOX_RUNS	2	Reference deterministic replay pair
MAX_DEPENDENCY_DEPTH	64	Maximum resolved artifact/lemma/registry dependency chain

8.1 Depth does not prove termination

A maximum stack depth prevents unbounded nesting but does not prevent a loop at one level. The reference profile additionally requires a finite operation list, bounded expansion, a decreasing fuel counter, and an acyclic dependency graph unless a registered well-founded recursion measure is supplied.

$$\text{fuel}_{(k+1)} = \text{fuel}_k - \text{cost}(\text{op}_k), \quad \text{cost}(\text{op}_k) \in \mathbb{N}_{\{>0\}}$$

$$\text{fuel}_k \geq 0 \text{ for every executed compiler/sandbox step}$$

8.2 Dependency graph

The compiler constructs a directed graph over operations, artifacts, schemas, evidence, and target prerequisites. Stable topological ordering uses lexicographic identifier tie-breaking. Unregistered cycles are rejected with the full cycle witness.

8.3 Resource outcomes

Exceeding a declared limit produces HELD_RESOURCE_LIMIT or REJECTED_LIMIT according to policy. It does not imply maliciousness and does not invoke FBS. Evidence is retained so the proposer may submit a smaller or differently scoped directive.

9. Sandbox, Static Analysis, Dry Run, and Deterministic Replay

9.1 Static analysis

- No undeclared file, registry, archive, route, network, process, credential, device, or privilege effects.
- No dynamic code loading, reflection, eval, shell expansion, templated execution, or host-language escape.
- No path traversal, wildcard target, implicit default target, or ambient authority.
- No target-owner mismatch or cross-module write outside the declared adapter.
- No use of revoked opcode, schema, issuer, artifact, registry, or policy version.
- No unbounded loop, unbounded allocation, uncontrolled concurrency, wall-clock dependency, or unseeded randomness.

9.2 Sandbox requirements

- Read-only frozen target copy; no production credentials, sockets, devices, or host namespace writes.
- Deterministic virtual clock, locale, filesystem order, environment, random seed, and concurrency schedule.
- Resource limits enforced outside the adapter process.
- Observed read/write/effect trace compared exactly against EffectManifest.
- Postconditions checked by a verifier distinct from the adapter that proposed the result.
- Sandbox timeout or crash yields a typed fault; it is not evidence of a system catastrophe.

9.3 Replay rule

For the same CompilationManifest and canonical DirectiveEnvelope, conforming implementations must reproduce the same AST digest, resolved registry records, operation sequence, static-analysis result, effect manifest, rollback plan, sandbox output digest, package digest, and terminal compilation outcome. Performance telemetry may differ only when explicitly excluded from canonical artifacts.

10. Authorization, Target Execution, Rollback, and Receipts

10.1 EXEC_DIRECTIVE_BOUND semantics

EXEC_DIRECTIVE_BOUND is retained as a legacy-visible label for PLAN_READY plus independent package verification. It means the directive package is complete and bound to exact evidence, registry, policy, target, effects, and rollback data. It does not grant execution authority and does not permit a kernel write or privilege escalation.

10.2 External authorization

- Permit ID, issuer, signature, issued-at and expiry cycle.
- Exact package digest and target-state snapshot digest.
- Exact granted capability and effect sets; no wildcard authority.
- Target owner and target adapter version.
- Single-use consumption token and expected state version.
- Review class, required countersignatures, and rollback requirement.

10.3 Target-owned two-phase execution

1. Target owner verifies permit, package, policy, registry, target snapshot, and unconsumed token.
2. PREPARE: target creates an isolated prepared state and validates forward preconditions.
3. COMMIT: target uses compare-and-swap from expected state version to prepared state.
4. READ-BACK: independent target verifier recomputes state digest and postconditions.
5. RECEIPT: target emits ExecutionReceipt and atomically consumes the permit.
6. ROLLBACK: on postcondition failure, the target executes the registered inverse and emits RollbackReceipt; Module 18 does not perform it.

10.4 Time-of-check/time-of-use protection

The permit is bound to the exact target-state digest and expected version. Any intervening state change invalidates the prepare or commit and requires recompilation or reauthorization. A stale package cannot be silently rebased.

11. Archive, Event, Recovery, and Cross-Module Boundaries

11.1 Module 17 evidence boundary

A Module 17 proof certificate may justify a precondition or artifact claim only when the exact claim and certificate digests are imported into BuildDirectiveEvidencePacket. The certificate does not authorize execution. Module 18 independently validates syntax, registry binding, capabilities, effects, policy, target ownership, resources, rollback, and package verification.

11.2 Psi and RMX boundary

Module 18 emits ArtifactArchiveRequest for the directive package, verifier record, authorization decision, and execution/rollback receipts. Psi owns address allocation, RMX resolution, archive commit, and archive receipts. Archive failure does not change compile, authorization, or target-execution status.

11.3 Module 12 event boundary

Compilation, verification, authorization, and rejected directives do not increment persistent Phi. After a target owner has durably committed and Psi has accepted the corresponding receipt, the target may emit one Module18ChangeCompletionEventReceipt. Module 12 decides whether and how to record it exactly once.

11.4 Guardian and network boundary

Remote directive ingress and remote artifact egress require Guardian authorization. QGM/GlyphNet and the host network stack transport opaque, authenticated packets only. Module 18 does not choose routes, endpoints, IPv4 addresses, or sockets.

11.5 ROVER, FBS, and RTSOM boundary

Stack overflow, parse rejection, type failure, policy denial, sandbox crash, resource exhaustion, and target contention are ordinary outcomes. Module 18 may report a catastrophic incident only for persisted control-plane corruption such as conflicting signed registry roots or contradictory committed receipts. ROVER decides whether to issue an FBS activation permit; FBS executes recovery; RTSOM accepts only registered, receipted recovery residue.

12. Security Model and Registry Governance

12.1 Threat model

Threat	Control
Prompt/directive injection	Only canonical DSL/AST enters the authoritative compiler; free-form source never executes
Hash-to-semantics confusion	Digest is identity only; exact opcode lookup is mandatory
Registry poisoning	Signed monotonic snapshots, publisher allow-list, revocation checks, offline validation, atomic epoch swap
Confused deputy	Capabilities and effects bound to target owner and external permit
Privilege escalation	No wildcard effects; exact <code>C_required</code> = <code>C_granted</code> at execution
TOCTOU	Permit binds target state digest/version; target uses compare-and-swap
Replay	Nonce, issuer sequence, expiry, parent digest, and single-use token ledger
Compiler self-modification	Ordinary directives cannot target active compiler/registry/trust roots
Dependency substitution	Every artifact, schema, evidence packet, and adapter is digest-bound
Nondeterminism	Frozen environment and repeated sandbox/replay checks
Rollback gap	Mutating operation requires registered inverse or immutable-additive proof
Receipt forgery	Signed target receipt plus package, permit, state, and sequence reconciliation

12.2 Registry publication

1. Validate candidate registry offline, including schemas, ownership, capabilities, effects, costs, pre/postconditions, and rollback definitions.
2. Assign a monotonic registry version and compute the canonical root digest.
3. Sign with an authorized Registry Authority key and publish revocation metadata.
4. Activate through an atomic epoch swap; in-flight compilations continue using their frozen snapshot.
5. Retain the prior registry and migration rules for deterministic replay and rollback.

12.3 Key and trust governance

Module 18 assumes an external key-management and identity system. Key creation, rotation, revocation, threshold signatures, operator identity, and emergency trust-root recovery are outside Module 18. A cryptographic signature is accepted only after trust-chain, purpose, time, revocation, and algorithm-policy validation.

13. Outcomes, Invariants, Faults, and Catastrophic Incidents

13.1 Normative invariants

Invariant	Requirement
M18-I01	Every compilation has one frozen CompilationManifest before parsing.
M18-I02	Identical canonical inputs and manifest produce identical compilation artifacts and outcome.
M18-I03	Free-form text is never executed directly.
M18-I04	Opcode resolution uses exact registered identifiers, never digest similarity or modulo mapping.
M18-I05	SHA-256 binds canonical bytes but never supplies semantic or authorization authority.
M18-I06	Every mutating operation has exact target ownership, capability, effect, precondition, postcondition, and rollback metadata.
M18-I07	Declared and inferred effects are identical for PLAN_READY packages.
M18-I08	No average score overrides a failed mandatory obligation.
M18-I09	MAX_STACK_DEPTH = 8 is enforced and does not replace fuel/dependency termination checks.
M18-I10	Compilation and package verification do not mutate the target state.
M18-I11	Module 18 cannot issue its own ChangeExecutionPermit.
M18-I12	A permit is single-use and bound to exact package, target, state version, capabilities, effects, and expiry.
M18-I13	Only the target owner may prepare, commit, verify, or roll back target state.
M18-I14	The active compiler, registry, trust roots, and Change Control Authority are not ordinary directive targets.
M18-I15	Archive status is independent of compilation and execution status.
M18-I16	Persistent Phi changes only through a Module 12-owned, exactly-once event transaction after committed change.
M18-I17	Ordinary rejection, denial, limits, timeout, or stack overflow never triggers FBS.
M18-I18	Module 18 never mutates QGM/GlyphNet, network, Psi, RMX, RTSOM, or FBS state.
M18-I19	Rejected evidence and packages are retained or receipted according to policy; no silent deletion.
M18-I20	Every terminal outcome emits a canonical result record with a no-target-mutation assertion.

13.2 Typed outcomes and faults

Code / outcome	Meaning
PLAN_READY	Package complete and independently verified; eligible for external authorization
AUTH_REQUESTED	ChangeAuthorizationRequest emitted
COMPILE_REJECTED	A hard semantic, registry, type, capability, effect, precondition, or rollback check failed
HELD_RESOURCE_LIMIT	Declared resource limit reached with evidence retained
HELD_STALE_TARGET	Target snapshot is stale or unavailable
HELD_AUTHORITY_UNAVAILABLE	Independent authorization service unavailable
M18_PARSE_ERROR	Canonical source does not parse under declared grammar

Code / outcome	Meaning
M18_SCHEMA_ERROR	Operand or output schema mismatch
M18_REGISTRY_MISSING	Opcode, schema, adapter, or extension absent
M18_REGISTRY_REVOKED	Registry item or publisher revoked
M18_TYPE_ERROR	Static type mismatch
M18_CAPABILITY_ERROR	Required capability exceeds request or policy ceiling
M18_EFFECT_MISMATCH	Declared and inferred effects differ
M18_PRECONDITION_FAILED	Frozen target does not satisfy required precondition
M18_DEPENDENCY_CYCLE	Unregistered cycle or missing well-founded measure
M18_STACK_LIMIT	Nested directive depth exceeds 8
M18_FUEL_LIMIT	Deterministic fuel exhausted
M18_ROLLBACK_INVALID	Mutating operation lacks valid rollback
M18_SANDBOX_FAULT	Sandbox crash, timeout, or prohibited effect
M18_NONDETERMINISTIC	Replay produced different canonical output
M18_PACKAGE_VERIFY_FAILED	Independent package verifier rejected
M18_REPLAY	Nonce, sequence, or permit token already consumed
M18_SIGNATURE_INVALID	Issuer, registry, evidence, permit, or receipt signature invalid
M18_RECEIPT_MISMATCH	Execution/rollback receipt does not reconcile with package and permit

13.3 Catastrophic incidents

Incident	Trigger
M18_CATASTROPHIC_REGISTRY_SPLIT	Two different valid signed registry roots claim the same active epoch
M18_CATASTROPHIC_POLICY_SPLIT	Two different valid policy roots claim the same active epoch
M18_CATASTROPHIC_DOUBLE_COMMIT	Conflicting durable target receipts claim the same package/permit token
M18_CATASTROPHIC_PERMIT_FORGERY	A forged permit was accepted by the durable target execution path
M18_CATASTROPHIC_LEDGER_CORRUPTION	Acknowledged compilation or consumption ledger cannot be replayed
M18_CATASTROPHIC_SANDBOX_ESCAPE	Directive adapter escaped the declared isolation boundary

For a catastrophic incident, Module 18 freezes affected work, emits `CatastrophicIncidentPacket` with no-mutation evidence, and notifies ROVER. It does not execute FBS, Cantor, Collatz, reseeding, rollback, or RTSOM transfer.

14. Metrics, Dashboard, and Reconciliation

Dashboard group	Required measurements
Throughput	envelopes_received, compilations_started, packages_ready, verifications_completed, auth_requests

Dashboard group	Required measurements
Outcomes	rejected, held, faulted, incident_reported by exact code
Parsing	bytes, AST nodes, parse latency, ambiguous/unknown field counts
Registry	version, publisher, resolved opcodes, missing/revoked entries
Capabilities/effects	required/requested/policy/granted counts and mismatches
Resources	stack depth, dependency depth, operations, fuel, sandbox CPU/memory/bytes
Telemetry	Φ_{ins} distribution and legacy hash-similarity distribution, clearly non-authoritative
Sandbox	required, completed, deterministic, prohibited-effect, timeout, crash counts
Authorization	requested, granted, denied, expired, consumed, stale-target
Execution	prepared, committed, verified, rolled back, receipt mismatch
Security	replay, signature, registry split, policy split, sandbox escape
Versions	compiler, parser, serializer, registry, policy, verifier, sandbox image, target adapter

14.1 Reconciliation equations

$$\begin{aligned}
 \text{received} &= \text{completed} + \text{in_flight} \\
 \text{completed} &= \text{plan_ready} + \text{rejected} + \text{held} + \text{faulted} + \text{incident_reported} \\
 \text{plan_ready} &= \text{auth_requested} + \text{compile_only_completed} \\
 \text{auth_requested} &= \text{granted} + \text{denied} + \text{expired} + \text{auth_pending} \\
 \text{granted} &= \text{permit_consumed} + \text{permit_unconsumed} + \text{permit_revoked} \\
 \text{permit_consumed} &= \text{committed} + \text{rolled_back} + \text{execution_failed_after_consumption} \\
 \text{sandbox_required} &= \text{sandbox_completed} + \text{sandbox_pending} + \text{sandbox_faulted}
 \end{aligned}$$

Dashboards must distinguish compile success, package verification, authorization, target prepare, target commit, postcondition verification, rollback, archive acceptance, and Module 12 event consumption. These are not interchangeable success counters.

15. Conformance, Property, Fuzz, Concurrency, and Stress Tests

15.1 Deterministic conformance suite - 60 tests

Test	Expected result
M18-T01	Reject invalid envelope signature.
M18-T02	Reject expired envelope.
M18-T03	Reject duplicate canonical map keys.
M18-T04	Normalize permitted Unicode to NFC and preserve canonical digest.
M18-T05	Reject unknown grammar version.
M18-T06	Reject host-language escape token.
M18-T07	Reject directive larger than MAX_DIRECTIVE_BYTES.

Test	Expected result
M18-T08	Identical semantic envelope serializes to identical bytes.
M18-T09	Different effect manifest changes h_dir.
M18-T10	Repeated identical submission is idempotent.
M18-T11	Exact opcode ID/version resolves registered record.
M18-T12	Unknown opcode is rejected; no modulo fallback.
M18-T13	Revoked opcode is rejected.
M18-T14	Registry order does not change resolution.
M18-T15	Registry signature failure is rejected.
M18-T16	Operand type mismatch is rejected.
M18-T17	Missing required operand is rejected.
M18-T18	Unknown operand is rejected.
M18-T19	Target-owner mismatch is rejected.
M18-T20	Active compiler/registry target is rejected under ordinary profile.
M18-T21	Required capability subset passes.
M18-T22	Capability above policy ceiling is rejected.
M18-T23	Wildcard capability is rejected.
M18-T24	Declared and inferred effects match.
M18-T25	Hidden write effect is rejected.
M18-T26	Network effect without registered capability is rejected.
M18-T27	Archive write effect is routed to Psi request, not direct mutation.
M18-T28	Precondition true on frozen target passes.
M18-T29	Precondition false produces M18_PRECONDITION_FAILED.
M18-T30	Stale target version produces HELD_STALE_TARGET.
M18-T31	Stack depth 8 is accepted.
M18-T32	Stack depth 9 produces M18_STACK_LIMIT without FBS.
M18-T33	Fuel reaches exactly zero after final valid step and passes.
M18-T34	Fuel underflow produces M18_FUEL_LIMIT.
M18-T35	Acyclic dependency graph topologically sorts deterministically.
M18-T36	Unregistered cycle produces exact cycle witness.
M18-T37	Registered well-founded recursion requires decreasing measure.
M18-T38	AST node limit boundary is exact.

Test	Expected result
M18-T39	Operation count limit boundary is exact.
M18-T40	Artifact dependency depth limit boundary is exact.
M18-T41	Mutating opcode with valid inverse passes rollback check.
M18-T42	Mutating opcode without rollback is rejected.
M18-T43	Immutable-additive operation passes only with registered proof.
M18-T44	Sandbox observed effects exactly match declared effects.
M18-T45	Sandbox prohibited socket attempt produces M18_SANDBOX_FAULT.
M18-T46	Two sandbox runs produce identical canonical outputs.
M18-T47	Nondeterministic adapter produces M18_NONDETERMINISTIC.
M18-T48	Independent verifier reproduces h_plan.
M18-T49	Tampered package is rejected by verifier.
M18-T50	Φ_{ins} below 0.80 does not independently determine outcome.
M18-T51	Φ_{ins} above 0.80 cannot override one failed hard check.
M18-T52	PLAN_READY reference package has all hard flags true.
M18-T53	Module 17 certificate is evidence only and cannot create permit.
M18-T54	ChangeAuthorizationRequest contains exact package/capability/effect bindings.
M18-T55	Module 18 rejects self-issued ChangeExecutionPermit.
M18-T56	Target CAS rejects stale expected version.
M18-T57	Duplicate permit consumption is rejected idempotently.
M18-T58	Postcondition failure yields target-owned rollback receipt.
M18-T59	Archive failure leaves compile and target status unchanged.
M18-T60	Ordinary stack/resource/sandbox failure emits no FBS or RTSOM request.

15.2 Property and metamorphic tests

- Canonicalization idempotency: $\text{canonical}(\text{canonical}(x)) = \text{canonical}(x)$.
- Permutation of registry storage order does not change resolution or package digest.
- Adding an unused registry entry does not change an existing package.
- Narrowing policy capabilities cannot turn a rejected privilege request into an accepted one.
- Adding a mandatory failed obligation cannot increase PLAN_READY or authorization eligibility.
- Changing any target, effect, capability, evidence, rollback, policy, registry, or adapter field changes h_plan.
- Compile-only and verify-package profiles agree on every canonical package field.
- Repeated deterministic sandbox execution yields the same observed-effect and output digests.

15.3 Fuzz, concurrency, crash, and stress requirements

- Fuzz parsers with malformed UTF-8, Unicode confusables, duplicate keys, deeply nested arrays, huge integers, invalid tags, and embedded NUL bytes.
- Fuzz schemas, registry references, capability/effect combinations, dependency graphs, rollback definitions, signatures, and receipt bindings.

- Run concurrent compilations during registry and policy epoch publication; each run must remain on one frozen snapshot.
- Crash after manifest freeze, package creation, verifier acceptance, authorization request, target prepare, target commit, receipt publication, and archive request; recovery must be idempotent.
- Execute 100,000 compilation attempts with mixed valid, invalid, stale, replayed, resource-limited, and sandboxed directives while reconciling every dashboard equation.
- Perform differential tests across at least two independent canonical serializers/package verifiers for all reference fixtures.

16. Reference Pseudocode and ASCII System Flow

16.1 Compile-only pseudocode

```
function compile_directive(envelope, context):
    manifest = freeze_manifest(
        compiler=context.compiler,
        parser=context.parser,
        serializer=context.serializer,
        registry=context.registry_snapshot,
        policy=context.policy_snapshot,
        target=context.target_snapshot,
        evidence=context.evidence_snapshots,
        limits=context.resource_limits,
        sandbox=context.sandbox_image,
        verifier=context.package_verifier
    )

    require verify_envelope_signature(envelope)
    require not_expired(envelope)
    require replay_guard_accepts(envelope.issuer, envelope.nonce)

    canonical_bytes = canonicalize(envelope)
    h_dir = sha256(canonical_bytes)
    ast = parse_typed_directive(canonical_bytes, manifest.parser)

    resolved = resolve_exact_registry_ids(ast, manifest.registry)
    require type_check(resolved)
    required_caps = infer_capabilities(resolved)
    inferred_effects = infer_effects(resolved, manifest.target)
    require required_caps <= envelope.requested_capabilities
    require envelope.requested_capabilities <= manifest.policy.capability_ceiling
    require inferred_effects == envelope.declared_effects
    require check_preconditions(resolved, manifest.target)

    graph = build_dependency_graph(resolved)
    require bounded_and_terminating(graph, ast, manifest.limits)
    rollback = build_and_verify_rollback(resolved, manifest.registry)

    static_record = make_static_analysis_record(...)
    sandbox_evidence = run_required_deterministic_sandbox(
        resolved, manifest.target, inferred_effects, manifest.sandbox
    )

    plan = build_directive_plan(
        manifest, h_dir, ast, resolved, required_caps,
        inferred_effects, static_record, rollback, sandbox_evidence
    )
    plan.h_plan = sha256(canonical(plan without h_plan))

    verification = independent_verify(plan, manifest.package_verifier)
    require verification.accepted

    return PLAN_READY(plan, verification, no_target_mutation=true)
```

16.2 Prepare-change pseudocode

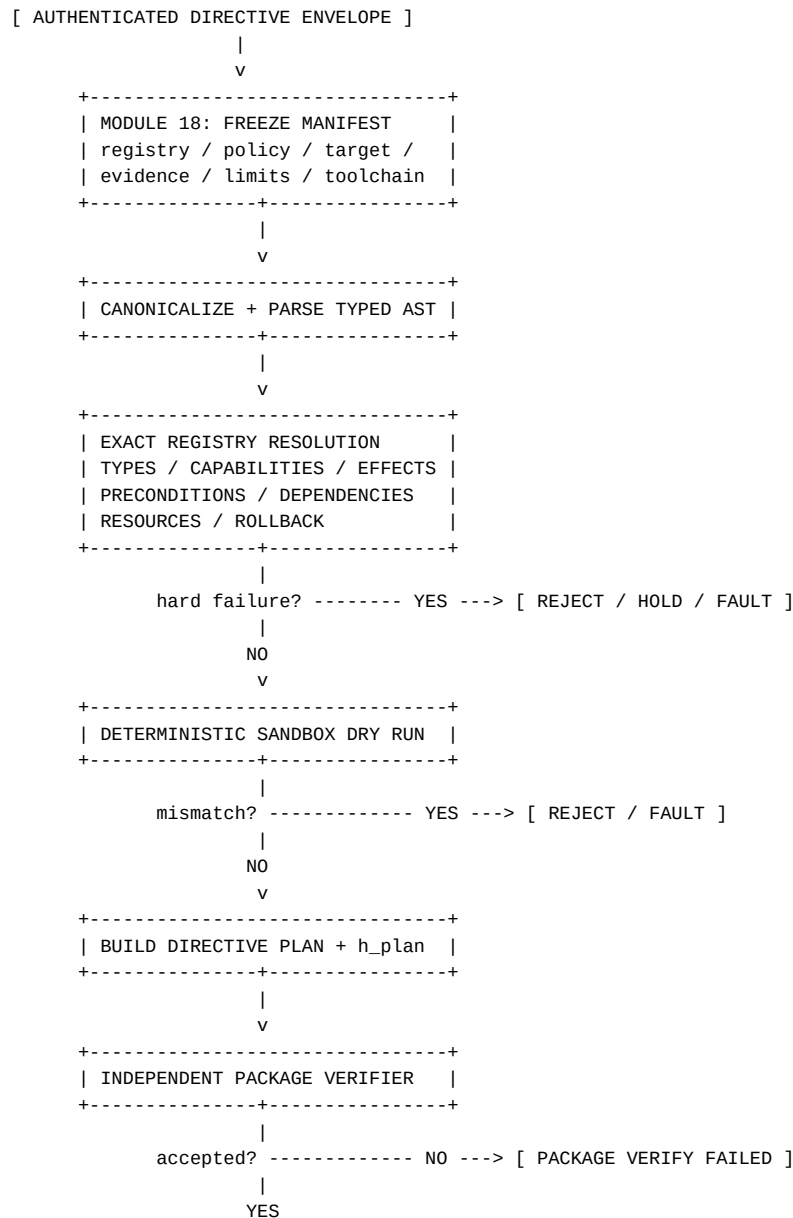
```

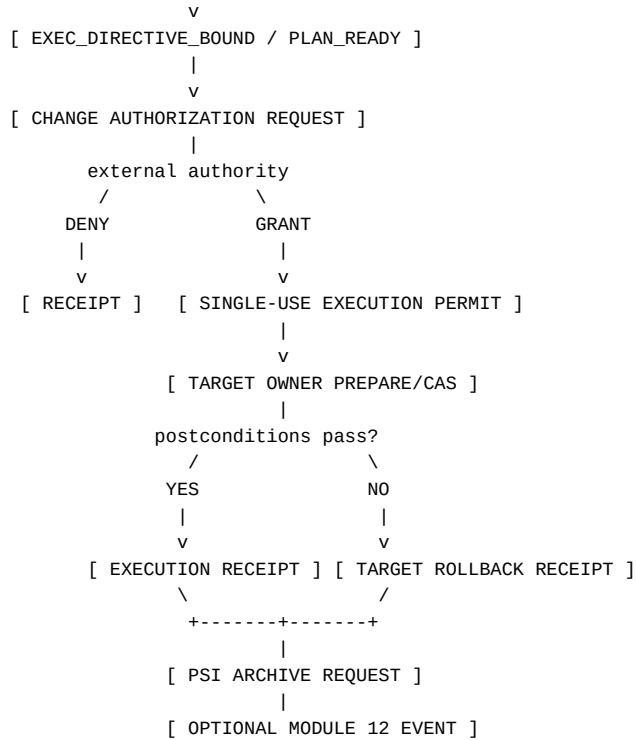
function prepare_change(plan_ready, authority):
  require plan_ready.verification.accepted
  request = ChangeAuthorizationRequest(
    package_digest=plan_ready.plan.h_plan,
    target=plan_ready.plan.target,
    requested_capabilities=plan_ready.plan.required_capabilities,
    requested_effects=plan_ready.plan.effect_manifest,
    target_state_digest=plan_ready.plan.target_snapshot.digest,
    expiry=bounded_expiry()
  )
  return authority.review(request) # external owner; Module 18 cannot grant

function target_execute(plan, permit, target_owner):
  require target_owner.verify_single_use_permit(permit, plan)
  prepared = target_owner.prepare(plan)
  committed = target_owner.compare_and_swap(prepared)
  checked = target_owner.read_back_and_verify(committed, plan.postconditions)
  if checked:
    return target_owner.execution_receipt(committed, permit)
  return target_owner.rollback_and_receipt(plan.rollback, permit)

```

16.3 ASCII architecture flow





Catastrophic control corruption only:

[MODULE 18 INCIDENT] -> [ROVER] -> optional FBS permit -> [FBS]

17. Rosetta Stone, Glossary, Revision Ledger, and Checklist

17.1 Cleaned Rosetta Stone

Project term	Formal meaning	Boundary
Gödel-Gödel Instruction Set	Typed directive compiler and change-plan builder	Not hardware microcode and not a proof of completeness
AI Build Directive	Authenticated declarative change proposal	Not executable free-form language
Microcode registry	Signed versioned opcode/schema/capability/effect registry	Not a vector chosen by hash similarity
H_dir	SHA-256 content digest of canonical directive bytes	Identity/integrity only; not semantics or uniqueness proof
Φ_ins	Diagnostic hard-obligation coverage ratio	Not permission, safety, truth, or consciousness
INSTRUCTION_FLOOR 0.80	Legacy migration telemetry threshold	Never overrides mandatory checks
MAX_STACK_DEPTH 8	Reference nested-call cap	Does not by itself prove termination
EXEC_DIRECTIVE_BOUND	Verified immutable package eligible for external review	Not a kernel write or privilege grant
E_LOCK_SIGIL	Legacy label for digest-bound package seal	No cryptographic or execution authority by itself
INDETERMINATE_LOOP_SHUNT	Typed reject/hold with evidence retained	No stack flush, deletion, or automatic FBS
FSB runaway	Corrected to ROVER-authorized FBS recovery outside Module 18	Not triggered by ordinary compiler errors

17.2 Glossary

Term	Definition
Capability	Named authority required to perform a declared class of effect.
Change Control Authority	Independent owner that grants or denies a package-bound execution permit.
CompilationManifest	Frozen record of toolchain, registry, policy, target, evidence, limits, and sandbox inputs.
DirectiveAST	Typed canonical syntax tree containing only registered declarative operations.
DirectivePlanPacket	Non-executable immutable plan containing operations, effects, pre/postconditions, rollback, snapshots, and digests.
EffectManifest	Exact declared set of reads, writes, creates, deletes, routes, network, privilege, and external effects.
OpcodeRecord	Signed registry entry defining one operation, operands, ownership, capabilities, effects, costs, and rollback.
Package verifier	Independent implementation that replays compilation evidence and validates h_plan.
Permit	Single-use external authorization bound to one package, target, state version, capability/effect set, and expiry.
RegistrySnapshot	Signed immutable view of allowed opcodes, schemas, adapters, and governance metadata.
Rollback	Target-owned inverse or registered recovery operation for a committed change.
Target owner	Module or service that exclusively owns the state being prepared, committed, verified, and rolled back.

17.3 Revision ledger

Source behavior	Formalized correction
Free-form directive string compiled into core	Replaced by declarative DSL/typed AST and non-executable DirectivePlanPacket
SHA-256 maps directive to registry	Digest binds canonical bytes; opcode resolution uses exact registered IDs/versions
“Un-aliased, non-colliding” hash mapping	Corrected to collision-resistant content addressing with explicit collision handling assumptions
Cosine digest/registry similarity authorizes execution	Removed from authority; Φ_{ins} is obligation-coverage telemetry only
INSTRUCTION_FLOOR = 0.80 pass gate	Retained as migration telemetry; all mandatory flags must pass
MAX_STACK_DEPTH guarantees termination	Added fuel, operation/node limits, dependency DAG, and well-founded recursion rules
PASS injects V_dir into core registers	Module 18 emits a plan; independent authority permits; target owner executes through prepare/CAS/read-back
E_LOCK_SIGIL confirms safety	Reframed as package digest seal with no independent authority
FAIL flushes registry stack	Replaced by typed reject/hold; active registry is immutable and evidence is retained
Stack overflow triggers direct FSB	Corrected FSB to FBS; ordinary overflow is a fault, catastrophic incidents go to ROVER
Direct Cantor/Collatz reseed	Removed from Module 18; FBS owns authorized recovery execution
Direct .frac serialization	Replaced by ArtifactArchiveRequest; Psi/RMX own archive addressing and commit
Proof output automatically precedes self-modification	Module 17 evidence is optional input only and cannot authorize a change
Privilege escalation approved analogue	Replaced by least-privilege, exact capability/effect permit semantics

17.4 Implementation checklist

- [] Implement the registered declarative DSL and reject all host-language execution paths.
- [] Freeze compiler, parser, serializer, registry, policy, target, evidence, limits, sandbox, and verifier before parsing.
- [] Use exact opcode/version lookup; never map hash bytes to semantic operations.
- [] Implement canonical serialization, h_dir, h_plan, signature validation, nonce/sequence replay protection, and single-use permit tracking.
- [] Enforce types, target ownership, capabilities, effects, preconditions, dependencies, resource bounds, deterministic behavior, and rollback.
- [] Keep Φ_{ins} and legacy hash similarity diagnostic only.
- [] Run required operations in an isolated deterministic sandbox and compare observed versus declared effects.
- [] Use an independent package verifier and an external Change Control Authority.
- [] Require target-owned prepare, compare-and-swap, read-back verification, receipts, and rollback.
- [] Keep Psi/RMX archive, Module 12 events, Guardian, QGM/GlyphNet, ROVER/FBS recovery, and RTSOM custody outside Module 18 ownership.
- [] Run all 60 conformance tests plus property, fuzz, concurrency, crash, and stress suites.
- [] Render, inspect, accessibility-audit, and independently verify the final DOCX/PDF artifacts.

Source Basis

Primary source: SPIRALCORE SPECIFICATION: MODULE 18 - GÖDEL-GÖDEL INSTRUCTION SET (AI BUILD DIRECTIVES). Architecture context: consolidated SpiralCore v14 and the formalized v14.1 ownership boundaries established through Modules 0-17, especially Module 17 evidence-only output, Psi/RMX archive ownership, Module 12 event ownership, Module 15 ROVER recovery authorization, Module 16 FBS execution, and Module 14 RTSOM custody.